

**SCOTT–VOGELIUS–NITSCHKE ON A POLYGONALLY APPROXIMATED
BOUNDARY: CONVERGENCE, THE MISSING PRESSURE TRACTION,
AND THE PENALTY THRESHOLD**

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ABSTRACT. We analyse the Scott–Vogelius–Nitsche method of Gjerde and Scott for two-dimensional Stokes flow whose no-slip curve is replaced by an inscribed polygon: continuous P_k velocities that are exactly divergence-free, with Dirichlet data imposed weakly on the polygon by Nitsche’s method. This answers Scott’s zero-gradient prize question, which asks whether the error behaves like $h_\Gamma^{3/2} + h_\Omega^k$ and, if not, why not.

For the method as printed, the answer is no as soon as the pressure is not constant on the boundary. The printed Nitsche form omits the traction $-pn$. We prove that the method converges, at the sharp rate h/μ in the H^1 seminorm and at the exact rate $(h/\mu)^{1/2}$ in the mesh-dependent energy norm. To leading order the error is an explicit field: the discrete velocity leaks through the wall with normal velocity $(h/\mu)(p - \bar{p})$, with constant exactly 1 in the slip and energy norms. The H^1 rate is 1 and not 1/2 only because incompressibility turns the normal part of the missing traction into a tangential derivative. For fixed γ and bounded ρ , (μ/h) times the error converges in H^1 to the Stokes flow driven by the leak as Dirichlet data; the resulting constant depends on the domain (2.7565 for our test, tending to $3/\sqrt{7}$ as the outer box grows). A penalty growing like $\mu \gtrsim h h_\Gamma^{-3/2}$ restores the H^1 rate 3/2 of the printed method, but no power-law scaling of the penalty lifts its energy rate above 1. When the wall pressure varies, its pressure converges in L^2 only at rate 1/2, because of an explicit boundary layer in the first row of triangles, and its drag error is, to leading order, $-(h/\mu)$ times an explicit reciprocity constant.

Adding the plain pressure term makes the square system singular. Subtracting the wall mean of the discrete pressure gives a well-posed, exactly divergence-free method; this mean-free correction is not new, since its velocity coincides with that of the zero-net-flux formulation of Frachon, Nilsson and Zahedi, and what we contribute is its analysis. We prove that it attains $h_\Gamma^{3/2} + h_\Omega^k$ for general Stokes data, for every fixed $k \geq 4$ on meshes satisfying (M0)–(M2) and every $k \geq 2$ on Clough–Tocher refinements. With flux-corrected outer data the H^1 bound holds uniformly in the penalty and with no relation between h_Γ and h_Ω ; its pressure then converges at the same rate and its drag at rate 2. Whenever the wall shear is not identically zero, we prove a matching lower bound $ch_\Gamma^{3/2}$ for the H^1 error, so the $h_\Gamma^{3/2}$ term of Scott’s rate is sharp. The lower bound holds for every $k \geq 4$; the local star condition it needs is proved with certified constants. In L^2 the method converges at the sharp rate h_Γ^2 , while the printed method has L^2 rate exactly 1. The leak limit of the printed method holds uniformly in the penalty at the exact rate $h_\Gamma^{1/2}$.

The Nitsche penalty must scale like $\rho = h_\Omega / \min_e |e|$. Sufficiency is proved on every mesh. Necessity is certified in exact rational arithmetic for every mesh whose shortest boundary edge carries a triangle with apex angle at least 35° and base angles at least 5° , and is proved with explicit constants on every mesh that is locally fine enough near one boundary edge; on every shape-regular mesh the scaling is proved, with explicit but impractically small constants. A purely local necessity statement is false, by exact counterexamples. The convergence theory extends to general smooth curved boundaries in two dimensions and to steady Navier–Stokes on nonsingular solution branches, with the Oseen leak flow, K_{NS} and rate-2 drag for the corrected method.

We also analyse strongly imposed no-slip on the same polygon. Under our mesh condition it already attains $h_\Gamma^{3/2} + h_\Omega^k$, two-sided; the zero-gradient lock is caused by wall vertices in only two triangles, each costing $h_\Gamma |\partial_n u|$, or by nearly singular wall stars, where the cost follows a needle-angle law. The advantages of Nitsche imposition on an inscribed polygon are therefore robustness to such meshes and a smaller constant, not a better rate. Finally, the printed method is not pressure-robust, the corrected method has a single pressure defect of order $h_\Gamma^{k+1/2}/\nu$, and a local BDM $_k$ load correction on the boundary triangles makes it exactly gradient-robust. The three-dimensional analysis is given in a companion paper [50].

A computational study up to $N = 256$ boundary chords, for $k = 4, 5, 6$ and for Clough–Tocher refinements with $k = 2, 3$, confirms the rates and constants. It includes a pure-pressure test that isolates the traction response exactly, pressure errors, a penalty study in which the threshold is linear in h_Ω/h_Γ , with $\mu^* h_\Gamma/h_\Omega \approx 20$ for $k = 4$, strong imposition with and without locked vertices, and a hydrostatic pressure-robustness test.

1. INTRODUCTION

Background. *Exactly divergence-free elements.* Scott and Vogelius [54] showed that continuous piecewise- P_k velocities with discontinuous piecewise- P_{k-1} pressures satisfy $\operatorname{div} V_h = \Pi_h$ for $k \geq 4$ on meshes without singular vertices; the inf-sup constant was later made explicit in terms of the local mesh geometry [37]. The discrete velocity is then divergence-free pointwise, not only weakly. This gives exact mass conservation. On fitted meshes with strongly imposed no-slip data it also gives pressure robustness: the velocity error does not depend on the pressure, which matters when pressure gradients are large or the viscosity is small [38]. Split meshes give similar pairs of lower degree [36]; see [47] for a review.

Curved walls. Exact incompressibility has a cost at a curved wall. A divergence-free field that is smooth at a polygon vertex and vanishes on both adjacent edges has zero gradient there, and piecewise-polynomial divergence-free fields inherit such gradient constraints at the vertices of a polygonal wall [53, 34]. If the wall is replaced by an inscribed polygon and no-slip is imposed strongly, the computed gradient can therefore be off by $O(1)$ at the polygon vertices, and the H^1 error of Scott–Vogelius elements then decays only like $h^{1/2}$ [53, 34], compared with $h^{3/2}$ for scalar problems [9, 56]. We show in Section 11 that this *zero-gradient lock* is a property of the wall mesh, not of strong imposition as such. Under the mesh condition (M2) used throughout this paper, strong imposition already attains the error $h_\Gamma^{3/2} + h^k$, with matching lower bound (Theorem S). The lock is caused by wall vertices lying in only two triangles, which (M2) excludes: each such vertex costs $h_\Gamma |\partial_n u(z)|$ in H^1 , and a positive fraction of them gives the rate $\frac{1}{2}$. With three or more triangles the same loss appears continuously as one triangle at the wall vertex degenerates. Scott’s observation that there is no constraint for more than two triangles [53] is thus correct for the rate under (M2), and the strong-imposition meshes of [34] must have contained locked or nearly singular wall stars.

One remedy is to fit the geometry, with isoparametric or Piola-mapped elements [48, 27]. Gjerde and Scott [34] keep the polygon and impose the wall condition weakly by Nitsche’s method [49], with penalty μ/h and no pressure term, and report errors orders of magnitude below those of strong imposition. By the above, the advantages of the Nitsche approach on an inscribed polygon are robustness to locked and nearly singular wall stars, on which GS and CNS* are unaffected (Section 16.8), and a smaller error constant (about 2.5 times smaller in our tests); under (M2) the rate is the same. Scott observed H^1 errors of order $h^{3/2}$ for this Scott–Vogelius–Nitsche method on a benchmark with constant pressure [53]. His zero-gradient prize [53] asks for a published proof that Scott–Vogelius–Nitsche converges, including a computational study of the influence of the Nitsche penalty. It states the expected error $h_\Gamma^{3/2} + h_\Omega^k$ and adds: *if this does not hold, why not?* We treat the question for general Stokes data.

Answer. For the method as printed in [34, (27)–(28)] and bounded γ , the expected estimate holds if and only if the pressure is constant on the wall. The printed Nitsche form contains no traction term $-pn$. When p varies along Γ , the penalty, not a traction, balances the pressure. The discrete velocity then leaks through the wall with normal velocity $(h/\mu)(p - \bar{p})$, and the error is dominated by this leak (Theorems A–C, Corollary B’ and Theorem B’). The mean-free correction CNS* (cf. Frachon, Nilsson and Zahedi [29]), well posed for $\gamma \geq \gamma_2$, restores the expected rate (Theorem D), and its $h_\Gamma^{3/2}$ term cannot be improved whenever the wall shear is not identically zero (Theorem D’). On every mesh whose shortest boundary edge carries a triangle with apex angle at least 35° and base angles at least 5° , the penalty threshold must grow with the ratio of bulk to boundary mesh size (Proposition E).

Main results. The notation is fixed in Section 2. Throughout, $h = h_\Omega$ is the bulk mesh size used in Scott’s penalty μ/h , h_Γ is the length of the longest boundary edge, $\rho = h / \min_e |e|$, $\gamma = \mu h_\Gamma / h$

and $G = \|p - \bar{p}\|_{L^2(\Gamma)}$. Constants do not depend on h, h_Γ, μ, γ or ρ . The solution (u, p) is assumed smooth on $\bar{\Omega}$; near Γ this follows from smooth data, but at the right-angled corners of the outer box it is a hypothesis on the solution, and every h^k and h^{k+1} term below relies on it (Remark 2.5).

Theorem A (Theorem 6.1). *The printed method GS converges, and its error in the mesh-dependent energy norm is*

$$\|\tilde{u} - u_h\| \asymp (h/\mu)^{1/2} G + O((1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h),$$

with matching upper and lower bounds.

Theorem C (Theorem 6.4). $\|\nabla(\tilde{u} - u_h)\| \leq C_p h/\mu + C((1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h)$. The rate 1 (rather than 1/2) holds because $\operatorname{div} v = 0$ turns the normal load into a tangential derivative (Lemma 6.3).

Theorem B (Theorem 6.6). *In the regime where the pressure term dominates, the slip satisfies $\|\tilde{u} - u_h\|_{L^2(\Gamma_h)} \asymp (h/\mu)G$ and $\|\nabla(\tilde{u} - u_h)\| \asymp h/\mu$. So Theorem C is sharp.*

Corollary B' (Corollary 6.7). $\tilde{u} - u_h = (h/\mu)v_\varphi + o((h/\mu)^{1/2})$ in the energy norm, where v_φ is an explicit divergence-free field with $v_\varphi \approx -(p - \bar{p})n$ on Γ_h . In particular

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{(h/\mu)^{1/2}G} \rightarrow 1.$$

Theorem B'' (Theorem 6.10, Corollary 6.11). *Let U be the Stokes flow in Ω with Dirichlet data $-(p - \bar{p})n$ on Γ and 0 on ∂R . For fixed γ and bounded ρ , $(\mu/h)(\tilde{u} - u_h) \rightarrow U$ in H^1 , so*

$$\frac{\|\nabla(\tilde{u} - u_h)\|}{(h/\mu)G} \rightarrow K := \frac{\|\nabla U\|_{L^2(\Omega)}}{G}.$$

The constant K is not universal: it depends on the domain. For the pressure of our test P it is $K = 2.7565157$ (series computation, inside a rigorous bracket $[2.0746, 3.0526]$), and $K \rightarrow 3/\sqrt{7}$ as the outer box grows. In the regime $\gamma h_\Gamma \rightarrow \infty$ the leak constant still converges to K (Corollary 7.48), while for general data the normalised GS constant diverges.

Theorem D (Theorems 7.2, 7.30 and 7.33). *The naive consistent form, which adds $\langle p_h, v \cdot n \rangle$, is singular; its L_0^2 variant is square but only gives $\operatorname{div} u_h \equiv c$, with $c \neq 0$ in general (Proposition 7.1). The mean-free correction CNS^* , which adds $\langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$ to the momentum row only, is well posed for $\gamma \geq \gamma_2$ and keeps u_h exactly divergence-free; its velocity coincides with that of the zero-net-flux formulation of [29] (Remark 7.3). With the flux-corrected outer data $\tilde{g}_I$ of Definition 2.2, for general Stokes data, every fixed $k \geq 4$ and every $\gamma \geq \gamma_2$,*

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k),$$

with C independent of γ and with no relation between h_Γ and h_Ω . This is Scott's expected rate. With the plain interpolant g_I the same holds under the proviso that the boundary edges are not too short relative to the bulk ($h_\Gamma \geq h_\Omega^4$ for even k , $h_\Gamma \geq h_\Omega^2$ for odd k); without it a single lower-order term, due to the net flux of the interpolated outer data, remains. On Clough-Tocher refinements the estimate of Theorem 7.2 holds for every $k \geq 2$ (Theorem F).

Theorem D' (Theorem 7.24, Corollary 7.25). *Suppose the wall shear $W = \partial_n u \cdot \tau$ is not identically zero. For CNS^* , and for GS with any G ,*

$$\|\nabla(\tilde{u} - u_h)\| \geq c\|W\|_{L^2(\Gamma)} h_\Gamma^{3/2} - Ch_\Gamma^2,$$

with c independent of γ, μ, ρ and the pressure. So, whenever $\varepsilon_h \leq Ch_\Gamma^{3/2}$ and γ is bounded (or the data are flux-corrected and $h_\Omega^k \leq h_\Gamma^{3/2}$), the rate $h_\Gamma^{3/2}$ of Theorem D (and of Theorem A at $G = 0$) is exact; it is the $h_\Gamma^{3/2}$ term, not the h_Ω^k term, that is shown to be sharp. This holds for every

$k \geq 4$: the local star condition (M3) used in the proof is proved for $k \geq 4$ once $h_\Gamma \leq 2 \sin(\theta_{\min}/4)$ (Theorem 7.8), with certified constants.

Proposition E (Proposition 10.1, Theorems 10.2, 10.13, Corollary 10.4). *The condition $\mu \geq 4\kappa C_I^2 \rho$, $\rho = h / \min_{e \in \mathcal{E}_h^\Gamma} |e|$, is sufficient for coercivity on every mesh. On every mesh, coercivity requires $\mu \geq \lambda_T h / |e|$ for each triangle T with an edge e on Γ_h (and not touching ∂R), with an explicit λ_T depending only on the shape of T . In exact rational arithmetic, $\lambda_T > 1$ whenever the apex angle is at least 35° and $\lambda_T > 9.3$ when it is at least 60° (base angles at least 5°). So $\mu \gtrsim \rho$ is necessary on every mesh whose shortest boundary edge carries a triangle with apex angle at least 35° and base angles at least 5° . On arbitrary (M0)–(M2) meshes, necessity holds with moderate explicit constants as soon as the mesh is locally fine enough near one boundary edge (Theorem 10.9), and on every shape-regular mesh satisfying (M0), $\mu^* \geq c_*(\theta_{\min}, c_0)\rho$ for small h , with explicit but astronomically small c_* (Theorem 10.13): the scaling is proved, a usable threshold is not. A local version for tall boundary triangles is false: there are (M0)–(M2) stars on which no field supported in the star is a witness (Proposition 10.6); on a global mesh built from them the threshold still scales exactly like ρ (numerically).*

Theorem A with $G = 0$ gives $\|\tilde{u} - u_h\| \leq C((1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h)$. This recovers, for the printed method, the constant-pressure benchmark studied numerically in [34, 53] and analysed in [20], whose abstract states the bound $\|u - u_h\|_{H^1} \leq C(h_\Omega^k + h_\Gamma^{3/2})$ for that benchmark with quartic velocities. Our proof does not use [20]. Theorems A–D' hold for every fixed $k \geq 4$ on meshes satisfying (M0)–(M2). All rates assume a solution that is smooth up to the boundary; at the corners of the box this is a hypothesis on the solution, not on the data (Remark 2.5).

Further results.

Theorem R (Theorems 7.30, 7.35, Corollaries 7.34, 7.36, Proposition 7.37). *The H^1 error of the printed method is bounded uniformly in the penalty: $\|\nabla(\tilde{u} - u_h)\| \leq C_p h_\Gamma / \gamma + C(h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2})$ for every $\gamma \geq \gamma_0$. So a growing penalty $\mu \gtrsim h h_\Gamma^{-3/2}$ ($\mu \gtrsim h^{-1/2}$ when $h \asymp h_\Gamma$) restores the H^1 rate $\frac{3}{2}$ of GS for general data. It cannot restore the energy norm: if $W \not\equiv 0$ the energy error is at least $c\|W\|\gamma^{1/2} h_\Gamma^{3/2} - C(h_\Gamma/\gamma)^{1/2}$ for large γ , so under a power-law penalty $\gamma \asymp h_\Gamma^{-\alpha}$ the energy rate of GS is $\min(\frac{1+\alpha}{2}, \frac{3-\alpha}{2}) \leq 1$ when $G > 0$ and $W \not\equiv 0$. The condition number of the velocity block grows by the factor $1 + \gamma$, that of the pressure Schur complement by the factor $1 + \mu/h$.*

Theorem P (Section 8). *Let $p^\circ = \tilde{p} - \bar{p}_\Gamma(\tilde{p})$. For CNS* with flux-corrected data and $\gamma \in [\gamma_2, \gamma_{\max}]$, $\|p^\circ - p_h\|_{L^2(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$, including the constant, with C depending on $\gamma_{\max}$ (Theorem 8.7). For GS the pressure is uniquely determined, but carries an explicit first-layer field ξ_b produced by the symmetric Nitsche term acting on the leak; for fixed γ , bounded ρ and $G > 0$ the L^2 pressure error converges at the exact rate $\frac{1}{2}$, and not at all pointwise in the first layer (Theorem 8.11). The variational drag, lift and torque coincide with the integral of the Nitsche flux. For GS with fixed γ and bounded ρ their error is $-(h/\mu) \int_\Gamma (p - \bar{p})(R_\psi - \bar{R}_\psi) + O(h^{3/2})$, with R_ψ the pressure of a dual Stokes problem, so the rate is 1 (Theorem 8.17); for CNS* with $\gamma \geq \max(1, \gamma_2)$ and $\gamma h_\Gamma \leq 1$ it is $O(h_\Gamma^2 + \varepsilon_h)$ (Theorem 8.19). Under a hypothesis (P) on a boundary pressure moment the CNS* pressure error is also bounded below by $ch_\Gamma^{3/2}$ in L^2 and ch_Γ in L^∞ (Theorem 8.34). A periodic first-layer cell problem shows that (P) holds exactly when an explicit cell density is not constant along Γ , that it fails on mirror-symmetric first layers, and gives the L^2 pressure and traction lower bounds without (P); these statements are proved modulo a cell-decay hypothesis (numerical) and a transplant lemma (not proved) (Section 8.6.5).*

Theorem U (Section 7.3). *For the leak of the printed method the convergence $(\mu/h)(\tilde{u} - u_h) \rightarrow U$ of Theorem B'' holds uniformly in the penalty, at the exact rate $\frac{1}{2}: cGh_\Gamma^{1/2} \leq \|\nabla((\mu/h)\delta_R - \tilde{U})\| \leq C(h_\Gamma^{1/2} + h^k)$ for every $\gamma \geq \gamma_0$, the Dirichlet limit included (Theorems 7.42, 7.44).*

Theorem L (Section 9). *For CNS* with flux-corrected data, $\|\tilde{u} - u_h\|_{L^2} \leq C(h_\Gamma^2 + hh_\Gamma^{3/2} + h^{k+1})$ uniformly in γ ; the Stokes H^2 regularity this needs is proved from the Kellogg–Osborn theorem on the convex box and smooth-domain regularity at Γ (Theorem 9.1). On graded meshes the bound improves to $C(h_\Gamma^2 + h^{k+1})$ (Theorem 9.14). For bounded γ the h_Γ^2 is sharp when the wall shear is not identically zero, without any regularity assumption. For GS with $G > 0$ the L^2 rate is exactly 1, with leading term $(h/\mu)\|U\|_{L^2}$.*

Theorem S (Section 11). *Strongly imposed no-slip on the same inscribed polygon satisfies $ch_\Gamma^{3/2} \leq \|\nabla(\tilde{u} - u_h)\| \leq C(h_\Gamma^{3/2} + h^k)$ under (M0)–(M2). If (M2) fails only at the set $\mathcal{L}$ of wall vertices with two triangles, then $\|\nabla(\tilde{u} - u_h)\| \asymp h_\Gamma^{3/2} + h_\Gamma(\sum_{z \in \mathcal{L}} |\partial_n u(z)|^2)^{1/2}$; the divergence inf-sup constant is then $\asymp \min_{\mathcal{L}} \phi_z \asymp h_\Gamma$, but it is uniform on the pressures that are continuous at the locked vertices, and a cheap filter recovers a convergent pressure. At a wall vertex whose star contains a needle of angle ε_z at the wall, the local cost is $h_\Gamma \min(1, \phi_z/\sqrt{\varepsilon_z})|\partial_n u(z)|$, two-sided, under a mesh hypothesis (H_K) that isolates each needle pair (Theorem 11.22). Needles come in pairs, and a straight pair has inf-sup constant $\lesssim \varepsilon$; the velocity does not see this, because the approximation argument never inverts the divergence on a needle.*

Theorem T (Section 15). *GS is not pressure-robust: the pressure part of its velocity error at viscosity ν is the leak flow times $h/(\nu\mu)$. CNS* has exactly one pressure-dependent defect, the projection error $p - \pi_h p$ on Γ_h , of H^1 size $\asymp \nu^{-1}\gamma^{-1}h_\Gamma^{k+1/2}$ for generic pressures on locally periodic meshes (every $k \geq 4$; the lower bound holds for every $\gamma > 0$), and of higher order for pressures whose k -th derivatives lie in an explicit invisible plane. A local BDM $_k$ correction of the load on the boundary triangles (CNS*-R) makes CNS* exactly gradient-robust, without changing its matrix or its rates.*

Related work and what is new. For Stokes, the consistent Nitsche form uses the full traction $\sigma(u, p)n$. It therefore contains the pressure term $\langle p, v \cdot n \rangle$ [30, 57, 8, 16], and this term is recognised as essential for consistency [18]. The Gjerde–Scott form omits it. On the geometric side, boundary-value correction [13] has been combined with Nitsche’s method for cut elements [17, 18] and for polygonal approximations of scalar problems [26]; the shifted boundary method [45] is a related approach. None of these works uses exactly divergence-free spaces.

With such spaces, the placement of the pressure term decides whether exact incompressibility survives. If the term also appears in the continuity equation, $\operatorname{div} u_h$ vanishes only away from an $O(h)$ boundary strip [43]. If it appears only in the momentum equation, the constant pressure mode must be handled separately. Frachon, Nilsson and Zahedi [29] show that fixing the pressure mean by a Lagrange multiplier makes $\operatorname{div} u_h$ equal to that multiplier, and avoid this by testing the momentum equation only with velocities of zero net flux. Our CNS* is this device in square form: its velocity coincides with theirs (Remark 7.3), and its wall-mean subtraction fixes the pressure constant automatically. The work closest to ours is that of Liu, Neilan and Otus [44], for Scott–Vogelius elements on Clough–Tocher splits of a mesh inside a curved domain. They add a mean-free boundary multiplier that approximates the wall pressure up to a constant, together with a zero-flux constraint and a Taylor boundary correction. Their velocity is exactly divergence-free and converges at the optimal order h^k ; without the constraint, they note, the method is ill posed [44, Remark 3.2]. Related multiplier formulations for cut elements are given in [19]. Exactly divergence-free methods also require discrete Dirichlet data with zero net flux [28]; our flux-corrected $\tilde{g}_I$ is a simple instance.

Our subject is the method as Gjerde and Scott printed it, and its minimal repair: no extra unknowns, no boundary correction, and the pressure itself on the wall. We do not claim the mean-free device as new. As far as we know, what is new is the analysis:

- the two-sided leak analysis of the printed method (Theorems A–C, Corollary B'), including the identification of its H^1 constant (Theorem B'') and of its pressure and drag defects (Theorem P);
- the proof of Scott's conjectured rate $h_\Gamma^{3/2} + h_\Omega^k$ for the mean-free correction on an inscribed polygon (Theorem D), with the penalty-uniform H^1 bound, and the proof that its $h_\Gamma^{3/2}$ term is sharp (Theorem D');
- the penalty threshold $\propto \rho$: sufficiency is a standard consequence of the trace-inverse inequality [57, 39], while the explicit necessity certificates of Proposition E appear to be new;
- the extensions: Clough–Tocher refinements, general smooth domains and steady Navier–Stokes (Theorems F–H), and the growing-penalty and penalty-uniform analyses (Theorems R and U);
- the L^2 theory (Theorem L), the two-sided strong-imposition theory with locked and nearly singular wall vertices (Theorem S), and the quantitative pressure-robustness analysis (Theorem T).

For strong imposition, pressure constraints that restore a shape-robust inf-sup constant at nearly singular vertices are due to Gräßle, Bohne and Sauter [35], and spurious pressures with robust velocities and pressure filters to Park [51] and Bohne, Gräßle and Sauter [11]; our constrained inf-sup result is essentially the wall-vertex case of theirs. Scott–Vogelius elements on anisotropic meshes obtained by barycentric and incenter refinement are analysed by Kean, Neilan and Schneier [40]; our needle-pair results concern single needles attached to the wall. What is new there is the two-sided velocity theory: at locked wall vertices the loss is an approximation loss caused by the zero trace, not a stability loss. For pressure robustness on curved domains with exact boundary conditions, load reconstruction with commuting interpolants is due to Neilan and Otus [48, Theorem 6.1] in two dimensions and Chen and Liu [21] in three; Chen and Liu attain the optimal order h^k with isoparametric Piola-mapped elements, so the inscribed-polygon Nitsche route studied here is the low-cost straight-sided alternative, whose best rate is $h_\Gamma^{3/2} + h^k$, not a competitor on rate. That the pressure term is needed for consistency is not new [18]; the mechanism of the leak is the classical consistency error of boundary penalty methods [6], here acting only on the pressure and with vanishing penalty parameter h/μ , which is why the method converges at all.

Scope and related variants. We analyse the symmetric Nitsche form of [34]. The leak comes from the missing pressure term, not from the symmetry of the viscous terms, so we expect it in nonsymmetric variants as well; we have not analysed these, nor penalty-free variants [30, 15, 12]. Nitsche's method with Robin-type data is treated in [39]. Multiplier and boundary-correction methods [44, 19, 13, 26] reach higher order than $h_\Gamma^{3/2}$, at the price of extra unknowns or Taylor corrections. A common alternative restricts the pressure to L_0^2 and adds the plain term $\langle p_h, v \cdot n \rangle$, testing the continuity equation only with mean-zero pressures. Because $\operatorname{div} V_h = \Pi_h$, this gives only $\operatorname{div} u_h \equiv c$ with $c|\Omega_h| = \int_{\partial\Omega_h} u_h \cdot n$ (Proposition 7.1; compare [29, (5.9)]). The constant vanishes exactly when the naive system is solvable, equivalently when the CNS* pressure has zero mean on Γ_h . With weak imposition nothing forces the flux through Γ_h to vanish, even for flux-corrected data, and in our computations $c \neq 0$. Our proofs use Scott–Vogelius elements with $k \geq 4$, and $k \geq 2$ on Clough–Tocher splits, in two dimensions. The three-dimensional analysis (inscribed polyhedra, Alfeld splits) is in [50]. Other divergence-free families, and cut or unfitted

meshes, are not treated. Gjerde and Scott also treat slip conditions by Nitsche's method [33]; we have not examined that method.

Extensions.

Theorem F (Theorem 12.8, Proposition 12.10). *On the Clough–Tocher refinement of any shape-regular mesh whose boundary edges have comparable lengths, Theorems A–D, Corollary B' and the sufficiency part of Proposition E hold for every $k \geq 2$, with no angle or singular-vertex condition. For example, CNS* satisfies $\|\tilde{u} - u_h\|_{H^1} \leq C(h_\Gamma^{3/2} + h^2)$ for $k = 2$ if $h_\Gamma \geq h^4$. The penalty necessity is recertified exactly on split stars.*

Theorem G (Section 13). *In two dimensions, Theorems A–D and Corollary B' extend to a bounded connected domain whose curved boundary consists of finitely many closed smooth curves, convex or not, with a nonempty polygonal component Σ on which the data are imposed strongly. The geometric lemmas use only C^3 curves; the results assume the regularity (R1) of the solution, which for generic f requires $\Gamma \in C^{k+1,\alpha}$ (and vanishing corner coefficients at the vertices of Σ), and the results that use the leak test field assume $\Gamma \in C^{k+3}$. With several curved components, the correct pressure constant is the global boundary mean. CNS* needs one global correction, and a per-component correction is inconsistent. Stokes H^2 regularity with L^2 data holds on such a domain if and only if Σ is a convex outer polygon (Theorem 13.6).*

Theorem H (Section 14). *For steady Navier–Stokes on a nonsingular solution branch, the discrete linearisation is uniformly stable for $h \leq h_0$, with h_0 independent of the penalty (explicit, but far too small to be useful). Then GS converges at H^1 rate h/μ with the same leak (slip and energy constants 1) and an H^1 constant $K_{NS} \geq K$ given by an Oseen leak flow, and CNS* attains $h_\Gamma^{3/2} + h_\Omega^k$; the CNS* drag, lift and torque converge at rate 2. No inflow/outflow correction of the convective term is needed for these statements.*

Computations. Section 16 reports computations with up to $N = 256$ boundary chords, mostly for $k = 4$, with $k = 5, 6$ and Clough–Tocher $k = 2, 3$ in Section 16.11. They show the following.

- The CNS* H^1 rate decreases monotonically to $3/2$, independently of the pressure, for $k = 4, 5, 6$.
- The printed method has rate 1 whenever $G > 0$.
- On a pure-pressure test that isolates the traction response exactly, the slip and energy constants of Corollary B' approach 1 for every element tested. For $k = 4$ the H^1 constants approach the K of Theorem B''; that they do so also for $k = 5, 6$ and on Clough–Tocher refinements is observed, not proved.
- In the penalty study, the coercivity threshold μ^* is linear in h/h_Γ over more than a decade, with $\gamma^* = \mu^* h_\Gamma / h \approx 18\text{--}21$ for $k = 4$. It is larger for higher k and on Clough–Tocher refinements.
- Strongly imposed no-slip has rate $\frac{3}{2}$ on our production meshes (every wall vertex in three triangles), rate 1 with one locked vertex and rate $\frac{1}{2}$ with alternating locks, as Theorem S predicts, while GS and CNS* are insensitive to the change of mesh (Sections 16.8 and 11.7).
- The pressure errors follow Theorem P: rate $\rightarrow \frac{3}{2}$ for CNS*, and rate $\rightarrow \frac{1}{2}$ for GS when $G > 0$, carried by the first layer (Section 16.9).
- A growing penalty behaves as Theorem R predicts, and flux-corrected data remove the flux floor exactly (Section 16.10).
- On a hydrostatic test GS is first order, CNS* has order $k + \frac{1}{2}$, and CNS*-R is at round-off (Section 15).

Most runs use the penalty $\mu = 100$, which is only about 1.4 times the coercivity threshold of GS and 1.13–1.27 times the stability threshold of CNS* (a margin that shrinks to about 7% as $h \rightarrow 0$);

the proofs' thresholds are not verified there, so the computations illustrate the theorems rather than being covered by them (Section 16). All code, raw outputs and logs are in the accompanying repository (Appendix A).

Organisation. Section 2 fixes the setting. Sections 3–5 collect geometric and discrete tools and the error equation. Section 6 treats the printed method and identifies its H^1 constant. Section 7 treats the pressure-consistent method, proves the lower bound (with the proof of (M3) in Section 7.1.3), and treats flux-corrected data, a growing penalty and the penalty-uniform leak limit. Section 8 treats the pressure, the boundary stress and drag, and Section 9 the L^2 error. Section 10 treats the penalty threshold, and Section 11 strong imposition. Sections 12–14 contain the extensions: Clough–Tocher refinements, general domains and Navier–Stokes. Section 15 treats pressure robustness, Section 16 reports the computations, and Section 17 summarises what is proved and what is open.

2. SETTING, METHODS AND ASSUMPTIONS

Domain. Let $R = (-L, L)^2$ with $L > 1$, let D be the open unit disk, $\Omega = R \setminus \overline{D}$ and $\Gamma = \partial D$. Let P_h be a convex polygon with vertices on Γ , $\Gamma_h = \partial P_h$ and $\Omega_h = R \setminus P_h$. Since $P_h \subset D$, we have $\Omega \subset \Omega_h$. The strip $S_h = \Omega_h \setminus \overline{\Omega}$ lies between each chord and its arc.

Mesh. $\mathcal{T}_h$ is a conforming triangulation of Ω_h , and $\mathcal{E}_h^\Gamma$ is its set of edges on Γ_h . We use the parameters

$$h = h_\Omega := \max_{T \in \mathcal{T}_h} \text{diam } T, \quad h_\Gamma := \max_{e \in \mathcal{E}_h^\Gamma} |e|, \quad \rho := \frac{h}{\min_{e \in \mathcal{E}_h^\Gamma} |e|}, \quad \gamma := \frac{\mu h_\Gamma}{h}.$$

Here h is the h in Scott's penalty μ/h , and γ is the effective boundary penalty, so that $\mu/h = \gamma/h_\Gamma$.

Assumption 2.1.

- (M0) Shape regularity, and $c_0 h_\Gamma \leq |e| \leq h_\Gamma$ for all $e \in \mathcal{E}_h^\Gamma$. Hence $\rho/h \leq 1/(c_0 h_\Gamma)$. No quasi-uniformity is assumed; $h_\Gamma \ll h$ is allowed.
- (M1) No vertex of $\mathcal{T}_h$ is singular. A vertex is singular if it is an interior vertex whose incident edges lie on two lines, a boundary vertex with one incident triangle, or a boundary vertex whose incident edges lie on two lines.
- (M1') Every vertex of Γ_h has at least three incident triangles. The angle of Ω_h at a vertex of Γ_h is $\pi + O(h_\Gamma)$, so with two triangles $\Theta(z) = O(h_\Gamma) \rightarrow 0$, contradicting (M2).
- (M2) $\Theta(z) := \max_j |\sin(\theta_j + \theta_{j+1})| \geq \Theta_0 > 0$ at every vertex z , in the notation of Guzmán–Scott [37]. At boundary vertices the wrap-around term is omitted.
- (D) $\tilde{f}$ is smooth and bounded on a fixed neighbourhood of $\overline{\Omega}$, $\tilde{f} = f$ on Ω , and $h \leq h_*$ is small (hence so is $h_\Gamma \leq h$).

Spaces. Fix $k \geq 4$ (Section 12 treats $k \geq 2$ on Clough–Tocher refinements). V_h is the space of continuous piecewise- P_k vector fields, $V_h^R := \{v \in V_h : v|_{\partial R} = 0\}$ and $V_h^0 := \{v \in V_h : v|_{\partial \Omega_h} = 0\}$. Π_h is the space of discontinuous piecewise- P_{k-1} functions and $Z_h := \{v \in V_h^R : \text{div } v = 0\}$. $W_h := \{z \in V_h : z|_{\partial R} = g_I, \text{div } z = 0\}$, where g_I is the P_k Lagrange interpolant of g .

Definition 2.2 (Flux-corrected outer data). Let ν be the outward unit normal of R , $m_R := \int_{\partial R} g_I \cdot \nu$, and

$$\tilde{g}_I := g_I - \frac{m_R}{8L^2} x \quad \text{on } \partial R,$$

where x is the position field. A method *with flux-corrected data* is the same method with g_I replaced by $\tilde{g}_I$, and $\tilde{W}_h := \{z \in V_h : z|_{\partial R} = \tilde{g}_I, \text{div } z = 0\}$.

Since x is linear, $\tilde{g}_I$ is again continuous and piecewise P_k on ∂R , and $\int_{\partial R} x \cdot \nu = \int_R \operatorname{div} x = 8L^2$, so $\int_{\partial R} \tilde{g}_I \cdot \nu = 0$ exactly. This is a simple instance of the flux-corrected Dirichlet interpolants that exactly divergence-free methods require [28]. We recommend it for CNS*: it removes the only term in Theorem 7.2 that needs a relation between h_Γ and h_Ω (Theorem 7.33). Gjerde and Scott use $k = 4$, as do the computations of Section 16 except those of Section 16.11 ($k = 5, 6$, and $k = 2, 3$ on Clough–Tocher refinements).

Forms. n is the outward unit normal of Ω_h on Γ_h (pointing into P_h), and $\langle \cdot, \cdot \rangle$ is the $L^2(\Gamma_h)$ pairing. Let

$$\begin{aligned} a_h(u, v) &= \frac{1}{2}(Du, Dv), \quad Du = \nabla u + (\nabla u)^\top, \\ N_h(u, v) &= a_h(u, v) - \langle \partial_n u, v \rangle - \langle u, \partial_n v \rangle + \frac{\mu}{h} \langle u, v \rangle, \quad \partial_n u := (\nabla u)n. \end{aligned}$$

Definition 2.3 (Method GS [34, (28)], as printed). Find $u_h \in V_h$ with $u_h|_{\partial R} = g_I$ and $p_h \in \Pi_h$ such that

$$N_h(u_h, v) - (p_h, \operatorname{div} v) = (\tilde{f}, v) \quad \forall v \in V_h^R, \quad (q, \operatorname{div} u_h) = 0 \quad \forall q \in \Pi_h.$$

Two details of the transcription. Gjerde and Scott write $b(v, p) = \int p \operatorname{div} v$ with a plus sign; our sign is the substitution $p_h \mapsto -p_h$. They also define V_h earlier as vanishing on Γ_h ; read literally, the Nitsche terms would then drop out of (28), so the evident intended test space is our V_h^R , which vanishes only on ∂R .

Definition 2.4 (Method CNS*: the mean-free correction). As GS, with momentum row

$$N_h(u_h, v) - (p_h, \operatorname{div} v) + \langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle = (\tilde{f}, v), \quad \bar{p}_\Gamma(q) := |\Gamma_h|^{-1} \int_{\Gamma_h} q.$$

The added term is the missing pressure traction with the wall mean of the discrete pressure removed. It appears in the momentum row only, so the continuity row, and with it the pointwise incompressibility of u_h , is unchanged. This device is not new: its velocity is that of the zero-net-flux formulation of Frachon, Nilsson and Zahedi [29], transcribed to our setting (Remark 7.3). What we add is its analysis for Scott–Vogelius elements on an inscribed polygon.

Exact solution and extensions. (u, p) is smooth with $-\Delta u + \nabla p = f$ and $\operatorname{div} u = 0$ in Ω , $u|_\Gamma = 0$ and $u|_{\partial R} = g$; note $-\operatorname{div} D(u) = -\Delta u$. The stream function ψ with $u = \operatorname{curl} \psi = (\partial_y \psi, -\partial_x \psi)$ is single-valued because $u \cdot n = 0$ on Γ . Let $\tilde{\psi}$ be a smooth extension of ψ to a neighbourhood of $\bar{\Omega}$ and $\tilde{u} := \operatorname{curl} \tilde{\psi}$, so $\operatorname{div} \tilde{u} = 0$. Let $\tilde{p}$ be a smooth extension of p . Then $F := \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$ vanishes on Ω and is supported in S_h .

Norms and constants. For v with traces on Γ_h ,

$$\|v\|^2 := \|\nabla v\|_{\Omega_h}^2 + \frac{\mu}{h} \|v\|_{\Gamma_h}^2 + \sum_{e \in \mathcal{E}_h^\Gamma} |e| \|\partial_n v\|_{L^2(e)}^2.$$

This is used for discrete fields and for errors $\tilde{u} - z$; since $\tilde{u}$ is smooth, every trace exists. We write $G := \|p - \bar{p}\|_{L^2(\Gamma)}$, with $\bar{p}$ the mean of p over Γ .

Remark 2.5 (Smoothness is a hypothesis on the solution: the corners of R). Near Γ smoothness follows from the data: if f is smooth up to Γ , so are u, p and ψ (Proposition 9.2), and the extensions above exist. At the four right-angled corners of R it does not. For the Stokes system with Dirichlet data in an angle ω the velocity singular exponents are the roots of $\sin^2(\lambda\omega) = \lambda^2 \sin^2 \omega$, $\lambda \neq 1$ [24]; for $\omega = \pi/2$ the first is $\lambda = 2.739593 \pm 1.119025 i$ (then $4.808251 \pm 1.463928 i, \dots$; computed with `mpmath, notes/v7/referee_reg/roots.py`). So for generic smooth, compatible data $u \in H^s$ near a corner of R only for $s < 3.7396$. The standing hypothesis that (u, p) is smooth on $\bar{\Omega}$, which (D)

and every h^k and h^{k+1} term below rely on, is therefore an assumption on the *solution*. It holds for the manufactured tests of Section 16. For data-driven problems on meshes that are quasi-uniform at the corners of R , the terms h^k in the H^1 bounds become $h^{\min(k, 2.7396)-\epsilon}$, and the terms h^{k+1} in the L^2 bounds (including Theorem 9.14, which needs $\tilde{u} \in H^{k+1} \supset H^5$) degrade likewise, unless the mesh is graded towards the corners. The dual problems of the L^2 arguments are not affected: they need only Theorem 9.1.

C and c depend on (M0)–(M2), k , L and norms of $(u, p, \tilde{\psi})$. They are independent of $h, h_\Gamma, \mu, \gamma, \rho$ unless displayed. C_p is a constant that may also depend on $\|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}$. Where G is tracked explicitly, “ δ small” means $\delta \leq \delta_0(C_p)$.

3. GEOMETRY

Lemma 3.1 (Chords). *Let e be a chord of length ℓ and $x \in e$. Then $\text{dist}(x, \Gamma) \leq \ell^2/8 + O(\ell^4) \leq \ell^2/4$. Let s be the signed arclength from the chord midpoint m_e , x^* the nearest point of Γ , and τ_e the unit tangent of e . Then, up to a fixed orientation sign,*

$$n(x^*) - n_e = -s \tau_e + O(\ell^2).$$

The arclength Jacobian of the radial projection $\Gamma_h \rightarrow \Gamma$ is $1 + O(\ell^2)$.

Proof. The sagitta is $1 - \sqrt{1 - \ell^2/4} = \ell^2/8 + \ell^4/128 + \dots$. The normal at x^* is the radial unit vector. Its angle differs from that of n_e by the central angle to x^* , which is $s + O(s^3)$. $\square$

Lemma 3.2 (Uniform trace, Friedrichs and Korn). *There are $C_{\text{tr}}, C_F, \kappa$, independent of h , such that every $v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$ satisfies*

$$\|v\|_{L^2(\Gamma_h)} \leq C_{\text{tr}} \|\nabla v\|, \quad \|v\| \leq C_F \|\nabla v\|, \quad \|\nabla v\|^2 \leq \kappa a_h(v, v).$$

Proof. (1) Write Γ_h in polar form as $r = \rho_h(\theta)$. On a chord with endpoint angles $\theta_m \pm \alpha$ we have $\rho_h = \cos \alpha / \cos(\theta - \theta_m)$. Hence $\|\rho_h - 1\|_\infty \leq Ch_\Gamma^2$ and $\|\rho'_h\|_\infty \leq Ch_\Gamma$.
(2) Let χ be smooth, with $\chi(1) = 1$ and $\chi \equiv 0$ outside a fixed collar in Ω . Define $\Phi_h(r, \theta) = (r + \chi(r)(\rho_h(\theta) - 1), \theta)$.
(3) Φ_h maps Ω onto Ω_h and Γ onto Γ_h , is the identity near ∂R , and is bi-Lipschitz with $\|D\Phi_h - I\|_\infty \leq Ch_\Gamma$ almost everywhere.
(4) For $w = v \circ \Phi_h$ we have $D(w) = D(v) \circ \Phi_h + E$ with $|E| \leq Ch_\Gamma |\nabla v \circ \Phi_h|$.
(5) On the fixed domain Ω , w vanishes on ∂R , a set of positive measure. So the first Korn inequality, the trace inequality and the Friedrichs inequality hold.
(6) Changing variables back gives $\|\nabla v\| \leq C(\|D(v)\| + h_\Gamma \|\nabla v\|)$; absorb the last term for $h_\Gamma \leq h_*$.
The other two inequalities transfer the same way. $\square$

Lemma 3.3 (Extension). $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$, so $\|\tilde{u}\|_{L^2(\Gamma_h)} \leq Ch_\Gamma^2$. For $v \in V_h^R$, $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$.

Proof. The first bound follows from Lemma 3.1 and $\tilde{u}|_\Gamma = 0$. For the second, apply a one-dimensional Poincaré inequality across strips of width at most Ch_Γ^2 , and use $|S_h| \leq Ch_\Gamma^2$, $\|F\|_\infty \leq C$ (by (D)) and Lemma 3.2. $\square$

Lemma 3.4 (Wall structure). *On Γ : $\nabla u = \partial_n u \otimes n$, $\partial_n u \cdot n = 0$ and $(\nabla u)^\top n = 0$.*

Proof. The tangential derivatives vanish, and $\text{div } u = n \cdot \partial_n u$. $\square$

Lemma 3.5 (Transposed-gradient term). *For $v \in V_h^R$,*

$$|((\nabla \tilde{u})^\top n, v)| \leq C \sum_{e \in \mathcal{E}_h^\Gamma} (|e|^2 \|v\|_{L^1(e)} + |e|^2 \|\partial_\tau v\|_{L^1(e)}) \leq Ch_\Gamma^{3/2} \|\nabla v\|.$$

Pointwise the transposed term is only $O(h_\Gamma)$ on Γ_h , not $O(h_\Gamma^2)$: although $\tilde{u} = O(h_\Gamma^2)$ there, the chord normal n_e is tilted by $O(h_\Gamma)$ against $n(x^*)$ (step 1 below). The rate $h_\Gamma^{3/2}$ comes from the odd symmetry of the leading term in s (step 2), not from the size of $\tilde{u}$. In the energy dual it can also be obtained from the penalty alone, $|\langle (\nabla \tilde{u})^\top n, v \rangle| \leq Ch_\Gamma \|v\|_{\Gamma_h} \leq C\gamma^{-1/2} h_\Gamma^{3/2} \|v\|$ (Lemma 8.4); in the plain H^1 dual, without odd symmetry, it would be only $O(h_\Gamma)$.

The rate $h_\Gamma^{3/2}$ is sharp on V_h^R : a first-layer field with odd trace on each chord attains it, and the dual-norm test of Section 16 gives 1.56–1.57. For the method itself, which is tested on Z_h , Theorem 7.24 proves the matching lower bound $ch_\Gamma^{3/2}$ for the H^1 error, by a different route, for every $k \geq 4$. The CNS* H^1 rates of Table 11 decrease monotonically toward 3/2. Lemma 3.5 is a sharper dual-norm statement than the upper bounds need. Lemma 5.2 and Theorem 7.2 also follow from the cruder energy-dual bound $C\gamma^{-1/2} h_\Gamma^{3/2} \|v\|$ above, which holds without the odd-in- s cancellation; likewise, in step 4 of Theorem 6.4 the $O(h_\Gamma)$ tangential load $(p - \bar{p})(n_e - n(x^*))$ can be bounded in the energy dual by $C_p h_\Gamma (h/\mu)^{1/2}$, which suffices because step 6 tests with r_h in $\|\cdot\|$. The cancellation matters for the plain H^1 duals, and in the companion three-dimensional analysis [50] it is not available.

Proof. (1) Taylor-expand about x^* and apply Lemmas 3.1 and 3.4:

$$(\nabla \tilde{u}(x))^\top n_e = n(x^*)(\partial_n u(x^*) \cdot (n_e - n(x^*))) + O(h_\Gamma^2) = s a_e + O(h_\Gamma^2),$$

where $a_e := n(m_e^*)(\partial_n u(m_e^*) \cdot \tau_e)$ is constant on e . The $O(h_\Gamma^2)$ remainder collects $|x - x^*|$, the $O(\ell^2)$ term of Lemma 3.1, and $s \cdot O(s)$.

(2) Since $\int_e s \, ds = 0$,

$$\left| \int_e s a_e \cdot v \right| = \left| \int_e s a_e \cdot (v - v(m_e)) \right| \leq \frac{|e|^2}{4} \|\partial_\tau v\|_{L^1(e)}.$$

(3) Use $\|\cdot\|_{L^1(e)} \leq |e|^{1/2} \|\cdot\|_{L^2(e)}$ and the inverse trace inequality $|e| \|\partial_\tau v\|_{L^2(e)}^2 \leq C_I^2 \|\nabla v\|_{T_e}^2$.

(4) Summing, $\sum_e |e|^{5/2} \|\partial_\tau v\|_{L^2(e)} \leq Ch_\Gamma^2 (\#\mathcal{E}_h^\Gamma)^{1/2} \|\nabla v\| \leq Ch_\Gamma^{3/2} \|\nabla v\|$. The zeroth-order term is handled the same way, finishing with Lemma 3.2. $\square$

4. DISCRETE TOOLS

Lemma 4.1 (Inverse traces). *For $v \in V_h$,*

$$\sum_{e \in \mathcal{E}_h^\Gamma} |e| \|\partial_n v\|_{L^2(e)}^2 \leq C_I^2 \|\nabla v\|^2, \quad \text{hence} \quad \|\partial_n v\|_{\Gamma_h} \leq C_I (\rho/h)^{1/2} \|\nabla v\|.$$

For $q \in \Pi_h$,

$$\|q\|_{\Gamma_h}^2 \leq C_I^2 \sum_{T \in \mathcal{T}_h^\Gamma} |e_T|^{-1} \|q\|_T^2 \leq C_I^2 (\rho/h) \|q\|^2.$$

Lemma 4.2 (Coercivity and continuity). *Let $\mu_0 := 4\kappa\rho C_I^2$. The condition $\gamma \geq \gamma_0 := 4\kappa C_I^2/c_0$ is sufficient for $\mu \geq \mu_0$. If $\mu \geq \mu_0$, then*

$$N_h(v, v) \geq c_1 \|v\|^2 \quad \forall v \in V_h^R, \quad |N_h(w, v)| \leq M \|w\| \|v\|,$$

with $c_1 = 1/\max(2\kappa(1 + C_I^2), 2)$ and $M = 3 + (\kappa C_I^2)^{-1/2}$. Both constants are independent of h , μ and ρ .

Proof. (1) By Lemma 4.1 and Young's inequality with $\varepsilon = 1/(2\kappa)$,

$$2|\langle \partial_n v, v \rangle| \leq 2C_I \|\nabla v\| (\rho/h)^{1/2} \|v\|_{\Gamma_h} \leq \frac{1}{2} a_h(v, v) + 2\kappa\rho C_I^2 h^{-1} \|v\|_{\Gamma_h}^2.$$

- (2) Hence $N_h(v, v) \geq \frac{1}{2}a_h(v, v) + (\mu - 2\kappa\rho C_I^2)h^{-1}\|v\|^2$. For $\mu \geq \mu_0$ this is at least $(2\kappa)^{-1}\|\nabla v\|^2 + (\mu/2h)\|v\|^2$.
- (3) Adding $\sum_e |e|\|\partial_n v\|^2 \leq C_I^2\|\nabla v\|^2$ gives the coercivity constant.
- (4) For continuity, $a_h(w, v) \leq 2\|\nabla w\|\|\nabla v\|$. Each ∂_n -term is at most $(\rho/\mu)^{1/2}\|w\|\|v\| \leq (4\kappa C_I^2)^{-1/2}\|w\|\|v\|$. $\square$

Lemma 4.3 (Divergence range). *Under (M1), $\operatorname{div} V_h^R = \Pi_h$ and $W_h \neq \emptyset$. The velocities of GS and CNS* are pointwise divergence-free.*

- Proof.* (1) $\operatorname{div} V_h^0 = \Pi_h \cap L_0^2$ by the inf-sup condition of Lemma 4.5 below (Guzmán–Scott [37], extending Scott–Vogelius [54]). This holds on the doubly connected domain Ω_h ; there is no circularity, since Lemma 4.5 does not use the present lemma.
- (2) The edge bubble $b = \beta_e n_e$ on one $e \subset \Gamma_h$ lies in V_h^R and has $\int \operatorname{div} b = \int_{\Gamma_h} b \cdot n = |e|/6 \neq 0$. This adds the constants, so $\operatorname{div} V_h^R = \Pi_h$.
- (3) For any $\mathcal{G} \in V_h$ with trace g_I , we have $\operatorname{div} \mathcal{G} \in \Pi_h = \operatorname{div} V_h^R$. Hence $\mathcal{G} - v \in W_h$ for a suitable $v \in V_h^R$.
- (4) For the discrete velocity, $u_h - \mathcal{G} \in V_h^R$, and the continuity row forces $\operatorname{div} u_h = 0$. $\square$

Remark 4.4. If a vertex of ∂R is singular (for example a corner with one triangle), then $A_z(\operatorname{div} \mathcal{G}) = 0$ is a condition on g_I that generic data violate, and $W_h = \emptyset$.

Lemma 4.5 (Uniform inf-sup constant). *There is $\beta > 0$, depending only on k , Θ_0 , shape regularity and L , such that for every $q \in \Pi_h \cap L_0^2(\Omega_h)$ there exists $v \in V_h^0$ with $\operatorname{div} v = q$ and $\|\nabla v\| \leq \beta^{-1}\|q\|$.*

Proof. Guzmán and Scott [37, Theorem 1] prove the inf-sup condition for $(V_h^0, \Pi_h \cap L_0^2)$ for $k \geq 4$ under (M1)–(M2). Their proof needs neither quasi-uniformity, nor small h , nor an interior vertex in each triangle. The discrete Fortin operator is built locally from vertex and edge patches, using shape regularity and (M2) only. The domain enters only through the continuous inf-sup constant (via the P_2 – P_0 Bernardi–Raugel pair) and a Poincaré constant. The Poincaré constant is uniform by Lemma 3.2, so it remains to show that the continuous inf-sup constant is uniform over the domains Ω_h .

- Cover Ω_h by the h -dependent pieces $W_j \cap \Omega_h$, where the W_j are fixed angular wedges of half-angle $\alpha < \pi/2$ about the origin. Each $W_j \cap R$ is convex, and $W_j \cap \Omega_h = (W_j \cap R) \setminus P_h$.
- Each $W_j \cap \Omega_h$ is star-shaped with respect to a ball $B(c_j, r_B) \subset W_j \cap R$ with $|c_j| = r_b$, provided every chord line meeting W_j leaves the ball on its outer side. This holds uniformly when $(r_b - r_B) \cos(\alpha + \arcsin(r_B/r_b) + h_\Gamma) > 1$, which can be arranged with fixed α, r_b, r_B because $P_h \subset D$.
- The pieces have diameters, ball radii and pairwise overlap measures bounded above and below independently of h .

The standard gluing of Bogovskiĭ operators over a finite union of such pieces (Lemma 13.14) then gives a Bogovskiĭ constant, and hence a continuous inf-sup constant, independent of h . Alternatively, Bernardi, Costabel, Dauge and Girault prove directly that the continuous inf-sup constant of an inscribed polygonal approximation differs from that of the curved domain by $O(h)$ [10, Theorem 4.6]. $\square$

Lemma 4.6 (Approximation in W_h). *Let $k \geq 4$. Let $\sigma_k := k + 2$ for even k and $\sigma_k := k + 1$ for odd k . Then*

$$E_h := \inf_{z \in W_h} \|\tilde{u} - z\| \leq C\varepsilon_h, \quad \varepsilon_h := (1 + (\gamma h_\Gamma)^{1/2}) h^k + h^{\sigma_k} (h_\Gamma^{-1/2} + (\mu/h)^{1/2}).$$

The second term comes from the net flux of the interpolated boundary data. It cannot be removed for the method as posed with g_I . Recall $m_R = \int_{\partial R} g_I \cdot \nu$ (Definition 2.2). Every

$z \in W_h$ (including the discrete velocity) has $\int_{\Gamma_h} z \cdot n = -m_R$, while $\int_{\Gamma_h} \tilde{u} \cdot n = 0$. Hence $\|\tilde{u} - z\| \geq (\mu/h)^{1/2} |m_R|/|\Gamma_h|^{1/2}$, and $|m_R| \asymp h^{\sigma_k}$ for generic g .

For bounded γ the term is at most Ch^k whenever $h_\Gamma \geq h^{2(\sigma_k-k)}$: that is, $h_\Gamma \geq h^4$ for even k and $h_\Gamma \geq h^2$ for odd k . If g_I is replaced by the flux-corrected $\tilde{g}_I$ of Definition 2.2 (Lemma 7.32), or for the test data of Section 16, where $g \cdot \nu$ is a polynomial of degree at most 5 on each side and the flux integrals over ∂R are exact, only the contribution of Γ_h to the net flux remains. By step 2 of the proof below it is $O(h_\Gamma^{\sigma_k})$, so the term reduces to $C(1 + \gamma^{1/2})h_\Gamma^{\sigma_k-1/2}$, which is below h^k for bounded γ .

Proof. (1) *Interpolation.* Let $w_h := I_h \tilde{u} \in V_h$ be the Lagrange interpolant of $\tilde{u}$, which is smooth on Ω_h . On ∂R , $w_h = g_I$. Boundary triangles have diameter at most Ch_Γ by shape regularity, so

$$\|\nabla(\tilde{u} - w_h)\| \leq Ch^k, \quad \|\tilde{u} - w_h\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^{k+1}, \quad \|\partial_n(\tilde{u} - w_h)\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^k.$$

Hence $\|\tilde{u} - w_h\| \leq C(h^k + (\mu/h)^{1/2}h_\Gamma^{k+1} + h_\Gamma^k) \leq C(1 + (\gamma h_\Gamma)^{1/2})h^k$. Moreover $\operatorname{div} w_h = \operatorname{div}(w_h - \tilde{u}) \in \Pi_h$, with $\|\operatorname{div} w_h\| \leq Ch^k$.

(2) *Net flux.* Let ν be the outward normal of R . Since $\tilde{u}|_{\partial R} = g$ and $\operatorname{div} \tilde{u} = 0$, we have $\int_{\Gamma_h} \tilde{u} \cdot n = -\int_{\partial R} g \cdot \nu = 0$. Therefore

$$m := \int_{\Omega_h} \operatorname{div} w_h = \int_{\partial R} (g_I - g) \cdot \nu + \int_{\Gamma_h} (w_h - \tilde{u}) \cdot n, \quad |m| \leq Ch^{\sigma_k}.$$

Here equispaced P_k Lagrange interpolation on an edge integrates like the closed $(k+1)$ -point Newton–Cotes rule. That rule is exact for polynomials of degree $\sigma_k - 1$, so each edge contributes $O(|e|^{\sigma_k+1})$.

(3) *Removing the mean.* Let $b := \sum_{e \in \mathcal{E}_h^\Gamma} \beta_e n_e$, where β_e is the quadratic edge bubble of the boundary triangle on e . Then $b \in V_h^R$, $\int \operatorname{div} b = \int_{\Gamma_h} b \cdot n = |\Gamma_h|/6$, and with $a := 6m/|\Gamma_h|$,

$$\|ab\| \leq C|a|(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) \leq Ch^{\sigma_k}(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) \leq C\varepsilon_h,$$

using $\#\mathcal{E}_h^\Gamma \leq C/h_\Gamma$. Also $\|\operatorname{div}(ab)\| \leq C|a|h_\Gamma^{-1/2} \leq C\varepsilon_h$.

(4) *Divergence correction.* The field $q := \operatorname{div}(w_h - ab) \in \Pi_h$ has mean zero and $\|q\| \leq C\varepsilon_h$. Lemma 4.5 gives $v \in V_h^0$ with $\operatorname{div} v = q$ and $\|\nabla v\| \leq \beta^{-1}\|q\| \leq C\varepsilon_h$. Since $v = 0$ on Γ_h , $\|v\|^2 = \|\nabla v\|^2 + \sum_e |e| \|\partial_n v\|_{L^2(e)}^2 \leq (1 + C_I^2) \|\nabla v\|^2$.

(5) Set $z := w_h - ab - v$. Then $z|_{\partial R} = g_I$ and $\operatorname{div} z = 0$, so $z \in W_h$, and

$$\|\tilde{u} - z\| \leq \|\tilde{u} - w_h\| + \|ab\| + \|v\| \leq C\varepsilon_h. \quad \square$$

Remark 4.7. Version 1 of this preprint proved Lemma 4.6 for $k = 4$ by matching Argyris stream-function data on ∂R . The proof above avoids stream functions and holds for every $k \geq 4$.

5. THE ERROR EQUATION

Lemma 5.1. For $v \in V_h^R$,

$$N_h(\tilde{u}, v) - (\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle = (\tilde{f}, v) + \mathcal{G}(v),$$

where

$$\mathcal{G}(v) := \langle (\nabla \tilde{u})^\top n, v \rangle - \langle \tilde{u}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{u}, v \rangle - (F, v).$$

Proof. The argument has four steps.

(1) By symmetry of $D(\tilde{u})$ and since $v = 0$ on ∂R ,

$$a_h(\tilde{u}, v) = (D(\tilde{u}), \nabla v) = -(\operatorname{div} D(\tilde{u}), v) + \langle D(\tilde{u})n, v \rangle.$$

- (2) $\operatorname{div} D(\tilde{u}) = \Delta \tilde{u}$.
- (3) $-\Delta \tilde{u} = \tilde{f} - F - \nabla \tilde{p}$, and $-(\nabla \tilde{p}, v) = (\tilde{p}, \operatorname{div} v) - \langle \tilde{p}, v \cdot n \rangle$.
- (4) $D(\tilde{u})n - \partial_n \tilde{u} = (\nabla \tilde{u})^\top n$.

□

Lemma 5.2. For $v \in V_h^R$, $|\mathcal{G}(v)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}\|v\|$. For $v \in V_h^0$, $|\mathcal{G}(v)| \leq Ch_\Gamma^{3/2}\|\nabla v\|$.

Proof. We bound the four terms of $\mathcal{G}(v)$ in turn.

- The transposed term is at most $Ch_\Gamma^{3/2}\|\nabla v\|$ by Lemma 3.5.
- $|\langle \tilde{u}, \partial_n v \rangle| \leq Ch_\Gamma^2 \cdot C_I(\rho/h)^{1/2}\|\nabla v\| \leq Ch_\Gamma^{3/2}\|\nabla v\|$.
- The penalty term is at most $(\mu/h)^{1/2}Ch_\Gamma^2\|v\| = C\gamma^{1/2}h_\Gamma^{3/2}\|v\|$.
- The F term is bounded by Lemma 3.3.

For $v \in V_h^0$, the transposed and penalty terms vanish. □

Corollary 5.3 (GS on Z_h). Let $v \in Z_h$. Then $(\cdot, \operatorname{div} v) = 0$ and $\int_{\Gamma_h} v \cdot n = 0$, so for every constant c ,

$$N_h(\tilde{u} - u_h, v) = \mathcal{R}(v) + \mathcal{G}(v), \quad \mathcal{R}(v) := -\langle \tilde{p} - c, v \cdot n \rangle.$$

6. THE PRINTED METHOD

Throughout this section $\gamma \geq \gamma_0$.

Test field. Let $\varphi_0(\theta) := \int_0^\theta (p - \bar{p})(\theta') d\theta'$, which is periodic, and $\varphi := \chi_1(r)\varphi_0(\theta)$, where $\chi_1 \equiv 1$ near $r = 1$ (so $\chi_1'(1) = 0$) and $\chi_1 \equiv 0$ near ∂R . Then $w := \operatorname{curl} \varphi$ has components $w_r = \partial_\theta \varphi / r$ and $w_\theta = -\partial_r \varphi$, and on Γ

$$w = (p - \bar{p})e_r = -(p - \bar{p})n.$$

Let Π_{k+1} be the C^1 piecewise- P_{k+1} interpolant of Morgan and Scott [46] (for $k = 4$, the Argyris interpolant [3]), which exists because $k+1 \geq 5$; its constants depend on the angles, hence on (M0) and (M2). Since $\varphi \equiv 0$ near ∂R and $h \leq h_*$, $v_\varphi := \operatorname{curl} \Pi_{k+1} \varphi \in Z_h$, and $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$.

Theorem 6.1 (Energy norm: exact rate 1/2). The error satisfies the upper bound

$$\|\tilde{u} - u_h\| \leq C \left[(h_\Gamma/\gamma)^{1/2}G + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h \right].$$

If $G > 0$ and $h \leq h_0(G)$, with h_0 independent of $\gamma \geq \gamma_0$, it also satisfies the lower bound

$$\|\tilde{u} - u_h\| \geq (2M)^{-1}(h_\Gamma/\gamma)^{1/2}G - C \left[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h \right].$$

In Scott's variables, $(h_\Gamma/\gamma)^{1/2} = (h/\mu)^{1/2}$.

Proof of the upper bound. For $z \in W_h$, $e_h := z - u_h \in Z_h$. By Corollary 5.3 and Lemma 4.2,

$$c_1\|e_h\|^2 \leq N_h(z - \tilde{u}, e_h) + \mathcal{R}(e_h) + \mathcal{G}(e_h).$$

Using the Jacobian bound of Lemma 3.1, $|\mathcal{R}(e_h)| \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2}\|e_h\|$. Then apply Lemmas 5.2 and 4.6. □

Proof of the lower bound. (1) We have $v_\varphi \cdot n_e = -(p - \bar{p})(x^*)n(x^*) \cdot n_e + O(h^k)$, $\mathcal{R} = -\langle \tilde{p} - \bar{p}, v \cdot n \rangle$ and $n(x^*) \cdot n_e = 1 + O(s^2)$. Hence

$$\mathcal{R}(v_\varphi) = \langle (\tilde{p} - \bar{p})(p - \bar{p})(x^*), n(x^*) \cdot n_e \rangle + O(h^k) = G^2 + O(h_\Gamma^2 + h^k).$$

(2) $\|v_\varphi\|^2 \leq (\mu/h)(G + Ch_\Gamma^2 + Ch^k)^2 + C$.

(3) By continuity, $M\|\tilde{u} - u_h\| \geq (|\mathcal{R}(v_\varphi)| - |\mathcal{G}(v_\varphi)|)/\|v_\varphi\|$, and Lemma 5.2 bounds $\mathcal{G}$.

(4) $|\mathcal{R}|/\|v_\varphi\| \geq (h/\mu)^{1/2}G(1 - C(h_\Gamma^2 + h^k)/G^2 - C(h_\Gamma^2 + h^k)/G - C(h_\Gamma/\gamma_0)^{1/2}/G)$, which is at least $\frac{1}{2}(h/\mu)^{1/2}G$ for $h \leq h_0(G)$. □

The lower bound is carried by the boundary parts of $\|\cdot\|$. By Theorem 6.4, $\|\nabla(\tilde{u} - u_h)\|$ is only $O(h/\mu)$.

Corollary 6.2. *If $G > 0$, $\gamma h_\Gamma \ll G$, $\gamma^{1/2} h_\Gamma \ll G$ and $\varepsilon_h \ll (h_\Gamma/\gamma)^{1/2} G$, then $\|\tilde{u} - u_h\| \asymp (h/\mu)^{1/2}$. The mesh-dependent energy rate is exactly $1/2$.*

The two conditions on γ make the terms $\gamma^{1/2} h_\Gamma^{3/2}$ and $h_\Gamma^{3/2}$ of Theorem 6.1 small against $(h_\Gamma/\gamma)^{1/2} G$. The second is implied by the first when $\gamma \geq 1$, which holds for $\gamma \geq \gamma_0$ as soon as $\gamma_0 \geq 1$.

Lemma 6.3 (Normal-load cancellation). *Let S be edgewise $W^{1,\infty}$ on Γ_h , continuous at the vertices of Γ_h , with $|S \cdot \tau_e| \leq Ch_\Gamma$ on every $e \in \mathcal{E}_h^\Gamma$. (For instance, S smooth with $S \cdot \tau = 0$ on Γ .) Then for $v \in Z_h$,*

$$|\langle S, \partial_n v \rangle| \leq C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2} \|\nabla v\|) \leq C \|\nabla v\|.$$

Proof. (1) *Tangential part.* By Lemma 4.1 and (M0), $|\langle (S \cdot \tau) \tau, \partial_n v \rangle| \leq Ch_\Gamma \|\partial_n v\|_{\Gamma_h} \leq Ch_\Gamma^{1/2} \|\nabla v\|$.

(2) *Normal part.* On e , v is a polynomial on T_e with $\operatorname{div} v = 0$, so $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$. With $\sigma_e := S \cdot n_e$,

$$\int_e \sigma_e \partial_n v \cdot n_e = \int_e \partial_\tau \sigma_e (v \cdot \tau_e) - [\sigma_e v \cdot \tau_e]_{\partial e}.$$

(3) At a polygon vertex x shared by e and e' , the endpoint terms combine to $v(x) \cdot (\sigma_{e'} \tau_{e'} - \sigma_e \tau_e)$, of size at most $C(|e| + |e'|)|v(x)|$.

(4) $\sum_e |e| \|v\|_{L^\infty(e)} \leq C \sum_e |e|^{1/2} \|v\|_{L^2(e)} \leq C |\Gamma_h|^{1/2} \|v\|_{\Gamma_h}$.

(5) Finish with Lemma 3.2. $\square$

Numerically (Section 16), the dual norm of $v \mapsto \langle S, \partial_n v \rangle$ stays bounded on Z_h for normal S . It grows like $h_\Gamma^{-1/2}$ on V_h^R (no divergence constraint), and also on Z_h for tangential S . Incompressibility is exactly what the proof uses.

Theorem 6.4 (H^1 seminorm: rate 1). *With $C_p := C(1 + \|p - \bar{p}\|_{W^{1,\infty}(\Gamma)})$,*

$$\|\nabla(\tilde{u} - u_h)\| \leq C_p (h_\Gamma/\gamma) + C \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

Proof. (1) *Split.* For $z \in W_h$, write $z - u_h = \delta_R + \delta_G$, where $\delta_R, \delta_G \in Z_h$ solve

$$N_h(\delta_R, v) = \mathcal{R}(v), \quad N_h(\delta_G, v) = \mathcal{G}(v) - N_h(\tilde{u} - z, v) \quad \forall v \in Z_h.$$

Lax–Milgram on Z_h (Lemma 4.2) gives existence and uniqueness.

(2) *Geometric part.* $\|\nabla \delta_G\| \leq \|\delta_G\| \leq c_1^{-1} (C(1 + \gamma^{1/2}) h_\Gamma^{3/2} + M E_h)$.

(3) *Ansatz.* Heuristically $\delta_R \approx -(h/\mu)(p - \bar{p})n = (h/\mu)w$ on Γ_h . Set $S_\varphi := v_\varphi \in Z_h$ and $r_h := \delta_R - (h/\mu)S_\varphi \in Z_h$. Then $N_h(r_h, v) = m(v) + \ell(v)$ with

$$\begin{aligned} m(v) &:= \mathcal{R}(v) - \langle S_\varphi, v \rangle = -\langle (\tilde{p} - \bar{p})n + S_\varphi, v \rangle, \\ \ell(v) &:= -(h/\mu) [a_h(S_\varphi, v) - \langle \partial_n S_\varphi, v \rangle - \langle S_\varphi, \partial_n v \rangle]. \end{aligned}$$

(4) *Bound on m .* On e ,

$$(\tilde{p} - \bar{p})n_e + S_\varphi = (p - \bar{p})(x^*)(n_e - n(x^*)) + O(h_\Gamma^2) + O(h^k) = (p - \bar{p})(m_e^*) s \tau_e + O(h_\Gamma^2) + O(h^k).$$

The leading term is odd in s with a constant coefficient. The argument of Lemma 3.5 handles it, and the $O(h^k)$ remainder contributes $Ch^k \|v\|_{L^1(\Gamma_h)}$. Hence $|m(v)| \leq C_p (h_\Gamma^{3/2} + h^k) \|\nabla v\|$.

(5) *Bound on ℓ .* $|a_h(S_\varphi, v)| + |\langle \partial_n S_\varphi, v \rangle| \leq C_p \|\nabla v\|$. By Lemma 6.3 applied to $S = w$,

$$|\langle S_\varphi, \partial_n v \rangle| \leq |\langle w, \partial_n v \rangle| + |\langle S_\varphi - w, \partial_n v \rangle| \leq C_p \|\nabla v\| + Ch^k h_\Gamma^{-1/2} \|\nabla v\|.$$

So $|\ell(v)| \leq (h/\mu)(C_p + Ch^k h_\Gamma^{-1/2}) \|\nabla v\|$.

(6) *Close.* Testing with $v = r_h$,

$$c_1 \|r_h\|^2 \leq [C_p(h/\mu + h_\Gamma^{3/2} + h^k) + C(h/\mu)h^k h_\Gamma^{-1/2}] \|\nabla r_h\|.$$

The last two terms are at most $C\varepsilon_h$, since $(h/\mu)h^k h_\Gamma^{-1/2} = \gamma^{-1}h^k h_\Gamma^{1/2} \leq Ch^k$. Hence

$$\|\nabla \delta_R\| \leq (h/\mu) \|\nabla S_\varphi\| + \|\nabla r_h\| \leq C_p(h_\Gamma/\gamma) + C(h_\Gamma^{3/2} + \varepsilon_h). \quad \square$$

Remark 6.5 (Why rate 1 and not 1/2). Without incompressibility, Lemma 6.3 would give only $Ch_\Gamma^{-1/2}$. The H^1 error would then be $(h/\mu)h_\Gamma^{-1/2}$: rate 1/2, with element-scale oscillation in the trace. The inconsistency $-pn$ is purely normal, and $\operatorname{div} v = 0$ converts $\partial_n v \cdot n$ into a tangential derivative. That is why GS still converges at rate 1 in H^1 .

Theorem 6.6 (Boundary slip; H^1 lower bound). *Let $\gamma \geq \gamma_0$ and $G > 0$, and put $X := C_p(h_\Gamma/\gamma) + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h$, the right-hand side of Theorem 6.4. Suppose that, for a fixed small δ ,*

$$X \leq \delta G \min(G, h_\Gamma^{1/2}), \quad h_\Gamma/\gamma \leq \delta G, \quad h_\Gamma \leq h_0(G).$$

All three hold for fixed γ and bounded ρ as $h_\Gamma \rightarrow 0$. Then

- (i) $c(h_\Gamma/\gamma)G \leq \|\tilde{u} - u_h\|_{L^2(\Gamma_h)} \leq C(h_\Gamma/\gamma)^{1/2} \|\tilde{u} - u_h\|$;
- (ii) $\|\nabla(\tilde{u} - u_h)\| \geq c(h_\Gamma/\gamma)G - Ch^{k+1/2}$.

Combined with Theorem 6.4, $\|\nabla(\tilde{u} - u_h)\| \asymp h/\mu$ in this regime.

Proof of (i). Let $e_u := \tilde{u} - u_h$ and $v := v_\varphi$. Corollary 5.3 rearranges to

$$\frac{\mu}{h} \langle e_u, v \rangle = \mathcal{R}(v) + \mathcal{G}(v) - a_h(e_u, v) + \langle \partial_n e_u, v \rangle + \langle e_u, \partial_n v \rangle.$$

We bound the terms, using that v is smooth up to $O(h^k)$.

- (1) $\mathcal{R}(v) = G^2 + O(h_\Gamma^2 + h^k)$.
- (2) $|\mathcal{G}(v)| \leq C(h_\Gamma^{3/2} + \gamma h_\Gamma^3 + \gamma h_\Gamma h^k)$. On the chord, $\tilde{u} = \pm d \partial_n u(x^*) + O(d^2)$ is tangential while $v \approx -(p - \bar{p})n$ is normal. Hence $\tilde{u} \cdot v = O(h_\Gamma^4 + h_\Gamma^2 h^k)$, the second term coming from $v_\varphi - w$.
- (3) $|a_h(e_u, v)| \leq C_p \|\nabla e_u\| \leq C_p X \leq C_p \delta G^2$, by Theorem 6.4.
- (4) By Theorem 6.1 and $h/\mu = h_\Gamma/\gamma \leq \delta G$,

$$|\langle e_u, \partial_n v \rangle| \leq C_p \|e_u\|_{\Gamma_h} \leq C_p (h/\mu)^{1/2} \|e_u\| \leq C_p [(h/\mu)G + X] \leq 2C_p \delta G^2.$$

(5) We have

$$|\langle \partial_n e_u, v \rangle| \leq \|\partial_n(\tilde{u} - z)\|_{\Gamma_h} \|v\| + C_I(\rho/h)^{1/2} (\|\nabla(z - \tilde{u})\| + \|\nabla e_u\|) \|v\|_{\Gamma_h}.$$

By Theorem 6.4 and $(\rho/h)^{1/2} \leq (c_0 h_\Gamma)^{-1/2}$, this is at most $Ch_\Gamma^{-1/2} XG \leq C\delta G^2$.

For δ small, $|\langle e_u, v \rangle| \geq \frac{1}{2}(h/\mu)G^2$. Dividing by $\|v\|_{\Gamma_h} \leq G + Ch_\Gamma^2$ gives the lower bound. The upper bound is $\|e_u\|_{\Gamma_h} \leq (h/\mu)^{1/2} \|e_u\|$. $\square$

Proof of (ii). Let $E \in H^1(\Omega_h)$ lift $(g - g_I)|_{\partial R}$, supported near ∂R , with $E|_{\Gamma_h} = 0$ and $\|E\|_{H^1} \leq C\|g - g_I\|_{H^{1/2}(\partial R)} \leq Ch^{k+1/2}$. By Lemma 3.2, $\|e_u\|_{\Gamma_h} = \|e_u - E\|_{\Gamma_h} \leq C_{\text{tr}}(\|\nabla e_u\| + \|\nabla E\|)$. $\square$

Corollary 6.7 (Exact leading term). *Let $G > 0$ and $h_\Gamma \rightarrow 0$ with $h/\mu \rightarrow 0$, $\gamma h_\Gamma \rightarrow 0$ and $\varepsilon_h/(h_\Gamma/\gamma)^{1/2} \rightarrow 0$. Then*

$$\tilde{u} - u_h = (h/\mu) v_\varphi + \vartheta_h, \quad \|\vartheta_h\| = o((h/\mu)^{1/2}), \quad \|\vartheta_h\|_{\Gamma_h} = o(h/\mu),$$

and consequently

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{(h/\mu)^{1/2}G} \rightarrow 1.$$

On Γ_h , $v_\varphi \approx -(p - \bar{p})n$. So to leading order the printed method replaces the no-slip condition by the leak condition $(\mu/h) u_h \cdot n \approx p - \bar{p}$: the penalty, not a traction, balances the pressure.

Proof. Write $\tilde{u} - u_h = (\tilde{u} - z) + \delta_R + \delta_G$ in the notation of Theorem 6.4, with $\delta_R = (h/\mu)S_\varphi + r_h$ and $S_\varphi = v_\varphi$.

(1) From step 6 of Theorem 6.4,

$$c_1 \|r_h\|^2 \leq [C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h] \|r_h\|.$$

So $\|r_h\| \leq C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h$ and $\|r_h\|_{\Gamma_h} \leq (h/\mu)^{1/2} \|r_h\|$.

(2) $\|\delta_G\| + \|\tilde{u} - z\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$, and the Γ_h norm carries an extra factor $(h/\mu)^{1/2}$.

(3) Relative to $(h/\mu)^{1/2}G$ (energy) and $(h/\mu)G$ (slip), steps 1–2 contribute

$$O((h/\mu)^{1/2}/G + (\gamma^{1/2} + \gamma)h_\Gamma/G + \varepsilon_h(\gamma/h_\Gamma)^{1/2}/G) \rightarrow 0.$$

(4) By Lemma 3.1 (Jacobian and displacement of Γ_h from Γ), $\|v_\varphi\|_{\Gamma_h} = G(1 + O(h_\Gamma^2)) + O(h^k)$ and $\|v_\varphi\|^2 = (\mu/h)\|v_\varphi\|_{\Gamma_h}^2 + O(1)$. $\square$

This gives the sharp answer to Scott's question for the printed method. The estimate $h_\Gamma^{3/2} + h^k$ fails exactly when p is not constant on Γ . The error is then the explicit field $(h/\mu)v_\varphi$, with the universal constant 1 in both the slip norm and the energy norm.

6.1. The H^1 leak constant. Corollary 6.7 identifies the slip and energy constants of GS (both $\rightarrow 1$), but not the H^1 constant $\|\nabla(\tilde{u} - u_h)\| \mu/(hG)$, because the remainder $r_h = \delta_R - (h/\mu)S_\varphi$ of Theorem 6.4 has the same H^1 order as $(h/\mu)S_\varphi$. We identify the limit of $(\mu/h)\delta_R$: it is the Stokes flow in Ω driven by the leak as *Dirichlet* data. The H^1 constant is $K = \|\nabla U\|/G$, a global quantity that depends on the outer box (it tends to $3/\sqrt{7}$ as $L \rightarrow \infty$ for the pressure of test P). For test P and $L = 2.5$ a converged series computation gives $K = 2.7565157$, inside the rigorous bracket $[2.0746, 3.0526]$ of Proposition 6.13.

6.1.1. *The limit problem.*

Definition 6.8 (Leak flow). Let (U, P) solve

$$-\Delta U + \nabla P = 0, \quad \operatorname{div} U = 0 \text{ in } \Omega, \quad U = (p - \bar{p})e_r = -(p - \bar{p})n \text{ on } \Gamma, \quad U = 0 \text{ on } \partial R,$$

and put $K := \|\nabla U\|_{L^2(\Omega)}/G$.

The data are compatible, since $\int_\Gamma U \cdot n = -\int_\Gamma (p - \bar{p}) = 0$. Hence (U, P) exists and is unique (P up to a constant), $U = \operatorname{curl} \Psi$ with a single-valued stream function, and $U|_\Gamma = w|_\Gamma$ with $w = \operatorname{curl} \varphi$ the test field above. U is smooth up to Γ (smooth boundary and data) and $U \in H^2(\Omega)$ (the corners of R are convex). Since $-\Delta U = -\nabla P$, for every divergence-free V with the same boundary values $\int \nabla U : \nabla(V - U) = -\int \nabla P \cdot (V - U) = 0$, so

$$\|\nabla U\|^2 = \min \left\{ \|\nabla V\|_{L^2(\Omega)}^2 : V \in H^1(\Omega)^2, \operatorname{div} V = 0, V|_\Gamma = (p - \bar{p})e_r, V|_{\partial R} = 0 \right\}. \quad (1)$$

In particular K depends on $p|_\Gamma$ and on the outer boundary: it is not a local quantity.

6.1.2. *Convergence.* Extend Ψ smoothly across Γ to a fixed neighbourhood and put $\tilde{U} := \text{curl } \tilde{\Psi}$ on Ω_h , so $\text{div } \tilde{U} = 0$; extend P smoothly to $\tilde{P}$. Then $F_U := -\Delta \tilde{U} + \nabla \tilde{P}$ vanishes on Ω , is bounded, and is supported in the strip S_φ .

Lemma 6.9. *There is $z_U \in Z_h$ with $\|\tilde{U} - z_U\| \leq E_U := C(h\|U\|_{H^2(\Omega)} + (1 + (\gamma h_\Gamma)^{1/2})h^k + (1 + \gamma^{1/2})h_\Gamma^{\sigma_k - 1/2})$. For bounded γ , $E_U \leq Ch$, since $\sigma_k - \frac{1}{2} \geq k + \frac{1}{2}$.*

Proof. Repeat the proof of Lemma 4.6 with $\tilde{u}$ replaced by $\tilde{U}$ and $g = 0$. Away from Γ only $\tilde{U} \in H^2$ is available, so the interpolation step gives $\|\nabla(\tilde{U} - I_h \tilde{U})\| \leq Ch\|U\|_{H^2(\Omega)}$. Near Γ , $\tilde{U}$ is smooth and the boundary terms are as before. The net flux $m = \int_{\Gamma_h} (I_h \tilde{U} - \tilde{U}) \cdot n$ is $O(h_\Gamma^{\sigma_k})$ because $\int_{\Gamma_h} \tilde{U} \cdot n = \int_{\Omega_h} \text{div } \tilde{U} = 0$. As in step 3 of that proof, the flux bubble then costs $C|m|(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) \leq C(1 + \gamma^{1/2})h_\Gamma^{\sigma_k - 1/2}$; the Bogovskii correction (Lemma 4.5) is unchanged. $\square$

Theorem 6.10 (The leak flow is the H^1 limit). *Let $\gamma \geq \gamma_0$ and let $\delta_R \in Z_h$ solve $N_h(\delta_R, v) = \mathcal{R}(v)$ for all $v \in Z_h$ (Theorem 6.4, step 1). Then*

$$\left\| \nabla \left(\frac{\mu}{h} \delta_R - \tilde{U} \right) \right\|_{\Omega_h} \leq \left\| \frac{\mu}{h} \delta_R - z_U \right\| + E_U \leq C_U \left[\min(\gamma h_\Gamma^{1/2}, (\gamma h_\Gamma)^{1/2}) + h_\Gamma^{1/2} + (h/\mu)^{1/2} + E_U \right],$$

where C_U depends on $(M0)$ – $(M2)$, k , L and on norms of p and U near Γ .

Proof. Put $\epsilon := \delta_R - (h/\mu)z_U \in Z_h$. For $v \in Z_h$,

$$N_h(\epsilon, v) = \mathcal{R}(v) - \frac{h}{\mu} N_h(\tilde{U}, v) + \frac{h}{\mu} N_h(\tilde{U} - z_U, v). \quad (2)$$

(1) *Green's formula for $\tilde{U}$.* As in Lemma 5.1, $\text{div } D(\tilde{U}) = \Delta \tilde{U}$, $\text{div } v = 0$, $v|_{\partial R} = 0$ and $\int_{\Gamma_h} v \cdot n = 0$ give, for any constant c ,

$$N_h(\tilde{U}, v) = (F_U, v) + \langle \sigma_U, v \rangle - \langle \tilde{U}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{U}, v \rangle, \quad \sigma_U := (\nabla \tilde{U})^\top n - (\tilde{P} - c)n.$$

(2) *The penalty terms cancel to leading order.* With $\mathcal{R}(v) = -\langle (\tilde{p} - \bar{p})n, v \rangle$, (2) becomes

$$N_h(\epsilon, v) = -\langle \zeta, v \rangle - \frac{h}{\mu} \left[(F_U, v) + \langle \sigma_U, v \rangle - \langle \tilde{U}, \partial_n v \rangle \right] + \frac{h}{\mu} N_h(\tilde{U} - z_U, v), \quad \zeta := (\tilde{p} - \bar{p})n + \tilde{U}.$$

(3) *The mismatch ζ .* On $e \in \mathcal{E}_h^\Gamma$, by Lemma 3.1 and $\tilde{U}(x^*) = -(p - \bar{p})(x^*)n(x^*)$,

$$\zeta = (p - \bar{p})(x^*)(n_e - n(x^*)) + O(h_\Gamma^2) = (p - \bar{p})(m_e^*) s \tau_e + O(h_\Gamma^2),$$

exactly the functional m of Theorem 6.4, step 4 (without the h^k term). Hence $|\langle \zeta, v \rangle| \leq Ch_\Gamma^{3/2} \|\nabla v\|$. Alternatively $\|\zeta\|_{\Gamma_h} \leq Ch_\Gamma$ gives $|\langle \zeta, v \rangle| \leq Ch_\Gamma(h/\mu)^{1/2} \|v\|$.

(4) *The $O(h/\mu)$ terms.* $|(F_U, v)| \leq Ch_\Gamma^2 \|\nabla v\|$ (Lemma 3.3); $|\langle \sigma_U, v \rangle| \leq C\|v\|_{\Gamma_h} \leq C(h/\mu)^{1/2} \|v\|$. On Γ_h , $\tilde{U}$ is edgewise smooth, continuous at the vertices, and $\tilde{U} \cdot \tau_e = -(p - \bar{p})(x^*)n(x^*) \cdot \tau_e + O(h_\Gamma^2) = O(h_\Gamma)$. So Lemma 6.3 applies with $S = \tilde{U}$: $|\langle \tilde{U}, \partial_n v \rangle| \leq C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2} \|\nabla v\|) \leq C((h/\mu)^{1/2} + h_\Gamma^{1/2}) \|v\|$. Finally $|N_h(\tilde{U} - z_U, v)| \leq ME_U \|v\|$ (Lemmas 4.2 and 6.9).

(5) *Close.* Test with $v = \epsilon$ and use Lemma 4.2:

$$c_1 \|\epsilon\| \leq C \left[\min(h_\Gamma^{3/2}, h_\Gamma(h/\mu)^{1/2}) + \frac{h}{\mu} (h_\Gamma^2 + (h/\mu)^{1/2} + h_\Gamma^{1/2} + E_U) \right].$$

Divide by h/μ , and use $h_\Gamma^{3/2} \mu/h = \gamma h_\Gamma^{1/2}$ and $h_\Gamma(\mu/h)^{1/2} = (\gamma h_\Gamma)^{1/2}$. The first inequality of the theorem is the triangle inequality with Lemma 6.9. $\square$

Corollary 6.11 (The H^1 constant). (i) For GS with general data, in the notation of Theorem 6.4,

$$\left\| \nabla \left(\frac{\mu}{h} (\tilde{u} - u_h) - \tilde{U} \right) \right\|_{\Omega_h} \leq (\text{bound of Theorem 6.10}) + C \frac{\mu}{h} \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

(ii) Consequently, if $h_\Gamma \rightarrow 0$ with $h/\mu \rightarrow 0$, $\gamma h_\Gamma \rightarrow 0$, $\gamma(1 + \gamma^{1/2}) h_\Gamma^{1/2} \rightarrow 0$ and $(\mu/h) \varepsilon_h \rightarrow 0$ (all of which hold for fixed γ and bounded ρ), then

$$\frac{\|\nabla(\tilde{u} - u_h)\|_{\Omega_h}}{(h/\mu)G} \rightarrow K = \frac{\|\nabla U\|_{L^2(\Omega)}}{G}.$$

(iii) On test P ($u = 0$, $f = \nabla p$) one has $\tilde{u} = 0$, $z = 0$ and $\mathcal{G} \equiv 0$, so $\tilde{u} - u_h = \delta_R$ exactly and Theorem 6.10 applies with no further condition.

(iv) The remainder of Theorem 6.4 converges: $(\mu/h) r_h \rightarrow U - w$ in H^1 . The limit is the Stokes correction of the cut-off field w : it vanishes on $\partial\Omega$ and solves $-\Delta(U - w) + \nabla P = \Delta w$. It depends on the cut-off χ_1 , whereas U does not.

Proof. (i) is $\tilde{u} - u_h = (\tilde{u} - z) + \delta_R + \delta_G$ with steps 2 of Theorem 6.4 and Lemma 4.6. (ii) Multiply out; $\|\nabla \tilde{U}\|_{\Omega_h}^2 - \|\nabla U\|_{\Omega}^2 = \|\nabla \tilde{U}\|_{S_h}^2 = O(h_\Gamma^2)$. (iii) $F = \tilde{f} - \nabla \tilde{p} = 0$ and all terms of $\mathcal{G}$ contain $\tilde{u}$. (iv) $r_h = \delta_R - (h/\mu)v_\varphi$ and $\|\nabla(v_\varphi - w)\| \leq Ch^k$. $\square$

Remark 6.12 (The Dirichlet-dominated regime). The hypothesis $\gamma h_\Gamma = \mu h_\Gamma^2/h \rightarrow 0$ is the one of Corollary 6.7. It fails for the large- μ runs of Table 13 ($\gamma h_\Gamma \approx 240$ at $N = 96$, $\mu = 10^4$). There the discrete problem is close to its $\mu \rightarrow \infty$ limit, a discrete Dirichlet problem whose data $-(p - \bar{p})n_e$ jumps in direction at every vertex of Γ_h . The computations below show that $(\mu/h)\delta_R \rightarrow U$ in that regime too, with relative H^1 distance d_h decaying roughly like $h_\Gamma^{0.6} - h_\Gamma^{0.8}$, the decay slowing as N grows. Theorem 6.10 does not cover it, but Section 7.3 does: $(\mu/h)\delta_R \rightarrow U$ in H^1 uniformly in $\mu \geq \mu_0$, and $d_h \asymp h_\Gamma^{1/2}$ for every penalty (Corollary 7.45); the observed exponents are pre-asymptotic.

6.1.3. *Rigorous bracket and exact asymptotics.* For $\varrho > 1$ let $\mathcal{A}_\varrho = \{1 < r < \varrho\}$ and let $\mathcal{E}(\varrho)$ be the minimum (1) over $\mathcal{A}_\varrho$ (zero data on $r = \varrho$). Extending by zero shows that the minimum decreases as the domain grows.

Proposition 6.13. *If $\mathcal{A}_{\varrho_1} \subset \Omega \subset \mathcal{A}_{\varrho_2}$, then $\mathcal{E}(\varrho_2) \leq \|\nabla U\|^2 \leq \mathcal{E}(\varrho_1)$. In the annulus the Fourier modes decouple: if $\Psi|_\Gamma = \sum_n (a_n \cos n\theta + b_n \sin n\theta)$, then $\mathcal{E}(\varrho) = \sum_n (a_n^2 + b_n^2) E_n(\varrho)$, where $E_n(\varrho) = \pi \int_1^\varrho [f''^2 + 2n^2((f/r)')^2 + (f'/r - n^2 f/r^2)^2] r dr$, and f is the combination of $r^n, r^{-n}, r^{n+2}, r^{2-n}$ ($r, r^{-1}, r^3, r \log r$ for $n = 1$) with $f(1) = 1$, $f'(1) = 0$, $f(\varrho) = f'(\varrho) = 0$.*

For test P, $p - \bar{p} = -\frac{3}{4} \sin \theta + \frac{1}{4} \sin 3\theta + \frac{1}{2} \cos \theta$ on Γ , so $\Psi|_\Gamma = \frac{3}{4} \cos \theta + \frac{1}{2} \sin \theta - \frac{1}{12} \cos 3\theta$ and $G^2 = 7\pi/8$. With $\varrho_1 = L = 2.5$ and $\varrho_2 = L\sqrt{2}$ (40-digit quadrature):

$$\boxed{2.0746004 \leq K \leq 3.0525769}$$

The same formulas give the dependence on the box. $E_3(\varrho) \rightarrow 45\pi$ (the exterior clamped solution is $f = \frac{3}{2}r^{-1} - \frac{1}{2}r^{-3}$). $E_1(\varrho) \rightarrow \pi$: the $n = 1$ data split into a potential dipole (exterior energy exactly π) and a rigid translation, whose energy vanishes as $\varrho \rightarrow \infty$ (the Stokes paradox). The annulus values $E_1(\varrho) - \pi = 2.47, 0.87, 0.38$ at $\varrho = 10, 10^2, 10^4$ are consistent with the classical $1/\log \varrho$ rate, which we observe numerically here rather than prove. Hence

$$K(L) \rightarrow \sqrt{\frac{(\frac{9}{16} + \frac{1}{4})\pi + \frac{45\pi}{144}}{7\pi/8}} = \frac{3}{\sqrt{7}} \approx 1.134 \quad (L \rightarrow \infty),$$

but only logarithmically: the annulus values are $K = 1.420, 1.242, 1.183$ at $\varrho = 10, 10^2, 10^4$. So the constant 2.7–2.8 of Table 13 belongs to the box $L = 2.5$, not to the method.

6.1.4. *The value for test P. Series solution.* Write $\Psi = \sum_{n \text{ odd} \leq n_{\max}} f_n(r) \cos n\theta$ for data $\cos \theta$ and $\cos 3\theta$ separately (the class of $\cos n\theta$, n odd, is invariant under both reflections of the square; the $\sin \theta$ datum is the rotation of the $\cos \theta$ one by $\pi/2$, and cross terms between the two reflection classes vanish). Imposing $\Psi = 0$ on ∂R , rather than an unknown constant, is justified by symmetry: for data $\cos n\theta$ with n odd, the reflection $x \mapsto -x$ maps Ω to itself and changes the sign of the data, so by uniqueness Ψ is odd under it, and its constant value on ∂R is zero; the $\sin \theta$ datum follows by rotation. The conditions on $r = 1$ are imposed exactly, mode by mode, and $\Psi = \partial_\nu \Psi = 0$ on ∂R by least squares at 16 000 Chebyshev-clustered points. The energy is integrated by Gauss quadrature on eight polar sectors. The value is stable to 8 digits under refinement of both the truncation $n_{\max}$ and the quadrature:

$n_{\max}$	max boundary residual	$\ \nabla U\ $	K
15	$4.4 \cdot 10^{-3}$	4.57012143	2.75644148
31	$6.1 \cdot 10^{-5}$	4.57024239	2.75651444
47	$5.1 \cdot 10^{-6}$	4.57024449	2.75651571
63	$1.1 \cdot 10^{-6}$	4.57024450	2.75651571
79	$1.8 \cdot 10^{-7}$	4.57024450	2.75651571

$$K_P = 2.7565157$$

This lies inside the rigorous bracket.

Computation: `code/h1_limit.py annulus`; `code/h1_limit.py mps`; output `logs/h1_limit_series.log`.

Finite element check. Our solver (SuperLU with refinement, as in Section 16) was run on test P, and the discrete field was compared *pointwise* with the series U at the quadrature points. Both the normalised H^1 error K_h and the relative distance $d_h := \|\nabla((\mu/h)(\tilde{u} - u_h) - \tilde{U})\|/\|\nabla U\|$ are shown below. The $\mu = 100$ and $N = 96$ values of K_h reproduce Table 13 to all printed digits.

N	$\mu = 10^2$		$\mu = 10^3$		$\mu = 10^4$		$\mu = 10^8$	
	K_h	d_h	K_h	d_h	K_h	d_h	K_h	d_h
16	2.4642	0.193	2.7844	0.318	3.0562	0.539	3.3254	0.717
24	2.5629	0.147	2.7913	0.259	2.9309	0.398	3.0452	0.497
32	2.6117	0.124	2.7880	0.222	2.8751	0.323	2.9339	0.385
48	2.6601	0.098	2.7808	0.179	2.8283	0.246	2.8514	0.277
64	2.6843	0.084	2.7759	0.154	2.8082	0.206	2.8208	0.226
96	2.7084	0.067	2.7701	0.124	2.7899	0.163	2.7959	0.175

Computation: `code/h1_limit.py fem`; output `logs/h1_limit_fem.log`.

The measured values 2.71–2.79 bracket the limit 2.7565 from both sides. The approach runs in two directions.

- At $\mu = 100$, K_h increases towards K and d_h decays like $h^{0.6}$ ($0.193 \rightarrow 0.067$ as h drops by a factor 5.3), with local rates $0.64 \rightarrow 0.54$. The asymptotic rate is exactly $\frac{1}{2}$ (Corollary 7.45); for fixed μ the terms $(h/\mu)^{1/2}$, $h_\Gamma^{1/2}$ and $(\gamma h_\Gamma)^{1/2}$ of Theorem 6.10 are all $\propto h^{1/2}$.
- At large μ (Remark 6.12), K_h decreases towards K from above. The excess is the energy of an element-scale oscillation that is nearly orthogonal to U : $K_h^2 = K^2(1 + d_h^2) + O(h_\Gamma^2)$ as $\gamma \rightarrow \infty$ (Corollary 7.47), for example $2.7565\sqrt{1 + 0.175^2} = 2.798$ against 2.796 at $N = 96$. So $K_h \rightarrow K$ at rate one although $d_h \asymp h_\Gamma^{1/2}$.

At fixed $N = 96$ the μ -dependence of Table 13 ($2.708 \rightarrow 2.770 \rightarrow 2.790$) is the crossover between the two regimes, not a sign of a different limit.

7. THE PRESSURE-CONSISTENT METHOD

Proposition 7.1 (The naive consistent form, and its L_0^2 variant). (a) Replace $\langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$ by $\langle p_h, v \cdot n \rangle$. The resulting linear system is square ($V_h^R \times \Pi_h$ tested by $V_h^R \times \Pi_h$). It has the kernel vector $(0, 1)$ and is therefore singular. Its left null space is therefore nontrivial as well, and the system is solvable only for data satisfying one linear compatibility condition. (b) The L_0^2 variant. Restrict the pressure to $\Pi_h \cap L_0^2(\Omega_h)$ and test the continuity row only with $q \in \Pi_h \cap L_0^2(\Omega_h)$, keeping the momentum row $N_h(u_h, v) - (p_h, \operatorname{div} v) + \langle p_h, v \cdot n \rangle = (\tilde{f}, v)$ for all $v \in V_h^R$. Any solution has $\operatorname{div} u_h \equiv c$, a constant, with

$$c |\Omega_h| = \int_{\partial R} g_I \cdot \nu + \int_{\Gamma_h} u_h \cdot n,$$

n being the outward normal of Ω_h . If $c = 0$, then (u_h, p_h) solves the naive system of (a).

(c) Let $\gamma \geq \gamma_2$ (Theorem 7.2). The naive system of (a) is solvable if and only if the CNS* pressure satisfies $\bar{p}_\Gamma(p_h^{\text{CNS}^*}) = 0$, and then its solutions are $(u_h^{\text{CNS}^*}, p_h^{\text{CNS}^*} + \text{const})$. If moreover the L_0^2 variant is uniquely solvable, then $c = 0$ if and only if the naive system is solvable.

Proof. (a) For $v \in V_h^R$,

$$-(1, \operatorname{div} v) + \langle 1, v \cdot n \rangle = - \int_{\partial \Omega_h} v \cdot n + \int_{\Gamma_h} v \cdot n = 0,$$

and the continuity row vanishes at $u = 0$. A square matrix with a nontrivial kernel has a nontrivial left kernel.

(b) The momentum rows of (a) and (b) coincide. Since $\operatorname{div} u_h \in \Pi_h$ (Lemma 4.3) and $(q, \operatorname{div} u_h) = 0$ for all mean-free $q \in \Pi_h$, $\operatorname{div} u_h$ is constant; integrate over Ω_h , using $u_h = g_I$ on ∂R . If $c = 0$, then $(q, \operatorname{div} u_h) = 0$ for all $q \in \Pi_h$.

(c) Write B^* as in the proof of Theorem 7.2. For every $r \in \Pi_h$ the naive pressure form satisfies $-(r, \operatorname{div} v) + \langle r, v \cdot n \rangle = B^*(r - \bar{p}_\Gamma(r), v)$, and it is invariant under $r \mapsto r + \text{const}$. If (u, p) solves the naive system, then $(u, p - \bar{p}_\Gamma(p))$ solves CNS*, so by uniqueness (Theorem 7.2) $u = u_h^{\text{CNS}^*}$ and $p - \bar{p}_\Gamma(p) = p_h^{\text{CNS}^*}$, which has zero mean on Γ_h . Conversely, if $\bar{p}_\Gamma(p_h^{\text{CNS}^*}) = 0$, the two pressure forms agree at $p_h^{\text{CNS}^*}$. For the last statement: if the naive system is solvable, shift its pressure into L_0^2 ; this gives a solution of the L_0^2 variant with $\operatorname{div} u_h = 0$, which by uniqueness is the solution, so $c = 0$. The converse is (b). $\square$

The singularity is the analogue, for our square Scott–Vogelius system, of the ill-posedness that Liu, Neilan and Otus note for their boundary-multiplier method when the zero-flux constraint is omitted [44, Remark 3.2]. The L_0^2 variant is the standard way to fix the constant; in the fully weak setting of Frachon, Nilsson and Zahedi the same computation shows that $\operatorname{div} u_h$ equals the multiplier of the mean constraint [29, (5.7)–(5.9)]. With weak imposition nothing forces $\int_{\Gamma_h} u_h \cdot n = - \int_{\partial R} g_I \cdot \nu$, even for flux-corrected data, so generically $c \neq 0$ and the L_0^2 variant is *not* exactly divergence-free. Unique solvability of the L_0^2 variant reduces to the nonvanishing of a single scalar (for zero data, write $u_h = u_0 + c w_c$ with $\operatorname{div} w_c \equiv 1$; then $(u_0, p_h - \bar{p}_\Gamma(p_h))$ is c times a fixed CNS* solution, and the constraint $\bar{p}_\Gamma(\cdot) = 0$ on its pressure must force $c = 0$). It is generic and held in every computation below, but we do not prove it.

Numerically ($k = 4$, meshes of Section 16), the kernel of (a) is exactly one-dimensional: in a dense SVD of the naive system for test B with $\mu = 20$ and an element-wise L^2 -orthonormal pressure basis, at $N = 8$ and 12 the smallest singular values are $8 \cdot 10^{-17}$ and $3 \cdot 10^{-16}$, the next ones $2.4 \cdot 10^{-2}$ and $8.1 \cdot 10^{-3}$. For the L_0^2 variant with $\mu = 100$ (SuperLU, relative residual about $6 \cdot 10^{-16}$), $\operatorname{div} u_h$ is constant to round-off. For test A at $N = 16$ it has $c = 9.69 \cdot 10^{-4}$, so $\|\operatorname{div} u_h\|_{L^2(\Omega_h)} = |c| |\Omega_h|^{1/2} = 4.54 \cdot 10^{-3}$, and the CNS* pressure has $\bar{p}_\Gamma(p_h) = 7.5 \cdot 10^{-2} \neq 0$. For

test B at $N = 16$ and 32 , $c = -5.98 \cdot 10^{-5}$ and $-2.94 \cdot 10^{-5}$; there the L_0^2 variant and CNS* have H^1 velocity errors that agree to four digits.

Earlier versions of this paper reported that an iterated-penalty solver and a bordered direct solve of the naive form returned discrete divergence “about $4.5 \cdot 10^{-3}$ ”, and read this as a sign of incompatibility. A bordered solve that fixes the pressure mean computes exactly the L_0^2 variant of (b), and the value coincides with $|c| |\Omega_h|^{1/2} = 4.54 \cdot 10^{-3}$ for test A, $N = 16$, $\mu = 100$. The reported divergence is therefore the constant c of (b), not a symptom of an incompatible system. We have not re-run the original solves, so the identification of the test case rests on this numerical coincidence.

Let β be the uniform inf-sup constant of Lemma 4.5.

Theorem 7.2 (CNS*: well posed, with Scott’s rate). *There is $\gamma_2 \asymp \max(\gamma_0, \beta^{-2})$ such that for $\gamma \geq \gamma_2$ the method CNS* has a unique solution and*

$$\|\tilde{u} - u_h\| \leq C \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

Consequently, for every fixed $k \geq 4$, for γ in a bounded range $[\gamma_2, \gamma_{\max}]$ and $h_\Gamma \geq h_\Omega^{2(\sigma_k - k)}$ (that is, $h_\Gamma \geq h_\Omega^4$ for even k and $h_\Gamma \geq h_\Omega^2$ for odd k),

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C (h_\Gamma^{3/2} + h_\Omega^k).$$

This condition ensures that the flux term of ε_h is at most Ch^k . Without it, that single extra term is kept, and by the remark after Lemma 4.6 it is then genuinely present for the plain interpolant g_I . With the flux-corrected data $\tilde{g}_I$ of Definition 2.2 no proviso is needed: $\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$ for every $\gamma \geq \gamma_2$, with C independent of γ and no relation between h_Γ and h_Ω (Theorem 7.33). This is the rate Scott conjectured, for general Stokes data. The L^2 part of the H^1 norm uses the lift E from the proof of Theorem 6.6(ii).

Proof. (1) *Consistency.* Write $B^*(q, v) := -(q, \operatorname{div} v) + \langle q - \bar{p}_\Gamma(q), v \cdot n \rangle$.

- For a constant c , $B^*(q + c, v) = B^*(q, v) - c \int_{\Gamma_h} v \cdot n$.
- $B^*(\tilde{p}, v)$ equals the pressure form of Lemma 5.1 minus $\bar{p}_\Gamma(\tilde{p}) \int_{\Gamma_h} v \cdot n$.
- Hence $p^* := \tilde{p} - \bar{p}_\Gamma(\tilde{p})$ satisfies $N_h(\tilde{u}, v) + B^*(p^*, v) = (\tilde{f}, v) + \mathcal{G}(v)$ for all $v \in V_h^R$.

With $e_u := \tilde{u} - u_h$ and $e_p := p^* - p_h$ this gives

$$N_h(e_u, v) + B^*(e_p, v) = \mathcal{G}(v) \quad \forall v \in V_h^R. \quad (3)$$

(2) *Split.* Write $e_p = \eta + \xi$ with $\eta := p^* - \pi_h p^*$ and $\xi \in \Pi_h$. For $v \in V_h^0$, $(\eta, \operatorname{div} v) = 0$ because $\operatorname{div} V_h^0 \subset \Pi_h$.

(3) *Pressure.* For $v \in V_h^0$, (3) gives

$$(\xi - \bar{\xi}, \operatorname{div} v) = N_h(e_u, v) - \mathcal{G}(v) = a_h(e_u, v) - \langle e_u, \partial_n v \rangle - \mathcal{G}(v).$$

Using $(\rho/\mu)^{1/2} \leq C$ and Lemma 5.2, the right-hand side is at most $C(\|e_u\| + h_\Gamma^{3/2}) \|\nabla v\|$.

Lemma 4.5 then gives $\beta \|\xi - \bar{\xi}\| \leq C(\|e_u\| + h_\Gamma^{3/2})$.

(4) *Velocity.* For $z \in W_h$, $e_h := z - u_h \in Z_h$. Test (3) with $v = e_h$. Since $(e_p, \operatorname{div} e_h) = 0$ and $\langle \bar{p}_\Gamma(e_p), e_h \cdot n \rangle = 0$,

$$c_1 \|e_h\|^2 \leq M E_h \|e_h\| + |\mathcal{G}(e_h)| + |\langle \eta, e_h \cdot n \rangle| + |\langle \xi - \bar{\xi}, e_h \cdot n \rangle|,$$

where

$$|\langle \eta, e_h \cdot n \rangle| \leq C h_\Gamma^k (h/\mu)^{1/2} \|e_h\|,$$

$$|\langle \xi - \bar{\xi}, e_h \cdot n \rangle| \leq C_I (\rho/h)^{1/2} \|\xi - \bar{\xi}\| (h/\mu)^{1/2} \|e_h\| \leq C(c_0 \gamma)^{-1/2} \|\xi - \bar{\xi}\| \|e_h\|.$$

- (5) *Absorb.* Insert step 3 and $\|e_u\| \leq E_h + \|e_h\|$. If $\gamma \geq \gamma_2$ with $C(c_0\gamma_2)^{-1/2}\beta^{-1} \leq c_1/2$, the pressure term is absorbed.
- (6) *Uniqueness.* Apply steps 3–5 to the homogeneous problem ($g_I = 0$, so $z = 0 \in W_h$). This gives $u_h = 0$, and then $p_h = c$ is constant. Now (3) reads $-c \int_{\Gamma_h} v \cdot n = 0$ for all v , and the edge bubble of Lemma 4.3 gives $c = 0$. The system is square, so uniqueness gives existence. $\square$

Remark 7.3 (Relation to the zero-flux formulation of Frachon, Nilsson and Zahedi). Let $V_h^{R,b} := \{v \in V_h^R : \int_{\Gamma_h} v \cdot n = 0\}$, which has codimension one (edge bubble, Lemma 4.3). Transcribed to our setting, the device of [29, Section 5.1] tests the momentum row with the naive form $-(p_h, \operatorname{div} v) + \langle p_h, v \cdot n \rangle$ only for $v \in V_h^{R,b}$, keeps the full continuity row, and fixes the pressure constant separately. On $V_h^{R,b}$ the naive and the CNS* pressure forms coincide, because $\langle \bar{p}_\Gamma(p_h), v \cdot n \rangle = \bar{p}_\Gamma(p_h) \int_{\Gamma_h} v \cdot n = 0$; and $B^*(p + c, w) = B^*(p, w) - c \int_{\Gamma_h} w \cdot n$ for the remaining direction w . Hence, for $\gamma \geq \gamma_2$: the CNS* solution solves the zero-flux formulation; conversely, if (u, p) solves the homogeneous zero-flux formulation, then $(u, p + c)$ solves homogeneous CNS* for the c that satisfies the w -row, so $u = 0$. So the two formulations have the *same velocity*, and pressures that differ by a constant. CNS* is a square packaging of the same device whose constant is fixed automatically, with $p_h \approx \tilde{p} - \bar{p}_\Gamma(\tilde{p})$.

7.1. A matching lower bound: the rate $h_\Gamma^{3/2}$ is sharp. Theorem 7.2 gives $\|\nabla(\tilde{u} - u_h)\| \leq \|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$ for CNS*, and Theorem 6.1 gives the same bound for GS when $G = 0$. Here we prove the reverse inequality in the H^1 seminorm whenever the wall shear is not identically zero.

Status of the hypothesis. The proof uses a local condition (M3) on the boundary stars (Assumption 7.4). For $k \geq 7$ a single-triangle field gives it on every mesh satisfying (M0)–(M2) (Lemma 7.5). For every $k \geq 4$ it is proved in Section 7.1.3 (Theorem 7.8) under (M0)–(M2) and the smallness condition (S): $h_\Gamma \leq 2\sin(\theta_{\min}/4)$, with an explicit P_4 witness on three triangles of each boundary star, a positive constant κ_0 by compactness, and certified numerical values for two angle bounds. So the lower bound is unconditional for every $k \geq 4$. (Earlier versions of this paper proved it only for $k \geq 7$ and verified (M3) numerically for $k = 4$.) The proof does *not* go through the transposed-gradient term of Lemma 3.5; see Section 7.1.6 for why that route fails for CNS*.

7.1.1. Notation. On Γ let n be the unit normal pointing into D and τ the unit tangent, oriented so that $\tau(x^*) \cdot \tau_e > 0$ for $x \in e$ (τ_e is n_e rotated in the same sense). By Lemma 3.4, $\partial_n u = W\tau$ on Γ , where

$$W := \partial_n u \cdot \tau \quad (\text{the wall shear}).$$

For $e \in \mathcal{E}_h^\Gamma$, T_e is the triangle with edge e , s is arclength on e from its midpoint m_e , and $W_e := W(m_e^*)$. No triangle has two edges on Γ_h : two such edges would share a vertex z of Γ_h , and the angle of Ω_h at z is $\pi + O(h_\Gamma) > \pi$. Define

$$Y_h := Z_h \cap V_h^0, \quad Y_h^1 := \left\{ v \in Y_h : \int_e \partial_n v = 0 \quad \forall e \in \mathcal{E}_h^\Gamma \right\}.$$

7.1.2. The local assumption. Write $\mathcal{M}(v) := -\frac{1}{2} \sum_{e \in \mathcal{E}_h^\Gamma} \int_e s^2 \partial_n v \cdot \tau_e ds$. For a vertex z of Γ_h , let e_z be the edge of Γ_h that follows z counterclockwise; $z \mapsto e_z$ is a bijection between the vertices and the edges of Γ_h .

Assumption 7.4 (Boundary test fields). **(M3)** There are constants $\kappa_0 > 0$ and K_{ov}, C_s such that every vertex z of Γ_h carries a field $v_z \in Y_h^1$ with the following properties:

- (i) $\|\nabla v_z\| = 1$;
- (ii) v_z is supported in a patch ω_z of triangles within distance $C_s h_\Gamma$ of z , and every triangle lies in at most K_{ov} patches;

(iii) $\mathcal{M}(v_z) \geq \kappa_0 |e_z|^2$, with e_z as above.

Condition (iii) is invariant under scaling, since $\|\nabla v\|$ is scale invariant in two dimensions. Only the sign of v_z is chosen in (iii): $\mathcal{M}(-v) = -\mathcal{M}(v)$.

Lemma 7.5 ((M3) for $k \geq 7$). *If $k \geq 7$, (M3) holds with $\omega_z = T_{e_z}$. Here $K_{\text{ov}} = 1$ and κ_0 depends only on (M0).*

Proof. Let $T = T_{e_z}$, and let $\lambda_1, \lambda_2, \lambda_3$ be its barycentric coordinates with $\lambda_3 = 0$ on $e := e_z$. Put $b := \lambda_1 \lambda_2 \lambda_3$ and $\sigma := 2s/|e| = \lambda_1 - \lambda_2$ on e (up to sign), and set

$$\phi := b^2((\lambda_1 - \lambda_2)^2 - \frac{1}{7}) \in P_8, \quad v := \text{curl } \phi \in P_7^2.$$

- ϕ vanishes to second order on ∂T , so $v = 0$ on ∂T . Extended by zero, v is continuous, lies in $V_h^0 \subset V_h^R$ (because $k \geq 7$), and satisfies $\text{div } v = 0$. Hence $v \in Y_h$.
- On e , $\partial_n v \cdot \tau_e = \pm 2|\nabla \lambda_3|^2(\lambda_1 \lambda_2)^2(\sigma^2 - \frac{1}{7}) = \pm \frac{1}{8}|\nabla \lambda_3|^2(1 - \sigma^2)^2(\sigma^2 - \frac{1}{7})$.
- Since $\int_{-1}^1 (1 - \sigma^2)^2 = \frac{16}{15}$ and $\int_{-1}^1 \sigma^2(1 - \sigma^2)^2 = \frac{16}{105}$, this has mean zero. With $\partial_n v \cdot n_e = 0$ (Lemma 7.18 below), $\int_e \partial_n v = 0$, so $v \in Y_h^1$.
- Its second moment is $\int_{-1}^1 \sigma^2(1 - \sigma^2)^2(\sigma^2 - \frac{1}{7}) d\sigma = \frac{16}{315} - \frac{16}{735} = \frac{64}{2205} \neq 0$.

The other two edges of T are interior edges and do not enter $\mathcal{M}$. Hence $\mathcal{M}(v) \neq 0$. The ratio $|\mathcal{M}(v)|/(|e|^2 \|\nabla v\|)$ is invariant under similarity. It is a continuous, nonvanishing function of the shape of T , and the shapes allowed by (M0) form a compact set, so the ratio is bounded below by some $\kappa_0 > 0$. Normalise and choose the sign. $\square$

Remark 7.6 ($k = 4$). For $k = 4$ no field in Y_h^1 can be supported in a single triangle, or in the two triangles of one first-layer quadrilateral; for the production meshes the latter space is $\{0\}$ (Section 7.1.7). The smallest natural patch is the boundary star ω_z (the triangles at z).

Write $v = \text{curl } \phi$ with ϕ a C^1 piecewise quintic; since $v = 0$ on $\partial \omega_z$ and on Γ_h , we may take $\phi = 0$ and $\nabla \phi = 0$ on $\partial \omega_z$ and on the two edges e, e' of Γ_h at z . On such an edge e , $\partial_n v \cdot \tau_e = \pm \partial_{n_e n_e} \phi|_{T_e}$ is a cubic with zero mean ($v \in Y_h^1$). At the far endpoint of e the gradient of $\phi|_{T_e}$ vanishes along two independent lines, so its Hessian vanishes there. If also $\partial_{n_e n_e} \phi|_{T_e}(z) = 0$, the cubic vanishes at both ends of e and has zero mean, so it is odd about the midpoint and $\int_e s^2 \partial_n v \cdot \tau_e = 0$. Hence (M3) requires $\partial_{n_e n_e} \phi|_{T_e}(z) \neq 0$ for at least one of the two edges e at z , the value being taken from the triangle T_e . This is possible only because C^1 quintics can have Hessians that jump across the interior edges at z : an Argyris (C^2 -vertex) stream function has a single Hessian at z , which vanishes on both boundary directions and hence is zero. Such jumps exist at a boundary vertex with at least three triangles, which is (M1').

Section 7.1.3 proves (M3) for $k = 4$ (hence every $k \geq 4$) by constructing such a field explicitly. As a check, the floating-point star computation of the earlier versions gives:

- for every boundary vertex of every production mesh ($N = 8, 12, 16, 20, 24, 32, 48, 64, 96, 128, 192, 256$) and also for $N = 512$ and 1024 (Table 1): $\dim Y^1(\omega_z) = 1$, $\min_z \kappa_z \geq 0.00288$ in every case, and $\min_z \kappa_z$ converges to about 0.0029 as N grows;
- for randomly generated three- and four-triangle boundary stars: $\dim Y^1(\omega_z) = 2m - 5$ for m triangles, and $\kappa_z > 0$ in every sample.

(M3) is monotone in the degree: a field in Y_h^1 for degree k is continuous piecewise $P_{k'}$ for every $k' \geq k$, and $\mathcal{M}$ and $\|\nabla \cdot\|$ do not depend on k . So (M3) for $k = 4$ on a mesh implies (M3) for every $k \geq 4$ on the same mesh. The exact star constants of Section 7.1.3 reproduce the table below to all printed digits.

7.1.3. *Proof of (M3) for every $k \geq 4$.* This subsection removes the hypothesis (M3) of Theorem 7.24 for $4 \leq k \leq 6$. The construction was first drafted in version 4 of this work; the two

geometric steps marked *exact* below replace “for h_Γ small” arguments of that draft, and the witness identities were re-verified independently (`notes/v6/m3/verify_witness.py`). Throughout, z is a vertex of Γ_h , and the triangles at z form a fan $T_j = (z, p_{j-1}, p_j)$, $j = 1, \dots, m$, listed counterclockwise, with $[z, p_0]$ and $[z, p_m]$ the two edges of Γ_h at z . We write α_j for the angle of T_j at z , β_j and γ_j for its angles at p_{j-1} and p_j , and $\Theta_z = \sum_j \alpha_j$ for the angle of Ω_h at z . Put $z = 0$, and for points a, b let $a \times b := a_1 b_2 - a_2 b_1$ and $D_{ij} := p_i \times p_j$.

Hypotheses. Besides (M1') we use:

(M0 $_\theta$) every triangle with a vertex on Γ_h has all its angles $\geq \theta_{\min}$, where $0 < \theta_{\min} \leq \pi/3$;

(S) $h_\Gamma \leq 2 \sin(\theta_{\min}/4)$.

(M0 $_\theta$) is the quantitative form of the shape regularity in (M0), and (S) is a smallness condition of the type allowed by (D). (M2) is used only through (M1').

Lemma 7.7 (Exact boundary geometry). *(a) At every vertex z of Γ_h , with ℓ_1, ℓ_2 the lengths of the two chords at z , $\Theta_z - \pi = \arcsin(\ell_1/2) + \arcsin(\ell_2/2)$. Under (S), $|\Theta_z - \pi| \leq \theta_{\min}/2$, so the condition*

(A $_\delta$) $|\Theta_z - \pi| \leq \delta$ at every vertex z of Γ_h holds with $\delta = \theta_{\min}/2 < \theta_{\min}$.

(b) On every mesh, the only edges of Γ_h contained in the star of z are $[z, p_0]$ and $[z, p_m]$:

(G) no other edge of the star lies on Γ_h .

Proof. (a) A chord of length ℓ of the unit circle subtends the central angle $2 \arcsin(\ell/2)$, and the angle between the chord and the tangent at either endpoint is half of it. The interior angle of $\Omega_h = R \setminus P_h$ at z is π plus the two tangent–chord angles. Under (S), $\arcsin(\ell_i/2) \leq \arcsin(\sin(\theta_{\min}/4)) = \theta_{\min}/4$.

(b) A triangle of the star with an edge on Γ_h not through z has all three vertices on ∂P_h , so it lies in the convex polygon $\overline{P_h}$; its interior would then be disjoint from Ω_h . An edge of Γ_h through z is one of the two chords at z . $\square$

Theorem 7.8 ((M3) holds for $k \geq 4$). *Assume (M0 $_\theta$), (M1') and (A $_\delta$) with $0 \leq \delta < \theta_{\min}$, and let $k \geq 4$. (By Lemma 7.7, (A $_\delta$) holds with $\delta = \theta_{\min}/2$ under (S), and (G) holds on every mesh.) Then (M3) holds with $K_{\text{ov}} = 3$, $C_s = (\sin \theta_{\min})^{-3}$ and*

$$\kappa_0 = \kappa_*(\theta_{\min}, \delta) > 0,$$

where κ_* is the minimum of the explicit continuous function κ_w of Lemma 7.12 over the compact families $F_3(\theta_{\min}, \delta)$ and $F_s(\theta_{\min}, \delta)$ of Definition 7.15. The patch ω_z is the star of z if $m = 3$, and the three triangles T_1, T_2, T_3 of the star next to e_z if $m \geq 4$. Moreover:

- (a) $\kappa_*(\theta_{\min}, \delta) \geq \kappa_{\text{el}}(\theta_{\min}, \delta)$, the closed-form expression of Corollary 7.17;
- (b) rigorous interval certificates give $\kappa_*(30^\circ, 10^\circ) \geq 0.0012$ and $\kappa_*(20^\circ, 15^\circ) \geq 0.0003$ (Section 7.1.3). The certificate parameters $(\theta_{\min}, \delta)$ are separate from (S): the value at $(30^\circ, 10^\circ)$ applies when $\theta_{\min} \geq 30^\circ$ and $\sup_z |\Theta_z - \pi| \leq 10^\circ$, e.g. for $h_\Gamma \leq 2 \sin 5^\circ$;
- (c) on every mesh of Section 16 ($N = 8, \dots, 1024$) every boundary star carries a field with $\kappa_z \geq 0.00287$, for either choice of e_z (Section 7.1.3); these are the exact star constants of Table 1.

Consequently Theorem 7.24 and Corollary 7.25 hold for every $k \geq 4$ under (M0)–(M2), (D) and (S), with $\kappa_0 = \kappa_*(\theta_{\min}, \theta_{\min}/2)$. For $k \geq 7$, Lemma 7.5 gives a single-triangle field as well. The witness below is a P_4 field, so by the monotonicity in Remark 7.6 it serves for all $k \geq 4$.

The idea. Remark 7.6 shows that a field in Y_h^1 can have $\mathcal{M} \neq 0$ only if the Hessian of its stream function at z jumps across the interior edges at z . On a fan of three triangles the homogeneous quadratic C^1 splines vanishing doubly on the two outer rays form a one-dimensional space, spanned by a spline whose Hessians are given by a classical Plücker-type identity; all its coefficients have

a sign when $\alpha_1 + \alpha_2 < \pi$ and $\alpha_2 + \alpha_3 < \pi$. We complete this quadratic part by a cubic part, in closed form, to a C^1 quintic stream function with zero boundary data and zero means on the chords. Its second moment is a sum of two terms of the same sign.

Stream functions. Let F be a fan of triangles at z as above, a simply connected polygon. A continuous piecewise- P_k field v on F with $v = 0$ on ∂F and $\operatorname{div} v = 0$ is $v = \operatorname{curl} \phi$ with $\phi \in C^1(F)$ piecewise P_{k+1} and $\phi = |\nabla \phi| = 0$ on ∂F , and conversely; moreover $\|\nabla v\|_F = \|D^2 \phi\|_F$. If $e = [z, p] \subset \partial F$, then $\partial_t \partial_n \phi = 0$ on e , so $\partial_n v = (\partial_{nn} \phi) Rn$ with R the rotation by $-\pi/2$. Since τ_e is n_e rotated in a fixed sense for all edges,

$$\partial_n v \cdot \tau_e = \sigma \partial_{nn} \phi \quad \text{on every edge } e \subset \partial F \cap \Gamma_h, \quad \sigma \in \{\pm 1\} \text{ the same for all edges.} \quad (4)$$

So $\int_e \partial_n v = 0$ if and only if $\int_e \partial_{nn} \phi = 0$ (the normal component vanishes by Lemma 7.18), and $\mathcal{M}(v) = -\frac{\sigma}{2} \sum_e \int_e s^2 \partial_{nn} \phi$.

C^1 quintics on a fan. On T_j let L_j be the barycentric coordinate of z ; it is affine, $L_j(z) = 1$, and $L_j = 0$ on the line $p_{j-1}p_j$. Put $N_j(x) := p_j \times x$, a linear form vanishing on the ray through p_j .

Lemma 7.9 (Gluing). *Let ϕ be piecewise quintic on the fan with $\phi = |\nabla \phi| = 0$ on its boundary, and $\phi_j := \phi|_{T_j}$. Then $\phi_j = L_j^2(Q_j + C_j)$ with Q_j, C_j homogeneous of degrees 2 and 3. On the boundary rays, $Q_1 = aN_0^2$, $C_1 = N_0^2 \ell_1$, $Q_m = a'N_m^2$, $C_m = N_m^2 \ell_m$ with $a, a' \in \mathbb{R}$ and ℓ_1, ℓ_m linear. Write $L_{j+1} - L_j = \beta_j N_j$ ($\beta_j = 0$ iff p_{j-1}, p_j, p_{j+1} are collinear). Then ϕ is C^1 across the ray through p_j if and only if*

- (G1) $Q_j - Q_{j+1} \in \mathbb{R}N_j^2$;
- (G2) $C_j(p_j) = C_{j+1}(p_j)$;
- (G3) $\nabla(C_j - C_{j+1})(p_j) = 2\beta_j Q_j(p_j) \nabla N_j$;
- (G4) $\beta_j(Q_j(p_j) + C_j(p_j)) = 0$.

Proof. ϕ_j vanishes to first order on the line $p_{j-1}p_j$, so L_j^2 divides it; the quotient q_j is a cubic, and $q_j(0) = 0$, $\nabla q_j(0) = 0$ because ϕ and $\nabla \phi$ vanish at z and $L_j(z) = 1$. On the boundary ray through p_0 , N_0^2 divides q_1 , hence each homogeneous part. For the gluing, $L_{j+1} - L_j$ is linear and vanishes at p_j , so it is a multiple of N_j . Modulo N_j^2 , $L_{j+1}^2 \equiv L_j^2 + 2\beta_j N_j L_j$, so

$$\phi_j - \phi_{j+1} \equiv L_j^2(q_j - q_{j+1}) - 2\beta_j N_j L_j q_{j+1} \pmod{N_j^2}.$$

This vanishes iff $q_j - q_{j+1} = N_j g$ and $L_j g - 2\beta_j q_{j+1} = 0$ on the ray. Write $g = G_1 + G_2$ (degrees 1, 2). On the ray, $x = tp_j$, $L_j = 1 - t$, and the second condition reads $tG_1(p_j) + t^2(G_2(p_j) - G_1(p_j) - 2\beta_j Q_{j+1}(p_j)) - t^3(G_2(p_j) + 2\beta_j C_{j+1}(p_j)) = 0$. Hence $G_1(p_j) = 0$, which with $N_j \mid Q_j - Q_{j+1}$ is (G1). $N_j \mid C_j - C_{j+1}$ is (G2). $G_2(p_j) = 2\beta_j Q_j(p_j)$ is (G3), because $\nabla(N_j G_2)(p_j) = G_2(p_j) \nabla N_j$ and $Q_j(p_j) = Q_{j+1}(p_j)$ by (G1). Finally $G_2(p_j) = -2\beta_j C_{j+1}(p_j)$ gives (G4) with (G2). Every step reverses. $\square$

The witness. From now on $F = (0; p_0, p_1, p_2, p_3)$ is a fan of three triangles with $D_{01}, D_{12}, D_{23} > 0$. Put

$$A := D_{13}D_{23}, \quad A_3 := D_{01}D_{02}.$$

If $p_i = r_i(\cos \vartheta_i, \sin \vartheta_i)$ with $\vartheta_0 = 0$, $\vartheta_j = \alpha_1 + \dots + \alpha_j$, then

$$A = r_1 r_2 r_3^2 \sin(\alpha_2 + \alpha_3) \sin \alpha_3, \quad A_3 = r_0^2 r_1 r_2 \sin \alpha_1 \sin(\alpha_1 + \alpha_2), \quad (5)$$

so $A, A_3 > 0$ whenever $\alpha_1 + \alpha_2 < \pi$ and $\alpha_2 + \alpha_3 < \pi$.

Definition 7.10 (Witness). Let ℓ_1 be the linear form with $\ell_1(p_0) = -4A$, $\ell_1(p_1) = -A$, and ℓ_3 the one with $\ell_3(p_3) = -4A_3$, $\ell_3(p_2) = -A_3$. Let

$$Q_1 = AN_0^2, \quad Q_2 = AN_0^2 - \frac{D_{02}D_{03}D_{23}}{D_{12}}N_1^2, \quad Q_3 = A_3N_3^2, \quad C_1 = N_0^2\ell_1, \quad C_3 = N_3^2\ell_3,$$

and let C_2 be the cubic form with polar values

$$\begin{aligned} C_2(p_1, p_1, p_1) &= -AD_{01}^2, & C_2(p_2, p_2, p_2) &= -A_3D_{23}^2, \\ C_2(p_1, p_1, p_2) &= \frac{AD_{01}}{3}(6D_{12} - 5D_{02} + 2D_{01}), & C_2(p_1, p_2, p_2) &= \frac{A_3D_{23}}{3}(6D_{12} - 5D_{13} + 2D_{23}). \end{aligned}$$

Set $\phi_j := L_j^2(Q_j + C_j)$ on T_j and $w := \text{curl } \phi$. In the barycentric coordinates $(\lambda_0, \lambda_1, \lambda_2)$ of T_j ($\lambda_0 = L_j$ at z , λ_1 at p_{j-1} , λ_2 at p_j),

$$\begin{aligned} \phi_1 &= AD_{01}^2 \lambda_0^2 \lambda_2^2 (\lambda_0 - 3\lambda_1), & \phi_3 &= A_3 D_{23}^2 \lambda_0^2 \lambda_1^2 (\lambda_0 - 3\lambda_2), \\ \phi_2 &= \lambda_0^2 \left[AD_{01}^2 \lambda_1^2 (\lambda_0 + \lambda_2) + A_3 D_{23}^2 \lambda_2^2 (\lambda_0 + \lambda_1) + 2AD_{01}D_{02} \lambda_1 \lambda_2 (\lambda_0 + \lambda_1 + \lambda_2) \right. \\ &\quad \left. + AD_{01}(6D_{12} - 5D_{02} + 2D_{01}) \lambda_1^2 \lambda_2 + A_3 D_{23}(6D_{12} - 5D_{13} + 2D_{23}) \lambda_1 \lambda_2^2 \right]. \end{aligned}$$

Lemma 7.11 (The witness is admissible). $\phi \in C^1(F)$ is piecewise quintic with $\phi = |\nabla \phi| = 0$ on ∂F . On both outer rays $\int \partial_{nn} \phi = 0$, and

$$\int_{[0, p_0]} s^2 \partial_{nn} \phi \, ds = \frac{|p_0|^5 A}{30}, \quad \int_{[0, p_3]} s^2 \partial_{nn} \phi \, ds = \frac{|p_3|^5 A_3}{30}.$$

Hence w is a continuous, divergence-free, piecewise P_4 field vanishing on ∂F . Extended by zero it lies in Y_h^1 , both when F is the whole star ($m = 3$) and when $F = T_1 \cup T_2 \cup T_3$ with $m \geq 4$. In the first case $|\mathcal{M}(w)| = |p_0|^5 A + |p_3|^5 A_3|/60$. In the second, $|\mathcal{M}(w)| = |p_0|^5 |A|/60$.

Proof. Boundary. The factor L_j^2 gives first-order vanishing on the outer edges, and $N_0^2 \mid q_1, N_3^2 \mid q_3$ on the outer rays.

Ray through p_1 . (G1) holds by construction, and (G2)+(G4) read $C_1(p_1) = C_2(p_1) = -Q_1(p_1) = -AD_{01}^2$. This holds since $\ell_1(p_1) = -A$ and $N_0(p_1) = D_{01}$. (G3) is an identity between two vectors, so we test it with p_1 and with p_2 .

- The p_1 -component. By Euler's identity it reads $3(C_1 - C_2)(p_1) = 2\beta_1 Q_1(p_1) N_1(p_1)$. Both sides vanish, by (G2) and because $N_1(p_1) = 0$.
- The p_2 -component. It reads $3C_1(p_1, p_1, p_2) - 3C_2(p_1, p_1, p_2) = 2\beta_1 AD_{01}^2 D_{12}$. Here $3C_1(p_1, p_1, p_2) = D_{01}^2 \ell_1(p_2) + 2D_{01}D_{02} \ell_1(p_1) = AD_{01}(4D_{12} - 3D_{02})$, using $p_2 = (D_{02}p_1 - D_{12}p_0)/D_{01}$. Also $\beta_1 = (D_{02} - D_{12} - D_{01})/(D_{01}D_{12})$, obtained by evaluating $L_2 - L_1$ at p_2 .

Substituting gives exactly the prescribed $C_2(p_1, p_1, p_2)$.

Ray through p_2 . This is symmetric. (G1) is the identity

$$AN_0^2 - \frac{D_{02}D_{03}D_{23}}{D_{12}}N_1^2 + \frac{D_{01}D_{03}D_{13}}{D_{12}}N_2^2 - A_3N_3^2 \equiv 0, \quad (6)$$

a three-term Plücker relation among the squares of four linear forms in two variables. In angular form it is $\sum_j N_j^2 / (r_j^2 \prod_{k \neq j} \sin(\vartheta_j - \vartheta_k)) = 0$.

Means and moments. On $[0, p_0]$ we have $\phi_1 = L_1^2 N_0^2 (A + \ell_1)$ and $N_0 = 0$. With n the unit normal, $\partial_{nn} \phi = 2|\nabla N_0|^2 L_1^2 (A + \ell_1) = 2|p_0|^2 L_1^2 (A + \ell_1)$. At $x = tp_0$ this is $2|p_0|^2 A(1-t)^2(1-4t)$, because $\ell_1(tp_0) = -4At$. Now $\int_0^1 (1-t)^2(1-4t) dt = 0$ and $\int_0^1 (t - \frac{1}{2})^2(1-t)^2(1-4t) dt = \frac{1}{60}$, and $ds = |p_0| dt$, $s = |p_0|(t - \frac{1}{2})$. The ray $[0, p_3]$ is symmetric. The formula for $\mathcal{M}$ follows from (4). By (G), the only chords meeting F are $[0, p_0]$ and, when $m = 3$, $[0, p_3]$; on all others the trace of $\partial_n w$ from T_e vanishes.

All identities in this proof are verified as identities of rational functions of the free coordinates of p_1, p_2, p_3 (with $p_0 = (1, 0)$, which is no loss by similarity) in `m3.certify.py` symbolic. The barycentric form is checked there as well. $\square$

Reflection. Reflect the fan in a line through z and reverse the order of the rays. Then $D_{ij} \mapsto D_{3-j,3-i}$, so $A \leftrightarrow A_3$, $N_1 \leftrightarrow N_3$ and N_2 is unchanged in Lemma 7.12 below. Hence the witness data are reflection invariant; this is checked exactly in the log, including a fan with $\beta_1 = 0$.

Lemma 7.12 (The norm). *For a polynomial f in the barycentrics of a triangle T with angles g_0, g_1, g_2 ,*

$$\|D^2 f\|_{L^2(T)}^2 = \frac{P_f(\cot g_0, \cot g_1, \cot g_2)}{2|T|}$$

with P_f a quadratic form with rational coefficients. This follows from $2|T|\nabla\lambda_a \cdot \nabla\lambda_a = \cot g_b + \cot g_c$, $2|T|\nabla\lambda_a \cdot \nabla\lambda_b = -\cot g_c$ and $\int_T \lambda^\alpha = 2|T|\alpha!/(|\alpha|+2)!$. Consequently

$$\begin{aligned} \|\nabla w\|^2 &= N_1 + N_2 + N_3, & N_1 &= A^2 D_{01}^3 P_1(\cot T_1), & N_3 &= A_3^2 D_{23}^3 P_3(\cot T_3), \\ N_2 &= \frac{k^\top K(\cot T_2) k}{D_{12}}, \end{aligned}$$

$$k = (AD_{01}^2, A_3 D_{23}^2, 2AD_{01}D_{02}, AD_{01}(6D_{12} - 5D_{02} + 2D_{01}), A_3 D_{23}(6D_{12} - 5D_{13} + 2D_{23})),$$

where $\cot T_j$ lists the cotangents at (z, p_{j-1}, p_j) . The forms are

$$P_1 = \frac{19}{70}k_0^2 + \frac{1}{10}k_0k_1 + \frac{17}{35}k_0k_2 + \frac{3}{70}k_1^2 + \frac{1}{7}k_1k_2 + \frac{3}{10}k_2^2,$$

$P_3 = P_1$ with $k_1 \leftrightarrow k_2$, and K is the 5×5 Gram matrix of the five bracketed terms of ϕ_2 printed in the log. For a star with $m = 3$ put

$$\kappa_w := \frac{|p_0|^5 A + |p_3|^5 A_3}{60 \max(|p_0|, |p_3|)^2 \sqrt{N_1 + N_2 + N_3}},$$

and for $m \geq 4$ put $\kappa_w := |p_0|^5 A / (60|p_0|^2 \sqrt{N_1 + N_2 + N_3})$. Then $w/\|\nabla w\|$, with sign chosen, satisfies (M3)(iii) with constant κ_w .

The formula is checked in the log against direct exact integration of $|D^2\phi|^2$ on random rational fans.

Lemma 7.13 (Uniqueness for $m = 3$). *Let $m = 3$, $\beta_1\beta_2 \neq 0$, and suppose no two of the rays through $p_0, \dots, p_3$ lie on one line, except possibly those through p_0 and p_3 . Then $Y^1(\omega_z) := \{v \in Y_h^1 : \text{supp } v \subset \omega_z\}$ is spanned by w . In particular the star constant is $\kappa_z = \kappa_w$ (normalised by $|e_z|$), and $\kappa_z = 0$ iff $|p_0|^5 A + |p_3|^5 A_3 = 0$.*

Proof. By Lemma 7.9, the quadratic parts satisfy $aN_0^2 + c_1N_1^2 + c_2N_2^2 - a'N_3^2 = 0$. The squares of three pairwise independent linear forms in two variables are linearly independent. So this relation is unique up to scale, and it is (6) (if $N_3 \parallel N_0$, it forces $c_1 = c_2 = 0$, $a = a'$). If the quadratic part vanishes, (G4) with $\beta_j \neq 0$ gives $C_{j+1}(p_j) = C_j(p_j) = 0$, so each C_j is divisible by the forms of both of its rays: $C_1 = \alpha N_0^2 N_1$, $C_2 = N_1 N_2 \mu$, $C_3 = \alpha_3 N_3^2 N_2$. The means on the two chords are $\propto \alpha N_1(p_0)$ and $\propto \alpha_3 N_2(p_3)$, so $\alpha = \alpha_3 = 0$. Then (G3) at p_1 and p_2 gives $\mu(p_1) = \mu(p_2) = 0$, so $\mu = 0$. Hence $\dim Y^1(\omega_z) \leq 1$, and $w \neq 0$. $\square$

The hypotheses of Lemma 7.13 are checked exactly for every star of the production meshes (Section 7.1.3). The exact dimension counts in the log agree: $\dim Y(\omega_z) = 2m - 3$ and $\dim Y^1(\omega_z) = 2m - 5$. Two cases behave differently: when $\beta_1 = 0$ the dimension jumps to 2, and the witness is still admissible; a fan of one or two triangles carries nothing.

Remark 7.14 (The hypothesis $\alpha_1 + \alpha_2, \alpha_2 + \alpha_3 < \pi$ is needed). If $\alpha_1 + \alpha_2 > \pi$, then $A_3 < 0 < A$ and the two terms of $\mathcal{M}(w)$ compete. The log exhibits a segment of $m = 3$ stars on which $|p_0|^5 A + |p_3|^5 A_3$ changes sign, with $\beta_1\beta_2 \neq 0$. At the zero, by Lemma 7.13, no field supported in the star satisfies (M3). Such stars have $\Theta_z > \pi + \alpha_3$, which (A_δ) excludes. The coarsest production mesh $N = 8$ has $\alpha_1 + \alpha_2 > \pi$ at every vertex ($\Theta_z = 225^\circ$), but there the sum is still positive: $\kappa_z \geq 0.00287$.

Qualitative statement and compactness.

Definition 7.15 (Families). $F_3(\theta_{\min}, \delta)$ is the set of three-triangle fans with all nine angles $\geq \theta_{\min}$ and $|\alpha_1 + \alpha_2 + \alpha_3 - \pi| \leq \delta$. $F_s(\theta_{\min}, \delta)$ is the set with all nine angles $\geq \theta_{\min}$ and $\alpha_1 + \alpha_2 + \alpha_3 \leq \pi + \delta - \theta_{\min}$. Both are taken modulo similarity, normalised by $|p_0| = 1$ and parametrised by $(\alpha_j, \beta_j)_{j=1}^3$. They are compact.

Under $(M0_\theta)$, $(M1')$, (A_δ) the star is in F_3 if $m = 3$. If $m \geq 4$, then $T_1 \cup T_2 \cup T_3$ (numbered from e_z , after a reflection if necessary) is in F_s , since the remaining $m - 3 \geq 1$ angles at z are $\geq \theta_{\min}$. On both families every pair sum $\alpha_j + \alpha_{j+1}$ lies in $[2\theta_{\min}, \pi + \delta - \theta_{\min}] \subset (0, \pi)$, so $A, A_3 > 0$ by (5). Hence $\mathcal{M}(w) \neq 0$, $w \neq 0$ and $\kappa_w > 0$. κ_w is a continuous function of (α_j, β_j) , since D_{01}, D_{12}, D_{23} stay positive. So

$$\kappa_*(\theta_{\min}, \delta) := \min\left(\min_{F_3} \kappa_w, \min_{F_s} \kappa_w\right) > 0.$$

By the reflection invariance, $\min_{F_3} \kappa_w = \min_{F_3 \cap \{|p_3| \leq |p_0|\}} \kappa_w$, and on the latter set κ_w is normalised by $|p_0|$.

Proof of Theorem 7.8, main part. Take $v_z = \pm w / \|\nabla w\|$ on ω_z . For $m = 3$, κ_w is normalised by the longer chord, so (iii) holds whichever chord is e_z . (i) holds by construction, and (iii) by Lemmas 7.11 and 7.12. For (ii), ω_z lies within $\max_j r_j \leq (\sin \theta_{\min})^{-3} |e_z|$ of z (law of sines, three triangles), and a triangle lies in at most three patches, one per vertex. $\square$

A decomposed bound. For the certificates and for (a) we bound κ_w from below by an expression whose factors each involve only one or two triangles. Normalise $|p_0| = 1$. Let $q_j := r_j / r_{j-1} = \sin \beta_j / \sin \gamma_j$, $s_j := \sin \alpha_j$, $s_{12} := \sin(\alpha_1 + \alpha_2)$, $s_{23} := \sin(\alpha_2 + \alpha_3)$ and $\rho := s_1 s_{12} / (s_{23} s_3)$. Write $\|u\|_K^2 = u^\top K(\cot T_2)u$.

Lemma 7.16 (Decomposition). *On $F_3 \cap \{r_3 \leq 1\}$ and on F_s ,*

$$\kappa_w \geq \frac{b}{\sqrt{3600(F_1 + (\sqrt{U} + \sqrt{W})^2 + \Phi_3)}},$$

with $b = 1 + r_3^3 \rho$ on F_3 and $b = 1$ on F_s , where

$$\begin{aligned} F_1 &= (q_1 s_1)^3 P_1(\cot T_1), & U &= \frac{q_1^2}{q_2 s_2} \|\tilde{u}\|_K^2, & W &= \frac{\rho^2 q_1^2 q_2}{s_2} \|\tilde{w}\|_K^2, \\ \Phi_3 &= \frac{\rho^2 (q_1 q_2)^2 s_3^3}{q_3} P_3(\cot T_3), & \tilde{u} &= (s_1^2, 0, 2s_1 q_2 s_{12}, s_1(6q_1 q_2 s_2 - 5q_2 s_{12} + 2s_1), 0), \\ \tilde{w} &= (0, q_2 s_3^2, 0, 0, s_3(6s_2/q_3 - 5s_{23} + 2q_2 s_3)). \end{aligned}$$

Proof. By (5), $A = r_1 r_2 X$ with $X = r_3^2 s_{23} s_3$ and $A_3 = r_1 r_2 X \rho / r_3^2$. So $|p_0|^5 A + |p_3|^5 A_3 = r_1 r_2 X (1 + r_3^3 \rho)$, and the common factor $(r_1 r_2)^2$ cancels between $\mathcal{M}^2$ and $N_1 + N_2 + N_3$. Dividing the latter by $(r_1 r_2 X)^2$ gives:

- N_1 gives F_1 ($D_{01} = q_1 s_1$) and N_3 gives Φ_3 .
- $N_2 = \|k\|_K^2 / D_{12}$. Substituting $D_{01} = q_1 s_1$, $D_{12} = q_1^2 q_2 s_2$, $D_{02} = q_1 q_2 s_{12}$, $D_{13} = r_1 r_3 s_{23}$, $D_{23} = r_2 r_3 s_3$ gives $k / (r_1 r_2 X) = q_1^2 (\tilde{u} + \rho q_2 \tilde{w})$. Minkowski's inequality in the seminorm $\|\cdot\|_K$ (K is a Gram matrix) then gives $\sqrt{N_2} / (r_1 r_2 X) \leq \sqrt{U} + \sqrt{W}$.

On $F_3 \cap \{r_3 \leq 1\}$ the normalising chord is $|p_0| = 1$. $\square$

Corollary 7.17 (Elementary bound). *Let $\sigma = \sin \theta_{\min}$, $c = \cot \theta_{\min}$, $\sigma_* = \min(\sin 2\theta_{\min}, \sin(\theta_{\min} - \delta))$, $Q = 1/\sigma$, $R = 1/(\sigma \sigma_*)$. For a polynomial p let $\|p\|_1$ be the sum of the absolute values of its coefficients, and $\kappa_{ij} := \|K_{ij}\|_1$. Then $\kappa_*(\theta_{\min}, \delta) \geq \kappa_{\text{el}} := (3600(\bar{F}_1 + (\sqrt{U} + \sqrt{W})^2 + \bar{\Phi}_3))^{-1/2}$,*

with

$$\begin{aligned}\bar{F}_1 &= Q^3 \|P_1\|_1 c^2, & \bar{U} &= Q^4 c^2 \sum_{i,j} \bar{u}_i \kappa_{ij} \bar{u}_j, \\ \bar{W} &= R^2 Q^4 c^2 \sum_{i,j} \bar{w}_i \kappa_{ij} \bar{w}_j, & \bar{\Phi}_3 &= R^2 Q^5 \|P_3\|_1 c^2, \\ \bar{u} &= (1, 0, 2Q, 6Q^2 + 5Q + 2, 0), & \bar{w} &= (0, Q, 0, 0, 8Q + 5).\end{aligned}$$

For instance $\kappa_{\text{el}}(30^\circ, 10^\circ) = 5.1 \cdot 10^{-5}$ and $\kappa_{\text{el}}(20^\circ, 15^\circ) = 2.4 \cdot 10^{-6}$ (log).

Proof. Every angle lies in $[\theta_{\min}, \pi - 2\theta_{\min}]$, so $\sin \geq \sigma$ and $|\cot| \leq c$. Hence $q_j \in [\sigma, 1/\sigma]$, and $\rho \leq R$ because the pair sums lie in $[2\theta_{\min}, \pi + \delta - \theta_{\min}]$. Also $|P(k)| \leq \|P\|_1 c^2$ for the quadratic forms involved. Insert these into Lemma 7.16 with $b \geq 1$. This uses only $A_3 \geq 0$, not $r_3 \leq 1$, so it bounds κ_w normalised by $|p_0|$ on all of F_3 and F_s ; for F_3 use reflection when $e_z = [z, p_3]$. $\square$

The elementary bound is valid for all $\theta_{\min} \leq \pi/3$ and $\delta < \theta_{\min}$. It is pessimistic: about 25 times below the certified value at $(30^\circ, 10^\circ)$ and about 125 times below at $(20^\circ, 15^\circ)$.

Interval certificates. For fixed $(\theta_{\min}, \delta)$ we certify $\kappa_w \geq \kappa_0$ on $F_3 \cap \{r_3 \leq 1\}$ and on F_s by branch and bound over boxes in $(\alpha_1, \alpha_2, S \text{ or } \alpha_3, \beta_1, \beta_2, \beta_3)$, where $S = \alpha_1 + \alpha_2 + \alpha_3$ for F_3 . Each box is first *contracted*: its upper and lower ends are moved inwards using the linear constraints of the family $(\alpha_j + \beta_j \leq \pi - \theta_{\min}; \theta_{\min} \leq S - \alpha_1 - \alpha_2 \leq \pi - 2\theta_{\min}$ for F_3 ; $\sum \alpha_j \leq \pi + \delta - \theta_{\min}$ for F_s). The new ends are rounded outward, so the contracted box still contains every point of the family that the original box contained. A box is then

- *discarded* if it provably misses the family: it is empty after contraction, or some $\gamma_j < \theta_{\min}$ on the whole box, or $\alpha_3 < \theta_{\min}$, or S out of range;
- *covered by symmetry* (F_3 only) if $r_3 > 1$ on the whole box, since its reflection has $r_3 < 1$;
- *certified* if an interval evaluation of Lemma 7.16 gives $b^2 \geq 3600\kappa_0^2 (F_1 + (\sqrt{U} + \sqrt{W})^2 + \Phi_3)$ on the whole box;
- *bisected* along its widest side otherwise.

Rigour rests on the following.

- All arithmetic is in IEEE double precision with every result widened outward by one ulp; basic operations are correctly rounded.
- $\sin$ and $\cos$ at box endpoints are Taylor polynomials of degree 41 evaluated in interval arithmetic, plus the Lagrange remainder. They are extended to intervals by monotonicity, including the extrema at $\pi/2$ and $3\pi/2$.
- $\cot$ is enclosed by monotonicity on $(0, \pi)$. q_j is enclosed by its exact corner values when γ_j stays on one side of $\pi/2$, where it is monotone in α_j and β_j .
- π is enclosed by the two doubles adjacent to it.
- The rational coefficients of P_1, P_3, K come from exact SymPy derivations and are enclosed in outward-rounded doubles.

Results (log, section 5):

family	certified $\kappa_w \geq$	boxes (cert./disc./sym./total)	float min.
$F_3(30^\circ, 10^\circ)$	0.0020	203150/154/6503/419613	0.002558
$F_s(30^\circ, 10^\circ)$	0.0012	209555/0/0/419109	0.001400
$F_3(20^\circ, 15^\circ)$	0.0005	1326711/9699/38333/2749485	0.000880
$F_s(20^\circ, 15^\circ)$	0.0003	452009/0/0/904017	0.000395

As a negative control, the same code with κ_0 set 5% above the float minimum of the bound on $F_s(30^\circ, 10^\circ)$ does not certify. It is refuted when a box of width $< 10^{-9}$ cannot be certified.

The production meshes. For the meshes of Section 16 we evaluate κ_w exactly. The double-precision vertex coordinates are read as exact rationals, D_{ij} and all cotangents are rational, and $|p_0|, |p_3|$ are enclosed by rational square roots. We check exactly that $D_{01}, D_{12}, D_{23} > 0$, $D_{02}D_{13} \neq 0$ and $\beta_1\beta_2 \neq 0$ at every boundary vertex. By Lemma 7.13 the resulting numbers are therefore the exact star constants. They reproduce the floating-point Table 1 to all printed digits, for example $\min_z \kappa_z = 0.002879$ ($N = 8$), 0.005551 ($N = 16$), 0.003663 ($N = 64$), 0.002876 ($N = 1024$). The minimum over both choices of e_z is ≥ 0.00287 on all levels.

Computation: `code/m3_certify.py all; notes/v6/m3/recert.py; output logs/m3_certify.log, notes/v6/m3/recert.30_10.log.`

The interval code is hand-written outward-rounded double arithmetic, not a verified library; Theorem 7.8 itself (positivity of κ_*) does not depend on it.

What is not covered. The certified numerical constants are for $(\theta_{\min}, \delta)$ as listed; for other values Theorem 7.8 gives $\kappa_* > 0$ by compactness and the explicit, pessimistic κ_{el} . The theorem needs $\delta < \theta_{\min}$, which (S) guarantees; coarse meshes with $\Theta_z - \pi \geq \min(\alpha_1, \alpha_3)$ need the star-by-star check of Section 7.1.3. By Remark 7.14 (M3) can fail on such stars for star-supported fields. For $m \geq 4$ we use only three of the m triangles, which is enough but not optimal.

7.1.4. Three lemmas.

Lemma 7.18. *If $v \in Y_h$ and $e \in \mathcal{E}_h^\Gamma$, then $\partial_n v \cdot n_e = 0$ on e .*

Proof. $v|_{T_e}$ is a polynomial that vanishes on e , so $\partial_\tau v = 0$ on e . Then $0 = \operatorname{div} v = \partial_\tau(v \cdot \tau_e) + \partial_n(v \cdot n_e)$ on e . $\square$

Lemma 7.19 (The error identity on Y_h). *Let u_h be the velocity of CNS* ($\gamma \geq \gamma_2$), or of GS ($\gamma \geq \gamma_0$, any G), and let $e_u := \tilde{u} - u_h$. Then for all $v \in Y_h$,*

$$a_h(e_u, v) - \langle e_u, \partial_n v \rangle = -\langle \tilde{u}, \partial_n v \rangle - (F, v).$$

Proof. For CNS* use (3): $B^*(e_p, v) = -(e_p, \operatorname{div} v) + \langle e_p - \bar{p}_\Gamma(e_p), v \cdot n \rangle = 0$, because $\operatorname{div} v = 0$ and $v = 0$ on Γ_h . For GS use Corollary 5.3: $\mathcal{R}(v) = -\langle \tilde{p} - c, v \cdot n \rangle = 0$. In both cases $N_h(e_u, v) = \mathcal{G}(v)$. Since $v = 0$ on Γ_h , $N_h(e_u, v) = a_h(e_u, v) - \langle e_u, \partial_n v \rangle$, and in $\mathcal{G}(v)$ the transposed and penalty terms vanish, leaving $\mathcal{G}(v) = -\langle \tilde{u}, \partial_n v \rangle - (F, v)$. $\square$

Neither the pressure error nor the penalty enters this identity. It is simply the discrete momentum equation $a_h(u_h, v) - \langle u_h, \partial_n v \rangle = (f, v)$ on Y_h , which $\tilde{u}$ violates by $-\langle \tilde{u}, \partial_n v \rangle$ (Nitsche's symmetric term applied to the nonzero trace of $\tilde{u}$ on the chords) and by (F, v) .

Lemma 7.20 (Trace control on Y_h^1). *Let C_{PT} be the constant in $\|w - \bar{w}_T\|_{L^2(e)} \leq C_{PT}|e|^{1/2}\|\nabla w\|_T$ for $e \subset \partial T$, where $\bar{w}_T$ is the mean of w over T ; it depends only on shape regularity. Then for $w \in H^1(\Omega_h)^2$ and $v \in Y_h^1$,*

$$|\langle w, \partial_n v \rangle| \leq C_{PT}C_I \|\nabla w\| \|\nabla v\|.$$

Proof. On each e , $\int_e \partial_n v = 0$, so $\int_e w \cdot \partial_n v = \int_e (w - \bar{w}_{T_e}) \cdot \partial_n v$. Hence

$$|\langle w, \partial_n v \rangle| \leq \sum_e C_{PT} \|\nabla w\|_{T_e} |e|^{1/2} \|\partial_n v\|_{L^2(e)} \leq C_{PT}C_I \sum_e \|\nabla w\|_{T_e} \|\nabla v\|_{T_e},$$

using the local form of Lemma 4.1. Finish with Cauchy–Schwarz; the T_e are distinct. $\square$

Corollary 7.21 (Abstract lower bound). *With $\Lambda_0 := 2 + C_{PT}C_I$ and $e_u = \tilde{u} - u_h$ as in Lemma 7.19,*

$$\Lambda_0 \|\nabla e_u\| \geq \sup_{0 \neq v \in Y_h^1} \frac{|\langle \tilde{u}, \partial_n v \rangle + (F, v)|}{\|\nabla v\|}.$$

Proof. $|a_h(e_u, v)| \leq 2 \|\nabla e_u\| \|\nabla v\|$; apply Lemmas 7.19 and 7.20. $\square$

Lemma 7.22 (Leading term). *For $v \in Y_h^1$,*

$$\left| \langle \tilde{u}, \partial_n v \rangle - \ell_W(v) \right| \leq Ch_\Gamma^{5/2} \|\nabla v\|, \quad \ell_W(v) := -\frac{1}{2} \sum_{e \in \mathcal{E}_h^\Gamma} W_e \int_e s^2 \partial_n v \cdot \tau_e ds,$$

and $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$.

Proof. (1) *Geometry.* Let $x \in e$ and let $x^* = x/|x|$ be its nearest point on Γ . Then $x = x^* + d n(x^*)$ with $d = 1 - |x|$. Put $\ell = |e|$. Since $|x|^2 = 1 - \ell^2/4 + s^2$,

$$d = \frac{1}{2} \left(\frac{\ell^2}{4} - s^2 \right) + O(\ell^4).$$

(2) *Taylor expansion.* Using $\tilde{u}(x^*) = 0$ and Lemma 3.4,

$$\tilde{u}(x) = d W(x^*) \tau(x^*) + O(d^2).$$

Moreover $\tau(x^*) \cdot \tau_e = 1 + O(s^2)$ and $W(x^*) = W_e + O(|s|)$. Hence

$$\tilde{u} \cdot \tau_e = \frac{1}{2} \left(\frac{\ell^2}{4} - s^2 \right) W_e + O(h_\Gamma^3) \quad \text{on } e.$$

(3) *Pairing on one edge.* By Lemma 7.18, $\tilde{u} \cdot \partial_n v = (\tilde{u} \cdot \tau_e)(\partial_n v \cdot \tau_e)$ on e . The constant part $\frac{\ell^2}{8} W_e$ integrates to zero against $\partial_n v \cdot \tau_e$, whose mean vanishes because $v \in Y_h^1$. Therefore

$$\int_e \tilde{u} \cdot \partial_n v = -\frac{1}{2} W_e \int_e s^2 \partial_n v \cdot \tau_e + O(h_\Gamma^3) \|\partial_n v\|_{L^1(e)}.$$

(4) *Summation.* $\sum_e \|\partial_n v\|_{L^1(e)} \leq \sum_e |e|^{1/2} \|\partial_n v\|_{L^2(e)} \leq C_I (\#\mathcal{E}_h^\Gamma)^{1/2} \|\nabla v\| \leq Ch_\Gamma^{-1/2} \|\nabla v\|$.

(5) *The F term.* This is Lemma 3.3. $\square$

Lemma 7.23 (Assembling the test field). *Assume (M3), $W \not\equiv 0$, and $h_\Gamma \leq h_1$, where h_1 depends on $\|W\|_{W^{1,\infty}(\Gamma)}/\|W\|_{L^2(\Gamma)}$, on (M0), on C_I , and on the constants $\kappa_0, C_s, K_{\text{ov}}$ of (M3). Then $v := \sum_z W(z) v_z \in Y_h^1$ satisfies*

$$\ell_W(v) \geq \frac{1}{2} \kappa_0 c_0^{5/2} K_{\text{ov}}^{-1/2} \|W\|_{L^2(\Gamma)} h_\Gamma^{3/2} \|\nabla v\|.$$

Proof. (1) *Norm.* By bounded overlap, $\|\nabla v\|^2 \leq K_{\text{ov}} \sum_z W(z)^2$.

(2) *Splitting ℓ_W .* Write $m_{z,e} := -\frac{1}{2} \int_e s^2 \partial_n v_z \cdot \tau_e$, so that $\sum_e m_{z,e} = \mathcal{M}(v_z)$. Then $|m_{z,e}| \leq \frac{h_\Gamma^2}{8} C_I$, and $m_{z,e} \neq 0$ only for the boundedly many edges e in ω_z . For those, $|W_e - W(z)| \leq Ch_\Gamma \|W\|_{W^{1,\infty}}$. Hence

$$\ell_W(v) = \sum_z W(z) \sum_e W_e m_{z,e} \geq \sum_z W(z)^2 \mathcal{M}(v_z) - Ch_\Gamma^3 \|W\|_{W^{1,\infty}} \sum_z |W(z)|.$$

(3) *Lower bound.* By (M3)(iii) and (M0), $\mathcal{M}(v_z) \geq \kappa_0 c_0^2 h_\Gamma^2$. Moreover $\sum_z |W(z)| \leq \#\mathcal{E}_h^\Gamma \|W\|_\infty \leq Ch_\Gamma^{-1} \|W\|_\infty$.

(4) *Riemann sum.* Let $\Sigma_W := \sum_z W(z)^2 |e_z|$. Since $z \mapsto e_z$ is a bijection onto the edges, this is a Riemann sum for $\int_\Gamma W^2$, and $|\Sigma_W - \|W\|_{L^2(\Gamma)}^2| \leq Ch_\Gamma \|W\|_{W^{1,\infty}}^2$; and $\sum_z W(z)^2 \in [\Sigma_W/h_\Gamma, \Sigma_W/(c_0 h_\Gamma)]$.

(5) Combining,

$$\ell_W(v) \geq \kappa_0 c_0^2 h_\Gamma \Sigma_W - Ch_\Gamma^2 \|W\|_{W^{1,\infty}}^2, \quad \|\nabla v\| \leq (K_{\text{ov}} \Sigma_W / (c_0 h_\Gamma))^{1/2}.$$

For $h_\Gamma \leq h_1$ we have $\Sigma_W \geq \frac{3}{4} \|W\|_{L^2(\Gamma)}^2$, and the negative term is at most a quarter of the positive one. $\square$

7.1.5. The theorem.

Theorem 7.24 (H^1 lower bound). *Assume (M0)–(M2), (D) and (M3). By Theorem 7.8, (M3) holds for every $k \geq 4$ when $h_\Gamma \leq 2 \sin(\theta_{\min}/4)$, with $\kappa_0 = \kappa_*(\theta_{\min}, \theta_{\min}/2)$, $K_{\text{ov}} = 3$ and $C_s = (\sin \theta_{\min})^{-3}$. Let $W = \partial_n u \cdot \tau \not\equiv 0$ on Γ . Let u_h be the velocity of CNS* with $\gamma \geq \gamma_2$, or of GS with $\gamma \geq \gamma_0$ and any G . Then, for $h_\Gamma \leq h_1$,*

$$\|\nabla(\tilde{u} - u_h)\|_{\Omega_h} \geq c_{\text{LB}} \kappa_0 \|W\|_{L^2(\Gamma)} h_\Gamma^{3/2} - C h_\Gamma^2, \quad c_{\text{LB}} := \frac{c_0^{5/2}}{2\Lambda_0 K_{\text{ov}}^{1/2}}.$$

Here c_{LB} depends on c_0 , on K_{ov} from (M3), and on C_{PT} and C_I (hence on (M0)–(M2)); h_1 is as in Lemma 7.23; and C depends on $\|\tilde{u}\|_{W^{2,\infty}}$, $\|W\|_{W^{1,\infty}}$ and $\|F\|_\infty$. Neither depends on γ , μ , ρ , h_Ω , the pressure, or ε_h .

Proof. Take v from Lemma 7.23 in Corollary 7.21, and use Lemma 7.22:

$$\Lambda_0 \|\nabla e_u\| \geq \frac{\ell_W(v)}{\|\nabla v\|} - C h_\Gamma^{5/2} - C h_\Gamma^2.$$

Lemma 7.23 bounds the first term from below. $\square$

Corollary 7.25 (Sharpness of Theorem 7.2 and of Theorem 6.1 at $G = 0$). *Assume (M0)–(M2) (so (M3) holds for small h_Γ by Theorem 7.8), $W \not\equiv 0$, $\gamma \in [\gamma_2, \gamma_{\max}]$ and $\varepsilon_h \leq C h_\Gamma^{3/2}$ (for instance $h_\Omega^k \leq h_\Gamma^{3/2}$ and the flux condition of Theorem 7.2). Then for CNS*, and for GS when p is constant on Γ ,*

$$c h_\Gamma^{3/2} \leq \|\nabla(\tilde{u} - u_h)\| \leq \|\tilde{u} - u_h\| \leq C h_\Gamma^{3/2}.$$

So the H^1 rate is exactly $3/2$. For GS with $G > 0$, Theorems 6.6 and 7.24 together give

$$\|\nabla(\tilde{u} - u_h)\| \geq c \max((h_\Gamma/\gamma)G, h_\Gamma^{3/2})$$

in the regime of Theorem 6.6.

Remark 7.26. (1) *Only the chordal bump matters.* The lower bound is carried by the part of $\tilde{u}|_{\Gamma_h}$ that oscillates on each chord, $-\frac{1}{2}s^2 W_e$ minus its mean. It has amplitude h_Γ^2 and wavelength h_Γ , hence $H^{1/2}(\Gamma_h)$ -size $h_\Gamma^{3/2}$. No discrete velocity that is Nitsche-consistent on Y_h can reproduce it. The mean part $\frac{\ell_e^2}{12} W_e$ is smooth along Γ_h and contributes only $O(h_\Gamma^2)$. (2) *The terms that do not carry it.* The transposed-gradient term and the penalty term of $\mathcal{G}$ vanish identically on Y_h , as does every pressure term. The lower bound therefore holds for CNS* even though its pressure error is itself $O(h_\Gamma^{3/2})$. (3) *A method-independent statement.* Corollary 7.21 holds for any discrete velocity with $a_h(u_h, v) - \langle u_h, \partial_n v \rangle = (\tilde{f}, v)$ for all $v \in Y_h$. Every Nitsche variant with this volume form and zero boundary data, tested on Z_h , has this property.

7.1.6. *The route through the transposed-gradient term.* The natural first attempt tests with a field whose trace is odd on each chord, and uses $\mathcal{G}(v) \approx \langle s a_e, v \rangle$. We record what that route gives.

- (1) *The relevant trace is normal, not tangential.* Since $a_e = n(m_e^*) W_e$, we have $\langle s a_e, v \rangle = \sum_e W_e \int_e s v \cdot n_e + O(h_\Gamma^2) \|v\|_{L^1}$. An odd *tangential* trace pairs to zero at leading order.
- (2) *Incompressibility allows an odd normal trace.* For $v = \text{curl } \phi$, $v \cdot n_e = -\partial_\tau \phi$, so $\int_e v \cdot n_e = 0$ just says that ϕ takes equal values at the two ends of e . The identity $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$ constrains the normal *derivative*, not the trace. So there is no obstruction there.
- (3) *The penalty term reinforces, at relative order γh_Γ .* On e ,

$$\tilde{u} \cdot n_e = d W(x^*) \tau(x^*) \cdot n_e + O(h_\Gamma^4) \quad \text{and} \quad (\nabla \tilde{u})^\top n_e \cdot n_e = W(x^*) \tau(x^*) \cdot n_e + O(h_\Gamma^2),$$

where $\tau(x^*) \cdot n_e = \pm s + O(s^3)$. So on the normal trace the penalty term multiplies the transposed term pointwise by $1 + (\mu/h)d(s)$, with $0 \leq (\mu/h)d \leq \gamma h_\Gamma/8$. It is lower order for fixed γ , of the same sign, and becomes leading once $\gamma \gtrsim h_\Gamma^{-1}$. On the tangential trace the penalty pairs $\frac{1}{2}(\frac{\ell^2}{4} - s^2)W_e$ with $v \cdot \tau_e$. This is not lower order unless $v \cdot \tau_e$ is odd or zero; choosing $\partial_n \phi = 0$ on Γ_h removes it.

- (4) *The Nitsche symmetric term is not lower order.* $-\langle \tilde{u}, \partial_n v \rangle$ pairs the chordal bump with $\partial_n v \cdot \tau_e$, which for a first-layer field of amplitude A is of size A/h_Γ . It has the same order $h_\Gamma A$ as the transposed term. This is the term Theorem 7.24 uses.
- (5) *Where the route fails: the pressure.* A test field with $v \cdot n \neq 0$ lies outside Y_h . For CNS* the error equation then contains $\langle \xi - \bar{\xi}, v \cdot n \rangle$, which by Theorem 7.2 (steps 3–4) is only bounded by

$$C(c_0\gamma)^{-1/2}\beta^{-1}(\|e_u\| + h_\Gamma^{3/2})\|v\|.$$

That is the same order as the term one is trying to bound below, so it can cancel the lower bound unless γ is large compared with constants we do not control.

- (6) *What the route gives when it works.* For GS with $G = 0$, where $|\mathcal{R}(v)| \leq Ch_\Gamma^2\|v\|_{L^1(\Gamma_h)}$ is lower order, the route gives only $\|\tilde{u} - u_h\| \gtrsim h_\Gamma^{3/2}/(1 + \gamma^{1/2})$ in the *energy* norm. That lower bound may be carried entirely by the slip part $(\mu/h)^{1/2}\|e_u\|_{\Gamma_h}$. Numerically this part dominates: for GS on test A at $N = 64$, $\mu = 100$, it is 0.0795 against $\|\nabla e_u\| = 0.0182$. So it says nothing about H^1 .

7.1.7. *Numerical checks.* All computations use $k = 4$ and the meshes of Section 16. The script is `code/lower_bound_tests.py`.

TABLE 1. (M3) on the production meshes: star constants $\kappa_z := \mathcal{M}(v_z)/|e_z|^2$, with $\|\nabla v_z\| = 1$ and v_z supported in the three triangles at z (floating-point star computation; the exact witness of Section 7.1.3 reproduces every entry). All production levels $N = 8, \dots, 256$ were checked; selected rows are shown.

N	$\dim Y^1(\omega_z)$	$\min_z \kappa_z$	mean	$\max_z \kappa_z$
8	1	0.00288	0.0055	0.0080
16	1	0.00555	0.0109	0.0179
64	1	0.00366	0.0145	0.0252
256	1	0.00303	0.0149	0.0260
1024	1	0.00288	0.0149	0.0261

Computation: `code/lower_bound_tests.py stars 8,12,...,1024`; output `logs/lower_bound_stars.log`.

The minimum is attained in the diagonal directions of the square, where the first-layer triangles are most elongated.

Computation: `code/lower_bound_tests.py field 16,...,512`; output `logs/lower_bound_field.log`.

On the full spaces, for test A with $\mu = 100$ and $N = 16 \rightarrow 64$ (`logs/lower_bound_global.log`):

- The exact dual norm of $v \mapsto \langle \tilde{u}, \partial_n v \rangle$ on Y_h^1 has rates 1.12, 1.27, 1.35, 1.40; its ratio to $h_\Gamma^{3/2}$ increases from 0.086 to 0.115.
- It is between 14% and 24% of the actual CNS* error $\|\nabla(\tilde{u} - u_h)\|$.
- On Z_h , the energy-dual norms of $\mathcal{G}$ and of its transposed part decay at rates $1.70 \rightarrow 1.54$ and $1.61 \rightarrow 1.51$. So Lemma 3.5 is sharp on Z_h as well as on V_h^R .

TABLE 2. The assembled field $v = \sum_z W(z)v_z$ of Lemma 7.23: $\langle \tilde{u}, \partial_n v \rangle / (\|\nabla v\| h_\Gamma^{3/2})$, with the leading term $\ell_W(v)$ in parentheses.

N	test A	rate	test B ₁	rate
32	0.0606 (0.0605)	–	0.0184 (0.0181)	–
64	0.0651 (0.0651)	1.39	0.0204 (0.0203)	1.35
128	0.0665 (0.0665)	1.47	0.0209 (0.0209)	1.46
256	0.0669 (0.0669)	1.49	0.0211 (0.0211)	1.49
512	0.0670 (0.0670)	1.50	0.0212 (0.0212)	1.50

- The $\|\nabla \cdot\|$ -dual norm of $\mathcal{G}$ on Z_h decays only like h_Γ^1 . It is dominated by the penalty part, which is of size γh_Γ in that norm. The $h_\Gamma^{3/2}$ upper bound genuinely needs the mesh-dependent norm.

7.2. Flux-corrected data, a γ -uniform H^1 bound, and a growing penalty. This subsection does three things. It proves an H^1 estimate that is uniform in the penalty (Theorem 7.30). With it, it removes the proviso of Theorem 7.2 for flux-corrected data (Theorem 7.33). And it asks whether a mesh-dependent penalty can rescue the printed method (Corollary 7.34, Theorem 7.35). The computations are in Section 16.10.

Throughout, μ may depend on the mesh. All constants below are independent of $\gamma \geq \gamma_0$ (or $\gamma \geq \gamma_2$ for CNS*) unless displayed; this is the uniformity of Lemma 4.2. When we speak of a *power-law penalty* we assume that h/h_Γ and ρ are bounded (as on the meshes of Section 16) and that

$$\gamma = \frac{\mu h_\Gamma}{h} \asymp h_\Gamma^{-\alpha}, \quad \text{equivalently} \quad \mu = \mu_0 \left(\frac{h_{\text{ref}}}{h} \right)^\alpha, \quad \alpha \geq 0.$$

Scott's normalisation, μ fixed, is $\alpha = 0$; the choice $\mu \propto h^{-1/2}$ is $\alpha = \frac{1}{2}$, i.e. $\mu/h \asymp h^{-3/2}$.

Remark 7.27 (Larger μ only helps stability). If $\mu' \geq \mu \geq \mu_0$ then, with $\|\cdot\|_\mu$ the norm of Section 2 at penalty μ ,

$$N_h^{\mu'}(v, v) = N_h^\mu(v, v) + \frac{\mu' - \mu}{h} \|v\|_{\Gamma_h}^2 \geq c_1 \|v\|_\mu^2 + \frac{\mu' - \mu}{h} \|v\|_{\Gamma_h}^2 \geq \min(c_1, 1) \|v\|_{\mu'}^2,$$

and the continuity constant M of Lemma 4.2 does not depend on μ . The absorption condition $\gamma \geq \gamma_2$ of Theorem 7.2 is also monotone. So coercivity, continuity and the CNS* pressure absorption are uniform for all $\mu \geq \mu_0$; a growing penalty never threatens stability. Its costs are in accuracy in the energy norm (Theorem 7.35) and in conditioning (Proposition 7.37).

7.2.1. A γ -uniform H^1 estimate. Let $Y_h = Z_h \cap V_h^0$ as in Section 7.1. Let $b = \sum_{e \in \mathcal{E}_h^\Gamma} \beta_e n_e$ be the flux bubble of Lemma 4.6, step 3, with $\int_{\Gamma_h} b \cdot n = |\Gamma_h|/6$, and recall that every $z \in W_h$ has $\int_{\Gamma_h} z \cdot n = -m_R$, $m_R = \int_{\partial R} g_I \cdot \nu$, $|m_R| \leq Ch^{\sigma_k}$ (remark after Lemma 4.6).

Lemma 7.28 (Divergence-free lift of a trace). *Let t be continuous and piecewise P_k on Γ_h , with $\int_{\Gamma_h} t \cdot n = 0$. There is $L \in Z_h$ with $L|_{\Gamma_h} = t$ and*

$$\|\nabla L\| \leq C_L h_\Gamma^{-1/2} \|t\|_{L^2(\Gamma_h)}.$$

Proof. (1) Let $L_0 \in V_h$ take the nodal values of t at the Lagrange nodes on Γ_h and vanish at all other nodes. It is supported in the triangles that meet Γ_h ; for $h \leq h_*$ these do not meet ∂R , so $L_0 \in V_h^R$.

- (2) Each such triangle T has diameter $\asymp |e|$ for an adjacent $e \in \mathcal{E}_h^\Gamma$ (shape regularity, (M0)). The nodal values on e are bounded by $C|e|^{-1/2}\|t\|_{L^2(e)}$ (equivalence of norms on $P_k(e)$), and $\|\nabla\phi_i\|_{L^2(T)} \leq C$ for each nodal basis function (scale invariance in two dimensions). Summing over the boundedly overlapping triangles, $\|\nabla L_0\| \leq Ch_\Gamma^{-1/2}\|t\|_{\Gamma_h}$ (using $|e| \geq c_0 h_\Gamma$).
- (3) $q := \operatorname{div} L_0 \in \Pi_h$, $\|q\| \leq \sqrt{2}\|\nabla L_0\|$, and $\int_{\Omega_h} q = \int_{\Gamma_h} L_0 \cdot n = \int_{\Gamma_h} t \cdot n = 0$. Lemma 4.5 gives $v \in V_h^0$ with $\operatorname{div} v = q$, $\|\nabla v\| \leq \beta^{-1}\|q\|$. Put $L := L_0 - v$. $\square$

Lemma 7.29 (The discrete polygonal Dirichlet problem). *Put $a := -6m_R/|\Gamma_h|$ and $W_h^D := \{z \in W_h : z|_{\Gamma_h} = ab\}$. Then $W_h^D \neq \emptyset$, there is a unique $u^D \in W_h^D$ with*

$$a_h(u^D, v) = (\tilde{f}, v) \quad \forall v \in Y_h,$$

and

$$\|\nabla(\tilde{u} - u^D)\| \leq C(h_\Gamma^{3/2} + h^k + |m_R| h_\Gamma^{-1/2}).$$

Proof. (1) A member of W_h^D close to $\tilde{u}$. Let $t := I_h \tilde{u}|_{\Gamma_h} - ab$ and let L_0 be the nodal extension of t from step 1 of Lemma 7.28. By Lemma 3.3, $\|I_h \tilde{u}\|_{\Gamma_h} \leq \|\tilde{u}\|_{\Gamma_h} + Ch_\Gamma^{k+1} \leq Ch_\Gamma^2$, and $|a||b|_{\Gamma_h} \leq C|m_R|$, so $\|\nabla L_0\| \leq Ch_\Gamma^{-1/2}(h_\Gamma^2 + |m_R|)$. The field $z_0 := I_h \tilde{u} - L_0$ has trace g_I on ∂R and ab on Γ_h , and

$$\int_{\Omega_h} \operatorname{div} z_0 = \int_{\partial R} g_I \cdot \nu + \int_{\Gamma_h} ab \cdot n = m_R + a|\Gamma_h|/6 = 0.$$

Since $\|\operatorname{div} z_0\| \leq \|\operatorname{div}(I_h \tilde{u} - \tilde{u})\| + \sqrt{2}\|\nabla L_0\| \leq C(h^k + h_\Gamma^{3/2} + |m_R|h_\Gamma^{-1/2})$, Lemma 4.5 gives $v_B \in V_h^0$ with $\operatorname{div} v_B = \operatorname{div} z_0$ and the same bound. Then $z := z_0 - v_B \in W_h^D$ and $\|\nabla(\tilde{u} - z)\| \leq C(h^k + h_\Gamma^{3/2} + |m_R|h_\Gamma^{-1/2})$.

- (2) *Existence.* W_h^D is an affine space with direction Y_h , and a_h is coercive on Y_h (Lemma 3.2); so u^D exists and is unique.
- (3) *Galerkin orthogonality.* For $v \in Y_h$, as in Lemma 5.1 but with $v = 0$ on all of $\partial\Omega_h$ and $\operatorname{div} v = 0$: $a_h(\tilde{u}, v) = (-\Delta \tilde{u}, v) = (\tilde{f} - F - \nabla \tilde{p}, v) = (\tilde{f} - F, v)$. Hence $a_h(\tilde{u} - u^D, v) = -(F, v)$.
- (4) *Céa.* With z from step 1, $y := z - u^D \in Y_h$ and $a_h(y, y) = a_h(z - \tilde{u}, y) - (F, y)$, so by Lemmas 3.2 and 3.3, $\|\nabla y\| \leq \kappa(2\|\nabla(z - \tilde{u})\| + Ch_\Gamma^2)$. $\square$

Theorem 7.30 (H^1 error, uniformly in the penalty). *Let $\gamma \geq \gamma_0$, with no upper bound. Then GS satisfies, with C_p as in Theorem 6.4,*

$$\|\nabla(\tilde{u} - u_h)\| \leq C_p \frac{h_\Gamma}{\gamma} + C(h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2}).$$

For CNS* with any $\gamma \geq \gamma_2$,

$$\|\nabla(\tilde{u} - u_h)\| \leq C(h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2}).$$

Compared with Theorems 6.4 and 7.2, the terms $\gamma^{1/2}h_\Gamma^{3/2}$, $(\gamma h_\Gamma)^{1/2}h^k$ and $h^{\sigma_k}(\mu/h)^{1/2}$ are gone. In particular the H^1 statement of Theorem 7.2 needs no upper bound $\gamma_{\max}$, and its proviso $h_\Gamma \geq h^{2(\sigma_k - k)}$ is exactly the condition $h^{\sigma_k} h_\Gamma^{-1/2} \leq h^k$, independent of μ .

Proof. GS.

- (1) *Remove the traction defect.* Fix $z \in W_h$ with $\|\tilde{u} - z\| \leq C\varepsilon_h$ (Lemma 4.6) and split $z - u_h = \delta_R + \delta_G$ as in Theorem 6.4. Steps 3–6 of that proof use only coercivity, continuity and Lemma 6.3, all uniform in $\gamma \geq \gamma_0$, and give

$$\|\nabla \delta_R\| \leq C_p(h_\Gamma/\gamma + h_\Gamma^{3/2} + h^k).$$

Put $u'_h := u_h + \delta_R = z - \delta_G \in W_h$. By construction $N_h(\tilde{u} - u'_h, v) = \mathcal{G}(v)$ for all $v \in Z_h$.

- (2) *The identity on Y_h .* For $v \in Y_h$ the penalty and transposed terms of N_h and $\mathcal{G}$ vanish (Lemma 7.19), leaving $a_h(\tilde{u} - u'_h, v) - \langle \tilde{u} - u'_h, \partial_n v \rangle = -\langle \tilde{u}, \partial_n v \rangle - (F, v)$. With step 3 of Lemma 7.29,

$$a_h(u'_h, v) - \langle u'_h, \partial_n v \rangle = (\tilde{f}, v) = a_h(u^D, v) \quad \forall v \in Y_h.$$

- (3) *The trace of u'_h is small.* By Lemma 3.3 and step 2 of Theorem 6.4 ($\|\tilde{u} - u'_h\| \leq \|\tilde{u} - z\| + \|\delta_G\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$),

$$\|u'_h\|_{\Gamma_h} \leq \|\tilde{u}\|_{\Gamma_h} + (h/\mu)^{1/2} \|\tilde{u} - u'_h\| \leq C(h_\Gamma^2 + (h_\Gamma/\gamma)^{1/2} \varepsilon_h),$$

because $(h/\mu)^{1/2} = (h_\Gamma/\gamma)^{1/2}$ and $(h_\Gamma/\gamma)^{1/2} \gamma^{1/2} h_\Gamma^{3/2} = h_\Gamma^2$. Moreover $h_\Gamma^{-1/2} (h_\Gamma/\gamma)^{1/2} \varepsilon_h = \gamma^{-1/2} \varepsilon_h \leq C(h^k + h^{\sigma_k} h_\Gamma^{-1/2})$: the two γ -growing parts of ε_h are exactly compensated, since $\gamma^{-1/2} (\gamma h_\Gamma)^{1/2} = h_\Gamma^{1/2}$ and $\gamma^{-1/2} (\mu/h)^{1/2} = h_\Gamma^{-1/2}$.

- (4) *Compare with u^D .* $w := u'_h - u^D \in Z_h$ (both lie in W_h), with $w|_{\Gamma_h} = u'_h - ab$. Let L be the lift of Lemma 7.28 of this trace (its flux vanishes because $w \in Z_h$), so $\|\nabla L\| \leq C h_\Gamma^{-1/2} (\|u'_h\|_{\Gamma_h} + |m_R|)$. Then $y := w - L \in Y_h$ and, by step 2, $a_h(y, y) = \langle u'_h, \partial_n y \rangle - a_h(L, y)$. With Lemma 4.1 and $(\rho/h)^{1/2} \leq (c_0 h_\Gamma)^{-1/2}$,

$$\|\nabla w\| \leq \|\nabla L\| + \|\nabla y\| \leq C h_\Gamma^{-1/2} (\|u'_h\|_{\Gamma_h} + |m_R|) \leq C(h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2}).$$

- (5) *Assemble.* $\tilde{u} - u_h = (\tilde{u} - u'_h) + \delta_R = (\tilde{u} - u^D) - w + \delta_R$; apply Lemma 7.29 and steps 1 and 4. CNS*. Lemma 7.19 gives step 2 for u_h itself, and Theorem 7.2 ($\gamma \geq \gamma_2$, any size) gives step 3 for u_h . Steps 4–5 are unchanged, with $\delta_R = 0$. $\square$

The mechanism is the classical one for Nitsche's method as $\mu \rightarrow \infty$: on the fields that vanish on Γ_h the method is exactly the strongly imposed discrete Dirichlet problem, and the only coupling to the penalty is through the trace of the discrete solution, which the energy estimate makes small by the factor $(h/\mu)^{1/2}$. The H^1 seminorm never sees the penalty directly.

7.2.2. Flux-corrected outer data: Theorem 7.2 without the proviso. Let $\tilde{g}_I$ and $\tilde{W}_h$ be as in Definition 2.2. Then $\int_{\partial R} \tilde{g}_I \cdot \nu = 0$ exactly (numerically, $|\int_{\partial R} \tilde{g}_I \cdot \nu| \leq 1.3 \cdot 10^{-15}$ in every run of Section 16.10), and $\|\tilde{g}_I - g\|_{L^\infty(\partial R)} \leq C h^{k+1} + C |m_R| \leq C h^{k+1}$, since $|m_R| \leq C h^{\sigma_k}$ and $\sigma_k \geq k+1$; so the accuracy of the boundary data is unchanged. Every $z \in \tilde{W}_h$ has $\int_{\Gamma_h} z \cdot n = 0 = \int_{\Gamma_h} \tilde{u} \cdot n$: the obstruction of the remark after Lemma 4.6 is removed.

Proposition 7.31 (The flux floor). *For GS and CNS* with uncorrected data g_I and any μ ,*

$$\|\tilde{u} - u_h\| \geq \left(\frac{\mu}{h}\right)^{1/2} \|\tilde{u} - u_h\|_{\Gamma_h} \geq \left(\frac{\mu}{h}\right)^{1/2} \frac{|m_R|}{|\Gamma_h|^{1/2}},$$

with constant exactly 1. With flux-corrected data, $\int_{\Gamma_h} (\tilde{u} - u_h) \cdot n = 0$ and the second inequality is void.

Proof. $u_h \in W_h$, so $\int_{\Gamma_h} (\tilde{u} - u_h) \cdot n = 0 + m_R$; apply Cauchy–Schwarz. $\square$

Lemma 7.32 (Approximation with flux-corrected data). *Let $k \geq 4$. Then $\tilde{W}_h \neq \emptyset$ and*

$$\inf_{z \in \tilde{W}_h} \|\tilde{u} - z\| \leq C \tilde{\varepsilon}_h, \quad \tilde{\varepsilon}_h := (1 + (\gamma h_\Gamma)^{1/2}) h^k + (1 + \gamma^{1/2}) h_\Gamma^{\sigma_k - 1/2} + |m_R|.$$

Proof. Follow Lemma 4.6 with one change in step 1. Let ζ be a smooth cut-off equal to 1 near ∂R and 0 for $r \leq (1 + L)/2$, and put $w_h := I_h \tilde{u} - c I_h(\zeta x)$, $c := m_R/(8L^2)$. Then $w_h|_{\partial R} = \tilde{g}_I$,

$w_h = I_h \tilde{u}$ near Γ_h , and $\|c I_h(\zeta x)\| = |c| \|\nabla I_h(\zeta x)\| \leq C|m_R|$ (no boundary terms on Γ_h). In step 2 the net flux becomes

$$\tilde{m} := \int_{\Omega_h} \operatorname{div} w_h = \int_{\partial R} \tilde{g}_I \cdot \nu + \int_{\Gamma_h} I_h \tilde{u} \cdot n = \int_{\Gamma_h} (I_h \tilde{u} - \tilde{u}) \cdot n,$$

using $\int_{\Gamma_h} \tilde{u} \cdot n = -\int_{\partial R} g \cdot \nu = 0$. On each chord $\tilde{u}$ is smooth and $I_h \tilde{u}$ is its equispaced P_k interpolant, so the Newton–Cotes argument of step 2 gives $O(|e|^{\sigma_k+1})$ per chord and $|\tilde{m}| \leq Ch_\Gamma^{\sigma_k}$ (not h^{σ_k} : only the chords of Γ_h contribute). Step 3 then costs $C|\tilde{m}|(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) = C(1 + \gamma^{1/2})h_\Gamma^{\sigma_k-1/2}$, and $\|\operatorname{div} w_h\| \leq C(h^k + |m_R|)$. Steps 4–5 are unchanged. $\square$

The remaining term $(1 + \gamma^{1/2})h_\Gamma^{\sigma_k-1/2}$ is the flux of the *interpolation error on Γ_h* , and it is below $(1 + \gamma^{1/2})h_\Gamma^{k+1/2}$ since $\sigma_k \geq k + 1$. The method itself has no flux defect left (its velocity and $\tilde{u}$ both carry zero flux through Γ_h), so this term is an artefact of the comparison field, not a floor.

Theorem 7.33 (Theorem 7.2 without the proviso). *Let $\gamma \geq \gamma_2$. CNS* with flux-corrected data satisfies*

$$\|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \tilde{\varepsilon}_h], \quad \|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h^k),$$

the second for every $\gamma \geq \gamma_2$, with C independent of γ , and every $h_\Gamma \leq h$ (no relation between h_Γ and h is needed). For bounded γ also $\|\tilde{u} - u_h\| \leq C(h_\Gamma^{3/2} + h^k)$. The same replacement $\varepsilon_h \rightarrow \tilde{\varepsilon}_h$ holds in Theorems 6.1–6.4 for GS, and in Theorem 7.30 the term $h^{\sigma_k}h_\Gamma^{-1/2}$ becomes $h_\Gamma^{\sigma_k-1/2} + |m_R|$.

Proof. The consistency identities (Lemma 5.1, (3)) involve only test functions vanishing on ∂R and are unaffected. The proof of Theorem 7.2 uses the data only through E_h , now bounded by Lemma 7.32. For bounded γ , $\tilde{\varepsilon}_h \leq C(h^k + h_\Gamma^{k+1/2} + h^{k+1})$. For the H^1 bound uniform in γ , repeat Theorem 7.30 with $\tilde{g}_I$ in place of g_I . In Lemma 7.29 now $a = 0$ (u^D vanishes on Γ_h), the comparison field of step 1 is $z_0 = I_h \tilde{u} - c I_h(\zeta x) - L_0$ with L_0 the nodal extension of $I_h \tilde{u}|_{\Gamma_h}$, and $\int_{\Omega_h} \operatorname{div} z_0 = \int_{\partial R} \tilde{g}_I \cdot \nu = 0$; so $\|\nabla(\tilde{u} - u^D)\| \leq C(h_\Gamma^{3/2} + h^k + |m_R|)$. In step 3 of Theorem 7.30, $\gamma^{-1/2}\tilde{\varepsilon}_h \leq C(h^k + h_\Gamma^{\sigma_k-1/2} + |m_R|)$, and in step 4 the term $|m_R|h_\Gamma^{-1/2}$ is absent. Altogether $\|\nabla(\tilde{u} - u_h)\| \leq C(h_\Gamma^{3/2} + h^k + h_\Gamma^{\sigma_k-1/2} + |m_R|) \leq C(h_\Gamma^{3/2} + h^k)$. The L^2 part uses the lift of $(g - \tilde{g}_I)|_{\partial R}$, of size $C(h^{k+1/2} + |m_R|)$. $\square$

With Theorem 7.24 (which is indifferent to the outer data), Corollary 7.25 now holds with no flux condition: for $W \not\equiv 0$ and small h_Γ ((M3) by Theorem 7.8), $ch_\Gamma^{3/2} \leq \|\nabla(\tilde{u} - u_h)\| \leq Ch_\Gamma^{3/2}$ whenever $h^k \leq h_\Gamma^{3/2}$, for every $h_\Gamma \leq h$.

7.2.3. A growing penalty for the printed method. What Theorems 6.1 and 6.4 give. Insert $\gamma \asymp h_\Gamma^{-\alpha}$ into Theorem 6.4: $\|\nabla(\tilde{u} - u_h)\| \lesssim h_\Gamma^{1+\alpha} + h_\Gamma^{3/2-\alpha/2} + \varepsilon_h$. The two terms balance at $\alpha = \frac{1}{3}$, so Theorem 6.4 by itself gives at best the H^1 rate $\frac{4}{3}$, and it *loses* rate for $\alpha > \frac{1}{3}$. The culprit is the term $\gamma^{1/2}h_\Gamma^{3/2}$, which Theorem 6.4 inherits from the energy estimate of the geometric part δ_G (Lemma 5.2, penalty term). In the energy norm this term is genuine (Theorem 7.35); in the H^1 seminorm it is not (Theorem 7.30).

Corollary 7.34 (Rescue of the printed method in H^1). *Let $\gamma \geq \gamma_0$ and $\gamma \geq ch_\Gamma^{-1/2}$, that is,*

$$\mu \geq ch h_\Gamma^{-3/2} \quad (\text{for } h \asymp h_\Gamma : \mu \gtrsim h^{-1/2}, \mu/h \gtrsim h^{-3/2}).$$

Then GS satisfies $\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C_p(h_\Gamma^{3/2} + h^k + h^{\sigma_k}h_\Gamma^{-1/2})$, and with flux-corrected data $\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C_p(h_\Gamma^{3/2} + h^k)$. Under the hypotheses of Theorem 7.24 ($W \not\equiv 0$; (M3) holds by

Theorem 7.8, $\|\nabla(\tilde{u} - u_h)\| \geq ch_\Gamma^{3/2}$ as well, so for $h^k \leq h_\Gamma^{3/2}$ (and, for uncorrected data, $h_\Gamma \geq h^{2(\sigma_k - k)}$) the H^1 rate of GS is exactly $\frac{3}{2}$, for general Stokes data. The L^2 part of the H^1 norm is handled by the lift E of Theorem 6.6(i).

Proof. Theorems 7.30 and 7.33 with $h_\Gamma/\gamma \leq c^{-1}h_\Gamma^{3/2}$; the lower bound is Theorem 7.24, which holds for GS with any $\gamma \geq \gamma_0$ and any G . $\square$

The energy norm cannot be rescued. Let $W = \partial_n u \cdot \tau$ be the wall shear (Section 7.1).

Theorem 7.35 (The penalty consistency error is real in the energy norm). *Let u_h be the velocity of GS ($\gamma \geq \gamma_0$) or of CNS* ($\gamma \geq \gamma_2$), and $W \not\equiv 0$. There are γ_1 and h_1 , depending on $\|W\|_{L^2(\Gamma)}$ and on the data, such that for $\gamma \geq \gamma_1$, $h_\Gamma \leq h_1$ and $h^k \leq h_\Gamma$,*

$$\|\tilde{u} - u_h\| \geq \left(\frac{\mu}{h}\right)^{1/2} \|\tilde{u} - u_h\|_{\Gamma_h} \geq c_E \|W\|_{L^2(\Gamma)} \gamma^{1/2} h_\Gamma^{3/2} - C(h_\Gamma/\gamma)^{1/2}, \quad c_E := \frac{c_0^2}{72}.$$

Proof. Let $e := \tilde{u} - u_h$.

- (1) *Tangential test field.* Let $\varphi_W := \pm \chi_1(r)(r-1)W(\theta)$, with χ_1 as in Section 6 and the sign chosen so that $w_W := \text{curl } \varphi_W = W\tau$ on Γ . As for v_φ , $v_W := \text{curl } \Pi_{k+1}\varphi_W \in Z_h$ and $\|v_W - w_W\|_{W^{1,\infty}} \leq Ch^k$. On a chord, $|x - x^*| \leq Ch_\Gamma^2$, so $v_W = W(x^*)\tau(x^*) + O(h_\Gamma^2 + h^k)$ and $v_W \cdot n_e = O(h_\Gamma)$; also $\|v_W\|_{\Gamma_h} \leq 2\|W\|_{L^2(\Gamma)}$ for small h .
- (2) *The discrete trace is small along v_W .* Test the discrete momentum equation with v_W ; the pressure term $(p_h, \text{div } v_W)$ vanishes. For GS,

$$\frac{\mu}{h} \langle u_h, v_W \rangle = (\tilde{f}, v_W) - a_h(u_h, v_W) + \langle \partial_n u_h, v_W \rangle + \langle u_h, \partial_n v_W \rangle.$$

Here $\|\nabla u_h\| \leq C + \|e\|$, $\|\partial_n u_h\|_{\Gamma_h} \leq \|\partial_n \tilde{u}\|_{\Gamma_h} + (c_0 h_\Gamma)^{-1/2} \|e\|$ (the last part of $\|\cdot\|$), and $\|u_h\|_{\Gamma_h} \leq Ch_\Gamma^2 + \|e\|_{\Gamma_h}$. Hence $\frac{\mu}{h} |\langle u_h, v_W \rangle| \leq C(1 + h_\Gamma^{-1/2} \|e\|)$. For CNS* the extra term $\langle p_h - \bar{p}_\Gamma(p_h), v_W \cdot n \rangle$ is at most $C_I(\rho/h)^{1/2} \|p_h\| \cdot Ch_\Gamma |\Gamma_h|^{1/2} \leq Ch_\Gamma^{1/2}$, since $\|p_h\|$ is bounded by Theorem 7.2.

- (3) *The exact trace along v_W .* By Lemma 7.22, step 2, $\tilde{u} = dW(x^*)\tau(x^*) + O(d^2)$ on e with $d = \frac{1}{2}(|e|^2/4 - s^2) + O(|e|^4) \geq 0$, and $\int_e d ds = |e|^3/12 + O(|e|^5)$. Hence

$$\langle \tilde{u}, v_W \rangle = \sum_e W_e^2 \frac{|e|^3}{12} + O(h_\Gamma^3 + h_\Gamma^2 h^k) \geq \frac{c_0^2 h_\Gamma^2}{12} \sum_e W_e^2 |e| - Ch_\Gamma^3 \geq \frac{c_0^2 h_\Gamma^2}{12} (\|W\|_{L^2(\Gamma)}^2 - Ch_\Gamma),$$

using $|e| \geq c_0 h_\Gamma$, the Riemann sum of Lemma 7.23, step 4, and $h^k \leq h_\Gamma$.

- (4) *Close.* $\langle e, v_W \rangle = \langle \tilde{u}, v_W \rangle - \langle u_h, v_W \rangle$, and $\|e\| \geq (\mu/h)^{1/2} \|e\|_{\Gamma_h} \geq (\mu/h)^{1/2} \langle e, v_W \rangle / \|v_W\|_{\Gamma_h}$. With steps 2–3, $(\mu/h)^{1/2} h_\Gamma^2 = \gamma^{1/2} h_\Gamma^{3/2}$ and $(h/\mu)^{1/2} h_\Gamma^{-1/2} = \gamma^{-1/2}$,

$$\|e\| \geq \frac{c_0^2}{24} \|W\| \gamma^{1/2} h_\Gamma^{3/2} - C \gamma^{1/2} h_\Gamma^{5/2} - \frac{C}{\|W\|} \left((h_\Gamma/\gamma)^{1/2} + \gamma^{-1/2} \|e\| \right).$$

For $\gamma \geq \gamma_1 := (2C/\|W\|)^2$ the last term is at most $\frac{1}{2}\|e\|$; move it to the left, and take $h_\Gamma \leq h_1$ to absorb $C\gamma^{1/2} h_\Gamma^{5/2}$ into half of the first term. $\square$

Combined with Theorem 6.1 (upper bound, and lower bound $(2M)^{-1}(h_\Gamma/\gamma)^{1/2}G - C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$) this pins down the energy error of GS for $G > 0$, $W \not\equiv 0$:

$$\|\tilde{u} - u_h\| \asymp (h_\Gamma/\gamma)^{1/2} G + \gamma^{1/2} h_\Gamma^{3/2} \|W\| \quad \text{whenever } \gamma h_\Gamma \leq \lambda_0 \text{ or } \gamma h_\Gamma \geq \Lambda_0,$$

with λ_0 small and Λ_0 large, both depending on the data. In the window $\lambda_0 < \gamma h_\Gamma < \Lambda_0$ the upper bound is $\asymp h_\Gamma$ and we have no matching lower bound (the leak is normal and the geometric slip tangential, so we expect none is lost, but the constant in step 2 above is not controlled finely

enough to exclude a partial cancellation). The same theorem shows that for CNS^* , and for GS with $G = 0$, the energy error grows like $\gamma^{1/2}h_\Gamma^{3/2}$: bounded γ is optimal in the energy norm.

Corollary 7.36 (Rates of GS under a power-law penalty). *Let h/h_Γ and ρ be bounded, $\gamma \asymp h_\Gamma^{-\alpha}$ with $\alpha \geq 0$, $G > 0$, $W \neq 0$, $h^k \leq h_\Gamma^{3/2}$. Then, as $h_\Gamma \rightarrow 0$:*

	H^1 seminorm	energy norm $\ \cdot\ $
$0 \leq \alpha < \frac{1}{2}$	$\asymp h_\Gamma^{1+\alpha}$	$\asymp h_\Gamma^{(1+\alpha)/2}$
$\frac{1}{2} \leq \alpha < 1$	$\asymp h_\Gamma^{3/2}$	$\asymp h_\Gamma^{(1+\alpha)/2}$
$\alpha = 1$	$\asymp h_\Gamma^{3/2}$	$\lesssim h_\Gamma$ ($\asymp h_\Gamma$ unless γh_Γ lies in (λ_0, Λ_0))
$\alpha > 1$	$\asymp h_\Gamma^{3/2}$	$\asymp h_\Gamma^{(3-\alpha)/2}$

So the H^1 rate is $\min(1+\alpha, \frac{3}{2})$, optimal for every $\alpha \geq \frac{1}{2}$, and the energy rate is $\min(\frac{1+\alpha}{2}, \frac{3-\alpha}{2}) \leq 1$, with maximum 1 at $\alpha = 1$. No penalty scaling gives GS the energy rate $\frac{3}{2}$ of CNS^* .

Proof. H^1 , upper: Theorem 7.30. H^1 , lower: for $\alpha < \frac{1}{2}$, Theorem 6.6(ii); its hypotheses hold as $h_\Gamma \rightarrow 0$ because $X \asymp h_\Gamma^{1+\alpha} + h_\Gamma^{3/2-\alpha/2} + \varepsilon_h = o(h_\Gamma^{1/2})$ for $\alpha < 2$ and $h_\Gamma/\gamma = o(1)$. For $\alpha \geq \frac{1}{2}$, Theorem 7.24. Energy: Theorem 6.1 (both bounds; its lower bound is effective when $\gamma^{1/2}h_\Gamma^{3/2} = o((h_\Gamma/\gamma)^{1/2})$, i.e. $\alpha < 1$) and Theorem 7.35 (effective for $\alpha > 1$). The flux part of ε_h is $h^{\sigma_k}(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) = o(h_\Gamma^{3/2})$ for $\alpha < 2\sigma_k - 4$. $\square$

The price, in the energy norm, is the product $(h_\Gamma/\gamma)^{1/2} \cdot \gamma^{1/2}h_\Gamma^{3/2} = h_\Gamma^2$: the leak and the penalty consistency error cannot both be below h_Γ . In the slip norm $\|\cdot\|_{L^2(\Gamma_h)}$ the growing penalty is harmless: $\|\tilde{u} - u_h\|_{\Gamma_h} \lesssim (h_\Gamma/\gamma)G + h_\Gamma^2$.

7.2.4. *The price in conditioning.* Work in the Lagrange nodal basis of V_h (Euclidean norm $|\cdot|$ on the free coefficients) and an L^2 -orthonormal basis of Π_h on each element. Let A_μ be the matrix of N_h on V_h^R , B that of $(q, \text{div } v)$, $S_\mu := BA_\mu^{-1}B^\top$ the pressure Schur complement, and

$$K_\mu = \begin{pmatrix} A_\mu & B^\top \\ B & 0 \end{pmatrix}$$

the (symmetric form of the) GS saddle matrix. Then $q^\top S_\mu q = \sup_v (q, \text{div } v)^2 / N_h(v, v)$.

Proposition 7.37 (Conditioning). *Let $\mu \geq \mu_0$ and let $\mathcal{T}_h$ be quasi-uniform, $h_{\min} \asymp h$.*

- (i) $c(1+\gamma) \leq \lambda_{\max}(A_\mu) \leq C(1+\gamma)$ and $ch^2 \leq \lambda_{\min}(A_\mu) \leq Ch^2$, uniformly in μ . Hence $\kappa_2(A_\mu) \asymp (1+\gamma)h^{-2}$.
- (ii) $\lambda_{\max}(S_\mu) \leq 2/c_1$, and

$$\frac{c \min(\beta^2, 1)}{1 + \mu/h} \leq \lambda_{\min}(S_\mu) \leq \frac{|\Gamma_h|}{c_1 |\Omega_h|} \min\left(C_{\text{tr}}^2, \frac{h}{\mu}\right).$$

The small eigenvalue belongs to the constant pressure; on L_0^2 the spectrum stays above $c\beta^2$.

Hence $\kappa_2(S_\mu) \asymp 1 + \mu/h = 1 + \gamma/h_\Gamma$.

- (iii) *Every eigenvalue λ of K_μ satisfies $\min(\lambda_{\min}(A_\mu), \frac{1}{2}\lambda_{\min}(S_\mu)) \leq |\lambda| \leq \lambda_{\max}(A_\mu) + \|B\|$, so*

$$\kappa_2(K_\mu) \leq C(1+\gamma) \max(h^{-2}, 1 + \mu/h).$$

Proof. (i) $|v|^{-2}\|v\|^2$ is bounded by $C(1+\gamma)$: $\|\nabla v\|^2 + \sum_e |e| \|\partial_n v\|_e^2 \leq C|v|^2$ by scale invariance in two dimensions, and $(\mu/h)\|v\|_{\Gamma_h}^2 \leq (\mu/h)C \sum_e |e| \|v_e\|^2 \leq C\gamma|v|^2$. With $v^\top A_\mu v \leq M\|v\|^2$ this is the upper bound for $\lambda_{\max}$; the nodal function at a vertex of Γ_h gives the lower bound $v^\top A_\mu v \geq c_1(\mu/h)\|v\|_{\Gamma_h}^2 \geq c\gamma$. For $\lambda_{\min}$: $v^\top A_\mu v \geq c_1\|\nabla v\|^2 \geq c_1C_F^{-2}\|v\|^2 \geq ch^2|v|^2$; an interior

nodal function (vanishing on Γ_h , so the penalty does not see it) gives the upper bound by the standard argument for the stiffness matrix.

(ii) $(q, \operatorname{div} v)^2 \leq 2\|q\|^2 \|\nabla v\|^2 \leq (2/c_1)\|q\|^2 N_h(v, v)$. Upper bound for $\lambda_{\min}$: take $q \equiv |\Omega_h|^{-1/2}$; then $(q, \operatorname{div} v) = |\Omega_h|^{-1/2} \int_{\Gamma_h} v \cdot n$ (as $v = 0$ on ∂R), and $(\int_{\Gamma_h} v \cdot n)^2 \leq |\Gamma_h| \|v\|_{\Gamma_h}^2 \leq |\Gamma_h| \min(C_{\text{tr}}^2 \|\nabla v\|^2, (h/\mu) \|v\|^2)$. Lower bound: write $q = q_0 + \bar{q}$ with $\bar{q}$ the mean. Lemma 4.5 gives $v_0 \in V_h^0$ with $\operatorname{div} v_0 = q_0$, $N_h(v_0, v_0) = a_h(v_0, v_0) \leq \beta^{-2} \|q_0\|^2$, and $(\bar{q}, \operatorname{div} v_0) = 0$. Let $v_c \in V_h^R$ be a fixed smooth field (interpolated) with $\int_{\Omega_h} \operatorname{div} v_c = |\Omega_h|$, corrected by Lemma 4.5 so that $\operatorname{div} v_c \equiv 1$; then $N_h(v_c, v_c) \leq C(1 + \mu/h)$. Test with $v = v_0 + tv_c$, $t = \bar{q}/(1 + \mu/h)$: with $X = \|q_0\|^2$ and $Y = \|\bar{q}\|^2/(1 + \mu/h)$ we get $(q, \operatorname{div} v) = X + Y$ and $N_h(v, v) \leq C'(X + Y)$, where C' depends on β^{-2} and $|\Omega_h|$. So $q^T S_\mu q \geq (X + Y)/C' \geq \|q\|^2/(C'(1 + \mu/h))$ and, on L_0^2 , $\geq \|q_0\|^2/C'$.

(iii) Positive eigenvalues of a symmetric saddle matrix with A_μ positive definite are at least $\lambda_{\min}(A_\mu)$. For $\lambda < 0$ with eigenvector (x, y) , $y \neq 0$ and $B(A_\mu - \lambda)^{-1} B^T y = -\lambda y$. Since $(A_\mu - \lambda)^{-1} \succeq (1 + |\lambda|/\lambda_{\min}(A_\mu))^{-1} A_\mu^{-1}$, either $|\lambda| \geq \lambda_{\min}(A_\mu)$ or $|\lambda| \geq \frac{1}{2} \lambda_{\min}(S_\mu)$. The upper bound is $\|K_\mu\|$, and $\|B\| \leq C$ in these bases. $\square$

Consequences. Relative to a fixed γ , the penalty $\gamma \asymp h_\Gamma^{-\alpha}$ multiplies $\kappa(A_\mu)$ by $h_\Gamma^{-\alpha}$ (so $h^{-2} \rightarrow h^{-5/2}$ for the H^1 -optimal $\alpha = \frac{1}{2}$) and $\kappa(S_\mu)$ by the same factor ($h^{-1} \rightarrow h^{-3/2}$). For $\alpha \leq 1$ the bound (iii) for the saddle matrix is $\asymp (1 + \gamma)h^{-2}$ asymptotically; for $\alpha > 1$ the constant-pressure mode dominates and $\kappa(K_\mu) \lesssim \gamma^2/h$, quadratic in the penalty. On the meshes of Section 16 the pressure-mode branch is already the active one (Section 16.10), because $\lambda_{\min}(S_\mu) \approx (|\Gamma_h|/|\Omega_h|) h/\mu$ with constant ≈ 1 while $\lambda_{\min}(A_\mu) \approx 0.03h^2$.

7.3. The leak limit uniformly in the penalty. Theorem 6.10 identifies the limit $(\mu/h)\delta_R \rightarrow U$ only when $\gamma h_\Gamma \rightarrow 0$, and Remark 6.12 left the Dirichlet-dominated regime open. Here we close it: the convergence holds uniformly in $\mu \geq \mu_0$, at the *exact* H^1 rate $\frac{1}{2}$, and the lower bound holds for every penalty for which the discrete problem is solvable. We also prove that the H^1 constant converges at rate one, which was an empirical observation in Section 6.1.

Notation: $\lambda := \mu/h = \gamma/h_\Gamma$, $X := \lambda \delta_R$ with $\delta_R \in Z_h$ the solution of $N_h(\delta_R, v) = \mathcal{R}(v)$ on Z_h (Theorem 6.4, step 1), $W := X - \tilde{U}$, and $d_h := \|\nabla W\|_{\Omega_h}/\|\nabla U\|_\Omega$. On each $e \in \mathcal{E}_h^\Gamma$ put $g := -(\tilde{p} - \bar{p})n_e$; since $\int_{\Gamma_h} v \cdot n = 0$ on Z_h , $\mathcal{R}(v) = \langle g, v \rangle$ there. Put $\zeta := \tilde{U} - g = (\tilde{p} - \bar{p})n_e + \tilde{U}$; by step 3 of the proof of Theorem 6.10,

$$\zeta|_e = c_e s \tau_e + r_e, \quad c_e = \pm(p - \bar{p})(m_e^*), \quad |r_e| \leq Ch_\Gamma^2, \quad (7)$$

with a fixed orientation sign, so $\|\zeta\|_{\Gamma_h} \leq Ch_\Gamma$: the smooth leak flow, which is normal to the *arc*, has a tangential sawtooth of amplitude $O(h_\Gamma)$ on every chord. Let $\mathsf{T}_h := \{t \in C^0(\Gamma_h)^2 : t|_e \in P_k(e)^2, \int_{\Gamma_h} t \cdot n = 0\}$ and Π_T the $L^2(\Gamma_h)$ -orthogonal projection onto T_h . Finally

$$E_U^0 := h\|U\|_{H^2(\Omega)} + h^k + h_\Gamma^{\sigma_k - 1/2}.$$

The field $z_U \in Z_h$ of Lemma 6.9 satisfies the penalty-free bounds $\|\nabla(\tilde{U} - z_U)\| \leq CE_U^0$ and $\|\tilde{U} - z_U\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^{k+1}$: on Γ_h , $z_U = I_h \tilde{U} - ab$ with $|a| \leq Ch_\Gamma^{\sigma_k}$, and the γ -dependent parts of E_U come only from the penalty weight. In this subsection C depends on (M0)–(M2), k , L , $\|p - \bar{p}\|_{W^{1,\infty}}$ near Γ and norms of U ; never on $h, h_\Gamma, \mu, \gamma, \rho$. (M2) is used at *every* vertex: it is what makes the trace lift of Lemma 7.28 uniform. At a wall vertex in only two triangles the lift constant grows like h_Γ^{-1} (Section 11).

7.3.1. The mechanism.

Lemma 7.38 (Traces of Z_h ; the projection identity). $\{v|_{\Gamma_h} : v \in Z_h\} = \mathsf{T}_h$. With $t^* := \Pi_\mathsf{T} g$, $N_h(X, v) = \lambda \langle t^*, v \rangle$ for all $v \in Z_h$.

Proof. For $v \in Z_h$, $v|_{\Gamma_h}$ is continuous piecewise P_k with $\int_{\Gamma_h} v \cdot n = \int_{\Omega_h} \operatorname{div} v = 0$. Conversely Lemma 7.28 lifts every $t \in \mathbf{T}_h$ into Z_h . Hence $\langle g - \Pi_{\mathbf{T}} g, v \rangle = 0$ for $v \in Z_h$. $\square$

So for *every* μ the printed method imposes on Z_h the projection $\Pi_{\mathbf{T}} g$ of the edgewise leak data; the penalty never sees the part $g - \Pi_{\mathbf{T}} g$ that Z_h cannot represent. The vertex-gradient constraints of Section 11 restrict full gradients at wall vertices, not traces, and play no role here.

Lemma 7.39. *For $v \in Z_h$, $|\langle g, \partial_n v \rangle| \leq C \|v\|_{\Gamma_h}$; for $v \in Y_h$, $\langle g, \partial_n v \rangle = 0$.*

Proof. g is exactly normal on each edge, with $\sigma_e := g \cdot n_e = -(\tilde{p} - \bar{p})$. On T_e , $\operatorname{div} v = 0$ gives $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$, so $\int_e \sigma_e \partial_n v \cdot n_e = \int_e \partial_\tau \sigma_e (v \cdot \tau_e) - [\sigma_e v \cdot \tau_e]_{\partial e}$. If $v = 0$ on Γ_h everything vanishes. Otherwise, at a vertex x shared by e, e' the endpoint terms combine to $\sigma(x) v(x) \cdot (\tau_{e'} - \tau_e)$ ($\tilde{p}$ is continuous), of size $\leq Ch_\Gamma |v(x)|$, and $\sum_x h_\Gamma |v(x)| \leq C \|v\|_{\Gamma_h}$ by (M0). $\square$

Unlike Lemma 6.3 there is no $h_\Gamma^{1/2} \|\nabla v\|$ term, because g has no tangential part.

Lemma 7.40. $\|t^* - g\|_{\Gamma_h} \leq Ch_\Gamma$ and $\|t^* - \tilde{U}\|_{\Gamma_h} \leq Ch_\Gamma$.

Proof. $z_U|_{\Gamma_h} \in \mathbf{T}_h$, so $\|\Pi_{\mathbf{T}} g - g\| \leq \|z_U - g\| \leq \|z_U - \tilde{U}\|_{\Gamma_h} + \|\zeta\|_{\Gamma_h} \leq Ch_\Gamma$; and $t^* - \tilde{U} = (t^* - g) - \zeta$. $\square$

Let $X^D \in Z_h$ be the discrete Stokes–Dirichlet field with $X^D|_{\Gamma_h} = t^*$ and $a_h(X^D, v) = 0$ for all $v \in Y_h$. It exists and is unique: the affine set $\{z \in Z_h : z|_{\Gamma_h} = t^*\}$ is nonempty (Lemma 7.38), its direction is Y_h , and a_h is coercive on Y_h (Lemma 3.2).

Lemma 7.41. $\|\nabla(X^D - \tilde{U})\| \leq C(h_\Gamma^{1/2} + E_U^0)$.

Proof. $W_0 := X^D - z_U \in Z_h$ has trace in $\mathbf{T}_h$ of norm $\leq \|t^* - \tilde{U}\|_{\Gamma_h} + \|\tilde{U} - z_U\|_{\Gamma_h} \leq Ch_\Gamma$. Its lift L (Lemma 7.28) has $\|\nabla L\| \leq Ch_\Gamma^{1/2}$, and $y := W_0 - L \in Y_h$ satisfies $a_h(y, y) = -a_h(z_U, y) - a_h(L, y)$. Green’s formula on Ω_h gives $a_h(\tilde{U}, y) = (F_U, y)$ for $y \in Y_h$, with $|(F_U, y)| \leq Ch_\Gamma^2 \|\nabla y\|$ (Lemma 3.3). By Korn, $\|\nabla y\| \leq \kappa(2\|\nabla(z_U - \tilde{U})\| + Ch_\Gamma^2 + 2\|\nabla L\|)$. $\square$

7.3.2. *Two-sided bounds, uniform in the penalty.*

Theorem 7.42 (Uniform upper bound). *Let $\gamma \geq \gamma_0$, with no upper bound, and $D := X - X^D \in Z_h$. Then*

$$\|D\| \leq C \left[\left(\frac{h_\Gamma}{\gamma} \right)^{1/2} + \gamma^{-1/2} E_U^0 + h_\Gamma^{1/2} \right], \quad \left\| \nabla \left(\frac{\mu}{h} \delta_R - \tilde{U} \right) \right\|_{\Omega_h} \leq C(h_\Gamma^{1/2} + E_U^0),$$

and $\|D\|_{\Gamma_h} \leq Ch_\Gamma \gamma^{-1/2} (1 + h_\Gamma^{-1/2} E_U^0)$. So $(\mu/h) \delta_R \rightarrow U$ in H^1 uniformly in $\mu \geq \mu_0$.

Proof. By Lemma 7.38 and $X^D|_{\Gamma_h} = t^*$, for $v \in Z_h$ the two penalty terms cancel *exactly*:

$$N_h(D, v) = \lambda \langle t^*, v \rangle - N_h(X^D, v) = -a_h(X^D, v) + \langle \partial_n X^D, v \rangle + \langle t^*, \partial_n v \rangle =: \ell(v).$$

- (1) Let $L_v \in Z_h$ lift $v|_{\Gamma_h}$ (Lemma 7.28); $v - L_v \in Y_h$, so $a_h(X^D, v) = a_h(\tilde{U}, L_v) + a_h(X^D - \tilde{U}, L_v)$. Green’s formula ($\operatorname{div} L_v = 0$, $L_v|_{\partial R} = 0$, $\int_{\Gamma_h} L_v \cdot n = 0$) gives $a_h(\tilde{U}, L_v) = (F_U, L_v) + \langle (D(\tilde{U}) - (\tilde{P} - c))n, v \rangle$, of size $\leq C \|v\|_{\Gamma_h}$. By Lemma 7.41, $|a_h(X^D - \tilde{U}, L_v)| \leq C(1 + h_\Gamma^{-1/2} E_U^0) \|v\|_{\Gamma_h}$.
- (2) $\|\partial_n(X^D - I_h \tilde{U})\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2} \|\nabla(X^D - I_h \tilde{U})\|$ (Lemma 4.1, (M0)) and $\|\partial_n I_h \tilde{U}\|_{\Gamma_h} \leq C$; so $|\langle \partial_n X^D, v \rangle| \leq C(1 + h_\Gamma^{-1/2} E_U^0) \|v\|_{\Gamma_h}$.
- (3) $\langle t^*, \partial_n v \rangle = \langle g, \partial_n v \rangle + \langle t^* - g, \partial_n v \rangle$; Lemma 7.39 bounds the first by $C \|v\|_{\Gamma_h}$, and Lemma 7.40 with $\|\partial_n v\|_{\Gamma_h} \leq (c_0 h_\Gamma)^{-1/2} \|v\|$ bounds the second by $Ch_\Gamma^{1/2} \|v\|$.

With $\|v\|_{\Gamma_h} \leq (h_\Gamma/\gamma)^{1/2} \|v\|$ and coercivity (Lemma 4.2, c_1 independent of μ), test with $v = D$. Then $\|\nabla W\| \leq \|D\| + \|\nabla(X^D - \tilde{U})\|$. $\square$

The terms $\min(\gamma h_\Gamma^{1/2}, (\gamma h_\Gamma)^{1/2})$ and $(h/\mu)^{1/2}$ of Theorem 6.10 are artefacts of estimating $\lambda\langle\zeta, v\rangle$ directly instead of using Lemma 7.38.

For the lower bound we test with fields that *vanish* on Γ_h . On Y_h the penalty and the data drop out, and the bound is carried by Nitsche's symmetric term acting on the sawtooth (7): the first-moment analogue of the mechanism of Theorem 7.24.

Lemma 7.43 (A three-triangle field with odd normal derivative). *Assume (M0), (M1') and $k \geq 4$. Let $e = [x_0, x_1] \in \mathcal{E}_h^\Gamma$, $T_e = [x_0, x_1, z]$, and let $T'_0 = [x_0, z, y_0]$, $T'_1 = [x_1, z, y_1]$ be the triangles across $[x_0, z]$ and $[x_1, z]$; they are distinct, and each meets Γ_h only in x_0 , resp. x_1 . With $\lambda_0, \lambda_1, \lambda_2$ the barycentrics of T_e (vertices x_0, x_1, z) and $\mu^{(i)}$ those of T'_i , put*

$$\psi_e = \lambda_0 \lambda_1 \lambda_2^2 (\lambda_1 - \lambda_0) \text{ on } T_e, \quad \psi_e = \alpha_i (\mu_{x_i}^{(i)})^2 (\mu_z^{(i)})^2 \mu_{y_i}^{(i)} \text{ on } T'_i,$$

$\psi_e = 0$ elsewhere, with $\alpha_0 \nabla \mu_{y_0}^{(0)} = -\nabla \lambda_1$ and $\alpha_1 \nabla \mu_{y_1}^{(1)} = \nabla \lambda_0$. Then $\psi_e \in C^1$ is piecewise P_5 , $v_e := \text{curl} \psi_e \in Y_h$, and on e : $\partial_n v_e \cdot n_e = 0$, $\partial_n v_e \cdot \tau_e = \pm 2H_e^{-2} \lambda_0 \lambda_1 (\lambda_1 - \lambda_0)$ (odd about m_e ; in particular $v_e \in Y_h^1$), and $\int_e s \partial_n v_e \cdot \tau_e = \pm |e|^2 / (30H_e^2)$. There is $\kappa_1 > 0$, depending only on the minimum angle in (M0), with $m_e := |\int_e s \partial_n v_e \cdot \tau_e| \geq \kappa_1 |e| \|\nabla v_e\|$.

Proof. Combinatorics. $z \notin \Gamma_h$ (a triangle with three vertices on the convex polygon lies in $\overline{P_h}$). By (M1'), x_0 has at least three triangles, so T'_0 is a middle triangle of the fan at x_0 : $y_0 = x_{-1}$ would leave only T_{e-1} and T_e at x_0 . So T'_0, T'_1 contain no edge of Γ_h and are distinct.

C^1 . On T_e , $\psi_e = \lambda_0 \lambda_1^2 \lambda_2^2 - \lambda_0^2 \lambda_1 \lambda_2^2$; each term is the Argyris edge bubble $\lambda_a^2 \lambda_b^2 \lambda_E$ of one edge ($[x_1, z]$, resp. $[x_0, z]$). It vanishes to second order on the other two edges and to third order at the vertices; on its edge it vanishes with its tangential derivative and has normal derivative $\lambda_a^2 \lambda_b^2 \partial_n \lambda_E$. The pieces on T'_i are the bubbles of the same edges from the other side, and the choice of α_i matches the normal derivatives (the gradients are normal to the common edge). All other edges carry second-order zeros from both sides.

Traces and moments. ψ_e and $\nabla \psi_e$ vanish on e and at x_0, x_1 , the only points of Γ_h in the support, so $v_e \in Y_h$. On e , $\nabla \psi_e \equiv 0$, so $\partial_n v_e = \pm \partial_{nn} \psi_e \tau_e$ and $\partial_{nn}(f \lambda_2^2) = 2f/H_e^2$. With $\sigma = 2s/|e|$, $\lambda_0 \lambda_1 = (1 - \sigma^2)/4$, $\lambda_1 - \lambda_0 = \sigma$, and $\frac{|e|^2}{16} \int_{-1}^1 \sigma^2 (1 - \sigma^2) d\sigma = |e|^2/60$.

κ_1 . $m_e/(|e| \|\nabla v_e\|)$ is similarity-invariant, continuous and positive on the compact set of normalised five-vertex configurations allowed by the minimum angle. $\square$

The field has odd $\partial_n v \cdot \tau_e$, so its *second* moment vanishes; it does not give (M3), and the star fields of (M3) are not needed here.

Theorem 7.44 (Uniform lower bound). *Assume (M0)–(M2), (D), $k \geq 4$ and $G > 0$, and let $\mu > 0$ be any penalty for which δ_R exists. N_h on Z_h is an affine pencil in μ , invertible for $\mu \geq \mu_0$; so its determinant is a nonzero polynomial and δ_R exists (uniquely) for all μ outside a finite set on each mesh; at an exceptional μ the statement holds for any solution. Then*

$$\left\| \nabla \left(\frac{\mu}{h} \delta_R - \tilde{U} \right) \right\|_{\Omega_h} \geq c_b G h_\Gamma^{1/2} - C h_\Gamma^{3/2}, \quad c_b := \frac{\kappa_1 c_0^{1/2}}{\sqrt{2} \Lambda_0},$$

with $\Lambda_0 = 2 + C_{PT} C_I$ of Corollary 7.21 and κ_1 of Lemma 7.43. Neither c_b nor C depends on μ, γ, h, ρ ; C depends on G^{-1} .

Proof. Identity. For $v \in Y_h$, $N_h(\delta_R, v) = \mathcal{R}(v) = 0$ and $N_h(\delta_R, v) = a_h(\delta_R, v) - \langle \delta_R, \partial_n v \rangle$; so $a_h(X, v) = \langle X, \partial_n v \rangle$ for every such μ . With $a_h(\tilde{U}, v) = (F_U, v)$ this gives

$$a_h(W, v) - \langle W, \partial_n v \rangle = \langle \tilde{U}, \partial_n v \rangle - (F_U, v) \quad \forall v \in Y_h.$$

Sawtooth. On e , $|x - x^*| \leq h_\Gamma^2/4$ and $e_r(x^*) \cdot \tau_e = \epsilon s/|x|$, so $\tilde{U} \cdot \tau_e = c_e s + E_e$ with $c_e = \epsilon(p - \bar{p})(m_e^*)$, $|E_e| \leq C_E h_\Gamma^2$, and $\Sigma := \sum_e c_e^2 |e| = G^2 + O(h_\Gamma)$.

Test field. Normalise $\|\nabla v_e\| = 1$ with the sign making $\int_e s \partial_n v_e \cdot \tau_e = m_e \geq \kappa_1 |e|$, and put $v := \sum_e c_e v_e \in Y_h^1$. Every triangle lies in at most two patches, so $\|\nabla v\|^2 \leq 2 \sum_e c_e^2 \leq 2\Sigma/(c_0 h_\Gamma)$; on e only v_e is nonzero in T_e . By Lemma 7.18,

$$\langle \tilde{U}, \partial_n v \rangle = \sum_e c_e \int_e (c_e s + E_e) \partial_n v_e \cdot \tau_e \geq \kappa_1 \Sigma - C_E C_I h_\Gamma^2 \sum_e |c_e| \geq \kappa_1 \Sigma - C h_\Gamma,$$

and $|(F_U, v)| \leq C h_\Gamma^2 \|\nabla v\|$. By Lemma 7.20 and $|a_h(W, v)| \leq 2 \|\nabla W\| \|\nabla v\|$, $\Lambda_0 \|\nabla W\| \|\nabla v\| \geq \langle \tilde{U}, \partial_n v \rangle - |(F_U, v)|$. If $\kappa_1 \Sigma > C h_\Gamma$, divide by the bound for $\|\nabla v\|$ and insert $\Sigma = G^2 + O(h_\Gamma)$; otherwise $h_\Gamma \geq c G^2$ and the claim is trivial after enlarging C . $\square$

Corollary 7.45 (Exact rate $\frac{1}{2}$ for every penalty). *For all $\gamma \in [\gamma_0, \infty)$, the Dirichlet limit $\mu = \infty$ included, and $h_\Gamma \leq h_1(G)$,*

$$\frac{1}{2} c_b G h_\Gamma^{1/2} \leq \|\nabla((\mu/h)\delta_R - \tilde{U})\| \leq C(h_\Gamma^{1/2} + E_U^0).$$

On test P ($\tilde{u} = 0$), $d_h \asymp h_\Gamma^{1/2}$ uniformly in γ . The observed $h_\Gamma^{0.6} - h_\Gamma^{0.8}$ of Section 6.1 are pre-asymptotic.

Proof. Theorems 7.44 and 7.42; the Dirichlet limit X^∞ satisfies the same identity on Y_h . $\square$

Corollary 7.46 (General data). *For GS with general data and every $\gamma \geq \gamma_0$,*

$$\begin{aligned} \left\| \nabla \left(\tilde{u} - u_h - \frac{h}{\mu} \tilde{U} \right) \right\| &\leq C \frac{h}{\mu} (h_\Gamma^{1/2} + E_U^0) + C(h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2}), \\ \left\| \nabla \left(\frac{\mu}{h} (\tilde{u} - u_h) - \tilde{U} \right) \right\| &\geq c_b G h_\Gamma^{1/2} - C(1 + \gamma) h_\Gamma^{3/2}. \end{aligned}$$

Hence $\|\nabla(\tilde{u} - u_h)\| \mu/(hG) \rightarrow K$ whenever $h/\mu \rightarrow 0$, $E_U^0 \rightarrow 0$ and $\frac{\mu}{h}(h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2}) \rightarrow 0$; the geometric part of the last condition is $\gamma h_\Gamma^{1/2} \rightarrow 0$, which replaces $\gamma h_\Gamma \rightarrow 0$ and $\gamma(1 + \gamma^{1/2}) h_\Gamma^{1/2} \rightarrow 0$ in Corollary 6.11(ii). For test P no condition is needed. (With flux-corrected data, $h^{\sigma_k} h_\Gamma^{-1/2}$ becomes $h_\Gamma^{\sigma_k - 1/2} + |m_R|$.)

Proof. Upper bound: in the proof of Theorem 7.30, $\tilde{u} - u_h = (\tilde{u} - u'_h) + \delta_R$ with $\|\nabla(\tilde{u} - u'_h)\| \leq C(h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2})$ uniformly in γ ; apply Theorem 7.42. Lower bound: Lemma 7.19 gives $a_h(e_u, v) - \langle e_u, \partial_n v \rangle = -\langle \tilde{u}, \partial_n v \rangle - (F, v)$ on Y_h ; multiply by μ/h and subtract the identity for $\tilde{U}$. For the test field above, $\tilde{u} \cdot \tau_e$ is even in s up to $O(h_\Gamma^3)$ (Lemma 7.22) and $\partial_n v_e \cdot \tau_e$ is exactly odd, so $|\langle \tilde{u}, \partial_n v \rangle| \leq C h_\Gamma^2$; and $|(F, v)| \leq C h_\Gamma^3$. Times $\mu/h = \gamma/h_\Gamma$ the numerator loses at most $C\gamma h_\Gamma$. $\square$

The condition $\gamma h_\Gamma^{1/2} \rightarrow 0$ cannot be dropped for general data: if $\gamma h_\Gamma^{1/2} \rightarrow \infty$ the normalised constant diverges (Corollary 7.48(b)).

Corollary 7.47 (The H^1 constant converges at rate one). *On test P let $K_h := \|\nabla X\|_{\Omega_h}/G$. For every $\gamma \geq \gamma_0$,*

$$|K_h^2 - K^2(1 + d_h^2)| \leq C(h_\Gamma^2 + h_\Gamma \gamma^{-1/2}(1 + h_\Gamma^{-1/2} E_U^0)),$$

hence $|K_h - K| \leq C(h_\Gamma + (E_U^0)^2)$ uniformly in γ , although $d_h \asymp h_\Gamma^{1/2}$.

Proof. $\|\nabla X\|^2 = \|\nabla \tilde{U}\|_{\Omega_h}^2 + 2(\nabla \tilde{U}, \nabla W) + \|\nabla W\|^2$, and $\|\nabla \tilde{U}\|_{\Omega_h}^2 - \|\nabla U\|_{\Omega}^2 = O(h_\Gamma^2)$. Since $\operatorname{div} W = 0$, $W|_{\partial R} = 0$ and $\int_{\Gamma_h} W \cdot n = 0$, Green gives $(\nabla \tilde{U}, \nabla W) = (F_U, W) + \langle \sigma, W \rangle$ with $\sigma := \partial_n \tilde{U} - (\tilde{P} - c)n$, edgewise smooth with $O(h_\Gamma)$ jumps at vertices, so $\|\sigma - \Pi_\Gamma \sigma\|_{\Gamma_h} \leq C h_\Gamma$. On Γ_h , $W = D + (\Pi_\Gamma \tilde{U} - \tilde{U}) - \Pi_\Gamma \zeta$. Then $|\langle \sigma, \Pi_\Gamma \zeta \rangle| \leq |\langle \sigma, \zeta \rangle| + C h_\Gamma^2 \leq C h_\Gamma^2$ by (7) and $\int_e s ds = 0$; $|\langle \sigma, \Pi_\Gamma \tilde{U} - \tilde{U} \rangle| \leq C h_\Gamma^{k+1}$; and $|\langle \sigma, D \rangle| \leq C \|D\|_{\Gamma_h}$ is bounded by Theorem 7.42. $\square$

Corollary 7.48 (The leak constant for every penalty). *Let $h_\Gamma, h \rightarrow 0$ with $\mu \geq \mu_0$ arbitrary, in particular with $\gamma h_\Gamma \rightarrow \infty$.*

- (a) *On test P, $K_h := \|\nabla(\tilde{u} - u_h)\| \mu / (hG) = \|\nabla X\| / G$ satisfies $|K_h - K| \leq C(h_\Gamma + (E_U^0)^2)$, with C independent of μ, γ and ρ . The same holds for the leak part $(\mu/h)\delta_R$ of general data.*
- (b) *For GS with general data, $\partial_n u \cdot \tau \not\equiv 0$ on Γ and $G > 0$, the normalised constant $K_h^{\text{gen}} := \|\nabla(\tilde{u} - u_h)\| \mu / (hG)$ tends to K if $\gamma h_\Gamma^{1/2} \rightarrow 0$ (and the data terms of Corollary 7.46 vanish), and to ∞ if $\gamma h_\Gamma^{1/2} \rightarrow \infty$, in particular if $\gamma h_\Gamma \rightarrow \infty$.*

Proof. (a) Corollary 7.47 and $d_h \leq C(h_\Gamma^{1/2} + E_U^0)$ (Theorem 7.42) give $|K_h^2 - K^2| \leq C(h_\Gamma + (E_U^0)^2 + h_\Gamma^{1/2} E_U^0)$, and $K > 0$. For δ_R use the decomposition of Corollary 6.11(i). (b) The first case is Corollary 7.46. In the second, Theorem 7.24, valid for GS with every $\gamma \geq \gamma_0$, gives $\|\nabla(\tilde{u} - u_h)\| \geq c \|\partial_n u \cdot \tau\|_{L^2(\Gamma)} h_\Gamma^{3/2} - C h_\Gamma^2$; multiply by $\mu / (hG) = \gamma / (h_\Gamma G)$. $\square$

So the convergence of the leak constant in the regime $\gamma h_\Gamma \rightarrow \infty$ holds in the only sense in which it can: for the leak itself. The $\mu = 10^3, 10^4$ columns of Table 13 are test P and are covered. For the total GS error the normalised constant diverges in that regime, because the geometric error $\asymp h_\Gamma^{3/2}$ then exceeds the leak.

Remark 7.49 (What the constants are, and what is not proved). With the sawtooth removed from the data (Dirichlet data $\Pi_\Gamma \tilde{U}$ instead of t^*) the discrete Dirichlet field is accurate to $5 \cdot 10^{-5}$ on the production meshes, so the $h_\Gamma^{1/2}$ error is entirely the sawtooth. At $\mu = 10^8$, $d_h \approx 0.55 h_\Gamma^{1/2} (1 + 0.96 h_\Gamma + 3.0 h_\Gamma^2)$ (fit residual 10^{-3}), with predicted local rates 0.58 at $N = 192$ and 0.53 at $N = 384$; at $\mu = 100$, $d_h \approx 0.225 h_\Gamma^{1/2} (1 + 0.39 h_\Gamma + 0.30 h_\Gamma^2)$, local rates $0.64 \rightarrow 0.54$. The ratio 0.41 of the two amplitudes is the fraction of the sawtooth that the Robin problem at $\gamma \approx 30$ captures. At $\mu = 10^8$ the paper's table gives $K_h^2 - K^2(1 + d_h^2) \approx -1.87 h_\Gamma^2$ from $N = 48$ on, the $\gamma \rightarrow \infty$ form of Corollary 7.47. A periodic one-edge cell problem reproduces these capture fractions (0.47, 0.89, 0.99 at $\gamma = 30, 300, 3000$ for first-layer aspect 1, against 0.43, 0.86, 0.98 measured) and shows that $d_h/h_\Gamma^{1/2}$ is *not* monotone in μ : its minimum lies near the coercivity threshold. The certificate of Theorem 7.44, evaluated with all remainders, gives $d_h \geq 0.012\text{--}0.014 h_\Gamma^{1/2}$ for every μ on the production meshes ($N \leq 128$), about 5% of the truth. (NUMERICAL: `notes/v6/unifmu`, `notes/v6/unifmu2`.) Not proved: that $d_h/h_\Gamma^{1/2}$ converges on a given mesh family. The minimum star constant κ_1 of Lemma 7.43 drifts slowly downward on the production family ($0.0199 \rightarrow 0.0123$ for $N = 16 \rightarrow 128$, with shrinking decrements), so we use only that the family is uniformly shape regular.

8. PRESSURE, BOUNDARY STRESS AND DRAG

All error theorems so far bound the velocity. This section adds pressure, boundary-traction and drag/lift/torque estimates for both methods; drag is what Gjerde and Scott use the method for [34]. In brief: the CNS* pressure converges at the velocity rate $h_\Gamma^{3/2} + h_\Omega^k$, including its constant, while the GS pressure carries an explicit first-layer boundary layer and converges in L^2 only at the exact rate $\frac{1}{2}$. The variational (Babuška–Miller) drag equals the integral of the Nitsche flux; for GS its error is, to leading order, $-(h/\mu)$ times a reciprocity constant, for CNS* it is $O(h_\Gamma^2)$. The computations are in Section 16.9.

8.1. Setting, notation and summary.

8.1.1. *Setting.* Everything is in the setting of Section 2: $\Omega = R \setminus \overline{D}$, inscribed polygon Γ_h , (M0)–(M2), (D), $k \geq 4$, the forms a_h , N_h , the norm $\|\cdot\|$, the extensions $\tilde{u} = \text{curl } \tilde{\psi}$, $\tilde{p}$, and $F = \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$ (supported in $\overline{S_h}$, $S_h = \Omega_h \setminus \overline{\Omega}$). Section 8.7 treats Navier–Stokes and Remark 8.45 general domains. Throughout, $\gamma \geq \gamma_0$ for GS and $\gamma \geq \gamma_2$ for CNS* (Theorem 7.2). We use $\theta = 0$ for GS and $\theta = 1$ for CNS*, so both methods read

$$N_h(u_h, v) - (p_h, \text{div } v) + \theta(p_h - \bar{p}_\Gamma(p_h), v \cdot n) = (\tilde{f}, v) \quad \forall v \in V_h^R, \quad (q, \text{div } u_h) = 0 \quad \forall q \in \Pi_h.$$

Further notation:

- $e_u := \tilde{u} - u_h$; $q := p - \bar{p}$ on Γ ; $G = \|q\|_{L^2(\Gamma)}$; C_p is as in Theorem 6.4; $x^* = x/|x|$ is the nearest point of Γ to $x \in \Gamma_h$.
- T_e is the triangle with edge $e \in \mathcal{E}_h^\Gamma$ (distinct for distinct e , Section 7.1), $H_e := 2|T_e|/|e|$ its height, and $L_1 := \bigcup_e T_e$ the *first layer*. By (M0), $\text{diam } T_e \leq Ch_\Gamma$, $c \leq |e|/H_e \leq C$ and $ch_\Gamma \leq |L_1| \leq Ch_\Gamma$.
- $p^\circ := \tilde{p} - \bar{p}_\Gamma(\tilde{p})$, where $\bar{p}_\Gamma(\cdot)$ is the mean over Γ_h (this is the p^* of Theorem 7.2), and $e_p := p^\circ - p_h$.
- $m_R := \int_{\partial R} g_I \cdot \nu$. Then $\int_{\Gamma_h} u_h \cdot n = -m_R$ and $\int_{\Gamma_h} \tilde{u} \cdot n = 0$, so $\int_{\Gamma_h} e_u \cdot n = m_R$; with flux-corrected data (Definition 2.2) the same holds with m_R replaced by $\int_{\partial R} \tilde{g}_I \cdot \nu = 0$; and $m_R = 0$ for the test data of Section 16.
- $B_h(y, w) := N_h(y, w) - a_h(y, w) = -\langle \partial_n y, w \rangle - \langle y, \partial_n w \rangle + \frac{\mu}{h} \langle y, w \rangle$.
- The error sizes

$$Y := (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h, \quad X_1 := \frac{h}{\mu} + Y.$$

By Theorem 7.2, $\|e_u\| \leq CY$ for CNS*. By Theorems 6.1 and 6.4, $\|e_u\| \leq C[(h/\mu)^{1/2}G + Y]$ and $\|\nabla e_u\| \leq C_p X_1$ for GS.

8.1.2. *Summary.* The rates below are for fixed γ and bounded ρ (so $h \asymp h_\Gamma$ and $h/\mu \asymp h_\Gamma$), $k \geq 4$, and $\varepsilon_h \leq Ch_\Omega^k$ (or flux-corrected data); each entry states the upper bound, and “sharp” means a matching lower bound is proved.

quantity	GS	CNS*
$\ p^\circ - p_h\ _{L^2(\Omega_h)}$	$\asymp (h/\mu)h_\Gamma^{-1/2}G$: rate $\frac{1}{2}$, sharp (Thm 8.11)	$O(h_\Gamma^{3/2})$ (Thm 8.7)
$\ p^\circ - p_h\ _{L^2(\Omega_h \setminus L_1)}$	$O(h/\mu)$	$O(h_\Gamma^{3/2})$
$\ p^\circ - p_h - c\ _{L^2(\Omega_h)}$, lower bound	sharp (above)	$\geq ch_\Gamma^{3/2}$ under (P) (Thm 8.34), or modulo (Dc), (TL) (Cor. 8.40)
$\ p^\circ - p_h - c\ _{L^\infty(L_1)}$, any c	$\geq cG/\gamma$: no convergence	$\geq ch_\Gamma$ under (P) (Thm 8.34)
drag/lift/torque: Nitsche flux = volume formula with v_ψ	$-(h/\mu)K_\psi + O(h^{3/2})$: rate 1 (Thm 8.17)	$-\Theta_f - \Lambda_h - \frac{h}{\mu}P_h(\rho_\psi) + O(h_\Gamma^{5/2})$: rate 2 (Thms 8.19, 8.42)
drag/lift (translations), GS volume formula $\omega_h(\chi_h)$	$-2(h/\mu)K_e + O(h^{3/2})$ (Prop. 8.21)	$J_h^N + O(h_\Gamma^2)$
torque, GS volume formula $\omega_h(\chi_h)$	rate-1 upper bound; leading term $-(h/\mu)J_{x^\perp}$ as $\gamma \rightarrow \infty$ (numerical)	$J_h^N - (h/\mu)J_{x^\perp}(1 + o(1))$ as $\gamma \rightarrow \infty$ (Prop. 8.22)
drag, pointwise traction $\sigma(u_h, p_h)n$	$\Xi_h(\psi) = O(1/\gamma)$: no convergence (Props. 8.23, 8.24)	$O(h_\Gamma)$; $\geq ch_\Gamma$ under (P), or modulo (Dc), (TL)
Nitsche traction in $L^2(\Gamma_h)$	$O(h_\Gamma^{1/2})$	$O(h_\Gamma)$

8.2. Tools.

Lemma 8.1 (Pressure identity on V_h^0). *For both methods and all $v \in V_h^0$,*

$$(p^\circ - p_h, \text{div } v) = a_h(e_u, v) + \langle u_h, \partial_n v \rangle + (F, v).$$

Proof. For $v \in V_h^0$ we have $v = 0$ on Γ_h , so $N_h(w, v) = a_h(w, v) - \langle w, \partial_n v \rangle$ and the θ -term vanishes. The method gives $a_h(u_h, v) - \langle u_h, \partial_n v \rangle - (p_h, \operatorname{div} v) = (\tilde{f}, v)$. On V_h^0 , Lemma 5.1 reads $a_h(\tilde{u}, v) - \langle \tilde{u}, \partial_n v \rangle - (\tilde{p}, \operatorname{div} v) = (\tilde{f}, v) + \mathcal{G}(v)$ with $\mathcal{G}(v) = -\langle \tilde{u}, \partial_n v \rangle - (F, v)$, i.e. $a_h(\tilde{u}, v) - (\tilde{p}, \operatorname{div} v) = (\tilde{f} - F, v)$. Subtract, and note $(c, \operatorname{div} v) = 0$ for constants c . $\square$

The term $\langle u_h, \partial_n v \rangle$ is the *Nitsche coupling*: the symmetric Nitsche term applied to the discrete trace. For $v \in V_h^0$ and $e \in \mathcal{E}_h^\Gamma$, $v|_{T_e}$ vanishes on e , so $\nabla v = \partial_n v \otimes n_e$ there and

$$(\operatorname{div} v)|_{T_e} = \partial_n v \cdot n_e \quad \text{on } e. \quad (8)$$

Hence the *normal* part of the discrete trace acts on the pressure exactly like a boundary source $\langle u_h \cdot n, \operatorname{div} v \rangle_{\Gamma_h}$.

Lemma 8.2 (Edge representers). *For $e \in \mathcal{E}_h^\Gamma$ and $g \in L^\infty(e)$ let $\xi_{e,g} \in P_{k-1}(T_e)$ be defined by $(\xi_{e,g}, r)_{T_e} = \int_e g r$ for all $r \in P_{k-1}(T_e)$.*

(a) *For constant $g \equiv 1$, $\xi_{e,1}(x) = \frac{|e|}{2|T_e|} \hat{r}_k(F_e^{-1}x)$, where F_e is an affine map from the reference triangle $\hat{T} = \{(0,0), (1,0), (0,1)\}$ onto T_e with $F_e(\hat{e}) = e$, $\hat{e} = [0,1] \times \{0\}$, and $\hat{r}_k \in P_{k-1}(\hat{T})$ is the representer of $\hat{r} \mapsto \int_{\hat{e}} \hat{r}$. Consequently*

$$\|\xi_{e,1}\|_{T_e}^2 = \int_e \xi_{e,1} = \hat{c}_k \frac{|e|}{H_e}, \quad \|\xi_{e,1}\|_{L^2(e)}^2 = \hat{d}_k \frac{|e|}{H_e^2},$$

with $\hat{c}_k := \int_{\hat{e}} \hat{r}_k$ and $\hat{d}_k := \|\hat{r}_k\|_{L^2(\hat{e})}^2$ positive and depending only on k .

(b) *$\|\xi_{e,g} - g_e \xi_{e,1}\|_{T_e} \leq C \|g - g_e\|_{L^\infty(e)}$ and $\|\xi_{e,g} - g_e \xi_{e,1}\|_{L^2(e)} \leq C |e|^{-1/2} \|g - g_e\|_{L^\infty(e)}$ for every constant g_e .*

(c) *For every $k \geq 1$, $\hat{r}_k(\hat{x}, \hat{y}) = 2P'_k(1 - 2\hat{y})$ with P_k the Legendre polynomial on $[-1, 1]$; hence $\hat{c}_k = k(k+1)$, $\hat{d}_k = \hat{c}_k^2 = k^2(k+1)^2$, and $\xi_{e,1}|_e \equiv k(k+1)/H_e$. In particular $\hat{c}_4 = 20$.*

Proof. (a) With $x = F_e \hat{x}$, $\int_{T_e} \xi r = 2|T_e| \int_{\hat{T}} \hat{\xi} \hat{r}$ and $\int_e r = |e| \int_{\hat{e}} \hat{r}$, which gives the formula. Then $\|\xi_{e,1}\|_{T_e}^2 = \int_e \xi_{e,1}$ (take $r = \xi_{e,1}$), and the scalings follow from $|e|^2/(2|T_e|) = |e|/H_e$. (b) The map $g \mapsto \xi_{e,g}$ is linear, and $\|\xi_{e,g}\|_{T_e} = \sup_r \int_e g r / \|r\|_{T_e} \leq \|g\|_{L^\infty(e)} |e|^{1/2} \sup_r \|r\|_{L^2(e)} / \|r\|_{T_e} \leq C \|g\|_{L^\infty(e)}$ by the inverse trace inequality $\|r\|_{L^2(e)} \leq C |e|^{-1/2} \|r\|_{T_e}$ on P_{k-1} ; the trace bound follows from the same inequality.

(c) For $r \in P_{k-1}(\hat{T})$ let $\bar{r}(\hat{y}) := (1 - \hat{y})^{-1} \int_0^{1-\hat{y}} r(\hat{x}, \hat{y}) d\hat{x} \in P_{k-1}$. Then $\int_{\hat{e}} r = \bar{r}(0)$ and $(q, r)_{\hat{T}} = \int_0^1 (1 - \hat{y}) q \bar{r} d\hat{y}$ for $q = q(\hat{y})$. With $g(\hat{y}) := 2P'_k(1 - 2\hat{y})$, $t = 1 - 2\hat{y}$ and $s(t) := \bar{r}((1-t)/2)$,

$$(g, r)_{\hat{T}} = \frac{1}{2} \int_{-1}^1 P'_k(t) (1+t) s(t) dt = \frac{1}{2} [P_k(1+t) s]_{-1}^1 - \frac{1}{2} \int_{-1}^1 P_k((1+t)s)' dt = s(1) = \int_{\hat{e}} r,$$

since $P_k(1) = 1$ and $((1+t)s)' \in P_{k-1} \perp P_k$. So $\hat{r}_k = g$, which is constant ($= 2P'_k(1) = k(k+1)$) on $\hat{e}$. The physical formula follows because $\hat{y} \circ F_e^{-1} = \operatorname{dist}(x, \ell_e)/H_e$. (An exact rational check for $k \leq 12$ is `notes/v6/misc/chat_symbolic.py`; the constant is also the sharp P_{k-1} edge trace-inverse constant [59].) $\square$

Lemma 8.3 (Normal-load cancellation without incompressibility). *Let S be as in Lemma 6.3: edgewise $W^{1,\infty}$ on Γ_h , continuous at the vertices, with $|S \cdot \tau_e| \leq Ch_\Gamma$. Then for every $\delta \in V_h^R$,*

$$|\langle S, \partial_n \delta \rangle| \leq C \left(\|\nabla \delta\| + \sum_{e \in \mathcal{E}_h^\Gamma} \|(\operatorname{div} \delta)|_{T_e}\|_{L^1(e)} \right).$$

Proof. Repeat Lemma 6.3. The only change is in step 2: on e , $\partial_n \delta \cdot n_e = (\operatorname{div} \delta)|_{T_e} - \partial_\tau(\delta \cdot \tau_e)$. This adds the term $\int_e \sigma_e (\operatorname{div} \delta)|_{T_e}$, which is bounded by $\|S\|_\infty \|(\operatorname{div} \delta)|_{T_e}\|_{L^1(e)}$. $\square$

Lemma 8.4 ($\mathcal{G}$ on non-discrete fields). *For every $w \in H^1(\Omega_h)^2$ that is piecewise H^2 on $\mathcal{T}_h$,*

$$|\mathcal{G}(w)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}\|w\|.$$

Proof. $|\langle \nabla \tilde{u}^\top n | \leq Ch_\Gamma$ on Γ_h (step 1 of Lemma 3.5), so $|\langle \langle \nabla \tilde{u}^\top n, w \rangle | \leq Ch_\Gamma \|w\|_{\Gamma_h} \leq Ch_\Gamma (h/\mu)^{1/2} \|w\| = C\gamma^{-1/2} h_\Gamma^{3/2} \|w\|$. Next, $|\langle \tilde{u}, \partial_n w \rangle| \leq Ch_\Gamma^2 \|\partial_n w\|_{\Gamma_h} \leq Ch_\Gamma^2 (c_0 h_\Gamma)^{-1/2} \|w\|$ and $\frac{\mu}{h} |\langle \tilde{u}, w \rangle| \leq C\gamma^{1/2} h_\Gamma^{3/2} \|w\|$. Finally, the proof of Lemma 13.11 gives $|(F, w)| \leq Ch_\Gamma^2 (\|w\|_{L^1(\Gamma_h)} + h_\Gamma \|\nabla w\|) \leq Ch_\Gamma^2 \|w\|$. $\square$

Lemma 8.5 (B_h is bounded). $|B_h(y, w)| \leq (1 + 2(c_0 \gamma)^{-1/2}) \|y\| \|w\|$ for all y, w for which the norms are finite.

Proof. $\|\partial_n y\|_{\Gamma_h} \leq (c_0 h_\Gamma)^{-1/2} \|y\|$ and $\|w\|_{\Gamma_h} \leq (h/\mu)^{1/2} \|w\|$, and $(h/(\mu c_0 h_\Gamma))^{1/2} = (c_0 \gamma)^{-1/2}$. $\square$

Lemma 8.6 (Normal test fields). *Let $\chi_0 \in C^\infty(\mathbb{R}^2)$ with $\chi_0 = 1$ on $\{|x| \leq r_1\}$ and $\chi_0 = 0$ on $\{|x| \geq r_2\}$, $1 < r_1 < r_2 < L$. For $\phi \in C^{k+1}(\Gamma)$ put $\hat{n}(x) := -x/|x|$ and*

$$w_\phi := I_h(\chi_0(\phi \circ \pi)\hat{n}), \quad \pi(x) := x/|x|,$$

with I_h the P_k Lagrange interpolant. For $h \leq h_$: $w_\phi \in V_h^R$, $\|w_\phi\|_{W^{1,\infty}} \leq C_\phi$, and on each $e \in \mathcal{E}_h^\Gamma$*

$$|w_\phi - \phi(x^*)n(x^*)| \leq C_\phi h_\Gamma^{k+1}, \quad |w_\phi \cdot \tau_e| \leq C_\phi h_\Gamma, \quad |w_\phi \cdot n_e - \phi(x^*)| \leq C_\phi h_\Gamma^2,$$

and $|\int_{\Gamma_h} w_\phi \cdot n - \int_\Gamma \phi| \leq C_\phi h_\Gamma^2$. For $\phi \equiv 1$ also $(\operatorname{div} w_1)|_{T_e} = -1 + O(h_\Gamma^2)$ on e .

Proof. The triangles meeting Γ_h lie in $\{|x| \leq 1 + Ch_\Gamma\} \subset \{\chi_0 = 1\}$, and those meeting ∂R lie in $\{\chi_0 = 0\}$. On Γ_h , $\hat{n} = n \circ \pi$, and the interpolation error is $O(h_\Gamma^{k+1})$ in L^∞ and $O(h_\Gamma^k)$ in $W^{1,\infty}$ on boundary triangles. By Lemma 3.1, $|n(x^*) \cdot \tau_e| \leq Ch_\Gamma$ and $n(x^*) \cdot n_e = 1 - \frac{1}{2}|n(x^*) - n_e|^2 = 1 + O(h_\Gamma^2)$. The arclength Jacobian of $\pi|_{\Gamma_h}$ is $1 + O(h_\Gamma^2)$. Finally $\operatorname{div} \hat{n} = -1/|x| = -1 + O(h_\Gamma^2)$ on Γ_h . $\square$

8.3. The pressure of CNS*.

Theorem 8.7 (CNS* pressure). *Let $\gamma \geq \gamma_2$. The CNS* pressure p_h (unique by Theorem 7.2) satisfies, with $\bar{e}_p$ the mean of $e_p = p^\circ - p_h$ over Ω_h ,*

$$\|e_p - \bar{e}_p\|_{L^2(\Omega_h)} \leq CY, \\ |\bar{e}_p| \leq C[(1 + (\gamma h_\Gamma)^{1/2})Y + (\mu/h)|m_R|].$$

Hence $\|p^\circ - p_h\|_{L^2(\Omega_h)} \leq C[(1 + (\gamma h_\Gamma)^{1/2})Y + (\mu/h)|m_R|]$. With flux-corrected data (Definition 2.2) the same bounds hold with the term $(\mu/h)|m_R|$ deleted and ε_h replaced by $\tilde{\varepsilon}_h$ of Lemma 7.32 in Y ; consequently, for $\gamma \in [\gamma_2, \gamma_{\max}]$,

$$\|p^\circ - p_h\|_{L^2(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k),$$

with no relation between h_Γ and h_Ω required. Here C depends on $\gamma_{\max}$: unlike the H^1 bound of Theorem 7.33, this pressure bound is not uniform in γ (Remark 8.8). Moreover $|\bar{p}_\Gamma(p_h)| \leq C[h_\Gamma^{-1/2}Y + (1 + (\gamma h_\Gamma)^{1/2})Y + (\mu/h)|m_R|]$.

Proof. Mean-free part. Write $e_p = \xi + \eta$ with $\eta := p^\circ - \pi_h p^\circ$, so $\|\eta\| \leq Ch^k$ and $\|\eta\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^k$. Step 3 of Theorem 7.2 gives $\beta\|\xi - \bar{\xi}\| \leq C(\|e_u\| + h_\Gamma^{3/2}) \leq CY$.

Constant. Let $B^*(r, v) := -(r, \operatorname{div} v) + \langle r - \bar{p}_\Gamma(r), v \cdot n \rangle$. Then $B^*(c, v) = -c \int_{\Gamma_h} v \cdot n$ for constants c , and the error equation (3) gives, for $v := w_1$ of Lemma 8.6,

$$\bar{e}_p \int_{\Gamma_h} w_1 \cdot n = N_h(e_u, w_1) + B^*(e_p - \bar{e}_p, w_1) - \mathcal{G}(w_1). \quad (9)$$

Here $\int_{\Gamma_h} w_1 \cdot n \geq |\Gamma_h|/2$. We bound the three terms on the right.

- $B^*(e_p - \bar{e}_p, w_1) = -(e_p - \bar{e}_p, \operatorname{div} w_1) + \langle e_p - \bar{p}_\Gamma(e_p), w_1 \cdot n - 1 \rangle$, because $\langle e_p - \bar{p}_\Gamma(e_p), 1 \rangle = 0$. The first term is at most $C\|e_p - \bar{e}_p\|$. The second is at most $Ch_\Gamma^2\|e_p - \bar{e}_p\|_{L^1(\Gamma_h)} \leq Ch_\Gamma^{3/2}\|\xi - \bar{\xi}\| + Ch_\Gamma^{k+2}$, by Lemma 8.6 and the inverse trace inequality Lemma 4.1.
- $|\mathcal{G}(w_1)| \leq C(h_\Gamma^2 + \gamma h_\Gamma^3)$. The transposed term is $\sum_e \int_e (sa_e + O(h_\Gamma^2)) \cdot w_1$ with $a_e \parallel n(m_e^*)$; since $sa_e \cdot w_1 = sW_e(1 + O(s^2))$ and $\int_e s ds = 0$, it is $O(h_\Gamma^2)$. Next, $|\langle \tilde{u}, \partial_n w_1 \rangle| \leq Ch_\Gamma^2$. On e , $\tilde{u}(x) = d \partial_n u(x^*) + O(d^2)$ with $\partial_n u(x^*) \perp n(x^*)$, so $|\tilde{u} \cdot w_1| \leq Ch_\Gamma^4$ and $\frac{\mu}{h} |\langle \tilde{u}, w_1 \rangle| \leq C\gamma h_\Gamma^3$. Finally $|(F, w_1)| \leq Ch_\Gamma^2$.
- In $N_h(e_u, w_1) = a_h(e_u, w_1) - \langle \partial_n e_u, w_1 \rangle - \langle e_u, \partial_n w_1 \rangle + \frac{\mu}{h} \langle e_u, w_1 \rangle$:
 - $|a_h(e_u, w_1)| \leq C\|\nabla e_u\|$ and $|\langle e_u, \partial_n w_1 \rangle| \leq C\|e_u\|_{\Gamma_h} \leq C(h/\mu)^{1/2}\|e_u\|$.
 - Write $e_u = (\tilde{u} - I_h \tilde{u}) + \delta$ with $\delta := I_h \tilde{u} - u_h \in V_h^R$. Then $\operatorname{div} \delta = \operatorname{div}(I_h \tilde{u} - \tilde{u}) = O(h_\Gamma^k)$ on L_1 , and $\|\nabla \delta\| \leq \|\nabla e_u\| + Ch^k$. Lemma 8.3 with $S = w_1|_{\Gamma_h}$ gives $|\langle \partial_n e_u, w_1 \rangle| \leq C(\|\nabla e_u\| + h^k)$.
 - $\frac{\mu}{h} \langle e_u, w_1 \rangle = \frac{\mu}{h} [m_R + \langle e_u \cdot n, w_1 \cdot n - 1 \rangle + \langle e_u \cdot \tau, w_1 \cdot \tau \rangle]$, and the last two terms are at most $\frac{\mu}{h} Ch_\Gamma \|e_u\|_{L^1(\Gamma_h)} \leq C(\gamma h_\Gamma)^{1/2} \|e_u\|$.

Collect, using $\|e_u\| \leq CY$, $\gamma h_\Gamma^3 \leq (\gamma h_\Gamma)^{1/2} Y$ and $h^k \leq \varepsilon_h$.

Flux-corrected data. Take $z \in \tilde{W}_h$ from Lemma 7.32, so $\|e_u\| \leq CY$ with $\tilde{\varepsilon}_h$ in Y (Theorem 7.33). In the second bullet replace $I_h \tilde{u}$ by the field $w_h = I_h \tilde{u} - c I_h(\zeta x)$ from the proof of Lemma 7.32: it equals $I_h \tilde{u}$ near Γ_h and has trace $\tilde{g}_I$ on ∂R , so $\delta := w_h - u_h \in V_h^R$, $\operatorname{div} \delta = \operatorname{div}(I_h \tilde{u} - \tilde{u})$ on L_1 , and $\|\nabla \delta\| \leq \|\nabla e_u\| + C(h^k + |m_R|)$ with $|m_R| \leq Ch^{\sigma_k}$. In the third bullet $\int_{\Gamma_h} e_u \cdot n = 0$, so the term $(\mu/h)m_R$ is absent. Finally $h^k \leq \tilde{\varepsilon}_h$, and for bounded γ , $Y \leq C(h_\Gamma^{3/2} + h^k)$. For the last claim, $\bar{p}_\Gamma(p^\circ) = 0$, so $|\bar{p}_\Gamma(p_h)| = |\bar{p}_\Gamma(e_p)| \leq |\bar{e}_p| + |\Gamma_h|^{-1/2} \|e_p - \bar{e}_p\|_{\Gamma_h}$, and $\|\xi - \bar{\xi}\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2} \|\xi - \bar{\xi}\|$. $\square$

Remark 8.8 (Where the pressure error sits). The $h_\Gamma^{3/2}$ is produced by the right-hand side of Lemma 8.1: $a_h(e_u, \cdot)$ and the Nitsche coupling $\langle u_h, \partial_n \cdot \rangle$, where $\|u_h\|_{\Gamma_h} \leq \|\tilde{u}\|_{\Gamma_h} + (h/\mu)^{1/2} \|e_u\| \leq C(1 + \gamma^{-1/2})h_\Gamma^2 + C(h/\mu)^{1/2} \varepsilon_h$ and $\|\partial_n v\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2} \|\nabla v\|$. A $h_\Gamma^{3/2}$ lower bound for the pressure is not proved. The factor $(1 + \gamma^{1/2})$ in Y is real: at fixed N the computed pressure error grows with μ (Section 16.9).

8.4. The pressure of GS.

Proposition 8.9 (The GS pressure is fully determined). *For $\gamma \geq \gamma_0$, GS has a unique solution $(u_h, p_h) \in W_h \times \Pi_h$. No normalisation of p_h is needed or possible: the constant is fixed by the flux-carrying test functions in V_h^R .*

Proof. The velocity is the unique Lax–Milgram solution on W_h , because N_h is coercive on Z_h Lemma 4.2. The functional $v \mapsto N_h(u_h, v) - (f, v)$ vanishes on $Z_h = \ker(\operatorname{div}|_{V_h^R})$. Since $\operatorname{div} V_h^R = \Pi_h$ Lemma 4.3 and Π_h is finite-dimensional, there is exactly one $p_h \in \Pi_h$ with $(p_h, \operatorname{div} v)$ equal to that functional on V_h^R . Adding a constant c to p_h changes the momentum row by $-c \int_{\Gamma_h} v \cdot n \neq 0$ (edge bubble). $\square$

Definition 8.10 (The Nitsche pressure layer). With $g := (h/\mu)q \circ \pi$ on Γ_h , let $\xi_b \in \Pi_h$ be $\xi_{e,g}$ (Lemma 8.2) on each T_e and 0 outside L_1 . Equivalently,

$$(\xi_b, r) = (h/\mu) \langle q \circ \pi, r|_{T_e} \rangle_{\Gamma_h} \quad \forall r \in \Pi_h.$$

Theorem 8.11 (GS pressure). *Let $\gamma \geq \gamma_0$ and $\rho_h := p^\circ - p_h - \xi_b$. Then:*

(a) $\|\rho_h - \bar{\rho}_h\| \leq C_p X_1$ and

$$\|\rho_h\|_{L^2(\Omega_h)} \leq C_p [(1 + (\gamma h_\Gamma)^{1/2}) X_1 + (\mu/h) |m_R|] =: C_p Q.$$

$$(b) \quad \|\xi_b\|^2 = \left(\frac{h}{\mu}\right)^2 \left[\hat{c}_k \sum_{e \in \mathcal{E}_h^\Gamma} \frac{|e|}{H_e} q(m_e^*)^2 + O(C_p) \right]. \quad \text{In particular } c(h/\mu)h_\Gamma^{-1/2}G \leq \|\xi_b\| \leq$$

$$C(h/\mu)h_\Gamma^{-1/2}(G + C_p h_\Gamma^{1/2}), \text{ the lower bound holding for } h_\Gamma \leq h_0(G).$$

(c) For every constant c ,

$$c_1 \frac{h}{\mu} h_\Gamma^{-1/2} G - C_p Q \leq \|p^\circ - p_h - c\|_{L^2(\Omega_h)}$$

$$\text{and } \|p^\circ - p_h\|_{L^2(\Omega_h)} \leq C(h/\mu)h_\Gamma^{-1/2}G + C_p Q.$$

$$(d) \quad \|p^\circ - p_h\|_{L^2(\Omega_h \setminus L_1)} \leq C_p Q.$$

$$(e) \quad \text{For every constant } c, \|p^\circ - p_h - c\|_{L^\infty(L_1)} \geq c_2 G/\gamma - C_p h_\Gamma^{-1/2} Q.$$

For fixed γ and bounded ρ with $G > 0$: the L^2 pressure error converges at the exact rate $1/2$. It is carried by ξ_b on the first layer, with $\|p^\circ - p_h\|_{L^2(L_1)}/\|\xi_b\| \rightarrow 1$. It is $O(h/\mu)$ elsewhere, and in the first layer it does not converge pointwise. If $G = 0$, then $\xi_b = 0$ and the CNS* bounds of Theorem 8.7 hold for GS with C in place of C_p .

Proof. Step 1: the equation for ρ_h . By Lemma 8.1, (8) and Definition 8.10, for $v \in V_h^0$,

$$(\rho_h, \operatorname{div} v) = a_h(e_u, v) + \langle u_h - g n, \partial_n v \rangle + (F, v).$$

Step 2: the trace $u_h - g n$. As in the proof of Corollary 6.7, $e_u = (h/\mu)v_\varphi + \vartheta$ with $\vartheta := (\tilde{u} - z) + \delta_G + r_h$ and $\|\vartheta\| \leq C_p X_1$. (By Theorem 6.4, steps 2 and 6: $\|\delta_G\| + \|\tilde{u} - z\| \leq CY$ and $\|r_h\| \leq C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h$.) On e , $v_\varphi = w + O(h^k)$ and $w(x) = -q(x^*)n(x^*) + O(h_\Gamma^2)$, so

$$v_\varphi + (q \circ \pi)n_e = q(x^*)(n_e - n(x^*)) + O(h_\Gamma^2 + h^k) = O_p(h_\Gamma + h^k).$$

Hence

$$\|u_h - g n\|_{\Gamma_h} \leq \|\tilde{u}\|_{\Gamma_h} + \frac{h}{\mu} C_p(h_\Gamma + h^k) + \|\vartheta\|_{\Gamma_h} \leq C h_\Gamma^2 + C_p \frac{h}{\mu} h_\Gamma + (h/\mu)^{1/2} C_p X_1.$$

Step 3: mean-free part. With $\|\partial_n v\|_{\Gamma_h} \leq C h_\Gamma^{-1/2} \|\nabla v\|$, $\|\nabla e_u\| \leq C_p X_1$ Theorem 6.4 and $|(F, v)| \leq C h_\Gamma^2 \|\nabla v\|$, the right-hand side of Step 1 is at most $C_p[X_1 + h_\Gamma^{3/2} + (h/\mu)h_\Gamma^{1/2} + \gamma^{-1/2}X_1]\|\nabla v\| \leq C_p X_1 \|\nabla v\|$. Now $\rho_h - \eta \in \Pi_h$ with $\eta = p^\circ - \pi_h p^\circ$, and $(\eta, \operatorname{div} v) = 0$. Lemma 8.12 below (Lemma 4.5) gives $\|\rho_h - \bar{\rho}_h\| \leq C_p X_1 + C h^k \leq C_p X_1$.

Step 4: constant. The GS error equation on V_h^R is $N_h(e_u, v) - (p^\circ - p_h, \operatorname{div} v) + \langle p^\circ, v \cdot n \rangle = \mathcal{G}(v)$; the constant in p° cancels because $\int_{\Omega_h} \operatorname{div} v = \int_{\Gamma_h} v \cdot n$. With $v = w_1$,

$$\bar{\rho}_h \int_{\Gamma_h} w_1 \cdot n = N_h(e_u, w_1) + \langle p^\circ, w_1 \cdot n \rangle - \mathcal{G}(w_1) - (\xi_b, \operatorname{div} w_1) - (\rho_h - \bar{\rho}_h, \operatorname{div} w_1).$$

We bound each term.

- $\langle p^\circ, w_1 \cdot n \rangle = \langle p^\circ, w_1 \cdot n - 1 \rangle = O(h_\Gamma^2)$, because $\langle p^\circ, 1 \rangle = 0$.
- $|\mathcal{G}(w_1)| \leq C(h_\Gamma^2 + \gamma h_\Gamma^3)$, as in Theorem 8.7.
- $\operatorname{div} w_1 \in \Pi_h$, so $(\xi_b, \operatorname{div} w_1) = (h/\mu)\langle q \circ \pi, -1 + O(h_\Gamma^2) \rangle = O_p((h/\mu)h_\Gamma^2)$, since $\int_\Gamma q = 0$ and the Jacobian is $1 + O(h_\Gamma^2)$.
- $|\langle \rho_h - \bar{\rho}_h, \operatorname{div} w_1 \rangle| \leq C_p X_1$.
- $N_h(e_u, w_1)$ is treated as in Theorem 8.7, except that $\frac{\mu}{h} \langle e_u \cdot \tau, w_1 \cdot \tau \rangle$ is split along $e_u = (h/\mu)v_\varphi + \vartheta$. On e , $v_\varphi \cdot \tau_e$ and $w_1 \cdot \tau_e$ are both $O_p(h_\Gamma)$, so the first part is $O_p(h_\Gamma^2)$. The second is at most $\frac{\mu}{h} C h_\Gamma (h/\mu)^{1/2} \|\vartheta\| \leq C(\gamma h_\Gamma)^{1/2} C_p X_1$. The term $\frac{\mu}{h} \langle e_u \cdot n, w_1 \cdot n - 1 \rangle$ is at most $C \gamma^{1/2} h_\Gamma^{3/2} \|e_u\| \leq C_p(h_\Gamma^2 + (\gamma h_\Gamma)^{1/2} Y)$, by Theorem 6.1.

Hence $|\bar{\rho}_h| \leq C_p[(1 + (\gamma h_\Gamma)^{1/2})X_1 + (\mu/h)|m_R|]$, which proves (a).

(b) Apply Lemma 8.2 on each T_e with $g_e = (h/\mu)q(m_e^*)$ and $\|g - g_e\|_{L^\infty(e)} \leq C_p(h/\mu)|e|$. Expanding the square, the cross terms are $O_p((h/\mu)^2 \sum_e |e|)$. The two-sided bound uses $c \leq |e|/H_e \leq C$ and the Riemann sum $\sum_e q(m_e^*)^2 |e| = G^2 + O_p(h_\Gamma)$, with $h_\Gamma^{-1} \sum_e q_e^2 |e| \leq \sum_e q_e^2 \leq (c_0 h_\Gamma)^{-1} \sum_e q_e^2 |e|$.

(c) $p^\circ - p_h - c = \xi_b - c + \rho_h$. The L^2 -closest constant to ξ_b is its mean, $|\Omega_h|^{-1}(\xi_b, 1) = |\Omega_h|^{-1}(h/\mu)\langle q \circ \pi, 1 \rangle = O_p((h/\mu)h_\Gamma^2)$. Hence $\|\xi_b - c\| \geq \|\xi_b\| - C_p(h/\mu)h_\Gamma^2$. Use (a) and (b).

(d) $\xi_b = 0$ off L_1 .

(e) $\|\cdot\|_{L^\infty(L_1)} \geq |L_1|^{-1/2} \|\cdot\|_{L^2(L_1)}$ with $|L_1| \leq Ch_\Gamma$. The mean of ξ_b over L_1 is $|L_1|^{-1}(h/\mu)\langle q \circ \pi, 1 \rangle = O_p((h/\mu)h_\Gamma)$, so, as in (c), $\|\xi_b - c\|_{L^2(L_1)} \geq \|\xi_b\| - C_p(h/\mu)h_\Gamma^{3/2}$. Then use (a), (b) and $(h/\mu)h_\Gamma^{-1} = 1/\gamma$.

The regime statement. For fixed γ and bounded ρ , $Q = O(h_\Gamma)$ while $(h/\mu)h_\Gamma^{-1/2} \asymp h_\Gamma^{1/2}$. If $G = 0$, then $g = 0$, and Theorems 6.1 and 6.4 give $\|e_u\| + \|\nabla e_u\| \leq CY$; Steps 1–4 then run with X_1 replaced by Y . $\square$

Lemma 8.12 (Lemma 4.5, restated). *For $r \in \Pi_h$, $\beta\|r - \bar{r}\| \leq \sup_{0 \neq v \in V_h^0}(r, \operatorname{div} v)/\|\nabla v\|$.*

Remark 8.13 (Mechanism). The leak $u_h \cdot n \approx (h/\mu)(p - \bar{p})$ (Corollary 6.7) enters the pressure only through the Nitsche coupling $\langle u_h, \partial_n v \rangle$. By (8), that term is a boundary source $\langle u_h \cdot n, \operatorname{div} v \rangle_{\Gamma_h}$, which the discontinuous pressure resolves exactly, as the first-layer field ξ_b . Its L^2 norm is $(h/\mu)|e|^{-1/2}$ per unit length of boundary, and its pointwise size is $(h/\mu)/H_e \asymp 1/\gamma$. The penalty scaling μ/h is what makes it independent of the mesh at fixed γ . It is a discrete artefact: it has no counterpart in the leak flow (U, P) of Definition 6.8, whose pressure enters only through $\rho_h \approx (h/\mu)P$.

8.5. Boundary stress and drag.

8.5.1. *Functionals.* Let $\operatorname{RM} := \{\psi(x) = a + bx^\perp : a \in \mathbb{R}^2, b \in \mathbb{R}\}$, $x^\perp = (-x_2, x_1)$. Every $\psi \in \operatorname{RM}$ has $\operatorname{div} \psi = 0$, $D(\psi) = 0$ and $\nabla \psi$ skew, and $\int_{\Gamma_h} \psi \cdot n = \int_\Gamma \psi \cdot n = 0$. For $\psi \in \operatorname{RM}$ put

$$J_\psi := \int_\Gamma \sigma(u, p)n \cdot \psi \, ds, \quad \sigma(u, p) = D(u) - pI,$$

with n the outward normal of Ω (pointing into D). J_ψ does not depend on the pressure constant. With $\psi = e_1, e_2, x^\perp$ it is Gjerde–Scott’s $\beta(\chi)$ [34, (3), (7)], $\chi = \psi$ on Γ : drag, lift and torque, up to their sign convention.

Definition 8.14 (Discrete functionals). Let $\phi_\psi(x) := a_1 x_2 - a_2 x_1 - \frac{b}{2}|x|^2$, so that $\operatorname{curl} \phi_\psi = \psi$. Let χ_0 be as in Lemma 8.6, and put

$$V_\psi := \operatorname{curl}(\chi_0 \phi_\psi), \quad v_\psi := \operatorname{curl} \Pi_{k+1}(\chi_0 \phi_\psi) \in Z_h,$$

where Π_{k+1} is the C^1 interpolant of Section 6. For $w \in V_h^R$ put

$$\omega_h(w) := a_h(u_h, w) - (p_h, \operatorname{div} w) - (\tilde{f}, w)_{\Omega_h}.$$

We consider:

- (i) the *Nitsche-flux (variational) functional* $J_h^N(\psi) := \omega_h(v_\psi)$;
- (ii) the *pointwise traction functional* $J_h^\beta(\psi) := \int_{\Gamma_h} (D(u_h)n - p_h n) \cdot \psi$;
- (iii) for translations $\psi = e$, the *Gjerde–Scott volume functional* $J_h^\chi(e) := \omega_h(\chi_h)$. Here $\chi_h \in Z_h$ is the same method’s discrete Stokes solution with zero body force, zero data on ∂R and Nitsche data e on Γ_h : for GS, $N_h(\chi_h, w) - (\pi_h, \operatorname{div} w) = \ell_e(w)$, with $\ell_e(w) := -\langle e, \partial_n w \rangle + \frac{\mu}{h} \langle e, w \rangle$.

Let $\Theta_f(\psi) := \int_{S_h} \tilde{f} \cdot \psi$ be the body force carried by the strip.

Gjerde–Scott compute (ii), with χ_h in place of ψ on Γ_h , and (iii), with the pressure term dropped because $\operatorname{div} \chi_h = 0$. Their inconsistency parameter is $\varepsilon = (\omega - \beta)/\omega$ [34, (10)].

Lemma 8.15 (Flux identity and error identity). *Let $t_h := \partial_n u_h - \theta(p_h - \bar{p}_\Gamma(p_h))n - \frac{\mu}{h}u_h$ be the Nitsche flux. For both methods:*

(a) $\omega_h(w) = \langle t_h, w \rangle + \langle u_h, \partial_n w \rangle$ for all $w \in V_h^R$. Consequently $v_\psi = \psi$ on L_1 ,

$$J_h^N(\psi) = \int_{\Gamma_h} t_h \cdot \psi + \langle u_h, \partial_n \psi \rangle,$$

and $J_h^N(\psi) = \omega_h(w)$ for every $w \in V_h^R$ with $w = \psi$ on L_1 . For GS, $J_h^N(\psi)$ contains no pressure at all.

(b) $J_h^N(\psi) - J_\psi = -a_h(e_u, v_\psi) - \Theta_f(\psi)$.

(c) $\Theta_f(\psi) = \sum_e \frac{|e|^3}{12} (f \cdot \psi)(m_e^*) + O(h_\Gamma^3)$, and $\Theta_f = 0$ if $f = 0$ near Γ .

Proof. (a) Subtract the method equation from the definition of ω_h : $\omega_h(w) = a_h(u_h, w) - N_h(u_h, w) - \theta(p_h - \bar{p}_\Gamma(p_h), w \cdot n)$. The triangles of L_1 lie in $\{\chi_0 = 1\}$, where $\chi_0 \phi_\psi = \phi_\psi \in P_2$ is reproduced by Π_{k+1} , so $v_\psi = \psi$ on L_1 and $\partial_n v_\psi = \partial_n \psi$ on Γ_h (traces from T_e). If $w = \psi$ on L_1 , then $w - v_\psi$ has zero trace and zero normal derivative on Γ_h , so $\omega_h(w - v_\psi) = 0$.

(b) For $w \in V_h^R$ let $\tilde{\omega}(w) := a_h(\tilde{u}, w) - (p^\circ, \operatorname{div} w) - (\tilde{f}, w)$. Steps 1–3 of Lemma 5.1, keeping $D(\tilde{u})n$, give $\tilde{\omega}(w) = \langle \sigma(\tilde{u}, p^\circ)n, w \rangle - (F, w)$. The segment between each chord and its arc lies in T_e (the arc leaves the chord at angle $O(h_\Gamma)$, while the base angles of T_e are bounded below), so $v_\psi = \psi$ on S_h . Since $\sigma(\tilde{u}, p^\circ)$ is symmetric and $\nabla \psi$ is skew, $\operatorname{div}(\sigma \psi) = (\operatorname{div} \sigma) \cdot \psi = (F - \tilde{f}) \cdot \psi$ on S_h . The divergence theorem on S_h (outward normal n on Γ_h and $-n$ on Γ ; constants in the pressure drop because $\int \psi \cdot n = 0$) gives $\langle \sigma(\tilde{u}, p^\circ)n, \psi \rangle_{\Gamma_h} = J_\psi + (F - \tilde{f}, \psi)_{S_h}$. Hence $\tilde{\omega}(v_\psi) = J_\psi - \Theta_f(\psi)$. Finally $\omega_h(v_\psi) - \tilde{\omega}(v_\psi) = -a_h(e_u, v_\psi) + (e_p, \operatorname{div} v_\psi) = -a_h(e_u, v_\psi)$.

(c) The segment over e has area $\frac{|e|^3}{12} + O(|e|^5)$ and width $O(|e|^2)$, and $\tilde{f}$ is smooth. $\square$

So the Babuška–Miller volume drag, with any discrete test field equal to ψ on the first layer, is the Nitsche-flux drag, exactly. By (b), its error is the energy error tested against a fixed field that is rigid near the wall. Since $D(v_\psi) = D(\psi) = 0$ on $\{\chi_0 = 1\}$, $a_h(e_u, v_\psi)$ only sees e_u away from Γ .

8.5.2. Dual problems and reciprocity. For $\psi \in \operatorname{RM}$ let (Z_ψ, R_ψ) solve the Stokes problem in Ω with $Z_\psi = \psi$ on Γ and $Z_\psi = 0$ on ∂R . Then $Z_\psi \in H^2(\Omega)$ (the corners of R are convex), Z_ψ is smooth up to Γ , and $Z_\psi = \operatorname{curl} \Phi_\psi$ with Φ_ψ single-valued; extend Φ_ψ, R_ψ smoothly across Γ to $\tilde{Z}_\psi, \tilde{R}_\psi$. Let (U, P) be the leak flow of Definition 6.8: $U = -qn$ on Γ , $U = 0$ on ∂R .

Lemma 8.16 (Reciprocity). $J_\psi(U) := \int_\Gamma \sigma(U, P)n \cdot \psi = \int_\Gamma (p - \bar{p})(R_\psi - \bar{R}_\psi) ds =: K_\psi$.

Proof. For Stokes pairs with zero body force, $\int_{\partial\Omega} \sigma(U, P)n \cdot Z = \frac{1}{2} \int_\Omega D(U) : D(Z) = \int_{\partial\Omega} \sigma(Z, R)n \cdot U$, using $\operatorname{div} U = \operatorname{div} Z = 0$ and the $H^2 \times H^1$ regularity. Both fields vanish on ∂R . On Γ , $Z_\psi - \psi$ is divergence-free and zero, so $\partial_n(Z_\psi - \psi) \cdot n = 0$ Lemma 3.4. Also $(\nabla \psi)n \cdot n = 0$ by skewness. Hence $\sigma(Z_\psi, R_\psi)n \cdot n = 2\partial_n Z_\psi \cdot n - R_\psi = -R_\psi$. Since $U = -qn$, the right-hand side equals $\int_\Gamma q R_\psi$; and $\int_\Gamma q = 0$. $\square$

8.5.3. The printed method.

Theorem 8.17 (GS drag, lift and torque). *Let $\gamma \geq \gamma_0$ and $\psi \in \operatorname{RM}$. Then*

$$J_h^N(\psi) = J_\psi - \frac{h}{\mu} K_\psi - \Theta_f(\psi) + \mathcal{R}_h, \quad |\mathcal{R}_h| \leq C \frac{h}{\mu} B_U + C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h],$$

where $B_U := C_U[\min(\gamma h_\Gamma^{1/2}, (\gamma h_\Gamma)^{1/2}) + h_\Gamma^{1/2} + (h/\mu)^{1/2} + E_U]$ is the bound of Theorem 6.10. Without that theorem, $|J_h^N(\psi) - J_\psi| \leq C_p(h/\mu) + C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$. For fixed γ and bounded ρ , $J_h^N(\psi) - J_\psi = -(h/\mu)K_\psi + O(h^{3/2})$. So the rate is exactly 1 whenever $K_\psi \neq 0$, and the defect is the force that the leak flow exerts on the body.

Proof. By Lemma 8.15(b), the claim is about $a_h(e_u, v_\psi)$. Split $e_u = \delta_R + [\delta_G + (\tilde{u} - z)]$ as in Theorem 6.4. The bracket contributes at most $2\|\nabla v_\psi\| C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$. Next, $a_h(\delta_R, v_\psi) = \frac{h}{\mu}a_h(\tilde{U}, v_\psi) + a_h(\delta_R - \frac{h}{\mu}\tilde{U}, v_\psi)$, and the second term is at most $C\frac{h}{\mu}B_U$ by Theorem 6.10. By Green's formula with $F_U := -\Delta\tilde{U} + \nabla\tilde{P}$ (supported in $\overline{S_h}$), $\operatorname{div} v_\psi = 0$ and $v_\psi = \psi$ on $L_1 \supset S_h$,

$$a_h(\tilde{U}, v_\psi) = (F_U, \psi)_{S_h} + \langle \sigma(\tilde{U}, \tilde{P})n, \psi \rangle_{\Gamma_h} = (F_U, \psi)_{S_h} + J_\psi(U) - (F_U, \psi)_{S_h} = J_\psi(U).$$

The middle equality is the divergence theorem on S_h , as in Lemma 8.15(b). Conclude with Lemma 8.16. The unconditional bound is $|a_h(e_u, v_\psi)| \leq C\|\nabla e_u\|$ together with Theorem 6.4. $\square$

Remark 8.18 (The leak drag is the pressure–pressure correlation). The leak $u_h \cdot n \approx (h/\mu)q$ makes the Nitsche flux reproduce the missing traction at leading order: $-\frac{\mu}{h}u_h \approx -qn$. That is why J_h^N converges at all, although the GS flux contains no pressure. The residual $-(h/\mu)K_\psi$ couples the flow's wall pressure with the wall pressure of the dual problem. In Gjerde–Scott's cylinder application, $f = 0$ (so $\Theta_f = 0$) and both pressures are dipole-like, so $K_e \neq 0$ generically. With their $\mu = 10^6$ the defect has relative size about $10^{-6}h$ and is invisible.

8.5.4. The pressure-consistent method.

Theorem 8.19 (CNS* drag, lift and torque). *Let $\gamma \geq \max(1, \gamma_2)$, $\gamma h_\Gamma \leq 1$ and $\psi \in \operatorname{RM}$. Define*

$$\Lambda_h(\psi) := \langle \tilde{u}, \partial_n(\tilde{Z}_\psi - \psi) \rangle_{\Gamma_h} = \sum_{e \in \mathcal{E}_h^\Gamma} \frac{|e|^3}{12} (\partial_n u \cdot \partial_n(Z_\psi - \psi))(m_e^*) + O(h_\Gamma^3),$$

and $E_Z := \inf_{z \in Z_h} \|\tilde{\zeta} - z\|$, where $\tilde{\zeta} := V_\psi - \tilde{Z}_\psi$. Then:

- (a) $|J_h^N(\psi) - J_\psi + \Theta_f(\psi)| \leq C\|\nabla e_u\| \leq CY$;
- (b) $|J_h^N(\psi) - J_\psi + \Theta_f(\psi) + \Lambda_h(\psi)| \leq C[\gamma^{-1/2}h_\Gamma^2 + \gamma h_\Gamma^3 + \gamma^{1/2}h_\Gamma^{3/2}E_Z + \varepsilon_h]$.

Consequently $|J_h^N(\psi) - J_\psi| \leq C(h_\Gamma^2 + \varepsilon_h)$: rate 2. As $\gamma \rightarrow \infty$ with $\gamma h_\Gamma \rightarrow 0$, $J_h^N(\psi) - J_\psi = -\Theta_f(\psi) - \Lambda_h(\psi) + o(h_\Gamma^2) + O(\varepsilon_h)$. Since $E_Z \leq Ch$ (argument of Lemma 6.9), the E_Z term is $o(h_\Gamma^2)$ whenever $h \leq Ch_\Gamma$.

As $\gamma \rightarrow \infty$ (with $\gamma h_\Gamma \rightarrow 0$) the leading term is *geometric* and method-independent. With uniform chords of length ℓ ,

$$\Theta_f + \Lambda_h \approx \frac{\ell^2}{12} \int_\Gamma (f \cdot \psi + \partial_n u \cdot \partial_n(Z_\psi - \psi)) ds.$$

This is the shape derivative of J_ψ under the inward normal displacement $\frac{1}{2}(\ell^2/4 - s^2)$ of the polygon. As $\gamma \rightarrow \infty$ (with $\gamma h_\Gamma \rightarrow 0$) only the polygonal geometry is left. At *fixed* γ there is an additional term $-\frac{h}{\mu}P_h(\rho_\psi)$ of order h_Γ^2 , the weighted normal slip of the discrete solution (Theorem 8.42); numerically it is genuinely of order h_Γ^2 , so the h_Γ^2 coefficient is not purely geometric at fixed γ .

Proof. (a) is Lemma 8.15(b). For (b), write $e_u = (\tilde{u} - z) + e_h$, with $z \in W_h$ attaining Lemma 4.6 and $e_h := z - u_h \in Z_h$.

Step 1: the auxiliary problem. Let $(\zeta_h, \pi_h) \in Z_h \times \Pi_h$ solve

$$N_h(\zeta_h, w) + B^*(\pi_h, w) = a_h(v_\psi, w) \quad \forall w \in V_h^R. \quad (10)$$

This is the CNS* Stokes system for the smooth-near- Γ solution $(\tilde{\zeta}, \Pi)$ with $\Pi := -\tilde{R}_\psi$, since $-\Delta\tilde{\zeta} + \nabla\Pi = -\Delta V_\psi$ on Ω , $\tilde{\zeta} = 0$ on $\Gamma \cup \partial R$ and $\operatorname{div} \tilde{\zeta} = 0$. Indeed, Lemma 5.1 for $(\tilde{\zeta}, \Pi)$ gives $N_h(\tilde{\zeta}, w) + B^*(\Pi^*, w) = (-\Delta V_\psi, w) + \mathcal{G}_\zeta(w)$, where $\Pi^* := \Pi - \bar{p}_\Gamma(\Pi)$ and $\mathcal{G}_\zeta$ is $\mathcal{G}$ with $\tilde{u} \rightarrow \tilde{\zeta}$ and $F \rightarrow F_\zeta := \Delta\tilde{Z}_\psi - \nabla\tilde{R}_\psi$ (supported in $\bar{S}_h$). Moreover $a_h(v_\psi, w) = (-\Delta V_\psi, w) + a_h(v_\psi - V_\psi, w)$, because $D(V_\psi) = 0$ near Γ_h , and $\|\nabla(v_\psi - V_\psi)\| \leq Ch^k$. Existence and uniqueness, and the bounds

$$\|\tilde{\zeta} - \zeta_h\| \leq CY_Z, \quad \beta\|(\Pi^* - \pi_h) - c_\Pi\| \leq CY_Z, \quad Y_Z := (1 + \gamma^{1/2})h_\Gamma^{3/2} + E_Z + h^k,$$

with c_Π the appropriate mean, follow from the proof of Theorem 7.2. That proof uses the regularity only near Γ (in Lemma 5.2 and in $\|\eta\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^k$), and ε_h enters only through E_h , here E_Z . Also $\|\zeta_h\| + \|\nabla\zeta_h\| \leq C$, since $\|\tilde{\zeta}\| \leq C(1 + \gamma^{1/2}h_\Gamma^{3/2})$.

Step 2: the error representation. Test (10) with $w = e_h$ (so $\operatorname{div} e_h = 0$, and $B^*(\pi_h, e_h) = \langle \pi_h - \bar{p}_\Gamma(\pi_h), e_h \cdot n \rangle$), and the error equation (3) with $w = \zeta_h$. By symmetry of N_h ,

$$\begin{aligned} a_h(e_u, v_\psi) &= T_1 + \mathcal{G}(\zeta_h) - T_4 + T_5, & T_1 &:= a_h(v_\psi - \zeta_h, \tilde{u} - z) - B_h(\tilde{u} - z, \zeta_h), \\ T_4 &:= \langle e_p - \bar{p}_\Gamma(e_p), \zeta_h \cdot n \rangle, & T_5 &:= \langle \pi_h - \bar{p}_\Gamma(\pi_h), e_h \cdot n \rangle. \end{aligned}$$

Hence $J_h^N - J_\psi + \Theta_f + \Lambda_h = -T_1 - (\mathcal{G}(\zeta_h) - \Lambda_h) + T_4 - T_5$.

Step 3: T_1 . $|T_1| \leq C\varepsilon_h$, by $\|\nabla(v_\psi - \zeta_h)\| \leq C$, $\|\zeta_h\| \leq C$ and Lemma 8.5.

Step 4: $\mathcal{G}(\zeta_h)$. By Lemma 8.4, $|\mathcal{G}(\zeta_h) - \mathcal{G}(\tilde{\zeta})| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}Y_Z$. On Γ_h , $|\tilde{\zeta}| \leq Ch_\Gamma^2$, $|(\nabla\tilde{u})^\top n| \leq Ch_\Gamma$ and $|\tilde{u}| \leq Ch_\Gamma^2$, and near Γ_h , $\partial_n\tilde{\zeta} = \partial_n\psi - \partial_n\tilde{Z}_\psi$. Hence

$$\mathcal{G}(\tilde{\zeta}) = \langle \tilde{u}, \partial_n(\tilde{Z}_\psi - \psi) \rangle + O(h_\Gamma^3 + \gamma h_\Gamma^3 + h_\Gamma^4) = \Lambda_h + O((1 + \gamma)h_\Gamma^3).$$

For the expansion of Λ_h : on e , $\tilde{u}(x) = d\partial_n u(x^*) + O(d^2)$ with $d = 1 - |x| = \frac{1}{2}(\frac{|e|^2}{4} - s^2) + O(|e|^4)$. Also $\nabla\tilde{Z}_\psi(x)n_e = \partial_n(Z_\psi - \psi)(x^*) + \partial_n\psi + O(h_\Gamma^2)$, by Lemma 3.4 applied to $Z_\psi - \psi$ and $n(x^*) \cdot n_e = 1 + O(h_\Gamma^2)$. Finally $\int_e d = \frac{|e|^3}{12} + O(|e|^5)$.

Step 5: T_4 . By step 3 of Theorem 7.2 and Lemma 4.1, $\|e_p - \bar{p}_\Gamma(e_p)\|_{\Gamma_h} \leq \|\xi - \bar{\xi}\|_{\Gamma_h} + \|\eta\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}Y$. By Lemma 3.4 applied to ζ , $\tilde{\zeta} \cdot n_e = d\partial_n\zeta(x^*) \cdot n_e + O(h_\Gamma^4) = O(h_\Gamma^3)$. So $\|\zeta_h \cdot n\|_{\Gamma_h} \leq Ch_\Gamma^3 + (h/\mu)^{1/2}CY_Z$, and $|T_4| \leq C(h_\Gamma^{5/2}Y + \gamma^{-1/2}Y Y_Z)$.

Step 6: T_5 . Since $\int_{\Gamma_h} e_h \cdot n = 0$, constants may be subtracted from π_h freely. Then

$$|T_5| \leq |\langle (\Pi - \bar{\Pi}) \circ \pi, e_u \cdot n \rangle| + C\|\tilde{u} - z\|_{\Gamma_h} + Ch_\Gamma^2\|e_h\|_{\Gamma_h} + C(h_\Gamma^{-1/2}Y_Z + h_\Gamma^k)\|e_h\|_{\Gamma_h},$$

where $\bar{\Pi}$ is the mean of Π on Γ and $\|e_h\|_{\Gamma_h} \leq (h/\mu)^{1/2}CY$. The first term is bounded by Lemma 8.20 below, with $\phi = \Pi - \bar{\Pi}$.

Step 7: collect. Insert $Y \leq C(\gamma^{1/2}h_\Gamma^{3/2} + \varepsilon_h)$ and $Y_Z \leq C(\gamma^{1/2}h_\Gamma^{3/2} + E_Z + h^k)$, and use $\gamma \geq 1$, $\gamma h_\Gamma \leq 1$ and $|\bar{e}_p| \leq C$ (Theorem 8.7). Every term is bounded by the right-hand side of (b); the $\gamma^{-1/2}h_\Gamma^2$ comes from the term $\frac{h}{\mu}h_\Gamma^{-1/2}Y$ of Lemma 8.20. $\square$

Lemma 8.20 (Weighted normal slip of CNS*). *Let $\gamma \geq \gamma_2$ and $\phi \in C^{k+1}(\Gamma)$ with $\int_\Gamma \phi = 0$. Then*

$$|\langle \phi \circ \pi, e_u \cdot n \rangle_{\Gamma_h}| \leq C_\phi \left[\frac{h}{\mu} (h_\Gamma^{-1/2}Y + h^k + h_\Gamma^2 + \gamma h_\Gamma^3 + h_\Gamma^2|\bar{e}_p|) + h_\Gamma \left(\frac{h}{\mu} \right)^{1/2} Y \right].$$

Proof. Test (3) with w_ϕ of Lemma 8.6:

$$\frac{\mu}{h} \langle e_u, w_\phi \rangle = \mathcal{G}(w_\phi) - a_h(e_u, w_\phi) + \langle \partial_n e_u, w_\phi \rangle + \langle e_u, \partial_n w_\phi \rangle - B^*(e_p, w_\phi).$$

On the left, $w_\phi - (\phi \circ \pi)n = O_\phi(h_\Gamma)$ on Γ_h , so $\frac{\mu}{h} \langle e_u, w_\phi \rangle = \frac{\mu}{h} \langle e_u \cdot n, \phi \circ \pi \rangle + O_\phi((\mu/h)^{1/2}h_\Gamma\|e_u\|)$. On the right:

- $|\mathcal{G}(w_\phi)| \leq C_\phi(h_\Gamma^2 + \gamma h_\Gamma^3)$, exactly as for w_1 in Theorem 8.7.

- $|a_h(e_u, w_\phi)| \leq C_\phi Y$.
- $|\langle \partial_n e_u, w_\phi \rangle| \leq C_\phi(Y + h^k)$, by Lemma 8.3 as in Theorem 8.7.
- $|\langle e_u, \partial_n w_\phi \rangle| \leq C_\phi(h/\mu)^{1/2} Y$.
- $B^*(e_p, w_\phi) = -(e_p - \bar{e}_p, \operatorname{div} w_\phi) - \bar{e}_p \int_{\Gamma_h} w_\phi \cdot n + \langle e_p - \bar{p}_\Gamma(e_p), w_\phi \cdot n \rangle$ is at most $C_\phi(Y + |\bar{e}_p| h_\Gamma^2 + h_\Gamma^{-1/2} Y)$, by Lemma 8.6 and $\int_\Gamma \phi = 0$.

Multiply by h/μ . □

8.5.5. Gjerde–Scott’s volume functional.

Proposition 8.21 (J_h^χ). *Let $\psi = e$ be a translation.*

- (a) (CNS*, $\gamma \geq \max(1, \gamma_2)$, $\gamma h_\Gamma \leq 1$.) $|J_h^\chi(e) - J_h^N(e)| \leq C[h_\Gamma^2 + (h/\mu)^{1/2}(Y + Y_Z)(1 + h_\Gamma^{-1/2} Y_Z)]$, which is $O(h_\Gamma^2 + (h/\mu)^{1/2} E_Z)$.
- (b) (GS, $\gamma \geq \gamma_0$ fixed, bounded ρ , and $E_Z + E_U \leq Ch$.)

$$J_h^\chi(e) = J_e - 2\frac{h}{\mu} K_e - \Theta_f(e) + O(h^{3/2}).$$

Gjerde–Scott’s own drag formula thus has twice the leak defect of the Nitsche flux, because the discrete test field χ_h leaks as well.

Proof. For translations, $\ell_e(w) = N_h(v_e, w) - a_h(v_e, w)$, since $v_e = e$ on L_1 . Hence $\zeta_h := v_e - \chi_h \in Z_h$ satisfies $N_h(\zeta_h, w) - \theta B^*(\pi_h, w) = a_h(v_e, w)$ for all $w \in V_h^R$ (for GS, for all $w \in Z_h$). By Lemma 8.15(a), $J_h^\chi - J_h^N = -\omega_h(\zeta_h)$.

(a) For CNS*, ζ_h is the solution of (10) (with $\pi_h \rightarrow -\pi_h$). Then $|\omega_h(\zeta_h)| \leq \|t_h\|_{\Gamma_h} \|\zeta_h\|_{\Gamma_h} + \|u_h\|_{\Gamma_h} \|\partial_n \zeta_h\|_{\Gamma_h}$. Here $\|t_h\|_{\Gamma_h} \leq C$ (Proposition 8.23(a) and $\gamma h_\Gamma \leq 1$), $\|\zeta_h\|_{\Gamma_h} \leq Ch_\Gamma^2 + (h/\mu)^{1/2} CY_Z$, $\|u_h\|_{\Gamma_h} \leq Ch_\Gamma^2 + (h/\mu)^{1/2} CY$ and $\|\partial_n \zeta_h\|_{\Gamma_h} \leq C + Ch_\Gamma^{-1/2} Y_Z$.

(b) For GS, write $\omega_h(\chi_h) = \tilde{\omega}(\chi_h) - a_h(e_u, \chi_h)$ (the pressure terms vanish since $\operatorname{div} \chi_h = 0$). With $\tilde{\omega}(\chi_h) = \langle \sigma(\tilde{u}, p^\circ) n, \chi_h \rangle - (F, \chi_h)$, $\chi_h = v_e - \zeta_h$ and Lemma 8.15(b),

$$J_h^\chi - J_h^N = \langle \sigma(\tilde{u}, p^\circ) n, \chi_h - e \rangle - (F, \chi_h - e) + a_h(e_u, \zeta_h).$$

By Corollary 5.3 with $c = \bar{p}_\Gamma(\tilde{p})$, $a_h(e_u, \zeta_h) = -\langle p^\circ, \zeta_h \cdot n \rangle + \mathcal{G}(\zeta_h) - B_h(e_u, \zeta_h)$, and $-\langle p^\circ, \zeta_h \cdot n \rangle$ cancels the pressure part of the first term. Hence

$$J_h^\chi - J_h^N = \langle D(\tilde{u}) n, \chi_h - e \rangle - (F, \chi_h - e) + \mathcal{G}(\zeta_h) - B_h(e_u, \zeta_h).$$

The dual error $e_Z := \tilde{Z}_e - \chi_h$ satisfies $N_h(e_Z, w) = -\langle \tilde{R}_e - c, w \cdot n \rangle + \mathcal{G}_Z(w)$ on Z_h , where $\mathcal{G}_Z$ is $\mathcal{G}$ with $\tilde{u} \rightarrow \tilde{Z}_e - e$. This is the structure of Corollary 5.3, so Theorems 6.1, 6.4 and 6.10 and Corollary 6.7 hold for it, with $p \rightarrow R_e$ and $\varepsilon_h \rightarrow E_Z + Ch^k$. In particular $e_Z = (h/\mu)v_\varphi^R + \vartheta^Z$, with $v_\varphi^R \approx -(R_e - \bar{R}_e)n$ on Γ_h and $\|\vartheta^Z\| \leq CX_1^Z := C(h/\mu + Y_Z)$. On Γ_h , $\zeta_h = (e - \tilde{Z}_e) + e_Z$, where $e - \tilde{Z}_e = -d \partial_n Z_e(x^*) + O(h_\Gamma^4)$ is tangential. Then:

- The penalty part of $-B_h(e_u, \zeta_h)$ is $-\frac{\mu}{h} \langle (h/\mu)v_\varphi + \vartheta, (e - \tilde{Z}_e) + (h/\mu)v_\varphi^R + \vartheta^Z \rangle$. Its leading term is $-\frac{h}{\mu} \langle v_\varphi, v_\varphi^R \rangle = -\frac{h}{\mu} K_e + O((h/\mu)(h_\Gamma^2 + h^k))$. The other five products are at most $C(h_\Gamma^4 + h_\Gamma^2 h^k)$, $C(h/\mu)^{1/2} X_1^Z$, $C\gamma^{1/2} h_\Gamma^{3/2} X_1$, $C(h/\mu)^{1/2} X_1$ and $CX_1 X_1^Z$, using $v_\varphi \perp (e - \tilde{Z}_e)$ up to $O(h_\Gamma^2 + h^k)$.
- $|\langle \partial_n e_u, \zeta_h \rangle| \leq Ch_\Gamma^{-1/2} X_1(h_\Gamma^2 + h/\mu + (h/\mu)^{1/2} X_1^Z)$.
- $\langle e_u, \partial_n \zeta_h \rangle = \frac{h}{\mu} \langle v_\varphi, \partial_n \zeta_h \rangle + \langle \vartheta, \partial_n \zeta_h \rangle$. The first part is $O((h/\mu)(\|\zeta_h\|_{\Gamma_h} + h_\Gamma^{1/2} + h^k h_\Gamma^{-1/2}))$, by Lemma 6.3 applied to $w|_{\Gamma_h}$ and $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$. For the second, Theorem 6.10 gives $r_h = (h/\mu)(z_U - v_\varphi) + \epsilon$ with $\|z_U - v_\varphi\|_{\Gamma_h} \leq (h/\mu)^{1/2} E_U + C(h_\Gamma^2 + h^k)$ and $\|\epsilon\|_{\Gamma_h} \leq (h/\mu)^{3/2} B'_U$ (B'_U the middle bound of that theorem). With $\|\partial_n \zeta_h\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}$ this part is $O((h/\mu)\gamma^{-1/2}(E_U + B'_U) + (h/\mu)(h_\Gamma^{3/2} + h^k h_\Gamma^{-1/2}) + \gamma^{-1/2} Y)$.

- $|\langle D(\tilde{u})n, \chi_h - e \rangle| \leq C(h_\Gamma^2 + (h/\mu)h_\Gamma + (h/\mu)^{1/2}X_1^Z)$, because $D(\tilde{u})n \cdot n_e = 2\partial_n \tilde{u} \cdot n_e = O(h_\Gamma)$ and the normal part of ζ_h is $O(h/\mu)$.
- $|\langle F, \chi_h - e \rangle| \leq Ch_\Gamma^3$ and $|\mathcal{G}(\zeta_h)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}$.

In the stated regime every remainder is $O(h^{3/2})$. Combine with Theorem 8.17. $\square$

8.5.6. *The torque of the volume functional.* For $\psi = a + bx^\perp \in \text{RM}$ let $\chi_h \in Z_h$ solve the same method with Nitsche data $\ell_\psi(w) = -\langle \psi, \partial_n w \rangle + \frac{\mu}{h} \langle \psi, w \rangle$ (zero body force, zero data on ∂R), and $J_h^X(\psi) := \omega_h(\chi_h)$. Since $v_\psi = \psi$ on L_1 and $D(\psi) = 0$,

$$\ell_\psi(w) = N_h(v_\psi, w) - a_h(v_\psi, w) + \langle \partial_n \psi, w \rangle, \quad \partial_n \psi = bJn \quad (J = \text{rotation by } \frac{\pi}{2}).$$

The extra term vanishes for translations only.

Proposition 8.22 (Torque of J_h^X). *Let ζ^0 be the solution of (10), and let $(\zeta^1, \pi^1) \in Z_h \times \Pi_h$ solve $N_h(\zeta^1, w) + \theta B^*(\pi^1, w) = -\langle \partial_n \psi, w \rangle$ for all $w \in V_h^R$ (for GS, on Z_h). Then*

$$J_h^X(\psi) = J_h^N(\psi) - \omega_h(\zeta^0) - \omega_h(\zeta^1).$$

For CNS* with $\gamma \geq \max(1, \gamma_2)$ and $\gamma h_\Gamma \leq 1$:

- (i) $\omega_h(\zeta^0)$ is bounded as in Proposition 8.21(a).
- (ii) $\|\zeta^1\| \leq C|b|(h/\mu)^{1/2}$, $\|\zeta^1\|_{\Gamma_h} \leq C|b|h/\mu$ and $|\omega_h(\zeta^1)| \leq C|b|(h/\mu + h_\Gamma^2\gamma^{-1/2})$.
- (iii) $-\omega_h(\zeta^1) = \frac{h}{\mu} \left[\int_\Gamma \sigma(u, p)n \cdot \partial_n \psi \, ds + O(|b|(\gamma^{-1/2} + (1 + \gamma)h_\Gamma)) \right]$.

On the unit circle $\partial_n \psi = -bx^\perp$ (with $n = -x$), so $J_h^X(x^\perp) = J_h^N(x^\perp) - \frac{h}{\mu} J_{x^\perp} (1 + O(\gamma^{-1/2} + \gamma h_\Gamma)) + O(h_\Gamma^2 + (h/\mu)^{1/2}E_Z)$. So, as $\gamma \rightarrow \infty$ with $\gamma h_\Gamma \rightarrow 0$ and $J_{x^\perp} \neq 0$, the torque from Gjerde-Scott's volume functional is only of first order in h/μ even for CNS*, unlike drag and lift (Prop. 8.21(a)); at fixed γ only the rate-1 upper bound is proved, and the first-order lower bound is numerical.

Proof. With $\zeta := v_\psi - \chi_h$, the displayed identity for ℓ_ψ gives $N_h(\zeta, w) - \theta B^*(\pi, w) = a_h(v_\psi, w) - \langle \partial_n \psi, w \rangle$. By linearity $\zeta = \zeta^0 + \zeta^1$ (pressures up to sign), and $J_h^X = \omega_h(v_\psi) - \omega_h(\zeta)$.

(ii) The load vanishes on V_h^0 , so $\beta\|\pi^1 - \bar{\pi}^1\| \leq C\|\zeta^1\|$. Test with ζ^1 : $B^*(\pi^1, \zeta^1) = \langle \pi^1 - \bar{\pi}_\Gamma(\pi^1), \zeta^1 \cdot n \rangle$ is absorbed exactly as in steps 4–5 of Theorem 7.2, and $|\langle \partial_n \psi, \zeta^1 \rangle| \leq |b|\|\Gamma_h\|^{1/2}(h/\mu)^{1/2}\|\zeta^1\|$. Next, $\omega_h(\zeta^1) = \langle t_h, \zeta^1 \rangle + \langle u_h, \partial_n \zeta^1 \rangle$ (Lemma 8.15(a)), with $\|t_h\|_{\Gamma_h} \leq C$ and $\|\partial_n \zeta^1\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}\|\nabla \zeta^1\|$.

(iii) Let $w_\sigma \in V_h^R$ interpolate $\chi_0 \cdot (\sigma(u, p)n) \circ \pi$. First, $|\bar{\pi}^1| \leq C|b|(h_\Gamma + (h/\mu)^{1/2})$: test with w_1 as in Theorem 8.7, using $Jn \cdot n = 0$, $\int_{\Gamma_h} \zeta^1 \cdot n = 0$ and Lemma 8.3. Then testing with w_σ gives

$$\frac{\mu}{h} \langle \zeta^1, w_\sigma \rangle = -\langle \partial_n \psi, w_\sigma \rangle + O(|b|(\gamma^{-1/2} + h_\Gamma)).$$

Here $|a_h(\zeta^1, w_\sigma)| \leq C(h/\mu)^{1/2}$, $|\langle \partial_n \zeta^1, w_\sigma \rangle| \leq Ch_\Gamma^{-1/2}\|\nabla \zeta^1\| \leq C\gamma^{-1/2}$, $|\langle \zeta^1, \partial_n w_\sigma \rangle| \leq Ch/\mu$, and $|B^*(\pi^1, w_\sigma)| \leq C(h_\Gamma^{-1/2}\|\pi^1 - \bar{\pi}^1\| + |\bar{\pi}^1|)$. Finally

$$\omega_h(\zeta^1) = \langle w_\sigma, \zeta^1 \rangle + \langle t_h - w_\sigma, \zeta^1 \rangle + \langle u_h, \partial_n \zeta^1 \rangle.$$

The second term is at most $\|t_h - \sigma^\circ n\|_{\Gamma_h}\|\zeta^1\|_{\Gamma_h} + Ch_\Gamma h/\mu \leq C(1 + \gamma)h_\Gamma h/\mu$ (Proposition 8.23(a); the chord/arc normal mismatch is $O(h_\Gamma)$). The third is at most $Ch_\Gamma^2\gamma^{-1/2} = \frac{h}{\mu} C\gamma^{1/2}h_\Gamma$. Also $\langle \partial_n \psi, w_\sigma \rangle = \int_\Gamma \sigma n \cdot \partial_n \psi + O(h_\Gamma)$. $\square$

Computation: `notes/v6/misc/misc_numerics.py` (columns `-omega(zeta1)_rot`, `pred_rot`); output `notes/v6/misc/tables.log`.

NUMERICAL, and the fixed- γ lower bound is numerical only (it would need $J_{x^\perp} \neq 0$ and the leading term at fixed γ , where the remainder $O(\gamma^{-1/2} + \gamma h_\Gamma)$ does not vanish; every run in the table has $\gamma h_\Gamma \geq 3.7$, outside the hypothesis). For $\gamma \gtrsim 300$ the leading term is confirmed to

TABLE 3. CNS*: $-\omega_h(\zeta^1)$ divided by the predicted $\frac{h}{\mu} \int_{\Gamma} \sigma n \cdot \partial_n \psi$, for $\psi = x^\perp$. Here $\gamma := \mu h_\Gamma / h$.

test	μ	N	γ	ratio
S, B ₁ , A, Q	100	16, 32, 64	≈ 30	1.55–1.83, 2.21–2.30, 2.58–2.60
S, B ₁ , Q	10^3	32, 64	≈ 300	1.02, 1.05 (S, B ₁)
Q	$10^4, 10^5$	32	$\approx 3 \cdot 10^3, 3 \cdot 10^4$	0.976, 0.974

3–5%. At $\gamma \approx 30$ the ratio is 1.6–2.6. It is the *same* for all four flows at a given N , so it is an amplification by the mesh and the penalty, not by the flow. It grows with N as the first-layer triangles become more elongated (min w_e drops from 0.51 to 0.42). Our reading, a heuristic, is the near-threshold tangential slip mode of Section 10: the effective Robin constant acquires a factor of about $(1 - \gamma_{\text{loc}}/\gamma)^{-1}$. The rigorous remainder $(1 + \gamma)h_\Gamma$ in (iii) is pessimistic: at $\gamma \approx 300$ it is not small, yet the ratio is 1.05. For translations, $J_h^X - J_h^N$ is $O(h_\Gamma^2)$ as Proposition 8.21(a) states. GS gives the same $\omega_h(\zeta^1)$ to within 5%.

8.5.7. Pointwise traction versus Nitsche flux.

Proposition 8.23 (Traction on Γ_h). *Let $\sigma^\circ := \sigma(\tilde{u}, p^\circ)$ and $T_h^\beta := D(u_h)n - p_h n$.*

(a) CNS* ($\gamma \geq \gamma_2$):

$$\begin{aligned} \|t_h - \sigma^\circ n\|_{\Gamma_h} &\leq C[(1 + \gamma^{1/2})h_\Gamma^{-1/2}Y + (1 + \gamma)h_\Gamma], \\ \|T_h^\beta - \sigma^\circ n\|_{\Gamma_h} &\leq C[h_\Gamma^{-1/2}Y + |\bar{e}_p| + h_\Gamma^k], \\ |J_h^\beta(\psi) - J_\psi| &\leq C(h_\Gamma^{-1/2}Y + h_\Gamma^2). \end{aligned}$$

(b) GS ($\gamma \geq \gamma_0$):

$$\begin{aligned} \|t_h - \sigma^\circ n\|_{\Gamma_h} &\leq C_p[(1 + \gamma^{1/2})h_\Gamma^{-1/2}X_1 + (1 + \gamma)h_\Gamma], \\ cG/\gamma - C_p h_\Gamma^{-1/2}(Q + X_1) &\leq \|T_h^\beta - \sigma^\circ n\|_{\Gamma_h} \leq cG/\gamma + C_p h_\Gamma^{-1/2}(Q + X_1), \end{aligned}$$

and

$$J_h^\beta(\psi) = J_\psi + \Xi_h(\psi) + O_p(h_\Gamma^{-1/2}X_1 + h_\Gamma^2), \quad \Xi_h(\psi) := \langle \xi_b, \psi \cdot n \rangle_{\Gamma_h},$$

where, for translations,

$$\Xi_h(e) = \frac{h}{\mu} \hat{c}_k \sum_{e' \in \mathcal{E}_h^\Gamma} \frac{|e'|}{H_{e'}} q(m_{e'}^*) n_{e'} \cdot e + O_p(h/\mu),$$

so $|\Xi_h| \leq C_p/\gamma$. At fixed γ the pointwise traction of GS does not converge, and J_h^β converges to J_ψ only up to the $O(1/\gamma)$ defect Ξ_h . The Nitsche flux of GS converges in $L^2(\Gamma_h)$ at rate at least $1/2$.

Proof. On Γ_h , $\sigma^\circ n = \partial_n \tilde{u} + (\nabla \tilde{u})^\top n - p^\circ n$ with $|(\nabla \tilde{u})^\top n| \leq Ch_\Gamma$ and $|\tilde{u}| \leq Ch_\Gamma^2$. For the full gradient, $\|\nabla(\tilde{u} - u_h)\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}(\|\nabla e_u\| + h^k) + Ch_\Gamma^k$ (the inverse trace applied to $I_h \tilde{u} - u_h$).

(a) $t_h - \sigma^\circ n = -\partial_n e_u - (\nabla \tilde{u})^\top n + (e_p - \bar{p}_\Gamma(e_p))n - \frac{\mu}{h}(\tilde{u} - e_u)$. Use $\|e_p - \bar{p}_\Gamma(e_p)\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}Y$ and $\frac{\mu}{h}\|e_u\|_{\Gamma_h} \leq (\mu/h)^{1/2}\|e_u\| = \gamma^{1/2}h_\Gamma^{-1/2}\|e_u\|$. The bound for T_h^β is similar, and the constant part of e_p is Theorem 8.7. For J_h^β , use $\int \psi \cdot n = 0$ (so only $e_p - \bar{e}_p$ enters) and $\langle \sigma^\circ n, \psi \rangle - J_\psi = -\Theta_f + (F, \psi) = O(h_\Gamma^2)$ (proof of Lemma 8.15).

(b) For t_h : $-\frac{\mu}{h}u_h + p^\circ n = -\frac{\mu}{h}\tilde{u} + (v_\varphi + (q \circ \pi)n) + O(h_\Gamma^2) + \frac{\mu}{h}\vartheta$, and $v_\varphi + (q \circ \pi)n = O_p(h_\Gamma + h^k)$ (Step 2 of Theorem 8.11). Then $\frac{\mu}{h}\|\vartheta\|_{\Gamma_h} \leq (\mu/h)^{1/2}C_p X_1$. For T_h^β : $T_h^\beta - \sigma^\circ n = -D(e_u)n + (\xi_b + \rho_h)n$.

Lemma 8.2 gives $\|\xi_b\|_{L^2(e)}^2 = g_e^2 \hat{d}_k |e|/H_e^2 + O_p((h/\mu)^2)$, so $\|\xi_b\|_{\Gamma_h} \asymp (h/\mu) h_\Gamma^{-1} G = G/\gamma$ by the Riemann-sum argument of Theorem 8.11(b). Also $\|\rho_h\|_{\Gamma_h} \leq C h_\Gamma^{-1/2} \|\rho_h - \eta\| + C h_\Gamma^k$. For J_h^β : only $\rho_h - \bar{\rho}_h$ enters, $\|\rho_h - \bar{\rho}_h\|_{\Gamma_h} \leq C_p h_\Gamma^{-1/2} X_1$, and $\int_{e'} \xi_b = \int_{e'} g \xi_{e',1} = g_{e'} \hat{c}_k |e'|/H_{e'} + O_p((h/\mu)|e'|)$. $\square$

By Proposition 8.23(b) and Lemma 8.2(c), with $w_{e'} := |e'|/H_{e'}$ and $h/\mu = h_\Gamma/\gamma$,

$$\Xi_h(e) = \frac{k(k+1)}{\gamma} h_\Gamma \sum_{e' \in \mathcal{E}_h^\Gamma} w_{e'} q(m_{e'}^*) n_{e'} \cdot e + O_p(h/\mu).$$

Proposition 8.24 (Lower bounds for Ξ_h). *Let $G > 0$, $\gamma \geq \gamma_0$, bounded ρ , and let $c_w \leq w_{e'} \leq C_w$ (from (M0)).*

(a) Sign-definite case, every mesh. *If $q(n \cdot e) \geq 0$ on Γ (or ≤ 0) and $q(n \cdot e) \not\equiv 0$, then for h small*

$$|\Xi_h(e)| \geq \frac{k(k+1)c_w}{\gamma} \left(\int_\Gamma |q n \cdot e| ds - C h_\Gamma \right) - C_p h/\mu.$$

This covers a dipole wall pressure: for $q = a \cos \theta$, $q n \cdot e_1 = -a \cos^2 \theta$.

(b) Asymptotically regular first layer. *If there are $\omega, \lambda \in C(\Gamma)$, $\lambda > 0$, with $w_{e'} = \omega(m_{e'}^*) + o(1)$ and $|e'| = h_\Gamma \lambda(m_{e'}^*)(1 + o(1))$ uniformly, then*

$$\Xi_h(e) = \frac{k(k+1)}{\gamma} \int_\Gamma \frac{\omega}{\lambda} q n \cdot e ds + o(1/\gamma),$$

so $|\Xi_h(e)| \geq c/\gamma$ whenever the integral is nonzero. For a uniform first layer (ω/λ constant) the integral is $-(\omega/\lambda) J_e^p$, the pressure drag.

(c) No general lower bound. *If $q(n \cdot e)$ takes both signs on sets of positive measure, and the admissible weight range satisfies $C_w/c_w \geq \max(I_+/I_-, I_-/I_+)$ with $I_\pm := \int_\Gamma (q n \cdot e)_\pm$, then for all small h there are first layers satisfying (M0) on which the leading sum vanishes up to $O(h_\Gamma)$, so $\Xi_h(e) = O_p(h/\mu)$.*

Proof. (a) $\sum_{e'} w_{e'} |q n_{e'} \cdot e|(m_{e'}^*) \geq c_w h_\Gamma^{-1} \sum_{e'} |e'| |q n \cdot e|(m_{e'}^*)$, because $|e'| \leq h_\Gamma$. Here $|n_{e'} - n(m_{e'}^*)| \leq C h_\Gamma$, and the sum is a Riemann sum for $\int_\Gamma |q n \cdot e|$ with error $O(h_\Gamma)$. (b) Write $w_{e'} = |e'| \cdot w_{e'}/|e'|$ and take the Riemann sum. (c) Choose the apex of $T_{e'}$ (for instance on the perpendicular bisector) so that $w_{e'} = a$ where $q n \cdot e > 0$ and $w_{e'} = b$ elsewhere, with $a I_+ = b I_-$. The first-layer heights can be set independently, since distinct $T_{e'}$ are distinct triangles and the rest of the mesh can be adjusted within (M0). $\square$

Production meshes (NUMERICAL, layer_weights.py): $w_{e'} \in [0.40, 1.33]$; the weights are exactly C_4 -invariant ($w(\theta + \pi/2) = w(\theta)$ to 10^{-16}), reflection-symmetric up to $O(h_\Gamma)$, and the sums converge. For $q = \cos \theta$, $h_\Gamma \sum w q n \rightarrow (-3.564, 0)$ ($N = 256$), so $\Xi_h(e_1) \rightarrow -71.3/\gamma$ per unit dipole amplitude at $k = 4$. By C_4 symmetry the first Fourier mode of q couples only to the modes $0, 4, 8, \dots$ of ω/λ , so (b) gives a nonzero limit for every q whose weighted mode-1 content does not cancel.

Computation: notes/v6/misc/layer_weights.py 16,32,64,128,256; output notes/v6/misc/layer_weights.log.

Remark 8.25 (Gjerde-Scott's inconsistency parameter). The quantity $\beta_h(\chi_h) = \int_{\Gamma_h} T_h^\beta \cdot \chi_h$ differs from $J_h^\beta(e)$ by $O(h/\mu)$ for GS. So Proposition 8.23(b) and Proposition 8.21(b) give

$$\varepsilon_{\text{GS}} = \frac{\omega - \beta}{\omega} = -\frac{\Xi_h(e)}{J_e} + O(h/\mu + h^{3/2}).$$

At fixed μ , and with h/h_Γ fixed, ε_{GS} tends to a nonzero limit of size $\approx \hat{c}_k(h/\mu h_\Gamma)(|e|/H_e)|J_e^p|/|J_e|$, where $J_e^p = -\int_\Gamma q n \cdot e$ is the pressure drag. It measures the Nitsche pressure layer, not the drag

error. With $\mu = 10^6$, $\hat{c}_4 = 20$ and moderate h/h_Γ , this is of order 10^{-5} – 10^{-4} . That is the order of the $\varepsilon \approx 8 \cdot 10^{-5}$ in [34, Table 1b]. We have not checked this against their meshes, which we do not have.

Remark 8.26 (Removing the GS drag defect). To leading order $u_h \cdot n \approx (h/\mu)q$ and $(\chi_h - e) \cdot n \approx (h/\mu)(R_e - \bar{R}_e)$ on Γ_h , so $\frac{\mu}{h} \langle u_h \cdot n, (\chi_h - e) \cdot n \rangle \approx \frac{h}{\mu} K_e$. Moreover J_h^N is affine in $1/\mu$ at leading order, so two solves with different μ allow extrapolation. We have *not* proved rates for either corrected functional. CNS* needs no correction.

8.6. Lower bounds for the CNS* pressure, traction and drag. Theorem 8.7 and Proposition 8.23(a) give upper bounds for the CNS* pressure and traction, and Theorem 8.19 for its drag. Here we identify their structure and reduce all three lower bounds to a single scalar hypothesis (P) on a smooth boundary moment of the pressure error. Section 8.6.5 then derives the boundary moment from a periodic first-layer cell problem, which shows when (P) holds, and proves the lower bounds without it, modulo two hypotheses on the cell problem. Throughout, the method is CNS* with $\gamma \geq \gamma_2$, and the error equation is (3). “Fixed γ ” means $\gamma \in [\gamma_2, \gamma_{\max}]$, bounded ρ , and $\varepsilon_h \leq Ch^k$ (or flux-corrected data). The computations are in `notes/v6/misc`.

8.6.1. Pressure error = distance to the unconstrained Nitsche solution.

Proposition 8.27 (Pressure error and the unconstrained Nitsche solution). *Let $u_h^L \in g_I + V_h^R$ be the Nitsche discretisation of the vector problem $-\operatorname{div} D(u) = f - \nabla p$ in the full velocity space, with no incompressibility constraint and the exact pressure as data:*

$$N_h(u_h^L, v) = (\tilde{f}, v) - B^*(p^\circ, v) \quad \forall v \in V_h^R$$

(uniquely solvable by Lemma 4.2). Put $\delta := u_h - u_h^L \in V_h^R$. Then $\operatorname{div} u_h^L = -\operatorname{div} \delta$, and

$$\beta \|\xi - \bar{\xi}\| \leq (2 + C_I(c_0\gamma)^{-1/2}) \|\delta\|, \quad \|\delta\| \leq C(\|\xi - \bar{\xi}\| + h^k + (h_\Gamma/\gamma)^{1/2} |\bar{e}_p|).$$

In particular, at fixed γ with flux-corrected data (so that $|\bar{e}_p| \leq Ch_\Gamma^{3/2}$ by Theorem 8.7), $c \|u_h - u_h^L\| - Ch_\Gamma^2 \leq \|e_p - \bar{e}_p\| \leq C \|u_h - u_h^L\| + Ch^k$ and

$$\|e_p - \bar{e}_p\| \geq c \|\operatorname{div} u_h^L\|_{L^2(\Omega_h)} - C(h_\Gamma^2 + h^k).$$

Proof. By Lemma 5.1 and step 1 of Theorem 7.2, $N_h(\tilde{u}, v) + B^*(p^\circ, v) = (\tilde{f}, v) + \mathcal{G}(v)$, so $N_h(\tilde{u} - u_h^L, v) = \mathcal{G}(v)$. Subtracting (3) gives $N_h(\delta, v) = B^*(e_p, v)$ for all $v \in V_h^R$. The CNS* velocity satisfies $\operatorname{div} u_h = 0$ pointwise ($\operatorname{div} V_h \subset \Pi_h$), so $\operatorname{div} u_h^L = -\operatorname{div} \delta$.

Lower bound for δ . For $v \in V_h^0$, $B^*(e_p, v) = -(\xi - \bar{\xi}, \operatorname{div} v)$, because $(\eta, \operatorname{div} v) = 0$ and constants integrate to zero against $\operatorname{div} v$. Also $|N_h(\delta, v)| = |a_h(\delta, v) - \langle \delta, \partial_n v \rangle| \leq 2 \|\nabla \delta\| \|\nabla v\| + (h/\mu)^{1/2} \|\delta\| C_I(c_0 h_\Gamma)^{-1/2} \|\nabla v\|$. Apply Lemma 8.12.

Upper bound for δ . Coercivity gives $c_1 \|\delta\|^2 \leq B^*(e_p, \delta) = B^*(e_p - \bar{e}_p, \delta) - \bar{e}_p \int_{\Gamma_h} \delta \cdot n$. The first term is at most $\sqrt{2} \|e_p - \bar{e}_p\| \|\nabla \delta\| + \|e_p - \bar{e}_p\|_{\Gamma_h} \|\delta\|_{\Gamma_h}$. By Lemma 4.1 on ξ and $\|\eta\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^k$, $\|e_p - \bar{e}_p\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2} \|\xi - \bar{\xi}\| + Ch^k$. Also $\|\delta\|_{\Gamma_h} \leq (h_\Gamma/\gamma)^{1/2} \|\delta\|$ and $|\int \delta \cdot n| \leq |\Gamma_h|^{1/2} (h_\Gamma/\gamma)^{1/2} \|\delta\|$. The last claims use Theorem 8.7 for $|\bar{e}_p|$ and $\|\operatorname{div} \delta\| \leq \sqrt{2} \|\nabla \delta\|$. $\square$

Remark 8.28. So the CNS* pressure error measures exactly how far the unconstrained Nitsche solution, which sees the same chordal bump $\tilde{u}|_{\Gamma_h}$, is from being solenoidal. A continuous analogue: for a tangential boundary layer $b(s)e^{-\kappa t}$ the vector-harmonic extension has $\operatorname{div} = \partial_s b e^{-\kappa t} \neq 0$, while the Stokes layer has pressure $\pm 2\kappa b e^{-\kappa t}$. The equivalence is NUMERICALLY tight, with $\|e_p - \bar{e}_p\|/\|u_h - u_h^L\| \in [1.08, 1.41]$ and $\|e_p - \bar{e}_p\|/\|\operatorname{div} u_h^L\| \in [6.9, 8.2]$ on tests S and A at $\mu = 100$, $N = 8$ – 64 (Table 4).

8.6.2. Pressure identities in which the velocity error drops out. Let $\Phi_h := \{\psi \in C^1(\overline{\Omega_h}) : \psi|_T \in P_{k+1} \forall T, \nabla \psi = 0 \text{ on } \partial\Omega_h, \psi = 0 \text{ on } \Gamma_h\}$. Then $\nabla \Phi_h = \{v \in V_h^0 : \text{rot } v = 0\}$: a curl-free $v \in V_h^0$ has zero circulation around Γ_h because it vanishes there. Also $\Delta \Phi_h = \text{div}(\nabla \Phi_h) \subset \Pi_h$, and $\|\Delta \psi\| = \|\nabla^2 \psi\|$ for $\psi \in \Phi_h$.

Lemma 8.29 (Curl-free test fields). *For all $\psi \in \Phi_h$,*

$$(e_p, \Delta \psi) = \langle u_h \cdot n, \Delta \psi|_{T_e} \rangle_{\Gamma_h} + (F, \nabla \psi).$$

Hence, with P_Δ the L^2 projection onto $\Delta \Phi_h$,

$$\|P_\Delta \xi\| \leq Ch_\Gamma^{-1/2} \|u_h \cdot n\|_{\Gamma_h} + Ch_\Gamma^3 \leq C(c_0 \gamma)^{-1/2} \|e_u\| + Ch_\Gamma^{5/2}.$$

Proof. Apply Lemma 8.1 with $v = \nabla \psi \in V_h^0$. Since $v \in H_0^1$, $\int \partial_j e_i \partial_i v_j = (\text{div } e_u, \text{div } v) = 0$. Since $\nabla v = \nabla^2 \psi$ is symmetric, $a_h(e_u, v) = 2 \int \partial_j e_i \partial_i v_j = 0$. On e , $\nabla \psi \equiv 0$ forces $\nabla^2 \psi|_{T_e} = \partial_{nn} \psi n_e \otimes n_e$, so $\partial_n v = (\Delta \psi) n_e$. The bound uses Lemma 4.1 on $\Delta \psi|_{T_e} \in P_{k-1}$, $|(F, \nabla \psi)| \leq Ch_\Gamma^3 \|\nabla^2 \psi\|$ (proof of Lemma 8.4, $\nabla \psi = 0$ on Γ_h), $|\tilde{u} \cdot n_e| \leq Ch_\Gamma^3$ and $\|e_u\|_{\Gamma_h} \leq (h_\Gamma/\gamma)^{1/2} \|e_u\|$. $\square$

So the part of the pressure error in $\Delta \Phi_h$ is driven only by the normal slip, and it is $O(\gamma^{-1/2} Y)$. Whatever carries the $h_\Gamma^{3/2}$ lives in the “discrete harmonic” complement $\Pi_h \ominus \Delta \Phi_h$, which is paired with $\text{rot } e_u$ in every remaining identity. This is the discrete Cauchy–Riemann coupling of pressure and vorticity.

Lemma 8.30 (Smooth moments). *For $g \in C^\infty(\overline{R})$,*

$$|(e_p - \bar{e}_p, g)_{\Omega_h}| \leq C_g (\|e_u\|_{L^2(\Omega)} + \|u_h\|_{L^2(\Gamma)} + (h + h_\Gamma^{1/2})Y + h_\Gamma^2(1 + hh_\Gamma^{-1/2}) + h^k).$$

With $\|e_u\|_{L^2} \leq C(h_\Gamma^2 + hh_\Gamma^{3/2} + h^{k+1})$ (Theorem 9.6), on quasi-uniform meshes at fixed γ , $|(e_p - \bar{e}_p, g)| \leq C_g(h_\Gamma^2 + h^k)$. This is half an order better than $\|e_p - \bar{e}_p\| \asymp h_\Gamma^{3/2}$, so the pressure error is an oscillating first-layer field.

Proof. Let $V_0 \in H_0^1(\Omega) \cap C^\infty(\overline{\Omega})$ solve $\text{div } V_0 = g - \bar{g}_\Omega$ (a Stokes problem on the smooth domain Ω), extended by 0 to S_h . Then $V_0 \in H_0^1(\Omega_h)$, and $\text{div } V_0 = (g - \bar{g}_\Omega) \mathbf{1}_\Omega$. Let $v_h := \Pi_F V_0 \in V_h^0$ be a Fortin interpolant: $\text{div } v_h = \pi_h \text{div } V_0$ and $\|\nabla(V_0 - v_h)\| \leq C \inf_\chi \|\nabla(V_0 - \chi)\| \leq C(h + h_\Gamma^{1/2})$ (the kink of V_0 along Γ costs $h_\Gamma^{1/2}$). Then $(\xi, \text{div } V_0) = (\xi, \text{div } v_h)$, so

$$(e_p, \text{div } V_0) = (e_p, \text{div } v_h) + (\eta, \text{div } V_0) = a_h(e_u, v_h) + \langle u_h, \partial_n v_h \rangle + (F, v_h) + O(h^k)$$

by Lemma 8.1. Next, $a_h(e_u, V_0) = -(e_u, \text{div } DV_0)_\Omega + \int_{\partial\Omega} e_u \cdot DV_0 \nu$ is at most $C(\|e_u\|_{L^2(\Omega)} + \|u_h\|_{L^1(\Gamma)} + h^{k+1})$ (on Γ , $e_u = -u_h$), and $|a_h(e_u, v_h - V_0)| \leq 2\|\nabla e_u\| \|\nabla(v_h - V_0)\|$. Also $\|\partial_n v_h\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2} \|\nabla v_h\|_{L_1} \leq C(1 + hh_\Gamma^{-1/2})$ with $\|u_h\|_{\Gamma_h} \leq Ch_\Gamma^2$, and $|(F, v_h)| \leq Ch_\Gamma^3$. Finally, $(e_p - \bar{e}_p, g \mathbf{1}_{S_h}) = O(h_\Gamma^{3/2} \|\xi - \bar{\xi}\| + h^k)$, by the inverse estimate $\|\xi\|_{L^\infty(T_e)} \leq Ch_\Gamma^{-1} \|\xi\|_{T_e}$ and $|S_h \cap T_e| \leq Ch_\Gamma^3$. $\square$

NUMERICAL: for test S the moment $(e_p - \bar{e}_p, y)$ decays at rates 1.91, 1.94, 1.96, 1.96, 1.97 ($N = 8 \rightarrow 64$), while $\|e_p - \bar{e}_p\|$ decays at $1.79 \rightarrow 1.66$ (Table 4).

8.6.3. Boundary moments and normal slip. For $\phi \in C^{k+1}(\Gamma)$ with $\int_\Gamma \phi = 0$ define

$$\mathbf{P}_h(\phi) := \langle e_p - \bar{p}_\Gamma(e_p), \phi \circ \pi \rangle_{\Gamma_h}, \quad \mathbf{S}_h(\phi) := \langle \phi \circ \pi, e_u \cdot n \rangle_{\Gamma_h}.$$

Proposition 8.31 (Boundary moments and normal slip). *For $\gamma \geq \gamma_2$,*

$$\left| \frac{\mu}{h} \mathbf{S}_h(\phi) + \mathbf{P}_h(\phi) \right| \leq C_\phi [(1 + (\gamma h_\Gamma)^{1/2})Y + h^k + h_\Gamma^2 + \gamma h_\Gamma^3 + h_\Gamma^2 |\bar{e}_p|], \quad |\mathbf{P}_h(\phi)| \leq C_\phi (h_\Gamma^{-1/2} Y + h^k).$$

At fixed γ the first right-hand side is $O(h_\Gamma^{3/2})$ and the second is $O(h_\Gamma)$.

Proof. This is the proof of Lemma 8.20 with the pressure boundary term kept instead of estimated. Test (3) with w_ϕ :

$$\frac{\mu}{h} \langle e_u, w_\phi \rangle = \mathcal{G}(w_\phi) - a_h(e_u, w_\phi) + \langle \partial_n e_u, w_\phi \rangle + \langle e_u, \partial_n w_\phi \rangle - B^*(e_p, w_\phi).$$

Split $B^*(e_p, w_\phi) = -(e_p - \bar{e}_p, \operatorname{div} w_\phi) - \bar{e}_p \int_{\Gamma_h} w_\phi \cdot n + P_h(\phi) + \langle e_p - \bar{p}_\Gamma(e_p), w_\phi \cdot n - \phi \circ \pi \rangle$, and $\frac{\mu}{h} \langle e_u, w_\phi \rangle = \frac{\mu}{h} S_h(\phi) + \frac{\mu}{h} \langle e_u \cdot n, w_\phi \cdot n - \phi \circ \pi \rangle + \frac{\mu}{h} \langle e_u \cdot \tau, w_\phi \cdot \tau \rangle$. The bounds are those of Lemma 8.20 and Theorem 8.7. The tangential penalty term is at most $\frac{\mu}{h} C_\phi h_\Gamma \|e_u\|_{L^1(\Gamma_h)} \leq C_\phi (\gamma h_\Gamma)^{1/2} \|e_u\|$. The two $O(h_\Gamma^2)$ mismatches are lower order, and $|\int_{\Gamma_h} w_\phi \cdot n| \leq C_\phi h_\Gamma^2$. The bound on P_h is Lemma 4.1 applied to $\xi - \bar{\xi}$. $\square$

NUMERICAL: $(\mu/h)S_h(\cos \theta) + P_h(\cos \theta)$ decays at rates $1.49 \rightarrow 1.85$ (tests S and B₁, $\mu = 100$), and it is about 3% of P_h at $N = 64$.

Lemma 8.32 (Normal traction). *Let $T_h^\beta = D(u_h)n - p_h n$ (Proposition 8.23). For ϕ as above and fixed γ ,*

$$\langle (T_h^\beta - \sigma^\circ n) \cdot n, \phi \circ \pi \rangle_{\Gamma_h} = P_h(\phi) + O_\phi(h_\Gamma^2 + h^k).$$

Proof. $(T_h^\beta - \sigma^\circ n) \cdot n = -2\partial_n e_u \cdot n + e_p$. Also $\langle e_p, \phi \circ \pi \rangle = P_h(\phi) + \bar{p}_\Gamma(e_p) \int_{\Gamma_h} \phi \circ \pi$, where $|\int_{\Gamma_h} \phi \circ \pi| \leq Ch_\Gamma^2$ and $|\bar{p}_\Gamma(e_p)| \leq Ch_\Gamma$ (Theorem 8.7). On e , taken from T_e , $\operatorname{div} e_u = 0$ gives $\partial_n e_u \cdot n_e = -\partial_\tau(e_u \cdot \tau_e)$. Integrate by parts along each chord:

$$\langle \partial_n e_u \cdot n, \phi \circ \pi \rangle = \sum_e \int_e (e_u \cdot \tau_e) \partial_\tau(\phi \circ \pi) - \sum_z \phi(z) e_u(z) \cdot (\tau_{e_-(z)} - \tau_{e_+(z)}).$$

The first sum is at most $C\|e_u\|_{L^1(\Gamma_h)} \leq C(h_\Gamma/\gamma)^{1/2} Y$. In the vertex sum, $e_u(z) = -u_h(z)$ ($\tilde{u} = 0$ on Γ), $|\tau_{e_-} - \tau_{e_+}| \leq Ch_\Gamma$ and $|u_h(z)| \leq C|e|^{-1/2} \|u_h\|_{L^2(e)}$, so the sum is at most $C\|u_h\|_{\Gamma_h} \leq Ch_\Gamma^2$. $\square$

8.6.4. The lower bounds.

Assumption 8.33 ((P)). There are $\phi \in C^{k+1}(\Gamma)$ with $\int_\Gamma \phi = 0$, and $c_P, h_P > 0$, such that $|P_h(\phi)| \geq c_P h_\Gamma$ for $h_\Gamma \leq h_P$. Equivalently (Prop. 8.31), the weighted normal slip satisfies $|S_h(\phi)| \geq c'_P(h/\mu)h_\Gamma$.

(P) is observed on the production meshes (NUMERICAL): $P_h(\cos \theta)/h_\Gamma \approx 0.82$ on test B₁ and ≈ 2.0 on test S at $N = 64$, $\mu = 100$, both still slowly increasing (Table 4); the cell limits of Section 8.6.5 are 0.84 and 2.11. Section 8.6.5 shows that (P) *as stated fails* on mirror-symmetric first layers, identifies exactly when it holds (Corollary 8.39), and proves the lower bounds without it (Corollary 8.40), all modulo two hypotheses on the cell problem. Test A and a rotational P_3 flow have $P_h(\cos \theta) = P_h(\sin \theta) = 0$ to round-off: the flow is rotation invariant and the mesh is C_4 -symmetric, so only the modes $\equiv 0 \pmod{4}$ survive.

Theorem 8.34 (Lower bounds under (P)). *At fixed γ , for every ϕ as above and every constant c :*

- (i) $\|p^\circ - p_h - c\|_{L^2(\Omega_h)} \geq c_1 h_\Gamma^{1/2} |P_h(\phi)| / \|\phi\|_{L^2(\Gamma)} - Ch^k$;
- (ii) $\|p^\circ - p_h - c\|_{L^\infty(L_1)} \geq |P_h(\phi)| / (\|\phi\|_{L^1(\Gamma)} + Ch_\Gamma^2)$;
- (iii) $\|T_h^\beta - \sigma^\circ n\|_{\Gamma_h} \geq (|P_h(\phi)| - C_\phi(h_\Gamma^2 + h^k)) / (2\|\phi\|_{L^2(\Gamma)})$.

Hence, under (P) and with $h^k \leq h_\Gamma^{3/2}$: $\|p^\circ - p_h - c\| \geq ch_\Gamma^{3/2}$, $\|p^\circ - p_h - c\|_{L^\infty(L_1)} \geq ch_\Gamma$ and $\|T_h^\beta - \sigma^\circ n\|_{\Gamma_h} \geq ch_\Gamma$. With Theorem 8.7 and Proposition 8.23(a), all three rates $(\frac{3}{2}, 1, 1)$ are then exact.

Proof. (i) $|\mathbf{P}_h(\phi)| = |\langle e_p - \bar{e}_p, \phi \circ \pi - \bar{p}_\Gamma(\phi \circ \pi) \rangle| \leq \|e_p - \bar{e}_p\|_{\Gamma_h} \|\phi \circ \pi\|_{\Gamma_h}$, and $\|e_p - \bar{e}_p\|_{\Gamma_h} \leq C_I(c_0 h_\Gamma)^{-1/2} \|\xi - \bar{\xi}\| + Ch^k$. Also $\|\xi - \bar{\xi}\| \leq \|e_p - \bar{e}_p\| + Ch^k \leq \|e_p - c\| + Ch^k$. (ii) The traces of e_p on Γ_h are taken from L_1 . Moreover $\langle e_p - c, \phi \circ \pi \rangle = \mathbf{P}_h(\phi) + (\bar{p}_\Gamma(e_p) - c) \int_{\Gamma_h} \phi \circ \pi$ and $|\bar{p}_\Gamma(e_p) - c| \leq \|e_p - c\|_{L^\infty(L_1)}$. (iii) Use Lemma 8.32 and Cauchy–Schwarz. $\square$

Remark 8.35 (Why (P) is not proved: the precise obstruction). (1) *Only circular pressure-only identities.* A test field in V_h^0 eliminates e_u from the pressure identity of Lemma 8.1 for every divergence-free e_u only if it is curl-free, i.e. lies in $\nabla \Phi_h$ (write $a_h(e_u, v) - \langle e_u, \partial_n v \rangle = (\text{rot } e_u, \text{rot } v) \mp \langle e_u \cdot \tau, \text{rot } v \rangle - \langle e_u \cdot n, \text{div } v \rangle$ on V_h^0). For those fields Lemma 8.29 expresses e_p through $u_h \cdot n$, which Prop. 8.31 expresses through $\mathbf{P}_h$ again. Every other field involves $(\text{rot } e_u, \text{rot } v)$, which is of the same order $h_\Gamma^{3/2} \|\nabla v\|$ as the term to be bounded below. This is the mirror image of the velocity lower bound (Section 7.1), where Y_h eliminates the pressure exactly.

(2) *(P) is a discrete first-layer effect* (made precise in Section 8.6.5). In the flat periodic model of one chord (bump $\frac{1}{2}(\ell^2/4 - s^2)W$, even about the midpoint), reflection $s \mapsto -s$ makes the layer pressure odd about each midpoint. So its chord means vanish, and $\mathbf{P}_h = o(h_\Gamma)$; the mean bump $\ell^2 W/12$ only drives the smooth $O(h_\Gamma^2)$ flow. On the production meshes every T_e is close to right-angled with its apex over the counterclockwise endpoint: the apex offset is $+0.53|e|$ to $+0.82|e|$, always of the same sign (`apex_offsets.log`). The discrete cell is therefore not reflection symmetric, and an $O(h_\Gamma)$ chord mean is expected. Section 8.6.5 carries out the two-scale description: $\mathbf{P}_h(\phi) = h_\Gamma \int_\Gamma \mathcal{F} \phi + O(h_\Gamma^2)$ with an explicit cell density $\mathcal{F}$, which vanishes on mirror-symmetric columns; the oscillating part of e_p then gives (i) and (iii) directly.

TABLE 4. CNS*, $\mu = 100$ ($\gamma \approx 30$), $k = 4$. Test S: $u = \text{curl}((r^2 - 1)^2(1 + x)/4) \in P_4$, $p = x^2 y - y + x/2 \in P_3$; the interior discretisation is exact, so every error comes from Γ_h . Rates are in h_Γ .

N	h_Γ	$\ e_p - \bar{e}_p\ $ (rate)	$\frac{\ e_p - \bar{e}_p\ }{\ \text{div } u_h\ }$	$\frac{\ e_p - \bar{e}_p\ }{\ u_h - u_h^T\ }$	$(e_p - \bar{e}_p, y)$ (rate)	$\mathbf{P}_h(\cos \theta)/h_\Gamma$	$\frac{\mu}{h} \mathbf{S}_h + \mathbf{P}_h$ (rate)
8	.765	1.21	8.18	1.41	4.47	1.03	1.7e-1
16	.460	4.84e-1 (1.79)	8.07	1.33	1.69 (1.91)	1.48	8.0e-2 (1.49)
24	.320	2.61e-1 (1.71)	8.04	1.29	8.39e-1 (1.94)	1.71	4.5e-2 (1.62)
32	.244	1.65e-1 (1.71)	7.94	1.26	4.94e-1 (1.96)	1.84	2.8e-2 (1.70)
48	.165	8.48e-2 (1.69)	7.75	1.20	2.28e-1 (1.96)	1.97	1.4e-2 (1.78)
64	.124	5.30e-2 (1.66)	7.62	1.17	1.31e-1 (1.97)	2.04	8.2e-3 (1.85)

Test B₁ (paper), same μ : $\mathbf{P}_h(\cos \theta)/h_\Gamma = 0.59, 0.68, 0.73, 0.79, 0.82$ at $N = 16, 24, 32, 48, 64$; test S at $\mu = 1000$: 5.2, 4.3 ($N = 32, 64$).

Computation: `notes/v6/misc/misc_numerics.py S1 100 8,16,32 (and 24,48,64; B1; A; S1 1000)`; output `notes/v6/misc/run_*.jsonl`, collected by `tables.py` into `tables.log`.

Test A (Scott’s shear flow) and test Q (rotational P_3 flow) have $\mathbf{P}_h(\cos \theta) = \mathbf{P}_h(\sin \theta) = 0$ to round-off: the flow is rotation invariant and the mesh is C_4 -symmetric, so only the modes $\equiv 0 \pmod{4}$ survive.

8.6.5. *The first-layer cell problem.* This subsection replaces (P) by a statement about a periodic cell problem for the first boundary layer. Everything is at fixed γ ; the computations use $k = 4$, and the propositions hold for every $k \geq 1$ with $k(k+1)$ in place of 20 (`notes/v7/P`).

Geometry. For a wall edge e with midpoint m_e , unit tangent τ_e and unit normal e_{in} into the fluid, let $\Phi_e(x) := |e|^{-1} R_e(x - m_e) + (\frac{1}{2}, 0)$, with R_e the rotation taking $\tau_e \mapsto (1, 0)$, $e_{\text{in}} \mapsto (0, 1)$; so $\Phi_e(e) = [0, 1] \times \{0\}$. Let κ be the curvature of Γ , $W := \partial_{e_{\text{in}}}(u \cdot \tau_e)|_\Gamma$ the wall shear rate and

$\hat{\ell} := |e|/h_\Gamma$ on e . A *column lattice* $\mathbf{L}$ is a 1-periodic strip triangulation with rows $L_j = (x_j, y_j)$, $0 = y_0 < y_1 < \dots$, right vertices $L_j + (1, 0)$ and a diagonal per row; the *sheared lattice* has $L_j = (j\delta, ja)$ with forward diagonals, aspect a and apex offset $o = \delta + \frac{1}{2}$ ($o = 0$: isosceles, mirror symmetric; $o = -\frac{1}{2}$: right-angled). A mesh family is *locally periodic in columns* (LPC) if there are $M_h \geq C_0 |\log h_\Gamma|$ and a Lipschitz map $x \mapsto \mathbf{L}(x)$ on Γ such that for every e the first M_h rows above e , mapped by Φ_e , coincide with those of $\mathbf{L}(m_e)$ up to vertex displacements $\leq Ch_\Gamma$. The annulus meshes of `notes/v7/P` are LPC with constant $\mathbf{L}$. The production meshes have first-layer offsets tending to $-\frac{1}{2}$ and aspects $a \in [0.75, 2.5]$; whether their deeper rows are LPC was not checked.

Definition 8.36 (Cell problem). Let $V_\mathcal{C}$ be the continuous piecewise- P_k fields on $\mathbf{L} \cap \{0 \leq y \leq y_J\}$, 1-periodic in x , with no Dirichlet condition, and $Q_\mathcal{C}$ the discontinuous piecewise- P_{k-1} functions. On the wall $e_0 = [0, 1] \times \{0\}$, $n = (0, -1)$; the top is traction free. With the forms of Section 2 and penalty $\lambda = \gamma \hat{\ell}$,

$$N(U, v) = \frac{1}{2}(DU, Dv) - \langle \partial_n U, v \rangle_{e_0} - \langle U, \partial_n v \rangle_{e_0} + \lambda \langle U, v \rangle_{e_0}, \quad b(P, v) = -(P, \operatorname{div} v) + \langle P, v \cdot n \rangle_{e_0},$$

find (U, P) with $N(U, v) + b(P, v) = F(v)$, $(q, \operatorname{div} U) = 0$. The loads are, with $\sigma := x - \frac{1}{2}$ and $\beta(\sigma) := \frac{1}{2}(\frac{1}{4} - \sigma^2)$, the *bump* $F_1(v) = \langle \sigma e_{\text{in}}, v \rangle - \langle g, \partial_n v \rangle + \lambda \langle g, v \rangle$, $g = -\beta\tau$, and the *normal traction* $F_c(v) = \langle n, v \rangle$. Write $A := \int_{e_0} P$ (trace from the wall triangle), $V := \int_{e_0} U \cdot n$, A_1, A_c for the two loads, $P_\mathcal{C}, T_\mathcal{C}$ for the bump pressure and wall traction $D(U)n - Pn$, and $\varsigma := k(k+1)/a$ with a the height of the wall triangle.

The loads are the pull-backs of the CNS* error data: on e , $\tilde{u}|_e = -\kappa W|e|^2 \beta(\sigma) \tau_c + O(h_\Gamma^3)$ is the Nitsche datum of e_u , and the transposed term $\langle (\nabla \tilde{u})^\top n_e, v \rangle$ of $\mathcal{G}$ (Lemma 3.5) is, with $\tau_c \cdot n_e = \kappa|e|\sigma + O(h_\Gamma^2)$, the odd normal load $\kappa W|e|\sigma e_{\text{in}}$. The forms are dilation invariant in two dimensions, so the column of e carries $\kappa W|e|^2$ times the cell velocity and $\kappa W|e|$ times the cell pressure. The transposed load exists because the Nitsche flux is $(\nabla u)n$ while the volume form is $\frac{1}{2}(Du, Dv)$; with the flux $D(u)n$ it would disappear, and A_1 would drop from $+0.182$ to -0.020 at $a = 1$, $o = -\frac{1}{2}$, $\lambda = 30$. All constants below are tied to the paper's choice $(\nabla u)n$.

Proposition 8.37 (The cell problem). (a) Well-posedness. *There is λ_0 , depending only on the shape of the wall triangle and k , such that for $\lambda > \varsigma + \lambda_0$ the cell problem (truncated at any J , or on the half-infinite column in the energy space) has a unique solution with $\|\nabla U\| + \|P\| \leq C\|F\|_{V'_\mathcal{C}}$.*

(b) Normal traction, closed form. *Let $\hat{r} \in P_{k-1}$ on the wall triangle (extended by 0) represent $q \mapsto \int_{e_0} q$; its trace is $\hat{r}|_{e_0} \equiv \varsigma$ (Lemma 8.2). For $\lambda \neq \varsigma$, $(U, P) = ((\lambda - \varsigma)^{-1}n, (\varsigma - \lambda)^{-1}\hat{r})$ solves the traction problem, so $A_c = \varsigma/(\varsigma - \lambda)$, independently of the rows $j \geq 1$. At $\lambda = \varsigma$ the cell operator is singular with kernel $(n, -\hat{r})$.*

(c) Flux identity. *For every load, $\lambda V + A = F(n)$; so $\lambda V_1 + A_1 = 0$ and $\lambda V_c + A_c = 1$.*

(d) Reflection. *If $\mathbf{L}^*$ is the mirror image of $\mathbf{L}$ under $(x, y) \mapsto (1 - x, y)$, then $A_1(\mathbf{L}^*) = -A_1(\mathbf{L})$ while $A_c, \|P_\mathcal{C}\|$ and $\|T_\mathcal{C}\|$ are unchanged. In particular $A_1 = 0$ on every mirror-symmetric column, and for sheared lattices $A_1(-o) = -A_1(o)$.*

Proof. (a) On the divergence-free fields only the wall term $\langle P, v \cdot n \rangle$ of b survives, and eliminating the wall-triangle pressure layer turns the form into N with normal penalty reduced by at most ς (Lemma 15.13); the trace-inverse bound gives coercivity for $\lambda - \varsigma > \lambda_0$, and inf-sup holds for the Scott–Vogelius pair on one period of the shape-regular lattice. (b) Try $U = Vn$, $P = \beta'\hat{r}$ with constants V, β' . U is constant, so $N(U, v) = -V\langle n, \partial_n v \rangle + \lambda V\langle n, v \rangle$. For $v \in V_\mathcal{C}$, $(\hat{r}, \operatorname{div} v) = \int_{e_0} \operatorname{div} v = \int_{e_0} \partial_n v \cdot n$, because $\int_{e_0} \partial_\tau v_\tau = 0$ by periodicity; so $b(\beta'\hat{r}, v) = -\beta'\langle n, \partial_n v \rangle + \beta'\varsigma\langle n, v \rangle$, and the momentum equation holds iff $-V - \beta' = 0$ and $\lambda V + \beta'\varsigma = 1$. (c) Test with the constant field n : $\operatorname{div} n = 0$, $\partial_n n = 0$, $b(P, n) = A$, and $\int_{e_0} \partial_n U \cdot n = -\int_{e_0} \partial_\tau(U \cdot \tau) = 0$ since $\operatorname{div} U = 0$

exactly; $F_1(n) = \int \sigma = 0$. (d) $v \mapsto Sv \circ \varrho$, $S = \text{diag}(-1, 1)$, preserves divergence, D -products, n , ∂_n and wall integrals; it maps $g \mapsto -g$ (β even, τ flipped) and $\sigma e_{\text{in}} \mapsto -\sigma e_{\text{in}}$, so $F_1 \mapsto -F_1$, and fixes F_c . $\square$

NUMERICAL (notes/v7/P): $A_c = 20/(20 - \lambda)$ to 10^{-13} for $\lambda \in [5, 10^6]$; the flux identity holds to 10^{-14} ; truncation $J = 6, \dots, 20$ changes A_1 by $< 10^{-10}$, and the row norms of P and ∇U decay by factors 6–7 per row. On all 344 asymmetric sheared cells scanned ($a \in [\frac{1}{2}, 3]$, $0.1 \leq |o| \leq 0.75$, $\lambda/\varsigma \in [1.001, 10^4]$), $A_1 \neq 0$ with $\text{sign } A_1 = -\text{sign } o$ and a simple pole at $\lambda = \varsigma$; for example $A_1 = 0.1824$ at $a = 1$, $o = -\frac{1}{2}$, $\lambda = 30$. A_1 is dominated by, but not a function of, the wall triangle (an isosceles wall triangle over rectangular rows gives -0.028), so the relevant symmetry is that of the column.

The two-scale statement needs two hypotheses on the cell problem.

- (D_c) (cell decay; NUMERICAL) For $\mathbf{L}$ in the range of the LPC map and $\lambda \in [\gamma \min \hat{\ell}, \gamma \max \hat{\ell}] \subset (\varsigma + \lambda_0, \infty)$, the bump and traction cell solutions satisfy $\|\nabla(U - U_\infty)\|_{\text{row } j} + \|P\|_{\text{row } j} \leq Cq^j$ with $q < 1$ uniform and U_∞ constant.
- (TL) (transplant lemma; not proved) On an LPC family, the glued field (E_u, E_p) below, transplanted to the mesh through the $O(h_\Gamma)$ vertex perturbation and cut off after M_h rows, has CNS* residual (in V'_h , plus the divergence defect in L^2) at most $Ch_\Gamma^{5/2}$.

(D_c) is called (D) in notes/v7/P; it is unrelated to the data hypothesis (D) of Section 2. (TL) is local and algebraic: the relative size of the column-to-column mismatch is the Lipschitz modulus of $(\kappa W, \hat{\ell}, \mathbf{L})$ times h_Γ , and (D_c) localises every mismatch to $O(1)$ rows; what is missing is a written estimate for the transplant of P_k/P_{k-1} fields between two lattices at $O(h_\Gamma)$ vertex distance, including the $O(h_\Gamma)$ divergence defect.

Glued field. On the column of e , $E_u = \kappa W|e|^2 U_1 \circ \Phi_e + t_0|e|U_c \circ \Phi_e + u_s$ and $E_p = \kappa W|e|P_1 \circ \Phi_e + t_0 P_c \circ \Phi_e + p_s$, with κ, W at m_e , cell parameters $(\mathbf{L}(m_e), \gamma \hat{\ell})$, and (u_s, p_s) the CNS* solution of the smooth problem driven by the far-field slip $-\kappa W|e|^2 \tau/12$. The constant t_0 is fixed by the exact constraint $\int_{\Gamma_h} u_h \cdot n = 0$; by Proposition 8.37(c), $t_0 = h_\Gamma \bar{t}_0 + O(h_\Gamma^2)$ with $\bar{t}_0 := \int_\Gamma \hat{\ell} \kappa W A_1 ds / \int_\Gamma (1 - A_c) ds$.

Theorem 8.38 (Two-scale expansion; proved modulo (D_c), (TL)). *On an LPC family at fixed γ with $\gamma \min \hat{\ell} > \varsigma + \lambda_0$ and $h^k \leq h_\Gamma^{5/2}$, let*

$$\mathcal{F}(x) := \hat{\ell} \kappa W A_1(\mathbf{L}(x), \gamma \hat{\ell}(x)) + \bar{t}_0 A_c(\mathbf{L}(x), \gamma \hat{\ell}(x)),$$

$Q_x := \hat{\ell} \kappa W P_1 + \bar{t}_0 P_c$ and $\Theta_x := \hat{\ell} \kappa W T_1 + \bar{t}_0 T_c$ (cell pressure and wall traction of the load $\hat{\ell} \kappa W F_1 + \bar{t}_0 F_c$ at $(\mathbf{L}(x), \gamma \hat{\ell}(x))$). Then for $\phi \in C^1(\Gamma)$ with $\int_\Gamma \phi = 0$:

- (i) $P_h(\phi) = h_\Gamma \int_\Gamma \mathcal{F} \phi ds + O_\phi(h_\Gamma^2)$;
- (ii) $\inf_c \|e_p - c\|_{L^2(\Omega_h)}^2 = h_\Gamma^3 \int_\Gamma \hat{\ell} \|Q_x\|_{L^2(C)}^2 ds + O(h_\Gamma^{7/2})$;
- (iii) $\|T_h^\beta - \sigma^\circ n\|_{\Gamma_h}^2 = h_\Gamma^2 \int_\Gamma \|\Theta_x\|_{L^2(e_0)}^2 ds + O(h_\Gamma^{5/2})$.

On mirror-symmetric columns $\bar{t}_0 = 0$ and $\mathcal{F} \equiv 0$.

Proof. Step 1. On each column the leading part of the Nitsche datum and of $\mathcal{G}$ is $\kappa W|e|^2$ times the pull-back of F_1 , up to relative $O(h_\Gamma)$. The glued field solves the column problems, the flux constraint and the smooth problem, so by (TL) $(R_u, R_p) := (e_u - E_u, e_p - E_p)$ satisfies the CNS* equations with residual $O(h_\Gamma^{5/2})$. *Step 2.* Coercivity on Z_h (with the effective penalty of Lemma 15.13) and Lemma 8.12 give $\|\nabla R_u\| + (\mu/h)^{1/2} \|R_u\|_{\Gamma_h} + \inf_c \|R_p - c\| \leq Ch_\Gamma^{5/2}$. The constant part of R_p is controlled by the edge-bubble row, as the constant of e_p in the proof of Theorem 8.7: testing with an edge-bubble field of nonzero flux gives $|\bar{p}_\Gamma(R_p)| \leq Ch_\Gamma^{-1/2} \cdot h_\Gamma^{5/2} = Ch_\Gamma^2$, which is what (iii) needs,

since (iii) is not invariant under constants. *Step 3.* $P_h(\phi) = \langle E_p - \bar{p}_\Gamma(E_p), \phi \circ \pi \rangle + O(h_\Gamma^{-1/2} h_\Gamma^{5/2})$ by the inverse trace inequality. Column e contributes $|e| \kappa W |e| A_{1,e} \phi(m_e) + t_0 |e| A_{c,e} \phi(m_e) + O(h_\Gamma^3)$, the smooth part $O(h_\Gamma^2)$ (Lemma 8.30), the constant $O(h_\Gamma^2)$; a Riemann sum gives (i). *Steps 4–5.* On column e , $E_p - p_s = h_\Gamma Q_{m_e} \circ \Phi_e$ and the column traction is $h_\Gamma \Theta_{m_e} \circ \Phi_e$; sum over columns $(\sum_e |e|^2 f(m_e) = h_\Gamma \int \hat{\ell} f + O(h_\Gamma^2))$, and bound the cross terms with R by Step 2 and inverse estimates, and the smooth parts and the σ° mismatch by $O(h_\Gamma^2)$ (Lemma 8.32). $\square$

Corollary 8.39 ((P) corrected; proved modulo (D_c), (TL)). *On LPC families, (P) holds for some ϕ if and only if $\mathcal{F}$ is not constant on Γ ; then $P_h(\phi)/h_\Gamma \rightarrow \int \mathcal{F} \phi$ and one may take $\phi = \mathcal{F} - \bar{\mathcal{F}}$. (P) fails on mirror-symmetric columns, where $\mathcal{F} \equiv 0$ and $P_h(\phi) = O(h_\Gamma^2)$, and in rotation-invariant situations ($\kappa W \hat{\ell}$ and $\mathbf{L}$ constant, as for tests A and Q on the C_4 -symmetric production meshes), where $\mathcal{F}$ is constant.*

On the isosceles annulus, $P_h(\sin \theta)/h_\Gamma^2 = 1.714, 1.618, 1.589, 1.591$ for $N = 16, \dots, 128$: the rate is exactly 2, and the h_Γ^2 coefficient is not a cell quantity (it changes to 21.5 when the outer radius is 1.4), so the second-order remainder is genuinely global.

Corollary 8.40 (Pressure and traction lower bounds without (P); proved modulo (D_c), (TL)). *On every LPC family at fixed admissible γ , mirror-symmetric columns included:*

- (a) *if $Q_x \neq 0$ on a set of positive measure (which holds when $\kappa W \neq 0$ and $P_1 \neq 0$ in the rows $j \geq 1$, where $P_c = 0$), then $\inf_c \|p^\circ - p_h - c\| \geq ch_\Gamma^{3/2}$;*
- (b) *if Θ_x is not a.e. a constant multiple of n , then $\|T_h^\beta - \sigma^\circ n\|_{\Gamma_h} \geq ch_\Gamma$;*

with the sharp constants of Theorem 8.38(ii), (iii). With the upper bounds of Theorem 8.7 and Proposition 8.23(a), the rates $\frac{3}{2}$ and 1 are then exact.

The hypotheses of the corollary are NUMERICAL: $\|P_c\| \in [0.19, 0.41]$ and $\|T_c\| \in [0.39, 1.12]$ on the cells of Table 5, with minima 0.026 and 0.16 over 135 scanned cells, and the odd part of the wall pressure (which is not a multiple of n) has norm 0.83 at $a = 1$, $o = -\frac{1}{2}$, $\lambda = 30$.

Corollary 8.41 (Drag at fixed γ ; proved modulo (D_c), (TL)). *By Theorems 8.42 and 8.38(i), with $h/\mu = h_\Gamma/\gamma$,*

$$J_h^N(\psi) - J_\psi = h_\Gamma^2 \mathcal{D}(\gamma) + o(h_\Gamma^2), \quad \mathcal{D}(\gamma) = -G_0 - \gamma^{-1} \int_\Gamma \mathcal{F}_\gamma \rho_\psi ds,$$

where $G_0 = \lim h_\Gamma^{-2}(\Theta_f + \Lambda_h) = \frac{1}{12} \int_\Gamma \hat{\ell}^2(f \cdot \psi + \partial_n u \cdot \partial_n(Z_\psi - \psi)) ds$ is geometric. Rate 2 is sharp at fixed γ iff $\mathcal{D}(\gamma) \neq 0$. If (D_c) holds uniformly for λ in a complex neighbourhood of the admissible range, $\mathcal{D}$ is real-analytic there (on the truncated cell A_1 is rational in λ without poles for $\lambda > \varsigma + \lambda_0$) and tends to $-G_0$ as $\gamma \rightarrow \infty$; so sharpness then holds outside a discrete (possibly empty) set of γ whenever $G_0 \neq 0$ or $\mathcal{D}$ is not constant. On mirror-symmetric columns the h_Γ^2 coefficient is the geometric $-G_0$ at every fixed γ .

TABLE 5. Annulus meshes (exactly LPC), test S₁ ($\kappa = 1$, $W = 2(1 + \cos \theta)$): cell values and first-order Richardson extrapolations $2f(128) - f(64)$ divided by the prediction of Theorem 8.38 (notes/v7/P/compare_ann.log).

cell	A_1	$\ P_c\ $	$\ T_c\ $	$P_h(\cos)/h_\Gamma$ at $N=128$	extrap./pred.	$\ e_p\ /h_\Gamma^{3/2}$	$\ T - \sigma n\ /h_\Gamma$
$a = 1, \lambda = 30, o = -\frac{1}{2}$	0.1824	0.2357	0.6017	1.320	1.023	1.009	1.005
$a = 2, \lambda = 30, o = -\frac{1}{2}$	0.0291	0.2970	0.6205	0.256	0.979	1.006	1.000
$a = 1, \lambda = 100, o = -\frac{1}{2}$	0.4413	0.4069	1.1219	3.199	1.020	1.010	1.005
$a = 1, \lambda = 30, o = 0$ (sym.)	0	0.1881	0.3874	0 (round-off)	–	1.010	1.001

On the production meshes the prediction uses, edge by edge, the rectangular lattice with the measured aspect and $\lambda_e = \mu|e|/h$, including the t_0 term (without it the $\cos 4\theta$ moment is off by a factor 2). For test S_1 at $\mu = 1000$ the linear extrapolation of $P_h(\cos)/h_\Gamma$ from $N = 64, 96$ is 3.13 against the cell limit 3.06; at $\mu = 100$ it is 2.25 against 2.11 (S_1) and 0.91 against 0.84 (B_1). At $\mu = 100$ the production cells lie within 7% of the resonance $\lambda = \varsigma$ ($\lambda/\varsigma \geq 1.07$), possibly below the proved well-posedness range $\varsigma + \lambda_0$, and $A_1 \propto (\lambda - \varsigma)^{-1}$ amplifies every $O(h_\Gamma)$ perturbation there.

Computation: `notes/v7/P/cell.py`, `signscan.py`, `rhoscan.py`, `mixed.py`, `annulus.py`, `annpred.py`, `compare_ann.py`, `predict.py`, `compare.py`; output `notes/v7/P/cell_scan.log`, `signscan.log`, `rhoscan.log`, `mixed.log`, `annpred.log`, `compare_ann.log`, `compare.log`.

8.6.6. *Drag at fixed γ : the weighted normal slip.*

Theorem 8.42 (CNS* drag at fixed γ). *Assume the hypotheses of Theorem 8.19 at fixed γ , with $E_Z \leq Ch$ and $\varepsilon_h \leq Ch^k$. Let $\rho_\psi := (R_\psi - \bar{R}_\psi)|_\Gamma$. Then*

$$J_h^N(\psi) - J_\psi = -\Theta_f(\psi) - \Lambda_h(\psi) - \langle \rho_\psi \circ \pi, u_h \cdot n \rangle_{\Gamma_h} + O(h_\Gamma^{5/2}) = -\Theta_f(\psi) - \Lambda_h(\psi) - \frac{h}{\mu} P_h(\rho_\psi) + O(h_\Gamma^{5/2}).$$

Proof. In the proof of Theorem 8.19, at fixed γ with $E_Z \leq Ch$: $|T_1| \leq C\varepsilon_h$; $\mathcal{G}(\zeta_h) - \Lambda_h = O(h_\Gamma^3 + h_\Gamma^{3/2} Y_Z) = O(h_\Gamma^{5/2})$; $|T_4| \leq C(h_\Gamma^{5/2} Y + Y Y_Z) = O(h_\Gamma^{5/2})$. In step 6 every term except the first is $O(h_\Gamma^{5/2})$ or $O(\varepsilon_h)$. For instance, $(h_\Gamma^{-1/2} Y_Z) \|e_h\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2} h \cdot h_\Gamma^2$ and $\|\tilde{u} - z\|_{\Gamma_h} \leq C(h/\mu)^{1/2} \varepsilon_h$. Hence $T_5 = S_h(\Pi - \bar{\Pi}) + O(h_\Gamma^{5/2})$ with $\Pi = -\tilde{R}_\psi$, i.e. $T_5 = -S_h(\rho_\psi) + O(h_\Gamma^{5/2})$, and

$$J_h^N - J_\psi + \Theta_f + \Lambda_h = -T_5 + O(h_\Gamma^{5/2}) = S_h(\rho_\psi) + O(h_\Gamma^{5/2}).$$

Since $|\tilde{u} \cdot n_e| \leq Ch_\Gamma^3$, $S_h(\rho_\psi) = -\langle \rho_\psi \circ \pi, u_h \cdot n \rangle + O(h_\Gamma^3)$. Prop. 8.31 gives $S_h(\rho_\psi) = -\frac{h}{\mu} P_h(\rho_\psi) + O(\frac{h}{\mu} h_\Gamma^{3/2})$. $\square$

Remark 8.43 (Consequences). (1) The formula mirrors Theorem 8.17. There the leak $u_h \cdot n \approx (h/\mu)q$ gives $-\langle \rho_\psi, u_h \cdot n \rangle \approx -(h/\mu)K_\psi$. For CNS* the wall pressure q is replaced by the boundary trace of the pressure *error*. The drag defect is, for both methods, the dual wall pressure weighted by the normal slip.

(2) $|(h/\mu)P_h(\rho_\psi)| \leq C\gamma^{-1/2}h_\Gamma^2$: this is exactly the $\gamma^{-1/2}h_\Gamma^2$ of Theorem 8.19(b), now identified. Since $P_h \asymp h_\Gamma$ numerically (Table 4), the term is *genuinely* of order h_Γ^2 at fixed γ . The h_Γ^2 coefficient is $-(\Theta_f + \Lambda_h)/h_\Gamma^2$ plus a γ -dependent discrete-layer correction. It is purely geometric only as $\gamma \rightarrow \infty$ (as Theorem 8.19 states).

(3) *Sharpness at fixed γ* holds if and only if $\liminf h_\Gamma^{-2}|\Theta_f + \Lambda_h + (h/\mu)P_h(\rho_\psi)| > 0$. Corollary 8.41 identifies the limit of P_h/h_Γ modulo (D_c) , (TL) , and hence the h_Γ^2 coefficient.

(4) The first form is computable a posteriori, given the dual wall pressure R_ψ : subtracting $-\langle \rho_\psi \circ \pi, u_h \cdot n \rangle$ leaves the geometric term plus $O(h_\Gamma^{5/2})$. We have not tested this numerically; we did not compute R_ψ .

8.7. **Navier–Stokes.** Assume (NS1)–(NS2) of Section 14, with (L) for GS and (L*) for CNS*, and use the convective form c of the scheme also in the drag functional:

$$\omega_h(w) := \nu a_h(u_h, w) + c(u_h; u_h, w) - (p_h, \operatorname{div} w) - (\tilde{f}, w), \quad J_\psi := \int_\Gamma (\nu D(u) - pI)n \cdot \psi.$$

Put $X := h/\mu + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h$ as in Section 14, and write $e := \tilde{u} - u_h$.

Proposition 8.44. (a) (CNS*, pressure.) *Theorem 8.7 holds verbatim, with C depending also on ν , N , C_1 and β_* .*

- (b) (GS, pressure.) Theorem 8.11 holds with $X_1 \rightarrow X$ and the same ξ_b , i.e. with $g = (h/\mu)q$ and q the Navier–Stokes pressure. In the convention νN_h of Remark 14.2 the layer does not depend on ν . With the unscaled penalty $\hat{\mu} = \nu\mu$ it is $g = \nu(h/\hat{\mu})q$.
- (c) (Drag.) Lemma 8.15(a) holds with t_h multiplied by ν . Lemma 8.15(b) becomes

$$J_h^N(\psi) - J_\psi = -\nu a_h(e, v_\psi) - c(e; \tilde{u}, v_\psi) - c(\tilde{u}; e, v_\psi) + c(e; e, v_\psi) - \Theta(\psi) + O(h_\Gamma^4),$$

where $\Theta(\psi) := (\tilde{f} - (\tilde{u} \cdot \nabla)\tilde{u}, \psi)_{S_h} = \Theta_f(\psi) + O(h_\Gamma^4)$. Hence $|J_h^N - J_\psi| \leq CX$ for GS (rate 1) and $\leq CY$ for CNS* (rate 3/2).

Proof. (a),(b): Lemma 8.1 acquires $c(\tilde{u}; \tilde{u}, v) - c(u_h; u_h, v) + \kappa_c(v)$ on the right. By Lemma 14.4 this is at most $C\|e\|_{H^1}\|\nabla v\|$ plus $Ch_\Gamma^4\|\nabla v\|$. The same convective terms appear when testing with w_1 . The decomposition $e = (h/\nu\mu)v_\varphi + \vartheta$ with $\|\vartheta\| \leq CX$, and $\|\nabla e\| \leq CX$, are Theorem 14.8 and the proof of Corollary 14.9. The layer source is $\nu\langle u_h, \partial_n v \rangle$, whose normal part is $\nu(h/\nu\mu)q$. (c): As in Lemma 8.15, now with $\text{div } \sigma(\tilde{u}, \tilde{p}) = F - \tilde{f} + (\tilde{u} \cdot \nabla)\tilde{u}$; the c_1/c_s difference on v_ψ is $\frac{1}{2}\langle (\tilde{u} \cdot n)\tilde{u}, \psi \rangle = O(h_\Gamma^4)$, and $|\tilde{u}| \leq Ch_\Gamma^2$ on S_h . The bounds use Lemma 14.4. $\square$

Sharper Navier–Stokes drag statements are proved in Section 14, using the Stokes regularity of Theorem 9.1 that we also use for U and Z_ψ . For GS, $J_h^N(\psi) = J_\psi - \frac{h}{\nu\mu}K_\psi^\dagger - \Theta_f(\psi) + o(h/\mu)$ with $K_\psi^\dagger = \int_\Gamma (p - \bar{p})(R_\psi^\dagger - \bar{R}_\psi^\dagger)$ and $(Z_\psi^\dagger, R_\psi^\dagger)$ the *adjoint* linearised (Oseen) problem with data ψ on Γ (Corollary 14.22). For CNS* the rate is 2, with leading term $-\Theta_f - \Lambda_h^\dagger$ as $\gamma \rightarrow \infty$ (Theorem 14.25); the adjoint stability it needs is $(L^\dagger)$ (Theorem 14.23). Proposition 8.23 transfers verbatim for the Stokes part.

Remark 8.45 (General domains). Theorems 8.7, 8.11 and Proposition 8.23(a) use only lemmas that Section 13 re-proves, together with $|L_1| \asymp h_\Gamma$. They therefore hold on the general domains of Section 13, with the global $\bar{p}$, with $\hat{n} := n \circ \pi$ in Lemma 8.6, and with $\text{div } \hat{n} = \varkappa + O(h_\Gamma^2)$ replacing -1 (only $\int_\Gamma q = 0$ is used). In the drag identities S_h must be replaced by the signed region $\Omega_h \triangle \Omega$: $\Theta_f = (\tilde{f}, \psi)_{\Omega_h \setminus \Omega} - (f, \psi)_{\Omega \setminus \Omega_h}$, and Λ_h carries the factor $\varkappa$ through $d = \frac{1}{2}\varkappa(\ell^2/4 - t^2)$. Theorem 8.17 uses Theorem 6.10, which Section 13 does not cover, so on general domains only its unconditional $O(h/\mu)$ bound is proved.

8.8. What is not proved.

- The cell hypotheses (D_c) (numerical) and (TL) (not proved) of Section 8.6.5. Modulo them, the CNS* pressure and traction lower bounds hold on every LPC family without (P) (Corollary 8.40), and (P) holds exactly when the cell density $\mathcal{F}$ is not constant (Corollary 8.39); it fails on mirror-symmetric columns. That $A_1 \neq 0$ for every asymmetric column is numerical; whether the deeper production rows are LPC was not checked.
- Sharpness of the CNS* drag rate 2 at *fixed* γ : modulo (D_c), (TL) it holds iff $\mathcal{D}(\gamma) \neq 0$ (Corollary 8.41), which is generic but not proved for a given γ .
- A general lower bound for the GS pointwise-traction defect Ξ_h : none exists (Proposition 8.24(c)); it holds for sign-definite $qn \cdot e$ on every mesh and on asymptotically regular first layers.
- The fixed- γ lower bound for the torque of J_h^χ : the leading term $-\frac{h}{\mu}J_{x^\perp}$ is proved only as $\gamma \rightarrow \infty$ with $\gamma h_\Gamma \rightarrow 0$ (Proposition 8.22); at fixed γ the first-order behaviour is numerical.
- Rates for the corrected functionals of Remark 8.26.

9. THE L^2 ERROR

The H^1 theory of Sections 6–8 leaves open the L^2 error of the velocity. For CNS* it is of order h_Γ^2 , and this rate is sharp; for GS it is of order h/μ , with an explicit leading term. The pressure term of CNS* is not adjoint consistent (Remark 9.7), so the duality arguments below test only

with fields in $Y_h = Z_h \cap V_h^0$ (upper bound) or with a dual problem whose pressure vanishes (lower bound). The dual problem is posed on Ω , not on Ω_h : the vertices of Γ_h are re-entrant corners of Ω_h with Stokes singular exponent below 1, so there is no h -uniform H^2 bound on Ω_h .

9.1. Stokes regularity on Ω . The dual problems below, the leak flow U (Definition 6.8), the drag dual Z_ψ (Section 8.5) and the Navier–Stokes duality of Section 14 all use $H^2 \times H^1$ regularity of the Stokes problem on $\Omega = R \setminus \bar{D}$. We prove it from two classical results.

- (K1) *Convex polygons* [41, Theorem 2]. If P is a convex polygon, $F \in L^2(P)^2$ and $(w, \varpi) \in H_0^1(P)^2 \times L^2(P)$ solves $-\Delta w + \nabla \varpi = F$, $\operatorname{div} w = 0$, then $\|w\|_{H^2(P)} + \|\nabla \varpi\|_{L^2(P)} \leq c(P)\|F\|$.
- (K2) *Smooth domains* (Cattabriga; [58, Ch. I, Prop. 2.2], [31, Thm. IV.6.1]). If A is a bounded domain of class C^{m+2} , $F \in H^m(A)^2$ and $(w, \varpi) \in H_0^1(A)^2 \times L^2(A)$ solves the same system, then $\|w\|_{H^{m+2}(A)} + \|\varpi - \bar{\varpi}\|_{H^{m+1}(A)} \leq c(A, m)\|F\|_{H^m(A)}$.

We use (K2) only on annuli bounded by two circles.

Theorem 9.1 ((R): Stokes regularity on Ω). *There is $C_R = C_R(L)$ such that for every $g \in L^2(\Omega)^2$ the weak solution $(\varphi, \pi) \in H_0^1(\Omega)^2 \times L^2(\Omega)$ of $-\Delta \varphi + \nabla \pi = g$, $\operatorname{div} \varphi = 0$ satisfies*

$$\|\varphi\|_{H^2(\Omega)} + \|\pi - \bar{\pi}\|_{H^1(\Omega)} \leq C_R \|g\|_{L^2(\Omega)}.$$

With viscosity ν , $\|\varphi\|_{H^2} \leq C_R \nu^{-1} \|g\|$ and $\|\pi - \bar{\pi}\|_{H^1} \leq C_R \|g\|$.

Proof. Energy and stream function. $\|\nabla \varphi\| \leq C_F \|g\|$, and the continuous inf-sup condition on the Lipschitz domain Ω bounds $\|\pi - \bar{\pi}\|$; replace π by $\pi - \bar{\pi}$. Since $\varphi = 0$ on $\partial\Omega$ its flux through Γ vanishes, so $\varphi = \operatorname{curl} \zeta$ with a single-valued $\zeta \in H^2(\Omega)$, $\int_\Omega \zeta = 0$, $\|\zeta\|_{H^2} \leq C \|\varphi\|_{H^1}$.

Partition. Fix $1 < r_1 < r_2 < \rho_A < L$ and a radial $\chi_1 \in C^\infty$ with $\chi_1 = 0$ for $|x| \leq r_1$, $\chi_1 = 1$ for $|x| \geq r_2$, and put $\chi_2 := 1 - \chi_1$. Then $\varphi = w_1 + w_2$ with $w_i := \operatorname{curl}(\chi_i \zeta) = \chi_i \varphi + \zeta \operatorname{curl} \chi_i$, and each w_i is exactly divergence free.

Near ∂R . Extended by 0, $w_1 \in H_0^1(R)^2$, and in the weak sense on R

$$-\Delta w_1 + \nabla(\chi_1 \pi) = \chi_1 g + \mathcal{R}_1, \quad \mathcal{R}_1 := -2(\nabla \varphi) \nabla \chi_1 - \varphi \Delta \chi_1 - \Delta(\zeta \operatorname{curl} \chi_1) + \pi \nabla \chi_1,$$

with $\|\mathcal{R}_1\| \leq C(\|\varphi\|_{H^1} + \|\zeta\|_{H^2} + \|\pi\|) \leq C\|g\|$. R is a convex polygon, so (K1) applies; by uniqueness (pressures modulo constants) its pressure is $\chi_1 \pi + c$, and $\|w_1\|_{H^2(R)} + \|\nabla(\chi_1 \pi)\| \leq C\|g\|$.

Near Γ . On the annulus $A := \{1 < |x| < \rho_A\}$, $w_2 \in H_0^1(A)^2$ solves the same system with χ_2 in place of χ_1 , and (K2) with $m = 0$ gives $\|w_2\|_{H^2(A)} + \|\nabla(\chi_2 \pi)\|_{L^2(A)} \leq C\|g\|$. Add the two pieces and use Poincaré for $\pi - \bar{\pi}$. The viscous form follows by scaling $\pi \mapsto \pi/\nu$. $\square$

The stream-function cut-off keeps both pieces exactly divergence free, so no Bogovskii operator is needed. The proof uses only that ∂R and Γ are disjoint, that R is a convex polygon and that Γ is C^2 (here C^∞).

Proposition 9.2 (Local higher regularity at Γ). *Let (φ, π) be as in Theorem 9.1, $0 < r_1 < L - 1$, $N_s := \{1 < |x| < 1 + sr_1\}$, and $g \in H^m(N_1)^2$, $m \geq 0$. Then*

$$\|\varphi\|_{H^{m+2}(N_{1/2})} + \|\pi - \bar{\pi}\|_{H^{m+1}(N_{1/2})} \leq C(m, r_1, L)(\|g\|_{H^m(N_1)} + \|g\|_{L^2(\Omega)}).$$

In particular, if f is smooth up to Γ then u , p and ψ are smooth up to Γ .

Proof. Induction on $j = 0, \dots, m$ for $\varphi \in H^{j+2}(N_{s_j})$, $\pi \in H^{j+1}(N_{s_j})$, $s_j := 1 - j/(2m)$; $j = 0$ is Theorem 9.1. Given step $j < m$, take a radial cut-off χ equal to 1 on $N_{s_{j+1}}$ and 0 for $|x| \geq 1 + s_j r_1$, and $w := \operatorname{curl}(\chi \zeta) \in H_0^1(N_{s_j})^2$. As above, $-\Delta w + \nabla(\chi \pi) = \chi g + \mathcal{R}$ with $\mathcal{R}$ supported where $\nabla \chi \neq 0$ and built from $\nabla \varphi$, φ , π and at most second derivatives of ζ ($\nabla \zeta$ is φ rotated), so $\mathcal{R} \in H^{j+1}$ by the induction hypothesis, and $\chi g \in H^{j+1}$ since $j+1 \leq m$. (K2) with $m \rightarrow j+1$ gives the step. For the last claim apply this to $u - G$ with a divergence-free lift G of the outer data supported near ∂R , and use Sobolev embedding. $\square$

This covers the regularity used for U (whose Dirichlet datum $-(p - \bar{p})n$ on Γ is lifted by the smooth divergence-free field w of Section 6), for Z_ψ and for the dual problems below. At the corners of R the solution of a data-driven problem is less regular; see Remark 2.5.

9.2. Tools.

Lemma 9.3 (Strip). *Let N be a fixed collar of Γ containing S_h . For $w \in H^1(\Omega_h \cup \Omega)$:*

- (a) $\|w\|_{S_h} \leq C(h_\Gamma \|w\|_{\Gamma_h} + h_\Gamma^2 \|\nabla w\|_{S_h})$ and $\|w\|_{\Gamma_h} \leq C\|w\|_{H^1(N)}$;
- (b) if $w \in H^2(N)$ and $w|_\Gamma = 0$, then $\|w\|_{\Gamma_h} \leq Ch_\Gamma^2 \|w\|_{H^2(N)}$ and $\|w\|_{\Gamma_h} \leq Ch_\Gamma \|\nabla w\|_{S_h} \leq Ch_\Gamma^{3/2} \|w\|_{H^2(N)}$;
- (c) if $w|_{\Gamma_h} = 0$, then $\|w\|_{S_h} \leq Ch_\Gamma^2 \|\nabla w\|_{S_h}$.

Proof. Write Γ_h in polar form $r = \rho_h(\theta)$ with $0 \leq 1 - \rho_h \leq Ch_\Gamma^2$ (Lemma 3.2) and integrate along radial segments. For (b), $|\nabla w| \leq Ch_\Gamma^2 \|w\|_{W^{2,\infty}}$ is not available, but $\|\nabla w\|_{S_h} \leq |S_h|^{1/4} \|\nabla w\|_{L^4(N)} \leq Ch_\Gamma^{1/2} \|w\|_{H^2(N)}$. $\square$

Lemma 9.4 (Extension of the dual). *Every $\varphi \in H_0^1(\Omega)^2 \cap H^2(\Omega)^2$ with $\operatorname{div} \varphi = 0$ and every $\pi \in H^1(\Omega)$ have extensions $\tilde{\varphi} \in H^2(\Omega \cup N)^2$, $\tilde{\pi} \in H^1(\Omega \cup N)$ with $\operatorname{div} \tilde{\varphi} = 0$, $\|\tilde{\varphi}\|_{H^2} \leq C_E \|\varphi\|_{H^2(\Omega)}$ and $\|\tilde{\pi}\|_{H^1} \leq C_E \|\pi\|_{H^1(\Omega)}$.*

Proof. $\int_\Gamma \varphi \cdot n = 0$, so $\varphi = \operatorname{curl} \zeta$ with ζ single valued near Γ , $\zeta|_\Gamma = 0$, $\nabla \zeta|_\Gamma = 0$ and $\|\zeta\|_{H^3} \leq C\|\varphi\|_{H^2}$ on a collar. Extend ζ across $r = 1$ by a Hestenes reflection of order 3 in the radial variable, and put $\tilde{\varphi} := \operatorname{curl} \tilde{\zeta}$; extend π by a first-order reflection. $\square$

Lemma 9.5 (Approximation in Y_h). *Let φ and $\tilde{\varphi}$ be as in Lemma 9.4. There is $y_h \in Y_h$ with*

$$\|\nabla(\tilde{\varphi} - y_h)\|_{\Omega_h} \leq C_Y(h + h_\Gamma)\|\varphi\|_{H^2}, \quad \|\partial_n y_h\|_{\Gamma_h} \leq C_Y(1 + hh_\Gamma^{-1/2})\|\varphi\|_{H^2}, \quad \|\nabla y_h\| \leq C_Y\|\varphi\|_{H^2},$$

with C_Y depending only on C_E , k , (M0)–(M2) and β .

Proof. (1) By Lemma 9.3(b), $\|\tilde{\varphi}\|_{\Gamma_h} \leq Ch_\Gamma^{3/2} \|\varphi\|_{H^2}$; the scaled trace inequality on T_e gives $\|\tilde{\varphi} - I_h \tilde{\varphi}\|_{L^2(e)} \leq C|e|^{3/2} |\tilde{\varphi}|_{H^2(T_e)}$. So $\|I_h \tilde{\varphi}\|_{\Gamma_h} \leq Ch_\Gamma^{3/2} \|\varphi\|_{H^2}$.
(2) Let $L_0 \in V_h$ take the nodal values of $I_h \tilde{\varphi}$ at the Lagrange nodes on Γ_h and vanish at all other nodes. By step 2 of Lemma 7.28, $\|\nabla L_0\| \leq Ch_\Gamma^{-1/2} \|I_h \tilde{\varphi}\|_{\Gamma_h} \leq Ch_\Gamma \|\varphi\|_{H^2}$, and $z_0 := I_h \tilde{\varphi} - L_0 \in V_h^0$.
(3) $\operatorname{div} z_0 \in \Pi_h$ has zero mean and $\|\operatorname{div} z_0\| \leq C(h + h_\Gamma)\|\varphi\|_{H^2}$; Lemma 4.5 gives $v_B \in V_h^0$ with $\operatorname{div} v_B = \operatorname{div} z_0$. Put $y_h := z_0 - v_B$.
(4) $\|\partial_n \tilde{\varphi}\|_{\Gamma_h} \leq C\|\tilde{\varphi}\|_{H^2}$ (uniform trace from the collar), $\|\nabla(I_h \tilde{\varphi} - \tilde{\varphi})\|_{L^2(e)} \leq C|e|^{1/2} |\tilde{\varphi}|_{H^2(T_e)}$, and $L_0 + v_B$ is discrete, so Lemma 4.1 gives $\|\partial_n(L_0 + v_B)\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}(h + h_\Gamma)\|\varphi\|_{H^2}$. $\square$

9.3. Upper bound: rate 2 for CNS*.

Theorem 9.6 (L^2 error of CNS*). *Let CNS* use flux-corrected data $\tilde{g}_I$ and $\gamma \geq \gamma_2$ (no upper bound), and let $e := \tilde{u} - u_h$. Then*

$$\|e\|_{L^2(\Omega_h)} \leq C(h_\Gamma^2 + h h_\Gamma^{3/2} + h^{k+1}),$$

with C independent of γ . On quasi-uniform meshes ($h \leq Ch_\Gamma$) this is $C(h_\Gamma^2 + h^{k+1})$. The same bound holds for GS when $G = 0$.

Proof. Put $\phi := e|_\Omega$, let (z, r) solve the dual Stokes problem of Theorem 9.1 with $g = \phi$, and extend by Lemma 9.4. On Ω , $-\Delta \tilde{z} + \nabla \tilde{r} = \phi$; on S_h the left side is some L^2 field of norm

$\leq C\|z\|_{H^2}$. Since $\operatorname{div} e = 0$ pointwise, $-\Delta\tilde{z} = -\operatorname{div} D(\tilde{z})$ and $e = g - \tilde{g}_I$ on ∂R , integration by parts on Ω_h gives

$$\|\phi\|^2 = a_h(e, \tilde{z}) - \langle \sigma(\tilde{z}, \tilde{r})n, e \rangle_{\Gamma_h} - \langle \sigma(\tilde{z}, \tilde{r})\nu, g - \tilde{g}_I \rangle_{\partial R} - (e, -\Delta\tilde{z} + \nabla\tilde{r})_{S_h}.$$

Let $y_h \in Y_h$ be as in Lemma 9.5. By Lemma 7.19, $a_h(e, y_h) = \langle e, \partial_n y_h \rangle - \langle \tilde{u}, \partial_n y_h \rangle - (F, y_h) = -\langle u_h, \partial_n y_h \rangle - (F, y_h)$. Hence

$$\|\phi\|^2 = a_h(e, \tilde{z} - y_h) - \langle u_h, \partial_n y_h \rangle - (F, y_h) - \langle \sigma n, e \rangle_{\Gamma_h} - \langle \sigma\nu, g - \tilde{g}_I \rangle_{\partial R} - (e, -\Delta\tilde{z} + \nabla\tilde{r})_{S_h}.$$

Two inputs, both uniform in γ : $\|\nabla e\| \leq C(h_\Gamma^{3/2} + h^k)$ (Theorem 7.33), and $\|e\|_{\Gamma_h} \leq (h_\Gamma/\gamma)^{1/2}\|e\| \leq C(h_\Gamma^2 + h_\Gamma^{1/2}h^k)$ (Theorem 7.33 and $(h_\Gamma/\gamma)^{1/2}\tilde{\varepsilon}_h \leq C(h_\Gamma^{1/2}h^k + h_\Gamma^{\sigma_k})$). So $\|u_h\|_{\Gamma_h} \leq \|\tilde{u}\|_{\Gamma_h} + \|e\|_{\Gamma_h} \leq C(h_\Gamma^2 + h_\Gamma^{1/2}h^k)$. The six terms are bounded, in order, by $C\|z\|_{H^2}$ times

- (1) $h(h_\Gamma^{3/2} + h^k)$ (Lemma 9.5);
- (2) $(h_\Gamma^2 + h_\Gamma^{1/2}h^k)(1 + hh_\Gamma^{-1/2})$;
- (3) h_Γ^3 (F is bounded on S_h , $|S_h| \leq Ch_\Gamma^2$, and Lemma 9.3(c) for y_h);
- (4) $h_\Gamma^2 + h_\Gamma^{1/2}h^k$ (uniform trace of $\sigma \in H^1$);
- (5) h^{k+1} ($|g - \tilde{g}_I| \leq Ch^{k+1} + |m_R|$);
- (6) $h_\Gamma(\|e\|_{\Gamma_h} + h_\Gamma\|\nabla e\|) \leq C(h_\Gamma^3 + h_\Gamma^{3/2}h^k)$ (Lemma 9.3(a)).

With $\|z\|_{H^2} \leq C_R\|\phi\|$ and Young's inequality $h_\Gamma^{1/2}h^k \leq h_\Gamma^2 + h^{4k/3} \leq h_\Gamma^2 + Ch^{k+1}$ ($k \geq 3$), $\|e\|_{L^2(\Omega)} \leq C(h_\Gamma^2 + hh_\Gamma^{3/2} + h^{k+1})$. Finally $\|e\|_{S_h} \leq Ch_\Gamma(\|e\|_{\Gamma_h} + h_\Gamma\|\nabla e\|)$.

For GS with $G = 0$, apply the same argument to $u'_h := u_h + \delta_R$ of the proof of Theorem 7.30, which satisfies the identity of Lemma 7.19 on Y_h ; the proof of Theorem 7.30 supplies both inputs, and $\|\delta_R\| \leq Ch_\Gamma^2$ when $|\bar{p} - \bar{p}| \leq Ch_\Gamma^2$ on Γ_h . $\square$

Remark 9.7 (Adjoint inconsistency). The pressure term of CNS^* , with the Γ_h -mean removed, is not adjoint consistent. The algebraic transpose of the CNS^* saddle system has the plain GS pressure form in its momentum row, and relative to a smooth dual pair its consistency defect is the GS leak functional $v \mapsto \langle r - \bar{r}, v \cdot n \rangle$ driven by the dual pressure (Corollary 5.3). The proof above avoids it by testing only with $y_h \in Y_h$, on which every pressure and penalty term vanishes. The same issue, and the same cure, appear in the Navier–Stokes drag analysis (Section 14.10).

9.4. Lower bound: the rate 2 is sharp.

Theorem 9.8 (Sharpness of the L^2 rate). *Let $W = \partial_n u \cdot \tau \neq 0$, and let u_h be the CNS^* velocity with flux-corrected data, or the GS velocity with $G = 0$. There are $\delta_0, c > 0$ such that if $(1 + \gamma)h_\Gamma \leq \delta_0$ (in particular for bounded γ and small h_Γ) and $h^k \leq \delta_0 h_\Gamma^2$, then*

$$\|\tilde{u} - u_h\|_{L^2(\Omega_h)} \geq c\|W\|_{L^2(\Gamma)}^2 h_\Gamma^2 / C_W, \quad C_W := \|\Delta(\chi(r-1)^2 W)\|_{H^1}.$$

No (M3) and no regularity theory are needed: the dual problem is explicit.

Proof. Dual. Let $\zeta := \pm \frac{1}{2}\chi(r)(r-1)^2 W(\theta)$, with χ a smooth cut-off equal to 1 near Γ and 0 near ∂R , the sign chosen so that $z := \operatorname{curl} \zeta$ has $\partial_n z \cdot \tau = W$ on Γ . Then z is smooth on $\overline{\Omega_h}$, $z|_\Gamma = 0$, $z|_{\partial R} = 0$, $\operatorname{div} z = 0$, and with $\tilde{\phi} := -\Delta z$ the pair $(z, 0)$ is an exact Stokes pair on Ω_h : $-\Delta z + \nabla 0 = \tilde{\phi}$. Let $(z_h, r_h) \in Z_h \times \Pi_h$ be the CNS^* (resp. GS) solution with body force $\tilde{\phi}$ and zero data. By Lemma 5.1 applied to $(z, 0)$ (with $F \equiv 0$), $\|z - z_h\| \leq CY_z$, $Y_z := (1 + \gamma^{1/2})h_\Gamma^{3/2} + h^k$, and, since the dual pressure vanishes, step 3 of Theorem 7.2 gives $\beta\|r_h - \bar{r}_h\| \leq CY_z$.

Identity. Let $Z \in \tilde{W}_h$ attain Lemma 7.32 and $e_h := Z - u_h \in Z_h$. Testing the dual problem with e_h and the error equation (3) with z_h gives, since $\operatorname{div} e_h = \operatorname{div} z_h = 0$ and N_h is symmetric,

$$(e, \tilde{\phi}) = \underbrace{(\tilde{u} - Z, \tilde{\phi}) - N_h(\tilde{u} - Z, z_h)}_{T_1} + \mathcal{G}(z_h) - \langle e_p - \bar{p}_\Gamma(e_p), z_h \cdot n \rangle + \langle r_h - \bar{r}_h, e_h \cdot n \rangle.$$

Bounds. $|T_1| \leq Ch^k$. By Lemma 8.4, $|\mathcal{G}(z_h) - \mathcal{G}(z)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}Y_z$. On Γ_h , $|z| \leq Ch_\Gamma^2$, $|(\nabla \tilde{u})^\top n| \leq Ch_\Gamma$, and $\tilde{u}$ and z are both tangential to leading order ($\tilde{u} \cdot z = O(h_\Gamma^4)$), so $\mathcal{G}(z) = -\langle \tilde{u}, \partial_n z \rangle + O((1 + \gamma)h_\Gamma^3)$. For the third term, $\|e_p - \bar{p}_\Gamma(e_p)\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}Y$ (step 3 of Theorem 7.2 and Lemma 4.1) and $\|z_h \cdot n\|_{\Gamma_h} \leq \|z \cdot n\|_{\Gamma_h} + \|z_h - z\|_{\Gamma_h} \leq Ch_\Gamma^3 + (h_\Gamma/\gamma)^{1/2}CY_z$; the product is $O((1 + \gamma)h_\Gamma^3 + h^k)$. (Pointwise $z \cdot n = O(h_\Gamma^3)$, but $z_h \cdot n$ is only $O(h_\Gamma^3)$ in $L^2(\Gamma_h)$; it is the $h_\Gamma^{-1/2}$ trace factor of the pressure that makes the product small.) For the last term, $\|r_h - \bar{r}_h\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2}Y_z$ and $\|e_h \cdot n\|_{\Gamma_h} \leq (h_\Gamma/\gamma)^{1/2}CY$, so it is $O(\gamma^{-1/2}Y Y_z) = O((1 + \gamma)h_\Gamma^3 + h^k)$. For GS with $G = 0$ the leak term of the dual vanishes because its pressure is zero, and there are no pressure terms.

Leading term. On e , $\tilde{u} = dW(x^*)\tau(x^*) + O(d^2)$ with $d = \frac{1}{2}(|e|^2/4 - s^2) + O(|e|^4)$ (Lemma 7.22), $\partial_n z = W\tau + O(h_\Gamma)$ and $\int_e d = |e|^3/12 + O(|e|^5)$. So $\langle \tilde{u}, \partial_n z \rangle = \sum_e \frac{|e|^3}{12} W^2(m_e^*) + O(h_\Gamma^3) \geq \frac{c_0^2}{12} h_\Gamma^2 (\|W\|_{L^2(\Gamma)}^2 - Ch_\Gamma)$ by the Riemann sum of Lemma 7.23. Altogether $|(e, \tilde{\phi})| \geq \frac{c_0^2}{12} h_\Gamma^2 \|W\|^2 - C((1 + \gamma)h_\Gamma^3 + h^k)$; divide by $\|\tilde{\phi}\| \leq C_W$. $\square$

The proof shows more: to leading order e is the Stokes flow driven by the chord-mean slip $\frac{|e|^2}{12} \partial_n u$, a smooth tangential boundary datum of size h_Γ^2 , plus a normal-slip remainder of size $O(\gamma^{-1/2}h_\Gamma^2)$. Numerically (test A, $N = 96$, $\mu = 100$), $(e, \tilde{\phi})/(-\Lambda_h) = 0.980$ for CNS* and 0.987 for GS, where $\Lambda_h = \sum_e \frac{|e|^3}{12} W^2(m_e^*)$, and the difference settles at a constant times h_Γ^3 .

9.5. The printed method: rate exactly 1.

Theorem 9.9 (L^2 error of GS). *For GS with fixed $\gamma \geq \gamma_0$ and bounded ρ ,*

$$\|\tilde{u} - u_h\|_{L^2(\Omega_h)} = \frac{h}{\mu} \|U\|_{L^2(\Omega)} (1 + O(h^{1/2})) + O(h_\Gamma^2 + hh_\Gamma^{3/2} + h^{k+1}),$$

where U is the leak flow of Definition 6.8. Hence the L^2 rate of GS is exactly 1 whenever $G > 0$. Uniformly in $\gamma \geq \gamma_0$, $\|\tilde{u} - u_h\|_{L^2} \leq C_p h_\Gamma/\gamma + C(h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2})$.

Proof. Write $\tilde{u} - u_h = (\tilde{u} - u'_h) + \delta_R$ as in the proof of Theorem 7.30. By Friedrichs' inequality (Lemma 3.2; $\delta_R = 0$ on ∂R) and Theorem 6.10, $\|(\mu/h)\delta_R - \tilde{U}\|_{L^2(\Omega_h)} \leq C_F \|\nabla((\mu/h)\delta_R - \tilde{U})\| \leq Ch^{1/2}$ at fixed γ and bounded ρ , and $\|\tilde{U}\|_{S_h} \leq Ch_\Gamma$. The part $\tilde{u} - u'_h$ satisfies the identity of Lemma 7.19 on Y_h and the bounds of steps 1–4 of Theorem 7.30, so the proof of Theorem 9.6 bounds it by $C(h_\Gamma^2 + hh_\Gamma^{3/2} + h^{k+1})$. The uniform bound is Theorem 7.30 with Friedrichs' inequality. $\square$

For test P, $\|U\|_{L^2(\Omega)}/G = 1.2736341$ (series solution of Section 6.1). The ratio $\|\tilde{u} - u_h\|/((h/\mu)\|U\|)$ at $N = 96$ is 0.980, 0.997 and 0.999 for $\mu = 10^2, 10^3, 10^4$, and $\|(\mu/h)e - U\|_{L^2}/\|U\|$ decays like h at $\mu = 100$ (notes/v6/12/leak.log).

9.6. Graded meshes. The term $hh_\Gamma^{3/2}$ of Theorem 9.6 matters only when $h \gg h_\Gamma^{1/2}$, i.e. on strongly graded meshes. It arises twice: in $\langle u_h, \partial_n y_h \rangle$, through the global divergence correction of y_h , and in $a_h(e, \tilde{z} - y_h)$, through the global product $\|\nabla e\| \|\nabla(\tilde{z} - y_h)\| \leq h_\Gamma^{3/2} \cdot h$. Both are removed by localisation: a local right inverse of the divergence (Lemma 9.10) and a local energy estimate in which the pressure never appears (Theorem 9.12). The grading hypothesis is

(G_κ) $h_T \leq \max(C_g h_\Gamma, \kappa \text{dist}(T, \Gamma_h))$ for all $T \in \mathcal{T}_h$.

Lemma 9.10 ((LF): local divergence correction). *Assume (M0)–(M2), $k \geq 4$. There is a linear map $\mathcal{L} : \{q \in \Pi_h : \int_T q = 0 \ \forall T\} \rightarrow V_h^0$ with $\text{div } \mathcal{L}q = q$,*

$$\|\nabla \mathcal{L}q\|_T \leq C_{\text{loc}} \|q\|_{\omega_T}, \quad (\mathcal{L}q)|_T = 0 \text{ if } q = 0 \text{ on } \omega_T,$$

where ω_T is the union of the stars of the vertices of T and C_{loc} depends only on k , shape regularity and Θ_0 .

Proof. The Guzmán–Scott construction [37] has three steps: a global P_2 step for element means, a vertex step (their Lemma 6: for each vertex z a field v^z supported in the star of z , with $\text{div } v^z|_T(z) = q|_T(z)$ for $T \ni z$, $\text{div } v^z = 0$ at the other vertices, zero element means of $\text{div } v^z$, and $\|\nabla v^z\| \leq C(\Theta(z)^{-1} + 1)\|q\|_{\text{star}(z)}$; boundary vertices use the boundary Θ), and an element step (their Proposition 2: for $p_T \in P_{k-1}(T)$ with zero mean vanishing at the vertices, $v_T \in P_k(T)^2 \cap H_0^1(T)^2$ with $\text{div } v_T = p_T$, $\|v_T\|_{H^1} \leq \alpha_2\|p_T\|$). The global step is needed only for element means, which q already has. Put $v_2 := \sum_z v^z$; then $q - \text{div } v_2$ vanishes at every vertex and has zero element means, and $\mathcal{L}q := v_2 + \sum_T v_T$ with v_T for $q - \text{div } v_2$. On T , $\|\nabla v_2\|_T \leq C\|q\|_{\omega_T}$ by (M2) and $\|\nabla v_T\| \leq \alpha_2(\|q\|_T + \sqrt{2}\|\nabla v_2\|_T)$. All fields vanish on T if $q = 0$ on ω_T . (The lemma numbers are those of arXiv:1705.00020.) $\square$

Lemma 9.11 (Local divergence-free interpolant). *For a continuous, piecewise H^{k+1} field v on Ω_h with $\text{div } v = 0$ elementwise, let $y := I_h v + \sum_E \alpha_E b_E n_E$ over all edges E (b_E the quadratic edge bubble, $\alpha_E := \int_E (v - I_h v) \cdot n_E / \int_E b_E$), and $\Pi_h^{\text{div}} v := y - \mathcal{L}(\text{div } y)$. Then $\Pi_h^{\text{div}} v \in V_h$, $\text{div } \Pi_h^{\text{div}} v = 0$,*

$$\|\nabla(v - \Pi_h^{\text{div}} v)\|_T + h_T^{-1}\|v - \Pi_h^{\text{div}} v\|_T \leq C \sum_{T' \subset \omega_T^2} h_{T'}^k |v|_{H^{k+1}(T')},$$

with ω_T^2 the two-layer patch; $\Pi_h^{\text{div}} v \in V_h^0$ if $v = 0$ on $\partial\Omega_h$, and $\Pi_h^{\text{div}} v = 0$ outside the two-layer neighbourhood of $\text{supp } v$.

Proof. b_E vanishes on the other edges, so $\int_E (y - v) \cdot n_E = 0$ for every E and $\int_T \text{div } y = \int_T \text{div } v = 0$; Lemma 9.10 applies to $\text{div } y = \text{div}(y - v)$. The bound follows from $|\alpha_E| \leq C|E|^{-1/2}\|v - I_h v\|_{L^2(E)}$, the scaled trace inequality, one more layer for $\mathcal{L}$, and Poincaré on patches for the L^2 part. $\square$

For $x_0 \in \overline{\Omega_h}$ and $r > 0$ put $B(r) := B(x_0, r) \cap \Omega_h$.

Theorem 9.12 ((IE): a pressure-free local energy estimate). *Assume (M0)–(M2), $k \geq 4$, $d > 0$, and either (I) $B(x_0, 2d) \subset R$ or (B) $x_0 \in \partial R$ and $2d \leq L$; in both cases $\text{dist}(B(x_0, 2d), \Gamma_h) \geq d$ and $h_T \leq \kappa_0 d$ for every T meeting $B(x_0, 2d)$. Let $U_h \in V_h$ satisfy (i) $\text{div } U_h = 0$; (ii) $a_h(\tilde{u} - U_h, v) = 0$ for every $v \in Z_h$ supported in $B(x_0, 2d)$; (iii) in case (B), $U_h = \Pi_h^{\text{div}} \tilde{u}$ on $\partial R \cap B(x_0, 2d)$. There are $\kappa_{\text{IE}} > 0$ and C_{IE} , depending only on k , shape regularity and Θ_0 , such that for $\kappa_0 \leq \kappa_{\text{IE}}$, with $\eta := \tilde{u} - \Pi_h^{\text{div}} \tilde{u}$,*

$$\|\nabla(\tilde{u} - U_h)\|_{B(d)} \leq C_{\text{IE}}(d^{-1}\|\tilde{u} - U_h\|_{B(2d)} + d^{-1}\|\eta\|_{B(2d)} + \|\nabla \eta\|_{B(2d)}).$$

Hypothesis (ii) holds for the CNS* and the GS velocities, with any outer data and any γ for which they exist: for $v \in Z_h$ supported at distance $\geq (1 - \kappa_0)d$ from Γ_h , v and $\partial_n v$ vanish on Γ_h and $(F, v) = 0$, so the error equation (3) (resp. Corollary 5.3) reduces to $a_h(\tilde{u} - u_h, v) = 0$.

Proof. Setup. $E := \tilde{u} - U_h = \eta + \psi$ with $\psi := \Pi_h^{\text{div}} \tilde{u} - U_h$, $\text{div } \psi = 0$, and in case (B) $\psi = 0$ on $\partial R \cap B(x_0, 2d)$. Let $B' := B(x_0, 2d) \cap R$; it is convex, and in case (B) $\partial R \cap B(x_0, 2d)$ is connected ($2d \leq L$ excludes two corners). Let φ be the stream function, $\text{curl } \varphi = \psi$; it is C^1 and piecewise P_{k+1} . Normalise $\int_{B'} \varphi = 0$ in case (I) and $\varphi = 0$ on $\partial R \cap B(x_0, 2d)$ in case (B) ($\psi \cdot \nu = 0$ there). The domains B' are convex and, after scaling by d , range over a compact family (in case (B) they contain a boundary segment of length $\geq 2d$ on which φ vanishes), so Poincaré, resp. Friedrichs, holds with a uniform constant: $\|\varphi\|_{B'} \leq C_P d \|\psi\|_{B'}$.

Fix $\omega \in C_c^\infty(B(x_0, \frac{3}{2}d))$, $0 \leq \omega \leq 1$, $\omega = 1$ on $B(x_0, d)$, $|\nabla^j \omega| \leq C_j d^{-j}$, put $\sigma := \omega^2$, $w := \text{curl}(\sigma^2 \varphi) = \sigma^2 \psi + \varphi \text{curl}(\sigma^2)$ and $v := \Pi_h^{\text{div}} w$. Then w is continuous, divergence free, in $H_0^1(\Omega_h)$, and $v \in Z_h$ is supported in $B(x_0, 2d)$ once $C\kappa_0 \leq \frac{1}{2}$ (Lemma 9.11). By (ii), $a_h(E, v) = 0$; since $\text{div } \psi = 0$ and $w, w - v \in H_0^1$, $a_h(\psi, \cdot) = (\nabla \psi, \nabla \cdot)$ on them, so

$$(\nabla \psi, \nabla w) = (\nabla \psi, \nabla(w - v)) - a_h(\eta, v).$$

Put $M := d^{-1}\|\psi\|_{B'} + d^{-2}\|\varphi\|_{B'} \leq (1 + C_P)d^{-1}\|\psi\|_{B'}$.

(A) *Coercivity*. Every term of $\nabla(\varphi \operatorname{curl} \sigma^2)$ carries a factor σ or $|\nabla \sigma|^2 \leq Cd^{-2}\sigma$, so $(\nabla \psi, \nabla w) \geq \|\sigma \nabla \psi\|^2 - C\|\sigma \nabla \psi\|M$ and $\|\nabla w\| \leq \|\sigma \nabla \psi\| + CM$.

(B) *Superapproximation*. On an element T' , $\varphi \in P_{k+1}$, so by Lemma 9.11, Leibniz and inverse estimates,

$$h^k|w|_{H^{k+1}(T')} \leq C \sum_{j=1}^k (h/d)^j \bar{\omega}^{(4-j)+} \|\nabla \psi\|_{T'} + C(h/d)^k (d^{-1}\|\psi\|_{T'} + d^{-2}\|\varphi\|_{T'}),$$

with $\bar{\omega}$ the maximum of ω on the three-layer patch, $\bar{\omega} \leq \omega + Ch/d$ there. Young's inequality $(h/d)^j \bar{\omega}^{4-j} \leq \epsilon \omega^4 + C_\epsilon (h/d)^4$ and the inverse inequality $(h_T/d)^4 \|\nabla \psi\|_T^2 \leq C\kappa_0^2 d^{-2} \|\psi\|_T^2$ give

$$|(\nabla \psi, \nabla(w - v))| \leq \frac{1}{4} \|\sigma \nabla \psi\|^2 + C(d^{-2}\|\psi\|_{B'}^2 + M^2), \quad \|\nabla(w - v)\| \leq C(\|\sigma \nabla \psi\| + d^{-1}\|\psi\|_{B'}).$$

No norm of $\nabla \psi$ outside the weight survives: where ω is small the factor h/d is present and the inverse inequality turns $h_T \|\nabla \psi\|_T$ into $\|\psi\|_T$. This is the device of Demlow, Guzmán and Schatz for the Laplacian [25], applied to $\sigma^2 = \omega^4$.

(C) $|a_h(\eta, v)| \leq C\|\nabla \eta\|_{B(2d)}(\|\sigma \nabla \psi\| + d^{-1}\|\psi\|_{B'})$. Inserting (A)–(C), $\|\nabla \psi\|_{B(d)} \leq \|\sigma \nabla \psi\| \leq C(d^{-1}\|\psi\|_{B'} + \|\nabla \eta\|_{B(2d)})$; finally $\|\psi\| \leq \|E\| + \|\eta\|$. $\square$

Arnold and Liu prove local estimates for Stokes under abstract approximation, superapproximation, inverse and inf-sup hypotheses at fixed scale [4]; Scott–Vogelius is not among their examples. Theorem 9.12 combines the Demlow–Guzmán–Schatz weighting with an exactly divergence-free test field, so that for Z_h -Galerkin orthogonality neither a local inf-sup condition nor a pressure norm is needed. We claim no novelty beyond this combination.

Lemma 9.13 (Data shift). *Let CNS^* use flux-corrected data and $\gamma \in [\gamma_2, \gamma_{\max}]$, let $u_h^\sharp$ be the CNS^* velocity with the same $\tilde{f}$ and outer data $\hat{g} := (\Pi_h^{\operatorname{div}} \tilde{u})|_{\partial R}$, and $\ell_h := u_h^\sharp - u_h$. If $\tilde{u} \in W^{k+1, \infty}$ near ∂R , then $\|\ell_h\| \leq Ch^{k+1/2}$ and $\|\ell_h\|_{L^2(\Omega_h)} \leq Ch^{k+1}$.*

Proof. ℓ_h is the CNS^* velocity with $\tilde{f} = 0$ and data $\epsilon := \hat{g} - \tilde{g}_I = c_R x + \sum_{E \subset \partial R} \alpha_E b_E \nu$, where $|c_R| = |m_R|/(8L^2) \leq Ch^{k+1}$ and $|\alpha_E| \leq Ch_E^{k+1} \|\tilde{u}\|_{W^{k+1, \infty}}$; the data have zero flux. *Energy.* A divergence-free $\mathcal{E} \in V_h$ with $\mathcal{E}|_{\partial R} = \epsilon$, vanishing on the triangles touching Γ_h , is built from $c_R I_h(\hat{\chi}x)$ ($\hat{\chi}$ a cut-off equal to 1 near ∂R), the P_k extensions of $\alpha_E b_E \nu$, and a V_h^0 correction of the (mean-zero) divergence by Lemma 4.5; $\|\nabla \mathcal{E}\| \leq C(|c_R| + (\sum_E \alpha_E^2)^{1/2}) \leq Ch^{k+1/2}$ since $\sum_E h_E^{2k+2} \leq Ch^{2k+1}$. The energy argument of Theorem 7.2 with exact solution $(0, 0)$ ($\mathcal{G} = 0$, $\eta = 0$) and $z := \mathcal{E}$ gives $\|\ell_h\| \leq C\|\mathcal{E}\| \leq Ch^{k+1/2}$. L^2 . Repeat the proof of Theorem 9.6 with $e := -\ell_h$; since $\tilde{f} = 0$, Lemma 7.19 reads $a_h(\ell_h, y) = \langle \ell_h, \partial_n y \rangle$ on Y_h . The six terms are bounded by $C\|z\|_{H^2}$ times: $h\|\ell_h\|$; $\|\ell_h\|_{\Gamma_h}(1 + hh_\Gamma^{-1/2})$ with $\|\ell_h\|_{\Gamma_h} \leq (h_\Gamma/\gamma)^{1/2}\|\ell_h\|$; 0 (no strip force); $\|\ell_h\|_{\Gamma_h}$; $\|\epsilon\|_{L^2(\partial R)} \leq Ch^{k+1}$; and the strip term $\|\ell_h\|_{S_h} \leq Ch_\Gamma(\|\ell_h\|_{\Gamma_h} + h_\Gamma \|\nabla \ell_h\|) \leq Ch_\Gamma h^{k+1/2}$ (Lemma 9.3(a)). Each is $\leq Ch^{k+1}\|z\|_{H^2}$, and Theorem 9.1 closes the argument. $\square$

Theorem 9.14 (Graded meshes). *Let CNS^* use flux-corrected data, $\gamma \in [\gamma_2, \gamma_{\max}]$, and assume (M0)–(M2), (D), $\tilde{u} \in H^{k+1}(\Omega_h) \cap W^{k+1, \infty}$ near ∂R , and (G_κ) with $\kappa \leq \kappa_*$ (κ_* depending only on k , shape regularity, Θ_0 , L , C_g). Then*

$$\|\tilde{u} - u_h\|_{L^2(\Omega)} \leq C(h_\Gamma^2 + h^{k+1}).$$

Proof. By Lemma 9.13 it suffices to bound $e := \tilde{u} - u_h^\sharp$, whose inputs ($\|\nabla e\| \leq C(h_\Gamma^{3/2} + h^k)$, $\|e\|_{\Gamma_h}, \|u_h^\sharp\|_{\Gamma_h} \leq C(h_\Gamma^2 + h_\Gamma^{1/2} h^k)$) are those of $\tilde{u} - u_h$ up to $Ch^{k+1/2}$. Run the proof of Theorem 9.6 with a local dual approximant: start from $y^0 := I_h \tilde{z} - L_0 \in V_h^0$ (L_0 the nodal lift of $I_h \tilde{z}|_{\Gamma_h}$, $\|\nabla L_0\| \leq Ch_\Gamma \|z\|_{H^2}$) and add $\alpha_E b_E n_E$ on each interior edge so that $\int_E (y^1 - \tilde{z}) \cdot n_E = 0$. On every $e \in \mathcal{E}_h^\Gamma$, $\int_e \tilde{z} \cdot n_e = -\int_e \partial_\tau \tilde{\zeta} = 0$ exactly, because both endpoints of e lie on Γ , where

TABLE 6. L^2 velocity errors, $\mu = 100$, $k = 4$ (notes/v6/l2/existing_l2.txt). Rates between consecutive runs of the full sequence.

N	CNS*, test A				CNS*, test B ₁			GS, test B ₁	
	$\ e\ $	rate	$\ e\ /h_T^2$		$\ e\ $	rate	$\ e\ /h_T^2$	$\ e\ $	$\ e\ \mu/h$
16	$4.58 \cdot 10^{-2}$	–	0.217		$1.32 \cdot 10^{-2}$	–	0.0625	$2.94 \cdot 10^{-2}$	1.950
32	$1.24 \cdot 10^{-2}$	2.03	0.208		$2.95 \cdot 10^{-3}$	2.05	0.0494	$1.58 \cdot 10^{-2}$	1.943
64	$3.21 \cdot 10^{-3}$	1.99	0.208		$7.87 \cdot 10^{-4}$	1.95	0.0510	$8.51 \cdot 10^{-3}$	2.015
128	$8.15 \cdot 10^{-4}$	1.99	0.209		$2.04 \cdot 10^{-4}$	1.97	0.0523	$4.44 \cdot 10^{-3}$	2.060
256	$2.05 \cdot 10^{-4}$	1.99	0.210		$5.18 \cdot 10^{-5}$	1.98	0.0531	$2.27 \cdot 10^{-3}$	2.085

$\zeta = 0$. Hence $\int_T \operatorname{div} y^1 = 0$ for every T , and $y_h := y^1 - \mathcal{L}(\operatorname{div} y^1) \in Y_h$ (Lemma 9.10) satisfies $\|\nabla(\tilde{z} - y_h)\|_T \leq C(h_T|\tilde{z}|_{H^2(\omega_T^2)} + \|\nabla L_0\|_{\omega_T^2})$ and $\|\partial_n y_h\|_{\Gamma_h} \leq C\|z\|_{H^2}$. Term 5 becomes $\langle \sigma\nu, g - \hat{g} \rangle_{\partial R} = O(h^{k+1})\|z\|_{H^2}$, terms 2–6 give $C(h_T^2 + h^{k+1} + h_T^{1/2}h^k)\|z\|_{H^2}$, and $|\text{term 1}| \leq C(\|h_T \nabla e\| + h_T \|\nabla e\|)\|z\|_{H^2}$.

Near zone. On elements with $\operatorname{dist}(T, \Gamma_h) < d_0 := C_g h_T / \kappa$, $h_T \leq C_g h_T$, contributing $\leq C h_T (h_T^{3/2} + h^k)$ to $\|h_T \nabla e\|$.

Far zone. Let $r(x) := \min(\operatorname{dist}(x, \Gamma_h), (L-1)/8)$. Points with $\operatorname{dist}(x, \partial R) \geq r(x)/16$ get the interior ball $B(x, r(x)/32)$; the others are within $r/16$ of some $x' \in \partial R$ and get the boundary ball $B(x', r(x)/8)$. In both cases the doubled ball is at distance $\geq \frac{11}{16}r(x) \geq d$ from Γ_h , d its radius. Besicovitch's covering theorem applied to $\{B(x, r(x)/32)\}$, and the Lipschitz continuity of r , give balls $B_i = B(x_i, d_i)$ covering $\{\operatorname{dist} \geq d_0\}$ whose doubled balls have bounded overlap and satisfy $h_T \leq C\kappa d_i$ on $2B_i$ by (G_κ) . So Theorem 9.12 applies with $\kappa_0 = C\kappa$, with (iii) guaranteed by the choice of $\hat{g}$, and gives $\|h_T \nabla e\|_{B_i} \leq C\kappa\|e\|_{2B_i} + C\kappa\|\eta\|_{2B_i} + C \max_{2B_i} h_T \|\nabla \eta\|_{2B_i}$. Summing with Lemma 9.11,

$$\|h_T \nabla e\| \leq C(h_T^{5/2} + h_T h^k + h^{k+1} + \kappa\|e\|_{L^2(\Omega)}).$$

Absorb. With $\|z\|_{H^2} \leq C_R\|e\|_{L^2(\Omega)}$, $h_T^{1/2}h^k \leq h_T^2 + h^{k+1}$ and $h_T h^k \leq h_T^2 + h^{2k}$, $\|e\|_{L^2(\Omega)} \leq C(h_T^2 + h^{k+1}) + C\kappa\|e\|_{L^2(\Omega)}$; absorb for $\kappa \leq \kappa_*$ and add $\|\ell_h\| \leq Ch^{k+1}$. $\square$

The theorem needs $\tilde{u} \in H^{k+1}$ up to ∂R , which is the smooth-solution hypothesis of Remark 2.5; for data-driven problems the term h^{k+1} degrades at the corners of R unless the mesh is graded there. The same statement holds for GS with $G = 0$ by the device $u'_h = u_h + \delta_R$ of Theorem 9.6; we have not written it out. With only global energy information the product $h h_T^{3/2}$ cannot be improved, since both factors are attained; the localisation of Theorem 9.12 is what removes it. We did not test graded meshes numerically (notes/v7/reg).

9.7. Computations. The stored runs of Section 16 contain the L^2 errors (Table 6). For CNS*, $\|e\|_{L^2}/h_T^2$ is constant to two digits from $N = 32$ on, independently of μ and of the pressure (B₁, B₁₀₀ and C agree). For GS, the L^2 rate is 2 on test A ($G = 0$) and 0.95–0.98 on P, B₁ and C, and $\|e\| \mu/h$ on B₁ tends to 2.085 at $N = 256$, close to $\|U\|_{L^2} = 2.112$ for the common wall pressure of B₁ and P.

Computation: notes/v6/l2/collect_l2.py; notes/v6/l2/l2_numerics.py; output notes/v6/l2/existing_l2.txt, leak.log, pair.log, slip.log.

10. THE PENALTY THRESHOLD SCALES WITH ρ

Proposition 10.1.

(a) Sufficiency. $\mu \geq \mu_0 = 4\kappa C_I^2 \rho$ implies coercivity (Lemma 4.2).

- (b) Necessity, with an exact certificate. Let z be a vertex of a straight part of the Nitsche boundary, and let ℓ_z be the length of its boundary edges. Suppose its star, dilated by the factor $1/\ell_z$, is congruent to one of the reference stars $\mathcal{S}_3$ or $\mathcal{S}'_4$ below. If $\mu/(h/\ell_z) < 9$, then N_h is indefinite on Z_h .

The same holds, by continuity, at a vertex of Γ_h with interior angle $\pi + O(h_\Gamma)$ whose star is a small perturbation of $\mathcal{S}_3$ or $\mathcal{S}'_4$. This uses that both reference stars are non-singular: every vertex, including the rim vertices, has non-collinear incident edges. The dimension of the divergence-free subspace therefore equals the generic count and is locally constant. This fails at the singular $m = 2$ configuration, which $(M1')$ excludes.

So, for meshes that contain such a boundary star at a vertex with $\rho \asymp h/\ell_z$, the condition $\mu_0 \asymp \rho$ is both necessary and sufficient. Theorem 10.2 and Corollary 10.4 below replace the star hypothesis by a condition on a single boundary triangle. The threshold is local: what is required is $\mu/h \gtrsim c/\ell_z$. The ρ -scaling is the price of Scott's normalisation of the penalty by the global h .

Proof of (b). (1) In two dimensions, $a(v, v)$ and $\langle \partial_n v, v \rangle$ are invariant under dilation, and $\|v\|_{L^2(\Gamma_h)}^2$ scales linearly.

- (2) A witness v on the reference star, extended by zero, lies in Z_h : it is continuous, vanishes on the star's outer chords and on ∂R , and is divergence-free.
- (3) It satisfies $N_h(v, v) = a(v, v) - 2\langle \partial_n v, v \rangle + (\mu\ell_z/h)\|v\|_{\text{ref}}^2$.
- (4) By the certificate below, $2B - A - 9C > 0$ on each reference star, so $N_h(v, v) < 0$ when $\mu\ell_z/h < 9$. $\square$

The exact certificate. The reference stars are fans at $O = (0, 0)$ with rational rim points:

- $\mathcal{S}_3$: rim points $(1, 0), (1, 1), (-1, 1), (-1, 0)$;
- $\mathcal{S}'_4$: rim points $(1, 0), (1, 1), (0, \frac{6}{5}), (-1, 1), (-1, 0)$.

The apex of $\mathcal{S}'_4$ is moved off the line $y = 1$. With apex $(0, 1)$ the rim vertex is singular, the kernel dimension jumps from 16 to 17, and the perturbation argument does not apply.

The Nitsche boundary is $\{y = 0\}$, and the rim chords carry zero Dirichlet data. The velocity is represented on each triangle by P_4 polynomials in global coordinates with rational coefficients. The computation is done in exact rational arithmetic (SymPy):

- Continuity across the spokes, vanishing on the rim, and pointwise $\text{div } v = 0$ are imposed as polynomial identities. This gives 80 and 110 linear constraints and kernel dimensions 14 and 16, the generic counts.
- The quadratic forms $A = \sum_T \int \frac{1}{2} Dv : Dv$, $B = \int (\partial_n v) \cdot v$ and $C = \int |v|^2$ are integrated exactly.
- A rational witness in the kernel satisfies $2B - A - 9C > 0$ *exactly* on each star. Its exact Rayleigh quotients $(2B - A)/C$ are 9.97854... ($\mathcal{S}_3$) and 9.26747... ($\mathcal{S}'_4$); these are lower bounds for the largest Rayleigh quotient λ^* of each star.

Computation: `code/lamstar_exact.py`, `code/lamstar_exact_v2.py`; output `logs/lamstar_exact_*.log`.

The witnesses are continuous piecewise- P_4 and divergence-free, so they lie in Z_h for every $k \geq 4$. The necessity statement therefore holds for all $k \geq 4$; for $k > 4$ the true local threshold is at least as large.

In floating point, the largest Rayleigh quotient λ^* is about 15 on uniform fans with $m = 3, 4$ and ranges over 14.8–38.5 on random 4-fans (`code/lamstar.py`). As a global check, the measured threshold $\gamma^* = \mu^* h_\Gamma / h$ stays in 18.1–20.1 as h/h_Γ varies over 3.3–44 on two mesh levels. Under refinement at fixed $h/h_\Gamma \approx 3.3$ it lies in 14.7–20.8, with increments that roughly halve per doubling (Table 14). Section 10.1 removes the reference-star hypothesis and explains the gap between the certified 9 and the measured 18–21.

10.1. Penalty necessity beyond two reference stars. Proposition 10.1(b) certifies indefiniteness of N_h on Z_h for $\mu\ell_z/h < 9$ only at vertices whose star is congruent to (or a perturbation of) $\mathcal{S}_3$ or $\mathcal{S}'_4$. We replace it by a statement about *every* boundary triangle, with an explicit closed-form constant and an exact certificate over a two-parameter family of shapes. We then explain the gap between the local constant and the global $\gamma^* \approx 18$ –21 of Table 14.

10.1.1. A witness supported in one boundary triangle. Let $T \in \mathcal{T}_h$ have an edge $e \subset \Gamma_h$, and let T not meet ∂R . Rotate, translate and dilate by $1/|e|$ so that $e = [(0, 0), (1, 0)]$, the fluid lies above and the apex is (a, b) , $b > 0$. Let $\lambda_0, \lambda_1, \lambda_2$ be the barycentric coordinates of the apex and of the base vertices $(0, 0)$, $(1, 0)$. For $c \in \mathbb{R}^3$ put

$$\psi_c := \lambda_1^2 \lambda_2^2 (c_0 \lambda_0 + c_1 \lambda_1 + c_2 \lambda_2) \in P_5, \quad v_c := \text{curl } \psi_c \in P_4^2.$$

Since ψ_c vanishes to second order on the two interior edges $\{\lambda_1 = 0\}$ and $\{\lambda_2 = 0\}$, v_c vanishes there. Extended by zero, v_c is continuous, zero on ∂R and pointwise divergence-free. So $v_c \in Z_h$ for every $k \geq 4$ and every mesh containing T . No condition on the vertices (singular or not), on the other triangles, or on the straightness of Γ_h beyond the edge e itself is needed.

Let $A_T = \int_T \frac{1}{2} Dv : Dv$, $B_T = \int_e (\partial_n v) \cdot v$ and $C_T = \int_e |v|^2$, with $n = (0, -1)$, as quadratic forms in c . They are computed exactly (SymPy) and, writing $\varpi := 4a^2 + 4b^2$, are

$$C_T = \frac{1}{1260 b^2} \begin{pmatrix} 2 & a-6 & -(a+5) \\ a-6 & 2(\varpi-3a+9) & \varpi-4a+15 \\ -(a+5) & \varpi-4a+15 & 2(\varpi-5a+10) \end{pmatrix},$$

$$B_T = \frac{1}{840 b^3} \begin{pmatrix} 12 & -(\varpi-6a+27) & -(\varpi-2a+25) \\ -(\varpi-6a+27) & 6(\varpi-3a+9) & 3(\varpi-4a+15) \\ -(\varpi-2a+25) & 3(\varpi-4a+15) & 6(\varpi-5a+10) \end{pmatrix},$$

and $A_T = (\alpha_{ij})/b^3$ with

$$\begin{aligned} 630 \alpha_{00} &= 3a^4 - 6a^3 + 6a^2b^2 + 12a^2 - 6ab^2 - 9a + 3b^4 + 4b^2 + 9, \\ 420 \alpha_{01} &= 3a^4 + 6a^2b^2 - 5a^2 + 6a + 3b^4 - 7b^2 - 9, \\ 420 \alpha_{02} &= 3a^4 - 12a^3 + 6a^2b^2 + 13a^2 - 12ab^2 - 8a + 3b^4 - b^2 - 5, \\ 70 \alpha_{11} &= 3a^4 - 3a^3 + 6a^2b^2 + 5a^2 - 3ab^2 - 3a + 3b^4 + 7b^2 + 3, \\ 140 \alpha_{12} &= a^4 - 2a^3 + 2a^2b^2 + 5a^2 - 2ab^2 - 4a + b^4 + 7b^2 + 5, \\ 70 \alpha_{22} &= 3a^4 - 9a^3 + 6a^2b^2 + 14a^2 - 9ab^2 - 10a + 3b^4 + 10b^2 + 5. \end{aligned}$$

$\det C_T = (3a^4 - 6a^3 + 7a^2b^2 + 3a^2 - 7ab^2 + 4b^4 + 5b^2)/(83\,349\,000 b^6) > 0$. (The forms were checked against an independent quadrature/finite-difference evaluation, relative difference $4.5 \cdot 10^{-9}$, and λ_T below against our penalty-threshold code on single-triangle patches of the Section 16 meshes: 10.48912 vs. 10.48912.)

Theorem 10.2 (Single-triangle necessity). *Let λ_T be the largest eigenvalue of the pencil $(2B_T - A_T, C_T)$; it depends only on the shape of T . If $\mu|e|/h < \lambda_T$, then N_h is indefinite on Z_h for every $k \geq 4$. Consequently every mesh satisfies the necessary condition*

$$\mu \geq h \max_{e \in \mathcal{E}_h^\Gamma} \frac{\lambda_{T_e}}{|e|}.$$

Proof. In two dimensions A and B are dilation invariant and C scales like $|e|$. So for the eigenvector c , $N_h(v_c, v_c) = A_T - 2B_T + (\mu|e|/h)C_T = (\mu|e|/h - \lambda_T) C_T(c) < 0$. $\square$

Proposition 10.3 (Exact certificates over shape families). *For $\varphi \in (0, \pi)$ let Q_φ be the set of apex positions (a, b) for which the apex angle is at least φ and $b \geq s \max(a, 1 - a)$, with $s = 0.087 < \tan 5^\circ$. This contains every triangle with apex angle $\geq \varphi$ and base angles $\geq 5^\circ$. Then*

$$\lambda_T > 9.3 \text{ on } Q_{60^\circ}, \quad \lambda_T > 4 \text{ on } Q_{45^\circ}, \quad \lambda_T > 1 \text{ on } Q_{35^\circ}.$$

The certificate proves only the three strict lower bounds. As numerical observations, not certified (`logs/lamstar_family_levels.log`): the first bound is nearly sharp, since a scan gives the minimum 9.31381 over Q_{60° , at the equilateral triangle; along the level sets of the apex angle the scanned minima are 6.07 (50°), 4.47 (45°), 2.83 (40°), 1.12 (35°) and -0.73 (30°), each at the isosceles shape.

Proof (computer-assisted, exact). Q_φ is the part of the disk through $(0, 0)$, $(1, 0)$ with inscribed angle φ that lies above the two lines. Cover a rational box containing Q_φ by dyadic sub-boxes. A box is discarded only if it provably misses Q_φ , using outward-rounded rational enclosures of the disk. On every other box, take a rational vector q (the eigenvector at the box centre, rounded to denominators ≤ 1000) and verify

$$\Phi_q(a, b) := 2520 b^3 q^\top (2B_T - A_T - cC_T)q > 0$$

on the whole box. Φ_q is a polynomial with rational coefficients (the denominators clear exactly), and the lower bound is evaluated monomial-wise in exact rational interval arithmetic. The boxes are split until this succeeds.

- $c = 9$ and $c = 9.3$ on Q_{60° : 323 and 716 boxes, depth ≤ 11 .
- $c = 4$ on Q_{45° : 259 boxes.
- $c = 1$ on Q_{35° : 297 boxes.

As negative controls, the same procedure fails for $c = 9.4$ on Q_{60° and for $c = 4.5$ on Q_{45° , as it must. $\square$

Computation: `code/lamstar_family.py bubble`; `code/lamstar_family.py certify`; output `logs/lamstar_family_bubble.log`, `logs/lamstar_family_certify.log`.

Corollary 10.4 (ρ -scaling is necessary on a broad class). *Let e_* be a shortest edge of Γ_h and suppose its triangle T_{e_*} has apex angle $\geq 35^\circ$ and base angles $\geq 5^\circ$, and does not meet ∂R (automatic for small h , since $\text{diam } T_{e_*} \leq Ch_\Gamma < L - 1$). Then coercivity of N_h on Z_h requires $\mu > \rho$. If the apex angle is $\geq 45^\circ$ (resp. $\geq 60^\circ$) it requires $\mu > 4\rho$ (resp. $\mu > 9.3\rho$). Together with Lemma 4.2, $\mu_0 \asymp \rho$ is necessary and sufficient on every such mesh, with no star-congruence or perturbation hypothesis.*

Remark 10.5 (What the single triangle cannot do). $\lambda_T < 0$ for tall triangles, for example every right triangle with the right angle at a base vertex and apex angle $\leq 30^\circ$ (numerically $\lambda_T \approx -0.2500$ at apex 30° , decreasing to -15.7 at 10°). A uniform statement under (M0)–(M2) alone would need witnesses on several triangles. In the Section 16 meshes such tall boundary triangles occur near the corner directions; at $N = 64$ the smallest value is $\lambda_T = -3.8$, with scaled apex about $(-0.08, 2.38)$ (`logs/lamstar_family_levels.log`). There the vertex stars and first-layer windows still have positive λ^* . Earlier versions conjectured that $\lambda_{\text{star}}^* \geq c(\Theta_0) > 0$ for every nonsingular boundary star. That conjecture is false (Proposition 10.6).

10.1.2. *Stars need not carry a witness.*

Proposition 10.6 (Exact counterexamples). *Let $\mathcal{S}^{(H)}$ be the three-triangle column star at $O = (0, 0)$ with rim points $(1, 0)$, $(1, H)$, $(0, H)$, $(-1, 0)$ and Nitsche boundary $\{y = 0\}$; it is the star of a boundary vertex of a mesh of columns of width 1 and height H split by the forward diagonal. It satisfies (M1), (M1') and (M2) with $\Theta = 1$ at every vertex; its minimum angle is $\arctan(1/H)$.*

In exact rational arithmetic (LDL^T without pivoting, all pivots positive), on the divergence-free space of continuous piecewise- P_k fields on the patch vanishing on the rim:

- (a) $k = 4$: $A - 2B - C \succ 0$ on $\mathcal{S}^{(5)}$ (minimum angle 11.3°) and $A - 2B - \frac{1}{2}C \succ 0$ on $\mathcal{S}^{(9/2)}$;
- (b) $k = 4$, the union of the two stars of one boundary edge: $A - 2B - \frac{1}{10}C \succ 0$ at $H = 6$ and $A - 2B - C \succ 0$ at $H = 8$;
- (c) the star fails for $k = 5$ at $H = 8$ and $k = 6$ at $H = 12$; on Clough–Tocher split stars it fails for $k = 2$ at $H = 2$ (macro minimum angle 26.6°), $k = 3$ at $H = 4$ and $k = 4$ at $H = 8$.

In each case no field supported in the patch is a witness for any $\mu \geq 0$. Wider windows still work on the same meshes: three consecutive stars at $H = 5$ give $2B - A - C > 0$ and four stars at $H = 8$ give $2B - A - \frac{1}{4}C > 0$, so $\mu \gtrsim \rho$ still holds there.

Proof. The forms and spaces are those of Section 10 ($A = \sum_T \int \frac{1}{2} Dv : Dv$, $B = \int (\partial_n v) \cdot v$ with $n = (0, -1)$, $C = \int |v|^2$), with continuity, vanishing on the rim and $\operatorname{div} v = 0$ imposed as polynomial identities over $\mathbb{Q}$ and the null space computed exactly. The code reproduces the witness quotient 9.978 of $\mathcal{S}_3$, and negative controls ($-\frac{6}{5}$ at $H = 5$, 0 on $\mathcal{S}_3$ and on $\mathcal{S}^{(4)}$, $-\frac{3}{20}$ for the $H = 6$ edge patch) fail as they must. $\square$

Computation: `notes/v6/penalty/certs.py`; `notes/v6/penalty/exact.py`; output `notes/v6/penalty/cert_all_k.log`, `cert_edge6.log`, `cert_high.log`.

The conclusions persist under small perturbations, including bending the boundary: for fixed combinatorics, all vertices of these patches are non-singular, so the dimension of the divergence-free space equals the generic count and the space depends continuously on the vertex positions (as in step 4 of Proposition 12.10); the strict positivity $A - 2B - cC \succ 0$ is then an open condition. A global (M0)–(M2) mesh containing such stars is constructed in Section 10.1.4; on it the threshold still scales like ρ . The counterexamples refute $\lambda_{\text{star}}^* \geq c(\Theta_0)$ as stated, because the minimum angle there is small. They do *not* refute a bound $c(\Theta_0, \theta_{\min})$ for $\theta_{\min} \gtrsim 20^\circ$: adversarial floating-point searches found star positivity failing at minimum angle 15° and never at 20° .

10.1.3. *Necessity from a locally quasi-uniform region, with explicit constants.* The failure of star witnesses shows that necessity must come from wider patches. The witness below is the curl of the Argyris interpolant of a fixed profile of width Y ; on the exact half-plane its Rayleigh quotient is $Y(2B - A)/C = 15.487$. Here θ is the minimum angle of $\mathcal{T}_h$, and for $v \in Z_h$

$$N_h(v, v) = \frac{C(v)}{Y} \left[\frac{\mu Y}{h} - \mathcal{Q}(v) \right], \quad \mathcal{Q}(v) := \frac{Y(2B(v) - A(v))}{C(v)}.$$

Profile. $g(t) = t(1-t)^6$ for $-\frac{1}{8} \leq t \leq 1$, $g = 0$ for $t \geq 1$; $\phi(s) = (1-s^2)^6$ on $[-1, 1]$, 0 outside. Both are C^5 with piecewise-polynomial, bounded sixth derivative. With $X = 2Y$,

$$F(x, y) = Y \phi(x/X) g(y/Y), \quad w = \operatorname{curl} F = (F_y, -F_x),$$

in coordinates in which e lies on $\{y = 0\}$, $x_0 = 0$, and the fluid is locally $\{y > 0\}$. On the part of Ω_h below $y = -Y/8$ (the far side of P_h) set $F := 0$. That part and the part near x_0 are separated by P_h , so no triangle meets both (Lemma 10.8).

Scaled derivative bounds. (`notes/v6/consts/argyris_consts.py`, exact critical points). Let $m_j := Y^{j-1} \sup |D^j F|$ over $\{y \geq -Y/8\}$. Bounding each partial derivative termwise, with $\max_{|u|=1} |u_1|^i |u_2|^{j-i}$ in place of the monomials,

$$m_1 \leq 3.6635, \quad m_2 \leq 32.187, \quad m_3 \leq 214.88, \quad m_6 \leq 18918.2.$$

Half-plane values. (exact, SymPy; $Y = 1$): $\hat{A} = \frac{4717730594816}{558289907175} = 8.45032$, $\hat{B} = \frac{201326592}{16900975} = 11.91213$, $\hat{C} = \frac{16777216}{16900975} = 0.992677$. So $\hat{Q} = (2\hat{B} - \hat{A})/\hat{C} = 15.48734$, which agrees with the independent value 15.487 of `notes/v6/penalty/continuum.check.py`.

Lemma 10.7 (Explicit local bound). *Let T be a triangle with minimum angle $\geq \theta$ and diameter h_T , and let $f \in C^5(T)$ have $D^5 f$ Lipschitz, with $\text{ess sup}_T |D^6 f| \leq M_6$. Let I be the Argyris (C^1 - P_5) interpolant. Its DOFs are the values, gradients and Hessians at the vertices and the unit-normal derivatives at the edge midpoints. Then on T , for $j = 1, 2$,*

$$|D^j(f - If)| \leq E_j(\theta) M_6 h_T^{6-j},$$

where

$$E_j(\theta) = \frac{1}{(6-j)!} + \left(\frac{\sqrt{2}}{\sin \theta s(\theta)} \right)^j \left[P_j + \left(\frac{1}{60} + 2c_H \right) N_{0,j} + \left(\frac{1}{3840} + \frac{\sqrt{2}c_H}{2\sin \theta} \right) (N_{1,j} + N_{2,j}) \right].$$

Here $s(\theta) = \min(\sqrt{3}/2, \sin 2\theta)$ and $c_H = \frac{29}{1920}$. The reference constants are

$$P_1 \leq 0.19597, \quad P_2 \leq 1.40308,$$

$$(N_{0,1}, N_{1,1}, N_{2,1}) \leq (1.8857, 3.7713, 3.7713), \quad (N_{0,2}, N_{1,2}, N_{2,2}) \leq (16.866, 33.731, 33.731).$$

Numerically: $E_1 = 17.9, 6.54, 3.37, 1.47, 0.933, 0.724$ and $E_2 = 3606, 597, 179, 39.0, 17.3, 10.9$ for $\theta = 10^\circ, 15^\circ, 20^\circ, 30^\circ, 45^\circ, 60^\circ$.

Proof. 1. *Taylor.* Let a_0 be the vertex with the largest angle $\alpha \in [\pi/3, \pi - 2\theta]$, so $\sin \alpha \geq s(\theta)$.

Let p be the degree-5 Taylor polynomial of f at a_0 and $r = f - p$. Since $Ip = p$, $f - If = r - Ir$. Along each segment $[a_0, x] \subset T$, $D^5 f$ is Lipschitz, so the integral remainder gives

$$|D^m r(x)| \leq M_6 |x - a_0|^{6-m} / (6-m)! \leq M_6 h_T^{6-m} / (6-m),$$

and all derivatives of r of order ≤ 5 vanish at a_0 .

2. *Reference map.* Let $\Phi(\hat{x}) = J\hat{x} + a_0$ map $\hat{T} = \text{conv}\{(0,0), (1,0), (0,1)\}$ onto T . The columns of J are the edges E_1, E_2 at a_0 , of lengths ℓ_1, ℓ_2 . By the law of sines, $h_T \sin \theta \leq \ell_i \leq h_T$, and the longest edge E_0 (opposite a_0) has length h_T . Hence

$$\sigma_{\max}(J) \leq \sqrt{2}h_T, \quad \|J^{-1}\| = \sigma_{\max}/\det J \leq \frac{\sqrt{\ell_1^{-2} + \ell_2^{-2}}}{\sin \alpha} \leq \frac{\sqrt{2}}{h_T \sin \theta s(\theta)}.$$

3. *Pulled-back interpolant.* Let $\hat{q} = (Ir) \circ \Phi \in P_5$. Its vertex values, gradients and Hessians are those of $\hat{r} = r \circ \Phi$, because these DOF sets are affine-equivalent. Only the edge DOFs differ. On reference edge $\hat{e}$ with edge vector $\hat{t}$ (the reference DOF being $\nabla \hat{q}(\hat{m}) \cdot \hat{\nu}$, with $\hat{\nu} \in \{(1,1), (-1,0), (0,-1)\}$), write $J^{-1}n_E = a\hat{\nu} + b\hat{t}$. Then $\gamma_E := \nabla \hat{q}(\hat{m}) \cdot \hat{\nu} = [n_E \cdot \nabla r(m_E) - b(Hf_E)'(\frac{1}{2})]/a$. Here $f_E(s) = \hat{r}(\hat{p} + s\hat{t})$, and Hf_E is its quintic Hermite interpolant from f, f', f'' at $s = 0, 1$, which is the restriction of $\hat{q}$. Exactly, $(Hf)'(\frac{1}{2}) = \frac{15}{8}(f_1 - f_0) - \frac{7}{16}(f'_0 + f'_1) + \frac{1}{32}(f''_1 - f''_0)$.

4. *Coefficient bounds.* Since $\det(J^{-1}n_E, \hat{t}) = \pm \ell_E / \det J$,

$$|a|^{-1} = \det J |\det(\hat{\nu}, \hat{t})| / \ell_E, \quad |b/a| \leq |\hat{\nu}| \sigma_{\max}(J) / \ell_E.$$

This gives:

- on the hypotenuse (E_0): $|a|^{-1} \leq 2h_T$ and $|b/a| \leq 2$;
- on the legs: $|a|^{-1} \leq h_T$ and $|b/a| \leq \sqrt{2}/\sin \theta$.

With $f_E^{(m)} = D^m r[E^m]$ and step 1:

- $|\gamma_{E_0}| \leq M_6 h_T^6 (\frac{2}{120} + 2c_H)$;
- $|\gamma_{E_{1,2}}| \leq M_6 h_T^6 (\frac{2^{-5}}{120} + \frac{\sqrt{2}}{\sin \theta} \frac{c_H}{2})$, since the data at a_0 vanish.

The vertex DOFs at a_1, a_2 are $D^m r[E_{i_1}, \dots, E_{i_m}]$, of size $\leq M_6 h_T^6 / (6-m)!$.

5. *Reference basis.* Expand $\hat{q}$ in the exact reference Argyris basis (21×21 inverse over $\mathbb{Q}$). Bound each basis function's partial derivatives of order j on $\hat{T}$ by the largest absolute Bernstein coefficient, which is a rigorous sup bound. Combine them into operator-norm bounds via the Frobenius norm. This gives $|\hat{D}^j \hat{q}| \leq M_6 h_T^6 [P_j + \dots]$, the bracket in E_j .
6. *Back to T .* $|D^j(Ir)| \leq \|J^{-1}\|^j |\hat{D}^j \hat{q}|$. Add $|D^j r|$ from step 1. $\square$

Check (`notes/v6/consts/validate_interp.py`, (NUMERICAL)). Physical Argyris interpolation of F on 600 random triangles ($\theta \in [10^\circ, 53^\circ]$, $h_T \in [0.01, 0.2]$, random positions in $\text{supp } F$): the ratio of actual error to the bound is at most $1.8 \cdot 10^{-5}$ ($j = 1$) and $1.4 \cdot 10^{-5}$ ($j = 2$). The bound holds, and is very loose.

Lemma 10.8 (Local geometry of Γ_h). *Suppose every boundary edge meeting $B(x_0, 3Y)$ has length $\leq \kappa Y$, with $\kappa \leq 1$ and $Y \leq 10^{-2}$. Put $X' = (2 + \kappa)Y$, $\omega_X = \arcsin(X'/(1 - \kappa^2 Y^2/4))$,*

$$\delta = \frac{1}{4}\kappa^2 Y^2 + \frac{1}{2}\omega_X^2, \quad \sigma = \tan(\omega_X + \arcsin(\kappa Y/2)).$$

Then on $|x| \leq X'$ the part of Γ_h near x_0 is a graph $y = \varpi(x)$ with $-\delta \leq \varpi \leq 0$ and $|\varpi'| \leq \sigma$, and $\Omega_h \cap \{|x| \leq X', y > -Y/8\} = \{y > \varpi(x)\}$ (this uses $\delta \leq Y/8$, true for $Y \leq 10^{-2}$).

Proof. • $\varpi \leq 0$. P_h is convex and e is one of its edges, so P_h lies in $\{y \leq 0\}$.

- *Polar description.* Let $O = (0, -d)$ be the centre. A boundary point on a chord of length $\ell' \leq \kappa Y$ is $O + r(\sin \omega, \cos \omega)$, with $r \geq \sqrt{1 - \ell'^2/4} \geq 1 - \ell'^2/4$.
- *Depth.* So $y \geq -\ell'^2/4 - (1 - \cos \omega)$, while $|x| \leq X'$ forces $|\omega| \leq \omega_X$.
- *Slope.* A chord between polar angles ω_1, ω_2 has direction angle $(\omega_1 + \omega_2)/2$, and $|\omega_2 - \omega_1| \leq 2 \arcsin(\ell'/2)$.
- *Fluid side.* The region between ϖ and $y = -Y/8$ lies in P_h by convexity. $\square$

Theorem 10.9 (Theorem A with explicit constants). *Let $e \in \mathcal{E}_h^\Gamma$ have midpoint x_0 , let $0 < Y \leq Y_0 := 10^{-6}$ with $4Y < L - 1$, let $0 < \kappa \leq \kappa_0(\theta)$ (table below), and suppose every $T \in \mathcal{T}_h$ meeting $B(x_0, 3Y)$ has $\text{diam } T \leq \kappa Y$. Let $v := \text{curl } IF$, with I the global Argyris interpolant. Then $v \in Z_h$ for every $k \geq 4$ and $\mathcal{Q}(v) \geq 15$. Hence N_h is indefinite on Z_h whenever $\mu < 15h/Y$. The values of $\kappa_0(\theta)$ are*

θ	10°	15°	20°	30°	45°	60°
$\kappa_0(\theta)$	0.00379	0.00594	0.00803	0.0117	0.0143	0.0161

(bisection roots of $\mathcal{Q}_{\text{lb}} = 15$ below, rounded down; the constants are computed in exact rational arithmetic and the final assembly in floating point, with rounding far below the stated digits).

Proof. 1. *Membership in Z_h .*

- IF is globally C^1 and piecewise P_5 , so v is continuous, piecewise $P_4 \subset P_k$, and pointwise divergence-free.
- $IF|_T = 0$ on every T not meeting $\text{supp } F \cap \overline{\Omega_h} \subset \{|x| \leq 2Y, -\delta \leq y \leq Y\} \subset B(x_0, \sqrt{5}Y + \delta)$.
- So $\text{supp } v \subset S := \bigcup \{T : T \cap \text{supp } F \neq \emptyset\} \subset \{|x| \leq X', \varpi \leq y \leq (1 + \kappa)Y\}$, which misses ∂R .
- Every $T \subset S$ lies in $\{y \geq -\delta\}$, where F is given by the profile formula, so Lemma 10.7 applies with $M_6 = m_6 Y^{-5}$ and $h_T \leq \kappa Y$.

Write $e_F = F - IF$, $|S| \leq 2X'((1 + \kappa)Y + \delta)$ and $\Lambda := |\Gamma_h \cap S| \leq 2X'\sqrt{1 + \sigma^2}$.

2. *Continuum field on the true geometry.* By Lemma 10.8, $\Omega_h \supset \{0 < y < Y, |x| < X\}$, so $A_{\Omega_h}(w) = \hat{A} + A_{\text{sl}}$ with $0 \leq A_{\text{sl}} \leq 4 \sup |D^2 F|^2 \cdot 2X\delta = 16m_2^2\delta/Y$.

- *Boundary term.* Using $n ds = (\varpi', -1) dx$ and moving from $y = 0$ to $y = \varpi$: $|B_{\Gamma_h}(w) - \hat{B}| \leq \varepsilon_B := 4[\sigma m_1 m_2 + (\delta/Y)(m_1 m_3 + m_2^2)]$.
- *Trace term.* $|C_{\Gamma_h}(w) - Y\hat{C}| \leq \varepsilon_C := 4Y[m_1^2(\sqrt{1 + \sigma^2} - 1) + 2\sqrt{1 + \sigma^2} m_1 m_2 \delta/Y]$.

3. *Interpolation.* $v - w = \text{curl } e_F$, so $|\nabla(v - w)|_F = |D^2 e_F|_F \leq \sqrt{2}|D^2 e_F|$ and $|v - w| = |\nabla e_F|$. By Lemma 10.7, $|D^2 e_F| \leq E_2 m_6 \kappa^4 / Y$ and $|\nabla e_F| \leq E_1 m_6 \kappa^5$ on S . With $A(z)^{1/2} \leq \sqrt{2}\|\nabla z\|$:

$$A(v)^{1/2} \leq (\hat{A} + A_{\text{sl}})^{1/2} + \eta_A, \quad \eta_A = 2E_2 m_6 \kappa^4 \sqrt{|S|}/Y;$$

$$C(v)^{1/2} \leq (Y\hat{C} + \varepsilon_C)^{1/2} + \eta_C, \quad \eta_C = E_1 m_6 \kappa^5 \sqrt{\Lambda};$$

$$|B(v) - B_{\Gamma_h}(w)| \leq \varepsilon'_B := \frac{E_2 m_6 \kappa^4 \sqrt{\Lambda}}{Y} ((Y\hat{C} + \varepsilon_C)^{1/2} + \eta_C) + \frac{m_2 \sqrt{\Lambda}}{Y} \eta_C,$$

using $B(v) - B(w) = \langle \partial_n(v - w), v \rangle + \langle \partial_n w, v - w \rangle$.

4. *Assembly.*

$$\mathcal{Q}(v) \geq \mathcal{Q}_{\text{lb}} := \frac{Y[2(\hat{B} - \varepsilon_B - \varepsilon'_B) - ((\hat{A} + A_{\text{sl}})^{1/2} + \eta_A)^2]}{((Y\hat{C} + \varepsilon_C)^{1/2} + \eta_C)^2}.$$

Every error term is nondecreasing in κ and in Y . $\mathcal{Q}_{\text{lb}}$ is dimensionless: $|S|/Y^2$, Λ/Y , δ/Y and σ depend only on κ and Y through the formulas above. The table value $\kappa_0(\theta)$ is the bisection root of $\mathcal{Q}_{\text{lb}}(\kappa, \theta, Y_0) = 15$, rounded down (`notes/v6/consts/assemble.py`, `assemble.log`). A grid check below (κ_0, Y_0) confirms $\min \mathcal{Q}_{\text{lb}} \geq 15.0000$. As $\kappa, Y \rightarrow 0$, $\mathcal{Q}_{\text{lb}} \rightarrow 15.48734$. $\square$

Corollary 10.10 (ρ -scaling). *Assume (M0). Suppose that for some edge $e \in \mathcal{E}_h^\Gamma$, with midpoint x_0 , every triangle meeting $B(x_0, 3C_q|e|/\kappa_0)$ has diameter $\leq C_q|e|$, where $C_q \geq 1$ and $C_q|e|/\kappa_0 \leq Y_0$. Then*

$$\mu^* \geq \frac{15\kappa_0(\theta)}{C_q} \frac{h}{|e|} \geq \frac{15c_0\kappa_0(\theta)}{C_q} \rho,$$

and the factor c_0 can be dropped when e is a shortest edge of Γ_h .

Proof. Take $Y = C_q|e|/\kappa_0$ in Theorem 10.9. For the second inequality, $|e| \leq h_\Gamma \leq \min |e'|/c_0$ by (M0). $\square$

The smallness $Y_0 = 10^{-6}$ is crude but explicit: it forces $|e| \lesssim \kappa_0 Y_0 / C_q \approx 10^{-8}$, so the corollary is a qualitative statement about the scaling $\mu^* \gtrsim \rho$, not a practical threshold. At $Y_0 = 3 \cdot 10^{-6}$ the values of κ_0 drop by about 15%; for $Y \gtrsim 5 \cdot 10^{-6}$ the bound falls below 15 even as $\kappa \rightarrow 0$. The smallness comes almost entirely from the sliver and curvature terms A_{sl} and ε_B , which are $O(Y)$ with large profile constants.

Theorem 10.9 needs no inf-sup constant, no compactness and none of (M1), (M1'), (M2); the only analytic inputs are Taylor's formula, the exact reference Argyris basis and elementary triangle geometry. For Clough–Tocher refinements the same proof applies with the HCT reference basis (`notes/v6/penalty`), but we have not computed those constants.

How far the explicit κ_0 is from the truth. (NUMERICAL) (`realistic_kappa.py`, `.log`). On half-plane meshes (squares of side s split by a diagonal, optionally with random vertex perturbation $\pm s/4$), $\mathcal{Q}(\text{curl } IF)$ with $Y = 1$ is:

$\kappa = \text{diam } T/Y$	1.41	1.13	0.85	0.71	0.42	0.14	0.07
uniform (45°)	12.15	14.06	15.06	15.28	15.46	15.487	15.487

The perturbed meshes (min angle 9–27°) give 14.79 at $\kappa = 1.06$ and 15.28 at $\kappa = 0.88$. So the true threshold is $\kappa \approx 0.85$ –1, against the proved 0.004–0.016. The loss comes from the Taylor/sup-norm bound ($\approx 7 \cdot 10^4$ in E_j , entering as a fourth root) and from the $\sin \theta$ factors.

Graded meshes. The hypothesis of Theorem 10.9 is not implied by (M0)–(M2): shape regularity only gives $\text{diam } T \leq C(\theta)(|e| + \text{dist}(T, x_0))$ with $C(\theta) \geq 1$, and this growth is attained.

Proposition 10.11 (Obstruction; half-plane model). *Take the dyadic boundary layer:*

- row 0: unit squares on $y \in [0, 1]$, split by the forward diagonal;

- row $j \geq 1$: squares of side 2^j on $y \in [2^j - 1, 2^{j+1} - 1]$, each over two row- $(j-1)$ squares, cut into three triangles from the hanging bottom midpoint.

This mesh has minimum angle $\arctan \frac{1}{2} = 26.57^\circ$, equal boundary edges, boundary stars $\cong \mathcal{S}_3$, and $\Theta \geq \sin 45^\circ$ at every vertex; the worst vertex is a coarse-square corner, with consecutive angles 90° and 45° . For every x_0 on the boundary and every $Y > 0$, $B(x_0, 3Y)$ contains a triangle of diameter $\geq Y$. So the hypothesis of Theorem 10.9 fails for every $\kappa < 1$.

Proof. • $Y < 1$. The boundary triangle at x_0 has diameter ≥ 1 .

- $Y \geq 1$. The point $x_0 + (0, 2Y)$ lies in row j with $2^j - 1 \leq 2Y < 2^{j+1} - 1$. Its triangle has diameter $\geq 2^j > (2Y + 1)/2 \geq Y$. $\square$

The witness still works on graded meshes. (NUMERICAL) (`graded_check.py`, `graded_check3.py`). The boundary edge is 1, and "max κ " is the largest $\text{diam } T/Y$ over the support.

Y/h_Γ	1	1.5	2	3	4	8	16	32
dyadic (ratio 2)	12.36	14.82	15.28	15.44	15.46	15.485	15.488	15.488
max κ	1.41	1.49	1.12	0.75	1.12	1.12	1.12	1.12
uniform	12.36	14.82	15.28	15.44	15.47	15.486	15.487	15.487
triadic (ratio 3, 15.3°), $Y = 1, 2, 3, 9, 27$	12.36,	15.24,	15.42,	15.49,	15.48	(max κ up to 1.8)		

On the dyadic layer, $\mathcal{Q}$ tends to the continuum value *although* $\kappa \approx 1.1$ at every scale. The witness's quality is set by the fine cells near Γ_h . The coarse cells carry little energy, and the true interpolation error there is far below the worst-case bound. So on graded meshes the conclusion $\mu^* \gtrsim h/h_\Gamma$ appears to survive. Proving it this way needs a *sharp* interpolation bound on coarse cells, with $\kappa \sim 1$, which a Taylor/sup-norm argument cannot deliver. Two routes would close the general case:

- a computer-assisted, shape-uniform bound of the Argyris error functional for this specific F ;
- the half-plane-limit positivity of Theorem B in `notes/v6/penalty`, restricted to the explicit family $\{\text{curl } I_A F_Y\}$, which is compact under (M0).

Neither is needed: Theorem 10.13 below proves the scaling with a different witness.

What is gained unconditionally. $\mu^* \geq 15 h/Y_*$, where Y_* is the smallest $Y \leq Y_0$ such that some ball $B(x_0, 3Y)$ centred at an edge midpoint meets only triangles of diameter $\leq \kappa_0 Y$. Since the boundary triangle has diameter $\geq c_0 h_\Gamma$, $\kappa_0 Y_* \geq c_0 h_\Gamma$; equality up to constants holds if and only if about $3/\kappa_0$ quasi-uniform layers exist somewhere along Γ_h . A uniform necessity statement under (M0) alone, with non-practical constants, is Theorem 10.13 below.

10.1.4. *Necessity on every (M0) mesh.* Theorem 10.9 needs about $3/\kappa_0$ quasi-uniform layers near one boundary edge, and Proposition 10.11 shows that shape regularity alone does not provide them. The following witness needs no resolution at all. Its stream function is quadratic in the normal variable on every triangle that touches Γ_h , so the boundary forms carry no $O(1)$ interpolation error, and its non-polynomial part lives in two transition bands that carry a vanishing share of the energy (`notes/v7/penalty`).

Let θ be the minimum angle, $m_\theta := \lceil \pi/\theta \rceil$, $d_1 := (\sin \theta)^{-(m_\theta+1)}$ and $c_g := (\sin \theta)^{m_\theta+2}$.

Lemma 10.12 (Wall triangles and growth). *Assume shape regularity with minimum angle θ , $|e| \leq h_\Gamma$ on $\mathcal{E}_h^\Gamma$, and h_Γ so small that the exterior angle of P_h at each vertex is $< \theta$. Then (i) every T meeting Γ_h has $\text{diam } T \leq d_1 h_\Gamma$; (ii) every T meeting neither Γ_h nor ∂R has $\text{dist}(z, \Gamma_h) \geq c_g \text{diam } T$ for each of its vertices z .*

Proof. Edge-adjacent triangles have diameter ratio $\leq 1/\sin \theta$, and in a triangle the shortest edge is at least $\sin \theta$ times the longest. (i) A boundary vertex has fewer than $(\pi + \theta)/\theta$ triangles, each reached from the triangle of a boundary edge in at most m_θ steps. (ii) An interior vertex z of

T has at most $2\pi/\theta$ triangles, each reached from T in at most m_θ steps, so its edges at z are $\geq (\sin \theta)^{m_\theta+1} \text{diam } T$ and the star of z contains $B(z, c_g \text{diam } T)$, which misses Γ_h . $\square$

Profile. Let $S(t) = 35t^4 - 84t^5 + 70t^6 - 20t^7$ on $[0, 1]$ (C^3 smoothstep, $\|S^{(4)}\|_\infty = 840$), $S = 0$ for $t < 0$, $S = 1$ for $t > 1$. For $\Lambda \geq 1$ put $\beta := 1/\Lambda$, $T_1 := \Lambda^2$ and let $\omega(t)$ be 1 on $[0, 1]$, $1 - (1 - \beta)S(t - 1)$ on $[1, 2]$, β on $[2, T_1]$, $\beta(1 - S(t/T_1 - 1))$ on $[T_1, 2T_1]$ and 0 beyond. For $y_1 > 0$ let $q(y) = 1 - s \int_0^y \omega(\tau/y_1) d\tau$ with s fixed by $q(\infty) = 0$, and $\Psi(y) = \int_0^y q$. Then $\int_0^\infty \omega = \frac{3}{2}(1 + \Lambda - \beta)$, $\int_0^\infty \omega^2 \leq 2 + 2\beta^2 T_1 = 4$, and

$$\epsilon := sy_1 = \frac{2}{3(1 + \Lambda - \beta)} \leq \frac{2}{3\Lambda}, \quad \int_0^\infty q'^2 \leq 4s\epsilon.$$

Ψ is exactly quadratic on $(-\infty, y_1]$ and on $[2y_1, T_1 y_1]$, constant on $[2T_1 y_1, \infty)$, $\Psi \in C^5$ with $\Psi^{(5)}$ Lipschitz, and $|\Psi^{(6)}| \leq 840 sy_1^{-4}$ on $[y_1, 2y_1]$, $\leq 840 s\beta(T_1 y_1)^{-4}$ on $[T_1 y_1, 2T_1 y_1]$, $\Psi^{(6)} = 0$ elsewhere.

Witness. For a boundary edge e_* , use coordinates with $e_* \subset \{y = 0\}$, midpoint 0, $P_h \subset \{y \leq 0\}$. With the cut-offs $\Phi(t) = (1 - t^2)_+^6$ and $\chi \in C^5$, $\chi = 1$ on $(-\frac{1}{2}, 1]$, $\chi = 0$ on $(-\infty, -1] \cup [2, \infty)$ (the part $y < 0$ only separates the far side of P_h , as in Lemma 10.8), let

$$F(x, y) = \Phi(x/X)\chi(y/X)\Psi(y), \quad v := \text{curl } I_A F,$$

with I_A the global Argyris interpolant, and fix

$$y_1 := 2d_1 h_\Gamma, \quad K_A := 840 E_2(\theta)(3/c_g)^4, \quad K_1 := 2 + K_A \sqrt{40/c_g}, \quad \Lambda := \max(1, K_1^2),$$

with $E_2(\theta)$ from Lemma 10.7.

Theorem 10.13 (Penalty necessity on every (M0) mesh). *Assume (M0) with minimum angle θ and constant c_0 . There are $r(\theta)$ and $h_*(\theta, L) > 0$ such that, with $X := rh_\Gamma$ and $h \leq h_*$, $v \in Z_h$ for every $k \geq 4$ and*

$$\frac{2B(v) - A(v)}{C(v)} \geq \frac{s}{2} = \frac{\epsilon}{4d_1 h_\Gamma}, \quad \text{hence} \quad \mu^* \geq \frac{\epsilon}{4d_1} \frac{h}{h_\Gamma} \geq c_* \rho, \quad c_* := \frac{c_0}{6d_1(1 + \Lambda)}.$$

Neither (M1) nor (M2) is used, nor any quasi-uniformity or star or layer structure, and e_* may be any boundary edge. The same holds on Clough-Tocher refinements for every $k \geq 2$, modulo the standard HCT estimate $|f - I_{\text{HCT}} f|_{W^{2,\infty}(T)} \leq C_{\text{HCT}}(\theta) h_T^2 |f|_{W^{4,\infty}(T)}$.

Proof. All forms are per unit length of $\{|x| < X\}$ (divided by $\int \Phi^2$). The limit regime is: first fix $r = X/h_\Gamma$ large (the χ -band and the finite- X corrections depend only on r), then take h_Γ so small that $r^2 h_\Gamma$ is small (the curvature and sliver terms are of order $\epsilon r^2 h_\Gamma$). Every $o(1)$ below has an explicit bound in these quantities, which fixes r and h_* ; neither depends on c_0 except through the final conversion.

1. *Membership.* $I_A F$ is globally C^1 and piecewise P_5 , so v is continuous, piecewise P_4 and divergence free. By Lemma 10.12(ii) the triangles meeting $\text{supp } F \cap \overline{\Omega_h} \subset \{|x| \leq X, y \leq 2X\}$ have diameter $\leq 3X/c_g$, so $v = 0$ on ∂R once $3X/c_g < L - 1$.
2. *The wall.* Every triangle meeting Γ_h lies in $\{-\delta \leq y \leq d_1 h_\Gamma = y_1/2\}$ (Lemma 10.12(i); $\delta, \sigma = O(X^2)$, $O(X)$ are the depth and slope bounds of Lemma 10.8). There $F = \Phi(x/X)\Psi(y)$ with Ψ quadratic, so every term of $D^6 F$ carries at least four derivatives of Φ : $|D^6 F| \leq C_\Phi(sX^{-4} + X^{-5} + y_1 X^{-6})$. Lemma 10.7 gives $|D^j(F - I_A F)| \leq E_j(d_1 h_\Gamma)^{6-j} C_\Phi(sX^{-4} + X^{-5} + y_1 X^{-6})$, an error of relative size $O((h_\Gamma/X)^4) = O(r^{-4})$ against sh_Γ^{1-j} , $j = 1, 2$. So $B(v) = B_{\Gamma_h}(w) + o(1)s$ and $C(v) = C_{\Gamma_h}(w)(1 + o(1))$, $w := \text{curl } F$. As in step 2 of Theorem 10.9, $B_{\Gamma_h}(w) = s(1 + O(s\delta) + O(\sigma/(sX)))$ and $C_{\Gamma_h}(w) = 1 + O(s\delta + \sigma^2)$, and $s\delta \sim \epsilon r^2 h_\Gamma$, $\sigma/(sX) \sim h_\Gamma/\epsilon$ are $o(1)$.

3. *Bands.* A triangle meeting $[y_1, 2y_1]$ does not meet Γ_h ; by Lemma 10.12(ii) its diameter is $\leq 3y_1/c_g$ and it lies in $\{y \leq 5y_1/c_g\}$, so $|D^2(F - I_A F)| \leq K_A s(1 + o(1))$ there. The same holds on the second band with $y_1 \rightarrow T_1 y_1$, $s \rightarrow s\beta$; no triangle meets both since $T_1 \geq 2 + 3/c_g$. Elsewhere Ψ is quadratic and the error is $o(1)$; on the χ -region $[X, 2X]$ it costs $O(\Psi_\infty^2/X^3) = o(1)$ per unit length. With $A(\text{curl } z) \leq 4\|D^2 z\|^2$ and $\beta^2 T_1 = 1$,

$$A(\text{curl}(F - I_A F)) \leq 4K_A^2 s^2 (5y_1/c_g + \beta^2 5T_1 y_1/c_g) + o(1) = 40K_A^2 s\epsilon/c_g + o(1).$$

4. *Energy of w .* The separable identity $A(w) = 2 \int \Phi'^2 \int G'^2 + \int \Phi^2 \int G''^2 + \int \Phi''^2 \int G^2$ ($G = \chi\Psi$, $\Psi(0) = 0$) gives, per unit length, $\int q'^2 + o(1) \leq 4s\epsilon + o(1)$; the sliver $\{\varpi < y < 0\}$ adds $o(1)s$.
5. *Budget.* By Minkowski, $A(v) \leq s\epsilon K_1^2 + o(1)$, so $2B - A \geq 2s - s\epsilon K_1^2 - o(1)s \geq s(1 - o(1))$ because $\epsilon \leq 2/(3\Lambda) \leq K_1^{-2}$. With $C = 1 + o(1)$, $(2B - A)/C \geq s/2$, so $N_h(v, v) < 0$ whenever $\mu/h < s/2$; finally $h/h_\Gamma \geq c_0\rho$ by (M0).

For Clough–Tocher replace I_A by the HCT interpolant, which reproduces the quadratic Ψ ; the band bound becomes $C_{\text{HCT}}(3y_1/c_g)^2 \|\Psi^{(4)}\|$, the wall error is $O((h_\Gamma/X)^2)$, and v is continuous, piecewise P_2 on the refinement and divergence free. $\square$

The constants are explicit but astronomically small: $\log_{10}(c_*/c_0) \approx -67, -39, -23$ at $\theta = 20^\circ, 30^\circ, 45^\circ$, and the support height $2T_1 y_1$ is about $10^{75} h_\Gamma$ at 30° , so h_* is correspondingly tiny. The losses are the worst-case growth factor c_g^{-4} and the Taylor-based E_2 , both entering squared through Λ . The theorem settles the *scaling* $\mu^* \gtrsim \rho$ on every shape-regular mesh; it is not a usable threshold. The support has size $\leq C(\theta, c_0)h_\Gamma$, so bounded patches do suffice for necessity; what fails is small patches (Proposition 10.6).

Tall boundary triangles at every mesh size. On a layered boundary mesh, exact witnesses exist without any interpolation.

Proposition 10.14 (Stacked columns). *Let the half-plane mesh consist of the cells $[i, i+1] \times [jH, (j+1)H]$ ($H > 0$) split by the forward diagonal; every boundary star is $\mathcal{S}^{(H)}$ of Proposition 10.6, with apex angle $\arctan(1/H)$. Let Φ be the uniform cubic B-spline with knot spacing $d \geq 4H$ on vertical mesh lines, and G the C^1 piecewise quadratic with $G = y - y^2/(2H)$ on $[0, H]$, $G'' = -1/(8H)$ on $[H, 3H]$, $G'' = +1/(8H)$ on $[3H, 5H]$ and $G = 0$ beyond. Then $F = \Phi(x)G(y)$ is C^1 and piecewise P_5 on the mesh, so $v = \text{curl } F \in Z_h$ for every $k \geq 4$ exactly, and with $D := d/H$,*

$$H \cdot \frac{h_\Gamma(2B - A)}{C} = \frac{2265D^4 - 2800D^2 - 6944}{2416D^4} \geq 0.8538 \quad (D \geq 4).$$

Hence $\mu^* \geq 0.85 h/(Hh_\Gamma)$ on every mesh that contains this configuration over the witness support $(4d \times 5H)$, at every mesh size.

Proof. For $F = \Phi G$ with $G(0) = 0$ and compact support, $B = H^{-1} \int \Phi^2$, $C = \int \Phi^2$ and $A = 2 \int \Phi'^2 \int G'^2 + \int \Phi^2 \int G''^2 + \int \Phi''^2 \int G^2$ (the cross term integrates to zero by parts). The integrals are rational: $\int \Phi^2 = \frac{151d}{315}$, $\int \Phi'^2 = \frac{2}{3d}$, $\int \Phi''^2 = \frac{8}{3d^3}$, $\int G^2 = \frac{31H^3}{60}$, $\int G'^2 = \frac{5H}{12}$, $\int G''^2 = \frac{17}{16H}$. The expression is increasing in D and equals 0.8538 at $D = 4$. $\square$

Computation: `notes/v7/penalty/exact_columns.py`;
`notes/v7/referee_penalty/fe_check_columns.py`; output
`notes/v7/penalty/exact_columns.log`; `notes/v7/referee_penalty/fe_check_columns.log`
(curl F assembled by the production code: quotient agrees to 10^{-13} , discrete divergence 10^{-13}).

The proposition is a flat-model statement: exactness needs layer lines parallel to Γ_h . On the polar column meshes below the cells are not rectangles and F is not piecewise P_5 ; its transfer there, by a perturbation argument as for Proposition 10.6, is heuristic.

A global mesh containing the star counterexamples (NUMERICAL; `notes/v7/penalty/global_mesh.py`). Take P_h the regular N -gon with chord $\ell = 2 \sin(\pi/N)$; a first layer of radial columns of height $H\ell$ split on the forward diagonal (every boundary star is a perturbation of $\mathcal{S}^{(H)}$ of relative size $2\pi H/N$); then dyadic rings, each cell over two cells of the previous ring and cut into three triangles from the hanging midpoint; then a structured outer part. The code checks (M0) with $c_0 = 1$, (M1) (no singular vertex, every boundary vertex in three triangles) and (M2) ($\Theta_0 = 0.83$ and 0.77 at $N = 128, 256$ for $H = 5$; minimum angle 11.25°). Thresholds $\gamma^* = \mu^* h_\Gamma / h$ for fields supported in patches of consecutive stars, and globally (on Z_h , confirmed by a penalty sweep $r = 10^4, \dots, 10^8$ in the solver):

$H = 5$	1 star	2	3	4	8	16	layer 0	global
$N = 128$	none	0.432	1.110	1.459	2.096	2.624	3.053	3.0562
$N = 256$	none	0.497	1.177	1.520	2.121	2.641	3.127	3.1286

The single star carries no witness, as Proposition 10.6 certifies, while unions of two or more stars do. The global threshold is attained in the first layer, and it is the same to four digits for $L = 2.5, 5, 10$ ($\gamma^* = 3.0560$ – 3.0562 at $N = 128$, 3.1284 – 3.1286 at $N = 256$) while ρ ranges over 21.6 – 416 : $\mu^* = \gamma^* \rho$ scales exactly like ρ . This L -invariance is nearly automatic, since the mesh near the wall is the same for every L ; it shows that the threshold is set locally, as Theorem 10.13 predicts, and not more. For $H = 8$ (7.1°) there is no witness on 1–3 stars and $\gamma^* = 1.8527$ for every L . The star failure is certified only for $H = 5$ and $H = 8$.

The size of the threshold is governed by the first-layer height: on these meshes ($N = 128$, $L = 2.5$) $\gamma^* H = 16.0, 15.8, 15.6, 15.3, 14.8, 14.2$ for $H = 1, 2, 3, 5, 8, 12$, close to the half-plane constant 15.487 of Theorem 10.9, and the production meshes (layer height $\frac{3}{4}$, $\gamma^* \rightarrow 21.4$) give $\gamma^* H \approx 16$. So, numerically, $\mu^* \approx (15$ – $16) h / \delta_1$ with δ_1 the height of the first layer, and the decay of $c(\theta)$ with the apex angle in any bound $\mu^* \geq c(\theta) \rho$ is genuine. An explicit, moderate constant for general coarse meshes remains open.

Computation: `notes/v7/penalty/global_mesh.py`, `layer_witness.py`; output `notes/v7/penalty/global_mesh_H5_N128.log`, `global_mesh_H5_N256.log`, `global_mesh_H8_N128.log`, `global_mesh_Hscan.log`, `layer_witness.log`; `notes/v7/referee_penalty/global_r_check.log`.

10.1.5. *Why the certified constant is below the global threshold.* The global threshold is $\gamma^* = h_\Gamma \sup_{Z_h} (2B - A)/C$, where A is taken over Ω_h , B, C over Γ_h , and $h_\Gamma = \max |e|$. Every patch witness gives a lower bound. For the Section 16 meshes we computed the patch thresholds with the penalty code of Section 16 (bisection on the inertia of $A_\mu + rB^\top M^{-1}B$, fields vanishing outside the patch). We maximised over all boundary triangles and single-vertex stars. *winW* is the first layer restricted to W columns centred at $\theta = 0$, and the column J is the patch made of the first J full boundary layers, computed for $J \leq 2$ here (the log also has $J = 3, 4$, which change the values by at most 0.002). The *global* column is Table 14(a); for $N \leq 16$ the full mesh is $J = N/4 \leq 4$ layers and the log reproduces it.

N	triangle	star	win2	win4	win8	win16	$J = 1$	$J = 2$	global
8	6.13	13.71	13.91	14.65	–	–	14.68	14.74	14.74
16	8.93	16.58	16.89	17.91	17.94	–	17.95	18.09	18.09
32	10.08	17.48	17.82	19.54	19.69	19.69	19.69	19.85	19.85
64	10.49	17.70	18.03	20.29	20.68	20.69	20.69	20.80	20.80

Computation: `code/lamstar_family.py mesh`; output `logs/lamstar_family_mesh.log`.

The limit geometry. The meshes of Section 16 do not have equal chords. The square points are equispaced and projected radially, so $\max |e| / \min |e|$ grows from 1.43 ($N = 16$) to 1.97 ($N = 256$).

The longest chords are at the side midpoints $\theta = 0, \pi/2, \dots$. There the first layer, scaled by the chord, tends to columns $[i, i+1] \times [0, \frac{3}{4}]$ (layer height $(L-1)/2 = \frac{3}{4}$) split on the forward diagonal. The single-vertex star of that geometry is $\mathcal{S}_{3/4}$, with rim $(1, 0), (1, \frac{3}{4}), (0, \frac{3}{4}), (-1, 0)$. The general exact code (continuity, Dirichlet and divergence constraints over $\mathbb{Q}$, exact barycentric moments) gives:

	star	$W = 2$	$W = 4$	$W = 8$	$W = 16$	$W = 32$
1 layer	17.7064	17.9759	20.4872	21.1726	21.3358	21.3758
2 layers	–	18.1738	20.6781	21.2432	21.3675	21.3974
3 layers	–	18.1916	20.6811	21.2462	21.3708	–

Exact certificates in the limit geometry: $2B - A - 17C > 0$ on $\mathcal{S}_{3/4}$ (dimension 14, witness quotient 17.70640; also confirmed by `code/lamstar_exact.py`), and $2B - A - 20C > 0$ on the $W = 4$ window (dimension 41, witness quotient 20.48725).

Computation: `code/lamstar_family.py limit; lamstar_exact.run` on $\mathcal{S}_{3/4}$; output `logs/lamstar_family_limit.log`, `logs/lamstar_family_limitstar.log`.

The perturbation argument of Proposition 10.1(b) transfers these strict inequalities to the actual meshes near $\theta = 0$ for all sufficiently large N (non-explicit N_0). All patch vertices keep at least three distinct edge lines, or are Dirichlet corners where v and ∇v vanish, so the kernel dimension is locally constant. This gives $\gamma^*(N) > 20$ for $N \geq N_0$. The finite- N values in the first table agree: win4 is 20.29 at $N = 64$.

The gap between the certified 9 and the measured 18–21 is therefore accounted for, in order of size, by:

- (1) *Shape.* The reference stars $\mathcal{S}_3, \mathcal{S}'_4$ have layer height 1 per unit chord. The meshes have height $\frac{3}{4}$ at the longest chords, which is where γ^* is decided because γ is normalised by $\max |e|$. λ^* grows fast as the layer flattens: the exact witness quotients, which are lower bounds for λ^* , are 9.98 for $\mathcal{S}_3$ against 17.71 for $\mathcal{S}_{3/4}$.
- (2) *Collective boundary modes.* The optimal witness spreads over 4–8 boundary edges and adds about 3.7 ($17.7 \rightarrow 21.4$). Windows wider than about 16 edges add nothing.
- (3) *Depth.* In the limit geometry the second layer adds 0.20 ($W = 2$) and 0.19 ($W = 4$), decreasing to 0.02 at $W = 32$; the third layer adds 0.018 at $W = 2$ and at most 0.004 for $W \geq 4$. On the actual meshes the second layer adds 0.11–0.16 (compare $J = 1$ and $J = 2$). γ^* is a boundary-layer quantity (compare $J = 2$ with the global value).
- (4) *Finite N .* Away from the limit geometry (curvature, the varying chord and height along the layer) $\gamma^*(N)$ is lower. Richardson extrapolation of the window sequences (differences shrink by a factor of about 4 per doubling of W) gives $\gamma_\infty^* \approx 21.39$ (one layer) and 21.41 (two layers), so we expect $\gamma^*(N) \rightarrow \approx 21.4$ on this mesh family.

The limit. Scaled by the chord, the production mesh near the wall vertex at polar angle $\theta \in [0, \pi/4]$ converges to the rectangular lattice $\mathcal{L}_a$ (columns $[i, i+1]$, rows of height a , forward diagonals) with $a = a(\theta) = (L \sec \theta - 1)/(2 \cos^2 \theta)$, and the local penalty in edge units is $\gamma \cos^2 \theta$ (the other sides follow by the rotation symmetry). Let $\gamma_c(a)$ be the supremum of $(2B - A)/C$ over compactly supported, continuous, piecewise- P_4 , divergence-free fields on $\mathcal{L}_a$, and $\gamma_c(a, J, \phi)$ the same over ϕ -quasi-periodic fields on one column of J rows vanishing on row J . Put $\Gamma_\infty := \sup_\theta \gamma_c(a(\theta))/\cos^2 \theta$ and $\Gamma_\infty^{\text{cell}} := \sup_\theta \sup_{J, \phi} \gamma_c(a(\theta), J, \phi)/\cos^2 \theta$.

Proposition 10.15 (The limit of the threshold). $\Gamma_\infty \leq \Gamma_\infty^{\text{cell}}$ and $\liminf_{N \rightarrow \infty} \gamma^*(N) \geq \Gamma_\infty$ (proved). Modulo a discrete localisation inequality (LOC), $\limsup_N \gamma^*(N) \leq \Gamma_\infty^{\text{cell}}$; modulo a spline cut-off lemma (CUT), $\Gamma_\infty = \Gamma_\infty^{\text{cell}}$. Hence, modulo (LOC) and (CUT), $\gamma_\infty^* = \lim_N \gamma^*(N)$ exists and equals the periodic cell value.

Proof sketch. ($\Gamma_\infty \leq \Gamma_\infty^{\text{cell}}$.) A witness supported in W columns, periodised with period $W' \geq W$, is continuous and divergence free with the same forms per period; the discrete Fourier transform over $\mathbb{Z}_{W'}$ splits the supercell pencil into W' one-column pencils with phases $2\pi j/W'$. (liminf.) Take a lattice witness within ε of $\gamma_c(a(\theta_0))$ on a finite window; the corresponding patch of the production mesh converges to it, no vertex is singular, so the divergence-free space has constant dimension and depends continuously on the vertices (as in Proposition 10.1(b)). (limsup.) Window bounds hold uniformly along Γ by compactness in θ ; (LOC) asks that every $v \in Z_h$ splits as $v_0 + \sum_i v_i$ with $v_i \in Z_h$ supported in boundary windows, v_0 vanishing on Γ_h , at a relative cost ε in $A - 2B + \lambda C$ (an IMS localisation, exact for the boundary forms for smooth partitions; the missing step is to put $\chi_i v$ back into Z_h through a local quasi-interpolant of the stream function). Combined with coercivity at a high level (Lemma 4.2) by convexity in λ , it gives the bound. (Equality.) A Bloch witness's stream function is quasi-periodic after subtracting its constant value above row J ; its real part on W columns, cut off by (CUT) (a stable local basis for C^1 quintic splines, which exists on triangulations without singular vertices), has quotient tending to that of the Bloch witness as $W \rightarrow \infty$. Details: `notes/v7/nsconst`, Section 2. $\square$

NUMERICAL: the supremum is at $\theta = 0$ ($a = \frac{3}{4}$) and phase 0; $\gamma_c(\frac{3}{4}, J, 0) = 21.3860$ ($J = 1$), 21.4045 ($J = 2$) and 21.4080 ($J \geq 3$) after the exact $1/N$ curvature extrapolation. The global thresholds for fields in the first two layers, 19.851, 20.802, 21.165, 21.302, 21.358, 21.382, 21.394 for $N = 32, \dots, 2048$, extrapolate to 21.40452, the two-layer cell value; the $1/N$ coefficient 21.8 is predicted as 22.0 by a harmonic-well expansion in phase and angle plus the wall curvature. This also explains why Richardson extrapolation from $N \leq 64$ overshoots: the $1/N^2$ term is still large there.

Computation: `notes/v7/nsconst/cell_limit.py`; output `notes/v7/nsconst/run_table.log`, `run_theta.log`, `run_mesh.log`.

11. STRONG IMPOSITION, LOCKED VERTICES AND NEARLY SINGULAR STARS

The motivation for weak imposition in [34, 53] is the *zero-gradient lock*: at a polygon vertex a conforming divergence-free field that vanishes on both adjacent edges may be forced to have zero gradient, while the true gradient there is $\partial_n u \otimes n \neq 0$. This section analyses strongly imposed no-slip for the same Scott–Vogelius pair on the same inscribed polygon. The answer is sharper than the introduction of earlier versions suggested:

- under the paper's own mesh hypothesis (M2), strong imposition already has H^1 error $\asymp h_\Gamma^{3/2} + h^k$, the rate of Theorem 7.2 (Theorems 11.5 and 11.6);
- the lock is a property of wall vertices with only two triangles, which (M2) excludes. Each such vertex costs $h_\Gamma |\partial_n u(z)|$ in H^1 , and the rate is two-sided (Theorems 11.8 and 11.13);
- with three or more triangles the same loss appears continuously as a triangle at the wall vertex degenerates; the relevant parameter is the needle angle ε_z , not the Guzmán–Scott quantity $\Theta(z)$ (Section 11.5).

So the advantages of Nitsche's method on inscribed polygons are robustness to such meshes (Section 16.8: GS and CNS* are insensitive to locked vertices) and a smaller error constant (about 2.5 times smaller in our tests), not a better rate under (M2).

Prior art. The two-triangle lock and the statement that “there is no constraint for more than two triangles” (without proof or rate) are Scott's [53, Theorem 1]; Gjerde and Scott treat the two-triangle case [34, §3.4, App. B]. The scalar version of the lower bound of Theorem 11.6 is classical [9, 56]. For nearly singular vertices, pressure constraints that restore a shape-robust inf-sup constant are due to Gräßle, Bohne and Sauter [35], and spurious pressures with robust velocities, and postprocessing filters, to Park [51] and Bohne, Gräßle and Sauter [11]. Our Theorem 11.11 is essentially the boundary-vertex special case of the pressure-wired result, and our filter

(Theorem 11.15(iii)) is in the spirit of these works. What appears to be new is: the two-sided $h_\Gamma^{3/2}$ theory for strong imposition under (M2), the counting lower bound, the sharp two-sided characterisation of the divergence stability at locked wall vertices (Theorem 11.12), the resulting two-sided velocity rate with locks (Theorem 11.13, Corollary 11.14), and the needle-angle law of Section 11.5. In contrast to the interior nearly singular setting, where the velocity is robust, the loss here is an *approximation* loss caused by the zero trace, not a stability loss.

The method (S). Find $u_h \in V_h$ with $u_h|_{\partial R} = g_I$, $u_h|_{\Gamma_h} = 0$, and $p_h \in \Pi_h \cap L_0^2(\Omega_h)$ with

$$a_h(u_h, v) - (p_h, \operatorname{div} v) = (\tilde{f}, v) \quad \forall v \in V_h^0, \quad (q, \operatorname{div} u_h) = 0 \quad \forall q \in \Pi_h.$$

It needs the outer flux $m_R = \int_{\partial R} g_I \cdot \nu$ to vanish; this holds for the test data and for the flux-corrected data $\tilde{g}_I$ of Definition 2.2. Throughout, z is a vertex of Γ_h with triangles $T_1, \dots, T_m$ (counterclockwise), wall edges $e_1 \subset T_1$ and $e_2 \subset T_m$ with directions t_1, t_2 , interior edges $s_i = T_i \cap T_{i+1}$, and turning angle $\phi_z := \arcsin(|e_1|/2) + \arcsin(|e_2|/2) = \Theta_z - \pi$ (Lemma 7.7), so $ch_\Gamma \leq \phi_z \leq 1.1h_\Gamma$ by (M0). We write $a_z := |\partial_n u(z)|$; by Lemma 3.4, $\nabla \tilde{u}(z) = \partial_n u(z) \otimes n$.

11.1. Local structure at a wall vertex. For a divergence-free continuous piecewise- P_k field v vanishing on $e_1 \cup e_2$, put $G_i := \nabla v|_{T_i}(z)$. Then

$$\operatorname{tr} G_i = 0, \quad G_1 t_1 = 0, \quad G_m t_2 = 0, \quad (G_{i+1} - G_i) s_i = 0. \quad (11)$$

Let A_z be the set of tuples satisfying (11), and L_z the number of distinct lines among the edges at z .

Lemma 11.1 (Dimension of the attainable gradients). $\dim A_z = m + 1 - \min(3, L_z)$. At a wall vertex with distinct lines this is $\max(m - 2, 0)$; in particular $A_z = \{0\}$ (the lock) if and only if $m \leq 2$.

Proof. Put $H = J^{-1}G$ with J the rotation by $\pi/2$; trace-free G corresponds exactly to symmetric H ($\nabla \operatorname{curl} \psi = J D^2 \psi$). A symmetric K with $Ks = 0$ is a multiple of $s^\perp \otimes s^\perp$, so $H_1 = \alpha_1 t_1^\perp t_1^\perp$, $H_m = \alpha_m t_2^\perp t_2^\perp$ and $H_{i+1} - H_i = \beta_i s_i^\perp s_i^\perp$; the only remaining condition is the closure $\alpha_m t_2^\perp t_2^\perp - \alpha_1 t_1^\perp t_1^\perp - \sum_i \beta_i s_i^\perp s_i^\perp = 0$. For unit vectors at angles ψ_1, ψ_2, ψ_3 the determinant of $(\operatorname{vec} uu^\top)$ is $\pm \prod \sin(\psi_j - \psi_i)$, so the rank of the family $\{uu^\top\}$ is $\min(3, \#\text{lines})$. The map $(\alpha, \beta) \mapsto (H_i)$ is injective. $\square$

That the tuples of A_z are attained by actual P_k fields (with support in the star) was checked in exact rational arithmetic for $k = 4, 5, 6$ and $m \leq 5$ (notes/v6/strong/local-jets.py); nothing below uses it.

Lemma 11.2 (Θ at wall vertices). Let $\phi_z \leq \theta_{\min}/2$. Then $\Theta(z) = \sin \phi_z$ if $m = 2$, and

$$\Theta(z) \geq \min(\sin 2\theta_{\min}, \sin(\theta_{\min}/2)) \quad \text{if } m \geq 3.$$

So (M2) at the vertices of Γ_h is equivalent to (M1') for small h_Γ .

Proof. The angles at z sum to $\pi + \phi_z$. For $m = 2$ the only pair gives $\sin(\pi + \phi_z)$. For $m \geq 3$ every pair sum lies in $[2\theta_{\min}, \pi + \phi_z - \theta_{\min}]$. $\square$

Lemma 11.3 (The exact gradient is nearly attainable for $m \geq 3$). If $m \geq 3$ and all angles at z are $\geq \theta_{\min}$, there is $(G_i) \in A_z$ with $\max_i |G_i - \nabla \tilde{u}(z)| \leq C(\theta_{\min}) \phi_z a_z$. If $A_z = \{0\}$, the squared distance is $m a_z^2$.

Proof. In Hessian form $\nabla \tilde{u}(z)$ corresponds to $a n \otimes n$, and the chord normals are within ϕ_z of n . Set $\alpha_m = a$ and solve $x_1 t_1^\perp t_1^\perp + \beta_1 s_1^\perp s_1^\perp + \beta_2 s_2^\perp s_2^\perp = a(t_2^\perp t_2^\perp - t_1^\perp t_1^\perp)$, whose right side has size $\leq 2|a|\phi_z$; the lines of e_1, s_1, s_2 are separated by at least $\theta_{\min}/2$, so the 3×3 inverse is bounded by $C(\theta_{\min})$. Set the other $\beta_i = 0$. $\square$

Lemma 11.4 (Vertex gradients). *If $v|_e = 0$ on a chord e at z , then $|(\nabla v|_{T_e} - \nabla \tilde{u})(z) t_e| = a_z |e|/2$ exactly, for any field v , divergence-free or not.*

Proof. $\nabla v|_{T_e}(z) t_e = 0$ and $\nabla \tilde{u}(z) t_e = \partial_n u(z)(n \cdot t_e)$, with $|n \cdot t_e| = \sin(\arcsin(|e|/2))$. $\square$

This explains the observed vertex-gradient error $\asymp h$ on meshes with three triangles per wall vertex (Section 16.8).

11.2. Under (M2): the rate $h_\Gamma^{3/2}$, two-sided.

Theorem 11.5 (Upper bound under (M2)). *Assume (M0)–(M2), (D), $k \geq 4$ and $m_R = 0$. Then (S) is well posed, and with $p^\circ := \tilde{p} - |\Omega_h|^{-1} \int_{\Omega_h} \tilde{p}$,*

$$\|\nabla(\tilde{u} - u_h)\| + \|p^\circ - p_h\| \leq C(h^k + h_\Gamma^{3/2}).$$

No relation between h and h_Γ is needed.

Proof. (1) *Interpolant.* $w_h := I_h \tilde{u} - \sum_{x \in \mathcal{N}_\Gamma} \tilde{u}(x) \varphi_x$, with $\mathcal{N}_\Gamma$ the Lagrange nodes on Γ_h and φ_x the nodal basis functions. Then $w_h|_{\Gamma_h} = 0$ and $w_h|_{\partial R} = g_I$. By Lemma 3.3, $|\tilde{u}(x)| \leq Ch_\Gamma^2$; $\|\nabla \varphi_x\| \leq C$ on any shape-regular triangle (scale invariance in two dimensions); the supports overlap boundedly; and $\#\mathcal{N}_\Gamma \leq C/h_\Gamma$ by the lower bound $|e| \geq c_0 h_\Gamma$ of (M0). So $\|\nabla(\tilde{u} - w_h)\| \leq C(h^k + h_\Gamma^{3/2})$.

(2) *Divergence.* $\operatorname{div} w_h \in \Pi_h$, $\|\operatorname{div} w_h\| \leq \sqrt{2} \|\nabla(w_h - \tilde{u})\|$, and its mean is $|\Omega_h|^{-1} m_R = 0$.

(3) *Correction.* Lemma 4.5 gives $y \in V_h^0$ with $\operatorname{div} y = \operatorname{div} w_h$, $\|\nabla y\| \leq \beta^{-1} \|\operatorname{div} w_h\|$. Then $z := w_h - y$ is admissible and divergence-free, and $\|\nabla(\tilde{u} - z)\| \leq C(h^k + h_\Gamma^{3/2})$.

(4) *Consistency.* For $v \in V_h^0$, $a_h(\tilde{u}, v) = (-\Delta \tilde{u}, v) = (\tilde{f} - F - \nabla \tilde{p}, v)$, so $a_h(u_h - \tilde{u}, v) - (p_h - \tilde{p}, \operatorname{div} v) = (F, v)$ with $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$ (Lemma 3.3).

(5) *Velocity.* $\varepsilon := u_h - z \in V_h^0$ is divergence-free, so $a_h(\varepsilon, \varepsilon) = a_h(\tilde{u} - z, \varepsilon) + (F, \varepsilon)$ and Korn (Lemma 3.2) gives $\|\nabla \varepsilon\| \leq \kappa(2 \|\nabla(\tilde{u} - z)\| + Ch_\Gamma^2)$.

(6) *Pressure.* Let Π be the L^2 projection onto Π_h and $q := p_h - \Pi p^\circ$, which has mean zero. Lemma 4.5 gives $v \in V_h^0$ with $\operatorname{div} v = q$; since $(p^\circ - \Pi p^\circ, \operatorname{div} v) = 0$ and constants integrate to zero against $\operatorname{div} v$, $\|q\|^2 = (p_h - \tilde{p}, \operatorname{div} v) = a_h(u_h - \tilde{u}, v) - (F, v)$, so $\beta \|q\| \leq 2 \|\nabla(u_h - \tilde{u})\| + Ch_\Gamma^2$. Add $\|p^\circ - \Pi p^\circ\| \leq Ch^k$.

Well-posedness follows from Lemma 4.5 and Korn on the divergence-free subspace of V_h^0 . $\square$

With flux-corrected data add $-(m_R/8L^2)I_h(\chi x)$ as in Lemma 7.32; this costs $O(h^{\sigma_k})$. Without (M2), at a two-triangle vertex the inf-sup constant is $O(h_\Gamma)$ (Theorem 11.12), and step 3 only gives $h_\Gamma^{1/2}$, which matches the lock.

Theorem 11.6 (Lower bound). *Under (M0) and (M1), every $v \in H^1(\Omega_h)^2$ with $v|_{\Gamma_h} = 0$, divergence-free or not, satisfies*

$$\|\nabla(\tilde{u} - v)\| \geq c h_\Gamma^{3/2} \|\partial_n u\|_{L^2(\Gamma)} - C h_\Gamma^{5/2}.$$

Proof. (1) On each T_e , $|w|_{H^{1/2}(e)} \leq C(\theta_{\min})|w|_{H^1(T_e)}$: the one-dimensional Gagliardo seminorm is dilation invariant, and the trace theorem with Poincaré on the reference triangle applies. The T_e are distinct by (M1).

(2) On e let s be the coordinate from the midpoint and $\delta(s) := 1 - (1 - \ell^2/4 + s^2)^{1/2}$, the distance to Γ . Then $\tilde{u}|_e = \delta(s)a_e + r$ with $a_e := \partial_n u(m_e^*)$ and $|r'| \leq C\ell^2$, so $|r|_{H^{1/2}(e)} \leq C\ell^3$.

(3) $\delta(s) - \delta(t) = (t^2 - s^2)/(|x(t)| + |x(s)|)$ and $|x| \leq 1$, so $|\delta|_{H^{1/2}(e)}^2 \geq \frac{1}{4} \iint (s+t)^2 ds dt = \ell^4/24$.

- (4) Apply step 1 to $w = \tilde{u} - v$, whose trace on e is $\tilde{u}|_e$; with $(a - b)^2 \geq a^2/2 - b^2$ and a Riemann sum for $\sum_e \ell_e |a_e|^2$ (with $\ell_e \geq c_0 h_\Gamma$),

$$\|\nabla(\tilde{u} - v)\|^2 \geq c \sum_e (\ell_e^4 |a_e|^2 / 48 - C \ell_e^6) \geq c h_\Gamma^3 (\|\partial_n u\|_{L^2(\Gamma)}^2 - C h_\Gamma) - C h_\Gamma^5. \quad \square$$

Corollary 11.7 (Two-sided rate under (M2)). *Under (M0)–(M2), (D), $m_R = 0$, $h^k \leq h_\Gamma^{3/2}$ and $\partial_n u \not\equiv 0$ on Γ , strong imposition satisfies $ch_\Gamma^{3/2} \leq \|\nabla(\tilde{u} - u_h)\| \leq C h_\Gamma^{3/2}$.*

So, under the hypothesis used throughout this paper, strong imposition attains Scott’s expected rate, with the same exponent as CNS* (Theorem 7.2, Theorem 7.24). Theorem 11.5 is close to a corollary of the Guzmán–Scott inf-sup theory plus standard boundary approximation. The remark of [34] that the effect “will occur at the vertices of any polygonal domain” therefore fails under (M2); by Theorems 11.5 and 11.8, the strong-imposition meshes of [34] must have lost uniform (M2), either through a positive fraction of two-triangle wall vertices or through nearly singular wall stars (their star counts are not reported, so this is an inference).

11.3. Locked vertices: a counting lower bound. Let $\gamma_k^2 := \min\{\|1 - p\|_T^2 / |T| : p \in P_{k-1}(T), p(\text{a vertex}) = 0\}$; it is affine invariant and positive, and exactly $\gamma_k^2 = 4/(k^2(k+1)^2)$ for $k = 4, \dots, 8$ (exact computation). For a wall vertex z put

$$d_z^2 := \min_{(G_T) \in A_z} \sum_{T \ni z} |T| |\nabla \tilde{u}(z) - G_T|_F^2.$$

Theorem 11.8 (Counting lower bound). *Under (M0), (M1), for every divergence-free $v \in V_h$ vanishing on Γ_h ,*

$$\|\nabla(\tilde{u} - v)\| \geq \frac{\gamma_k}{\sqrt{2}} \left(\sum_z d_z^2 \right)^{1/2} - C h_\Gamma^{3/2}.$$

At a locked vertex ($m = 2$) $d_z^2 = |\text{star}(z)| a_z^2 \geq ch_\Gamma^2 a_z^2$, so the error is at least $ch_\Gamma (\sum_{z \in \mathcal{L}} a_z^2)^{1/2} - C h_\Gamma^{3/2}$, $\mathcal{L}$ the set of locked vertices. In particular: if $a_z \geq \delta$ on an arc A and the locked vertices fill a fraction $\geq \vartheta$ of A , the error is $\geq c\delta(\vartheta|A|)^{1/2} h_\Gamma^{1/2}$; n locked vertices with $a_z \geq \delta$ force an error $\geq c\delta\sqrt{n} h_\Gamma$, so a single lock already reduces the rate to 1. For $m \geq 3$, $d_z = O(h_\Gamma |\text{star}|^{1/2})$ (Lemma 11.3), which is harmless.

Proof. On each T with vertex z , $\nabla v|_T \in P_{k-1}$ componentwise, so $\|\nabla \tilde{u} - \nabla v\|_T \geq \gamma_k |T|^{1/2} |\nabla \tilde{u}(z) - \nabla v|_T(z)| - C h_T |T|^{1/2}$. The tuple $(\nabla v|_T(z))_T$ lies in A_z ; apply Minkowski over the star, and sum over z : a triangle has at most two vertices on Γ_h . The remainders sum to $C h_\Gamma^{3/2}$. $\square$

On the alternating mesh (a) of Section 16.8 the bound is positive from $N = 64$ on, $\gamma_4 (\sum d_z^2)^{1/2} = 0.119, 0.097, 0.084$ at $N = 64, 96, 128$ scales like $h^{1/2}$ to three digits, and the measured error is 3.0–3.15 times it (notes/v6/strong/lock_bound.log).

11.4. Locked vertices: stability and the matching upper bound. Hypothesis (H): (M0), (M1), $k \geq 4$, and (M2) at every vertex *except* vertices of Γ_h with two triangles (automatic at the others by Lemma 11.2). No condition on the number or arrangement of locked vertices; two locked neighbours may share the apex of their stars. Constants depend only on $k, \theta_{\min}, c_0, \Theta_0, L$, never on $h, h_\Gamma, \phi_z, \#\mathcal{L}$. Cited standard ingredients: the uniform continuous inf-sup constant (proof of Lemma 4.5); the P_2 Fortin operator with $\int_E \Pi_F w = \int_E w$ on edges, stable under shape regularity [32]; and the element bubble lemma of Scott and Vogelius [54] (for $q \in P_{k-1}(T)$ with zero mean vanishing at the vertices there is $v \in P_k(T)^2 \cap H_0^1(T)^2$ with $\text{div } v = q$, $\|\nabla v\| \leq C \|q\|$; checked exactly for $k = 3, \dots, 8$). None of them is applied on a locked star in a way that could depend on ϕ_z .

At $z \in \mathcal{L}$ let $T_1 = [z_-, z, w]$, $T_2 = [z, z_+, w]$, $s = [z, w]$, let λ_x be the hat function of vertex x , and $g_i := \nabla \lambda_w|_{T_i} = \hat{n}_i/H_i$ with $\hat{n}_i$ the unit normal of e_i into T_i and $H_i = \text{dist}(w, e_i) \asymp h_z := \text{diam}(T_1 \cup T_2)$. Since λ_w is linear on s from 0 to 1,

$$g_1 \cdot (w - z) = g_2 \cdot (w - z) = 1. \quad (12)$$

Let D_z be the 2×2 matrix with rows $g_1^\top, g_2^\top$.

- Lemma 11.9** (Locked star). (a) For every $v \in V_h^0$ there is a unique $c_v \in \mathbb{R}^2$ with $\nabla v|_{T_i}(z) = c_v \otimes g_i$; hence $(\text{div } v|_{T_1}(z), \text{div } v|_{T_2}(z)) = D_z c_v$, and $|c_v| \leq C \|\nabla v\|_{T_1}$.
(b) For $c \in \mathbb{R}^2$, $\delta \in \mathbb{R}$, $K_z(c, \delta) := c \lambda_w \lambda_z^2 + \delta n_s \lambda_z^2 \lambda_w^2$ (n_s the unit normal of s) lies in V_h^0 , is supported in $T_1 \cup T_2$, has $\nabla K_z = 0$ at every vertex other than z , $\nabla K_z|_{T_i}(z) = c \otimes g_i$, $\|\nabla K_z\| \leq C(|c| + |\delta|)$, and $\int_{T_1} \text{div } K_z = -\int_{T_2} \text{div } K_z = |s|(\frac{1}{12}c \cdot n_s + \frac{1}{30}\delta)$.
(c) $|\det D_z| = \sin \phi_z / (H_1 H_2)$.
(d) $D_z(w - z) = (1, 1)^\top$. So the vertex data (σ, σ) are realised by $\sigma(w - z) \lambda_w \lambda_z^2$, at cost $\leq C|\sigma|h_z$ and with zero flux through s .
(e) $|D_z^{-1}(1, 0)^\top| = H_1 / \sin \phi_z$ and $|g_1 - g_2| \leq C\phi_z/h_z$.

Proof. (a) $v = 0$ on e_1 , so $\nabla v|_{T_1}(z) = c \otimes g_1$; likewise on T_2 ; continuity across s and (12) give equal c . The bound uses the inverse inequality and $|g_1| \geq c/h_z$. (b) $\lambda_w \lambda_z$ vanishes outside $T_1 \cup T_2$; $\lambda_w = 0$ on e_1, e_2 and $\lambda_z = 0$ on $[z_\pm, w]$; at $x \neq z$, $\lambda_z(x) = 0 = \nabla(\lambda_z^2)(x)$. On s (parameter $\lambda_w = r$), $\int_s \lambda_w \lambda_z^2 = |s|/12$ and $\int_s \lambda_z^2 \lambda_w^2 = |s|/30$. (c) $g_1 \times g_2 = \sin \angle(\hat{n}_1, \hat{n}_2)/(H_1 H_2)$, and consecutive chord normals make the angle ϕ_z . (d) is (12), and $w - z \parallel s$ gives zero flux. (e) $D_z^{-1}(1, 0)^\top = (\det D_z)^{-1} g_2^\perp$; and with the angles $\alpha_1 + \alpha_2 = \pi + \phi_z$ at z and $H_i = |s| \sin \alpha_i$, $|H_1 - H_2| \leq |s|\phi_z$. $\square$

The key device is the exact identity (d): the radial field $(w - z) \lambda_w \lambda_z^2$ produces equal vertex divergences at a locked vertex at ϕ -independent cost.

Lemma 11.10 (Vertex lemma). Let $y \notin \mathcal{L}$ be a vertex with triangles $T_1, \dots, T_m$ and $\Theta(y) \geq \Theta_0$. For every $r \in \mathbb{R}^m$ there is $v_y \in V_h^0$ supported in the star of y with $\text{div } v_y|_{T_j}(y) = r_j$, $\nabla v_y(x) = 0$ at every vertex $x \neq y$, $\int_T \text{div } v_y = 0$ for every T , and $\|\nabla v_y\| \leq Ch_y |r|$.

Proof. Let $v_y = \lambda_y^2 \ell + \sum_E \delta_E n_E \lambda_y^2 \lambda_{x_E}^2$ with $\ell = \sum_j c_j \lambda_{x_j}$ over the neighbours ($c_j = 0$ on boundary edges) and the sum over the edges $E = [y, x_E]$ of the star not on $\partial\Omega_h$. The factor λ_y^2 gives the support and the vanishing gradients elsewhere. The tuples $(\nabla \ell|_{T_j})_j$ are all vertex-gradient tuples of continuous fields vanishing at y and on the boundary edges through y ; their divergence-free subset has dimension $m + 3 - \min(3, L_y)$ (interior) or $m + 1 - \min(3, L_y)$ (boundary) by the argument of Lemma 11.1, so $c \mapsto (\text{div } \ell|_{T_j})_j$ has rank $m - 3 + \min(3, L_y)$, onto iff $L_y \geq 3$, which $\Theta(y) \geq \Theta_0$ implies. Uniformity: the normalised admissible stars form a compact set on which the least singular value is continuous and positive. The edge bubbles move flux $\delta_E |E|/30$ between the two triangles at E ; the triangles of the star form a cycle or a path, so every zero-sum vector of triangle means is matched. $\square$

Theorem 11.11 (Uniform constrained inf-sup). Assume (H). Let

$$Q_{\text{lock}} := \{q \in \Pi_h \cap L_0^2(\Omega_h) : q|_{T_1(z)}(z) = q|_{T_2(z)}(z) \text{ for all } z \in \mathcal{L}\}.$$

For every $q \in Q_{\text{lock}}$ there is $y \in V_h^0$ with $\text{div } y = q$ and $\|\nabla y\| \leq C\|q\|$, with C independent of $h, h_\Gamma, (\phi_z)$ and of the number and arrangement of locked vertices. In particular, if $q|_{T_i(z)}(z) = 0$ for $i = 1, 2$ at every $z \in \mathcal{L}$, such y exists.

Proof. A (means). Take $w \in H_0^1(\Omega_h)^2$ with $\text{div } w = q$, $\|\nabla w\| \leq C\|q\|$, and its P_2 Fortin interpolant $v_A \in V_h^0$ ($\int_T \text{div } v_A = \int_T q$). Put $v'_A := v_A - \sum_{z \in \mathcal{L}} K_z(c_{v_A, z}, -\frac{5}{2}c_{v_A, z} \cdot n_{s_z})$; each subtracted field is flux-neutral on every triangle (Lemma 11.9(b)), kills the vertex gradient of v_A at its own z , and

leaves the other vertices alone. A triangle lies in at most two locked stars, so $\|\nabla v'_A\| \leq C\|\nabla v_A\|$. Then $q_1 := q - \operatorname{div} v'_A$ has zero element means and $q_1|_{T_i(z)}(z) = q|_{T_i(z)}(z) =: \sigma_z$ at locked z . B (locked vertices). $v_B := \sum_{z \in \mathcal{L}} \sigma_z (w_z - z) \lambda_{w_z} \lambda_z^2$ matches (σ_z, σ_z) by Lemma 11.9(d), with zero element means and $\|\nabla v_B\| \leq C\|q_1\|$. C (other vertices). Lemma 11.10 removes the vertex values of $q_2 := q_1 - \operatorname{div} v_B$ at all other vertices, keeping means and the locked values. D (elements). The remainder has zero element means and vanishes at all vertices; the element bubble lemma finishes. The only place a locked vertex enters is Lemma 11.9(a),(b),(d), whose constants do not involve ϕ_z . $\square$

Theorem 11.12 (Sharp stability without the constraint). *Assume (H). For $q \in \Pi_h \cap L_0^2$ put $\ell_z(q) := q|_{T_1(z)}(z) - q|_{T_2(z)}(z)$. Then*

$$cN(q) \leq \min\{\|\nabla v\| : v \in V_h^0, \operatorname{div} v = q\} \leq C N(q), \quad N(q) := \|q\| + \left(\sum_{z \in \mathcal{L}} \left(\frac{h_z |\ell_z(q)|}{\sin \phi_z} \right)^2 \right)^{1/2}.$$

So $\operatorname{div} V_h^0 = \Pi_h \cap L_0^2$ always, and the inf-sup constant satisfies $c \min(1, \min_{\mathcal{L}} \phi_z) \leq \beta_h \leq C \min_{\mathcal{L}} \phi_z$, i.e. $\beta_h \asymp h_\Gamma$ as soon as $\mathcal{L} \neq \emptyset$.

Proof. Upper. As in Theorem 11.11, with step B replaced by $v_B = \sum_z K_z(c_z, -\frac{5}{2}c_z \cdot n_s)$, $c_z = D_z^{-1}(q_1|_{T_1(z)}, q_1|_{T_2(z)})^\top$; split the data as $\sigma(1, 1) + \ell_z(q)(1, 0)$ and use Lemma 11.9(d),(e). *Lower.* $\|\nabla v\| \geq \|q\|/\sqrt{2}$, and by Lemma 11.9(a),(e), $c_v = \sigma(w - z) + \ell_z(q)D_z^{-1}(1, 0)^\top$, so $\|\nabla v\|_{T_1(z)} \geq c|c_v| \geq ch_z|\ell_z(q)|/\sin \phi_z - C\|q\|_{T_2(z)}$; square and sum. *Inf-sup.* Let $\rho_z := R_1 - R_2$ with $R_i \in P_{k-1}(T_i)$ the $L^2(T_i)$ representer of $p \mapsto p(z)$. Then $\int \rho_z = 0$, $(q, \rho_z) = \ell_z(q)$, $\|\rho_z\| \geq c/h_z$, and $(\rho_z, \operatorname{div} v) = c_v \cdot (g_1 - g_2)$, so $|(\rho_z, \operatorname{div} v)| \leq C\phi_z h_z^{-1} \|\nabla v\|$. $\square$

Theorem 11.13 (Upper bound with locked vertices). *Assume (H), (D) and $m_R = 0$. Then (S) is uniquely solvable and*

$$\|\nabla(\tilde{u} - u_h)\| \leq C \left(h^k + h_\Gamma^{3/2} + h_\Gamma \left(\sum_{z \in \mathcal{L}} a_z^2 \right)^{1/2} \right).$$

Proof. Take w_h of Theorem 11.5. At $z \in \mathcal{L}$, T_1, T_2 carry a wall edge, so $h_{T_i} \asymp h_\Gamma$; the interpolation error of ∇w_h at z is $O(h_\Gamma^k)$ and the nodal zeroing costs $Ch_\Gamma^2 \cdot h_\Gamma^{-1}$, so $|\nabla w_h|_{T_i}(z)| \leq a_z + Ch_\Gamma$ and $|c_{w_h, z}| \leq Ch_\Gamma(a_z + h_\Gamma)$. Put $w'_h := w_h - \sum_{z \in \mathcal{L}} K_z(c_{w_h, z}, 0)$: same traces, zero vertex gradient on both triangles at every locked z , and $\|\nabla(w'_h - w_h)\|^2 \leq Ch_\Gamma^2 \sum_{\mathcal{L}} a_z^2 + Ch_\Gamma^3$. The field $K_z(c, 0)$ is not flux-neutral, but divergence-freeness is re-established next: $q := \operatorname{div} w'_h$ has mean $m_R = 0$ and vanishes at every locked vertex on both triangles, so $q \in Q_{\text{lock}}$, and Theorem 11.11 gives $y \in V_h^0$ with $\operatorname{div} y = q$, $\|\nabla y\| \leq C\|q\| \leq C\|\nabla(w'_h - \tilde{u})\|$. Then $z_h := w'_h - y$ is admissible and divergence-free, and steps 4–5 of Theorem 11.5 finish. Existence and uniqueness follow from Theorem 11.12. $\square$

Corollary 11.14 (Two-sided rate with locks). *Under (H), (D), $m_R = 0$, $h^k \leq h_\Gamma^{3/2}$ and $\partial_n u \neq 0$ on Γ , for h_Γ small*

$$\|\nabla(\tilde{u} - u_h)\| \asymp h_\Gamma^{3/2} + h_\Gamma \left(\sum_{z \in \mathcal{L}} a_z^2 \right)^{1/2}$$

(Theorems 11.6, 11.8, 11.13). *One lock gives rate 1; a positive arc fraction of locks where $\partial_n u \neq 0$ gives rate $\frac{1}{2}$; n locks give $h_\Gamma^{3/2} + \sqrt{n} h_\Gamma$.*

Theorem 11.15 (Pressure with locks). *Assume (H), (D), $m_R = 0$. Let $\pi_h := p_h - \Pi p^\circ$, $R := \|\nabla(\tilde{u} - u_h)\| + Ch_\Gamma^2$, and P_{lock} the L^2 -orthogonal projection of $\Pi_h \cap L_0^2$ onto Q_{lock} (its complement is spanned by the ρ_z of Theorem 11.12). Then (i) $\|P_{\text{lock}} \pi_h\| \leq CR$; (ii) $\|\pi_h\| \leq CR/\min(1, \phi_{\min}) \leq CR/h_\Gamma$; (iii) the filtered pressure $p_h^f := P_{\text{lock}} p_h$, computable by subtracting $\sum_z \gamma_z \rho_z$ (a banded Gram solve), satisfies $\|p_h^f - \Pi p^\circ\| \leq C(R + h_\Gamma^{k+1/2})$.*

Proof. As in step 6 of Theorem 11.5, $|(\pi_h, \operatorname{div} v)| \leq CR \|\nabla v\|$ for $v \in V_h^0$. (i) Theorem 11.11 with data $P_{\text{lock}} \pi_h$. (ii) The same with Theorem 11.12. (iii) $p_h^f - \Pi p^\circ = P_{\text{lock}} \pi_h - (I - P_{\text{lock}}) \Pi p^\circ$, and $\|(I - P_{\text{lock}}) \Pi p^\circ\| \leq \|\Pi p^\circ - g^*\|$ for any $g^* \in Q_{\text{lock}}$; take $g^* = \Pi p^\circ - \sum_z \ell_z(\Pi p^\circ) e_z$ with $e_z \in P_{k-1}(T_1(z))$, $e_z(z) = 1$, vanishing at the other two vertices, zero mean, $\|e_z\| \leq Ch_z$; $|\ell_z(\Pi p^\circ)| \leq Ch_z^k$ and $\#\mathcal{L} \leq C/h_\Gamma$. $\square$

Bound (ii) is attained numerically: with one lock the raw pressure error does not converge ($\|\pi_h\| \rightarrow \approx 12$), with alternating locks it grows like $h_\Gamma^{-1/2}$, and in both cases it is almost entirely the $Q_{\text{lock}}^\perp$ component, while the filtered pressure converges at the velocity rate (Table 8). A rigorous lower bound for the raw pressure error was not attempted. Numerically the constrained constants $\beta_{Q_{\text{lock}}}$ equal the lock-free inf-sup constant to 4–5 digits on every mesh with locks and minimum angle $\geq 14^\circ$ (including two consecutive locks sharing an apex), and $\beta_{\text{full}}/\phi_{\min} \in [0.126, 0.159]$ (notes/v6/strong2/tab_infsup.tex).

11.5. Nearly singular wall stars. With a minimum angle, a wall star with $m \geq 3$ triangles is uniformly non-singular: for $\phi_z \leq \theta_{\min}/2$ some triple of lines at z has $V_z := \max |\sin(\psi_j - \psi_i) \sin(\psi_k - \psi_i) \sin(\psi_k - \psi_j)| \geq c \theta_{\min}^3$, and $\Theta(z) \geq \min(\sin(\theta_{\min} - \phi_z), \sin 2\theta_{\min})$. So Lemma 11.3 applies with $C(\theta_{\min}) \leq C \theta_{\min}^{-3}$, and there is nothing to add under (M0)–(M2). Near-singularity requires a small angle ε_z at z , and the paper’s $\Theta(z)$ does not detect it: for $m = 3$ there are stars with $\Theta(z) \geq 0.84$ whose attainable-gradient cost is that of a lock. The two relevant shapes are

I *needle at the wall*: the triangle next to a wall edge has angle ε at z ;

II *split edge*: the middle triangle has angle ε (s_1, s_2 nearly collinear), the lock of Section 11.4 with its interior edge doubled.

Theorem 11.16 (Area-weighted distance). *Let $m = 3$ with exactly one angle $\varepsilon_z \leq \varepsilon_0(\theta_0)$ (the needle), the other two $\geq \theta_0$, ray lengths in $[c_r h_z, h_z]$, and (M0). Then, with constants depending on θ_0, c_r, c_0 only, for the needle in either position (I or II),*

$$c h_z a_z \min\left(1, \frac{\phi_z}{\sqrt{\varepsilon_z}}\right) \leq d_z \leq C h_z a_z \min\left(1, \frac{\phi_z}{\sqrt{\varepsilon_z}}\right).$$

Proof. $\dim A_z = 1$ (Lemma 11.1), spanned by $x \in \ker[P_1 P_2 P_3 P_4]$ with $P_j = n_j n_j^\top$ for the lines e_1, s_1, s_2, e_2 : $x = (D_{234}, -D_{134}, D_{124}, -D_{123})$, $D_{ijk} = \det[\operatorname{vec} P_i, \operatorname{vec} P_j, \operatorname{vec} P_k]$. The tuples are $H_1 = t x_1 P_1$, $H_2 = t(x_1 P_1 + x_2 P_2)$, $H_3 = -t x_4 P_4$. With $x_1 = 1$ and γ the angle of T_1 , the sine formula for D_{ijk} gives exactly

$$\begin{aligned} \text{II: } |x_2| &= \frac{\sin(\gamma + \varepsilon) \sin \phi}{\sin \varepsilon \sin(\gamma - \phi)} = \frac{\phi}{\varepsilon} (1 + O(\phi + \varepsilon)), & |x_4 + 1| &= O(\phi); \\ \text{I: } |x_4 + 1| &= \frac{\sin \phi |\sin(\gamma - \varepsilon)|}{\sin(\gamma - \phi) \sin \varepsilon} = \frac{\phi}{\varepsilon} (1 + O(\phi + \varepsilon)), & |x_2| &= O(\phi). \end{aligned}$$

So with the regular triangle T_1 pinned ($t = a_z$) the regular triangles carry a defect $O(a_z \phi)$ and the needle $a_z \frac{\phi}{\varepsilon} (1 + o(1))$. *Upper*: $t = a_z$ gives $d_z^2 \leq C h_z^2 a_z^2 (\phi^2 + \varepsilon \phi^2 / \varepsilon^2)$, since $|T_*| \leq \varepsilon h_z^2$; $t = 0$ gives $d_z^2 \leq |\text{star}| a_z^2$. *Lower*: with $\eta := \min(\frac{1}{2}, \phi / (K \varepsilon))$, either $|t - a_z| \geq \eta a_z$, and $d_z^2 \geq |T_1| (\eta a_z / 2)^2$, or the needle defect is $\geq a_z \phi / (2\varepsilon)$ for K large, and $d_z^2 \geq |T_*| a_z^2 \phi^2 / (4\varepsilon^2) \geq c h_z^2 a_z^2 \phi^2 / \varepsilon$. In both cases $d_z^2 \geq c h_z^2 a_z^2 \min(1, \phi^2 / \varepsilon)$. $\square$

The square root comes from the needle’s area: the cheapest admissible tuple puts a defect $a_z \phi_z / \varepsilon_z$ on an area $\varepsilon_z h_z^2$. The max-norm defect is $a_z \min(1, \phi_z / \varepsilon_z)$ and overstates the H^1 cost. Numerically $d_z / (h_z a_z \min(1, \phi / \sqrt{\varepsilon})) \in [0.56, 0.92]$ over $\phi \in [0.005, 0.08]$, $\varepsilon \in [\phi^2 / 4, 0.5]$, and independently in 50-digit arithmetic in $[0.32, 1.0]$ down to $\varepsilon = \phi^4$.

Proposition 11.17 (Stability at a nearly singular star). *Let z be as in Theorem 11.16, T, T' its two regular triangles and $\rho_z := R_T - R_{T'}$ as in Theorem 11.12. Then $|(\rho_z, \operatorname{div} v)| \leq C(\phi_z + \sqrt{\varepsilon_z})h_z^{-1}\|\nabla v\|_{\operatorname{star}(z)}$ for $v \in V_h^0$, so $\beta_h \leq C \min_z(\phi_z + \sqrt{\varepsilon_z})$.*

Proof. $(\rho_z, \operatorname{div} v) = \operatorname{tr} G_T - \operatorname{tr} G_{T'}$, with the affine-invariant inverse inequality $|G_{T''}| \leq C|T''|^{-1/2}\|\nabla v\|_{T''}$ on every triangle, the needle included. In case II, $G_1 = c_1 \otimes g_1$, $G_3 = c_2 \otimes g_3$, continuity across the needle gives $|c_2 - c_1| \leq C\sqrt{\varepsilon}\|\nabla v\|$, and $|g_1 - g_3| \leq C(\phi + \varepsilon)/h$. In case I the needle has height $\asymp \varepsilon h$, so $|c_2| \leq C\sqrt{\varepsilon}\|\nabla v\|$, and $|g_1 - \nabla \lambda_{x_1}|_{T_2}| \leq C|\phi - \varepsilon|/h$. $\square$

Theorem 11.18 (Two-sided estimate at needles at the wall). *Hypothesis (H_N) : $(M0)$, $(M1)$, $k \geq 4$, maximum angle $\leq \theta_{\max} < \pi$; $\mathcal{N}$ is a set of wall vertices with stars of shape I or II (Theorem 11.16; for (ii), (H_K) of Section 11.6), $\mathcal{L}$ the locked wall vertices; all other vertices satisfy $(M2)$; all triangles other than the needles (and, for shape II, the partner needle across the short edge) have angles $\geq \theta_0$. Put $w_z := \min(1, \phi_z/\sqrt{\varepsilon_z})$ on $\mathcal{N}$ and $w_z := 1$ on $\mathcal{L}$. Assume (D) and $m_R = 0$.*

- (i) (Proved; shapes I and II.) *For every divergence-free $v \in V_h$ vanishing on Γ_h , $\|\nabla(\tilde{u} - v)\| \geq ch_\Gamma(\sum_{z \in \mathcal{N} \cup \mathcal{L}} w_z^2 a_z^2)^{1/2} - Ch_\Gamma^{3/2}$, and $\geq ch_\Gamma^{3/2}\|\partial_n u\|_{L^2(\Gamma)} - Ch_\Gamma^{5/2}$.*
- (ii) (Proved for shapes I and II under the hypothesis (H_K) of Section 11.6; Theorem 11.22.) $\|\nabla(\tilde{u} - u_h)\| \leq C(h^k + h_\Gamma^{3/2} + h_\Gamma(\sum_z w_z^2 a_z^2)^{1/2})$.
- (L_N) *For every $q \in \Pi_h \cap L_0^2$ with $q|_T(z) = 0$ for all $T \ni z$, $z \in \mathcal{N} \cup \mathcal{L}$, there is $y \in V_h^0$ with $\operatorname{div} y = q$ and $\|\nabla y\| \leq C\|q\|$, C independent of h, ϕ_z, ε_z .*

Proof. (i) is Theorem 11.8 (which needs only affine invariance, valid on needles) with Theorem 11.16 and Lemma 11.1, and Theorem 11.6. (ii) is Theorem 11.22. For shape I there is also a shorter argument under (L_N) and the anisotropic interpolation estimates [7, 2]: as in Theorem 11.13, with the locked-vertex step replaced by: at $z \in \mathcal{N}$ let $G^* \in A_z$ be the minimiser of Theorem 11.16 and $\Delta G_T := G_T^* - \nabla w_h|_T(z)$. Both tuples are continuous across the spokes and vanish on the wall edges, so $\Delta G = \nabla \ell$ for a piecewise-linear ℓ on the star with $\ell(z) = 0$ and $\ell = 0$ on the wall edges. $w'_h := w_h + \sum_z \lambda_z^2 \ell_z$ has vertex gradients G^* at z (so $\operatorname{div} w'_h$ vanishes there on all triangles) and unchanged vertex gradients elsewhere, at cost $\leq C(d_z + (\sum_T |T| \|\nabla w_h|_T(z) - \nabla \tilde{u}(z)\|^2)^{1/2})$; the interpolation part is $O(h^k)$ under the maximum angle condition [7, 2], and the nodal zeroing on a needle with a wall edge contributes $\leq Ch_\Gamma w_z(a_z + h_\Gamma)$. Then (L_N) gives the divergence corrector, and the energy argument of Theorem 11.5 is geometry-free. $\square$

For $|\partial_n u| \geq \delta > 0$ on the relevant arcs, one needle at the wall gives $h_\Gamma^{3/2} + h_\Gamma w_z$: rate $\frac{3}{2}$ for $\varepsilon \gtrsim h_\Gamma$ and rate 1 for $\varepsilon \lesssim h_\Gamma^2$; a positive fraction of wall vertices with $\varepsilon \asymp h_\Gamma^p$ gives rates $\frac{3}{2}, 1, \frac{1}{2}$ for $p = 0, 1, 2$. The stability hypothesis (L_N) is no longer needed for the velocity; it still controls the raw pressure (Theorem 11.15). For shape II it is false (Proposition 11.19); for shape I it is numerical: the constrained constants reproduce the lock-free value to five digits.

11.6. Needle pairs. A *needle* is a triangle with one angle $\leq \varepsilon_0$; the others are *regular* (angles in $[\theta_0, \theta_{\max}]$, $\theta_{\max} < \pi$). If a needle $N = [a, x_L, x_R]$ has angle ε at a , its short edge $e = [x_L, x_R]$ has length $\asymp \varepsilon h_N$; if $e \not\subset \partial\Omega_h$, the triangle across e has a short edge and diameter $\asymp h$, so by the sine rule and the maximum-angle condition it is again a needle with the same short edge. So needles come in *pairs* $K = N_1 \cup N_2$ (kites), with apexes a_i , short edge e of midpoint m , unit tangent τ , axes $A_i = m - a_i$ of length $L_i \asymp h$, $\varepsilon := |e|/\min L_i$, and *bend* $\vartheta := \angle(A_1, -A_2) \in [0, \pi)$. Shape II (split edge) is a straight pair, $\vartheta = 0$. In shape I the needle at the wall and its partner across the short edge meet at the angle α of a regular triangle, so $\vartheta = \pi - \alpha \geq \pi - \theta_{\max}$ (**notes/v7/needle**).

Proposition 11.19 (The cross-difference functional). *Let K be a needle pair whose angles at x_L, x_R lie in $[\pi - \theta_{\max}, \theta_{\max}]$. For $q \in \Pi_h$ let $\ell_K(q) := q_{N_1}(x_L) - q_{N_1}(x_R) - q_{N_2}(x_L) + q_{N_2}(x_R)$ and*

$\rho_K \in \Pi_h$ its L^2 representer (supported on K). Then $\int \rho_K = 0$, $\rho_K|_{N_i}(a_i) = 0$, $\|\rho_K\| \geq c|N_1|^{-1/2}$, and for every v continuous and piecewise P_k on K ,

$$|(\rho_K, \operatorname{div} v)_K| \leq C(\varepsilon + \vartheta) \|\rho_K\| \|\nabla v\|_K.$$

So $\beta_h \leq C(\varepsilon + \vartheta)$ for every pressure space containing ρ_K , in particular for the constrained space of (L_N) ; for straight pairs (shape II) $\beta_h \leq C\varepsilon$ and (L_N) is false.

Proof. Map $\hat{T} = \{0 \leq \hat{\sigma} \leq 1, |\hat{\tau}| \leq \hat{\sigma}\}$ onto N_i by $F_i(\hat{\sigma}, \hat{\tau}) = a_i + \hat{\sigma}A_i + \hat{\tau}\frac{|e|}{2}\tau$. Then $\partial_{\hat{\tau}}\hat{v} = \frac{|e|}{2}\partial_{\tau}v$ and $|N_i| \asymp L_i|e|$, so $\|\partial_{\hat{\tau}}\hat{v}\|_{\hat{T}} \leq C\sqrt{\varepsilon}\|\nabla v\|_{N_i}$. With the basis $(\hat{A}_i^*, \tau_i^*)$ dual to $(A_i/L_i, \tau)$, $\operatorname{div} v = \hat{A}_i^* \cdot \partial_{\hat{A}_i} v + \tau_i^* \cdot \partial_{\tau} v$ on N_i , the dual vectors are bounded, and $|\tau_1^* - \tau_2^*| \leq C\vartheta$. Writing $[g] := g(x_L) - g(x_R)$,

$$\ell_K(\operatorname{div} v) = \hat{A}_1^* \cdot [\partial_{\hat{A}_1} v_{N_1}] - \hat{A}_2^* \cdot [\partial_{\hat{A}_2} v_{N_2}] + (\tau_1^* - \tau_2^*) \cdot [\partial_{\tau} v],$$

because $\partial_{\tau}v$ is a tangential derivative on e and is the same from both sides. The axial terms are $L_i^{-1} \int_{-1}^1 \partial_{\hat{\tau}} \partial_{\hat{\sigma}} \hat{v}(1, \hat{\tau}) d\hat{\tau}$, bounded by $C\sqrt{\varepsilon}h^{-1}\|\nabla v\|_K$ by equivalence of norms on $P_{k-1}(\hat{T})$; the tangential term is $\leq C|N_1|^{-1/2}\|\nabla v\|_K$. On N_i , $\|\rho_K\|^2 \asymp |N_i|^{-1}$ with $|N_i| \asymp \varepsilon h^2$, so both bounds are $\leq C(\varepsilon + \vartheta)\|\rho_K\|\|\nabla v\|_K$. Each vertex representer has integral 1, so $\int \rho_K = 0$; the reflection $\hat{\tau} \mapsto -\hat{\tau}$ swaps the two short-edge corners and fixes the apex, so ρ_K vanishes at the apexes and satisfies every constraint imposed there. $\square$

Numerically $\beta \approx 0.5(\varepsilon + \vartheta)$ over four decades, a 45°-bent pair has the inf-sup constant of the unsplit mesh, and for straight pairs $\beta/\varepsilon = 0.54\text{--}0.56$; a lower bound $\beta \geq c\varepsilon$ is not proved. For comparison, Kean, Neilan and Schneier analyse Scott–Vogelius on anisotropic meshes obtained by barycentric and incenter refinement [40]; their setting is macro-split meshes, and we have not identified the pair/bend mechanism there.

The velocity does not need inf-sup at the needles. Let $\mathcal{K}$ be the set of wall needle pairs, $\Omega_S := \Omega_h \setminus \bigcup_K \bar{K}$, $\mathcal{T}_S$ the non-needle triangles and $V_S := \{v \in V_h^0 : v = 0 \text{ on } \bigcup \bar{K}\}$. A *corner* is a pair (T, x) , $T \in \mathcal{T}_S$, with both edges of T at x on $\partial\Omega_S$; a *slit lock* is a vertex in exactly two triangles of $\mathcal{T}_S$ sharing an interior edge, whose other edges at x lie on $\partial\Omega_S$. Put

$$Q_S := \{q \in \Pi_h \cap L_0^2 : q = 0 \text{ on needles, } q|_T(x) = 0 \text{ at corners, } q|_{T_1}(x) = q|_{T_2}(x) \text{ at slit locks}\}.$$

(H_K) (M0), (M1), (D), $k \geq 4$, $m_R = 0$; all angles $\leq \theta_{\max}$; non-needle triangles have angles $\geq \theta_0$; every needle belongs to a pair of shape I or II attached to a wall vertex $z \in \mathcal{N}$; the patches $\omega(K)$ (triangles meeting $\bar{K}$) are pairwise disjoint and disjoint from the stars of the locked vertices of $\mathcal{L}$; the slit vertices (x_L, x_R) for shape II, the wall end of the short edge for shape I) lie in at least three non-needle triangles; all other vertices satisfy (M2). This is a narrow hypothesis: every needle must sit in its own wall pair; it is not a general anisotropic-mesh condition.

Theorem 11.20 (Slit-domain right inverse). *Under (H_K), for every $q \in Q_S$ there is $y \in V_S$ with $\operatorname{div} y = q$ and $\|\nabla y\| \leq C\|q\|$, with C independent of h , h_Γ , the needle angles, the bends, ϕ_z and the number of pairs.*

Proof. Step 0. Take $w \in H_0^1(\Omega_h)^2$ with $\operatorname{div} w = q$, $\|\nabla w\| \leq C\|q\|$ (Lemma 4.5, continuous level). For each pair let D_K be the interior of $\omega(K)$ and χ_K a cut-off, 1 on $\bar{K}$, 0 on $\partial D_K \setminus \Gamma_h$, $|\nabla \chi_K| \leq C/h_K$; Friedrichs on D_K (with $w = 0$ on its wall part) gives $\|\nabla(\chi_K w)\| \leq C\|\nabla w\|_{D_K}$. Then $g_K := \operatorname{div}(\chi_K w)$ has zero mean on D_K and vanishes on K . $U_K := D_K \setminus \bar{K}$ is a John domain with constant depending only on $\theta_0, \theta_{\max}$ (its re-entrant corners and near-cracks are admissible), so the right inverse of Acosta, Durán and Muschietti [1] gives $c_K \in H_0^1(U_K)^2$, $\operatorname{div} c_K = g_K$, $\|\nabla c_K\| \leq C\|\nabla w\|_{D_K}$. Since the patches are disjoint, $w_S := w - \sum_K (\chi_K w - c_K) \in H_0^1(\Omega_S)^2$, $\operatorname{div} w_S = q$, $\|\nabla w_S\| \leq C\|q\|$. *Steps A–D.* Proceed as in Theorem 11.11 on the shape-regular mesh $\mathcal{T}_S$ of Ω_S , extending every field by 0 on the needles: the P_2 Fortin interpolant of w_S for

the element means; Lemma 11.9 at plain and slit locks (the two lock triangles contain no needle); Lemma 11.10 at the other vertices, with coefficients 0 on $\partial\Omega_S$ -edges, which is uniform because every slit star has at least three triangles with angles $\geq \theta_0$ filling an angle $\pi - O(\varepsilon)$ (at x_L, x_R) or $2\pi - O(\varepsilon)$ (at the far apex), and at corners nothing is to be done since q vanishes there; element bubbles on $\mathcal{T}_S$. No constant involves ε, ϑ or ϕ_z . $\square$

Lemma 11.21 (Divergence-free fields on a wall needle pair). *Under (H_K) , for $z \in \mathcal{N}$ with pair K let $G^* \in A_z$ be the minimiser of Theorem 11.16. There is $v_K \in V_h$ on $\omega(K)$, equal to the interpolant w_h of Theorem 11.5 on $\partial\omega(K)$ and to 0 on Γ_h , divergence free on $N_1 \cup N_2$, whose divergence vanishes at the corners and whose traces satisfy the slit-lock condition, with*

$$\|\nabla(\tilde{u} - v_K)\|_{\omega(K)} \leq C(d_z + h_z^2 + h_z^{k+1}).$$

Sketch. Shape II: on N_1 take $S_1 := P + \text{curl}(\omega\Phi_0)$, with P the (divergence-free) Taylor polynomial of $\tilde{u}$ at z , $\Phi_0(x) = \frac{1}{2}(x - z)^\top H(x - z)$ with $JH = G_{N_1}^* - \nabla\tilde{u}(z)$ (J the rotation by $-\pi/2$) and ω affine, $\omega(z) = 1$, $\omega = 0$ on the line of e ; on N_2 take $S_2 := P + \text{curl}(A(\xi) + tB(\xi))$ in oblique coordinates along e and the axis of N_2 , with A, B chosen so that curl matches $S_1 - P$ on e . Both are exactly divergence free, continuous across e , and $\nabla S_1(z) = G_{N_1}^*$. Because G^* is C^0 -compatible, a cone estimate gives $|(G_{N_1}^* - \nabla\tilde{u}(z))(x - z)| \leq Ch\eta$ on N_1 with $h\eta \leq Cd_z$, and $|N_1|^{1/2}|G_{N_1}^* - \nabla\tilde{u}(z)| \leq d_z$; all degrees stay $\leq 4 \leq k$. On the regular triangles of $\omega(K)$ the Lagrange nodal values are those of the needle fields on kite edges, 0 on Γ_h and $I_h\tilde{u}$ elsewhere, so the vertex gradients at z are G^* and the corner conditions hold. Shape I is analogous with $S_1 := \text{curl}(\frac{1}{2}\lambda^2 R)$, λ the distance to the wall line, and one extra scalar t^2 -term on N_2 fixing the divergence at the corner of T_2' . Details: [notes/v7/needle](#), Lemmas A1, A2. $\square$

Theorem 11.22 (Needles at the wall, strongly imposed method). *For the strongly imposed method of Section 11, under (H_K) (shapes I and II in any mixture, any $\varepsilon_z \in (0, \varepsilon_0]$),*

$$\|\nabla(\tilde{u} - u_h)\| \leq C\left(h^k + h_\Gamma^{3/2} + h_\Gamma\left(\sum_{z \in \mathcal{N} \cup \mathcal{L}} w_z^2 a_z^2\right)^{1/2}\right),$$

and with Theorem 11.18(i), for $h^k \leq h_\Gamma^{3/2}$, $\|\nabla(\tilde{u} - u_h)\| \asymp h_\Gamma^{3/2} + h_\Gamma(\sum_z w_z^2 a_z^2)^{1/2}$.

Proof. Let $v := w_h$ (modified at plain locks as in Theorem 11.13) outside $\bigcup \omega(K)$ and $v := v_K$ on each $\omega(K)$ (Lemma 11.21). Then $v \in V_h$, $v = 0$ on Γ_h , $v = g_I$ on ∂R , and $\|\nabla(\tilde{u} - v)\|^2 \leq C(h^k + h_\Gamma^{3/2})^2 + Ch_\Gamma^2 \sum_{\mathcal{L}} a_z^2 + C \sum_{\mathcal{N}} (d_z^2 + h_z^4)$, with $d_z \leq Ch_z a_z w_z$ (Theorem 11.16) and $\#\mathcal{N} \leq C/h_\Gamma$. Its divergence lies in Q_S , so Theorem 11.20 gives $y \in V_S$ with $\text{div } y = \text{div } v$ and $\|\nabla y\| \leq C\|\nabla(\tilde{u} - v)\|$, and $z_h := v - y$ is admissible and exactly divergence free. For the strongly imposed method the error equation on the divergence-free, wall-vanishing fields contains no pressure, so the energy argument of Theorem 11.5 (steps 4–5) needs only z_h and no inf-sup condition. $\square$

The proof never inverts the divergence on a needle, which is why $\beta \asymp \varepsilon$ on shape II is invisible in the velocity but visible in the raw pressure. It does not carry over to Nitsche imposition (GS, CNS*): there the term $\langle \xi - \tilde{\xi}, e_h \cdot n \rangle$ survives on Z_h and is absorbed through the inf-sup constant. The same cut-off argument shows that an interior needle pair (e.g. a split interior edge) leaves the best approximation in Z_h unchanged up to Ch^{k+1} per pair: numerically 0.0173285–0.0173698 for $\varepsilon = 10^{-1}, \dots, 10^{-4}$, against 0.0173699 unsplit.

Computation: [notes/v7/needle/probe2.py--probe9.py](#), [slit_lab.py](#); output [notes/v7/needle/probe*n8.log](#), [probe5.N16.log](#), [probe5.N24.log](#) (slit-domain constants 0.078--0.102, uniform in $\varepsilon = 10^{-2}, \dots, 10^{-5}$).

TABLE 7. Strong imposition on mesh (s) (every wall vertex in three triangles): H^1 error, observed rate, rate predicted by $E = ah_\Gamma^{3/2}(1 + \beta h_\Gamma)$, and $E/h_\Gamma^{3/2}$ (notes/v6/strong/fit_rates.log).

N	h_Γ	H^1 error	rate	predicted	$E/h_\Gamma^{3/2}$	vertex-gradient error
64	0.1243	$4.557 \cdot 10^{-2}$	1.657	1.653	1.040	0.307
96	0.0831	$2.384 \cdot 10^{-2}$	1.610	1.614	0.995	0.212
128	0.0624	$1.518 \cdot 10^{-2}$	1.576	1.584	0.974	0.162
160	0.0500	$1.073 \cdot 10^{-2}$	1.558	1.566	0.961	0.130
192	0.0416	$8.097 \cdot 10^{-3}$	1.547	1.555	0.953	0.109

TABLE 8. Strong imposition with locked vertices: H^1 velocity error, raw pressure error $\|\pi_h\|$ and filtered pressure error $\|\pi_h^f\|$ of Theorem 11.15 (notes/v6/strong2/tab_filter.tex).

mesh	$\#\mathcal{L}$	N	H^1 err	rate	$\ \pi_h\ $	rate	$\ \pi_h^f\ $	rate
one lock	1	32	$2.123 \cdot 10^{-1}$	1.36	7.291	-0.45	0.900	1.71
	1	64	$9.376 \cdot 10^{-2}$	1.16	9.160	-0.29	0.328	1.39
	1	128	$4.464 \cdot 10^{-2}$	1.06	10.447	-0.16	0.145	1.12
mesh (a)	16	32	$5.724 \cdot 10^{-1}$	0.70	19.747	-0.49	1.378	0.79
	32	64	$3.746 \cdot 10^{-1}$	0.60	26.384	-0.42	0.868	0.65
	64	128	$2.543 \cdot 10^{-1}$	0.55	35.653	-0.45	0.580	0.57

11.7. **Computations.** All runs use $k = 4$, test A, $\mu = 100$ for the Nitsche comparisons, and the code of Section 16.8.

The drift of the rate under (M2). On mesh (s) the strong-imposition H^1 rates decrease slowly (1.558 and 1.547 at $N = 160, 192$). The data fit $E \approx ah_\Gamma^{3/2}(1 + \beta h_\Gamma)$ with $\beta \approx 1.1$ – 1.3 and $a \approx 0.94$: the predicted local rates $1.5 + \beta h_\Gamma/(1 + \beta h_\Gamma)$ match the observed ones to 0.01 from $N = 48$ on, and a free-exponent fit $ah_\Gamma^p(1 + bh_\Gamma)$ gives $p = 1.457 \rightarrow 1.491$ (Table 7). The drift is a relative $O(h_\Gamma)$ correction, not a different rate. The ratio of the strong to the GS error settles near 2.48.

Locks. Table 8 shows a mesh with a single flipped first-layer diagonal (one locked vertex) and the alternating mesh (a). With one lock the rate falls to 1 (1.037 at $N = 192$; discrete divergence up to $3.2 \cdot 10^{-6}$ there), the raw pressure error does not converge, and the filtered pressure converges at the velocity rate. On mesh (a) the rate tends to $\frac{1}{2}$ (0.549 at $N = 128$), the raw pressure error grows like $h_\Gamma^{-1/2}$, and again the filtered pressure follows the velocity.

Needles. On meshes derived from mesh (a) by splitting each lock (shape II) or inserting a needle at the wall (shape I) with $\varepsilon = 0.3, h_\Gamma, h_\Gamma^2$, the rates approach $\frac{3}{2}, 1, \frac{1}{2}$ as predicted (1.56; 1.28 and 1.26; 0.68 and 0.70 at the finest levels), and $(\sum d_z^2)^{1/2}/\mathcal{I}$ with $\mathcal{I} := (\sum_z h_z^2 a_z^2 w_z^2)^{1/2}$ is constant to two or three digits along each sequence (0.53–0.71). The limits 1 and $\frac{1}{2}$ are not yet reached at $N \leq 96$.

Computation: notes/v6/strong/*.py; notes/v6/strong2/*.py; notes/v6/nearsing/*.py; output notes/v6/strong/fit_rates.log, lock_bound.log; notes/v6/strong2/filter.log, infsup*.log; notes/v6/nearsing/ns_bound.log, local_ns.log.

12. CLOUGH–TOCHER REFINEMENTS: EVERY $k \geq 2$

Sections 2–10 assume continuous P_k velocities with $k \geq 4$ on a mesh satisfying (M0)–(M2). Here we show that the results of Sections 3–10, through Theorem 7.2 and Proposition 10.1, hold

for all $k \geq 2$ on Clough–Tocher (barycentric, Alfeld) refinements, and that (M1), (M1') and (M2) can be dropped (the precise scope is Theorem 12.8 and Remark 12.13). Only four ingredients change: the inf-sup constant, the divergence range, the flux exponent in Lemma 4.6, and the C^1 test field v_φ . The penalty-necessity constants of Proposition 10.1 also change, and we recertify them exactly.

12.1. Setting. Let $\mathcal{T}_h^M$ be a conforming triangulation of Ω_h (the *macro mesh*). Its Clough–Tocher refinement $\mathcal{T}_h$ splits every $T \in \mathcal{T}_h^M$ at its barycentre g_T into three triangles. Macro edges are not split, so the edges of $\mathcal{T}_h$ on Γ_h and on ∂R are exactly the macro edges there, and $\mathcal{E}_h^\Gamma$ is unchanged. For $k \geq 2$, V_h is the space of continuous piecewise- P_k fields on $\mathcal{T}_h$ and Π_h the space of discontinuous piecewise- P_{k-1} functions on $\mathcal{T}_h$. V_h^R , V_h^0 , Z_h , W_h , the forms a_h and N_h , the methods GS and CNS* and the norm $\|\cdot\|$ are defined by the formulas of Section 2, applied to these spaces. We put $h := \max_{T \in \mathcal{T}_h^M} \text{diam } T$ and keep h_Γ , ρ , γ as before.

Assumption 12.1.

(CT0) $\mathcal{T}_h^M$ is shape regular: $\text{diam } T \leq \sigma r_T$ for every $T \in \mathcal{T}_h^M$, where r_T is the inradius. Moreover $c_0 h_\Gamma \leq |e| \leq h_\Gamma$ for all $e \in \mathcal{E}_h^\Gamma$.

Assumption (D) of Section 2 is kept. No analogue of (M1), (M1') or (M2) is assumed.

In this section C and c depend on σ , c_0 , k , L and norms of $(u, p, \tilde{\psi})$, and on nothing else.

Lemma 12.2 (Structure of the refined mesh). *Assume (CT0) and $h \leq h_*$.*

- (a) $\mathcal{T}_h$ is shape regular with a constant depending only on σ , and $\text{diam } K \geq c \text{diam } T$ for $K \subset T$.
- (b) Each $e \in \mathcal{E}_h^\Gamma$ is an edge of exactly one $K_e \in \mathcal{T}_h$, namely $K_e = \text{conv}(e \cup \{g_{T_e}\})$, where $T_e \in \mathcal{T}_h^M$ is the macro triangle on e . Its height over e is one third of that of T_e , and $\text{diam } K_e \leq C|e|$. No macro triangle has two edges on Γ_h . Every vertex of Γ_h lies in at least two macro triangles, hence in at least four triangles of $\mathcal{T}_h$.
- (c) $\mathcal{T}_h$ has no singular vertex in the sense of (M1). It satisfies (M2) with a constant $\Theta_0 = \Theta_0(\sigma) > 0$.

In particular, $\mathcal{T}_h$ satisfies (M0)–(M2) of Assumption 2.1, with constants depending only on σ and c_0 .

Proof. (a) For $K \subset T$ we have $|K| = |T|/3$ and $\text{diam } K \leq \text{diam } T$. Hence $\text{diam } K^2/|K| \leq 3 \text{diam } T^2/|T| \leq C(\sigma)$, which bounds the angles of K from below. Also $\text{diam } K \geq$ the length of the macro edge of K , which is $\geq c \text{diam } T$.

- (b) The barycentre lies at one third of each height, so K_e is the only refined triangle with an edge on e , and $\text{diam } K_e \leq \text{diam } T_e \leq C|e|$.

The interior angle of Ω_h at a vertex z of Γ_h is $\pi + O(h_\Gamma)$. Every macro angle is at most $\pi - 2\theta_{\min}(\sigma)$. So for $h \leq h_*$ at least two macro triangles meet at z . A macro triangle with two edges on Γ_h would have both edges at a common vertex z of Γ_h , and its angle at z would be that interior angle, which is impossible.

- (c) Consider a macro vertex z with incident macro triangles $T_1, \dots, T_m$. The refined edges at z are the macro edges and the medians zg_{T_i} , and each median lies strictly inside the angle of T_i at z .
 - If z is interior, $m \geq 3$, so z has $2m \geq 6$ refined edges. At most four rays from z lie on two lines, so z is not singular.
 - If z is a boundary vertex, it has $2m + 1 \geq 3$ refined edges. For $m \geq 2$ there are at least $5 > 4$ of them. For $m = 1$ the three rays are two macro edges, at an angle $< \pi$, and a median strictly between them, so no two are opposite.
 - A barycentre has three edges, and no two are opposite since g_T is interior.
 - Every vertex lies in at least two refined triangles.

For (M2), at a macro vertex the two refined angles α'_i, α''_i inside T_i are consecutive (never the wrap-around pair), and they add up to the macro angle $\alpha_i \in [\theta_{\min}, \pi - 2\theta_{\min}]$. Hence $\Theta(z) \geq \sin \theta_{\min}$.

At a barycentre, the angles ϕ_1, ϕ_2, ϕ_3 satisfy $\sum \phi_j = 2\pi$ and, by (a), $\phi_j \leq \pi - 2\theta'_{\min}(\sigma)$. Moreover $|\sin(\phi_j + \phi_{j+1})| = \sin \phi_{j+2}$ and $\max_j \phi_j \geq 2\pi/3$. Hence $\Theta(g_T) \geq \min(\sin \frac{2\pi}{3}, \sin 2\theta'_{\min})$. $\square$

By Lemma 12.2, for $k \geq 4$ the paper applies verbatim to $\mathcal{T}_h$. The new content is $k \in \{2, 3\}$, together with the fact that no angle or singular-vertex hypothesis on the macro mesh is needed for any k . The proofs below cover all $k \geq 2$ uniformly, except the C^1 test field of Lemma 12.7.

12.2. Divergence: local and global. For a triangle T with Clough–Tocher split T^{ct} , let $V_0(T)$ be the space of continuous piecewise- P_k fields on T^{ct} vanishing on ∂T , and $Q_0(T)$ the space of discontinuous piecewise- P_{k-1} functions on T^{ct} with mean zero on T .

Lemma 12.3 (Local surjectivity). *Let $k \geq 2$. Then $\text{div } V_0(T) = Q_0(T)$. For every $q \in Q_0(T)$ there is $v \in V_0(T)$ with $\text{div } v = q$ and $\|\nabla v\|_T \leq C_{\text{loc}} \|q\|_T$, where C_{loc} depends only on k and the shape regularity of T .*

Proof. (1) *Reference element.* Let $\hat{T}$ be the reference triangle, with its Clough–Tocher split. We show that $\text{div} : V_0(\hat{T}) \rightarrow Q_0(\hat{T})$ is onto.

- $k = 2$. Here $\dim V_0(\hat{T}) = 8 = \dim Q_0(\hat{T})$, so it suffices that div is injective. A divergence-free $v \in V_0(\hat{T})$ is $\text{curl } \psi$ with $\psi \in C^1(\hat{T})$ piecewise cubic on the split and $\psi = \nabla \psi = 0$ on $\partial \hat{T}$ (the boundary is connected). All twelve Hsieh–Clough–Tocher degrees of freedom of ψ [23, 22] then vanish, so $\psi = 0$.
- $k \geq 4$. The split is a three-triangle mesh of $\hat{T}$ with no singular vertex (Lemma 12.2(c) applied to one macro triangle), so the Scott–Vogelius theorem [54, 37] applies on the domain $\hat{T}$.
- $k = 3$ (and, as a check, $k = 2, 4, 5, 6$). An exact rational rank computation gives $\text{rank}(\text{div}|_{V_0(\hat{T})}) = \dim Q_0(\hat{T})$. The dimensions are $\dim V_0 = 8, 20, 38, 62, 92$, $\dim Q_0 = 8, 17, 29, 44, 62$ and $\dim \ker \text{div} = 0, 3, 9, 18, 30$.
- The statement for all $k \geq d = 2$ is also the local ingredient of Guzmán–Neilan [36]; for $k = 2$ it is Arnold–Qin [5].

A surjective linear map between finite-dimensional spaces has a bounded right inverse. Let $\hat{C}_k$ be its norm.

- (2) *Piola transfer.* Let $F(\hat{x}) = B\hat{x} + b$ map $\hat{T}$ onto T . Affine maps preserve barycentres, so F maps $\hat{T}^{\text{ct}}$ onto T^{ct} . For $q \in Q_0(T)$, put $\hat{q} := (\det B) q \circ F \in Q_0(\hat{T})$. Take $\hat{v}$ with $\widehat{\text{div}} \hat{v} = \hat{q}$ and $\|\widehat{\nabla} \hat{v}\| \leq \hat{C}_k \|\hat{q}\|$, and set $v := (\det B)^{-1} B \hat{v} \circ F^{-1}$. Then $v \in V_0(T)$, $\text{div } v = q$ and $\|\nabla v\|_T \leq \hat{C}_k \|B\| \|B^{-1}\| \|q\|_T$. $\square$

Computation: `code/lamstar_exact_ct.py` (Part 1); output `logs/lamstar_exact_ct.log`.

Lemma 12.4 (Uniform inf-sup constant, Clough–Tocher). *Let $k \geq 2$ and assume (CT0). There is $\beta > 0$, depending only on k , σ and L , such that for every $q \in \Pi_h \cap L_0^2(\Omega_h)$ there exists $v \in V_h^0$ with $\text{div } v = q$ and $\|\nabla v\| \leq \beta^{-1} \|q\|$.*

Proof. (1) *Continuous level.* By the second half of the proof of Lemma 4.5 (the Galdi wedge cover, which involves only Ω_h and not the mesh), there is $\beta_c > 0$ independent of h with the following property: for each $f \in L_0^2(\Omega_h)$ there is $w \in H_0^1(\Omega_h)^2$ with $\text{div } w = f$ and $\|\nabla w\| \leq \beta_c^{-1} \|f\|$.

- (2) *Macro Fortin step.* Let Π^{SZ} be the Scott–Zhang interpolant [55] onto continuous piecewise- P_1 fields on $\mathcal{T}_h^M$ that vanish on $\partial \Omega_h$. For an interior macro edge E , let $b_E = \lambda_a \lambda_b$ be its

quadratic macro bubble. Set

$$w_h := \Pi^{\text{SZ}} w + \sum_E c_E b_E, \quad c_E := \frac{6}{|E|} \int_E (w - \Pi^{\text{SZ}} w) \in \mathbb{R}^2.$$

Then w_h is continuous piecewise- P_2 on $\mathcal{T}_h^M$, hence in V_h^0 for $k \geq 2$, and $\int_E w_h = \int_E w$ on every macro edge (both vanish on $\partial\Omega_h$). By the divergence theorem, $\int_T \text{div } w_h = \int_T \text{div } w = \int_T q$ for every $T \in \mathcal{T}_h^M$. The trace inequality and the local estimates of Π^{SZ} give $|c_E| \leq C \|\nabla w\|_{\omega_E}$. Since $\|\nabla b_E\| \leq C$ in two dimensions, $\|\nabla w_h\| \leq C \|\nabla w\| \leq C \beta_c^{-1} \|q\|$.

- (3) *Local step.* On each $T \in \mathcal{T}_h^M$, $r := q - \text{div } w_h$ is piecewise P_{k-1} on T^{ct} (since $\text{div } w_h|_T \in P_1$) and has mean zero on T . Lemma 12.3 gives $v_T \in V_0(T)$ with $\text{div } v_T = r|_T$ and $\|\nabla v_T\|_T \leq C_{\text{loc}} \|r\|_T$. Extended by zero, $v_T \in V_h^0$.
- (4) Put $v := w_h + \sum_T v_T \in V_h^0$. Then $\text{div } v = q$ and $\|\nabla v\| \leq \|\nabla w_h\| + C_{\text{loc}}(\|q\| + \sqrt{2} \|\nabla w_h\|) \leq \beta^{-1} \|q\|$. $\square$

The domain enters only through β_c , in step 1. The transfer of the Galdi argument is therefore immediate: unlike Lemma 4.5, neither a Poincaré constant nor an angle condition is involved. For $k \geq 4$ one may instead apply Lemma 4.5 to $\mathcal{T}_h$, using Lemma 12.2(c).

Lemma 12.5 (Divergence range). *Let $k \geq 2$ and assume (CT0). Then $\text{div } V_h^R = \Pi_h$, and $W_h \neq \emptyset$ for every boundary datum g . The velocities of GS and CNS* are pointwise divergence-free.*

Proof. Follow the proof of Lemma 4.3, using Lemma 12.4 for $\text{div } V_h^0 = \Pi_h \cap L_0^2$, and the quadratic edge bubble $\beta_e n_e$ of the refined triangle K_e for the constants ($\int_e \beta_e = |e|/6$). The corner obstruction in the remark after Lemma 4.3 cannot occur, because by Lemma 12.2(c) no vertex of ∂R is singular in $\mathcal{T}_h$. $\square$

12.3. Approximation in W_h and the test field.

Lemma 12.6 (Approximation in W_h , $k \geq 2$). *Let $k \geq 2$, assume (CT0), and let $\sigma_k := k + 2$ for even k and $\sigma_k := k + 1$ for odd k . In particular $\sigma_2 = \sigma_3 = 4$. Then*

$$E_h = \inf_{z \in W_h} \|\tilde{u} - z\| \leq C \varepsilon_h, \quad \varepsilon_h = (1 + (\gamma h_\Gamma)^{1/2}) h^k + h^{\sigma_k} (h_\Gamma^{-1/2} + (\mu/h)^{1/2}).$$

The lower bound $\|\tilde{u} - z\| \geq (\mu/h)^{1/2} |m_R| / |\Gamma_h|^{1/2}$ for every $z \in W_h$, stated after Lemma 4.6, holds unchanged. The flux term is at most Ch^k for bounded γ whenever $h_\Gamma \geq h^{2(\sigma_k - k)}$: that is, $h_\Gamma \geq h^4$ for $k = 2$ and $h_\Gamma \geq h^2$ for $k = 3$. If $g \cdot \nu$ is a polynomial of degree at most $\sigma_k - 1$ on each side of R , the contribution of ∂R vanishes and the flux term reduces to a term below h^k (as in the remark after Lemma 4.6).

Proof. Steps 1–5 of the proof of Lemma 4.6 go through word for word, with three substitutions.

- (1) The Lagrange interpolant is taken on $\mathcal{T}_h$. Refined triangles meeting Γ_h have diameter at most Ch_Γ by Lemma 12.2(a,b), so the three interpolation bounds of step 1 hold for every $k \geq 1$.
- (2) In step 2, the trace of $I_h \tilde{u}$ on a boundary edge is its equispaced P_k Lagrange interpolant, and the edge is not split. Integrating it is the closed $(k + 1)$ -point Newton–Cotes rule, which is exact on $P_{\sigma_k - 1}$. For $k = 2$ this is Simpson’s rule and for $k = 3$ Simpson’s 3/8 rule, both exact on cubics. Each edge therefore contributes $O(|e|^{\sigma_k + 1})$, and $|m| \leq Ch^{\sigma_k}$.
- (3) Step 3 uses the bubbles β_e of the refined triangles K_e , and step 4 uses Lemma 12.4 in place of Lemma 4.5. $\square$

Lemma 12.7 (The C^1 test field). *Let $k \geq 2$, assume (CT0) and $h \leq h_*$, and let φ and $w = \text{curl } \varphi$ be as in Section 6. There is $v_\varphi \in Z_h$ with*

$$\|v_\varphi - w\|_{L^\infty(\Omega_h)} \leq Ch^k, \quad \max_{K \in \mathcal{T}_h} \|\nabla(v_\varphi - w)\|_{L^\infty(K)} \leq Ch. \quad (13)$$

Proof. • $k \in \{2, 3\}$. Let $\Pi^{\text{HCT}}\varphi$ be the Hsieh–Clough–Tocher interpolant on $\mathcal{T}_h^M$ [23]. It is C^1 on Ω_h and piecewise cubic on $\mathcal{T}_h$. Its degrees of freedom are the values and gradients at the macro vertices and the normal derivatives at the macro-edge midpoints, and it reproduces P_3 . The HCT interpolation estimate [22, Section 6.1] gives

$$|\varphi - \Pi^{\text{HCT}}\varphi|_{W^{j,\infty}(T)} \leq Ch_T^{4-j} |\varphi|_{W^{4,\infty}(T)}, \quad j \leq 2,$$

with C depending on σ . Put $v_\varphi := \text{curl} \Pi^{\text{HCT}}\varphi$. It is continuous and piecewise $P_2 \subset P_k$ on $\mathcal{T}_h$, and divergence-free. Since the interpolant is local and $\varphi \equiv 0$ near ∂R , $v_\varphi = 0$ near ∂R . So $v_\varphi \in Z_h$, with L^∞ error $Ch^3 \leq Ch^k$ and gradient error Ch^2 .

- $k \geq 4$. By Lemma 12.2, $\mathcal{T}_h$ satisfies (M0)–(M2), and the construction of Section 6 (the Morgan–Scott interpolant of degree $k+1$ [46] on $\mathcal{T}_h$) gives the errors Ch^{k+1} and Ch^k . $\square$

Section 6 writes $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$, but its proofs use only the two bounds in (13). The $O(h^k)$ terms in step 1 of the lower bound of Theorem 6.1, in steps 4–5 of Theorem 6.4, in steps 1–2 of Theorem 6.6(i) and in step 4 of Corollary 6.7 are all pointwise values of $v_\varphi - w$ on Γ_h . Everywhere else only boundedness of ∇v_φ is used: in $\|v_\varphi\|$, $a_h(S_h, \cdot)$, $\langle \partial_n S_h, \cdot \rangle$ and $\langle e, \partial_n v \rangle$.

12.4. The theorem.

Theorem 12.8 (Clough–Tocher version of the paper). *Let $k \geq 2$ be fixed, and let $\mathcal{T}_h$ be the Clough–Tocher refinement of a macro mesh satisfying (CT0). Let V_h, Π_h be as in this section, and assume (D).*

- (i) *The following hold as stated, with constants $C, c, C_I, \kappa, c_1, M, \beta, \gamma_0, \gamma_2, h_0, \delta_0$ depending on σ, c_0, k, L (and on the data as in Section 2) instead of on (M0)–(M2), k, L :*

- Lemmas 3.1–3.5, 4.1, 4.2, 5.1, 5.2 and 6.3, and Corollary 5.3;
- Lemma 4.3 in the form of Lemma 12.5, Lemma 4.5 in the form of Lemma 12.4, and Lemma 4.6 in the form of Lemma 12.6;
- Theorems 6.1, 6.4, 6.6 and 7.2, Corollaries 6.2 and 6.7, Proposition 7.1 and Proposition 10.1(a).

In Lemma 4.1 the triangle T_e is replaced by K_e ; no triangle has two edges on Γ_h (Lemma 12.2(b)).

- (ii) *In particular, for $\gamma \in [\gamma_2, \gamma_{\max}]$ the method CNS* satisfies*

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h^k) \quad \text{whenever } h_\Gamma \geq h^{2(\sigma_k - k)}.$$

This is $h_\Gamma^{3/2} + h^2$ for $k = 2$ with $h_\Gamma \geq h^4$, and $h_\Gamma^{3/2} + h^3$ for $k = 3$ with $h_\Gamma \geq h^2$. The printed method GS has the leading error $(h/\mu)v_\varphi$ of Corollary 6.7, with constant 1 in the slip and energy norms.

Proof. We go through the paper section by section and record every place where the mesh, the degree or the element enters.

- (1) *Section 3.* Lemmas 3.1–3.4 concern only Γ, Γ_h and Ω_h , which are unchanged. Lemma 3.5 uses, besides these, only the inverse trace inequality $|e| \|\partial_\tau v\|_{L^2(e)}^2 \leq C_I^2 \|\nabla v\|_{K_e}^2$ on the refined triangle K_e . This holds by Lemma 12.2(a,b) and a scaling argument.
- (2) *Section 4.* Lemma 4.1 holds on K_e for the same reason, and the K_e are distinct for distinct e . Lemma 4.2 uses only Lemmas 3.2 and 4.1, so its formulas for μ_0, γ_0, c_1, M are unchanged, with the new C_I . For the remaining three lemmas, see Lemmas 12.5, 12.4 and 12.6.
- (3) *Section 5.* Lemma 5.1 is an identity for smooth $\tilde{u}$ and any $v \in V_h^R$. Lemma 5.2 uses Lemmas 3.3, 3.5 and 4.1. Corollary 5.3 uses $\text{div } v = 0$ pointwise for $v \in Z_h$.
- (4) *Section 6.* Take v_φ from Lemma 12.7; by the discussion after it, (13) is all that Theorems 6.1, 6.4, 6.6 and Corollaries 6.2, 6.7 need. Lemma 6.3 uses only three facts: on e , $v \in Z_h$ is a

polynomial on K_e with $\operatorname{div} v = 0$ pointwise, so $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$ on e ; v is continuous at the vertices of Γ_h ; and Lemmas 4.1 and 3.2 hold. All three are available.

The remaining steps (the split $\delta_R + \delta_G$, Lax–Milgram on Z_h , the ansatz r_h , and the lower-bound arguments) are abstract: they use only Lemma 4.2, Lemma 4.6 through $E_h \leq C\varepsilon_h$, and Lemma 3.2. The lift E in the proof of Theorem 6.6(ii) uses $\|g - g_I\|_{H^{1/2}(\partial R)} \leq Ch^{k+1/2}$, which holds for every $k \geq 1$.

- (5) *Section 7.* Proposition 7.1 is algebraic. In Theorem 7.2, step 2 uses $\operatorname{div} V_h^0 \subset \Pi_h$ with π_h the $L^2(\Omega_h)$ projection onto Π_h (on $\mathcal{T}_h$). Step 3 uses Lemma 12.4. Step 4 uses $\|\eta\|_{L^\infty(K_e)} \leq C(\operatorname{diam} K_e)^k \leq Ch_\Gamma^k$. Step 6 uses the bubble $\beta_e n_e$ on K_e .
- (6) *Section 10(a)* is Lemma 4.2.

The rate in (ii) is Theorem 7.2 combined with Lemma 12.6. $\square$

Remark 12.9. For $k = 2$ the bulk term h^2 dominates $h_\Gamma^{3/2}$ exactly when $h_\Gamma \leq h^{4/3}$. So for quasi-uniform meshes the CNS* error is $O(h^{3/2})$ for every $k \geq 2$, and the degree matters only through ε_h . The constant-pressure benchmark of [34] is the case $G = 0$ of Theorem 6.1. For quartic velocities on unsplit meshes it is also analysed in [20]; that record does not mention Clough–Tocher splits, and the proof here does not use [20].

12.5. The penalty threshold on Clough–Tocher stars. The necessity half of Proposition 10.1 depends on the element, and we recertify it. A *Clough–Tocher star* at a boundary vertex z is the union of the macro triangles containing z , each split at its barycentre. We use the reference stars of Section 10, now split, together with two more. All are fans at $O = (0, 0)$ with Nitsche boundary $\{y = 0\}$:

- $\mathcal{S}_2$: rim points $(1, 0), (0, 1), (-1, 0)$;
- $\mathcal{S}_3$: rim points $(1, 0), (1, 1), (-1, 1), (-1, 0)$;
- $\mathcal{S}_4$: rim points $(1, 0), (1, 1), (0, 1), (-1, 1), (-1, 0)$;
- $\mathcal{S}'_4$: rim points $(1, 0), (1, 1), (0, \frac{6}{5}), (-1, 1), (-1, 0)$.

Without the split, $\mathcal{S}_2$ and $\mathcal{S}_4$ contain singular vertices, and Section 10 had to exclude them. After the split every vertex is non-singular (Lemma 12.2(c)).

Proposition 12.10 (Penalty necessity, Clough–Tocher). *Let z be a vertex of a straight part of the Nitsche boundary, and let ℓ_z be the length of its boundary edges. Suppose its Clough–Tocher star, dilated by the factor $1/\ell_z$, is congruent to the split star $\mathcal{S}$. Let $k \geq 2$, and let $\lambda_k(\mathcal{S})$ be the certified value in Table 9, taking $\lambda_k := \lambda_4$ for $k \geq 5$. If $\mu\ell_z/h < \lambda_k(\mathcal{S})$, then N_h is indefinite on Z_h .*

The same holds, with the same constants and by continuity, at every vertex of Γ_h with interior angle $\pi + O(h_\Gamma)$ whose split star is a small perturbation of a split reference star. No exclusion like $(M1')$ is needed; in particular the case of two macro triangles ($\mathcal{S}_2$), which is typical at vertices of Γ_h , is covered.

Computation: `code/lamstar_exact_ct.py` (Part 2); output `logs/lamstar_exact_ct.log`.

Proof. (1) The witnesses. Steps 1–4 of the proof of Proposition 10.1(b) apply verbatim. The witness is a continuous piecewise- P_k field on the split star, pointwise divergence-free, vanishing on the rim chords, which are macro edges. Extended by zero it therefore lies in Z_h .

- (2) Monotonicity in k . A witness for degree k is also continuous piecewise- $P_{k'}$ for $k' \geq k$, so λ^* is nondecreasing in k ; this justifies $\lambda_k := \lambda_4$ for $k \geq 5$. The witnesses of Section 10 (unsplit P_4 fans) are also piecewise P_4 on the split fans. For $k \geq 4$ they certify only 9, which Table 9 supersedes.

- (3) The certificate. The computation follows Section 10, with two changes. First, the per-element polynomials live on the split triangles, whose vertices (including barycentres) are rational.

		$\mathcal{S}_2$	$\mathcal{S}_3$	$\mathcal{S}_4$	$\mathcal{S}'_4$
$k = 2$	kernel dimension	6	7	8	8
	certified λ_2	5.71	7.54	7.54	7.52
	exact quotient	5.71715	7.54884	7.54467	7.52180
$k = 3$	kernel dimension	18	24	30	30
	certified λ_3	24.15	24.81	24.80	24.80
	exact quotient	24.15473	24.81227	24.80848	24.80696
$k = 4$	kernel dimension	36	50	64	64
	certified λ_4	48.38	48.78	48.78	48.77
	exact quotient	48.38946	48.78255	48.78181	48.77803

TABLE 9. Clough–Tocher split stars. *Kernel dimension*: the exact dimension of the divergence-free space of continuous piecewise- P_k fields on the split star vanishing on the rim. It equals $\dim V_{\mathcal{S}} - \dim \Pi_{\mathcal{S}}$ in every case, with $\text{rank div} = \dim \Pi_{\mathcal{S}}$. *Certified λ_k* : the rational λ_0 for which $2B - A - \lambda_0 C > 0$ is verified exactly on a rational witness. *Exact quotient*: the exact value of $(2B - A)/C$ at that witness. (Rounded to five decimals; the certified row is the rigorous statement.) It is a lower bound for the largest Rayleigh quotient $\lambda^*(\mathcal{S})$, and it agrees with a floating-point computation of $\lambda^*(\mathcal{S})$ to about 10^{-5} relative.

Second, the maximisation is reduced exactly. The kernel $\ker \text{div}$ and the splitting of it into $\ker C$ (fields with zero trace on the Nitsche boundary) and a complement are computed in exact rational arithmetic. On $\ker C$, $B = 0$ and $A > 0$, so the Schur complement of $2B - A$ onto the complement is formed exactly. Only the top eigenvector of this small exact pencil is computed in floating point, then rationalised.

A floating-point splitting misclassifies directions with tiny but non-zero trace. For $k = 3$ it overestimated λ^* (a spurious 25.11), which the exact check then rejected. We report only exactly verified values.

- (4) Continuity. For every Clough–Tocher star, not only the reference ones, $\text{div} : V_{\mathcal{S}} \rightarrow \Pi_{\mathcal{S}}$ is onto. Here $V_{\mathcal{S}}$ is the space of continuous piecewise- P_k fields vanishing on the rim, and $\Pi_{\mathcal{S}}$ the space of discontinuous piecewise- P_{k-1} functions. Indeed, the proof of Lemma 12.4 on the fixed Lipschitz polygon $\mathcal{S}$ (step 1 now being the continuous inf-sup condition on $\mathcal{S}$) gives $\text{div } V_0(\mathcal{S}) = \Pi_{\mathcal{S}} \cap L_0^2$, and the β_e bubble on a Nitsche edge adds the constants.

Hence $\dim \ker \text{div} = \dim V_{\mathcal{S}} - \dim \Pi_{\mathcal{S}}$ does not depend on the vertex positions, for fixed combinatorics. The divergence-free subspace is the kernel of a constraint matrix whose entries are polynomial in the vertex coordinates and whose rank is constant, so it depends continuously on them. The forms A , B , C (with edgewise normals) do as well, so the strict inequality $2B - A - \lambda_0 C > 0$ persists under small perturbations. This includes the rotation by $O(h_{\Gamma})$ of one Nitsche edge at a vertex of Γ_h and unequal edge lengths ℓ, ℓ' with ℓ'/ℓ near 1. $\square$

Remark 12.11 (Size of the threshold). The local thresholds on split stars are much larger than on unsplit ones: the exact witness quotients, which are lower bounds for λ^* , are 48.8 versus 9.98 on $\mathcal{S}_3$ for $k = 4$. They grow with k (7.5, 24.8, 48.8 for $k = 2, 3, 4$). This is consistent with the inverse-trace constant of the thinner boundary triangle K_e , whose height is one third of that of T_e , and with its growth in k . Sufficiency (Lemma 4.2) and necessity (Proposition 12.10) still both scale with ρ , so the conclusion $\mu_0 \asymp \rho$ of Section 10 holds for Clough–Tocher meshes whose shortest boundary edge carries such a star. The threshold $\gamma^* \approx 18$ –21 measured on unsplit $k = 4$

meshes does not transfer. On the Clough–Tocher refinements of the same meshes we measured $\gamma^* \approx 11.5$ ($k = 2$), 34.6 ($k = 3$) and 65.5 ($k = 4$) at the finest levels (Table 22), each above the corresponding local certificate of Table 9.

Remark 12.12 (Tall split stars). The local certificates of Table 9 are for layer height 1 per unit chord. On split column stars of height H they fail: in exact arithmetic no field supported in the split star is a witness at any $\mu \geq 0$ for $k = 2$ at $H = 2$ (macro minimum angle 26.6°), $k = 3$ at $H = 4$ and $k = 4$ at $H = 8$ (Proposition 10.6(c)). The explicit-constant necessity of Theorem 10.9 transfers to Clough–Tocher meshes with the HCT basis, but its constants have not been computed.

Remark 12.13 (Scope). Theorem 12.8 covers Sections 3–7 up to and including Theorem 7.2, and Proposition 10.1. We have not re-checked the identification of the H^1 constant (Section 6.1), the lower bound (Section 7.1), the single-triangle witness of Section 10.1, the uniform-in-penalty leak limit (Section 7.3), the L^2 theory (Section 9), strong imposition (Section 11), or Sections 13–15 on Clough–Tocher meshes, and make no claim about them there.

13. GENERAL SMOOTH DOMAINS

Sections 2–10 treat the exterior of the unit disk in a square. Here we extend them to a bounded connected domain whose no-slip boundary Γ consists of finitely many closed C^3 curves, convex or not, each approximated by an inscribed polygon, while the rest of the boundary, $\Sigma \neq \emptyset$, is polygonal and carries strong Dirichlet data. The main new points are:

- a curvature version of the chord lemma;
- a normal-coordinate map Φ_h that works when $\Omega \setminus \Omega_h$ and $\Omega_h \setminus \Omega$ are both nonempty;
- a chart-based star-shaped cover that gives a uniform Bogovskiĭ constant;
- with several curved components, the correct pressure constant is the *global* boundary mean. This forces a new test field (a divergence-free field without a global stream function) and a new discrete-curl interpolant.

The pressure-consistent method needs only one global correction, and its discrete velocity does not depend on how that constant is normalised; a per-component correction is inconsistent. The geometric lemmas use only $\Gamma \in C^3$. All results also assume the regularity (R1) of the exact solution ($u \in C^{k+1}(\bar{\Omega})$, $p \in C^k(\bar{\Omega})$), which for generic f requires $\Gamma \in C^{k+1,\alpha}$ and, at the vertices of Σ , vanishing corner coefficients (Remark 2.5). The results that use the leak test field (the lower bound of Theorem 6.1, Theorems 6.4 and 6.6, Corollary 6.7) are proved under the stronger assumption $\Gamma \in C^{k+3}$, $p|_\Gamma \in C^{k+1}$ (assumption (R2) below). The scope is Sections 3–10 up to and including Theorem 7.2 and Proposition 10.1; see Section 13.8. No computations have been run for general domains.

13.1. Setting and assumptions.

13.1.1. Domain.

Assumption 13.1 (Domain). $\Omega \subset \mathbb{R}^2$ is a bounded connected open set with $\partial\Omega = \Gamma \sqcup \Sigma$, where

- (a) $\Gamma = \Gamma_1 \sqcup \cdots \sqcup \Gamma_m$, $m \geq 1$, and each Γ_i is a whole connected component of $\partial\Omega$ and a closed embedded curve of class C^3 ;
- (b) $\Sigma \neq \emptyset$ is a finite disjoint union of closed simple polygons, each a whole component of $\partial\Omega$.

Which components are “inner” or “outer” is not restricted: the outer boundary may be curved (with Σ an inner polygonal hole) or polygonal. Components mixing curved and straight pieces are *not* covered (the junctions are corners of Γ).

For the domain of Section 2, $m = 1$, $\Gamma_1 = \partial D$, $\Sigma = \partial R$.

For each i fix an arclength parametrisation $\gamma_i : \mathbb{R}/|\Gamma_i|\mathbb{Z} \rightarrow \Gamma_i$, let $N_i(s)$ be the unit normal pointing into Ω , and orient γ_i so that (T_i, N_i) , $T_i := \gamma_i'$, is positively oriented. The signed curvature $\varkappa$ is defined by $T_i' = \varkappa N_i$ (so $N_i' = -\varkappa T_i$). Thus $\varkappa > 0$ where Ω is locally convex near Γ and $\varkappa < 0$ where it is locally concave; for the exterior of the unit disk $\varkappa \equiv -1$. The curvature is written $\varkappa$ because κ is the Korn constant of Lemma 3.2. Since $\Gamma \in C^3$, $\varkappa \in C^1(\Gamma)$.

On Γ we write $n := -N$ (the outward normal of Ω); on Γ_h , n is the outward normal of Ω_h , as in Section 2.

Fact 13.2 (Tubular neighbourhood; standard). *There is $r > 0$ with $r\|\varkappa\|_\infty \leq \frac{1}{2}$ such that, with $\mathcal{T}_r := \{x : \text{dist}(x, \Gamma) < r\}$:*

- (i) $\Psi(s, d) := \gamma_i(s) + dN_i(s)$ is a C^2 diffeomorphism from $\bigsqcup_i (\mathbb{R}/|\Gamma_i|\mathbb{Z}) \times (-r, r)$ onto $\mathcal{T}_r$, and d is the signed distance to Γ (positive in Ω), so $\mathcal{T}_r \cap \Omega = \{d > 0\}$;
- (ii) $d \in C^3(\mathcal{T}_r)$, the projection $\pi(x) = x - d(x)\nabla d(x)$ and the arclength coordinate $\sigma := s \circ \Psi^{-1}$ are C^2 , and

$$\nabla d = N \circ \pi, \quad \nabla \sigma = \frac{T \circ \pi}{J}, \quad J(x) := 1 - \varkappa(\pi(x)) d(x) \in [\frac{1}{2}, \frac{3}{2}], \quad dx = J dd ds;$$

- (iii) $\text{dist}(\mathcal{T}_r, \Sigma) > 0$, and $\text{dist}(\Gamma_i, \Gamma_j) > 2r$ for $i \neq j$.

In the orthogonal coordinates (s, d) (scale factors J and 1), a field $F N \circ \pi + G T \circ \pi$ has

$$\text{div}(F N \circ \pi + G T \circ \pi) = \frac{1}{J}(\partial_d(JF) + \partial_s G). \quad (14)$$

Fact 13.3 (Uniform local graphs; standard). *There is $a_0 \in (0, r/4]$ with $4\|\varkappa\|_\infty a_0 \leq \frac{1}{5}$ such that for every $y = \gamma_i(s_0) \in \Gamma$, in the Cartesian frame $(y; T_i(s_0), N_i(s_0))$ with coordinates (ξ, η) and with $Q'_y := (-2a_0, 2a_0)^2$,*

$$\Gamma \cap Q'_y = \{(\xi, f_y(\xi)) : |\xi| < 2a_0\}, \quad \Omega \cap Q'_y = \{\eta > f_y(\xi)\},$$

where $f_y \in C^3$, $f_y(0) = f_y'(0) = 0$, $f_y'' = \varkappa(1 + f_y'^2)^{3/2}$, $|f_y'(\xi)| \leq 2\|\varkappa\|_\infty|\xi| \leq \frac{1}{5}$, $|f_y(\xi)| \leq \|\varkappa\|_\infty \xi^2$ and $\|f_y'''\|_\infty \leq C_\Gamma$ (depending on $\|\varkappa\|_{C^1}$). Moreover $Q'_y \subset \mathcal{T}_r$.

Both facts concern the fixed curve only (tubular neighbourhood theorem, regularity of the distance function for C^3 curves, implicit function theorem and compactness); no h enters.

13.1.2. Inscribed polygons and mesh.

Assumption 13.4 (Inscribed polygons).

- (G1) For each i , $\Gamma_{h,i}$ is the closed polygon whose vertices $z_{i,0}, \dots, z_{i,N_i-1} \in \Gamma_i$ are in cyclic order along γ_i , consecutive vertices being joined by chords. For a chord e , Γ_e denotes the arc of Γ_i between its endpoints containing no other vertex, and $|\Gamma_e| \leq 2|e|$. $\Gamma_h := \bigcup_i \Gamma_{h,i}$.
- (G2) $h_\Gamma \leq h_*$, where $h_* \leq a_0/4$ is small in terms of Γ , Σ and the data (it is decreased finitely often below).

(G1) only excludes a chord that “wraps around” the long way; it is automatic if, e.g., the vertices are images of a quasi-uniform partition of the parameter.

The domain Ω_h is defined in Lemma 13.9. The mesh assumptions (M0)–(M2), (M1'), (D) of Assumption 2.1 are kept verbatim with ∂R replaced by Σ ; $\mathcal{T}_h$ is a conforming triangulation of Ω_h and $\mathcal{E}_h^\Gamma$ its edges on Γ_h . The spaces $V_h, V_h^R, V_h^0, \Pi_h, Z_h, W_h$, the forms a_h, N_h , the methods GS and CNS*, and the norm $\|\cdot\|$ are those of Section 2, with ∂R replaced by Σ . In CNS*,

$$\bar{p}_\Gamma(q) := |\Gamma_h|^{-1} \int_{\Gamma_h} q,$$

the mean over *all* of Γ_h (see Section 13.6 for why this is the right choice).

13.1.3. Exact solution, regularity and the pressure constant.

Assumption 13.5 (Regularity).

- (R1) $u \in C^{k+1}(\bar{\Omega})$, $p \in C^k(\bar{\Omega})$ solve $-\Delta u + \nabla p = f$, $\operatorname{div} u = 0$ in Ω , $u|_{\Gamma} = 0$, $u|_{\Sigma} = g$; $\tilde{f}$ as in (D).
 (R2) (Only for results that use the test field of Section 13.5: the lower bound of Theorem 13.22, Theorems 13.23 and 13.24, Corollary 13.25.) $\Gamma \in C^{k+3}$ and $p|_{\Gamma} \in C^{k+1}(\Gamma)$.

Since $u|_{\Gamma} = 0$ and $\operatorname{div} u = 0$, necessarily $\int_{\Sigma} g \cdot \nu = 0$ (ν the outward normal of Ω).

(R1) is a hypothesis on the solution. The $H^2 \times H^1$ regularity of the Stokes problem with L^2 data, which the duality arguments of Section 9 need, is a property of the domain, and it holds exactly for convex outer polygons.

Theorem 13.6 (Stokes regularity on general domains). *Let Ω satisfy Assumption 13.1 (for this statement $\Gamma_i \in C^2$ suffices). The estimate of Theorem 9.1 holds on Ω if and only if every vertex of Σ has interior angle $< \pi$ measured in Ω , that is, if and only if Σ is a single convex polygon forming the outer boundary of Ω (so $D = \bigcup_i D_i$ is any finite union of disjoint smooth obstacles in a convex polygonal box). Proposition 9.2 holds on a collar of each Γ_i for $g \in H^m$ there, provided $\Gamma_i \in C^{m+2}$.*

Proof. Sufficiency. A field $\varphi \in H_0^1(\Omega)$ has zero flux through every component of $\partial\Omega$, so its stream function ζ is single valued; normalise $\int_{\Omega} \zeta = 0$. Take a smooth partition of unity $\sum_{\ell} \chi_{\ell} = 1$ on $\bar{\Omega}$ subordinate to annular regions A_i between Γ_i and a smooth curve in the tube $\{r/4 < d < r/2\}$ of Fact 13.2, and to the convex polygon P bounded by Σ (with $\chi_{\ell} = 0$ on $\bigcup_i \{d < r/4\}$ for the latter). Then $\varphi = \sum_{\ell} \operatorname{curl}(\chi_{\ell} \zeta)$, and each piece is treated as in the proof of Theorem 9.1: (K1) on P , (K2) on A_i . The boundary values of ζ never enter, because $\operatorname{curl} \chi_{\ell} = 0$ near $\partial\Omega$. A polygonal hole has at least three vertices that are re-entrant for Ω , and an outer polygon has no re-entrant vertex iff it is convex, so the two descriptions agree.

Necessity. At a vertex of angle $\omega \in (\pi, 2\pi)$ the smallest Stokes exponent satisfies $\lambda(\omega) \in (\frac{1}{2}, 1)$ [24] (numerically $\lambda(3\pi/2) = 0.544484$). With the corresponding homogeneous solution $(S, Q) = (r^{\lambda} \Phi(\theta), r^{\lambda-1} \Psi(\theta))$, a cut-off χ at the vertex and a smooth compactly supported b with $\operatorname{div} b = \operatorname{div}(\chi S)$ (whose mean is zero, since $\nabla \chi \cdot S$ has zero flux), $\varphi := \chi S - b$ is the weak solution for a right-hand side in L^2 , and $\varphi \notin H^2$ because $|\nabla^2 S| \sim r^{\lambda-2}$. $\square$

So the L^2 duality arguments of Section 9 can be transferred to smooth obstacles in a convex polygonal box (with the tube-coordinate version of Lemma 9.4, which we have not re-checked), but not to domains with a polygonal hole, where the dual is only in H^{1+s} , $s < \lambda_{\min}$, near the re-entrant vertices.

Definition 13.7 (Pressure constant). $\bar{p} := |\Gamma|^{-1} \int_{\Gamma} p \, ds$ (the mean over all curved components), $q := p - \bar{p}$ on Γ , and

$$G := \|p - \bar{p}\|_{L^2(\Gamma)}, \quad \bar{q}_i := |\Gamma_i|^{-1} \int_{\Gamma_i} q \quad (\text{so } \sum_i |\Gamma_i| \bar{q}_i = 0).$$

$$C_p := C(1 + \|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}).$$

Proposition 13.26 shows that the per-component quantity $G_{\text{per}}^2 := \sum_i \|p - \bar{p}_i\|_{L^2(\Gamma_i)}^2$ ($\bar{p}_i$ the mean over Γ_i) is *not* the right one when $m \geq 2$. Note $G^2 = G_{\text{per}}^2 + \sum_i |\Gamma_i| \bar{q}_i^2$.

Constants. C, c depend on (M0)–(M2), k , the geometry ($\|\mathcal{K}\|_{C^1}$, r , a_0 , $|\Gamma|$, Σ , the fixed cover of Lemma 13.15), and norms of $(u, p, \tilde{u}, \tilde{p})$; they are independent of $h, h_{\Gamma}, \mu, \gamma, \rho$.

13.2. Geometry.

Lemma 13.8 (Chords; replaces Lemma 3.1). *Let $e \in \mathcal{E}_h^\Gamma$ be a chord of Γ_i of length ℓ , with midpoint m_e and unit tangent τ_e oriented like T_i along Γ_e . For $x \in e$ let $t \in [-\ell/2, \ell/2]$ be its coordinate along e from m_e , $x^* := \pi(x)$, and $\varkappa_e := \varkappa(m_e^*)$. Let N_e be the unit normal of e with $N_e \cdot N(x^*) > 0$ and $n_e := -N_e$. Then:*

- (a) (sagitta) $d(x) = \frac{1}{2}\varkappa(x^*)(\ell^2/4 - t^2) + O(\ell^3)$ and $|d(x)| \leq \|\varkappa\|_\infty \ell^2/8 + C\ell^4$. If $\varkappa \geq 0$ on Γ_e then $e \subset \overline{\Omega}$; if $\varkappa \leq 0$ on Γ_e then $e \cap \Omega = \emptyset$.
- (b) (normal rotation) $n(x^*) - n_e = \varkappa_e t \tau_e + O(\ell^2)$. In particular $n(x^*) \cdot n_e = 1 + O(\ell^2)$ and $|n(x^*) \cdot \tau_e| + |N(x^*) \cdot \tau_e| \leq C\ell$.
- (c) (Jacobian) $\pi|_e : e \rightarrow \Gamma_e$ is a C^2 bijection with $\frac{d(\sigma \circ \pi)}{dt} = \frac{T(x^*) \cdot \tau_e}{J(x)} = 1 + O(\ell^2)$; hence $|\Gamma_e| = \ell(1 + O(\ell^2))$.
- (d) (turning) If e, e' share a vertex, $|\tau_e - \tau_{e'}| + |n_e - n_{e'}| \leq 2\|\varkappa\|_\infty(|\Gamma_e| + |\Gamma_{e'}|)$.

The $O(\cdot)$ constants depend only on $\|\varkappa\|_{C^1}$. For $\varkappa \equiv -1$ this is Lemma 3.1 (there s plays the role of t).

Proof. By (G1), $|\Gamma_e| \leq 2h_\Gamma \leq a_0/2$, so Γ_e lies in Q'_z for its first endpoint z and is a graph there. By the mean value theorem there is $y_0 \in \Gamma_e$ with $T(y_0) \parallel e$. Work in the frame of Fact 13.3 at y_0 , with $f := f_{y_0}$ (the arc is still a graph there since $|\Gamma_e| \leq a_0/2$). The chord is the horizontal segment $\{(\xi, c) : \xi_a \leq \xi \leq \xi_b\}$ with $f(\xi_a) = f(\xi_b) = c$, $\xi_b - \xi_a = \ell$, $0 \in [\xi_a, \xi_b]$, and $\tau_e = e_1$, $N_e = e_2$. Put $\xi_m := (\xi_a + \xi_b)/2$, so $t = \xi - \xi_m$.

(a) Linear interpolation error: $c - f(\xi) = \frac{1}{2}f''(\zeta)(\xi - \xi_a)(\xi_b - \xi)$ for some $\zeta \in (\xi_a, \xi_b)$, and $(\xi - \xi_a)(\xi_b - \xi) = \ell^2/4 - t^2$. Since $|f'| \leq 2\|\varkappa\|_\infty \ell$ on $[\xi_a, \xi_b]$, $f'' = \varkappa(1 + f'^2)^{3/2} = \varkappa(1 + O(\ell^2))$, and $\varkappa$ at the graph point over ζ differs from $\varkappa(x^*)$ by $\|\varkappa'\|_\infty \cdot O(\ell)$. Next, d is C^2 with $d = 0$ on the graph and $\nabla d = N$, so $d(\xi, c) = (c - f(\xi))N_2 + O((c - f)^2)$ with $N_2 = (1 + f'^2)^{-1/2} = 1 + O(\ell^2)$. Hence $d(x) = \frac{1}{2}\varkappa(x^*)(\ell^2/4 - t^2) + O(\ell^3)$, and $|d(x)| \leq \frac{1}{2}\|\varkappa\|_\infty(1 + C\ell^2)\ell^2/4 + C\ell^4$. The sign statements follow because f'' has the sign of $\varkappa$ and $\Omega \cap Q'_z = \{\eta > f\}$.

(b) $N(x^*) = (1 + f'(\xi^*)^2)^{-1/2}(-f'(\xi^*), 1)$, so $N(x^*) - N_e = -f'(\xi^*)e_1 + O(\ell^2)$. The projection moves the abscissa by $|d||f'| = O(\ell^3)$, so $f'(\xi^*) = f'(\xi) + O(\ell^3) = f'(\xi_m) + f''(\cdot)t + O(\ell^3)$. The midpoint rule for divided differences gives $f'(\xi_m) = \frac{f(\xi_b) - f(\xi_a)}{\ell} + O(\ell^2\|f'''\|_\infty) = O(\ell^2)$, and $f''(\cdot) = \varkappa(x^*) + O(\ell) = \varkappa_e + O(\ell)$. Thus $N(x^*) - N_e = -\varkappa_e t \tau_e + O(\ell^2)$; multiply by -1 .

(c) By Fact 13.2(ii), $\frac{d}{dt}\sigma(\pi(x)) = \nabla\sigma \cdot \tau_e = T(x^*) \cdot \tau_e/J(x)$. Here $|T(x^*) - T(y_0)| \leq \|\varkappa\|_\infty|\Gamma_e|$, so $T(x^*) \cdot \tau_e = 1 + O(\ell^2)$, and $J = 1 + O(\ell^2)$ by (a). The derivative is positive, and the endpoints are fixed by π , so $\pi|_e$ is a bijection onto Γ_e .

(d) $\tau_e = T(y_0)$ and $\tau_{e'} = T(y'_0)$ with $y_0 \in \Gamma_e$, $y'_0 \in \Gamma_{e'}$, and $|T(y) - T(y')| \leq \|\varkappa\|_\infty \cdot (\text{arc distance})$; n_e is τ_e rotated by a fixed angle. $\square$

Lemma 13.9 (Γ_h is a normal graph; definition of Ω_h). *For $h_\Gamma \leq h_*$:*

- (a) $\Gamma_{h,i} = \{\gamma_i(s) + \eta_h(s)N_i(s)\}$ for a Lipschitz, piecewise C^2 function η_h with

$$\|\eta_h\|_\infty \leq \|\varkappa\|_\infty h_\Gamma^2/8 + Ch_\Gamma^4, \quad \|\eta'_h\|_\infty \leq Ch_\Gamma.$$

In particular the $\Gamma_{h,i}$ are simple closed polygons, pairwise disjoint and disjoint from Σ , and the vertex angle of each is $\pi + O(h_\Gamma)$ (this justifies (M1')).

- (b) Define

$$\Omega_h := (\Omega \setminus \mathcal{T}_{r/2}) \cup \{x \in \mathcal{T}_{r/2} : d(x) > \eta_h(\sigma(x))\}.$$

Then Ω_h is a bounded polygonal open set with $\partial\Omega_h = \Sigma \sqcup \Gamma_h$, it lies on the Ω -side of each chord (so the outward normal of Ω_h on e is n_e of Lemma 13.8), and

$$\Omega_h \setminus \Omega = \{\eta_h(\sigma) < d \leq 0\}, \quad \Omega \setminus \Omega_h = \{0 < d \leq \eta_h(\sigma)\}, \quad |\Omega \triangle \Omega_h| \leq Ch_\Gamma^2.$$

Both differences are nonempty as soon as $\varkappa$ takes both signs on Γ (for h_Γ small).

Proof. (a) By Lemma 13.8(c), π maps each chord bijectively onto its arc; the arcs Γ_e of consecutive chords tile Γ_i (G1). So $\pi|_{\Gamma_{h,i}} : \Gamma_{h,i} \rightarrow \Gamma_i$ is a bijection and $\eta_h(s) := d((\pi|_{\Gamma_{h,i}})^{-1}(\gamma_i(s)))$ represents $\Gamma_{h,i}$ as stated. The sup bound is Lemma 13.8(a). On a chord, $\frac{d\eta_h}{ds} = \nabla d \cdot \tau_e \frac{dt}{ds} = N(x^*) \cdot \tau_e (1 + O(\ell^2)) = O(\ell)$ by Lemma 13.8(b),(c). Since Ψ is a diffeomorphism and $|\eta_h| < r/2$, $\Gamma_{h,i}$ is embedded; disjointness follows from Fact 13.2(iii). The vertex angle is π plus the turning angle, which is $O(h_\Gamma)$ by Lemma 13.8(d).

(b) Openness and $\partial\Omega_h = \Sigma \sqcup \Gamma_h$ are immediate from the definition and Fact 13.2. The descriptions of the differences follow from $\Omega \cap \mathcal{T}_r = \{d > 0\}$; their measure is $\int_\Gamma \int |\eta_h| J dd ds \leq Ch_\Gamma^2$. If $\varkappa(y) < 0$ at some y , then for small h_Γ the chord e with $y \in \Gamma_e$ has $\varkappa < 0$ on Γ_e , and Lemma 13.8(a) gives $d < 0$ in its interior; similarly for $\varkappa > 0$. $\square$

Lemma 13.10 (Uniform trace, Friedrichs, Korn; replaces Lemma 3.2). *There are $C_{\text{tr}}, C_F, \kappa, C_P$, independent of h , such that every $v \in H^1(\Omega_h)^2$ with $v|_\Sigma = 0$ satisfies*

$$\|v\|_{L^2(\Gamma_h)} \leq C_{\text{tr}} \|\nabla v\|, \quad \|v\| \leq C_F \|\nabla v\|, \quad \|\nabla v\|^2 \leq \kappa a_h(v, v),$$

and every $v \in H^1(\Omega_h)$ satisfies $\inf_{c \in \mathbb{R}} \|v - c\|_{\Omega_h} \leq C_P \|\nabla v\|_{\Omega_h}$. Moreover Ω_h is connected.

Proof. (1) Let $\chi \in C^\infty(\mathbb{R})$, $0 \leq \chi \leq 1$, $\chi = 1$ on $(-\infty, r/8]$, $\chi = 0$ on $[r/4, \infty)$, $|\chi'| \leq 16/r$. Define Φ_h on $\bar{\Omega}$ by

$$\Phi_h(\Psi(s, d)) := \Psi(s, d + \chi(d)\eta_h(s)) \quad (0 \leq d < r), \quad \Phi_h := \text{id} \text{ on } \bar{\Omega} \setminus \mathcal{T}_r.$$

(2) For fixed s , $d \mapsto d + \chi(d)\eta_h(s)$ has derivative $1 + \chi'\eta_h \geq 1 - Ch_\Gamma^2/r \geq \frac{1}{2}$, equals $\eta_h(s)$ at $d = 0$ and d for $d \geq r/4$. So Φ_h is a homeomorphism of $\bar{\Omega}$ onto $\bar{\Omega}_h$ mapping Γ_i onto $\Gamma_{h,i}$, with $\Phi_h = \text{id}$ outside $\mathcal{T}_{r/4}$, in particular near Σ . Hence Ω_h is connected.

(3) In (s, d) coordinates $\hat{\Phi}_h(s, d) = (s, d + \chi\eta_h)$ has $D\hat{\Phi}_h = I + \begin{pmatrix} 0 & 0 \\ \chi'\eta_h & \chi'\eta_h \end{pmatrix} = I + O(h_\Gamma)$ a.e. Since $D\Psi \in C^1$, $D\Psi(s, d')D\Psi(s, d)^{-1} = I + O(|d' - d|) = I + O(h_\Gamma^2)$. Therefore $D\Phi_h = D\Psi \circ \hat{\Phi}_h \cdot D\hat{\Phi}_h \cdot (D\Psi)^{-1}$ satisfies $\|D\Phi_h - I\|_{L^\infty} + \|D\Phi_h^{-1} - I\|_{L^\infty} \leq Ch_\Gamma$, and $\Phi_h|_\Gamma$ has arclength Jacobian $1 + O(h_\Gamma^2)$ (it is the inverse of $\pi|_{\Gamma_h}$, Lemma 13.8(c)).

(4) The rest is Lemma 3.2, steps 4–6, verbatim: for $w = v \circ \Phi_h$, $D(w) = D(v) \circ \Phi_h + E$ with $|E| \leq Ch_\Gamma |\nabla v \circ \Phi_h|$; on the fixed connected Lipschitz domain Ω , w vanishes on Σ , which has positive length, so Korn's first inequality, the trace inequality on Γ and the Friedrichs inequality hold; transfer back and absorb $Ch_\Gamma \|\nabla v\|$ for $h_\Gamma \leq h_*$. The Poincaré inequality transfers the same way from Ω . $\square$

Here $\Sigma \neq \emptyset$ is used (Korn and Friedrichs with zero trace on Σ). The map Φ_h works for arbitrary signs of $\varkappa$: it moves points along normals in both directions.

Lemma 13.11 (Extension; replaces Lemma 3.3 and the extension paragraph of Section 2). *There are $\tilde{u} \in C^{k+1}$ and $\tilde{p} \in C^k$ on the open set $\mathcal{O} := \Omega \cup \mathcal{T}_r$ with $\tilde{u} = u$, $\tilde{p} = p$ on Ω and $\text{div } \tilde{u} = 0$ on $\mathcal{O}$. With $F := \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$ on Ω_h (so $F = 0$ on $\Omega_h \cap \Omega$ and $\text{supp } F \subset \Omega_h \setminus \bar{\Omega}$):*

$$\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2, \quad |(F, v)_{\Omega_h}| \leq Ch_\Gamma^2 \|\nabla v\| \quad \forall v \in V_h^R.$$

Proof. Construction. Let $U_i := \Psi(\Gamma_i \times (-r, r))$ and $U_i^+ := U_i \cap \Omega$, an annulus. The flux of u through $\Psi(\Gamma_i \times \{d\})$ is independent of $d \in (0, r)$ since $\text{div } u = 0$, and tends to $\int_{\Gamma_i} u \cdot N = 0$. So $u = \text{curl } \psi_i$ on U_i^+ with ψ_i single-valued and $\psi_i \in C^{k+2}(\overline{U_i^+})$. (A global stream function need not exist: when Σ has several components, g may have nonzero flux through each of them. The single-valued ψ of Section 2 is replaced by these local ones.) By Whitney's extension theorem ($\overline{U_i^+}$ has C^1 boundary, so C^{k+2} functions on it are Whitney jets), extend ψ_i to $\tilde{\psi}_i \in C^{k+2}(U_i)$,

and set $\tilde{u} := \operatorname{curl} \tilde{\psi}_i$ on $U_i \setminus \Omega$ and $\tilde{u} := u$ on Ω . On U_i , $\tilde{u} = \operatorname{curl} \tilde{\psi}_i$ on both sides, so $\tilde{u} \in C^{k+1}(\mathcal{O})$ and $\operatorname{div} \tilde{u} = 0$. $\tilde{p}$ is a Whitney extension of p .

Bounds. $\tilde{u} = 0$ on Γ and $|x - \pi(x)| = |d(x)| \leq Ch_\Gamma^2$ on Γ_h (Lemma 13.9) give the first bound. For the second, F is bounded on $\Omega_h \setminus \Omega = \{\eta_h(s) < d \leq 0\}$. For $x = \Psi(s, d)$ there, with $y_h(s) := \Psi(s, \eta_h(s)) \in \Gamma_h$,

$$|v(x)| \leq |v(y_h(s))| + \int_{\eta_h(s)}^0 |\nabla v(\Psi(s, t))| dt.$$

Integrate with $dx = J dd ds$ over the strip of width $|\eta_h(s)| \leq Ch_\Gamma^2$:

$$\int_{\Omega_h \setminus \Omega} |v| \leq Ch_\Gamma^2 \left(\int_{\Gamma} |v(y_h(s))| ds + \int_{\Omega_h \setminus \Omega} |\nabla v| \right) \leq Ch_\Gamma^2 (\|v\|_{L^1(\Gamma_h)} + h_\Gamma \|\nabla v\|),$$

using the Jacobian of Lemma 13.8(c) and $|\Omega_h \setminus \Omega|^{1/2} \leq Ch_\Gamma$. Finish with $\|v\|_{L^1(\Gamma_h)} \leq |\Gamma_h|^{1/2} \|v\|_{L^2(\Gamma_h)}$ and Lemma 13.10. $\square$

Remark 13.12 (The part $\Omega \setminus \Omega_h$). Where Γ is locally convex ($\varkappa > 0$) the chords cut off $\Omega \setminus \Omega_h$. Nothing needs to be extended there: all fields live on Ω_h and all integrations by parts are on Ω_h . The errors are measured on Ω_h as before, so $\Omega \setminus \Omega_h$ (area $O(h_\Gamma^2)$) is simply not approximated. On $\Omega_h \setminus \Omega$ the error $\tilde{u} - u_h$ depends on the extension; two C^2 extensions differ there by $O(h_\Gamma^2)$ in $W^{1,\infty}$ (their difference and its gradient vanish on Γ), hence by $O(h_\Gamma^3)$ in $H^1(\Omega_h)$, which is below every rate in this section.

Lemma 3.4 (wall structure) holds verbatim on any C^1 curve, with $n = -N$.

Lemma 13.13 (Transposed-gradient term; replaces Lemma 3.5). *The statement of Lemma 3.5 holds, with*

$$a_e := -\varkappa_e n(m_e^*) (\partial_n u(m_e^*) \cdot \tau_e).$$

Proof. Only step 1 changes. For $x \in e$, $\tilde{u} \in C^2$ gives $(\nabla \tilde{u}(x))^T n_e = (\nabla u(x^*))^T n_e + O(h_\Gamma^2)$. By Lemma 3.4, $(\nabla u)^T n(x^*) = 0$ and $(\nabla u)^T b = n(\partial_n u \cdot b)$ at x^* . With Lemma 13.8(b),

$$(\nabla u(x^*))^T n_e = n(x^*) (\partial_n u(x^*) \cdot (n_e - n(x^*))) = -\varkappa_e t n(x^*) (\partial_n u(x^*) \cdot \tau_e) + O(\ell^2) = t a_e + O(h_\Gamma^2),$$

since x^* and m_e^* are $O(\ell)$ apart. The coefficient a_e is constant on e , and $\int_e t dt = 0$. Steps 2–4 are unchanged; the sum over edges uses $\#\mathcal{E}_h^\Gamma \leq |\Gamma|/(c_0 h_\Gamma)$, since $|\Gamma_h| \leq |\Gamma|$. $\square$

For $\varkappa \equiv -1$ this is the a_e of Lemma 3.5. On straight parts of Γ ($\varkappa = 0$) the chords lie on Γ , and the leading term vanishes.

13.3. Discrete tools. Lemmas 4.1 and 4.2 are mesh-only and hold verbatim, with κ from Lemma 13.10. Lemma 4.3 and Remark 4.4 hold verbatim with $\partial R \rightarrow \Sigma$. The only domain property used in Lemma 4.3, step 1, is that Ω_h is connected (so that $\operatorname{div} V_h^0$ is the mean-zero part of Π_h), and this is Lemma 13.10. The edge bubble of step 2 may sit on any edge of any $\Gamma_{h,i}$.

13.3.1. Uniform continuous inf-sup constant.

Lemma 13.14 (Decomposition of the divergence equation). *Let $U = \bigcup_{j=1}^N U_j$ be a bounded open set, where each U_j is star-shaped with respect to every point of a ball B_j of radius $\geq r_*$ with $\overline{B_j} \subset U_j$, $\operatorname{diam} U_j \leq d_*$, and, for $j < N$, $|A_j| \geq m_*$ where $A_j := U_j \cap D_j$, $D_j := \bigcup_{l>j} U_l$. Then for every $f \in L_0^2(U)$ there is $v \in H_0^1(U)^2$ with $\operatorname{div} v = f$ and $\|\nabla v\| \leq C_B \|f\|$, where C_B depends only on N, r_*, d_*, m_* and $|U|$.*

Proof. Let $E_j := U_j \setminus D_j$ (so $E_N = U_N$ and the E_j partition U). Put $g_1 := f$ and, for $j = 1, \dots, N-1$,

$$f_j := g_j \mathbf{1}_{E_j} - \frac{\int_{E_j} g_j}{|A_j|} \mathbf{1}_{A_j}, \quad g_{j+1} := g_j - f_j, \quad f_N := g_N.$$

By induction, $\text{supp } g_j \subset E_j \cup D_j$ and $\int g_j = 0$. Hence $\text{supp } f_j \subset U_j$, $\int f_j = 0$ and $\text{supp } g_{j+1} \subset D_j$. Also $\|f_j\| \leq (1 + |U|^{1/2} m_*^{-1/2}) \|g_j\|$ and $\|g_{j+1}\| \leq \|g_j\| + \|f_j\|$. So $\|f_j\| \leq C(N, m_*, |U|) \|f\|$, and $f = \sum_j f_j$. By Bogovskiĭ's estimate on star-shaped domains [31, Lemma III.3.1] there is $v_j \in H_0^1(U_j)^2$ with $\text{div } v_j = f_j$ and $\|\nabla v_j\| \leq c_0(d_*/r_*)^2(1 + d_*/r_*) \|f_j\|$. Take $v := \sum_j v_j$, each extended by zero. $\square$

Lemma 13.15 (A uniform star-shaped cover of Ω_h). *There are $N, r_*, d_*, m_* > 0$, depending only on Ω , and h_* , such that for $h_\Gamma \leq h_*$, Ω_h admits a cover as in Lemma 13.14.*

Proof. Fix $a := a_0$ (Fact 13.3), so $\|\varkappa\|_\infty a \leq 1/20$.

Chart pieces. Choose $y_1, \dots, y_{N_1} \in \Gamma$ ($N_1 \leq 4|\Gamma|/a + m$) such that the arcs of length $a/2$ centred at the y_j cover Γ . In the frame of Fact 13.3 at y_j , let $Q_j := (-a, a)^2 \subset Q'_j := (-2a, 2a)^2$, $f := f_{y_j}$, and $U_j^h := \Omega_h \cap Q_j$.

- (1) $\Gamma_h \cap Q_j$ is the graph of $f_h := I_1 f$, the piecewise-linear interpolant of f at the abscissae of the vertices of Γ_h in Q'_j . A chord meeting Q_j has length $\leq h_\Gamma < a/4$, so both endpoints lie in $Q'_j \cap \Gamma$, i.e. on the graph of f . Chords of other components, and Σ , do not meet $Q'_j \subset \mathcal{T}_r$ (Fact 13.2(iii)). Moreover $|f'_h| \leq \sup_{|\xi| < 2a} |f'| \leq \frac{1}{5} =: L$ (slopes of $I_1 f$ are values of f'), and $|f_h| \leq \|\varkappa\|_\infty (a + h_\Gamma)^2 \leq a/5 =: b_0$.
- (2) $U_j^h = \{(\xi, \eta) \in Q_j : \eta > f_h(\xi)\}$. $Q_j \setminus \Gamma_h$ has exactly two components, above and below the graph. $\Omega_h \cap Q_j$ is relatively open and closed in $Q_j \setminus \Gamma_h$ (because $\partial\Omega_h \cap Q_j = \Gamma_h \cap Q_j$), so it is a union of components. The point $(0, a/2)$ lies in Ω at distance $> a/4$ from Γ , hence in Ω_h (as $\Omega \setminus \Omega_h \subset \{d \leq Ch_\Gamma^2\}$). The point $(0, -a/2)$ has $d = -a/2 < \eta_h$, so it is not in Ω_h .
- (3) U_j^h is star-shaped with respect to every point of $B_j := B((0, \frac{3}{4}a), \frac{a}{8})$. Let $y \in B_j$, $x \in U_j^h$ and $z_t := (1-t)y + tx$. The constraints $|\xi| < a$, $\eta < a$ are convex. Since $g := f_h$ is L -Lipschitz, $g(\xi_{z_t}) \leq g(\xi_x) + L(1-t)|\xi_x - \xi_y| \leq g(\xi_x) + L(1-t)\frac{9}{8}a$. Hence

$$\eta_{z_t} - g(\xi_{z_t}) \geq t(\eta_x - g(\xi_x)) + (1-t)\left(\frac{5}{8}a - b_0 - \frac{9}{8}La\right) = t(\eta_x - g(\xi_x)) + (1-t)\frac{a}{5} > 0.$$

Also $\overline{B_j} \subset U_j^h$, since $\frac{5}{8}a > b_0$ and $\frac{7}{8}a < a$. Finally $\text{diam } U_j^h \leq 2\sqrt{2}a$.

Interior pieces. Let $\Omega_* := \{x \in \Omega : \text{dist}(x, \Gamma) > a/4\}$. It is independent of h , and $\Omega_* \subset \Omega_h$ for $h_\Gamma \leq h_*$. It is connected: the map $\Psi(s, d) \mapsto \Psi(s, \frac{a}{4} + (1 - \frac{a}{4r})d)$ on $\{0 < d < r\}$, extended by the identity, is a homeomorphism $\Omega \rightarrow \Omega_*$. It is a bounded Lipschitz domain (its boundary is Σ together with the C^2 parallel curves $\{d = a/4\}$). By [31, Lemma II.1.3] it is a finite union $\Omega_* = \bigcup_{l=1}^{N_2} V_l$ of domains, each star-shaped with respect to every point of a ball. Since Ω_* is connected, the intersection graph of $\{V_l\}$ is connected. List the V_l in the reverse of a breadth-first order from a root; then each V_l except the last meets a later one in a nonempty open set. All of this is h -independent.

Covering. If $x \in \Omega_h$ and $\text{dist}(x, \Gamma) \geq a/2$, then $x \in \Omega_*$. If $\text{dist}(x, \Gamma) < a/2$, write $x = \pi(x) + dN$ with $\pi(x)$ within arclength $a/4$ of some y_j . In the frame at y_j , $|\xi_x| \leq \frac{a}{4} + \frac{a}{2} \cdot \frac{1}{5} < a$ and $|\eta_x| \leq \frac{a}{5} + \frac{a}{2} < a$, so $x \in U_j^h$. Hence $\Omega_h = \bigcup_j U_j^h \cup \bigcup_l V_l$.

Overlaps. Order the chart pieces first, then the V_l . Then $U_j^h \cap D_j \supset U_j^h \cap \Omega_* \supset y_j + \{|\xi| < \frac{a}{2}, \frac{a}{2} < \eta < a\}$. These points are above f_h , lie in Ω , and are at distance $\geq (\frac{a}{2} - b_0)/\sqrt{1 + L^2} > \frac{a}{4}$ from the graph part of Γ , and at distance $> \frac{a}{4}$ from $\Gamma \setminus Q'_j$. So $|A_j| \geq a^2/2$. The overlaps among the V_l are fixed positive numbers. Take $N = N_1 + N_2$, $r_* = \min(a/8, \text{radii of the } V_l)$, $d_* = \max(2\sqrt{2}a, \text{diam } V_l)$ and m_* the minimum overlap. $\square$

Lemma 13.16 (Uniform inf-sup constant; replaces Lemma 4.5). *The statement of Lemma 4.5 holds on Ω_h , with β depending only on k , Θ_0 , shape regularity and Ω .*

Proof. As in Lemma 4.5, the Guzmán–Scott argument [37] depends on the domain only through the continuous inf-sup (Bogovskii) constant and a Poincaré constant. The latter is uniform by Lemma 13.10, the former by Lemmas 13.14 and 13.15. The wedge cover and the hypothesis $P_h \subset D$ used in Lemma 4.5 are no longer needed. $\square$

13.3.2. *Approximation.* Lemma 4.6 and the remark after it hold verbatim with $\partial R \rightarrow \Sigma$, ν the outward normal of Ω on Σ , and $m_R := \int_{\Sigma} g_I \cdot \nu$. The only domain facts used are:

- $\tilde{u} \in W^{k+1,\infty}(\Omega_h)$ (Lemma 13.11);
- boundary triangles have diameter $\leq Ch_{\Gamma}$;
- $\int_{\Gamma_h} \tilde{u} \cdot n = \int_{\partial\Omega_h} \tilde{u} \cdot n - \int_{\Sigma} g \cdot \nu = \int_{\Omega_h} \operatorname{div} \tilde{u} - 0 = 0$;
- the edge-bubble field $b = \sum_{e \in \mathcal{E}_h^{\Gamma}} \beta_e n_e$ over all components;
- Lemma 13.16.

13.4. **Error equation and the flux structure.** Lemma 5.1 holds verbatim. Its proof integrates by parts on the polygon Ω_h , where $\tilde{u} \in C^2(\overline{\Omega_h})$, $\tilde{p} \in C^1(\overline{\Omega_h})$ and $v = 0$ on Σ . Nothing requires $\Omega \subset \Omega_h$. Lemma 5.2 holds verbatim by Lemmas 13.11 and 13.13.

For $v \in V_h^R$ put $\Phi_i(v) := \int_{\Gamma_{h,i}} v \cdot n$.

Lemma 13.17 (Fluxes).

- (a) For $v \in Z_h$, $\sum_{i=1}^m \Phi_i(v) = 0$.
- (b) If $m \geq 2$ and $h_{\Gamma}, h \leq h_*$ (with (R2) available for the construction, i.e. $\Gamma \in C^{k+3}$), then $v \mapsto (\Phi_i(v))_i$ maps Z_h onto $\{x \in \mathbb{R}^m : \sum_i x_i = 0\}$. In particular individual fluxes of Z_h functions do not vanish.

Proof. (a) $0 = \int_{\Omega_h} \operatorname{div} v = \int_{\Gamma_h} v \cdot n$, since $v = 0$ on Σ . (b) Given $\alpha \in \mathbb{R}^m$ with $\sum_i |\Gamma_i| \alpha_i = 0$, Lemma 13.19 below, applied with $q := \alpha_i$ on Γ_i (which is C^∞ on Γ , the components being disjoint, and has $\int_{\Gamma} q = 0$), and Lemma 13.20 give $v_\alpha \in Z_h$ with $v_\alpha = \alpha_i N + O(h_{\Gamma}^2 + h^k)$ on $\Gamma_{h,i}$. By Lemma 13.8(b) and (c), $\Phi_i(v_\alpha) = -|\Gamma_i| \alpha_i + O(h_{\Gamma}^2 + h^k) |\alpha|$. The map $\alpha \mapsto (\Phi_i(v_\alpha))_i$ is linear from an $(m-1)$ -dimensional space into the hyperplane, and it is a small perturbation of the isomorphism $\alpha \mapsto (-|\Gamma_i| \alpha_i)_i$. So it is onto for small h . $\square$

Corollary 13.18 (GS on Z_h ; replaces Corollary 5.3). *For $v \in Z_h$ and every constant $c \in \mathbb{R}$,*

$$N_h(\tilde{u} - u_h, v) = \mathcal{R}(v) + \mathcal{G}(v), \quad \mathcal{R}(v) := -\langle \tilde{p} - c, v \cdot n \rangle_{\Gamma_h}.$$

The freedom is one global constant. For piecewise constants c_i on $\Gamma_{h,i}$,

$$-\langle \tilde{p} - \sum_i c_i \mathbf{1}_{\Gamma_{h,i}}, v \cdot n \rangle = \mathcal{R}(v) + \sum_i (c_i - c) \Phi_i(v),$$

and the last sum is not zero on Z_h when $m \geq 2$ and the c_i are not all equal (Lemma 13.17(b)).

Proof. As in Corollary 5.3, using only Lemma 13.17(a). $\square$

13.5. The test field and the printed method.

13.5.1. *The test field.* The field $\varphi = \chi_1(r) \int_0^\theta (p - \bar{p})$ of Section 6 generalises, per component, to $\chi(d) \int_0^s (q - \bar{q}_i)$. This is single-valued only with the *per-component* mean. Since the correct constant is global (Corollary 13.18), w must carry the net fluxes $-|\Gamma_i| \bar{q}_i$ through the Γ_i . Such a w has no global stream function, and the construction below replaces φ .

Lemma 13.19 (Test field). *Assume (R2), and let $q \in C^{k+1}(\Gamma)$ with $\int_{\Gamma} q = 0$. There is $w \in C^{k+1}(\mathcal{O}')^2$ with:*

- (a) $w = 0$ on $\Omega \cap \mathcal{N}_\Sigma$ for a neighbourhood $\mathcal{N}_\Sigma$ of Σ ; extended by 0 to $\mathcal{N}_\Sigma \setminus \Omega$, w is defined on $\mathcal{O}' := \mathcal{O} \cup \mathcal{N}_\Sigma$ (an open neighbourhood of $\bar{\Omega}$) and $\operatorname{div} w = 0$ there;
- (b) $w = qN = -qn$ on Γ ;
- (c) $\|w\|_{W^{1,\infty}(\mathcal{T}_{r/16})} \leq C\|q\|_{W^{1,\infty}(\Gamma)}$, $\|\nabla w\|_{L^2(\Omega)} \leq C\|q\|_{W^{1,\infty}(\Gamma)}$, and $\|w\|_{W^{k+1,\infty}} \leq C\|q\|_{C^{k+1}(\Gamma)}$.

If $m = 1$, or more generally if all $\bar{q}_i = 0$, then $w = -\operatorname{curl}(\chi(d)Q_i(\sigma))$ near Γ_i , which is the construction of Section 6.

Proof. Let χ be as in Lemma 13.10 (so $\chi = 1$ on $(-r, r/8]$, $\chi = 0$ on $[r/4, r)$), and $Q_i(s) := \int_0^s (q - \bar{q}_i)$, which is $|\Gamma_i|$ -periodic. On $\Psi(\Gamma_i \times (-r, r))$ set

$$w_0 := \frac{\chi(d)q(\sigma)}{J} N \circ \pi - \chi'(d)Q_i(\sigma)T \circ \pi,$$

and $w_0 := 0$ elsewhere in $\mathcal{O}$. By (14),

$$\operatorname{div} w_0 = \frac{1}{J}(\chi'q - \chi'(q - \bar{q}_i)) = \frac{\chi'(d)\bar{q}_i}{J} =: g_0 \quad \text{on } U_i.$$

Here g_0 is supported in the bands $B_i := \Psi(\Gamma_i \times [r/8, r/4]) \subset \Omega$, and

$$\int g_0 = \sum_i \bar{q}_i |\Gamma_i| \int_0^r \chi' = - \sum_i |\Gamma_i| \bar{q}_i = - \int_\Gamma q = 0.$$

Write $g_0 = \sum_{i < m} \bar{q}_i g^{(i)}$ with $g^{(i)} := \chi'(d_i)/J - (|\Gamma_i|/|\Gamma_m|)\chi'(d_m)/J$, where d_j is the signed distance to Γ_j . These are $m-1$ fixed, h -independent, compactly supported mean-zero functions in C^{k+1} (since $\Gamma \in C^{k+3}$ gives $J \in C^{k+1}$).

Let $\Omega_{16} := \{x \in \Omega : \operatorname{dist}(x, \Gamma) > r/16\}$; it is connected, by the argument for Ω_* in Lemma 13.15. Let $K \Subset \Omega_{16}$ be a connected finite union of open balls with closures in Ω_{16} containing $\bigcup_i B_i$. It exists because $\bigcup B_i$ is compact and Ω_{16} is path-connected: join the balls covering the bands by finite chains of balls along paths. A tree argument on the intersection graph of the balls writes each $g^{(i)}$ as a finite sum of mean-zero functions in C_c^{k+1} of single balls: use a smooth partition of unity subordinate to the balls, and transfer masses along tree edges with fixed bumps in $C_c^\infty(B \cap B')$. Bogovskii's operator on each ball maps $W_0^{m,p}$ to $W_0^{m+1,p}$ for every integer $m \geq 0$ and $1 < p < \infty$ [31, Thm. III.3.3]; we need only $m = k+1$. This gives $w^{(i)} \in W_0^{k+2,p}(K) \subset C_c^{k+1}(K)$ ($p > 2$) with $\operatorname{div} w^{(i)} = g^{(i)}$.

Set $w := w_0 - \sum_{i < m} \bar{q}_i w^{(i)}$. Then:

- (a) holds, since w_0 vanishes outside the collars and $K \Subset \Omega$.
- (b): at $d = 0$, $\chi = 1$, $\chi' = 0$ and $J = 1$, and $w^{(i)} = 0$ near Γ .
- (c): on $\mathcal{T}_{r/16}$ only the first term of w_0 is present and $w^{(i)} = 0$ there, with $W^{1,\infty}$ norm $\leq C\|q\|_{W^{1,\infty}}$. On the bands $|Q_i| \leq |\Gamma_i| \|q\|_\infty$, and $|\bar{q}_i| \leq \|q\|_\infty$. The regularity count is: $N \circ \pi, T \circ \pi, \sigma \in C^{k+2}$, $J \in C^{k+1}$, $q \circ \sigma \in C^{k+1}$. $\square$

Lemma 13.20 (Discrete curl of a divergence-free field). *Let $\mathcal{O}'$ be an open neighbourhood of $\bar{\Omega}$, and let $w \in W^{k+1,\infty}(\mathcal{O}')^2$ be divergence-free and vanish on a neighbourhood of Σ . For $T \in \mathcal{T}_h$ let $B_T \subset \mathcal{O}'$ be an open disk of radius $2h$ containing $\bar{T}$. It exists for $h \leq h_*$, since $\bar{\Omega}_h \subset \bar{\Omega} \cup \{|d| \leq Ch_\Gamma^2\}$ lies in a fixed compact subset of $\mathcal{O}'$. Let φ_T be a stream function of w on B_T ($w = \operatorname{curl} \varphi_T$), and define*

$$v_w|_T := \operatorname{curl}(\Pi_{k+1}^T \varphi_T),$$

where Π_{k+1}^T is the element restriction of the C^1 piecewise- P_{k+1} nodal interpolant of [46] (Argyris for $k = 4$), whose degrees of freedom on T are functionals of the jet of the function on $\bar{T}$. Then v_w is well defined, $v_w \in Z_h$, and

$$\|v_w - w\|_{W^{1,\infty}(T)} \leq Ch_T^k |w|_{W^{k+1,\infty}(B_T)}.$$

If $w = \text{curl } \varphi$ globally, then $v_w = \text{curl } \Pi_{k+1} \varphi$ is the v_φ of Section 6.

Proof. Well defined. Two stream functions on B_T differ by a constant, and Π_{k+1}^T is linear and reproduces constants.

Continuity. Let T, T' share an edge e , and let $B \subset \mathcal{O}'$ be a disk containing $T \cup T'$ with stream function φ_B . On the connected set $B \cap B_T$, $\varphi_T - \varphi_B$ is constant, so $\text{curl } \Pi_{k+1}^T \varphi_T = \text{curl } \Pi_{k+1}^T \varphi_B$ on T , and likewise on T' . The global interpolant of φ_B on $T \cup T'$ is C^1 across e . So the two curls agree on e .

$v_w \in Z_h$. v_w is a curl elementwise, so $\text{div } v_w = 0$. If $\bar{T}$ meets Σ , then T lies in the neighbourhood where $w = 0$ (for h small), φ_T is constant and $v_w|_T = 0$. So $v_w = 0$ on Σ .

Error. $\|\text{curl}(\Pi_{k+1}^T \varphi_T - \varphi_T)\|_{W^{1,\infty}(T)} \leq Ch_T^k |\varphi_T|_{W^{k+2,\infty}(T)} = Ch_T^k |w|_{W^{k+1,\infty}(T)}$ by the local interpolation estimate for Π_{k+1} (constants depending on (M0), (M2) as in Section 6). $\square$

Definition 13.21 (General test field). Assume (R2). With $q = p - \bar{p}$ (Definition 13.7), let w be the field of Lemma 13.19 and $v_\varphi := v_w \in Z_h$ from Lemma 13.20. Then $\|v_\varphi - w\|_{W^{1,\infty}(\Omega_h)} \leq Ch^k$, and on Γ , $w = (p - \bar{p})N = -(p - \bar{p})n$, exactly as in Section 6.

The following pointwise facts on a chord e replace the uses of “ $n(x^*) = e_r$ ” in Section 6. They follow from Lemma 13.8, $|x - x^*| \leq Ch_\Gamma^2$, and $w \in W^{1,\infty}$:

$$\begin{aligned} v_\varphi \cdot n_e &= -q(x^*) + O(h_\Gamma^2 + h^k), \\ v_\varphi \cdot \tau_e &= O(\|q\|_\infty \|\varkappa\|_\infty h_\Gamma) + O(h_\Gamma^2 + h^k), \\ (\tilde{p} - \bar{p})(x) n_e + v_\varphi(x) &= -q(m_e^*) \varkappa_e t \tau_e + O(h_\Gamma^2 + h^k). \end{aligned} \quad (15)$$

For the third: $(\tilde{p} - \bar{p})(x) = q(x^*) + O(h_\Gamma^2)$, $v_\varphi(x) = q(x^*)N(x^*) + O(h_\Gamma^2 + h^k) = -q(x^*)n(x^*) + \dots$, then Lemma 13.8(b) and $q(x^*)\varkappa(x^*) = q(m_e^*)\varkappa_e + O(\ell)$. Also, by Lemma 13.8(c),

$$\|\tilde{p} - \bar{p}\|_{L^2(\Gamma_h)} \leq G(1 + Ch_\Gamma^2) + Ch_\Gamma^2, \quad \|v_\varphi\|_{L^2(\Gamma_h)} = G(1 + O(h_\Gamma^2)) + O(h^k). \quad (16)$$

13.5.2. The theorems.

Theorem 13.22 (Theorem 6.1 on general domains). *Under Assumptions 13.1, 13.4, (M0)–(M2), (D) and (R1), the upper bound of Theorem 6.1 holds with G of Definition 13.7. If in addition (R2) holds, the lower bound holds as stated. Corollary 6.2 follows as before.*

Proof. Upper bound. Verbatim, with $c = \bar{p}$ in Corollary 13.18 and (16): $|\mathcal{R}(e_h)| \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2} \|e_h\|$.

Lower bound. Use v_φ of Definition 13.21. Step 1 becomes: by (15) and Lemma 13.8(c),

$$\mathcal{R}(v_\varphi) = \langle \tilde{p} - \bar{p}, q \circ \pi \rangle_{\Gamma_h} + O(h_\Gamma^2 + h^k) = \int_\Gamma q^2 + O(h_\Gamma^2 + h^k) = G^2 + O(h_\Gamma^2 + h^k).$$

Step 2 uses (16) and $\|\nabla v_\varphi\| + (\sum_e |e| \|\partial_n v_\varphi\|_{L^2(e)}^2)^{1/2} \leq C$ (Lemma 13.19(c)). Steps 3–4 are verbatim. $\square$

Theorem 13.23 (Theorem 6.4 on general domains). *Under the assumptions of Theorem 13.22 including (R2), with $C_p = C(1 + \|p - \bar{p}\|_{W^{1,\infty}(\Gamma)})$ ($\bar{p}$ global),*

$$\|\nabla(\tilde{u} - u_h)\| \leq C_p(h_\Gamma/\gamma) + C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h].$$

Proof. Steps 1, 2 and 6 are verbatim. Step 3: $S_\varphi := v_\varphi$ of Definition 13.21.

Step 4: by the third line of (15), on each e ,

$$(\tilde{p} - \bar{p})n_e + S_\varphi = b_e t + O(h_\Gamma^2 + h^k), \quad b_e := -q(m_e^*)\varkappa_e \tau_e \text{ constant on } e, \quad |b_e| \leq \|q\|_\infty \|\varkappa\|_\infty.$$

The argument of Lemma 13.13 gives $|m(v)| \leq C_p(h_\Gamma^{3/2} + h^k) \|\nabla v\|$.

Step 5: $|a_h(S_\varphi, v)| + |\langle \partial_n S_\varphi, v \rangle| \leq C_p \|\nabla v\|$ by Lemma 13.19(c) and Lemma 13.10. For $\langle S_\varphi, \partial_n v \rangle$ apply Lemma 6.3 to $S = w|_{\Gamma_h}$. Its hypotheses hold:

- w is continuous, and edgewise $W^{1,\infty}$ with $\|\nabla w\|_{L^\infty(\mathcal{T}_{r/16})} \leq C\|q\|_{W^{1,\infty}}$;
- $|w \cdot \tau_e| \leq C\|q\|_{W^{1,\infty}} h_\Gamma$ by the second line of (15) (applied to w);
- in step 3 of Lemma 6.3, $|\sigma_{e'} \tau_{e'} - \sigma_e \tau_e| \leq |w(x)| (|n_{e'} - n_e| + |\tau_{e'} - \tau_e|) \leq C\|q\|_\infty \|\varkappa\|_\infty (|e| + |e'|)$ by Lemma 13.8(d);
- the other steps of Lemma 6.3 are local and use only $\operatorname{div} v = 0$ and (M0).

The remainder $\langle S_\varphi - w, \partial_n v \rangle$ is bounded as in Theorem 6.4. $\square$

Theorem 13.24 (Theorem 6.6 on general domains). *Under the assumptions of Theorem 13.23, Theorem 6.6 holds verbatim with the global G .*

Proof. In (i), item 1 is Theorem 13.22, step 1. For item 2, on e we have $\tilde{u}(x) = \tilde{u}(x^*) + d(x) \partial_n \tilde{u}(x^*) + O(d^2) = -d(x) \partial_n u(x^*) + O(h_\Gamma^4)$. This is tangential up to $O(h_\Gamma^4)$, by $\partial_n u \cdot n = 0$ (Lemma 3.4), while $v_\varphi(x) = q(x^*)N(x^*) + O(h_\Gamma^2 + h^k)$. Hence $\tilde{u} \cdot v_\varphi = O(h_\Gamma^4 + h_\Gamma^2 h^k)$ as in Theorem 6.6. Items 3–5 are verbatim. In (ii), the lift E of $(g - g_I)|_\Sigma$ is supported near Σ , where $\Omega_h = \Omega$. $\square$

Corollary 13.25 (Corollary 6.7 on general domains). *Under the assumptions of Theorem 13.23, Corollary 6.7 holds verbatim, with v_φ of Definition 13.21 and the global G . To leading order the printed method imposes the leak condition $(\mu/h) u_h \cdot n \approx p - \bar{p}$ with $\bar{p}$ the mean of p over all curved components. In particular, with n pointing out of Ω_h , the discrete flux through $\Gamma_{h,i}$ is $\Phi_i(u_h) \approx (h/\mu) |\Gamma_i| (\bar{p}_i - \bar{p})$ (Lemma 13.17).*

Proof. Step 4 of the proof of Corollary 6.7 is (16). The rest is verbatim. $\square$

Proposition 13.26 (The constant is global). *Let $m \geq 2$. No estimate $\|\tilde{u} - u_h\| \leq C[(h/\mu)^{1/2} G_{\text{per}} + (1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h]$ can hold. Take $u = 0$, $g = 0$, $\tilde{p}$ any smooth function with $\tilde{p}|_{\Gamma_i} \equiv c_i$ (not all equal), and $\tilde{f} := \nabla \tilde{p}$, so that $f = \nabla p$. Then $G_{\text{per}} = 0$, $F = 0$, $\mathcal{G} \equiv 0$ and $E_h = 0$, but for $h \leq h_0(G)$ (in particular for fixed γ and bounded ρ as $h \rightarrow 0$)*

$$\|u_h\| \geq (2M)^{-1} (h/\mu)^{1/2} G, \quad G^2 = \sum_i |\Gamma_i| (c_i - \bar{c})^2 > 0, \quad \bar{c} = \frac{\sum_i |\Gamma_i| c_i}{|\Gamma|}.$$

Proof. Theorem 13.22, lower bound, with $\tilde{u} = 0$ (so $E_h = 0$: take $z = 0 \in W_h$) and $\mathcal{G} \equiv 0$. Here q is piecewise constant, so (R2) reduces to $\Gamma \in C^{k+3}$. $\square$

13.6. The pressure-consistent method. Proposition 7.1 holds verbatim: for $v \in V_h^R$, $-(1, \operatorname{div} v) + \langle 1, v \cdot n \rangle_{\Gamma_h} = -\int_\Sigma v \cdot n = 0$.

More generally, for a linear functional ℓ on $\Pi_h + \mathbb{R}$ with $\ell(1) = 1$, let CNS_ℓ^* be CNS^* with $\bar{p}_\Gamma$ replaced by ℓ :

$$B_\ell(q, v) := -(q, \operatorname{div} v) + \langle q - \ell(q), v \cdot n \rangle_{\Gamma_h}.$$

$\text{CNS}^* = \text{CNS}_\ell^*$ with $\ell = \bar{p}_\Gamma$ (global boundary mean). Extend ℓ to $\tilde{p}$ by the same formula.

Theorem 13.27 (Theorem 7.2 on general domains). *Under Assumptions 13.1, 13.4, (M0)–(M2), (D) and (R1), and for any such ℓ , Theorem 7.2 holds verbatim for CNS_ℓ^* (with β from Lemma 13.16). In particular one global correction suffices for well-posedness and for the rate $h_\Gamma^{3/2} + h_\Omega^k$, for any number m of curved components.*

Proof. We go through the steps of Theorem 7.2.

- (1) *Consistency.* For $c \in \mathbb{R}$, $B_\ell(q + c, v) = B_\ell(q, v) - c \int_{\Gamma_h} v \cdot n$, because $\int_{\Omega_h} \operatorname{div} v = \int_{\Gamma_h} v \cdot n$ for $v \in V_h^R$. With $p^* := \tilde{p} - \ell(\tilde{p})$,

$$B_\ell(p^*, v) = -(\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle + \ell(\tilde{p}) \left(\int_{\Omega_h} \operatorname{div} v - \int_{\Gamma_h} v \cdot n \right) = -(\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle,$$

which is the pressure form of Lemma 5.1. So (3) holds for all $v \in V_h^R$.

- (2) *Split.* Verbatim.

- (3) *Pressure.* Verbatim: $v \in V_h^0$ vanishes on all of Γ_h , and Lemma 13.16 applies.

- (4) *Velocity.* $B_\ell(e_p, e_h) = \langle e_p, e_h \cdot n \rangle - \ell(e_p) \sum_i \Phi_i(e_h) = \langle e_p, e_h \cdot n \rangle$ by Lemma 13.17(a). Also $\langle \bar{\xi}, e_h \cdot n \rangle = \bar{\xi} \sum_i \Phi_i(e_h) = 0$. Only the *total* flux enters, never the individual Φ_i . The rest is verbatim.

- (5) *Absorb.* Verbatim.

- (6) *Uniqueness* (Proposition 13.28). $\square$

Proposition 13.28 (Kernel analysis). *For $\gamma \geq \gamma_2$, the homogeneous $\operatorname{CNS}_\ell^*$ system ($g_I = 0$, $\tilde{f} = 0$) has only the trivial solution. Hence $\operatorname{CNS}_\ell^*$ is uniquely solvable for every $m \geq 1$.*

Proof. Let (u_h, p_h) solve the homogeneous system. By Lemma 4.3 $u_h \in Z_h$, and steps 3–5 with $z = 0$, $\tilde{u} = 0$, $\tilde{p} = 0$ (so $\mathcal{G} = 0$, $E_h = 0$, $\eta = 0$) give $u_h = 0$. Step 3 then gives $\|p_h - \bar{p}_h\| = 0$, where $\bar{p}_h$ is the mean over Ω_h . Since Ω_h is connected (Lemma 13.10), $p_h \equiv c$. Then for all $v \in V_h^R$,

$$0 = B_\ell(c, v) = -c \int_{\Omega_h} \operatorname{div} v + (c - c \ell(1)) \int_{\Gamma_h} v \cdot n = -c \int_{\Gamma_h} v \cdot n.$$

The edge bubble $b = \beta_e n_e$ on any single $e \in \mathcal{E}_h^\Gamma$ has $\int_{\Gamma_h} b \cdot n \neq 0$, so $c = 0$. The system is square, so uniqueness gives existence. $\square$

Remark 13.29 (The velocity does not depend on ℓ). Let $B_0(q, v) := -(q, \operatorname{div} v) + \langle q, v \cdot n \rangle_{\Gamma_h}$ be the naive form of Proposition 7.1. Then $B_\ell(q, v) = B_0(q, v) - \ell(q) \int_{\Gamma_h} v \cdot n$, so $\operatorname{CNS}_\ell^*$ is the naive system plus the rank-one term $-\ell(p_h) \int_{\Gamma_h} v \cdot n$. For a constant c and $v \in V_h^R$, $B_0(c, v) = -c \int_{\partial\Omega_h} v \cdot n + c \int_{\Gamma_h} v \cdot n = 0$. Hence, if (u_h, p_h) solves $\operatorname{CNS}_\ell^*$ and ℓ' is another normalisation, then $(u_h, p_h + c)$ with $c := \ell(p_h) - \ell'(p_h)$ solves $\operatorname{CNS}_{\ell'}^*$, because $\ell'(p_h + c) = \ell(p_h)$ and the continuity row does not involve ℓ . So the discrete velocity and the scalar $\lambda_h := \ell(p_h)$ do not depend on the normalisation; the pressures differ by constants. The naive system is solvable if and only if $\lambda_h = 0$: if $\lambda_h = 0$, (u_h, p_h) solves it; conversely, if (u, p) solves it, then $(u, p - \ell(p))$ solves $\operatorname{CNS}_\ell^*$, and uniqueness gives $\lambda_h = \ell(p - \ell(p)) = 0$.

Remark 13.30 (Per-component correction is inconsistent). Let $m \geq 2$ and replace $\langle q - \bar{p}_\Gamma(q), v \cdot n \rangle$ by $\sum_i \langle q - \ell_i(q), v \cdot n \rangle_{\Gamma_{h,i}}$ with $\ell_i(q) := |\Gamma_{h,i}|^{-1} \int_{\Gamma_{h,i}} q$. This is *not* of the form $\operatorname{CNS}_\ell^*$. For $v \in V_h^R$,

$$B_{\text{per}}(q, v) = B_{\bar{p}_\Gamma}(q, v) - \sum_i (\ell_i(q) - \bar{p}_\Gamma(q)) \Phi_i(v).$$

Adding a constant to q does not change the sum, so no choice of pressure constant makes the exact pair consistent. The residual on Z_h is

$$\mathcal{I}(v) = \sum_i (\ell_i(\tilde{p}) - \bar{p}_\Gamma(\tilde{p})) \Phi_i(v) = \sum_i (\bar{p}_i - \bar{p}) \Phi_i(v) + O(h_\Gamma^2) \|v\|_{L^1(\Gamma_h)}.$$

Test with $v = v_\alpha$ from the proof of Lemma 13.17(b), $\alpha_i := \bar{p}_i - \bar{p}$, $A^2 := \sum_i |\Gamma_i| \alpha_i^2$. Then $\mathcal{I}(v_\alpha) = -A^2 + O(h_\Gamma^2 + h^k) A^2$, and $\|v_\alpha\| \leq (\mu/h)^{1/2} A(1 + o(1)) + CA$ (using $|\alpha| \leq CA$). Hence, if $A > 0$, the dual norm of $\mathcal{I}$ with respect to $\|\cdot\|$ is at least $c(h/\mu)^{1/2} A$ for $h \leq h_0$, because $h/\mu = h_\Gamma/\gamma \leq h_\Gamma/\gamma_0$ is small. We do *not* prove that the discrete error of the per-component

method is actually of this size. Unlike GS, its pressure does not drop out on Z_h , so the lower-bound argument of Theorem 13.22 does not apply directly. It is, however, not a consistent method of order $h_\Gamma^{3/2}$ when the component means of p differ. The per-component method is uniquely solvable for γ large: the extra term is absorbed in step 4 like $\langle \xi - \bar{\xi}, e_h \cdot n \rangle$, since $|\ell_i(\xi - \bar{\xi})| \leq |\Gamma_{h,i}|^{-1/2} \|\xi - \bar{\xi}\|_{\Gamma_{h,i}}$, and the kernel argument of Proposition 13.28 goes through.

13.7. The penalty threshold. Proposition 10.1 is local to a boundary star and uses only that the angle of Ω_h at a vertex of Γ_h is $\pi + O(h_\Gamma)$, which is Lemma 13.9(a). It holds verbatim. No computations have been run for general domains.

13.8. Summary of dependencies and scope.

- $\Gamma \in C^3$, $\Sigma \neq \emptyset$, Ω connected: Lemmas 13.8–13.13, Section 13.3, Lemmas 5.1 and 5.2, Theorem 13.22 (upper bound), Theorem 13.27.
- Additionally (R2) ($\Gamma \in C^{k+3}$, $p|_\Gamma \in C^{k+1}$): everything using v_φ , namely the lower bound in Theorem 13.22, Theorems 13.23 and 13.24, Corollary 13.25, Lemma 13.17(b), Proposition 13.26, and the lower bound in Remark 13.30.
- New geometric constants: $\|\varkappa\|_{C^1}$, the tube radius r , the graph scale a_0 , and the h -independent cover of Lemma 13.15. The discrete constants $C_I, c_1, M, \gamma_0, \gamma_2$ depend on these only through κ and β .
- Inherited without re-proof, exactly as in Sections 4 and 6: the reduction of the Guzmán–Scott discrete inf-sup constant to the continuous one plus a Poincaré constant, and the local nodal structure and interpolation estimate of the Morgan–Scott C^1 interpolant.
- Not covered: the identification of the H^1 constant (Section 6.1), the lower bound (Section 7.1), the single-triangle witness (Section 10.1), Clough–Tocher meshes (Section 12), and the case $\Sigma = \emptyset$ or components that mix curved and straight pieces. We make no claim about these on general domains.

14. STEADY NAVIER–STOKES

Gjerde and Scott’s application is cylinder drag, which is a Navier–Stokes computation [34]. This section extends Theorems 6.1–7.2 and Corollary 6.7 to

$$-\nu \Delta u + (u \cdot \nabla)u + \nabla p = f, \quad \operatorname{div} u = 0 \quad \text{in } \Omega, \quad u|_\Gamma = 0, \quad u|_{\partial R} = g.$$

All results of this section assume a discrete stability property of the linearisation at the exact solution, (L) for GS or (L*) for CNS*. They follow from a small-data condition (SD) (Lemma 14.7) and, more generally, from nonsingularity of the solution branch, with a threshold h_0 independent of the penalty (Theorem 14.16); with the Stokes regularity of Theorem 9.1, h_0 is explicit (Theorem 14.19), although far too small to be useful (Section 14.11). Sections 14.9–14.11 identify the Navier–Stokes leak flow and H^1 constant, give the GS and CNS* drag expansions, and make the leak limit uniform in the penalty. The Stokes estimates are reused, in particular steps 4–5 of the proof of Theorem 6.4. The only new ingredients are a fixed-point argument around the linearisation at the exact solution, and a check that the Stokes estimates are stable under lower-order perturbations that are bounded in the H^1 dual norm.

14.1. What Gjerde–Scott print. The arXiv version of [34] (arXiv:2306.12362) states the continuous problem $-\nu \Delta u + u \cdot \nabla u + \nabla p = 0$. The drag functional (their (7)) contains the standard convective term $(u \cdot \nabla u) \cdot v$. The discrete Nitsche system (27)–(28) is printed for the Stokes operator only. No discrete convective form is printed, and in particular there is no skew-symmetrisation and no inflow/outflow boundary term. The nonlinear solve is described only as the iterated penalty method. We therefore analyse both natural choices below, and show that the conclusions are the same for both.

14.2. Methods. Let $\langle \cdot, \cdot \rangle$, N_h , V_h^R , Z_h , W_h , Π_h and $\|\cdot\|$ be as in Section 2. For $w, u, v \in H^1(\Omega_h)^2$ define

$$c_1(w; u, v) := ((w \cdot \nabla)u, v), \quad c_s(w; u, v) := \frac{1}{2} [((w \cdot \nabla)u, v) - ((w \cdot \nabla)v, u)].$$

Here c denotes either one.

Definition 14.1 (GS and CNS* for Navier–Stokes). Find $u_h \in V_h$ with $u_h|_{\partial R} = g_I$ and $p_h \in \Pi_h$ such that, for all $v \in V_h^R$ and $q \in \Pi_h$,

$$\nu N_h(u_h, v) + c(u_h; u_h, v) - (p_h, \operatorname{div} v) [+ \langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle] = (\tilde{f}, v), \quad (q, \operatorname{div} u_h) = 0.$$

Without the bracket this is GS; with it, it is CNS*.

Remark 14.2 (Penalty convention). Multiplying N_h by ν makes the wall penalty $\nu\mu/h$. If instead one uses $\nu a_h - \nu \langle \partial_n u, v \rangle - \nu \langle u, \partial_n v \rangle + (\hat{\mu}/h) \langle u, v \rangle$ with an unscaled $\hat{\mu}$, this equals νN_h with $\mu = \hat{\mu}/\nu$. Every statement below holds with $\nu\mu$ replaced by $\hat{\mu}$. In particular the leak velocity below is then $(h/\hat{\mu})(p - \bar{p})$, and coercivity needs $\hat{\mu} \geq \nu\mu_0$.

14.3. Assumptions.

Assumption 14.3.

- (NS1) (u, p) is a smooth solution on $\bar{\Omega}$, $\tilde{f}$ is a smooth extension of f , and (M0)–(M2) and (D) hold.
 (NS2) The extensions are as in Section 2: $\tilde{u} = \operatorname{curl} \tilde{\psi}$ (so $\operatorname{div} \tilde{u} = 0$ on a neighbourhood of $\bar{\Omega}$) and $\tilde{p}$ is smooth. The defect

$$F := \tilde{f} + \nu \Delta \tilde{u} - (\tilde{u} \cdot \nabla) \tilde{u} - \nabla \tilde{p}$$

vanishes on Ω , is supported in $\bar{S}_h$, and $\|F\|_\infty \leq C$.

Stability is assumed in one of two forms, (L) or (L*) below. Lemma 14.7 shows that the standard small-data condition implies both. Let $\mathfrak{A}$ be the linearisation at $\tilde{u}$:

$$\mathfrak{A}(w, v) := \nu N_h(w, v) + c(w; \tilde{u}, v) + c(\tilde{u}; w, v).$$

Set $\|\ell\|_{*,X} := \sup_{0 \neq v \in X} \ell(v)/\|v\|$ for $X = Z_h$ or $X = V_h^R$.

- (L) There is $\beta_L > 0$, independent of h , with $\beta_L \|\delta\| \leq \|\mathfrak{A}(\delta, \cdot)\|_{*,Z_h}$ for all $\delta \in Z_h$.
 (L*) There is $\beta_* > 0$, independent of h , such that for every linear functional ℓ on V_h^R the problem

$$(\delta, \xi) \in Z_h \times \Pi_h : \quad \mathfrak{A}(\delta, v) + B^*(\xi, v) = \ell(v) \quad \forall v \in V_h^R,$$

with $B^*(q, v) = -(q, \operatorname{div} v) + \langle q - \bar{p}_\Gamma(q), v \cdot n \rangle$, has a unique solution, and $\beta_* \|\delta\| \leq \|\ell\|_{*,V_h^R}$.

(L) and (L*) are the discrete forms of “ u lies on a nonsingular branch” for GS and CNS* respectively; see Theorem 14.16.

14.4. Tools.

Lemma 14.4 (Uniform Sobolev constants). *There are C_1 and N , independent of h , such that*

- (a) $\|v\|_{H^1(\Omega_h)} \leq C_1 \|\nabla v\|$ for all $v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$;
 (b) $|c(w; u, v)| \leq N \|w\|_{H^1(\Omega_h)} \|u\|_{H^1(\Omega_h)} \|\nabla v\|$ for all $w, u \in H^1(\Omega_h)^2$ and all $v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$, for $c = c_1$ and $c = c_s$.

Proof. (a) is Lemma 3.2. For (b), let Φ_h be the bi-Lipschitz map of Lemma 3.2 with $\|D\Phi_h - I\|_\infty \leq Ch_\Gamma$. The embedding $H^1(\Omega) \subset L^4(\Omega)$ on the fixed Lipschitz domain Ω ($d = 2, 3$) transfers to $\|w\|_{L^4(\Omega_h)} \leq S \|w\|_{H^1(\Omega_h)}$ with S independent of h . Then $|c_1(w; u, v)| \leq \|w\|_{L^4} \|\nabla u\| \|v\|_{L^4}$ and $|((w \cdot \nabla)v, u)| \leq \|w\|_{L^4} \|\nabla v\| \|u\|_{L^4}$. Finish with (a). $\square$

Lemma 14.5 (Convective boundary identities). *Let $w \in H^1(\Omega_h)^2$ with $\operatorname{div} w = 0$, and $u, v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$. Then*

$$c_1(w; u, v) - c_s(w; u, v) = \frac{1}{2} \langle (w \cdot n) u, v \rangle, \quad c_1(w; v, v) = \frac{1}{2} \langle w \cdot n, |v|^2 \rangle, \quad c_s(w; v, v) = 0.$$

Proof. $((w \cdot \nabla)u, v) + ((w \cdot \nabla)v, u) = \int_{\Omega_h} w \cdot \nabla(u \cdot v) = \int_{\partial\Omega_h} (w \cdot n)(u \cdot v) - \int_{\Omega_h} (\operatorname{div} w)(u \cdot v)$. The last integral vanishes, and so does the part of the boundary integral on ∂R , since $v = 0$ there. The outward normal of Ω_h on Γ_h is n . $\square$

So $c_1(u_h; v, v)$ vanishes for all v only if $u_h \cdot n = 0$ on Γ_h . Nitsche's method never enforces this, and for GS the normal trace is the leak $(h/\nu\mu)(p - \bar{p})$ of Corollary 14.9. Section 14.7 shows that this term nevertheless does not affect any result.

Lemma 14.6 (Error equation). *For $v \in V_h^R$,*

$$\nu N_h(\tilde{u}, v) + c(\tilde{u}; \tilde{u}, v) - (\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle = (\tilde{f}, v) + \mathcal{G}_\nu(v) + \kappa_c(v),$$

where

$$\mathcal{G}_\nu(v) := \nu \left[\langle (\nabla \tilde{u})^\top n, v \rangle - \langle \tilde{u}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{u}, v \rangle \right] - (F, v), \quad \kappa_{c_1} = 0, \quad \kappa_{c_s}(v) = -\frac{1}{2} \langle (\tilde{u} \cdot n) \tilde{u}, v \rangle.$$

Moreover $|\mathcal{G}_\nu(v)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2} \|v\|$ on V_h^R , $|\mathcal{G}_\nu(v)| \leq Ch_\Gamma^{3/2} \|\nabla v\|$ on V_h^0 , and $|\kappa_{c_s}(v)| \leq Ch_\Gamma^4 \|\nabla v\|$.

Proof. Follow Lemma 5.1: $\nu a_h(\tilde{u}, v) = -\nu(\Delta \tilde{u}, v) + \nu \langle D(\tilde{u})n, v \rangle$ and $-\nu \Delta \tilde{u} = \tilde{f} - F - (\tilde{u} \cdot \nabla) \tilde{u} - \nabla \tilde{p}$. This gives the identity for c_1 . For c_s , apply Lemma 14.5 with $w = u = \tilde{u}$. The bound on $\mathcal{G}_\nu$ is Lemma 5.2, since F enters only through Lemma 3.3, which uses only $\|F\|_\infty \leq C$ and $\operatorname{supp} F \subset \overline{S_h}$. For κ_{c_s} , use $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$ (Lemma 3.3) and Lemma 3.2. $\square$

Lemma 14.7 (Small data implies (L) and (L*)). *Let $\gamma \geq \gamma_0$ and*

$$\lambda := \frac{NC_1 \|\tilde{u}\|_{H^1(\Omega_h)}}{\nu c_1} \leq \frac{1}{4}. \quad (\text{SD})$$

Then (L) holds with $\beta_L = \nu c_1/2$. There is $\gamma'_2 \geq \gamma_0$, independent of ν and h , such that for $\gamma \geq \gamma'_2$ (L*) holds with $\beta_* = \nu/C_*$, where C_* is independent of ν and h .

(SD) follows for small h from $NC_1 \|u\|_{H^1(\Omega)} < \nu c_1/4$, since $\|\tilde{u}\|_{H^1(\Omega_h)} \rightarrow \|u\|_{H^1(\Omega)}$. This is the usual small-data (low Reynolds number) condition. It is stated with the *discrete* Nitsche coercivity constant c_1 of Lemma 4.2, so it is more restrictive than the continuous one.

Proof. (1) (L). For $\delta \in V_h^R$, Lemma 14.4 gives $|c(\delta; \tilde{u}, \delta)| + |c(\tilde{u}; \delta, \delta)| \leq 2NC_1 \|\tilde{u}\|_{H^1} \|\nabla \delta\|^2 \leq 2\lambda \nu c_1 \|\delta\|^2$. With Lemma 4.2, $\mathfrak{A}(\delta, \delta) \geq \frac{1}{2} \nu c_1 \|\delta\|^2$.

(2) *Pressure bound for (L*)*. Let (δ, ξ) solve the problem in (L*) and $v \in V_h^0$. Then $B^*(\xi, v) = -(\xi - \bar{\xi}, \operatorname{div} v)$, so $(\xi - \bar{\xi}, \operatorname{div} v) = \mathfrak{A}(\delta, v) - \ell(v)$. On V_h^0 ,

$$N_h(\delta, v) = a_h(\delta, v) - \langle \delta, \partial_n v \rangle \leq C_N \|\delta\| \|\nabla v\|, \quad C_N := 2 + C_I(c_0 \gamma_0)^{-1/2},$$

the convective terms are at most $2\lambda \nu c_1 \|\delta\| \|\nabla v\|$, and $\|v\| \leq (1 + C_I^2)^{1/2} \|\nabla v\|$. Lemma 4.5 gives

$$\beta \|\xi - \bar{\xi}\| \leq \nu(C_N + c_1/2) \|\delta\| + (1 + C_I^2)^{1/2} \|\ell\|_{*, V_h^R}.$$

(3) *Velocity bound*. Test with $v = \delta \in Z_h$. Since $\operatorname{div} \delta = 0$ and $\int_{\Gamma_h} \delta \cdot n = 0$, we have $B^*(\xi, \delta) = \langle \xi - \bar{\xi}, \delta \cdot n \rangle$. As in step 4 of Theorem 7.2, its modulus is at most $C_I(c_0 \gamma)^{-1/2} \|\xi - \bar{\xi}\| \|\delta\|$. So

$$\frac{1}{2} \nu c_1 \|\delta\| \leq \|\ell\|_{*, V_h^R} + C_I(c_0 \gamma)^{-1/2} \beta^{-1} [\nu(C_N + c_1/2) \|\delta\| + (1 + C_I^2)^{1/2} \|\ell\|_{*, V_h^R}].$$

Choose γ'_2 with $C_I(c_0 \gamma'_2)^{-1/2} \beta^{-1} (C_N + c_1/2) \leq c_1/4$. The factor ν cancels, so γ'_2 does not depend on ν . This gives $\|\delta\| \leq C_* \nu^{-1} \|\ell\|_{*, V_h^R}$.

(4) *Existence and uniqueness*. If $\ell = 0$, then $\delta = 0$, then ξ is constant by step 2, and $\xi = 0$ by the edge bubble of Lemma 4.3. The system is square, because $\dim Z_h + \dim \Pi_h = \dim V_h^R$ by Lemma 4.3. $\square$

14.5. The printed method. Let $z \in W_h$ attain the bound of Lemma 4.6, so $\|\tilde{u} - z\| \leq C\varepsilon_h$. With the lift E from the proof of Theorem 6.6(ii) and Lemma 14.4(a),

$$\|\tilde{u} - z\|_{H^1(\Omega_h)} \leq C(\varepsilon_h + h^{k+1/2}) \leq C\varepsilon_h. \quad (17)$$

Put $\mathcal{R}(v) := -\langle \tilde{p} - \bar{p}, v \cdot n \rangle$, as in Corollary 5.3, where $\bar{p}$ is the mean over Γ of the Navier–Stokes pressure p . Let $v_\varphi \in Z_h$ be the field of Section 6, built from this p . Write $G = \|p - \bar{p}\|_{L^2(\Gamma)}$ and

$$X := h/\mu + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h.$$

Theorem 14.8 (GS for Navier–Stokes). *Assume (NS1)–(NS2), (L) and $\gamma \geq \gamma_0$, for either convective form. There are $c_* > 0$ and C , depending on ν , β_L , N , C_1 , the mesh constants and the solution (and on $\|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}$), such that if $X \leq c_*$, then:*

- (a) Existence and local uniqueness. GS has a solution (u_h, p_h) . It is the only one with $u_h \in z - (\delta_0 + \{\|\cdot\| \leq r\})$, where $\delta_0 \in Z_h$ is the linearised error defined in (19) and $r = CX$.
- (b) H^1 rate 1. $\|\nabla(\tilde{u} - u_h)\| \leq C[h/\mu + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$.
- (c) Energy rate 1/2. $\|\tilde{u} - u_h\| \leq \nu^{-1}(h/\mu)^{1/2}G + CX$.
- (d) Quadratic closeness to the linearisation. $\|(\tilde{u} - u_h) - (\tilde{u} - z) - \delta_0\| \leq CX^2$.

Proof. (1) *Reformulation.* Subtracting the GS equations from Lemma 14.6, and using $\operatorname{div} v = 0$ and $\int_{\Gamma_h} v \cdot n = 0$ for $v \in Z_h$, the error $e := \tilde{u} - u_h$ satisfies

$$\nu N_h(e, v) + c(\tilde{u}; \tilde{u}, v) - c(u_h; u_h, v) = \mathcal{R}(v) + \mathcal{G}_\nu(v) + \kappa_c(v) \quad \forall v \in Z_h.$$

By bilinearity, $c(\tilde{u}; \tilde{u}, v) - c(\tilde{u} - e; \tilde{u} - e, v) = c(e; \tilde{u}, v) + c(\tilde{u}; e, v) - c(e; e, v)$, so

$$\mathfrak{A}(e, v) = \mathcal{R}(v) + \mathcal{G}_\nu(v) + \kappa_c(v) + c(e; e, v).$$

Write $u_h = z - \delta$ with $\delta \in Z_h$, so $e = e(\delta) := (\tilde{u} - z) + \delta$. Then u_h solves the GS momentum equation on Z_h if and only if

$$\mathfrak{A}(\delta, v) = \ell_0(v) + c(e(\delta); e(\delta), v) \quad \forall v \in Z_h, \quad \ell_0 := \mathcal{R} + \mathcal{G}_\nu + \kappa_c - \mathfrak{A}(\tilde{u} - z, \cdot). \quad (18)$$

Given such $u_h \in W_h$, the functional $v \mapsto \nu N_h(u_h, v) + c(u_h; u_h, v) - (\tilde{f}, v)$ vanishes on $Z_h = \ker(\operatorname{div}|_{V_h^R})$. Since $\operatorname{div} V_h^R = \Pi_h$ (Lemma 4.3), there is a unique $p_h \in \Pi_h$ with $(p_h, \operatorname{div} v)$ equal to it on V_h^R . So solving (18) is equivalent to solving GS.

(2) *Linearised error.* By (L), $\mathfrak{A}$ is invertible on Z_h (a square system). Let

$$\delta_0 := \mathfrak{A}^{-1}\ell_0 = \delta_R + \delta_G, \quad \mathfrak{A}(\delta_R, \cdot) = \mathcal{R}, \quad \mathfrak{A}(\delta_G, \cdot) = \mathcal{G}_\nu + \kappa_c - \mathfrak{A}(\tilde{u} - z, \cdot) \quad \text{on } Z_h. \quad (19)$$

(3) *Bound on δ_G .* Since $|\mathfrak{A}(\tilde{u} - z, v)| \leq \nu M \|\tilde{u} - z\| \|v\| + 2N \|\tilde{u}\|_{H^1} \|\tilde{u} - z\|_{H^1} \|\nabla v\|$, Lemma 14.6 and (17) give $\|\delta_G\| \leq C\beta_L^{-1}[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$.

(4) *Ansatz for δ_R .* Set $\delta_R = (h/\nu\mu)v_\varphi + r$. Since $\nu N_h \cdot (h/\nu\mu) = (h/\mu)N_h$,

$$\mathfrak{A}(r, v) = m(v) + \ell(v) - \frac{h}{\nu\mu} [c(v_\varphi; \tilde{u}, v) + c(\tilde{u}; v_\varphi, v)],$$

where m and ℓ are exactly the functionals of step 3 of Theorem 6.4. Steps 4–5 of that proof give $|m(v)| + |\ell(v)| \leq [C(h/\mu + h_\Gamma^{3/2} + h^k) + C(h/\mu)h^k h_\Gamma^{-1/2}] \|\nabla v\|$. The convective term is at most $C(h/\mu) \|\nabla v\|$ by Lemma 14.4, since $\|v_\varphi\|_{H^1} \leq C$. Using $(h/\mu)h^k h_\Gamma^{-1/2} \leq Ch^k$, $\|\nabla v\| \leq \|v\|$ and (L),

$$\|r\| \leq C\beta_L^{-1}(h/\mu + h_\Gamma^{3/2} + \varepsilon_h).$$

(5) *Consequences for δ_0 .* Since $\|\nabla v_\varphi\| \leq C$ and $\|v_\varphi\| \leq (\mu/h)^{1/2}(G + Ch_\Gamma^2 + Ch^k) + C$ (step 2 of the lower bound in Theorem 6.1),

$$\|\nabla \delta_0\| \leq CX, \quad \|\delta_0\| \leq \nu^{-1}(h/\mu)^{1/2}G + CX. \quad (20)$$

(6) *Contraction.* Let $\Phi(\delta) := \mathfrak{A}^{-1}[\ell_0 + c(e(\delta); e(\delta), \cdot)]$ on Z_h . Set $A := \|\tilde{u} - z\|_{H^1} + C_1 \|\nabla \delta_0\|$, so $A \leq CX$ by (17) and (20). Consider the ball $B := \{\delta \in Z_h : \|\delta - \delta_0\| \leq A/C_1\}$. For $\delta \in B$, Lemma 14.4(a) gives $\|e(\delta)\|_{H^1} \leq A + C_1 \|\delta - \delta_0\| \leq 2A$.

- By (L) and Lemma 14.4(b), $\|\Phi(\delta) - \delta_0\| \leq \beta_L^{-1} N (2A)^2$, which is at most A/C_1 if $A \leq \beta_L/(4NC_1)$.
- For $\delta_1, \delta_2 \in B$ with $d := \delta_1 - \delta_2$, we have $c(e_1; e_1, v) - c(e_2; e_2, v) = c(d; e_1, v) + c(e_2; d, v)$. Hence

$$\|\Phi(\delta_1) - \Phi(\delta_2)\| \leq \beta_L^{-1} N \cdot 4A \cdot C_1 \|d\| \leq \frac{1}{2} \|d\| \quad \text{if } A \leq \beta_L/(8NC_1).$$

Both conditions follow from $X \leq c_*$. Banach's fixed-point theorem gives a unique $\delta \in B$ with $\Phi(\delta) = \delta$, which proves (a), and $\|\delta - \delta_0\| \leq 4\beta_L^{-1} N A^2 \leq CX^2$, which proves (d).

(7) (b) and (c). From $e = (\tilde{u} - z) + \delta_0 + (\delta - \delta_0)$, (20), (d), $\|\tilde{u} - z\| \leq C\varepsilon_h$ and $X^2 \leq X$. $\square$

Corollary 14.9 (Exact leading term; sharpness). *Assume in addition that $G > 0$, $h_\Gamma \rightarrow 0$, $h/\mu \rightarrow 0$, $\gamma h_\Gamma \rightarrow 0$ and $\varepsilon_h/(h_\Gamma/\gamma)^{1/2} \rightarrow 0$ (the hypotheses of Corollary 6.7). Then*

$$\tilde{u} - u_h = \frac{h}{\nu\mu} v_\varphi + \vartheta_h, \quad \|\vartheta_h\| = o((h/\mu)^{1/2}), \quad \|\vartheta_h\|_{\Gamma_h} = o(h/\mu),$$

and consequently

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\nu\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{\nu^{-1}(h/\mu)^{1/2}G} \rightarrow 1, \quad \|\nabla(\tilde{u} - u_h)\| \geq c \frac{h}{\nu\mu} G - Ch^{k+1/2} \text{ for small } h.$$

So $u_h \cdot n \approx (h/\nu\mu)(p - \bar{p})$ on Γ_h , with the Navier–Stokes pressure, and, if moreover μ is bounded (for instance fixed γ and bounded ρ), the H^1 rate 1 of Theorem 14.8(b) is sharp.

Proof. (1) Set $\vartheta_h := (\tilde{u} - z) + \delta_G + r + (\delta - \delta_0)$. Steps 3–4 and 6 of the proof of Theorem 14.8 give $\|\vartheta_h\| \leq C[h/\mu + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h] + CX^2 \leq CX$.

(2) Under the hypotheses, $X/(h/\mu)^{1/2} \rightarrow 0$. Indeed,

$$\frac{h/\mu}{(h/\mu)^{1/2}} \rightarrow 0, \quad \frac{(1 + \gamma^{1/2})h_\Gamma^{3/2}}{(h_\Gamma/\gamma)^{1/2}} = (\gamma^{1/2} + \gamma)h_\Gamma \rightarrow 0 \quad (\gamma \geq \gamma_0), \quad \frac{\varepsilon_h}{(h/\mu)^{1/2}} \rightarrow 0.$$

Hence $\|\vartheta_h\| = o((h/\mu)^{1/2})$ and $\|\vartheta_h\|_{\Gamma_h} \leq (h/\mu)^{1/2} \|\vartheta_h\| = o(h/\mu)$.

(3) Step 4 of the proof of Corollary 6.7 gives $\|v_\varphi\|_{\Gamma_h} = G(1 + o(1))$ and $\|v_\varphi\|^2 = (\mu/h)\|v_\varphi\|_{\Gamma_h}^2 + O(1)$. This yields the two limits.

(4) For the H^1 lower bound, use $\|e\|_{\Gamma_h} \leq C_{\text{tr}}(\|\nabla e\| + \|\nabla E\|)$ with $\|\nabla E\| \leq Ch^{k+1/2}$, where E is the lift from the proof of Theorem 6.6(ii). If μ is bounded, $h^{k+1/2} = o(h/\mu)$ and the term is absorbed. $\square$

Remark 14.10. The slip and energy constants are universal ($= 1$), exactly as for Stokes. The H^1 constant $\lim \|\nabla(\tilde{u} - u_h)\| \nu\mu/(hG)$, if it exists, is not universal. It now depends on the Oseen response: r solves the linearised problem, including $c(v_\varphi; \tilde{u}, \cdot) + c(\tilde{u}; v_\varphi, \cdot)$, which is of the same order as ∇v_φ . So the Stokes value $K = \|\nabla U\|/G$ of Corollary 6.11 does not carry over in general. The Navier–Stokes analogue is $K_{NS} = \|\nabla U_O\|/G$ with the Oseen leak flow U_O , and $K_{NS} \geq K_S$ (Corollary 14.21).

Remark 14.11 (Cylinder drag forces $G > 0$). Under (L) and the hypotheses of Corollary 14.9, the following holds. For flow past the cylinder, the pressure drag is $D_p = \int_\Gamma p n_1$, up to the orientation of n . Since $\int_\Gamma n_1 = 0$, $|D_p| = |\int_\Gamma (p - \bar{p}) n_1| \leq |\Gamma|^{1/2} G$, so $G \geq |D_p|/\sqrt{2\pi}$. In Gjerde–Scott's own application, the printed method therefore always leaks, at velocity $(h/\nu\mu)(p - \bar{p})$, and its asymptotic H^1 rate at fixed μ is 1, not 3/2. With their $\mu = 10^6$ the leak is small in absolute terms, which is consistent with their accurate drag values. For Stokes, the effect of the leak on

the computed drag is quantified in Section 8 (Theorem 8.17, Remark 8.25); for Navier–Stokes the leading term $-(h/\nu\mu)K_\psi^\dagger$ is identified in Corollary 14.22.

14.6. The pressure-consistent method.

Theorem 14.12 (CNS* for Navier–Stokes: Scott’s rate). *Assume (NS1)–(NS2) and (L*); by Lemma 14.7, (SD) and $\gamma \geq \gamma'_2$ suffice. Let $Y := (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h$. There are c_*, C such that if $Y \leq c_*$, then CNS* has a solution, whose velocity is unique in the $\|\cdot\|$ -ball of radius CY about $z - \delta_0$, where δ_0 is defined in step 2 below. It satisfies*

$$\|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h].$$

Hence, for fixed $k \geq 4$, $\gamma \in [\gamma'_2, \gamma_{\max}]$ and $h_\Gamma \geq h_\Omega^{2(\sigma_k - k)}$,

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k).$$

Proof. (1) *Error equation.* Let $p^* := \tilde{p} - \bar{p}_\Gamma(\tilde{p})$, $\eta := p^* - \pi_h p^*$ (with π_h the L^2 projection onto Π_h), and write $p_h = p^* - \eta - \xi$ and $u_h = z - \delta$. As in step 1 of Theorem 7.2 and step 1 above, (u_h, p_h) solves CNS* if and only if $(\delta, \xi) \in Z_h \times \Pi_h$ solves

$$\mathfrak{A}(\delta, v) + B^*(\xi, v) = \ell_0(v) + c(e(\delta); e(\delta), v) \quad \forall v \in V_h^R, \quad \ell_0 := \mathcal{G}_\nu + \kappa_c - \mathfrak{A}(\tilde{u} - z, \cdot) - B^*(\eta, \cdot).$$

The continuity row holds because $\operatorname{div} u_h = \operatorname{div} z - \operatorname{div} \delta = 0$.

(2) *Size of ℓ_0 .* We have $(\eta, \operatorname{div} v) = 0$ for all $v \in V_h^R$, because $\operatorname{div} V_h^R \subset \Pi_h$. Hence $|B^*(\eta, v)| \leq 2\|\eta\|_{\Gamma_h}(h/\mu)^{1/2}\|v\| \leq Ch_\Gamma^k\|v\|$. With Lemma 14.6 and (17), $\|\ell_0\|_{*, V_h^R} \leq CY$. Let (δ_0, ξ_0) be the solution of (L*) with data ℓ_0 ; then $\|\delta_0\| \leq \beta_*^{-1}CY$.

(3) *Fixed point.* Let $\Phi(\delta)$ be the Z_h -component of the (L*) solution with data $\ell_0 + c(e(\delta); e(\delta), \cdot)$. Since $\|c(e; e, \cdot)\|_{*, V_h^R} \leq N\|e\|_{H^1}^2$, step 6 of the proof of Theorem 14.8 applies verbatim, with β_L replaced by β_* and $A := \|\tilde{u} - z\|_{H^1} + C_1\|\delta_0\| \leq CY$.

(4) *Conclusion.* This gives the unique fixed point and $\|\tilde{u} - u_h\| \leq \|\tilde{u} - z\| + \|\delta_0\| + CY^2 \leq CY$. The H^1 statement follows as in Theorem 7.2, using the lift E for the L^2 part. $\square$

14.7. Is an inflow/outflow correction needed? The next lemma quantifies the convective boundary term.

Lemma 14.13 (Wall Péclet number). *Let $\mu \geq \mu_0$, $w \in V_h$ with $\operatorname{div} w = 0$, and $v \in V_h^R$. Then $\nu N_h(v, v) + c_s(w; v, v) \geq \nu c_1\|v\|^2$, and*

$$\nu N_h(v, v) + c_1(w; v, v) \geq (\nu c_1 - \tfrac{1}{2} \operatorname{Pe}_\Gamma(w))\|v\|^2, \quad \operatorname{Pe}_\Gamma(w) := \frac{h}{\mu} \|(w \cdot n)_-\|_{L^\infty(\Gamma_h)},$$

where $(w \cdot n)_- := \max(-w \cdot n, 0)$ is the inflow part. Here n points out of Ω_h (into P_h).

Proof. By Lemma 14.5, $c_1(w; v, v) = \frac{1}{2}\langle w \cdot n, |v|^2 \rangle \geq -\frac{1}{2}\|(w \cdot n)_-\|_\infty\|v\|_{\Gamma_h}^2$, and $\|v\|_{\Gamma_h}^2 \leq (h/\mu)\|v\|^2$. $\square$

Consequences:

- (1) *The results above need no correction, for either form.* (L) and (L*) linearise at $\tilde{u}$, whose normal trace on Γ_h is $O(h_\Gamma^2)$. For c_1 versus c_s , $\mathfrak{A}$ changes by $\frac{1}{2}\langle (\tilde{u} \cdot n)\delta, v \rangle + \frac{1}{2}\langle (\delta \cdot n)\tilde{u}, v \rangle = O(h_\Gamma^2)\|\delta\|_{\Gamma_h}\|v\|_{\Gamma_h}$ (Lemma 14.5). So (L) for one form implies (L) for the other with $\beta_L - Ch_\Gamma^2/h/\mu$. The discrete leak enters only through the quadratic term $c(e; e, v)$, which the contraction controls by continuity alone (Lemma 14.4); no sign is needed.
- (2) *The leak does make $c_1(u_h; u_h, u_h) \neq 0$.* For GS, Corollary 14.9 gives $u_h \cdot n = (h/\nu\mu)(p - \bar{p}) + o(h/\mu)$ in $L^2(\Gamma_h)$, with n pointing into the cylinder. The discrete wall is porous: fluid leaves Ω_h where $p > \bar{p}$ (the front stagnation region) and re-enters where $p < \bar{p}$. The leak law $(\nu\mu/h)u_h \cdot n \approx p - \bar{p}$ is a Darcy-type condition with permeability $h/\nu\mu$. If the leading term

also held in $L^\infty(\Gamma_h)$ (Corollary 14.9 is an L^2 statement; an L^∞ version would need an inverse estimate and is not proved here), the inflow part would give $\text{Per}(u_h) \approx (h/\mu)(h/\nu\mu)\|(p - \bar{p})_-\|_\infty = O((h/\mu)^2)$. The penalty would then dominate the inflow term by a factor of order $(h/\mu)^{-2}$. Formally, the kinetic-energy flux $\frac{1}{2} \int_{\Gamma_h} (u_h \cdot n) |u_h|^2$ is cubic in a boundary trace of size $h_\Gamma^2 + h/\mu$, which is negligible against the penalty dissipation $(\nu\mu/h)\|u_h\|_{\Gamma_h}^2 \approx (h/\nu\mu)G^2$. Point 1 is what the proofs actually use; point 2 is only the physical picture.

- (3) *When a correction would matter.* Lemma 14.13 degrades only when $\text{Per}(w) \gtrsim \nu c_1$ for iterates w far from $\tilde{u}$. This happens on coarse meshes, at large Reynolds number with fixed μ , or in Picard or iterated-penalty sweeps that start far from the solution. There c_s (for which $c_s(w; v, v) = 0$ identically), or an upwind inflow term $-\langle (w \cdot n)_- u, v \rangle$ in the style of Nitsche methods for Oseen problems (consistent to $O(h_\Gamma^4)$ at $\tilde{u}$), restores the energy inequality. For the asymptotic statements above neither is needed. c_s is also the natural choice for proving existence for arbitrary data, which we do not attempt: that would need a discrete Hopf extension of g_I inside W_h .

14.8. Nonsingular branches. This subsection removes the small-data condition: (L), (L*) and the adjoint and Dirichlet variants needed below follow from nonsingularity of the continuous solution, with a threshold h_0 independent of μ , γ and ρ . To avoid a clash with the convective form c_1 we write c_N for the coercivity constant of Lemma 4.2. “Admissible” means a mesh satisfying (M0)–(M2) with fixed constants and $\mu \geq \mu_0 = 4\kappa\rho C_I^2$ ($\gamma \geq \gamma_0$ suffices), with ρ unrestricted. We use repeatedly

$$\frac{h}{\mu} \leq \frac{\min_e |e|}{4\kappa C_I^2} \leq \frac{h_\Gamma}{4\kappa C_I^2}, \quad C_I \left(\frac{\rho}{\mu} \right)^{1/2} \leq (4\kappa)^{-1/2}. \quad (21)$$

Let $V := \{v \in H_0^1(\Omega)^2 : \text{div } v = 0\}$ and $\mathcal{V} := \{\varphi \in C_c^\infty(\Omega)^2 : \text{div } \varphi = 0\}$; $\mathcal{V}$ is dense in V since Ω is Lipschitz [58, Ch. I, Thm 1.6]. The Oseen operator $L : V \rightarrow V'$ is

$$\langle Lw, v \rangle := \nu(\nabla w, \nabla v)_\Omega + ((w \cdot \nabla)u, v)_\Omega + ((u \cdot \nabla)w, v)_\Omega.$$

On V , $c_1 = c_s$ (Lemma 14.5), so L is the common limit of both discrete linearisations.

(H_{ns}) L is injective on V .

Since $L = \nu L_0 + K_u$ with L_0 coercive and K_u compact, (H_{ns}) is equivalent to L being an isomorphism, i.e. to (u, p) being a nonsingular solution in the sense of Brezzi, Rappaz and Raviart [14], [32, Ch. IV]. Then $L^\top$, $\langle L^\top \varphi, w \rangle := \langle Lw, \varphi \rangle$, is also an isomorphism, with the same inf-sup constant $\beta_O := \|L^{-1}\|_{V' \rightarrow V}^{-1} > 0$ (the *Oseen inverse bound*). Besides (H_{ns}), the proofs use only $\tilde{u} \in W^{1,\infty}$ near $\bar{\Omega}$, $\text{div } \tilde{u} = 0$, and the wall bound $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$ of Lemma 3.3, which uses $u|_\Gamma = 0$. Without the last one, the term $C_e h_\Gamma^2 h/\mu$ below becomes $\|\tilde{u}\|_\infty h/\mu$, which still tends to 0. Write $K(w, v) := c(w; \tilde{u}, v) + c(\tilde{u}; w, v)$, so $\mathfrak{A} = \nu N_h + K$, and let $U_\infty := \max(\|\tilde{u}\|_{L^\infty}, \|\nabla \tilde{u}\|_{L^\infty})$ on Ω and a fixed collar of Γ , $C_K := (C_F + 3)U_\infty$.

Lemma 14.14 (The convective perturbation is L^2 -controlled). *For $w, v \in H^1(\Omega_h)^2$ vanishing on ∂R and either form,*

$$|K(w, v)| + |K(v, w)| \leq 2C_K \|w\| \|\nabla v\| + \frac{1}{2} U_\infty h_\Gamma^2 \|w\|_{\Gamma_h} \|v\|_{\Gamma_h},$$

and $|K_1(w, v)| \leq C_F(1 + C_F)U_\infty \|\nabla w\| \|\nabla v\|$ for the c_1 version K_1 . If $\text{div } w = 0$ and $v|_{\Gamma_h} = 0$, then $K(w, v) = K_1(w, v)$. In particular, for $\delta, v \in V_h^R$,

$$|K(\delta, v)| + |K(v, \delta)| \leq \left[2C_K \|\delta\| + U_\infty h_\Gamma^2 \frac{h}{\mu} \|\delta\| \right] \|v\|. \quad (22)$$

Proof. Terms that do not differentiate w are bounded by Hölder and Friedrichs. In the others integrate by parts with Lemma 14.5: $((\tilde{u} \cdot \nabla)w, v) = -((\tilde{u} \cdot \nabla)v, w) + \langle (\tilde{u} \cdot n)w, v \rangle$, and for c_s

also $((v \cdot \nabla)w, \tilde{u}) = -((v \cdot \nabla)\tilde{u}, w) - ((\operatorname{div} v)w, \tilde{u}) + \langle (v \cdot n)w, \tilde{u} \rangle$ with $\|\operatorname{div} v\| \leq \sqrt{2}\|\nabla v\|$. The boundary terms carry $|\tilde{u}| \leq \frac{1}{4}U_\infty h_\Gamma^2$ on Γ_h . For (22) use $\|\delta\|_{\Gamma_h}\|v\|_{\Gamma_h} \leq \frac{h}{\mu}\|\delta\|\|v\|$. $\square$

Lemma 14.15 (Strip, test fields and limits). (a) For $v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$: $\|v\|_{L^2(S_h)} \leq Ch_\Gamma^{1/2}\|\nabla v\|$ and $\|v\|_{L^2(\Gamma)} \leq 2\|v\|_{\Gamma_h} + Ch_\Gamma\|\nabla v\|_{S_h}$.
 (b) For $\varphi \in \mathcal{V}$ and $h < \operatorname{dist}(\operatorname{supp} \varphi, \partial\Omega)/2$ there is $\varphi_h = I_h\varphi - w_h \in Y_h = Z_h \cap V_h^0$ with $\|\nabla w_h\| \leq C\beta^{-1}h^k$ and $\|\varphi_h - \varphi\| \leq Ch^k$.
 (c) Let (mesh_j, μ_j) be admissible with $h_j \rightarrow 0$, $\delta_j \in Z_{h_j}$, $\|\delta_j\| \leq 1$. Along a subsequence there is $\delta \in V$ with $\delta_j|_\Omega \rightarrow \delta$ in $H^1(\Omega)$, $\|\delta_j\|_{L^2(\Omega_{h_j})} \rightarrow \|\delta\|_{L^2(\Omega)}$, and $\mathfrak{A}_j(\delta_j, \varphi_j) \rightarrow \langle L\delta, \varphi \rangle$, $\mathfrak{A}_j(\varphi_j, \delta_j) \rightarrow \langle L\varphi, \delta \rangle$ for every $\varphi \in \mathcal{V}$, for either form.

Proof. (a) $|S_h| \leq Ch_\Gamma^2$ and the uniform L^4 bound of Lemma 14.4; for the trace, integrate radially across $S_h = \{\rho_h(\theta) < r < 1\}$, $1 - \rho_h \leq Ch_\Gamma^2$, and use that arclength on Γ_h is at least $\rho_h d\theta \geq \frac{1}{2}d\theta$ (with $ds \geq \rho_h d\theta$ the constant 2 improves to $\sqrt{2}(1 + O(h_\Gamma^2))$). (b) Elements meeting $\operatorname{supp} \varphi$ do not touch $\partial\Omega_h$, so $I_h\varphi \in V_h^0$; $\operatorname{div} I_h\varphi = \operatorname{div}(I_h\varphi - \varphi)$ has zero mean and Lemma 4.5 gives w_h . (c) $\Omega \subset \Omega_{h_j}$ and Lemma 3.2 give $H^1(\Omega)$ bounds; Rellich and compactness of the trace give strong convergence in $L^2(\Omega)$ and $L^2(\partial\Omega)$; $\operatorname{div} \delta = 0$; by (a) and (21), $\|\delta_j\|_{L^2(\Gamma)} \leq 2(h_j/\mu_j)^{1/2} + Ch_\Gamma \rightarrow 0$, so $\delta \in V$, and (a) transfers the L^2 norms. Since $\varphi_j|_{\Gamma_h} = 0$, $N_h(\delta_j, \varphi_j) = a_h(\delta_j, \varphi_j) + \langle \delta_j, \partial_n w_{h_j} \rangle$ with $|\langle \delta_j, \partial_n w_{h_j} \rangle| \leq C_I(\rho/\mu)^{1/2}\|\nabla w_{h_j}\| \leq Ch_j^k$, and $a_h(\delta_j, \varphi) \rightarrow \langle \nabla \delta, \nabla \varphi \rangle$. $K(\delta_j, \varphi)$ and $K(\varphi, \delta_j)$ are integrals over $\operatorname{supp} \varphi$ of terms linear in $(\delta_j, \nabla \delta_j)$ with fixed coefficients (strong-weak convergence); the boundary terms of Lemma 14.5 vanish. $\square$

Theorem 14.16 ((L) and (L*) from nonsingularity). Assume (H_{ns}) and either convective form. There are $h_0, \beta_L > 0$, depending on u, ν, k, L and the mesh constants but not on μ, γ or ρ , such that (L), and its transpose on Z_h , hold for every admissible mesh with $h \leq h_0$. If also $\gamma \geq \gamma'_2$ (Lemma 14.7), there is $\beta_* > 0$ with the same independence such that (L*) holds for $h \leq h_0$.

Proof. For $\delta \in Z_h$, Lemma 4.2 and (22) give the Gårding inequality $\nu c_N \|\delta\| \leq \|\mathfrak{A}(\delta, \cdot)\|_{*, Z_h} + 2C_K \|\delta\| + U_\infty h_\Gamma^2 \frac{h}{\mu} \|\delta\|$. If (L) fails, there are admissible meshes with $h_j \rightarrow 0$ and $\delta_j \in Z_{h_j}$, $\|\delta_j\| = 1$, $\|\mathfrak{A}_j(\delta_j, \cdot)\|_* \rightarrow 0$; so $\liminf \|\delta_j\| \geq \nu c_N / (2C_K)$. Lemma 14.15(c) gives $\delta \in V$, $\delta \neq 0$, with $\langle L\delta, \varphi \rangle = \lim \mathfrak{A}_j(\delta_j, \varphi_j) = 0$ for all $\varphi \in \mathcal{V}$; by density $L\delta = 0$, a contradiction. The system on Z_h is square, so its inf-sup constant is its smallest singular value, which is also that of the transpose. For (L*): Lemma 14.7 with $\tilde{u}$ replaced by 0 gives, for $\gamma \geq \gamma'_2$, the Nitsche–Stokes bound $\nu \|\delta\| \leq C_S \|\ell\|_{*, V_h^R}$; a solution of the (L*) system solves it with data $\ell - K(\delta, \cdot)$. The test fields $\varphi_j \in Y_{h_j}$ give $B^*(\xi_j, \varphi_j) = 0$, and the contradiction argument is the same. For $\ell = 0$, $\delta = 0$ and then $\xi = 0$; the system is square (Lemma 4.3). $\square$

Theorem 14.16 replaces (SD) in Theorems 14.8, 14.12, Corollary 14.9 and Proposition 8.44, whose proofs use (SD) only through (L) and (L*). Local uniqueness improves to an h -independent ball: the contraction of step 6 of Theorem 14.8 works on $\{\|\delta - \delta_0\| \leq R_0\}$, $R_0 := \beta_L / (8NC_1^2)$, as soon as $\|\tilde{u} - z\|_{H^1} + C_1 \|\nabla \delta_0\| \leq \beta_L / (8NC_1)$; the same holds for CNS* with β_* .

Explicit thresholds. The constants of Theorem 14.16 are not explicit. A Schatz-type duality argument [52] makes them explicit, using the Stokes regularity of Section 9.1, which the paper also uses for U and Z_ψ :

(R_S) (a) For $g \in L^2(\Omega)^2$ the Stokes problem $-\Delta\varphi + \nabla\pi = g$, $\operatorname{div} \varphi = 0$, $\varphi|_{\partial\Omega} = 0$ has $\|\varphi\|_{H^2(\Omega)} + \|\pi - \bar{\pi}_\Omega\|_{H^1(\Omega)} \leq C_R \|g\|$. (b) If moreover $g \in H^m$ on a collar of Γ , then $(\varphi, \pi) \in H^{m+2} \times H^{m+1}$ on a smaller collar.

Both parts are proved: (a) is Theorem 9.1 and (b) is Proposition 9.2; we keep the name (R_S) for the pair. Put $\Lambda_O := 1 + (1 + C_F)C_F U_\infty / \beta_O$ and $\Theta_R := \max(1, C_R \Lambda_O)$. Then (Oseen regularity):

if $\varphi \in V$ solves $L^\top \varphi + \nabla \pi = g$ or $L\varphi + \nabla \pi = g$ with $g \in L^2$, then $\|\varphi\|_{H^2} \leq \Theta_R \nu^{-1} \|g\|$ and $\|\pi - \bar{\pi}\|_{H^1} \leq \Theta_R \|g\|$ (move the convective terms to the right-hand side and apply (R_S)(a)); and by bootstrapping with (R_S)(b) the Oseen leak flow (U_O, P_O) and the adjoint fields $(Z_\psi^\dagger, R_\psi^\dagger)$ below are smooth up to Γ (Lemma 14.17).

Lemma 14.17 (Oseen regularity). *Under (H_{ns}) the bounds just stated hold, and (U_O, P_O) of Section 14.9 and $(Z_\psi^\dagger, R_\psi^\dagger)$, $\psi \in \text{RM}$, lie in $H^2(\Omega) \times H^1(\Omega)$ and are C^∞ up to Γ .*

Proof. $\|\nabla \varphi\| \leq \beta_O^{-1} C_F \|g\|$; $(\varphi, \pi/\nu)$ solves Stokes with right-hand side g'/ν , $\|g'\| \leq \|g\| + U_\infty(\|\nabla \varphi\| + \|\varphi\|) \leq \Lambda_O \|g\|$. For the second part write $U_O = w + W$ with the smooth divergence-free w of Section 6 ($w|_\Gamma = -(p - \bar{p})n$) and $W \in V$, and $Z_\psi^\dagger = V_\psi - \zeta$ with V_ψ of Definition 8.14; W and ζ solve Oseen problems with smooth data. Bootstrap on nested collars with (R_S)(b). $\square$

The discrete dual approximant is that of Lemma 9.5, applied to the extension of Lemma 9.4. Let $s(\delta) := \sup_{0 \neq y \in Y_h} \mathfrak{A}(\delta, y) / \|\nabla y\|$, $s^\top(\delta)$ the same with $\mathfrak{A}(y, \delta)$, and

$$\omega(h) := \left(1 + \frac{U_\infty}{\nu}\right)h + \left(\frac{h}{\mu}\right)^{1/2}.$$

Lemma 14.18 (Duality for the convective perturbation). *Assume (H_{ns}) and an admissible mesh with $h \leq h_*$. There is $C_\sharp$, depending only on $(M0)-(M2)$, k , γ_0 , β , C_F and the extension and trace constants (not on $u, \nu, \mu, \gamma, \rho$), such that for $\delta \in Z_h$*

$$\|\delta\|_{L^2(\Omega_h)} \leq C_\sharp \Theta_R [\nu^{-1} s(\delta) + \omega(h) \|\delta\|], \quad \|\delta\|_{L^2(\Omega_h)} \leq C_\sharp \Theta_R [\nu^{-1} s^\top(\delta) + \omega(h) \|\delta\|].$$

Proof. Radial integration gives $\|\delta\|_{S_h} \leq Ch_\Gamma \|\delta\|$. Let (φ, π) solve $L^\top \varphi + \nabla \pi = \delta|_\Omega$ (for $s^\top$: $L\varphi + \nabla \pi = \delta|_\Omega$), $\Phi := \|\varphi\|_{H^2} \leq \Theta_R \nu^{-1} \|\delta\|$, $P := \|\pi\|_{H^1} \leq \Theta_R \|\delta\|$; extend by Lemma 9.4 and let $\tilde{G}$ be the extended residual, $\tilde{G} = \delta$ on Ω , $\|\tilde{G}\| \leq C(\nu + U_\infty)\Phi + CP$. Green's formula ($\text{div } \delta = 0$, $\delta|_{\partial R} = 0$, $\int_{\Gamma_h} \delta \cdot n = 0$) and insertion of y_h of Lemma 9.5 (for which $K(\delta, y_h) = K_1(\delta, y_h)$ by Lemma 14.14) give

$\|\delta\|_\Omega^2 = \mathfrak{A}(\delta, y_h) + \nu \langle \delta, \partial_n y_h \rangle + \nu a_h(\delta, \tilde{\varphi} - y_h) + K_1(\delta, \tilde{\varphi} - y_h) - \langle \sigma_\nu n, \delta \rangle - \langle (\tilde{u} \cdot n) \delta, \tilde{\varphi} \rangle - (\delta, \tilde{G})_{S_h}$, $\sigma_\nu := \nu D(\tilde{\varphi}) - \tilde{\pi} I$. The terms are bounded by $Cs(\delta)\Phi$; $C\nu[(h/\mu)^{1/2} + \gamma^{-1/2}h]\Phi \|\delta\|$ (Lemma 9.5 and $(h/\mu)^{1/2} h_\Gamma^{-1/2} = \gamma^{-1/2}$); $C(\nu + U_\infty)h\Phi \|\delta\|$; $C(\nu\Phi + P)(h/\mu)^{1/2} \|\delta\|$; $CU_\infty h_\Gamma^2 (h/\mu)^{1/2} \Phi \|\delta\|$; and $Ch_\Gamma \|\tilde{G}\| \|\delta\|$. Divide by $\|\delta\|_\Omega$ and add the strip. For $s^\top$ there is no convective boundary term and the second bound of Lemma 14.14 is used. $\square$

Besides (L) and (L*) we use:

- (L[†]) there is $\beta_\dagger > 0$, independent of h , such that for every linear functional ℓ on V_h^R the problem $(\zeta, \pi) \in Z_h \times \Pi_h$: $\mathfrak{A}(w, \zeta) + B^*(\pi, w) = \ell(w)$ for all $w \in V_h^R$, has a unique solution with $\beta_\dagger \|\zeta\| \leq \|\ell\|_{*, V_h^R}$;
- (L_D) there is $\beta_D > 0$, independent of h , with $\beta_D \|\nabla y\| \leq \sup_{0 \neq v \in Y_h} (\nu a_h(y, v) + K(y, v)) / \|\nabla v\|$ for all $y \in Y_h$.

Theorem 14.19 (Explicit thresholds). *Assume (H_{ns}) , either form, an admissible mesh with $h \leq h_*$, and put $C'_\sharp := (1 + C_I^2)^{1/2} C_\sharp$, $\Xi := 1 + 2C'_\sharp C_K \Theta_R / \nu$, $\vartheta(h) := 2C'_\sharp C_K \Theta_R \omega(h) + U_\infty h_\Gamma^2 h / \mu$. Then:*

- (a) if $\vartheta(h) \leq \frac{1}{2} \nu c_N$, (L) and its transpose hold with $\beta_L = \nu c_N / (2\Xi)$;
- (b) if $\gamma \geq \gamma'_2$ and $C_S \vartheta(h) \leq \frac{1}{2} \nu$, (L*) and (L[†]) hold with $\beta_* = \beta_\dagger = \nu / (2C_S \Xi)$;
- (c) if $2\kappa C'_\sharp C_K \Theta_R \omega(h) \leq \frac{1}{2} \nu$, (L_D) holds with $\beta_D = \nu / (2\kappa \Xi)$.

All three hold whenever

$$\left(1 + \frac{U_\infty}{\nu}\right)h + \left(\frac{h_\Gamma}{\gamma_0}\right)^{1/2} \leq c_{\sharp} \frac{\nu}{U_\infty \max(1, (1 + C_F)C_F) \max(1, C_R)(1 + U_\infty/\beta_O)}, \quad (23)$$

with $c_{\sharp} > 0$ depending only on the mesh and domain constants. The threshold is independent of $\mu \geq \mu_0$, γ and ρ .

Proof. (a) Gårding as in Theorem 14.16, and Lemma 14.18 with $s(\delta) \leq (1 + C_I^2)^{1/2} \|\mathfrak{A}(\delta, \cdot)\|_{*, Z_h}$: $2C_K \|\delta\| \leq (\Xi - 1) \|\mathfrak{A}(\delta, \cdot)\|_* + 2C_K' C_K \Theta_R \omega(h) \|\delta\|$; absorb. (b) As in Theorem 14.16 with the Stokes bound; testing with $y \in Y_h$ kills B^* , so $s(\delta) \leq (1 + C_I^2)^{1/2} \|\ell\|_{*, V_h^R}$. For $(L^\dagger)$ use that N_h is symmetric, the second bound of Lemma 14.14 and the second bound of Lemma 14.18. (c) $\nu \kappa^{-1} \|\nabla y\| \leq s(y) + 2C_K \|y\|$ on Y_h (Lemma 3.2) and Lemma 14.18. For (23): $\vartheta(h) \leq CU_\infty \Theta_R \omega(h)$, $\omega(h) \leq (1 + U_\infty/\nu)h + (h_\Gamma/\gamma_0)^{1/2}$, and $\Theta_R \leq \max(1, C_R) \max(1, (1 + C_F)C_F)(1 + U_\infty/\beta_O)$. $\square$

With $\text{Re} := U_\infty/\nu$ and $b_O := \beta_O/\nu$, (23) is the familiar Schatz scaling: the mesh must resolve the Reynolds number and the distance to the nearest singular point of the branch. The term $(h_\Gamma/\gamma_0)^{1/2}$ is the Nitsche contribution (the trace of δ enters the duality through $\langle \sigma_\nu n, \delta \rangle$); it disappears as $\gamma \rightarrow \infty$. For $U_\infty = 0$ no condition remains. Theorem 14.16 gives $(L^\dagger)$ and (L_D) without (R_S) by the same compactness argument (for $(L^\dagger)$ with $\mathfrak{A}_j(\varphi_j, \zeta_j)$ and Lemma 14.15(c)).

14.9. The Oseen leak flow. Let (U_O, P_O) solve the Oseen problem with the Stokes leak data

$$\begin{aligned} -\nu \Delta U_O + (U_O \cdot \nabla)u + (u \cdot \nabla)U_O + \nabla P_O &= 0, \quad \text{div } U_O = 0 \text{ in } \Omega, \\ U_O &= -(p - \bar{p})n \text{ on } \Gamma, \quad U_O = 0 \text{ on } \partial R, \end{aligned}$$

with the Navier–Stokes pressure p . Under (H_{ns}) it exists and is unique, and it is smooth up to Γ (Lemma 14.17). Extend it as in Definition 6.8; the defect F_O is bounded and supported in $\bar{S}_h$, and Lemma 6.9 gives $z_O \in Z_h$ with $\|\tilde{U}_O - z_O\| \leq E_O$.

Theorem 14.20 (Oseen analogue of Theorem 6.10). *Assume (NS1)–(NS2), (H_{ns}) , $\gamma \geq \gamma_0$, $h \leq h_0$, and either form. Let $\delta_R \in Z_h$ solve $\mathfrak{A}(\delta_R, v) = \mathcal{R}(v)$ on Z_h . Then*

$$\left\| \nabla \left(\frac{\nu \mu}{h} \delta_R - \tilde{U}_O \right) \right\|_{\Omega_h} \leq C_O \left[\nu \min(\gamma h_\Gamma^{1/2}, (\gamma h_\Gamma)^{1/2}) + (1 + \nu)(h_\Gamma^{1/2} + (h/\mu)^{1/2}) + E_O \right].$$

Proof. Let $\epsilon := \delta_R - \frac{h}{\nu \mu} z_O$. Green's formula with the Oseen equation gives, for $v \in Z_h$, $\mathfrak{A}(\tilde{U}_O, v) = (F_O, v) + \langle \sigma_O, v \rangle - \nu \langle \tilde{U}_O, \partial_n v \rangle + \nu \frac{\mu}{h} \langle \tilde{U}_O, v \rangle + \kappa_O(v)$, $\sigma_O := \nu(\nabla \tilde{U}_O)^\top n - (\tilde{P}_O - c)n$, with $\kappa_O = 0$ for c_1 and $\kappa_O(v) = O(h_\Gamma^2) \|v\|_{\Gamma_h}$ for c_s . Hence, with $\zeta := (\tilde{p} - \bar{p})n + \tilde{U}_O$,

$$\mathfrak{A}(\epsilon, v) = -\langle \zeta, v \rangle - \frac{h}{\nu \mu} \left[(F_O, v) + \langle \sigma_O, v \rangle - \nu \langle \tilde{U}_O, \partial_n v \rangle + \kappa_O(v) \right] + \frac{h}{\nu \mu} \mathfrak{A}(\tilde{U}_O - z_O, v).$$

Since $\tilde{U}_O|_\Gamma = \tilde{U}|_\Gamma$, steps 3–4 of Theorem 6.10 bound each term. Apply (L) and multiply by $\nu \mu / h$. $\square$

Corollary 14.21 (The Navier–Stokes H^1 constant). *Under the hypotheses of Corollary 6.11(ii), (H_{ns}) and $G > 0$, the GS solution satisfies*

$$\frac{\|\nabla(\tilde{u} - u_h)\|}{(h/\nu \mu)G} \rightarrow K_{NS} := \frac{\|\nabla U_O\|_{L^2(\Omega)}}{G}, \quad K_{NS}^2 = K_S^2 + \frac{\|\nabla(U_O - U_S)\|^2}{G^2} \geq K_S^2,$$

where U_S is the Stokes leak flow (Definition 6.8) with the same wall pressure and $K_S = \|\nabla U_S\|/G$. Equality holds iff $(U_S \cdot \nabla)u + (u \cdot \nabla)U_S$ is a gradient. For test P ($u = 0$), $K_{NS} = 2.7565157$ for every ν .

Proof. $\tilde{u} - u_h = (\tilde{u} - z) + \delta_R + \delta_G + (\delta - \delta_0)$ with the bounds of Theorem 14.8; $(\mu/h)X^2 \rightarrow 0$ under the hypotheses. For the identity, $W = U_O - U_S \in V$ and $(\nabla U_S, \nabla W) = 0$ (weak Stokes equation), cf. (1). $\square$

Corollary 14.22 (GS drag for Navier–Stokes). *For $\psi \in \text{RM}$ let $(Z_\psi^\dagger, R_\psi^\dagger)$ solve the adjoint Oseen problem*

$$-\nu \Delta Z - (u \cdot \nabla)Z + (\nabla u)^\top Z + \nabla R = 0, \quad \text{div } Z = 0 \text{ in } \Omega, \quad Z|_\Gamma = \psi, \quad Z|_{\partial R} = 0. \quad (24)$$

Under the hypotheses of Corollary 14.21, with J_h^N , J_ψ , Θ_f as in Section 8.7,

$$J_h^N(\psi) = J_\psi - \frac{h}{\nu\mu} K_\psi^\dagger - \Theta_f(\psi) + o(h/\mu), \quad K_\psi^\dagger = \int_\Gamma \sigma_\nu(U_O, P_O) n \cdot \psi = \int_\Gamma (p - \bar{p})(R_\psi^\dagger - \bar{R}_\psi^\dagger),$$

and the remainder is $O(h^{3/2})$ for fixed γ , bounded ρ and $\varepsilon_h \leq Ch^k$.

Proof. In Proposition 8.44(c) write $e = \frac{h}{\nu\mu} \tilde{U}_O + \vartheta$ with $\|\vartheta\|_{H^1} = o(h/\mu)$ (Corollary 14.21 and the lift E), and $|c(e; e, v_\psi)| \leq CX^2$. With $v_\psi = \psi$ on S_h , symmetry of σ and skewness of $\nabla\psi$, the divergence theorem on S_h gives $\nu a_h(\tilde{U}_O, v_\psi) + K(\tilde{U}_O, v_\psi) = J_\psi(U_O) + O(h_\Gamma^2)$. Reciprocity (Green's formula on Ω for both problems, with $((u \cdot \nabla)U, Z) = -((u \cdot \nabla)Z, U)$) gives $J_\psi(U_O) = \int_\Gamma \sigma_\nu(Z, R) n \cdot U_O$; on Γ , $U_O = -(p - \bar{p})n$ and $n \cdot D(Z)n = 2 \text{div}(Z - \psi) = 0$. $\square$

14.10. The CNS* drag for Navier–Stokes.

Which adjoint. The algebraic transpose of the linearised CNS* system carries the GS pressure form in its momentum row, and its constraint row makes $\text{div } \zeta$ a first-layer field (Lemma 8.2), so $\zeta \notin Z_h$; relative to a smooth adjoint its consistency defect is the GS functional with the dual pressure, of size $(h/\mu)^{1/2}$. This is the adjoint inconsistency of Remark 9.7. We never form that transpose. The drag representation needs only a field $\zeta_h \in Z_h$ that is exactly divergence-free and consistent with the smooth adjoint; the CNS* discretisation of the continuous adjoint problem, i.e. problem $(L^\dagger)$, provides it. For Stokes ($K = 0$), $(L^\dagger)$ is the primal CNS* system, because N_h is symmetric, and it is exactly the auxiliary problem (10).

Theorem 14.23 ($(L^\dagger)$ from nonsingularity). *Assume (H_{ns}) and $\gamma \geq \gamma'_2$. There are $h_0, \beta_\dagger > 0$, independent of μ, γ, ρ , such that $(L^\dagger)$ holds for $h \leq h_0$. Quantitatively, h_0 and $\beta_\dagger$ are as in Theorem 14.19(b).*

Proof. The quantitative statement is Theorem 14.19(b). Without (R_S) , repeat the proof of (L^*) in Theorem 14.16 with $\mathfrak{A}^\top$: $\nu N_h(\zeta, w) + B^*(\pi, w) = \ell(w) - K(w, \zeta)$ is the same Stokes system, and the limit $\zeta \neq 0$ satisfies $\langle L^\top \zeta, \varphi \rangle = 0$ for $\varphi \in \mathcal{V}$ (Lemma 14.15(c)), so $\zeta = 0$. $\square$

For $\psi \in \text{RM}$ put $\zeta := V_\psi - Z_\psi^\dagger \in V$, with $Z_\psi^\dagger$ from (24), $\Pi_c := -R_\psi^\dagger$ for c_1 and $\Pi_c := -R_\psi^\dagger - \frac{1}{2}u \cdot Z_\psi^\dagger$ for c_s (so $\Pi_c = -R_\psi^\dagger$ on Γ), extend by Lemma 9.4, and define

$$m_\psi(w) := \nu a_h(w, v_\psi) + K(w, v_\psi),$$

$$\Lambda_h^\dagger(\psi) := \nu \langle \tilde{u}, \partial_n(\tilde{Z}_\psi^\dagger - \psi) \rangle_{\Gamma_h} = \nu \sum_{e \in \mathcal{E}_h^\Gamma} \frac{|e|^3}{12} (\partial_n u \cdot \partial_n(Z_\psi^\dagger - \psi))(m_e^*) + O(h_\Gamma^3),$$

$E_Z := \inf_{z \in Z_h} \|\tilde{\zeta} - z\| \leq Ch$, and $Y_Z := (1 + \gamma^{1/2})h_\Gamma^{3/2} + E_Z + h^k$. The expansion of $\Lambda_h^\dagger$ is step 4 of Theorem 8.19.

Lemma 14.24 (The discrete dual). *Assume (H_{ns}) , $\gamma \geq \gamma'_2$ and $h \leq h_0$. The problem $(\zeta_h, \pi_h) \in Z_h \times \Pi_h$: $\mathfrak{A}(w, \zeta_h) + B^*(\pi_h, w) = m_\psi(w)$ for all $w \in V_h^R$, has a unique solution, with $\|\tilde{\zeta} - \zeta_h\| \leq CY_Z$, $\beta \|(\Pi^* - \pi_h) - c_\Pi\| \leq CY_Z$ for $\Pi^* := \tilde{\Pi}_c - \bar{p}_\Gamma(\tilde{\Pi}_c)$ and a constant c_Π , and $\|\zeta_h\| + \|\nabla \zeta_h\| \leq C$.*

Proof. Green's formula as in Lemma 5.1 (N_h symmetric), the adjoint equation, and $B^*(\Pi^*, w) = (\nabla \tilde{\Pi}_c, w)$ give $\mathfrak{A}(w, \tilde{\zeta}) + B^*(\Pi^*, w) = m_\psi(w) + \mathcal{G}^\dagger(w)$, where $\mathcal{G}^\dagger$ collects the strip defect, the boundary terms $\nu \langle (\nabla \tilde{\zeta})^\top n, w \rangle - \nu \langle \tilde{\zeta}, \partial_n w \rangle + \nu_h^\mu \langle \tilde{\zeta}, w \rangle$, $\langle (\tilde{u} \cdot n)w, \tilde{\zeta} - \psi \rangle$, $-\nu a_h + K](w, v_\psi - V_\psi)$ and an $O(h_\Gamma^2) \|w\|_{\Gamma_h}$ term; for c_s the difference $K_s - K_1$ is absorbed by the shift from Π_1 to Π_s . Since $\zeta|_\Gamma = 0$ and $\operatorname{div} \zeta = 0$, Lemma 3.4 gives $|(\nabla \tilde{\zeta})^\top n_e| \leq Ch_\Gamma$ and $|\tilde{\zeta}| \leq Ch_\Gamma^2$ on Γ_h , and as in Lemma 8.4, $|\mathcal{G}^\dagger(w)| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + h^k] \|w\|$. Then $(L^\dagger)$ and Lemma 4.5 give the bounds, as in step 3 of Theorem 7.2. $\square$

Theorem 14.25 (CNS* drag, lift and torque for Navier–Stokes). *Assume (NS1)–(NS2), (H_{ns}) , either form, $\gamma \geq \max(1, \gamma_2, \gamma'_2)$, $\gamma h_\Gamma \leq 1$, $h \leq h_0$ and $Y \leq c_*$ (Theorem 14.12), and let $\psi \in \text{RM}$. Then*

- (a) $|J_h^N(\psi) - J_\psi + \Theta_f(\psi)| \leq CY$;
- (b) $|J_h^N(\psi) - J_\psi + \Theta_f(\psi) + \Lambda_h^\dagger(\psi)| \leq C[\gamma^{-1/2}h_\Gamma^2 + \gamma h_\Gamma^3 + \gamma^{1/2}h_\Gamma^{3/2}E_Z + \varepsilon_h]$.

Consequently $|J_h^N(\psi) - J_\psi| \leq C(h_\Gamma^2 + \varepsilon_h)$: rate 2, as for Stokes. As $\gamma \rightarrow \infty$ (with $\gamma h_\Gamma \rightarrow 0$) the leading term $-\Theta_f - \Lambda_h^\dagger$ is geometric; it is the Stokes term of Theorem 8.19 with $\partial_n u$ the Navier–Stokes wall shear and Z_ψ replaced by $Z_\psi^\dagger$. At fixed γ the $\gamma^{-1/2}h_\Gamma^2$ term is genuinely of order h_Γ^2 (Remark 8.43).

Proof. (a) is Proposition 8.44(c). For (b) write $e = (\tilde{u} - z) + e_h$, $e_h := z - u_h \in Z_h$. Testing the discrete dual with $w = e_h$ and the error equation with $v = \zeta_h$ gives

$$J_h^N - J_\psi + \Theta + \Lambda_h^\dagger = -T_1 - (\mathcal{G}_\nu(\zeta_h) - \Lambda_h^\dagger) - \kappa_c(\zeta_h) + c(e; e, v_\psi - \zeta_h) + T_4 - T_5 + O(h_\Gamma^4),$$

with $T_1 := m_\psi(\tilde{u} - z) - \mathfrak{A}(\tilde{u} - z, \zeta_h)$, $T_4 := \langle e_p - \bar{p}_\Gamma(e_p), \zeta_h \cdot n \rangle$, $T_5 := \langle \pi_h - \bar{p}_\Gamma(\pi_h), e_h \cdot n \rangle$. Then $|T_1| \leq C\varepsilon_h$; $\mathcal{G}_\nu(\tilde{\zeta}) = \Lambda_h^\dagger + O((1 + \gamma)h_\Gamma^3)$ and $|\mathcal{G}_\nu(\zeta_h) - \mathcal{G}_\nu(\tilde{\zeta})| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}Y_Z$; $|\kappa_{c_s}(\zeta_h)| \leq Ch_\Gamma^4$; $|c(e; e, v_\psi - \zeta_h)| \leq CY^2$; and T_4, T_5 are steps 5–6 of Theorem 8.19 verbatim, with the dual pressure Π_c , Lemma 14.24, Proposition 8.44(a), and the weighted normal slip of Lemma 8.20, which holds for the Navier–Stokes CNS* solution (the new terms $K(e_u, w_\phi)$, $c(e_u; e_u, w_\phi)$ and $\kappa_c(w_\phi)$ are of the size of $a_h(e_u, w_\phi)$; this needs $\gamma \geq \gamma_2$). $\square$

14.11. The Oseen leak limit uniformly in the penalty. The Stokes results of Section 7.3 carry over. Let $g := -(\tilde{p} - \bar{p})n_e$ on each $e \in \mathcal{E}_h^\Gamma$ with the Navier–Stokes pressure, $t^* := \Pi_\Gamma g$, and let δ_R be as in Theorem 14.20; put $X := \frac{\nu}{h}\delta_R$ and $E_O^0 := h\|U_O\|_{H^2} + h^k + h_\Gamma^{\sigma_k-1/2}$. By Lemma 7.38, $\mathfrak{A}(X, v) = \nu_h^\mu \langle t^*, v \rangle$ for $v \in Z_h$. Constants depend on ν , U_∞ , β_L , β_D , the mesh constants and norms of p and U_O near Γ , never on $h, h_\Gamma, \mu, \gamma, \rho$.

Theorem 14.26 (Uniform two-sided bound). *Assume (H_{ns}) , $h \leq h_0$. Then for every $\gamma \geq \gamma_0$*

$$\|\nabla(X - \tilde{U}_O)\|_{\Omega_h} \leq C(h_\Gamma^{1/2} + E_O^0), \quad \text{so } \|\nabla \delta_R\| \leq Ch/\mu.$$

If moreover $k \geq 4$ and $G > 0$, then $\|\nabla(X - \tilde{U}_O)\| \geq c_O G h_\Gamma^{1/2} - C h_\Gamma^{3/2}$ with $c_O := \nu \kappa_1 c_0^{1/2} / (\sqrt{2}[\nu(2 + C_{PT}C_I) + C_F(1 + C_F)U_\infty])$, κ_1 of Lemma 7.43.

Proof. Upper. Let $X^D \in Z_h$ be the unique field with $X^D|_{\Gamma_h} = t^*$ and $\nu a_h(X^D, v) + K(X^D, v) = 0$ on Y_h (unique by (L_D)). As in Lemma 7.41, with $\nu a_h(\tilde{U}_O, v) + K(\tilde{U}_O, v) = (F_O, v)$ on Y_h (no boundary terms, $K = K_1$), $\|\nabla(X^D - \tilde{U}_O)\| \leq C(h_\Gamma^{1/2} + E_O^0)$. For $D := X - X^D$ the two penalty terms cancel exactly and $\mathfrak{A}(D, v) = -[\nu a_h + K](X^D, v) + \nu \langle \partial_n X^D, v \rangle + \nu \langle t^*, \partial_n v \rangle$; the proof of Theorem 7.42 (lift of $v|_{\Gamma_h}$, Green's formula with the Oseen equation, Lemma 7.39) bounds the right-hand side by $C(1 + h_\Gamma^{-1/2}E_O^0)\|v\|_{\Gamma_h} + Ch_\Gamma^{1/2}\|\nabla v\|$, and (L), uniform in μ , gives $\|D\| \leq C[(h_\Gamma/\gamma)^{1/2}(1 + h_\Gamma^{-1/2}E_O^0) + h_\Gamma^{1/2}]$. *Lower.* For $v \in Y_h$, $\nu a_h(W, v) + K(W, v) - \nu \langle W, \partial_n v \rangle = \nu \langle \tilde{U}_O, \partial_n v \rangle - (F_O, v)$ with $W := X - \tilde{U}_O$; $|K(W, v)| \leq C_F(1 + C_F)U_\infty\|\nabla W\|\|\nabla v\|$ and, on Y_h^1 ,

$|\langle W, \partial_n v \rangle| \leq C_{PT} C_I \|\nabla W\| \|\nabla v\|$ (Lemma 7.20). Since $\tilde{U}_O|_\Gamma = \tilde{U}|_\Gamma$, the test field of the proof of Theorem 7.44 applies verbatim. $\square$

Theorem 14.27 (GS for Navier–Stokes, uniformly in the penalty). *Assume (NS1)–(NS2), (H_{ns}) , $h \leq h_0$, either form, and put $\varepsilon_1 := h^k + h^{\sigma_k} h_\Gamma^{-1/2}$, $X_u := h/\mu + h_\Gamma^{3/2} + \varepsilon_1$. There are c_*, C independent of $\gamma \geq \gamma_0$ (and of μ, ρ) such that if $X_u \leq c_*$, GS has a locally unique solution with*

$$\|\nabla(\tilde{u} - u_h)\| \leq C X_u, \quad \left\| \nabla \left(\frac{\nu\mu}{h} (\tilde{u} - u_h) - \tilde{U}_O \right) \right\| \leq C (h_\Gamma^{1/2} + E_O^0) + C \frac{\mu}{h} (h_\Gamma^{3/2} + \varepsilon_1 + X_u^2).$$

The linearised geometric part is bounded uniformly in γ by $C(h_\Gamma^{3/2} + \varepsilon_1)$ using (L_D) in its quantitative form, Theorem 14.19(c) (the compactness proof of (L_D) is not written out).

Proof. The linearised geometric part δ_G ($\mathfrak{A}(\delta_G, v) = \mathcal{G}_\nu(v) + \kappa_c(v) - \mathfrak{A}(\tilde{u} - z, v)$ on Z_h) is bounded by the proof of Theorem 7.30, steps 2–5, with a_h replaced by $\nu a_h + K$ and Korn by (L_D) . Step 6 of Theorem 14.8 uses $\delta_0 = \delta_R + \delta_G$ only through $A = \|\tilde{u} - z\|_{H^1} + C_1 \|\nabla \delta_0\|$, which is now $\leq C X_u$ by Theorem 14.26; the contraction gives $\|\delta - \delta_0\| \leq C X_u^2$. $\square$

Corollary 14.28 (The Navier–Stokes H^1 constant without $\gamma h_\Gamma \rightarrow 0$). *Let $G > 0$. (a) If $h/\mu \rightarrow 0$, $E_O^0 \rightarrow 0$ and $\frac{\mu}{h}(h_\Gamma^{3/2} + \varepsilon_1) \rightarrow 0$ (i.e. $\gamma h_\Gamma^{1/2} \rightarrow 0$ for the geometric part), then $\|\nabla(\tilde{u} - u_h)\|/((h/\nu\mu)G) \rightarrow K_{NS}$. (b) Under the hypotheses of Theorem 14.27, $\|\nabla(\frac{\nu\mu}{h}(\tilde{u} - u_h) - \tilde{U}_O)\| \geq c_O G h_\Gamma^{1/2} - C(1 + \gamma)h_\Gamma^{3/2} - C \frac{\mu}{h} X_u^2$.*

So the two-sided $h_\Gamma^{1/2}$ law for the Oseen leak limit is uniform in γ for the linearised δ_R ; for the nonlinear GS solution the lower bound (b) still needs $\gamma h_\Gamma^{1/2} \rightarrow 0$ to be informative.

Leak constant for every penalty. Under the hypotheses of Theorem 14.26, $K_{NS,h} := \|\nabla X\|/G$ with $X = (\nu\mu/h)\delta_R$ satisfies $|K_{NS,h} - K_{NS}| \leq C(h_\Gamma^{1/2} + E_O^0)/G$ for every $\gamma \geq \gamma_0$ (triangle inequality). The rate-one argument of Corollary 7.47 uses the symmetry of a_h and does not transfer to the Oseen form verbatim.

The constants for the Navier–Stokes test. (NUMERICAL, notes/v7/nsconst). For test B of code/job_v4.py ns ($u = \text{curl}((r^2 - 1)^2(x + y^2/2)/10)$ on $(-2.5, 2.5)^2 \setminus \overline{B_1}$, $\nu \in \{1, 0.1\}$), computed on the exact domain with an isoparametric Taylor–Hood discretisation and convergence tables: $U_\infty = 175.6$ (the maximum of $|\nabla u|$, attained at the corners of the box, far from Γ ; near Γ it is below 1), $C_F = 0.5642$, $\beta_O = 0.8225$ ($\nu = 1$) and 0.0533 ($\nu = 0.1$), and $C_R \approx 3.5$ from the converged lower-bound sequence $C_R^{\text{quad}} \geq 2.477$ and $C_R \leq \sqrt{2} C_R^{\text{quad}}$; the contribution of the box corners (a quarter-plane problem) to C_R was not computed. These are discretisation-plus-extrapolation values, not enclosures. With them (23) requires $(1 + \text{Re})h + (h_\Gamma/\gamma_0)^{1/2} \leq 7.6 \cdot 10^{-6} c_\sharp$ ($\nu = 1$) and $\leq 4.9 \cdot 10^{-8} c_\sharp$ ($\nu = 0.1$), with $c_\sharp$ (through $C_\sharp$) not computed. The production meshes for this test have $h \geq 0.4$ ($N \leq 64$), so the explicit h_0 is hopelessly small: Theorem 14.19 is qualitative here. The discrete stability constant itself is harmless: $\beta_L/\nu = 0.021, 0.016, 0.014$ on $N = 16, 32, 48$ ($\nu = 1$; and $N = 16, 32$ for $\nu = 0.1$), equal to the Stokes value to 0.1–1%; its decrease with N tracks the margin of $\mu = 100$ above the coercivity threshold. It was not computed on $N = 24, 64$. What remains for Navier–Stokes. The constant $C_\sharp$ of Theorem 14.19, the box-corner part of C_R and rigorous enclosures of β_O ; sharpness of the CNS* drag rate 2 at fixed γ , a lower bound on the pairing capture fraction and the existence of $\lim d_h/h_\Gamma^{1/2}$, as for Stokes. The three-dimensional versions are in [50].

15. PRESSURE ROBUSTNESS

A Stokes discretisation is *pressure-robust* (or gradient-robust) if a change of the force by a gradient changes only the discrete pressure; equivalently, its velocity error does not depend on the pressure and does not grow like ν^{-1} as $\nu \rightarrow 0$ [38]. Divergence-free pairs such as Scott–Vogelius are

pressure-robust with strongly imposed boundary conditions on the true domain. Weak imposition of $u \cdot n$ destroys this: Liu, Neilan and Otus note it and mitigate it with a Lagrange multiplier for the normal trace [44], and Frachon, Nilsson and Zahedi observe the loss numerically for unfitted methods [29]. On curved domains with exactly imposed conditions, pressure robustness is restored by reconstructing the load with a commuting interpolant: Neilan and Otus in two dimensions [48, Theorem 6.1], Chen and Liu in three [21]. This section quantifies how far GS and CNS* are from pressure-robust on the inscribed polygon, and gives a one-line load correction (CNS*-R) that makes CNS* exactly gradient-robust. The correction is a Linke-type reconstruction localised to the boundary triangles; we claim no novelty for the device.

Throughout, the methods are those of Section 14 with the convective term removed and viscosity ν :

$$\nu N_h(u_h, v) + B_M(p_h, v) = (\tilde{f}, T_M v) \quad \forall v \in V_h^R, \quad (q, \operatorname{div} u_h) = 0 \quad \forall q \in \Pi_h,$$

with $B_{\text{GS}}(q, v) = -(q, \operatorname{div} v)$, $B_{\text{CNS}^*} = B_{\text{CNS}^*\text{-R}} = B^*$, $T_M = \operatorname{id}$ for GS and CNS* and $T_{\text{CNS}^*\text{-R}} = \mathcal{R}_b$ (Definition 15.8). The penalty is $\nu\mu/h$ (Remark 14.2), π_h is the elementwise L^2 projection onto Π_h , and $F := \tilde{f} + \nu\Delta\tilde{u} - \nabla\tilde{p}$ is the extension defect; it is smooth and vanishes on $\bar{\Omega}$. All numerical constants are at $\mu = 100$ ($\gamma \approx 30$), which is only 1.37–1.41 times the coercivity threshold of GS on Z_h (Table 14) and 1.13–1.27 times the stability threshold of CNS* on the production meshes (Remark 15.18); the thresholds $\gamma_0, \gamma_2, \gamma_3$ of the proofs are not shown to hold there, so the computations illustrate the theorems rather than being covered by them.

15.1. Splitting.

Lemma 15.1 (Scaling). *(u_h, p_h) solves method M at viscosity ν with data $(\tilde{f}, g_I)$ iff $(u_h, p_h/\nu)$ solves M at $\nu = 1$ with data $(\tilde{f}/\nu, g_I)$.*

For a smooth ϕ let $W_M(\phi)$ be the $\nu = 1$ velocity of M with data $(\nabla\phi, 0)$ (the *hydrostatic test*). M is gradient-robust iff $W_M \equiv 0$.

Theorem 15.2 (Splitting). *$\tilde{u} - u_h^{M,\nu} = E_M - \nu^{-1}W_M(\tilde{p})$, where $E_M := \tilde{u} - u_h^{M,1}[-\Delta\tilde{u} + F/\nu, g_I]$ is the $\nu = 1$ error for the problem with exact solution $(u, 0)$ and extension defect F/ν . E_M is independent of p and ν up to a term of size $Ch_\Gamma^4\|\nabla F\|_{L^\infty(S_h)}/\nu$ (Lemma 15.3), and exactly when $F = 0$.*

Proof. $\tilde{f} = -\nu\Delta\tilde{u} + \nabla\tilde{p} + F$; use linearity and Lemma 15.1. □

So the theorems of Sections 6–7 apply to E_M with $p \equiv 0$ ($G = 0$, $\mathcal{R} \equiv 0$), with constants depending only on u ; the whole pressure dependence is in W_M .

Lemma 15.3 (Strip defect).

$$|(F, v)| \leq Ch_\Gamma^4\|\nabla F\|_{L^\infty(S_h)}(\|v\|_{\Gamma_h} + h_\Gamma\|\nabla v\|) \leq Ch_\Gamma^4\|\nabla F\|_{L^\infty(S_h)}\|\nabla v\|.$$

This sharpens the h_Γ^2 bound of Lemma 3.3.

Proof. Radial segments in S_h have length $\leq h_\Gamma^2/4$ and $F = 0$ on Γ , so $|F| \leq \frac{1}{4}h_\Gamma^2\|\nabla F\|_\infty$ on S_h . In polar coordinates $\|v\|_{L^1(S_h)} \leq Ch_\Gamma^2(\|v\|_{L^1(\Gamma_h)} + \|\nabla v\|_{L^1(S_h)})$; then $|S_h| \leq Ch_\Gamma^2$, Cauchy–Schwarz and Lemma 3.2. □

Lemma 15.4 (Nitsche-harmonic fields). *If $w \in Z_h$ and $a_h(w, y) - \langle w, \partial_n y \rangle = 0$ for all $y \in Y_h$, then $\|w\|_{L^2(\Gamma)}/C_{\text{tr}} \leq \|\nabla w\| \leq C_Y h_\Gamma^{-1/2}\|w\|_{\Gamma_h}$, uniformly in μ , γ and ν . Every $W_M(\phi)$ satisfies the hypothesis.*

Proof. Let L be the lift of $w|_{\Gamma_h}$ (Lemma 7.28) and $y := w - L \in Y_h$. Then $a_h(y, y) = \langle w, \partial_n y \rangle - a_h(L, y)$; use Lemma 4.1 and Korn. For $v \in Y_h$, $(\nabla\phi, v) = 0$, $B_M(\cdot, v) = 0$ and $\mathcal{R}_b v = v$. □

15.2. The printed method is not pressure-robust.

Theorem 15.5 (GS). *Let $G_\phi := \|\phi - \bar{\phi}\|_{L^2(\Gamma)}$.*

- (a) $c_1 \|W_{\text{GS}}(\phi)\| \leq (h/\mu)^{1/2}(G_\phi + Ch_\Gamma^2)$ and $\|\nabla W_{\text{GS}}(\phi)\| \leq Ch_\Gamma^{-1/2}(h/\mu)(G_\phi + Ch_\Gamma^2)$, for every $\mu \geq \mu_0$.
- (b) $\|W_{\text{GS}}(\phi)\| \geq (2M)^{-1}(h/\mu)^{1/2}G_\phi$, and in the regime of Theorem 6.6, $\|\nabla W_{\text{GS}}(\phi)\| \geq c(h/\mu)G_\phi$.
- (c) In the regime of Corollaries 6.7 and 6.11, the normalised slip and energy tend to 1 and $\|\nabla W_{\text{GS}}\|/((h/\mu)G_\phi) \rightarrow K_\phi$.

Hence, at viscosity ν , $\tilde{u} - u_h = E_{\text{GS}} - (h/\nu\mu)v_{\bar{p}} + o(h/\nu\mu)$: the pressure part of the GS error is the leak flow divided by ν .

Proof. (a) is the proof of Theorem 6.1 with $\tilde{u} = z = 0$ and $\mathcal{G} = 0$, plus Lemma 15.4. (b) and (c) are Theorems 6.6, 6.4, Corollaries 6.7, 6.11 with $u = 0$. $\square$

With an unscaled penalty $\hat{\mu}/h$ ($\mu = \hat{\mu}/\nu$) the bounds of (a) become ν -uniform: $\nu^{-1}\|W\|_{\Gamma_h} \leq (h/\hat{\mu})(G + Ch_\Gamma^2)/c_1$. The leak law $(h/\hat{\mu})(p - \bar{p})$ itself is then not proved (it needs $\gamma h_\Gamma \rightarrow 0$, Remark 6.12); numerically the slip equals $(h/\hat{\mu})G$ to 10^{-3} at $N = 32, 64$.

15.3. The pressure-consistent method has one pressure defect.

Proposition 15.6 (Error identity). *On Z_h the CNS* error $e_u = \tilde{u} - u_h$ satisfies $\nu N_h(e_u, v) + \langle \eta, v \cdot n \rangle + \langle \xi - \bar{\xi}, v \cdot n \rangle = \mathcal{G}_\nu(v)$, with $\eta := p^* - \pi_h p^*$ on Γ_h and ξ the discrete pressure error of Theorem 7.2. There is no pressure traction mismatch: Lemma 5.1 integrates $\langle \tilde{p}, v \cdot n \rangle$ exactly with the chord normal n_e ; the tilt $n_e - n(x^*)$ appears only in the velocity term $\langle (\nabla \tilde{u})^\top n, v \rangle$ of $\mathcal{G}$, which does not involve p . Flux and mean terms vanish on Z_h , the strip enters only through F (Lemma 15.3), and ξ has no independent pressure source (step 3 of Theorem 7.2). So η is the only pressure-dependent term.*

Theorem 15.7 (CNS*: upper bounds). *For $\gamma \geq \gamma_2$ and $\phi \in W^{k,\infty}$ near Γ :*

- (a) $\|W_{\text{CNS}^*}(\phi)\| \leq (2/c_1)(h/\mu)^{1/2}\|\eta_\phi\|_{\Gamma_h} \leq C\gamma^{-1/2}h_\Gamma^{k+1/2}|\phi|_{W^{k,\infty}}$;
- (b) $\|\nabla W_{\text{CNS}^*}(\phi)\| \leq Ch_\Gamma^{-1/2}(h/\mu)\|\eta_\phi\|_{\Gamma_h} \leq C\gamma^{-1}h_\Gamma^{k+1/2}|\phi|_{W^{k,\infty}}$;
- (c) $W_{\text{CNS}^*}(\phi) = 0$ if $\phi|_T \in P_{k-1}$ on every boundary triangle.

Consequently $\|\tilde{u} - u_h\| \leq C_u[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h] + Ch_\Gamma^4\|\nabla F\|/\nu + (2/c_1)\nu^{-1}(h/\mu)^{1/2}\|\tilde{p} - \pi_h \tilde{p}\|_{\Gamma_h}$, with the analogous H^1 statement.

Proof. With $u = 0$, $F = 0$ and $v = W$ in Proposition 15.6: $c_1 \|W\|^2 \leq |\langle \eta, W \cdot n \rangle| + C(c_0\gamma)^{-1/2}\|\xi - \bar{\xi}\| \|W\|$ and $\beta \|\xi - \bar{\xi}\| \leq C \|W\|$; absorb for $\gamma \geq \gamma_2$. $\|\eta\|_{L^\infty(T)} \leq Ch_T^k |\phi|_{k,\infty}$ with $h_T \leq Ch_\Gamma$ on boundary triangles. (b) is Lemma 15.4. $\square$

Part (c) explains why the cubic pressure of test P gives errors at round-off for CNS* (Section 16). With an unscaled penalty $\nu^{-1}\|\nabla W_{\text{CNS}^*}\| \leq Ch_\Gamma^{k-1/2}(h/\hat{\mu})|p|$, which is ν -uniform.

15.4. An exactly gradient-robust variant.

Definition 15.8 (CNS*-R). For a boundary triangle T_e and $w \in P_k(e)$ let $b_e(w) \in \text{BDM}_k(T_e)$ satisfy $b \cdot n = w$ on e , $b \cdot n = 0$ on the other two edges, and $(\text{div } b, r)_{T_e} = \langle w, r \rangle_e$ for $r \in P_{k-1}(T_e)$ (for proofs fixed by Piola-mapping a reference right inverse). Put $\mathcal{R}_b v := v - \sum_{e \in \mathcal{E}_h^\Gamma} b_e(v \cdot n_e)$. CNS*-R is CNS* with the load $(\tilde{f}, \mathcal{R}_b v)$; the system matrix is unchanged.

The b_e exist by BDM unisolvence, satisfy $\int_{T_e} b \cdot \nabla r = 0$ for $r \in P_{k-1}$ (in particular $\int_{T_e} b = 0$), and $\|b\|_{T_e} \leq C|e|^{1/2}\|w\|_e$ by Piola scaling.

Theorem 15.9 (CNS*-R is gradient-robust). *Let $\gamma \geq \gamma_2$ and let the load be integrated exactly.*

- (a) *If $\tilde{f} = \nabla\phi$ on Ω_h (including the strip), the solution is $(0, \pi_h\phi - \bar{p}_\Gamma(\pi_h\phi))$. Hence $W_{\text{CNS}^*-R} \equiv 0$. For a non-gradient extension a residual $(F, \mathcal{R}_b v) = O(h_\Gamma^4)$ remains.*
- (b) *E_{CNS^*-R} satisfies the bounds of Theorems 7.2 and 7.30 and the lower bound of Theorem 7.24; the extra load $H(v) = \sum_e (-\Delta\tilde{u} + F/\nu, b_e(v \cdot n_e))$ satisfies $|H(v)| \leq C(h_\Gamma^2 + h_\Gamma^{7/2} \|\nabla F\|/\nu) \|v\|_{\Gamma_h}$ and vanishes on V_h^0 and on Y_h .*

Proof. (a) For $v \in V_h^R$, $(\nabla\phi, v) = -(\pi_h\phi, \text{div } v) + \langle \phi, v \cdot n \rangle$, since $\text{div } V_h \subset \Pi_h$ and $v = 0$ on ∂R . On T_e , $(\nabla\phi, b_e) = -(\phi, \text{div } b_e) + \langle \phi, v \cdot n \rangle_e = -\langle \pi_{T_e}\phi, v \cdot n \rangle_e + \langle \phi, v \cdot n \rangle_e$ (no triangle has two edges on Γ_h). So $(\nabla\phi, \mathcal{R}_b v) = -(\pi_h\phi, \text{div } v) + \langle \pi_h\phi, v \cdot n \rangle = B^*(\pi_h\phi - \bar{p}_\Gamma(\pi_h\phi), v)$. The continuity row holds for $u_h = 0$; uniqueness is Theorem 7.2. (b) $b_e \perp$ constants ($k \geq 2$) gives $\sum_e h_\Gamma^{5/2} \|v \cdot n\|_e \leq C h_\Gamma^2 \|v\|_{\Gamma_h}$; H vanishes where $v \cdot n = 0$ on Γ_h . $\square$

The same correction does nothing for GS: GS has no traction term, and $W_{\text{GS}-R}(\phi) = W_{\text{GS}}(\pi_h\phi)$ still leaks. The method needs $\tilde{f}$ on the boundary triangles, which the paper assumes anyway.

15.5. Sharpness for CNS*. The upper bounds of Theorem 15.7 are attained, but the structure is subtler than the bounds suggest: some directions of $\nabla^k\phi$ are invisible to the defect functional. Energy. Let V_R^0 be the zero-flux fields of V_h^R (not necessarily divergence-free).

Theorem 15.10 (Energy duality). *For $\mu \geq \mu_0$ and every $v \in V_R^0$, $|\langle \eta, v \cdot n \rangle| \leq K_1 \|W_{\text{CNS}^*}\| \|v\|$ with $K_1 := M + \beta^{-1}(2 + C_I c_0^{-1/2} \gamma^{-1/2})(\sqrt{2} + C_I c_0^{-1/2} \gamma^{-1/2})$. So $\|W_{\text{CNS}^*}\| \geq K_1^{-1} D_h^0$, $D_h^0 := \sup_{V_R^0} \langle \eta, v \cdot n \rangle / \|v\|$, with no absorption.*

Proof. For $v \in V_R^0$, $N_h(W, v) = \langle \eta, v \cdot n \rangle + \langle \xi - \xi_\Omega, v \cdot n \rangle - (\xi - \xi_\Omega, \text{div } v)$ (constants drop out since $\int_{\Gamma_h} v \cdot n = 0$). Testing with $v \in V_h^0$, $\text{div } v = \xi - \xi_\Omega$ (Lemma 4.5), gives $\|\xi - \xi_\Omega\| \leq \beta^{-1}(2\|\nabla W\| + C_I c_0^{-1/2} h_\Gamma^{-1/2} \|W\|_{\Gamma_h})$; use $h_\Gamma^{-1/2} \|W\|_{\Gamma_h} \leq \gamma^{-1/2} \|W\|$. $\square$

For every k , $D_h^0 \geq c(1 + \gamma)^{-1/2} h_\Gamma^{1/2} \|Y\|_{\Gamma_h}$ with Y the zero-flux projection of ηn ; and for even k an exact Rodrigues identity, $\langle \eta, L_k^e \rangle_e = |e|^{k+1} \frac{k!}{(2k+1)!} \text{avgBeta}(k+1, k+1) (\partial_{\tau_e}^k \phi)$ (valid for every k and independent of the triangle shape), yields $D_h^0 \geq c(\phi)(1 + \gamma)^{-1/2} h_\Gamma^{k+1/2}$ whenever $\nabla^k \phi[\tau^{\otimes k}] \not\equiv 0$ on Γ . For odd k the corresponding global test field does not exist (L_k^e has opposite signs at the two ends). So the energy order $\gamma^{-1/2} h_\Gamma^{k+1/2}$ is sharp for even k under that condition (details: [notes/v6/robust2](#)).

H^1 for every penalty. Call $v \in V_h^R$ *pressure-blind* if $B^*(q, v) = 0$ for all $q \in \Pi_h$; equivalently, $\int_{\Gamma_h} v \cdot n = 0$, $\text{div } v = 0$ on triangles without a Γ_h edge, and $(\text{div } v, r)_{T_e} = \langle v \cdot n, r \rangle_e$ for $r \in P_{k-1}(T_e)$ (the moment condition of Definition 15.8). For such v , exactly,

$$N_h(W_{\text{CNS}^*}(\phi), v) = \langle \eta, v \cdot n \rangle : \quad (25)$$

the pressure error disappears from the equation. If moreover, on a patch ω , (M) $\sum_{e \subset \partial\omega \cap \Gamma_h} \int_e v = 0$ and (D) $\sum_e \int_e \partial_n v = 0$, then $N_h(c, v) = 0$ for constants c , so $|N_h(w, v)| \leq \Phi_\omega^*(v) \|\nabla w\|_\omega$ with $\Phi_\omega^* \leq \Phi_\omega^0 + \gamma \Phi_\omega^1$ explicit.

Theorem 15.11 (Certificate for every $\gamma > 0$). *Let patches ω_j with overlap K and pressure-blind v_j satisfying (M), (D) on ω_j be given. Then for every $\gamma > 0$ and every CNS* solution velocity W of the hydrostatic problem, $\|\nabla W\|^2 \geq K^{-1} \sum_j \langle \eta, v_j \cdot n \rangle^2 / \Phi_{\omega_j}^*(v_j)^2$. With boundary stars and H_z the degree- k Taylor term of ϕ at z ,*

$$\|\nabla W\| \geq \frac{1}{2\sqrt{K}(1 + \gamma)} \left(\sum_z s_z^{2k+2} (\sigma_z(H_z) - C_R s_z |\phi|_{W^{k+1, \infty}})^2 \right)^{1/2},$$

where $s_z = |e_z|$ and σ_z is a scale-invariant seminorm on homogeneous degree- k polynomials, computable from the star.

Proof. By (25) and the patch bound, $|\langle \eta, v_j \cdot n \rangle| \leq \Phi_j^* \|\nabla W\|_{\omega_j}$; square and sum. For the second bound take v optimal for $\sigma_z(H_z)$; the Taylor remainder on each boundary triangle is $(I - \pi_T)(\phi - T_z^k \phi) = O(s^{k+1})$, the trace of v is controlled by Φ^1 , and $\Phi^* \leq \sqrt{2}(1 + \gamma)\hat{\Phi}$. $\square$

Only the momentum row is used: neither $W \in Z_h$, uniqueness, nor $\gamma \geq \gamma_0$ enters, and the bound holds below the coercivity threshold. Numerically ($k = 4$, $\phi = \sin 2x \cos y$, $\mu = 100$) the certificate times $\gamma/h_\Gamma^{4.5}$ converges to ≈ 0.015 ($N = 8, \dots, 256$), and $\|\nabla W\|$ is 3.2–4.6 times the certificate.

Invisible directions. For a boundary triangle T with Γ_h -edge $e = [L, R]$ and apex c , let ℓ_w be the linear part of the barycentric coordinate λ_w , $\text{Ker}_T := \text{span}\{\ell_c^k, \ell_L^k, \ell_R^k\}$ and $K_T := \text{span}\{\ell_c^k, \ell_L^k - \ell_R^k\}$. Then $(I - \pi_T)\lambda_w^k = J_k(\lambda_w)$ with J_k the monic Jacobi-type polynomial orthogonal to P_{k-1} with weight $1 - s$; the restriction of $H \mapsto ((I - \pi_T)H)|_e$, projected onto the zero-mean, zero-end part of $P_k(e)$, has kernel exactly Ker_T ; and for $H = \ell_L^k - \ell_R^k$ the edge defect d_k satisfies $\langle d_k, g \rangle_e = \beta_k |e| (g(L) - g(R))$ for all $g \in P_k(e)$, $\beta_k = (-1)^k J_k(0)/(k+1)$ ($\beta_4 = 1/630$). All three facts hold for every $k \geq 4$: the kernel statement follows from the Dubiner basis, whose restriction to e is $\pm$ the Legendre polynomials, and $\beta_k > 0$ because $\beta_k = (-1)^k J_k(0)/(k+1)$ and the zeros of J_k lie in $(0, 1)$ (notes/v6/robust2, robust4, notes/v7/robust; exact checks for $4 \leq k \leq 12$, independently on a physical triangle for $k = 4, 5, 7, 9$). Consequently a K_T -component of $\nabla^k \phi$ produces only an edge constant and a vertex-difference functional, and on a straight, translation-periodic boundary $\langle \eta_H, v \cdot n \rangle = \text{const} \cdot \int v \cdot n$ for every $v \in V_h$ and $H \in K_T$: no test field of any kind sees K_T . The production meshes converge to such a configuration. And the effect is real, not a limitation of the certificates: on the equispaced family UP (radial first layer of constant height ratio), the datum ϕ_B with $\nabla^4 \phi_B / 4! = \cos 4\theta (x^4 - (x - y/\rho)^4)$ on Γ (not purely normal) has $\|\nabla W_{\text{CNS}^*}(\phi_B)\| \leq C\gamma^{-1}h_\Gamma^{k+3/2}$ (proved), with observed order 5.45–5.48.

Theorem 15.12 (Order of the CNS* pressure part). *Let θ be the minimum angle, assume (M0)–(M2), quasi-uniformity along Γ_h , and that each Γ_h vertex has a neighbouring triangle without a Γ_h edge adjacent to the boundary triangle (condition (G)). Let K be the patch overlap and $W = W_{\text{CNS}^*}(\phi)$ with $\phi \in C^{k+3}$.*

- (a) (Proved; all meshes, $\gamma \geq \gamma_2$.) $\|\nabla W\| \leq C\gamma^{-1}h_\Gamma^{k+1/2}|\phi|_{W^{k,\infty}}$.
- (b) (Proved; all meshes, every $\gamma > 0$.)

$$\|\nabla W\| \geq \frac{\kappa_T(\theta)}{4\sqrt{K}(1+\gamma)} \left(\sum_z |e_z|^{2k+2} \text{dist}_F \left(\frac{\nabla^k \phi(z)}{k!}, \text{Ker}_{T_{e_z}} \right)^2 \right)^{1/2} - \frac{C|\phi|_{W^{k+1,\infty}} h_\Gamma^{k+3/2}}{1+\gamma},$$

with $\kappa_T(\theta) > 0$ for all $\theta \in (0, \pi/3]$ and every $k \geq 4$; for $k = 4$, certified $\kappa_T(30^\circ) \geq 2.1 \cdot 10^{-6}$ and $\kappa_T(20^\circ) \geq 5.6 \cdot 10^{-7}$.

- (c) (Proved; LP^+ families, $\gamma \geq \max(\gamma_2, 1)$.) Suppose there are Lipschitz maps $\mathcal{K}$ (normalised shape) and $\ell > 0$ on Γ with $|\text{shape}(T_e) - \mathcal{K}(x_e)| + ||e|/h_\Gamma - \ell(x_e)| \leq Ch_\Gamma$ (production and UP satisfy this). With $K(x) := K_{\mathcal{K}(x)}$ and $D(\phi)^2 := \int_\Gamma \ell^{2k+1} \text{dist}_F(\nabla^k \phi/k!, K(x))^2 ds$,

$$c\gamma^{-1}D(\phi) \leq \liminf_{h_\Gamma \rightarrow 0} \frac{\|\nabla W\|}{h_\Gamma^{k+1/2}} \leq \limsup_{h_\Gamma \rightarrow 0} \frac{\|\nabla W\|}{h_\Gamma^{k+1/2}} \leq C\gamma^{-1}D(\phi) + C\gamma^{-3/2}\|\phi\|_{C^{k+3}}.$$

- (d) (Proved; LP^+ , $D(\phi) = 0$.) $\|\nabla W\| \leq C\gamma^{-1}h_\Gamma^{k+1} + C\gamma^{-3/2}h_\Gamma^{k+1/2}$, and $\leq C\gamma^{-1}h_\Gamma^{k+3/2}$ if the edge-constant part of η is constant along Γ up to $O(h_\Gamma^{k+1})$.

Proof outline. (a) is Theorem 15.7(b) sharpened to $C\gamma^{-1}h_\Gamma^{1/2}\Xi$ with Ξ the norm of the edge-oscillating part of η . (b) is Theorem 15.11 with pressure-blind fields supported in the boundary

triangle and its neighbour across the spoke $[z, c]$: their normal traces are shape independent and span the zero-flux part of $P_k(e)$, their pairing with $\ell_z^j \ell_a^{4-j}$ is a constant rational matrix whose kernel is exactly Ker_T , and only the norms depend on the shape; κ_T is a certified lower bound of the resulting generalised Rayleigh quotient over all admissible shapes (interval arithmetic; negative controls at $2.17 \cdot 10^{-6}$ and $5.7 \cdot 10^{-7}$ are refuted). (c) The lower bound adds a vertex field v_z with $v_z(z) = \nu_z$, zero flux on each wall edge and bounded (M), (D) corrections uniformly through the flat locus; this makes the vertex functional of Ker_T / K_T visible, and a Riemann sum gives $D(\phi)$. The upper bound splits η into edge-oscillating, vertex and edge-constant parts; the last is a smooth trace of size h_Γ^k , handled by the bound $\|\nabla W^a\| \leq C\gamma^{-1}h_\Gamma + C\beta^{-1}\gamma^{-3/2}h_\Gamma^{1/2}\|a\|$ for smooth boundary data a , whose second term comes from the Nitsche pressure layer ξ_b of Lemma 8.2 with trace $k(k+1)(W \cdot n)/H_e$. (d) follows from the same split. Full details, scripts and certificates: [notes/v6/robust3-robust6](#). $\square$

Parts (b)–(d) hold for every $k \geq 4$. For $k = 4$ this is [notes/v6/robust6](#). For $k \geq 5$ every k -dependent ingredient has a structural proof ([notes/v7/robust](#)): the moment map on $X_k := \lambda_z \lambda_a P_{k-2}^2$ has rank $\dim P_{k-1} - 1$ (test with $\lambda_z \lambda_a \nabla r$); the pressure-blind fields have zero flux on the wall edge by a slicing identity with weight $(1-y)^3$, and their normal traces are exactly the zero-mean part of $\lambda_z \lambda_a P_{k-2}(e)$; the wall representer is a function of λ_c alone with trace $k(k+1)/H_e$ (a Jacobi Christoffel number); the kernel of the defect pairing is exactly Ker_T ; and the (M), (D) corrections use the single-triangle stream functions $\lambda_z^{k-2} \lambda_a^2 \lambda_c$ and $\lambda_z^{k-3} \lambda_a^2 \lambda_c^2$, whose curls give a triangular system, while the vertex field keeps the quintic Argyris stream function. Condition (G) is not needed for $k \geq 5$. All facts are checked in exact rationals for $4 \leq k \leq 12$, and the full vertex field for $k = 5$ (four stars) and $k = 6$ (three stars). The constants κ_T ($k \geq 5$) and κ_K are not explicit.

The Nitsche pressure layer is an effective penalty. The $\gamma^{-3/2}$ terms in (c) and (d) come from smooth wall data. For $\alpha \in C^3(\Gamma)$ with $\int_\Gamma \alpha = 0$ and $a := \alpha \circ \pi$, let (W^a, ξ) be the CNS* solution with load $v \mapsto \langle a, v \cdot n \rangle$. For $g \in L^2(e)$ let $\xi_b[g] \in P_{k-1}(T_e)$, extended by 0, satisfy $(\xi_b[g], r)_{T_e} = \langle g, r \rangle_e$ for all $r \in P_{k-1}(T_e)$ (Lemma 8.2), and put

$$m_h(u, v) := \sum_{e \in \mathcal{E}_h^\Gamma} \langle \xi_b[u \cdot n], v \cdot n \rangle_e = \sum_e (\xi_b[u \cdot n], \xi_b[v \cdot n])_{T_e}, \quad \tilde{\xi} := \xi + \xi_b[W^a \cdot n].$$

Lemma 15.13 (Effective penalty). (a) m_h is symmetric positive semidefinite, $m_h(u, u) \leq \sum_e \frac{k(k+1)}{H_e} \|u \cdot n\|_e^2$, and with $\bar{u}_e$ the edge mean and $\tilde{u}$ the fluctuation of $u \cdot n$ on e , $m_h(u, v) = \sum_e \frac{k(k+1)}{H_e} |e| \bar{u}_e \bar{v}_e + m_h(\tilde{u}, \tilde{v})$; the constant is attained on edge constants.

(b) For all $v \in Z_h$ and every constant c , $(N_h - m_h)(W^a, v) + \langle \tilde{\xi} - c, v \cdot n \rangle = \langle a, v \cdot n \rangle$.

(c) For all $v \in V_h^0$, $(\tilde{\xi}, \text{div } v) = a_h(W^a, v) - \langle W^a \cdot \tau, \partial_n v \cdot \tau \rangle$.

Proof. (a) $\|\xi_b[g]\|_{T_e}^2 = \langle g, \xi_b[g] \rangle_e \leq \|g\|_e \|\xi_b[g]\|_e$, and $\|r\|_e^2 \leq \frac{k(k+1)}{H_e} \|r\|_{T_e}^2$ on P_{k-1} [59]. The split holds because $\xi_b[1]$ has constant trace $k(k+1)/H_e$ (Lemma 8.2). (b) On Z_h , $\text{div } v = 0$ and the flux vanishes, so $B^*(\xi, v) = \langle \xi - c, v \cdot n \rangle$, and $\langle \xi_b[W^a \cdot n], v \cdot n \rangle = m_h(W^a, v)$. (c) For $v \in V_h^0$, $\text{div } v = \partial_n v \cdot n$ on Γ_h , so the layer absorbs the normal slip in the pressure identity ([notes/v6/robust5](#), Lemma layer). $\square$

So on Z_h , CNS* is GS with the normal penalty reduced by m_h , plus a pressure $\tilde{\xi}$ whose equation contains only the *tangential* slip, which is $O(h_\Gamma^2/\gamma)$ instead of $O(h_\Gamma/\gamma)$. Let $\gamma_m := 2k(k+1)h_\Gamma/(c_1 \min_e H_e)$, and let γ_S be the absorption threshold of the pressure bound (depending only on β, C_I, c_0).

Theorem 15.14 (Equal first-layer heights). Assume $k(k+1)h_\Gamma/H_e \equiv \bar{c}$ for all e (UP and every rotation-invariant mesh qualify), put $\gamma' := \gamma - \bar{c} \geq \gamma_0$, $\gamma \geq \max(\gamma_m, \gamma_S)$, and let $\delta' \in Z_h$ be the

GS leak solution at penalty γ'/h_Γ : $N_h^\gamma(\delta', v) = \langle a, v \cdot n \rangle$ on Z_h . Then

$$\|\|\nabla W^a\| - \|\nabla \delta'\|\| \leq C\beta^{-1}\gamma^{-1/2}\|\nabla \delta'\| + C_\alpha h_\Gamma^{3/2}, \quad \left\| \|\nabla \delta'\| - \frac{h_\Gamma}{\gamma'} \|\nabla \tilde{U}_\alpha\| \right\| \leq C_\alpha \frac{h_\Gamma}{\gamma'} (h_\Gamma^{1/2} + E_U^0),$$

with $\tilde{U}_\alpha$ the leak flow of Definition 6.8 with datum αn . So $\|\nabla W^a\| = (1 \pm C\beta^{-1}\gamma^{-1/2}) \frac{h_\Gamma}{\gamma'} \|\nabla \tilde{U}_\alpha\| + O(h_\Gamma^{3/2})$: two-sided up to the factor $1 \pm C\beta^{-1}\gamma^{-1/2}$, which does not tend to 1 as $h_\Gamma \rightarrow 0$.

Proof. With $\lambda = \gamma/h_\Gamma$ and $\lambda' = \gamma'/h_\Gamma$, Lemma 15.13(a) gives $N_h - m_h = N_h^{\gamma'} + f_h$ with $f_h(u, v) = \frac{\bar{c}}{h_\Gamma}(\langle u \cdot \tau, v \cdot \tau \rangle + \langle \tilde{u}, \tilde{v} \rangle) - m_h(\tilde{u}, \tilde{v})$, so $|f_h(u, v)| \leq \frac{2\bar{c}}{h_\Gamma}(\|u \cdot \tau\|_{\Gamma_h} + \|\tilde{u}\|_{\Gamma_h})\|v\|_{\Gamma_h}$. By Theorem 7.42 at penalty γ' , $\lambda'\delta'$ is within Ch_Γ of $\tilde{U}_\alpha$ on Γ_h in L^2 , and on chords $\tilde{U}_\alpha \cdot \tau_e = O(h_\Gamma)$ with $\tilde{U}_\alpha \cdot n$ varying by $O(h_\Gamma)$; so $\|\delta' \cdot \tau\|_{\Gamma_h} + \|\widetilde{\delta' \cdot n}\|_{\Gamma_h} \leq Ch_\Gamma^2/\gamma'$. The difference $\varepsilon := W^a - \delta' \in Z_h$ satisfies $(N_h - m_h)(\varepsilon, v) = -f_h(\delta', v) - \langle \tilde{\xi} - \tilde{\xi}, v \cdot n \rangle$. For $\gamma \geq \gamma_m$, $(N_h - m_h)(u, u) \geq \frac{c_1}{2}\|u\|^2$. By Lemma 15.13(c) and Lemma 4.5, $\beta\|\tilde{\xi} - \tilde{\xi}\| \leq 2\|\nabla W^a\| + C_I h_\Gamma^{-1/2}\|W^a \cdot \tau\|_{\Gamma_h} \leq 2\|\nabla \delta'\| + Ch_\Gamma^{3/2}/\gamma' + C\|\varepsilon\|$, and the inverse trace inequality bounds the pressure term by $C\gamma^{-1/2}\|\tilde{\xi} - \tilde{\xi}\|\|\varepsilon\|$. Absorbing for $\gamma \geq \gamma_S$ gives $\|\varepsilon\| \leq C\beta^{-1}\gamma^{-1/2}\|\nabla \delta'\| + Ch_\Gamma^{3/2}$. The second estimate is Theorem 7.42 divided by λ' . $\square$

Theorem 15.15 ((SLP)). *Let the family be LP^{++} , i.e. LP^+ (Theorem 15.12(c)) with C^3 maps $\mathcal{K}, \ell$. Then $k(k+1)h_\Gamma/H_e = \bar{c} + \varphi(x_e) + r_e$ with $\varphi \in C^3(\Gamma)$ of mean zero and $|r_e| \leq Ch_\Gamma$, and there is $\gamma_{LP} = \bar{c} + \gamma_0 + C\|\varphi\|_{C^3}$ (not explicit) such that for $\gamma \geq \max(\gamma_{LP}, \gamma_m, \gamma_S)$*

$$\|\nabla W^a\| \leq C\gamma^{-1}h_\Gamma\|\alpha\|_{C^3}.$$

Proof. Put $\gamma' := \gamma - \bar{c}$, $a_0 := a$ and

$$a_{j+1} := \varphi(a_j - \bar{a}_j)/\gamma', \quad \bar{a}_j := |\Gamma|^{-1} \int_\Gamma a_j,$$

and let δ_j be the GS leak solution at penalty γ' with data a_j . The mean is removed because the leak flow of Section 7.3 is driven by mean-free data, so the edge means of $\delta_j \cdot n$ are $(h_\Gamma/\gamma')(a_j - \bar{a}_j) + \rho_j$ with $\|\rho_j\|_{\Gamma_h} \leq Ch_\Gamma^2 q^j/\gamma'$; without the mean removal the telescoping would leave a residual $\langle (\bar{a}_j/\gamma')\varphi, v \cdot n \rangle$ of order q^j , which does not vanish on Z_h . The algebra bound in C^3 gives $\|a_j\|_{C^3} \leq q^j\|a\|_{C^3}$ with $q = C_3\|\varphi\|_{C^3}/\gamma' < 1$. With $W^b := \sum_j \delta_j$ and the extra term $-h_\Gamma^{-1}\langle (\varphi + r)\bar{u}, \bar{v} \rangle$ of $N_h - m_h$, the series telescopes: $\varepsilon := W^a - W^b$ satisfies $(N_h - m_h)(\varepsilon, v) = \langle \mathcal{R}, v \cdot n \rangle - \sum_j f_h(\delta_j, v) - \langle \tilde{\xi} - \tilde{\xi}, v \cdot n \rangle$ with $\|\mathcal{R}\|_{\Gamma_h} \leq Ch_\Gamma/\gamma'$ (from r_e and the Riemann sums). Finish as in Theorem 15.14; all constants are linear in the data. $\square$

Corollary 15.16 (Theorem 15.12(c), (d) without the $\gamma^{-3/2}$ terms). *On LP^{++} families, for $\gamma \geq \max(\gamma_{LP}, \gamma_m, \gamma_S)$ and every $k \geq 4$: in (c), $c\gamma^{-1}D(\phi) \leq \liminf \leq \limsup \leq C\gamma^{-1}D(\phi)$, so $\|\nabla W\| = o(h_\Gamma^{k+1/2})$ iff $D(\phi) = 0$; in (d), $\|\nabla W\| \leq C\gamma^{-1}h_\Gamma^{k+1}$ for every K -degenerate ϕ .*

Proof. In the split of (c), the edge-constant part is $h_\Gamma^k \alpha(x_e) + O(h_\Gamma^{k+1})$ with $\alpha \in C^3$ on LP^{++} ; its smooth part is handled by Theorem 15.15, and the rough part by the energy bound, $O(\gamma^{-1/2}h_\Gamma^{k+3/2})$. $\square$

This resolves the question left open in the previous version (whether the $\gamma^{-3/2}$ terms are real on smoothly varying LP^+ families) for $\gamma \geq \gamma_{LP}$; the window $\gamma_2 \leq \gamma < \gamma_{LP}$ remains open when the height map varies strongly. Off LP^+ the defect is real (NUMERICAL, exact Floquet reduction to $N = 4096$): on the 2-periodic mesh *alt* (first-layer heights $0.7|e|, 1.4|e|$), $\|\nabla(W^a - \delta)\| \approx 63\gamma^{-2}h_\Gamma^{1/2}$ at every $\gamma \in [100, 3000]$, so the amplitude is γ^{-2} , not $\gamma^{-3/2}$. On UP, W^a agrees with GS at penalty $\gamma - 20h_\Gamma/H$ to 10^{-4} relative, including at $\gamma = 30$.

Theorem 15.17 (Sharpness of the orders $k+1$ and $k+\frac{3}{2}$). (a) Let $\phi \in C^{k+3}$ be K -degenerate on an LP^{++} family with non-constant normalised edge constant α ($A_e = h_\Gamma^k \alpha(x_e) + O(h_\Gamma^{k+1})$). On equal-height families with $\gamma \geq \max(\gamma_m, \gamma_S)$, $\gamma' \geq \gamma_0$,

$$\|\nabla W\| \geq (1 - C\beta^{-1}\gamma^{-1/2}) \frac{h_\Gamma^{k+1}}{\gamma'} \|\nabla \tilde{U}_{\alpha-\bar{\alpha}}\| - Ch_\Gamma^{k+3/2},$$

and on general LP^{++} families the same holds for $\gamma \geq \gamma(\phi)$. With Corollary 15.16, $\|\nabla W\| \asymp \gamma^{-1}h_\Gamma^{k+1}$.

(b) On UP with first-layer ratio $\rho_1 > 0$ and $\phi = (r-1)^k$, $4 \leq k \leq 7$: $\|\nabla W\| \geq c(1+\gamma)^{-1}h_\Gamma^{k+3/2}$ for every $\gamma > 0$, and $\|\nabla W\| \leq C\gamma^{-1}h_\Gamma^{k+3/2}$ for $\gamma \geq \gamma_2$ (Theorem 15.12(d), A constant). So the order is exactly $k + \frac{3}{2}$.

Proof. (a) Split $W = W^A + W^{\text{rest}}$ with $\|\nabla W^{\text{rest}}\| \leq C\gamma^{-1}h_\Gamma^{k+3/2}$ and apply Theorem 15.14 (resp. Theorem 15.15, whose terms $j \geq 1$ are $\leq Cq(1-q)^{-1}h_\Gamma\|\alpha\|_{C^3}/\gamma'$) to the smooth part of A . (b) All boundary stars are congruent; apply Theorem 15.11 with one pressure-blind field per wall edge (overlap 1). Their stream parts have zero normal trace, so the pairing with η is proportional to $P_f = \int_e (I - \pi_{T_e})\phi t_f ds$ with t_f in the zero-mean trace space. Blowing up at a wall vertex, $(r-1)^k/\ell^k = \xi^k + \frac{k}{2}\ell\xi^{k-1}v^2 + O(\ell^2)$; the order- ℓ^{k+1} term vanishes ($\xi^k \in \text{Ker}_T$), and exact rational algebra gives $P_f/\ell^{k+2} \rightarrow c_f(\rho_1) \neq 0$: $\rho_1^3/900$ and $\rho_1^3/525$ for $k=4$, $-13\rho_1^4/33264$ for $k=5$, $\rho_1^5/8008$ for $k=6$, $-31\rho_1^6/823680$ for $k=7$. Summing $N \asymp h_\Gamma^{-1}$ stars with pairing $\asymp h_\Gamma^{k+2}$ gives $h_\Gamma^{k+3/2}$. $\square$

The $k=4$ coefficients were recomputed independently in 50-digit arithmetic with Richardson extrapolation to $N=4096$ (0.001111111073 and 0.00190476216 at $\rho_1=1$). Order $k+\frac{3}{2}$ for general constant- A data, including ϕ_B above, would need its explicit first-order pairing, which we did not evaluate.

Remark 15.18 (The production penalty $\mu=100$ for CNS*). Lemma 15.13 suggests comparing $\gamma - k(k+1)h_\Gamma/H_e$ (30 minus 10–27 on the production meshes) with the GS threshold $\gamma^* \approx 18$ –21, which would put CNS* at $\mu=100$ below its threshold. That comparison is wrong, and the direct computation settles it (notes/v7/mu100). CNS* is Petrov–Galerkin on Z_h , so its stability quantity is the largest $\mu_c^{\text{CNS*}}$ at which the square system is singular. On the production meshes $\mu_c^{\text{CNS*}} = 72.3, 78.9, 82.3, 84.4, 86.8, 88.1$ for $N=8, 16, 24, 32, 48, 64$, extrapolating to $\gamma_c \approx 26.5$, i.e. $\mu_c^{\text{CNS*}} \approx 93$; the effective-penalty form $A_\mu - C^\top M^{-1}C$ on Z_h has the same limit. So $\mu=100$ is stable on every production mesh, with margins 27% ($N=16$), 19% ($N=32$), 13% ($N=64$) that shrink to about 7% as $h \rightarrow 0$ ($\gamma(100) \approx 28.6$ against 26.5). For GS the limiting margin is about 34% (28.6 against 21.4). The two instabilities are different modes: the GS critical mode is tangential (its wall trace is 0.4–1.3% normal), on which m_h removes almost nothing, while the CNS* critical mode is about 99% normal and dominated by edge means (90–97%), exactly where m_h attains its sharp constant $k(k+1)/H_e$. It goes critical when the local effective penalty $(\mu/h - 20/H_e)|e|$ on the thinnest first-layer edges reaches about 0 to -1 , not when it reaches γ_{GS}^* .

The residual leak law. For large γ the CNS* pressure part is itself a leak flow: with $\varepsilon = h/\mu$ and Y as above, $\|W - \varepsilon Y\|_{\Gamma_h} \leq C_2\gamma^{-1}\|W\|_{\Gamma_h}$, and for $\gamma \geq \gamma_3 := \max(2C_2, 4C_2/\sigma)$ (σ the edge-oscillation fraction of Y) $\|\nabla W\| \asymp h_\Gamma^{1/2}\gamma^{-1}\|Y\|$ (proved, notes/v6/robust2). Numerically $C_2 \gtrsim 180$, so $\gamma_3 \gtrsim 10^3$ ($\mu \gtrsim 4000$): the law is not informative at $\mu=100$, where it indeed fails by a factor 1.1–4 in the slip. The H^1 order is unaffected; Theorems 15.11 and 15.12(b) cover every γ .

TABLE 10. Hydrostatic test, $k = 4$, $\mu = 100$, $\nu = 1$: $\|\nabla u_h\|$ (the exact velocity is 0), GS slip and energy normalised by their leading terms, and CNS* energy norm (notes/v6/robust/hydro_all.jsonl).

N	GS	slip ratio	energy ratio	CNS* (rate)	CNS* energy (rate)	CNS*-R
8	$1.074 \cdot 10^{-1}$	0.857	1.226	$2.29 \cdot 10^{-3}$	$7.07 \cdot 10^{-3}$	$6.3 \cdot 10^{-9}$
16	$6.733 \cdot 10^{-2}$	0.910	1.090	$1.23 \cdot 10^{-4}$ (5.73)	$3.86 \cdot 10^{-4}$ (5.70)	$2.4 \cdot 10^{-12}$
32	$3.805 \cdot 10^{-2}$	0.948	1.017	$5.34 \cdot 10^{-6}$ (4.97)	$1.88 \cdot 10^{-5}$ (4.79)	$6.9 \cdot 10^{-15}$
64	$2.027 \cdot 10^{-2}$	0.972	0.998	$2.22 \cdot 10^{-7}$ (4.71)	$9.53 \cdot 10^{-7}$ (4.41)	$1.3 \cdot 10^{-14}$

15.6. Computations. Table 10 shows the hydrostatic test $\tilde{f} = \nabla \phi$, $\phi = \sin 2x \cos y$, $\nu = 1$, $k = 4$, $\mu = 100$ on the production meshes. GS is first order with slip ratio tending to 1; CNS* has rate tending to $4.5 = k + \frac{1}{2}$ (the measured rates are reproduced to ± 0.2 by 4.5 plus the drift of $\|\eta\|_{\Gamma_h}/h_\Gamma^4$); CNS*-R is at the quadrature floor. For a cubic ϕ , CNS* and CNS*-R are $\leq 10^{-13}$ (Theorem 15.7(c)). With the ν -scaled penalty $\nu\|\nabla u_h\|$ is constant to 10^{-12} over $\nu \in \{1, 10^{-2}, 10^{-4}, 10^{-6}\}$ (a floating-point check of Lemma 15.1); with an unscaled penalty $\hat{\mu} = 100$ the GS and CNS* values are ν -uniform (2.0 – $2.1 \cdot 10^{-2}$ and 0.9 – $2.2 \cdot 10^{-7}$ at $N = 64$). On general data CNS*-R changes the error of test B₁ by at most 0.3%, and CNS*-R errors for two problems with the same velocity and different pressures agree to $3 \cdot 10^{-12}$ (for CNS* they differ by $7 \cdot 10^{-8}$). What is not proved. The leak law for the unscaled penalty as $\nu \rightarrow 0$; (S_{LP}) for γ between γ_2 and γ_{LP} , and explicit $\gamma_{LP}, \gamma_m, \gamma_S$; the failure of (S) off LP⁺ (numerical only); order $k + \frac{3}{2}$ for general constant- A data and on non-congruent stars; explicit κ_T for $k \geq 5$ and κ_K for any k ; exactness of CNS*-R under quadrature (numerical floor only); Navier–Stokes and Oseen versions.

Computation: notes/v6/robust/hydro.py, cert.py; notes/v6/robust2--6/*.py; output notes/v6/robust/hydro_all.jsonl, rates.jsonl; notes/v6/robust3/cert_sin*.jsonl; notes/v6/robust6/cert_fix*.log.

16. COMPUTATIONS

16.1. Code and meshes. Except in Section 16.11, we use P_4 Scott–Vogelius elements with Nitsche’s method on the inscribed polygon and strong Dirichlet data on the square $R = (-2.5, 2.5)^2$. Divergence is enforced exactly: all ten P_3 moments per triangle.

Saddle-point systems are solved by SuperLU [42] with partial pivoting and three steps of iterative refinement. Over the 148 production solves (74 cases, two methods each), the relative residual is at most $3.8 \cdot 10^{-15}$ and the discrete divergence at most $7 \cdot 10^{-11}$ (median $9 \cdot 10^{-12}$).

For the zero-pressure-gradient benchmark of [20], the code gives the coercivity threshold $\mu^* = 60.343$ at $N = 20$ (results/pod/thresh/N20_L2.5_m5.json), against the transition $60.33821 < \mu^* < 60.33831$ reported there.

The vertices of Γ_h are the radial projections of N equispaced points on the square, so the boundary chords are *not* equal (except for $N = 8$): $\max |e|/\min |e|$ is 1.43, 1.72, 1.87 and 1.97 at $N = 16, 32, 64$ and 256. The ratio $h/h_\Gamma = h_\Omega/\max |e|$ stays in 3.28–3.48, while $\rho = h/\min |e|$ of Section 2 is 4.70, 5.73 and 6.35 at $N = 16, 32$ and 64. h_Ω decreases from 1.505 to 0.109 for $N = 16 \rightarrow 256$; since h/h_Γ is nearly constant, rates in h_Ω and h_Γ coincide. Penalty thresholds are reported as $\gamma^* = \mu^* h_\Gamma/h$, which is the γ of Section 2. Rates in the tables are computed between consecutive runs of the full sequence $N = 16, 24, 32, 48, 64, 96, 128, 192, 256$, some of which are omitted from the tables.

The production penalty. Most runs use $\mu = 100$, i.e. $\gamma \approx 29$ –30. This is only 1.37–1.41 times the measured coercivity threshold on Z_h ($\mu^* = 50.3 \rightarrow 70.7$ for $N = 8 \rightarrow 64$, extrapolated limit ≈ 73 ; Table 14). For CNS*, which is Petrov–Galerkin on Z_h , the relevant quantity is the largest

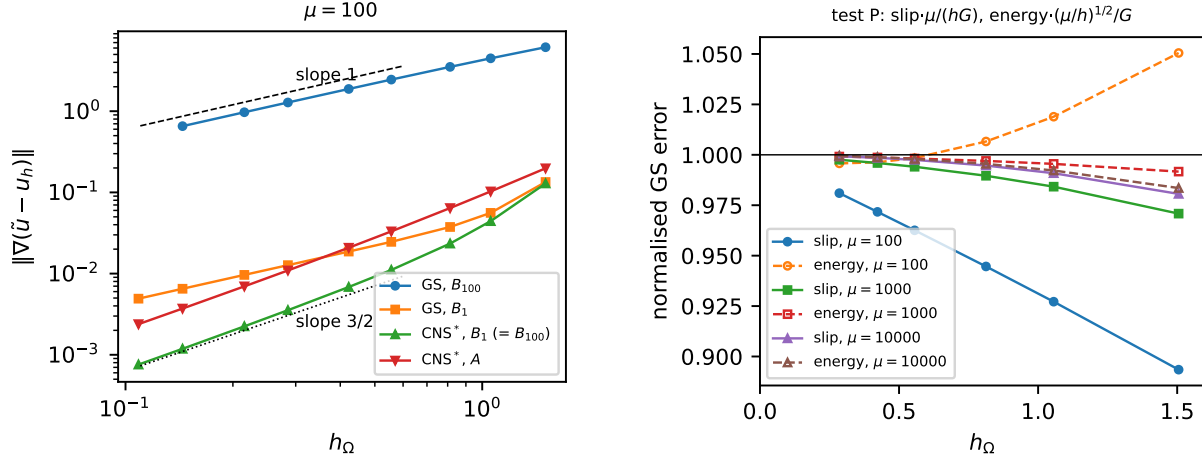


FIGURE 1. Left: H^1 errors at $\mu = 100$. Once $G > 0$, GS eventually has slope 1, while CNS* tends to slope 3/2 independently of the pressure. Right: the normalised slip and energy errors of GS on test P approach 1 (Corollary 6.7); at $N = 96$ the energy ratios are within 0.5% of 1 and the slip ratios within 2%.

μ at which the square system is singular: $\mu_c^{\text{CNS}^*} = 78.9, 84.4, 88.1$ at $N = 16, 32, 64$, extrapolating to about 93 ($\gamma_c \approx 26.5$). So $\mu = 100$ is stable for CNS* on every production mesh, but the margin shrinks from 27% ($N = 16$) to 19% ($N = 32$) and 13% ($N = 64$), and to about 7% as $h \rightarrow 0$; the critical mode is about 99% normal and edge-mean dominated, unlike the tangential GS mode (Remark 15.18). The thresholds γ_0, γ_2 required by the proofs are not computed, so the $\mu = 100$ runs illustrate the theorems rather than being covered by them. Near the threshold the discrete solution is amplified by a factor of order $(1 - \mu^*(N)/\mu)^{-1}$, which drifts from about 2.5 to 3.4 over $N = 16 \rightarrow 64$; on its own this can lower measured rates by up to about 0.2 per doubling at $N \leq 64$. Normalised constants that should tend to 1 or to a limit (slip ratios, drag ratios, H^1 constants) are therefore partly pre-asymptotic at $\mu = 100$; where $\mu = 300, 10^3$ or 10^4 data exist we quote them alongside (Tables 13, 23).

16.2. Test problems.

A Scott's benchmark: the shear flow $u = (1 - r^{-2})(-y, x)$ with $p \equiv 0$, so $G = 0$.

B_a $\psi = (r^2 - 1)^2(x + y^2/2)/10$ and $p = a(x^2y - y + x/2)$, so $G = a(7\pi/8)^{1/2}$: $G = 1.658$ for $a = 1$ and 165.8 for $a = 100$.

P Pure pressure: $u \equiv 0$, $f = \nabla p$, with p as in B_1 . CNS* reproduces this exactly ($u_h = 0$ and $p_h = p - \bar{p}_\Gamma(p)$, since $p \in \Pi_h$); its computed velocity is below 10^{-13} in H^1 . The GS output is therefore *exactly* the response to the missing traction.

C ψ as in B, with the non-polynomial pressure $p = e^{x/2} \cos y$. This exercises the term $p^* - \pi_h p^*$ of Theorem 7.2.

Tests A and B were run with $\mu \in \{100, 1000, 10000\}$ and $N \leq 192$, plus $N = 256$ for $(A, 100)$ and $(B_1, 100)$, on a 32-core machine (36 minutes in total). Test P (all three μ) and test C ($\mu = 100$) were run with $N \leq 96$ on a 2-core machine.

16.3. Theorem 7.2: CNS*. Table 11 gives the CNS* errors. At $N = 256$ the energy-norm rate is 1.52 and the slip rate is 2.02; the slip carries an extra factor $(h/\mu)^{1/2}$. The H^1 rates decrease monotonically toward 3/2, the geometric term of Lemma 3.5. The h_Ω^4 term is invisible at these levels because the geometric term dominates.

The error is independent of the pressure.

TABLE 11. CNS* attains Scott's rate $h_\Gamma^{3/2}$ (Theorem 7.2). H^1 error, $\mu = 100$.

N	h_Ω	A	rate	B_1	rate
32	0.813	$6.41 \cdot 10^{-2}$	1.80	$2.34 \cdot 10^{-2}$	2.43
64	0.422	$2.08 \cdot 10^{-2}$	1.69	$6.83 \cdot 10^{-3}$	1.76
96	0.285	$1.09 \cdot 10^{-2}$	1.64	$3.54 \cdot 10^{-3}$	1.68
128	0.215	$6.96 \cdot 10^{-3}$	1.60	$2.24 \cdot 10^{-3}$	1.63
192	0.144	$3.71 \cdot 10^{-3}$	1.58	$1.19 \cdot 10^{-3}$	1.60
256	0.109	$2.39 \cdot 10^{-3}$	1.56	$7.58 \cdot 10^{-4}$	1.57

TABLE 12. GS eventually loses Scott's rate once $G > 0$. H^1 rate between $N = 128$ and $N = 192$, test B_{100} .

μ	GS	CNS*
100	0.99	1.60
1000	1.00	1.56
10000	1.06	1.58

- For polynomial pressures this is structural ($p \in \Pi_h$). B_1 and B_{100} agree to all printed digits wherever both were run ($N \leq 192$, every μ).
- With the non-polynomial pressure of test C, the CNS* H^1 error differs from that of B_1 by a relative $7.9 \cdot 10^{-7}$, $3.9 \cdot 10^{-8}$ and $9.7 \cdot 10^{-9}$ at $N = 16$, 64 and 96. This is the size of the contribution $\langle p^* - \pi_h p^*, v \cdot n \rangle$, which is of higher order, as Theorem 7.2 predicts.

16.4. **Theorems 6.1–6.4: the printed method.** Table 12 compares the H^1 rates of GS and CNS* on B_{100} .

For the non-polynomial test C at $\mu = 100$, the GS H^1 rate from $N = 64$ to 96 is already 1.12, against 1.68 for CNS*. For the small pressure B_1 at $\mu = 1000$, GS still shows 1.51 at $N = 192$. The $(h/\mu)G$ term dominates only once $(h/\mu)G \gtrsim h_\Gamma^{3/2}$, which always happens eventually because $h_\Gamma^{3/2}$ decays faster. Larger μ or smaller G delays the crossover but does not remove it.

16.5. **Corollary 6.7: the leading term.** Table 13 and Figure 1 (right) show the normalised GS error on test P, and Figure 2 shows the leak field itself.

- At $N = 32$, $\mu = 100$, the normal trace of u_h on Γ_h matches the prediction $(h/\mu)(p - \bar{p})$ with relative $L^2(\Gamma_h)$ misfit 0.055. The maximum of the tangential trace is about 12% of that of the normal one, and the tangential trace oscillates at the element scale.
- The slip and energy ratios approach 1: at $N = 96$ the energy ratios are within 0.5% of 1 for every μ . The defect in the slip ratio shrinks with h/μ : at $N = 96$ it is 0.019, 0.0025 and 0.0007 for $\mu = 10^2$, 10^3 and 10^4 .
- The H^1 constant is not covered by the corollary. For $\mu = 100$ it rises from 2.46 to 2.71; for $\mu = 10^3$ and 10^4 it falls, from 2.78 to 2.77 and from 3.06 to 2.79. Section 6.1 identifies the limit as $K = \|\nabla U\|/G = 2.7565$ for this box, proves convergence to it as $h \rightarrow 0$ at fixed μ (Corollary 6.11; on these meshes h/h_Γ is nearly constant, so fixed μ means fixed γ and bounded ρ), and compares it pointwise with these runs. The $\mu = 100$ values approach it from below; the larger- μ values approach it from above, in the regime $\gamma h_\Gamma \gg 1$, which Corollary 7.48 covers: on test P, $|K_h - K| \leq C(h_\Gamma + (E_U^0)^2)$ for every penalty.
- On test B, GS itself (B_1 , $\mu = 100$, $N = 256$) gives slip and energy ratios 0.995 and 1.001.

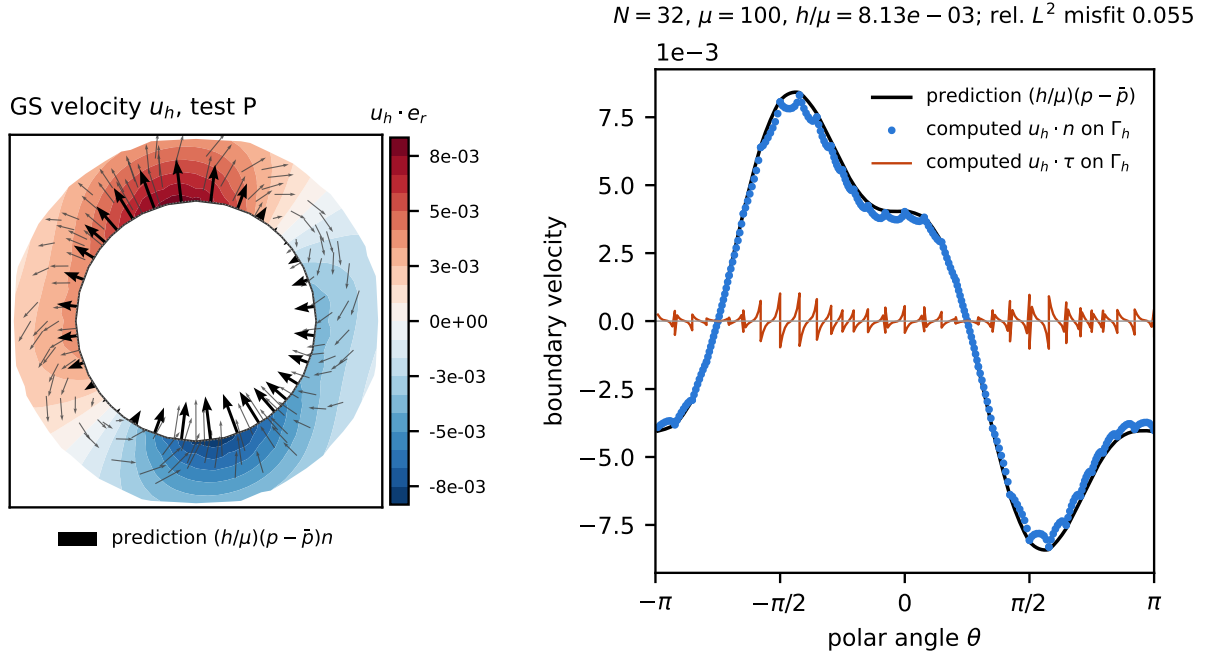


FIGURE 2. The leak of the printed method on test P ($u = 0$, so $u_h = -(\tilde{u} - u_h)$), $N = 32$, $\mu = 100$. Left: radial velocity $u_h \cdot e_r$ near the wall (colour) and u_h (thin arrows); thick arrows show the prediction $(h/\mu)(p - \bar{p})n$ of Corollary 6.7 at the chord midpoints. Right: normal and tangential traces of u_h along Γ_h against $(h/\mu)(p - \bar{p})$.

TABLE 13. The leading term of Corollary 6.7. Normalised GS error on test P ($G = 1.658$), where $e = \tilde{u} - u_h$.

N	μ	$\ \nabla e\ \mu / (hG)$	$\ e\ _{\Gamma_h} \mu / (hG)$	$\ e\ (\mu/h)^{1/2} / G$
16	100	2.464	0.893	1.050
32	100	2.612	0.945	1.007
64	100	2.684	0.972	0.996
96	100	2.708	0.981	0.996
96	1000	2.770	0.998	0.999
96	10000	2.790	0.9993	0.9994

- For $\mu = 10^4$ with the small pressure B_1 , $\gamma h_\Gamma \approx 120 \gg G$ at $N = 192$, so the hypothesis of the corollary fails. The GS energy ratio there is about 7.8, dominated by the geometric error rather than the traction response.
- The difference $u_{\text{GS}} - u_{\text{CNS}^*}$ is not a clean proxy for the traction response on test B. It also contains the difference between the two methods on the velocity part, which is $O(h_\Gamma^{3/2})$ and non-zero even for test A. Test P removes this.

16.6. Proposition 10.1: the penalty study. The coercivity threshold μ^* on Z_h is the smallest μ for which the unpivoted symmetric factorisation of $A_\mu + rB^\top M^{-1}B$ has only positive pivots; we locate it by bisection. Table 14 and Figure 3 show the results.

TABLE 14. Coercivity threshold μ^* and $\gamma^* = \mu^* h_\Gamma / h$ ($k = 4$). The row h/h_Γ is $h_\Omega / \max |e|$; the $\rho = h / \min |e|$ of Section 2 is larger by the factor $\max |e| / \min |e|$ (for example $\rho = 4.70, 5.73, 6.35$ at $N = 16, 32, 64$ in part (a)).

<i>(a) Refinement at fixed $h/h_\Gamma \approx 3.3$.</i>								
N	8	12	16	20	24	32	48	64
h/h_Γ	3.41	3.02	3.28	3.16	3.30	3.33	3.37	3.40
ρ	3.41	4.16	4.70	5.08	5.36	5.73	6.13	6.35
μ^*	50.3	51.8	59.3	60.3	63.3	66.0	69.1	70.7
γ^*	14.7	17.1	18.1	19.1	19.2	19.9	20.5	20.8

<i>(b) Varying h/h_Γ through the outer box $(-L, L)^2$.</i>							
	L	2.5	5	10	20	40	
$N = 16$	h/h_Γ	3.28	5.90	11.3	22.2	43.9	
	μ^*	59.3	112.1	208.0	411.5	809.3	
	γ^*	18.1	19.0	18.4	18.5	18.4	
$N = 32$	h/h_Γ	3.33	5.79	10.8	21.1	41.5	
	μ^*	66.0	113.9	218.0	422.2	835.7	
	γ^*	19.9	19.7	20.1	20.0	20.1	

Under refinement at fixed h/h_Γ , γ^* increases by 3.4 ($N = 8 \rightarrow 16$), 1.8 ($16 \rightarrow 32$) and 0.95 ($32 \rightarrow 64$). The increments roughly halve. The limit is the threshold of a periodic half-plane cell (Proposition 10.15): $\gamma_\infty^* = 21.408$ at full depth, 21.4045 for fields in the first two layers, and the two-layer global thresholds for $N = 128, \dots, 2048$ extrapolate to 21.40452, with $\gamma^*(N) \approx \gamma_\infty^* - 22/N$.

In part (b) of Table 14, h/h_Γ is varied by coarsening the far field while the vertices of Γ_h are held fixed and the boundary layer approximately fixed. Then μ^* is linear in h/h_Γ over more than a decade, with a constant slope of about 18–20. The vertices of Γ_h do not depend on L , so $\max |e| / \min |e|$ is fixed at fixed N , and μ^* is equally linear in ρ , with the slope divided by that ratio. The threshold is thus set locally at Γ_h ($\mu/h \gtrsim c/h_\Gamma$), and the scaling with h/h_Γ (hence with ρ) is the cost of normalising the penalty by the global h .

In practical terms:

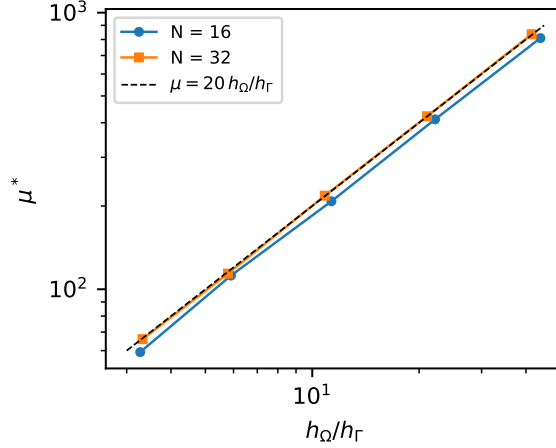
- for $k = 4$ a safe choice is $\gamma = \mu h_\Gamma / h \geq 25$, that is, $\mu \geq 25 h/h_\Gamma$; since $\rho \geq h/h_\Gamma$, $\mu \geq 25\rho$ is also safe;
- a fixed μ fails once the far field is coarsened beyond $h/h_\Gamma \approx \mu/20$;
- the threshold grows quickly with k , and is larger on Clough–Tocher refinements (Section 16.11), so these values do not transfer to other elements.

The exact certificate $\lambda \geq 9$ on $\mathcal{S}_3$ and $\mathcal{S}'_4$ is a local lower bound, consistent with the global $\gamma^* \approx 18$ –21; Section 10.1 accounts for the gap.

16.7. Lemma tests. We compute exact discrete dual norms, $\sup_v \ell(v) / \|\nabla v\|$ over Z_h or V_h^R , by a saddle-point solve with the pointwise divergence constraint. Table 15 gives the rates for $N = 48 \rightarrow 64$.

Computation: code/lemma_tests.py 16,24,32,48,64; output logs/lemma_tests.log.

16.8. Strong imposition and the $h^{1/2}$ phenomenon. Table 16 compares strongly imposed no-slip, $u_h = 0$ at every node of Γ_h with no Nitsche terms and the same P_4 – P_3^{disc} pair, with GS and CNS* on test A.

FIGURE 3. Coercivity threshold μ^* against h/h_Γ (Table 14(b)).TABLE 15. Exact discrete dual norms; rate in h_Γ for $N = 48 \rightarrow 64$ (positive means decay).

Functional	Space	Rate
$\langle S, \partial_n v \rangle, S$ normal	Z_h	+0.05 (bounded, ≈ 1.75)
$\langle S, \partial_n v \rangle, S$ normal	V_h^R	-0.52
$\langle S, \partial_n v \rangle, S$ tangential	Z_h	-0.52
$\langle (\nabla \tilde{u})^\top n, v \rangle$, tests A / B	V_h^R	1.57 / 1.56 (Lemma 3.5: 3/2)

Computation: `code/strong.bc.py`; `code/report_v5.py`; output `results/v5/strong/`,
`results/v5/letter_tables.txt` (T1, T1').

- *Mesh (a): the rate $h^{1/2}$ is reproduced.* On mesh (a) the first-layer diagonals alternate, so every other polygon vertex lies in only two triangles. The H^1 rate of strong imposition falls from 0.75 to 0.56 at $N = 128$, towards $\frac{1}{2}$. The largest vertex-gradient error $\max_{z, T \ni z} |\nabla u_h|_T(z) - \nabla u(z)|$ tends to 1.99, which is $|\partial_n u \cdot \tau| = 2$ at $r = 1$ for this flow: the discrete gradient vanishes at those vertices. The pressure norm grows like $h^{-1/2}$ ($\|p_h\| = 14.0 \rightarrow 35.7$, although $p \equiv 0$).
- *Mesh (s): no locking in the range computed.* On the mesh of the rest of this section every polygon vertex lies in three triangles. Strong imposition then converges with H^1 rates $1.93 \rightarrow 1.61$ ($N = 24 \rightarrow 128$), like the Nitsche methods and with an error about 2.5 times larger; the vertex-gradient error decays like h ($1.28 \rightarrow 0.16$), and $\|p_h\|$ decays. Imposing $u_h = \tilde{u}$ at the nodes of Γ_h instead of $u_h = 0$ gives rate ≈ 4 , so the defect of strong imposition is entirely the transfer of the no-slip value $u = 0$ from Γ to Γ_h .
- *GS and CNS* do not see the change of mesh.* On mesh (a) their H^1 errors at $N = 128$ are $6.00 \cdot 10^{-3}$ and $7.17 \cdot 10^{-3}$, against $6.11 \cdot 10^{-3}$ and $6.96 \cdot 10^{-3}$ on mesh (s); at every $N \leq 128$ the two meshes differ by at most 6%.

Remark 16.1 (How to read Table 16). Section 11 explains these observations. On mesh (s) (M2) holds, and strong imposition has H^1 error $\asymp h_\Gamma^{3/2}$ (Corollary 11.7); the slow drift of the observed rate (1.547 at $N = 192$) is a relative $O(h_\Gamma)$ correction, $E \approx ah_\Gamma^{3/2}(1 + \beta h_\Gamma)$ with $\beta \approx 1.1$ –1.3 (Table 7). The vertex-gradient error $\asymp h$ is exactly $a_z|e|/2$ on each chord (Lemma 11.4). On mesh (a) every other wall vertex is locked (two triangles), and the error is $\asymp h_\Gamma^{3/2} + h_\Gamma(\sum_{\mathcal{L}} a_z^2)^{1/2} \asymp h_\Gamma^{1/2}$ (Corollary 11.14); the counting lower bound of Theorem 11.8 is within a factor 3.0–3.15 of the

TABLE 16. Strong imposition against Nitsche. Test A, $k = 4$, $\mu = 100$: H^1 seminorm error $|\tilde{u} - u_h|_{1,\Omega_h}$. *Strong*: $u_h = 0$ at all nodes of Γ_h , no Nitsche terms, on mesh (s), the mesh of Section 16 (every polygon vertex in three triangles), and on mesh (a), the same mesh with alternating first-layer diagonals (polygon vertices in two or four triangles). GS and CNS* on mesh (s).

N	h	strong, mesh (a)		strong, mesh (s)		GS		CNS*	
		error	rate	error	rate	error	rate	error	rate
16	1.5052	$9.03 \cdot 10^{-1}$	–	$4.62 \cdot 10^{-1}$	–	$1.72 \cdot 10^{-1}$	–	$1.98 \cdot 10^{-1}$	–
24	1.0559	$6.91 \cdot 10^{-1}$	0.75	$2.33 \cdot 10^{-1}$	1.93	$8.87 \cdot 10^{-2}$	1.86	$1.03 \cdot 10^{-1}$	1.85
32	0.8127	$5.72 \cdot 10^{-1}$	0.72	$1.43 \cdot 10^{-1}$	1.86	$5.55 \cdot 10^{-2}$	1.79	$6.41 \cdot 10^{-2}$	1.80
48	0.5560	$4.44 \cdot 10^{-1}$	0.67	$7.29 \cdot 10^{-2}$	1.78	$2.88 \cdot 10^{-2}$	1.73	$3.31 \cdot 10^{-2}$	1.74
64	0.4224	$3.75 \cdot 10^{-1}$	0.62	$4.56 \cdot 10^{-2}$	1.71	$1.81 \cdot 10^{-2}$	1.68	$2.08 \cdot 10^{-2}$	1.69
96	0.2853	$2.98 \cdot 10^{-1}$	0.59	$2.38 \cdot 10^{-2}$	1.65	$9.57 \cdot 10^{-3}$	1.63	$1.09 \cdot 10^{-2}$	1.64
128	0.2153	$2.54 \cdot 10^{-1}$	0.56	$1.52 \cdot 10^{-2}$	1.61	$6.11 \cdot 10^{-3}$	1.59	$6.96 \cdot 10^{-3}$	1.60
192	0.1445	–	–	–	–	$3.27 \cdot 10^{-3}$	1.57	$3.71 \cdot 10^{-3}$	1.58
256	0.1087	–	–	–	–	$2.10 \cdot 10^{-3}$	1.55	$2.38 \cdot 10^{-3}$	1.56

measured error. A single locked vertex already gives rate 1, and nearly singular three-triangle stars give rates between $\frac{3}{2}$ and $\frac{1}{2}$ according to the needle angle (Section 11.5). Mesh (a) violates (M1') and (M2) at its two-triangle vertices, so it lies outside the hypotheses of our theorems for GS and CNS*; that these methods are unaffected on it is an observation. So the zero-gradient defect of strong imposition is a property of the wall mesh, and under (M2) the advantage of weak imposition is a smaller constant (about 2.5 here), not a better rate.

16.9. Pressure errors, stress and drag. All runs use $k = 4$ and $\mu = 100$, and measure $\|p^\circ - p_h\|_{L^2(\Omega_h)}$ with $p^\circ = \tilde{p} - \bar{p}_\Gamma(\tilde{p})$, the target of Theorems 8.7 and 8.11, for both methods. Removing the optimal constant instead changes these errors in the third digit at most; the discrete pressures are thus $\bar{p}_\Gamma$ -normalised up to a vanishing constant, even though $\bar{p}_\Gamma(p_h) \neq 0$ ($\bar{p}_\Gamma(p_h) = -4.6 \cdot 10^{-3} \rightarrow -9.7 \cdot 10^{-4}$ for CNS* on B_1 , $N = 16 \rightarrow 96$). The velocity H^1 errors of these runs reproduce the stored ones to a relative $2.1 \cdot 10^{-7}$. The results are in Tables 17–19.

Computation: `code/pressure_err.py`; `code/report_v5.py`; output `results/v5/pressure/`, `results/v5/letter_tables.txt` (T2, T4).

- CNS*. The pressure error does not depend on the pressure data: for B_1 , B_{100} and C it agrees to 10^{-6} at $N = 16$ and $3 \cdot 10^{-11}$ at $N = 96$. Its L^2 rates 3.0, 3.0, 2.6, 2.1, 1.8 decrease towards $\frac{3}{2}$, as Theorem 8.7 predicts; the first-layer part already has rate about 1.6, and the L^∞ error on the first layer has rate about 1 (0.98 at $N = 96$), for which we have no proof.
- GS with $G > 0$. The L^2 rate decreases towards $\frac{1}{2}$ (0.60, 0.62, 0.63 at $N = 96$ for B_1 , B_{100} , C). The first-layer part has rate 0.48–0.50 and the rest rate 0.9–1.0; the L^∞ error on the first layer does not converge (0.78 for B_1 , 77 for B_{100}). This is Theorem 8.11. For B_{100} the GS pressure error is 9.2 at $N = 96$, against $8 \cdot 10^{-3}$ for CNS*.
- GS with $G = 0$ (test A) behaves like CNS*: L^2 rate 1.68 at $N = 96$ and still decreasing, and the first layer converges.

Layer, drag and traction. On test P with GS at $N = 64$ and 96, the first-layer pressure error is 0.0958 and 0.0793, against $\|\xi_b\| = 0.0961$ and 0.0796 from Definition 8.10; the rest is 0.069 and 0.047. The Nitsche-flux drag defect $J_h^N - J_{e_1} + \Theta_f$ is 0.970 and 0.981 times the prediction $-(h/\mu)K_{e_1}$ of Theorem 8.17 ($K_{e_1} \approx 17.16$, $K_{e_2} \approx -25.36$); for Gjerde–Scott’s volume formula

TABLE 17. Test B₁₀₀ (large pressure variation, $p = 100(x^2y - y + x/2)$), $k = 4$, $\mu = 100$: velocity error $|\tilde{u} - u_h|_{1,\Omega_h}$ and pressure error $\|p^\circ - p_h\|_{L^2(\Omega_h)}$, $p^\circ = \tilde{p} - \bar{p}_\Gamma(\tilde{p})$.

N	h	GS, $ \cdot _1$		CNS*, $ \cdot _1$		GS, L^2 pressure		CNS*, L^2 pressure	
		error	rate	error	rate	error	rate	error	rate
16	1.5052	6.14	–	$1.29 \cdot 10^{-1}$	–	$2.75 \cdot 10^1$	–	$4.97 \cdot 10^{-1}$	–
24	1.0559	4.48	0.89	$4.42 \cdot 10^{-2}$	3.02	$2.17 \cdot 10^1$	0.67	$1.69 \cdot 10^{-1}$	3.04
32	0.8127	3.51	0.93	$2.34 \cdot 10^{-2}$	2.43	$1.82 \cdot 10^1$	0.68	$7.67 \cdot 10^{-2}$	3.03
48	0.5560	2.45	0.95	$1.11 \cdot 10^{-2}$	1.97	$1.41 \cdot 10^1$	0.67	$2.87 \cdot 10^{-2}$	2.59
64	0.4224	1.88	0.97	$6.83 \cdot 10^{-3}$	1.76	$1.18 \cdot 10^1$	0.65	$1.62 \cdot 10^{-2}$	2.08
96	0.2853	1.28	0.98	$3.54 \cdot 10^{-3}$	1.68	9.23	0.62	$8.01 \cdot 10^{-3}$	1.80
128	0.2153	$9.71 \cdot 10^{-1}$	0.98	$2.24 \cdot 10^{-3}$	1.63	–	–	–	–
192	0.1445	$6.54 \cdot 10^{-1}$	0.99	$1.18 \cdot 10^{-3}$	1.60	–	–	–	–

TABLE 18. Test B₁, $\mu = 100$: pressure error $\|p^\circ - p_h\|_{L^2(\Omega_h)}$ ($p^\circ = \tilde{p} - \bar{p}_\Gamma(\tilde{p})$); the off-layer column is the error on $\Omega_h \setminus L_1$ with the optimal constant removed.

N	h	GS		GS, off first layer		CNS*	
		error	rate	error	rate	error	rate
16	1.5052	$5.30 \cdot 10^{-1}$	–	$5.00 \cdot 10^{-1}$	–	$4.97 \cdot 10^{-1}$	–
24	1.0559	$2.47 \cdot 10^{-1}$	2.15	$1.96 \cdot 10^{-1}$	2.63	$1.69 \cdot 10^{-1}$	3.04
32	0.8127	$1.80 \cdot 10^{-1}$	1.22	$1.21 \cdot 10^{-1}$	1.84	$7.67 \cdot 10^{-2}$	3.03
48	0.5560	$1.37 \cdot 10^{-1}$	0.72	$8.05 \cdot 10^{-2}$	1.08	$2.87 \cdot 10^{-2}$	2.59
64	0.4224	$1.15 \cdot 10^{-1}$	0.61	$6.32 \cdot 10^{-2}$	0.88	$1.62 \cdot 10^{-2}$	2.08
96	0.2853	$9.13 \cdot 10^{-2}$	0.60	$4.45 \cdot 10^{-2}$	0.89	$8.01 \cdot 10^{-3}$	1.80

the ratio to $-2(h/\mu)K_{e_1}$ is 0.905 and 0.952 at $N = 32$ and 64 (Proposition 8.21). The pointwise-traction drag error does not converge: $J_h^\beta - J_{e_1} = -1.211$ and -1.225 at $N = 64$ and 96, against $\Xi_h = -1.192$ at $N = 64$ (Proposition 8.23). For CNS* on B₁, the lift defect divided by $-\Theta_f - \Lambda_h$ is 0.788, 0.895, 0.941, 0.955 for $N = 16$ –96, and on test A the torque ratio is 0.84, 0.92, 0.95 (Theorem 8.19). At fixed $N = 64$ the CNS* pressure error on B₁ grows with the penalty (0.016, 0.039, 0.059 for $\mu = 10^2, 10^3, 10^4$), which is the non-uniformity in γ of Remark 8.8.

Computation: notes/v4/pressure_drag_check.py; output notes/v4/pressure_drag_report.md, Section 3 (no log file).

16.10. A growing penalty and flux-corrected data. All runs use $k = 4$ on the meshes of Section 16 ($h/h_\Gamma \in [3.28, 3.43]$, so $\gamma \approx 0.29\mu$ and rates in h and h_Γ agree), SuperLU for $N \leq 64$ and PARDISO (Section 16.11) for $N = 96$; relative residuals are at most $1.8 \cdot 10^{-12}$ and the discrete divergence at most $1.6 \cdot 10^{-9}$. The $\mu = 100$ values for test P reproduce Table 13 ($\|\nabla e\| = 1.2810 \cdot 10^{-2}$ at $N = 96$).

Computation: code/rescue.py; code/rescue_run.sh; code/rescue_report.py; output logs/rescue_study.log, rescue_musweep.log, rescue_flux.log, rescue_cond.log; results/v4/rescue/.

Power-law penalty. We take $\mu = c(h_{16}/h)^\alpha$, $h_{16} = 1.505$. Table 20 gives the rates between $N = 64$ and 96. On the leak-only test P they are exactly those of Corollary 7.36 (H^1 : $1+\alpha$; energy: $(1+\alpha)/2$), with the normalised constants of Corollary 6.7 unchanged ($\|\nabla e\|_\mu/(hG) = 2.71$ –2.78 and $\|e\|(\mu/h)^{1/2}/G = 0.996$ –0.999 at $N = 96$). On B₁ the GS H^1 rate moves from 0.96 (fixed μ) to

TABLE 19. Test A ($p \equiv 0$), $\mu = 100$: pressure error $\|p^\circ - p_h\|_{L^2(\Omega_h)}$ ($p^\circ = \tilde{p} - \bar{p}_\Gamma(\tilde{p})$); the off-layer column is the error on $\Omega_h \setminus L_1$ with the optimal constant removed. Here $G = 0$, and GS behaves like CNS*.

N	h	GS		GS, off first layer		CNS*	
		error	rate	error	rate	error	rate
16	1.5052	$2.05 \cdot 10^{-1}$	—	$1.36 \cdot 10^{-1}$	—	$3.78 \cdot 10^{-1}$	—
24	1.0559	$1.05 \cdot 10^{-1}$	1.88	$7.03 \cdot 10^{-2}$	1.86	$1.95 \cdot 10^{-1}$	1.87
32	0.8127	$6.48 \cdot 10^{-2}$	1.85	$4.34 \cdot 10^{-2}$	1.84	$1.21 \cdot 10^{-1}$	1.81
48	0.5560	$3.29 \cdot 10^{-2}$	1.79	$2.20 \cdot 10^{-2}$	1.79	$6.23 \cdot 10^{-2}$	1.75
64	0.4224	$2.04 \cdot 10^{-2}$	1.73	$1.36 \cdot 10^{-2}$	1.74	$3.92 \cdot 10^{-2}$	1.69
96	0.2853	$1.06 \cdot 10^{-2}$	1.68	$6.99 \cdot 10^{-3}$	1.70	$2.05 \cdot 10^{-2}$	1.65

TABLE 20. GS (and CNS*) with $\mu = c(h_{16}/h)^\alpha$: rates between $N = 64$ and $N = 96$, and the predicted asymptotic rates (Corollary 7.36; for test P, where $\tilde{u} = 0$, only the leak is present and the rates are $1 + \alpha$ and $(1 + \alpha)/2$ for every α).

test	method	(α, c)	μ at $N = 96$	$\ \nabla e\ $ at $N = 96$	H^1 rate	energy rate	predicted
P	GS	(0, 100)	100	$1.28 \cdot 10^{-2}$	0.95	0.49	1, 0.5
P	GS	($\frac{1}{2}$, 100)	230	$5.64 \cdot 10^{-3}$	1.45	0.73	1.5, 0.75
P	GS	($\frac{1}{2}$, 1000)	2297	$5.72 \cdot 10^{-4}$	1.47	0.73	1.5, 0.75
P	GS	(1, 100)	528	$2.47 \cdot 10^{-3}$	1.94	0.97	2, 1
B ₁	GS	(0, 100)	100	$1.27 \cdot 10^{-2}$	0.96	0.56	1, 0.5
B ₁	GS	($\frac{1}{2}$, 100)	230	$6.42 \cdot 10^{-3}$	1.42	0.86	1.5, 0.75
B ₁	GS	($\frac{1}{2}$, 1000)	2297	$5.81 \cdot 10^{-3}$	1.52	1.28	1.5, 0.75
B ₁	GS	(1, 100)	528	$5.03 \cdot 10^{-3}$	1.47	1.06	1.5, 1
B ₁	CNS*	(0, 100)	100	$3.54 \cdot 10^{-3}$	1.64	1.53	1.5, 1.5
B ₁	CNS*	(1, 100)	528	$5.21 \cdot 10^{-3}$	1.45	1.16	1.5, 1
A	GS	(0, 100)	100	$9.57 \cdot 10^{-3}$	1.59	1.54	1.5, 1.5
A	GS	($\frac{1}{2}$, 100)	230	$1.21 \cdot 10^{-2}$	1.43	1.35	1.5, 1.25
A	GS	($\frac{1}{2}$, 1000)	2297	$1.86 \cdot 10^{-2}$	1.52	1.32	1.5, 1.25
A	GS	(1, 100)	528	$1.48 \cdot 10^{-2}$	1.34	1.16	1.5, 1

1.42–1.52, and the H^1 error at $N = 96$ halves ($1.27 \cdot 10^{-2} \rightarrow 6.4 \cdot 10^{-3}$, against $3.5 \cdot 10^{-3}$ for CNS* at $\mu = 100$); on B₁₀₀ it drops by a factor 22 at $c = 1000$. The energy rates follow Corollary 7.36: on B₁ with $\alpha = \frac{1}{2}$, $c = 1000$ the measured 1.28 is the geometric branch $(3 - \alpha)/2 = 1.25$ (the leak branch 0.75 takes over only later), and with $\alpha = 1$ it is 1.06.

The price when G is small. On test A ($G = 0$) a growing penalty only costs. Theorem 7.30 bounds the H^1 error uniformly, but at $N = 64$ the constant moves from its value at $\gamma \approx 30$ ($\|\nabla e\| = 1.82 \cdot 10^{-2}$) to the value of the Dirichlet limit $\mu \rightarrow \infty$ ($4.56 \cdot 10^{-2}$), 2.5 times larger. Along $\alpha = 1$ the ratio of the GS error to the Dirichlet-limit error climbs from 0.37 to 0.56 between $N = 16$ and 64; this transient is why the measured H^1 rate on A with $\alpha = 1$ is 1.34 rather than 1.5. The Dirichlet-limit error itself has rates 1.90, 1.80, 1.72, 1.66, decreasing towards $\frac{3}{2}$ as for CNS*. The energy error, by contrast, grows without bound: at $N = 64$, $\|e\|/(\gamma^{1/2}h_\Gamma^{3/2}) \rightarrow 0.339$ (A) and 0.105 (B₁) as $\mu \rightarrow \infty$, and these limits are exactly $\|\tilde{u}\|_{\Gamma_h}/h_\Gamma^2$: the discrete trace tends to 0 and the energy error becomes $(\mu/h)^{1/2}\|\tilde{u}\|_{\Gamma_h}$, the mechanism of Theorem 7.35. CNS* agrees with GS to four digits for $\mu \geq 10^6$.

TABLE 21. Test F, CNS*, uncorrected (g_I) against flux-corrected ($\tilde{g}_I$) outer data. Slip errors $\|\tilde{u} - u_h\|_{\Gamma_h}$ at $\mu = 10^6$; energy errors at $\mu = 10^8$, with the floor $(\mu/h)^{1/2}|m_R|/|\Gamma_h|^{1/2}$ of Proposition 7.31.

N	m_R	$\frac{ m_R }{ \Gamma_h ^{1/2}}$	$\ \tilde{u}\ _{\Gamma_h}$	slip, g_I	slip, $\tilde{g}_I$	$\ e\ $, g_I	$\ e\ $, $\tilde{g}_I$	floor
16	$2.04 \cdot 10^{-3}$	$8.19 \cdot 10^{-4}$	$2.40 \cdot 10^{-5}$	$8.204 \cdot 10^{-4}$	$2.396 \cdot 10^{-5}$	6.752	0.955	6.672
24	$1.54 \cdot 10^{-4}$	$6.14 \cdot 10^{-5}$	$5.21 \cdot 10^{-6}$	$6.172 \cdot 10^{-5}$	$5.210 \cdot 10^{-6}$	0.701	0.366	0.598
32	$2.56 \cdot 10^{-5}$	$1.02 \cdot 10^{-5}$	$1.70 \cdot 10^{-6}$	$1.038 \cdot 10^{-5}$	$1.697 \cdot 10^{-6}$	0.187	0.149	0.114
48	$2.14 \cdot 10^{-6}$	$8.55 \cdot 10^{-7}$	$3.42 \cdot 10^{-7}$	$9.213 \cdot 10^{-7}$	$3.422 \cdot 10^{-7}$	0.0372	0.0354	0.0115
64	$3.75 \cdot 10^{-7}$	$1.50 \cdot 10^{-7}$	$1.09 \cdot 10^{-7}$	$1.851 \cdot 10^{-7}$	$1.091 \cdot 10^{-7}$	0.0123	0.0121	0.0023

Conditioning (Proposition 7.37; $N = 32, 48, \mu = 10^2\text{--}10^5$). $\lambda_{\min}(A_\mu) \approx 0.03h^2$ does not depend on μ , and $\lambda_{\max}(A_\mu) \approx \max(102\text{--}118, 0.47\gamma)$, so for $\gamma \lesssim 200$ the P_4 stiffness part dominates and a moderate growth of μ is free. The smallest Schur eigenvalue is the constant-pressure mode and equals $(|\Gamma_h|/|\Omega_h|)h/\mu$ to within 2.5% at every N and μ ; the next one (0.007–0.010, the squared inf-sup constant on L_0^2) does not depend on μ . So $\kappa(K_\mu) \approx \lambda_{\max}(A_\mu)(|\Omega_h|/|\Gamma_h|)\mu/h$, from $4.9 \cdot 10^4$ ($N = 32, \mu = 10^2$) to $6.0 \cdot 10^9$ ($\mu = 10^5$): linear in μ while the stiffness part dominates $\lambda_{\max}$, quadratic after. For the H^1 -optimal $\alpha = \frac{1}{2}$ with $c = 100$ the whole study ($\mu \leq 230$) stays in the linear regime.

Flux correction. The test data above have $m_R = 0$ up to round-off ($|m_R| \leq 2 \cdot 10^{-13}$), either because $g \cdot \nu$ is a polynomial of low degree (B, P) or by symmetry (A). Test F uses $\psi = (r^2 - 1)^3 \sin(2x + 1) \cos(\frac{3}{2}y + \frac{7}{10})/400$ and $p = x^2y - y + x/2$: the cube makes $W \equiv 0$, so the geometric slip $\|\tilde{u}\|_{\Gamma_h} = O(h_\Gamma^4)$ is small, and the phase shift breaks the symmetries, so $m_R \neq 0$; it decays like $h^{6.6}$ ($\sigma_4 = 6$). Table 21 shows the floor of Proposition 7.31 directly. Without correction the CNS* slip error at large μ is $(|m_R|^2/|\Gamma_h| + \|\tilde{u}\|_{\Gamma_h}^2)^{1/2}$ to three digits: the uniform normal flux defect and the tangential geometric slip, orthogonal to each other. With the correction the slip error is $\|\tilde{u}\|_{\Gamma_h}$ to five digits, the flux through Γ_h is at most $2.5 \cdot 10^{-14}$, and the energy error loses the term $(\mu/h)^{1/2}|m_R|/|\Gamma_h|^{1/2}$ (a factor 7 at $N = 16, \mu = 10^8$). The H^1 errors with and without correction agree to a relative $3 \cdot 10^{-6}$ ($N = 16$) down to 10^{-9} ($N = 64$), as Theorem 7.30 predicts: the flux enters the H^1 error only as $|m_R|h_\Gamma^{-1/2}$, far below the interpolation error of this steep solution. For the companion test F2 ($(r^2 - 1)^2$ in place of $(r^2 - 1)^3$, so $W \not\equiv 0$) the geometric slip exceeds the floor by a factor 23 ($N = 16$) to $2 \cdot 10^4$ ($N = 64$), and the correction changes nothing visible.

16.11. Higher degree and Clough–Tocher refinements. We repeated the study with $k = 5, 6$ on the same meshes, and with $k = 2, 3$ on their Clough–Tocher refinements (Section 12), using a degree-general version of the solver (`code/svn.k.py`). The refinement leaves Γ_h , the square and $h = h_\Omega$ unchanged, so the h values are those of the standard meshes. At $k = 4$ with SuperLU the new code is bit-identical to `svn.py`, and agrees with the stored production records to $2.6 \cdot 10^{-10}$ relative. SuperLU runs out of memory at the target sizes, so these runs use MKL PARDISO. The saddle-point matrix is ill-conditioned, and for robustness of PARDISO we regularise the pressure block by 10^{-12} times the pressure mass matrix. The result agrees with SuperLU to $\leq 10^{-9}$ relative ($k = 4$ and $k = 6, N \leq 32$). Over the 260 solves (390 records, including 130 derived records for D) the relative residual of the unregularised system is at most $2.2 \cdot 10^{-12}$ and the discrete divergence at most 10^{-9} . One side effect: CNS* on test P returns $\|\nabla u_h\| \approx 4 \cdot 10^{-11}$, the regularisation floor, instead of round-off.

Thresholds. Table 22 gives γ^* . It grows quickly with k and is larger on the Clough–Tocher refinements, whose boundary triangles are three times thinner while h is the macro size. In every case the measured value exceeds the corresponding certified local constant (Proposition 10.1 for

TABLE 22. Coercivity threshold $\gamma^* = \mu^* h_\Gamma / h$ for higher degrees and on Clough-Tocher (CT) refinements. The $k = 4$ row is Table 14(a).

	$N = 8$	16	32	64	128
$k = 4$	14.7	18.1	19.9	20.8	–
$k = 5$	25.7	30.3	32.1	33.0	–
$k = 6$	–	45.2	47.4	48.3	–
CT, $k = 2$	5.7	9.1	10.5	11.2	11.5
CT, $k = 3$	–	31.6	33.9	34.6	–
CT, $k = 4$	–	61.9	65.5	–	–

TABLE 23. Normalised GS error on test P at the finest level run, $e = \tilde{u} - u_h$.

Element	μ	N	$\ \nabla e\ \mu / (hG)$	$\ e\ _{\Gamma_h} \mu / (hG)$	$\ e\ (\mu/h)^{1/2} / G$
$k = 4$	1000	96	2.770	0.998	0.999
$k = 5$	1000	128	2.768	0.9983	0.9994
$k = 6$	1000	96	2.769	0.9976	0.9993
CT, $k = 2$	100	256	2.737	0.9928	0.9974
CT, $k = 2$	1000	128	2.751	0.9982	0.9992
CT, $k = 3$	1000	128	2.763	0.9983	0.9996

$k \geq 4$, Proposition 12.10 on the refinements), which is consistent with those constants being local lower bounds. γ^* still drifts upward with N (+7% to +24% from $N = 16$ to 64). In particular $\mu = 100$ is *below* the threshold for $k = 5$ ($N \geq 32$), for $k = 6$ and for Clough-Tocher $k = 3$ (all N). The $\mu = 100$ runs in those cases are erratic, as they should be (for instance the CNS* H^1 error for $k = 5$ on B_1 jumps between 0.05 and 3.7), and they are not covered by the theory. We therefore report $\mu = 300$ and $\mu = 1000$ for them.

The leak (test P). Table 23 shows that the slip and energy constants of Corollary 6.7 tend to 1 for every element, with rates 1 (slip), 1/2 (energy) and 1 (H^1). The H^1 constants are close to the $K = 2.7565$ of Section 6.1, which does not depend on the element; this agreement is observed, not proved, for these elements. On test B_1 the leak part $D := u_{\text{GS}} - u_{\text{CNS}^*}$ has the same normalised slip and energy ratios (0.986–1.003 and 0.991–1.006 at the finest levels), H^1 rate 1.01–1.15 and energy rate 0.49–0.52. Its normalised H^1 constant is still drifting down at the finest levels (3.03 for $k = 5$, $\mu = 1000$).

Rates. Table 24 gives H^1 rates at the finest pair of levels.

- CNS* *tends to rate 3/2* for $k = 5, 6$ on B_1 (1.58–1.61, energy rate 1.53–1.54), non-increasing from $N = 32$ on. The CNS* errors for $k = 4, 5, 6$ are almost equal (for example $1.05 \cdot 10^{-2}$, $9.02 \cdot 10^{-3}$, $8.47 \cdot 10^{-3}$ at $N = 64$, $\mu = 1000$): the geometric term dominates. On Clough-Tocher $k = 3$, test A, the H^1 rate is 1.56 at $N = 128$.
- GS *tends to rate 1 when $G > 0$* . For $k = 5$, $\mu = 300$, the GS H^1 rate on B_1 decreases monotonically, 1.63, 1.58, 1.48, 1.38, 1.28, 1.20 for $N = 24, \dots, 128$, while its leak part D already has rate 1.03. At $\mu = 1000$ the crossover comes later (1.53 at $N = 128$), exactly as for $k = 4$.
- CNS* *is pressure-independent* for $k = 5, 6$: on test P it returns $u_h = 0$ to the regularisation floor at every N and every $\mu \geq \mu^*$. By linearity this is the statement that $u_h(B_a) = u_h(B_1)$ for every a .

Caveats for Clough-Tocher $k = 2, 3$.

TABLE 24. H^1 rates at the finest pair of levels (energy-norm rate of CNS* in parentheses). $D = u_{\text{GS}} - u_{\text{CNS}^*}$ is the leak part; it is not reported for test A, where $G = 0$.

Element	Test	μ	N	GS	CNS*	D
$k = 5$	B ₁	300	96 $\rightarrow$ 128	1.20	1.59 (1.53)	1.03
$k = 5$	B ₁	1000	96 $\rightarrow$ 128	1.53	1.58 (1.54)	1.13
$k = 6$	B ₁	300	64 $\rightarrow$ 96	1.27	1.61 (1.54)	1.04
$k = 6$	B ₁	1000	64 $\rightarrow$ 96	1.57	1.61 (1.54)	1.15
CT, $k = 3$	A	1000	96 $\rightarrow$ 128	1.56	1.56 (1.55)	–
CT, $k = 2$	A	100	192 $\rightarrow$ 256	1.78	1.78 (1.55)	–
CT, $k = 2$	B ₁	100	192 $\rightarrow$ 256	1.91	1.91 (1.91)	1.01
CT, $k = 3$	B ₁	1000	96 $\rightarrow$ 128	2.83	2.83 (1.64)	1.05

- (1) CNS* on test P is not exactly zero. The pressure $p = x^2y - y + x/2$ is cubic, so $p \notin \Pi_h$ for $k \leq 3$, and the term $\eta = p^* - \pi_h p^*$ of Theorem 7.2 is present. The spurious velocity is small and converges at a rate close to $k + \frac{1}{2}$: $\|\nabla u_h\| = 1.4 \cdot 10^{-6}$ at $N = 256$ for $k = 2$ (rate 2.56), and $4.7 \cdot 10^{-9}$ at $N = 128$ for $k = 3$ (rate 3.53). So for $k \leq 3$ the pressure-independence of CNS* holds only up to a term of order $h^{k+1/2}$ that scales with the amplitude of p . It is below $h_\Gamma^{3/2}$ asymptotically, but for B₁₀₀ it would be 100 times larger.
- (2) On test B₁ the rate 3/2 is not yet visible. The H^1 error is dominated by interpolation of the steep velocity (u grows like r^5 , since ψ grows like r^6 , on the square of side 5): for $k = 2$ it is 0.41 at $N = 256$, against $7.6 \cdot 10^{-4}$ for $k = 4$, with rate 1.91 still rising towards 2. For $k = 3$ the rate is 2.83, close to 3. The runs are pre-asymptotic and the crossover to $h_\Gamma^{3/2}$ is out of reach. The leak part D is on theory (rate 1.01, ratios 1.0009 and 1.0056 at $N = 256$ for $k = 2$).
- (3) On test A (bounded velocity) the energy-norm rate is already 1.55 for $k = 2$, while the H^1 rate (1.78 at $N = 256$) is still decreasing.

Memory limits (a shared 5.8 GB budget) prevented $k = 6$ at $N = 128$ and Clough–Tocher $k = 3$ at $N = 192$.

16.12. Further computations. Four sets of computations are reported next to the results they test: the L^2 velocity errors (Table 6); the strong-imposition drift fit, locked vertices with the pressure filter, and nearly singular stars (Tables 7, 8); the hydrostatic pressure-robustness test (Table 10); and the CNS* pressure lower bound and torque (Tables 4, 3). All use $k = 4$ and the meshes of this section unless stated, with the $\mu = 100$ caveat above.

17. WHAT IS PROVED AND WHAT IS OPEN

We list the status of the main claims. “Proved” means proved in this paper from the stated mesh hypotheses, the smooth-solution hypothesis (D) (see Remark 2.5 for what it means at the corners of R) and standard cited results. “Proved modulo X” means proved except for the named hypothesis X. “Numerical” means observed only.
Proved.

- The leak analysis of the printed method: Theorems A–C, Corollary B', Theorem B'' (fixed γ , bounded ρ), its penalty-uniform version at the exact rate $\frac{1}{2}$ (Theorem U), Theorem R, and the convergence of the leak constant for every penalty, including $\gamma h_\Gamma \rightarrow \infty$ (Corollary 7.48; for general data the normalised GS constant diverges when $\gamma h_\Gamma^{1/2} \rightarrow \infty$).

- Scott's rate $h_\Gamma^{3/2} + h^k$ for CNS* (Theorem D) and its sharpness for every $k \geq 4$ (Theorem D', with (M3) proved in Theorem 7.8 for $h_\Gamma \leq 2 \sin(\theta_{\min}/4)$).
- Stokes $H^2 \times H^1$ regularity on Ω (Theorem 9.1) and local higher regularity at Γ (Proposition 9.2); on general domains, (R) holds if and only if Σ is a convex outer polygon (Theorem 13.6).
- The L^2 theory: the upper bounds of Theorems 9.6, 9.9, the graded-mesh bound $C(h_\Gamma^2 + h^{k+1})$ with the local divergence correction (LF) and the pressure-free local energy estimate (IE) (Lemma 9.10, Theorems 9.12, 9.14), and the lower bound ch_Γ^2 (Theorem 9.8).
- The pressure bounds and the drag, lift and torque expansions of Section 8 (the CNS* drag leading term as $\gamma \rightarrow \infty$); the first-layer cell problem: well-posedness, the closed-form traction cell with its resonance, the flux identity and the mirror antisymmetry (Proposition 8.37).
- The penalty threshold: sufficiency, the single-triangle certificates, Theorem 10.9 with explicit constants, necessity $\mu^* \geq c_*(\theta_{\min}, c_0)\rho$ on every (M0) mesh with explicit but astronomically small constants (Theorem 10.13), the exact column-layer witness $\mu^* \geq 0.85 h/(Hh_\Gamma)$ at every mesh size (Proposition 10.14), the exact counterexamples to a local necessity statement (Proposition 10.6), and $\liminf_N \gamma^*(N) \geq \Gamma_\infty$ (Proposition 10.15).
- Strong imposition: the two-sided rate under (M2), the counting lower bound, uniform constrained inf-sup, the sharp stability at locked vertices, the two-sided rate with locks and the pressure filter; the needle-angle law as a two-sided estimate under (H_K) (Theorem 11.22), with the slit-domain right inverse (Theorem 11.20); the bound $\beta_h \leq C(\varepsilon + \vartheta)$ for needle pairs, so (L_N) is false for straight pairs (Proposition 11.19). The scope of (H_K) is narrow: each needle in its own wall pair.
- Pressure robustness: the splitting, non-robustness of GS, the single CNS* defect with its upper bounds, exact gradient robustness of CNS*-R, the energy duality, the certificate for every $\gamma > 0$, Theorem 15.12(a)–(d) for every $k \geq 4$, the effective-penalty identity (Lemma 15.13), the equal-height and LP^{++} bounds (Theorems 15.14, 15.15; the latter for $\gamma \geq \gamma_{LP}$, not explicit), and the orders $k + 1$ and $k + \frac{3}{2}$ (Theorem 15.17; the upper bound for $k + \frac{3}{2}$ needs $\gamma \geq \gamma_2$).
- Navier–Stokes on nonsingular branches: stability (Theorems 14.16, 14.23), the explicit thresholds, the Oseen leak flow, K_{NS} , the GS and CNS* drag and the penalty-uniform leak limit (under (H_{ns})); Clough–Tocher refinements and general domains (Theorems F, G).

Proved modulo a named hypothesis.

- The CNS* pressure lower bounds under the boundary moment hypothesis (P) (Theorem 8.34).
- Modulo the cell decay (D_c) (numerical) and the transplant lemma (TL) (not proved), on LPC mesh families: the two-scale expansion of the boundary moment, the L^2 pressure error and the traction error (Theorem 8.38); the corrected (P), which holds if and only if the cell density $\mathcal{F}$ is not constant, and *fails* on mirror-symmetric columns (Corollary 8.39); the pressure and traction lower bounds without (P), under the numerically verified non-degeneracy $Q_x \not\equiv 0$, resp. Θ_x not a.e. a constant multiple of n (Corollary 8.40); and the fixed- γ drag coefficient (Corollary 8.41; its analyticity in γ needs (D_c) on a complex neighbourhood). All constants are tied to the paper's Nitsche flux $(\nabla u)n$.
- The limit γ_∞^* of the coercivity threshold: $\limsup_N \gamma^*(N) \leq \Gamma_\infty^{\text{cell}}$ modulo (LOC), and $\Gamma_\infty = \Gamma_\infty^{\text{cell}}$ modulo (CUT) (Proposition 10.15).
- Penalty necessity on Clough–Tocher refinements of every (M0) mesh, modulo the standard HCT interpolation estimate (Theorem 10.13).

Numerical only.

- The existence of $\lim d_h/h_\Gamma^{1/2}$ on a given mesh family (a periodic cell formula predicts 0.2256 at $\mu = 100$ on the production meshes, against the fitted 0.225).

- (D_c); $A_1 \neq 0$ on asymmetric columns; the non-degeneracy of Q_x and Θ_x ; hypothesis (P) on the production meshes; the sign and size of the CNS* drag lower bound at fixed γ ; the fixed- γ torque lower bound.
- The value $\gamma_\infty^* = 21.408$ (21.4045 for the first two layers, matching the extrapolated global two-layer thresholds 21.40452).
- The penalty threshold on the global polar column meshes containing the star counterexamples: $\gamma^* = 3.06\text{--}3.13$ ($H = 5$), independent of the box, and $\gamma^*H \approx 15\text{--}16$ over $H = 1, \dots, 12$; the transfer of the column witness to these meshes is heuristic, and the star failure is certified only for $H = 5$ and 8.
- The production penalty $\mu = 100$: above the CNS* stability threshold on every production mesh, with a margin shrinking from 27% ($N = 16$) to 13% ($N = 64$) and to about 7% in the limit ($\mu_c^{\text{CNS}^*} \rightarrow \approx 93$).
- Strong imposition: (L_N) for shape I needles, and $\beta \geq c\varepsilon$ for straight needle pairs.
- The failure of (S_{LP}) off LP⁺ (amplitude $\propto \gamma^{-2}h_I^{1/2}$).
- The Navier–Stokes constants for the test case: $U_\infty = 175.6$, $\beta_O = 0.8225$ ($\nu = 1$), 0.0533 ($\nu = 0.1$), $C_R \approx 3.5$ (the box-corner contribution not computed); the discrete stability constant (L), checked on $N = 16, 32, 48$ only. The resulting explicit h_0 is hopelessly small.

Open.

- A proof of (TL), and of (D_c) and $A_1 \neq 0$ for all asymmetric columns; whether the deeper production rows are LPC; the second-order (global) coefficient of P_h on symmetric columns.
- Sharpness of the CNS* drag rate 2 at a given fixed γ ; the constant $C_\sharp$ of the Navier–Stokes threshold.
- An explicit, moderate constant in the penalty necessity for general coarse meshes (numerically $\mu^* \approx (15\text{--}16)h/\delta_1$ with δ_1 the first-layer height); proofs of (LOC) and (CUT).
- (S_{LP}) for $\gamma_2 \leq \gamma < \gamma_{\text{LP}}$ when the first-layer heights vary strongly; explicit κ_T ($k \geq 5$) and κ_K ; order $k + \frac{3}{2}$ for general constant- A data.
- Strong imposition on meshes whose needles do not come in isolated wall pairs.
- Nonsymmetric and penalty-free Nitsche variants; slip conditions; other divergence-free families; cut meshes.
- Three-dimensional versions of the results of this paper are treated separately [50].

APPENDIX A. REPRODUCIBILITY

All code, raw outputs and logs are in the repository that accompanies this paper. Paths below are relative to its root; commands are run from `code/`. `code/regen_logs.sh` regenerates every log in `logs/` with the commands below, and `make report` checks that all five reports (`report.txt`, `report_local.txt`, `report_v3.txt`, `letter_tables.txt` and `results/v4/rescue/summary.md`) are reproduced byte-for-byte from the shipped raw records.

Code (code/).

- `svn.py`: the solver. It contains the mesh generator, the P_4 space, assembly of N_h , B and the CNS* coupling, the SuperLU saddle-point solver (`solve.lu`), error norms, and the exact solutions for tests A, B, P and C.
- `job.py`: one study or threshold job per process.
- `run_all.py`: memory- and CPU-aware scheduler for the full campaign.
- `report.py` and `report_local.py`: produce the tables.
- `plots.py`: produces the figures.
- `penalty.py`: computes the coercivity threshold μ^* .
- `lemma_tests.py`: computes the dual norms of Table 15.
- `lamstar_exact.py` and `lamstar_exact_v2.py`: the exact certificates of Proposition 10.1.

- `lamstar.py`: the floating-point λ^* on other fans.

Added in this version:

- `svn.k.py`: degree-general solver (P_k velocities, $k \geq 2$; standard or Clough–Tocher mesh; SuperLU or PARDISO). `job.k.py` runs one study or threshold job, `penalty.k.py` computes μ^* , `report.v3.py` writes the tables, and `validate.k.py` checks $k = 4$ against `svn.py` and PARDISO against SuperLU. The scheduler is `results/v3/sched.py`, with job list `results/v3/jobs.txt`.
- `lamstar_exact.ct.py`: the exact local-surjectivity ranks and penalty certificates on Clough–Tocher stars (Section 12).
- `lamstar_family.py`: the single-triangle forms, the shape-family certificates and the patch thresholds (Section 10.1).
- `h1_limit.py`: the annulus bracket, the series solution and the finite element comparison for the H^1 constant (Section 6.1).
- `lower_bound_tests.py`: the (M3) star constants, the assembled test field and the dual norms (Section 7.1).
- `plot_leak.py`: Figure 2.

Added in this version (v4):

- `rescue.py`: the growing-penalty study, the μ -sweeps, the flux-corrected data $\tilde{g}_I$ (tests F and F2) and the spectra of Proposition 7.37 (Section 16.10); it reuses `svn.k.py` unchanged. `rescue_run.sh` is its batch driver; it writes the four logs `logs/rescue_*.log` and the records in `results/v4/rescue/`. From these records `rescue_report.py` writes the summary `summary.md` in the same directory.
- `pressure_err.py`: pressure errors of GS and CNS* against p° , with the first-layer split (Section 16.9); writes `results/v5/pressure/`.
- `strong_bc.py`: strongly imposed no-slip on Γ_h , and GS/CNS* on the alternating mesh (a) (Section 16.8); writes `results/v5/strong/`.
- `report.v5.py`: writes `results/v5/letter.tables.txt` from the JSONL records (and, with argument `thresh`, recomputes h/h_Γ and ρ for the threshold runs into `results/v5/thresh_rho.jsonl`). The job lists are `results/v5/jobs*.txt`; see `results/v5/README.md`.
- `notes/v4/pressure_drag_check.py`: the layer, drag and traction checks of Section 16.9; its output is recorded in `notes/v4/pressure_drag_report.md`, Section 3.

Where each number comes from.

Result	Command	Output
Tables 11, 12, 14	SVN_OUT=../results/pod python3 run_all.py, then python3 report.py ../results/pod	results/pod/report.txt
Table 13, test C	SVN_OUT=../results/local python3 job.py study N P1 MU and SVN_OUT=../results/local python3 job.py study N C 100, then python3 report_local.py	results/local/report_local.txt
Certificate for S_3	python3 lamstar_exact.py	logs/lamstar_exact_ S3.S4singular.log
Certificate for S'_4	python3 lamstar_exact.v2.py	logs/lamstar_exact.S4prime.log
Table 15	python3 lemma_tests.py 16,24,32,48,64	logs/lemma_tests.log

Result	Command	Output
Floating-point λ^*	<code>python3 -W ignore::DeprecationWarning lamstar.py</code>	<code>logs/lamstar_float.log</code>
Figures 1, 3	<code>python3 plots.py</code>	<code>paper/figures/h1_rates.pdf</code> etc.
Figure 2	<code>python3 plot_leak.py</code>	<code>paper/figures/leak_field.pdf</code>
Tables 22–24	<code>SVN_SOLVER=pardiso python3 ../results/v3/sched.py ../results/v3/jobs.txt LOG (runs job_k.py; output root results/v3 by default, or SVN_OUT), then python3 report_v3.py</code>	<code>results/v3/report_v3.txt</code>
Validation of <code>svn_k.py</code>	<code>python3 validate_k.py</code>	<code>results/v3/validation.txt</code>
Lemma 12.3 ($k = 3$), Table 9	<code>python3 lamstar_exact_ct.py</code>	<code>logs/lamstar_exact_ct.log</code>
Theorem 10.2, Proposition 10.3	<code>python3 lamstar_family.py bubble; ... certify</code>	<code>logs/lamstar_family_bubble.log, ..._certify.log</code>
Patch thresholds (Section 10.1)	<code>python3 lamstar_family.py mesh; ... limit; ... limitstar</code>	<code>logs/lamstar_family_mesh.log, ..._limit.log, ..._limitstar.log</code>
Level-set and tall-triangle values of λ_T (Section 10.1)	<code>python3 lamstar_family.py levels</code>	<code>logs/lamstar_family_levels.log</code>
H^1 constant (Section 6.1) (M3) and Tables 1, 2	<code>python3 h1_limit.py annulus and ... mps (series log); ... fem python3 lower_bound_tests.py with stars 8,12,...,1024, field 16,...,512, global 16,...,64</code>	<code>logs/h1_limit_series.log, ..._fem.log logs/lower_bound_stars.log, ..._field.log, ..._global.log</code>
Tables 20, 21; Section 16.10	<code>bash rescue_run.sh all, then python3 rescue_report.py</code>	<code>logs/rescue_*.log, results/v4/rescue/, .../summary.md</code>
Tables 16–19; Section 16.9	<code>the commands in results/v5/jobs*.txt (pressure_err.py, strong_bc.py), then python3 report_v5.py</code>	<code>results/v5/letter_tables.txt</code>
Layer, drag and traction checks (Section 16.9)	<code>python3 notes/v4/pressure_drag_check.py main xi K lam chi torque ... (from the root)</code>	<code>notes/v4/pressure_drag_report.md</code>
Section 7.1.3 ((M3) for $k \geq 4$)	<code>the certificate and witness scripts in notes/v4/ and notes/v6/m3/</code>	<code>notes/v6/m3/*.log</code>
Section 7.3	<code>notes/v6/unifmu/*.py, notes/v6/unifmu2/*.py</code>	<code>logs in the same directories</code>
Section 9, Table 6	<code>notes/v6/l2/*.py</code>	<code>notes/v6/l2/existing_l2.txt</code>
Section 8.6	<code>notes/v6/misc/*.py</code>	<code>logs in the same directory</code>
Section 10.1 (Theorem 10.9, Proposition 10.6)	<code>notes/v6/penalty/*.py</code>	<code>logs in the same directory</code>
Section 11, Tables 7, 8	<code>notes/v6/strong/*.py, notes/v6/strong2/*.py, notes/v6/nearsing/*.py</code>	<code>fit_rates.log, lock_bound.log, tab_filter.tex, tab_fe.tex</code>
Section 15, Table 10	<code>notes/v6/robust/hydro.py; notes/v6/robust2-robust6/*.py</code>	<code>hydro_all.jsonl, cert_fix_*.log</code>

Result	Command	Output
Remark 2.5 (corner exponents) Section 8.6.5, Table 5	python3 notes/v7/referee_reg/roots.py notes/v7/P/cell.py scan, signscan.py, rhoscan.py, mixed.py, annulus.py, annpred.py, compare_ann.py, predict.py, compare.py	notes/v7/referee_reg/roots.out logs in notes/v7/P/
Theorem 10.13 (numerical companion), Proposition 10.14, the global column mesh	notes/v7/penalty/ layer_witness.py, exact_columns.py, global_mesh.py H N [all star g]; notes/v7/ referee_penalty/ fe_check_columns.py, global_r_check.py	layer_witness.log, exact_columns*.log, global_mesh*.log; fe_check_columns.log, global_r_check.log
Proposition 10.15, γ_∞^* ; Navier–Stokes constants (Section 14.11)	notes/v7/nsconst/cell_limit.py; consts.py, discrete_L.py, halfplane.py, leak_cell.py	run_table.log, run_theta.log, run_mesh.log; run_beta.log, run_CR*.log, run_discreteL.log, run_predict*.log
Section 11.6	notes/v7/needle/slit_lab.py, probe1-probe9.py	notes/v7/needle/probe*.log
Theorems 15.12, 15.14, 15.17; Section 15 (all k , Floquet cell)	notes/v7/robust/allk_exact.py 12, vf_allk.py, cns_cell.py, smooth_S.py, pairing_exact.py k ; notes/v7/referee_robust/ ref_allk.py, ref_pairing.py	logs and *.jsonl in the same directories
Remark 15.18, Section 16 (CNS* margin at $\mu = 100$)	notes/v7/mu100/cns_crit.py, cns_crit_r.py, mu100_check.py, crit_modes.py	cns_crit*.log, thr.log, ldl.log, crit_modes.log; summary notes/v7/mu100/REPORT.txt

Resources. The full A/B campaign (74 study jobs up to $N = 256$ and 16 threshold jobs) took 36 minutes on 32 cores with 125 GB of memory. The largest single job ($N = 256$, both methods) takes about 23 minutes; the scheduler reserved up to 55 GB for it. Every job with $N \leq 96$ runs on a 2-core, 7 GB machine.

The runs of Section 16.11 used a 2-core machine with a 5.8 GB memory limit shared with other jobs; the largest single job used 3.6 GB (peak RSS is recorded in each record). Each job took at most about 2.5 minutes.

The v4 and v5 runs (Sections 16.8–16.10) were sequential single-thread jobs on the same 2-core machine; the largest used 2.2 GB (SuperLU, $N = 64$), and the $N \geq 64$ strong-imposition, mesh-(a) and $N = 96$ runs used PARDISO as in Section 16.11. Over the 83 v5 records the relative residual is at most $1.3 \cdot 10^{-13}$ and the discrete divergence at most $9.6 \cdot 10^{-7}$ (the nearly singular strong system on mesh (a) at $N = 128$; SuperLU at $N = 64$ agrees to five digits with divergence $1.5 \cdot 10^{-10}$).

The environment was Python 3.11 with NumPy 2.4, SciPy 1.17 (SuperLU) and SymPy 1.14. The runs of Section 16.11 also used MKL PARDISO through pypardiso.

USE OF GENERATIVE-AI TOOLS DECLARATION

The author used Anthropic’s Claude Opus 5.5 as a research-support tool. It assisted with the following:

- developing and checking the proofs;

- implementing, running and validating the finite element code and the exact-arithmetic certificates;
- independently reviewing the arguments as a referee;
- preparing the manuscript and the accompanying repository.

The author directed the project and retains authorship of the work. Responsibility for all mathematical statements, computations, references and conclusions rests with the author.

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